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(FF-Specific)			7	50	SYS PWR: NTCs (Bottom)	D52_AP_MASTER_3.47.0		8	63	CAMERA: Actuator Supply (Lex)	D52_AP_MASTER_3.47.0		9	65	CAMERA: Strobe Driver	D52_AP_MASTER_3.47.0		10	73	SENSORS: Accel / Gyro	D52_AP_MASTER_3.47.0		11	75	SENSORS: Jarvis (Bottom)	D52_AP_MASTER_3.47.0		12	122	B2B: LAT			13	125	MECHANICAL: Standoffs, Shields, etc.			14	128	TESTING: Probe Points			15	130	INTERPOSER: Symbol (1/2)			16	131	INTERPOSER: Symbol (2/2)			17	132	INTERPOSER: Aliases (1/4)	D53_MLB_TOP_3.39.1		18	133	INTERPOSER: Aliases (2/4)	D53_MLB_TOP_3.39.1		19	134	INTERPOSER: Aliases (3/4)	D53_MLB_TOP_3.39.1		20	135	INTERPOSER: Aliases (4/4)	D53_MLB_TOP_3.39.1		21	136	INTERPOSER: Misc. Aliases			22	150	HIER: WiFi			23	160	HIER: Radio			24	1	FRONT PAGE			25	2	BOMS AND CONSTRAINTS		07/30/2019	26	4	PMIC: BUCKs		07/30/2019	27	5	PMIC: LD0s		07/30/2019	28	6	PMIC: CLOCKS & CONTROL		07/30/2019	29	7	BB: POWER		08/22/2019	30	8	BB: GND		07/30/2019	31	9	BB: GPIOs & QLINK		07/30/2019	32	11	XCVR POWER_1		07/30/2019	33	12	XCVR POWER_2		07/30/2019	34	13	XCVR DIGITAL & GND		07/30/2019	35	14	XCVR RF		07/30/2019	36	16	QET_DISCRETE_0		07/30/2019	37	17	QET_DISCRETE_1		07/30/2019	38	18	DCDC2		07/30/2019	39	19	DCDC1		07/30/2019	<table><tr><th>PAGE</th><th>CSA</th><th>CONTENTS</th><th>SYNC</th><th>DATE</th></tr><tr><td>40</td><td>21</td><td>HB PAD</td><td></td><td>07/30/2019</td></tr><tr><td>41</td><td>25</td><td>LB DSM</td><td></td><td>07/30/2019</td></tr><tr><td>42</td><td>26</td><td>HB TXDSM</td><td></td><td>07/30/2019</td></tr><tr><td>43</td><td>27</td><td>DSM UHB</td><td></td><td>07/30/2019</td></tr><tr><td>44</td><td>28</td><td>MIMO DSM LOWER</td><td></td><td>07/30/2019</td></tr><tr><td>45</td><td>29</td><td>MIMO DSM UPPER</td><td></td><td>07/30/2019</td></tr><tr><td>46</td><td>30</td><td>COUPLER LOWER</td><td></td><td>07/30/2019</td></tr><tr><td>47</td><td>31</td><td>COUPLER UPPER</td><td></td><td>07/30/2019</td></tr><tr><td>48</td><td>32</td><td>GNSS_L1</td><td></td><td>07/30/2019</td></tr><tr><td>49</td><td>33</td><td>GNSS_L5</td><td></td><td>07/30/2019</td></tr><tr><td>50</td><td>35</td><td>LOWER ANTENNA FEEDS_ANT3</td><td></td><td>07/30/2019</td></tr><tr><td>51</td><td>37</td><td>LOWER ANTENNA FEEDS_ANT8</td><td></td><td>07/30/2019</td></tr><tr><td>52</td><td>38</td><td>LOWER ANTENNA FEEDS_ANT9</td><td></td><td>07/30/2019</td></tr><tr><td>53</td><td>41</td><td>PSIM</td><td></td><td>07/30/2019</td></tr><tr><td>54</td><td>43</td><td>UAT SAWT00TH</td><td></td><td>07/30/2019</td></tr><tr><td>55</td><td>44</td><td>LAT SAWT00TH</td><td></td><td>07/30/2019</td></tr><tr><td>56</td><td>46</td><td>DEBUG & TEST POINTS</td><td></td><td>07/30/2019</td></tr><tr><td>57</td><td>53</td><td>SYNONYMS</td><td></td><td>07/30/2019</td></tr><tr><td>58</td><td>57</td><td>MLB_ADJUSTABLES</td><td></td><td>07/30/2019</td></tr><tr><td>59</td><td>58</td><td>MMW</td><td></td><td>07/30/2019</td></tr><tr><td>60</td><td>59</td><td>TANGIER SIP</td><td></td><td>07/30/2019</td></tr><tr><td>61</td><td>60</td><td>BG1 ARRAY</td><td></td><td>07/30/2019</td></tr><tr><td>62</td><td>1</td><td>Table of Contents</td><td></td><td></td></tr><tr><td>63</td><td>2</td><td>THELONIOUS</td><td>D52_WIFI_MASTER_0.21.0</td><td></td></tr><tr><td>64</td><td>5</td><td>5G rFEM (LAT)</td><td>D52_WIFI_MASTER_0.21.0</td><td></td></tr></table>				PAGE	CSA	CONTENTS	SYNC	DATE	40	21	HB PAD		07/30/2019	41	25	LB DSM		07/30/2019	42	26	HB TXDSM		07/30/2019	43	27	DSM UHB		07/30/2019	44	28	MIMO DSM LOWER		07/30/2019	45	29	MIMO DSM UPPER		07/30/2019	46	30	COUPLER LOWER		07/30/2019	47	31	COUPLER UPPER		07/30/2019	48	32	GNSS_L1		07/30/2019	49	33	GNSS_L5		07/30/2019	50	35	LOWER ANTENNA FEEDS_ANT3		07/30/2019	51	37	LOWER ANTENNA FEEDS_ANT8		07/30/2019	52	38	LOWER ANTENNA FEEDS_ANT9		07/30/2019	53	41	PSIM		07/30/2019	54	43	UAT SAWT00TH		07/30/2019	55	44	LAT SAWT00TH		07/30/2019	56	46	DEBUG & TEST POINTS		07/30/2019	57	53	SYNONYMS		07/30/2019	58	57	MLB_ADJUSTABLES		07/30/2019	59	58	MMW		07/30/2019	60	59	TANGIER SIP		07/30/2019	61	60	BG1 ARRAY		07/30/2019	62	1	Table of Contents			63	2	THELONIOUS	D52_WIFI_MASTER_0.21.0		64	5	5G rFEM (LAT)	D52_WIFI_MASTER_0.21.0	
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Multi-Vendor Criticals

Capacitors

In descending order of value, then package size

CRITICAL PART#	COMMENT
138S0979	CAP, CER, XSR, 18UF, 20%, 18V, 0402, H=0.65MM
138S0652	CAP, CER, XSR, 4.7UF, 20%, 6.3V, H=0.65MM, 0402
138S00070	CAP, XSR, 4.7UF, 20%, 25V, 0402
132S0663	CAP, CER, XSR, 1UF, 10%, 25V, 0402
138S0683	CAP, CER, XSR, 1UF, 10%, 25V, 0402
138S0692	CAP, CER, XSR, 1UF, 20%, 6.3V, 0201
138S00014	CAP, CER, 1UF, 20%, 16V, XSR, 0201, H=0.39MM
132S0304	CAP, CER, XSR, 0.22UF, 20%, 6.3V, 0201
132S00000	CAP, CER, 0.1UF, 10%, 50V, XTR, 0402
132S0288	CAP, CER, XSR, 0.1UF, 10%, 16V, 0201
132S0288	CAP, CER, XSR, 0.1UF, 10%, 16V, 0201
132S0534	CAP, CER, XSR, 0.1UF, 10%, 25V, 0201
132S0664	CAP, CER, 0.647UF, 10%, 25V, XSR, 0201
132S00025	CAP, CER, XSR, 0.047UF, 20%, 6.3V, 01005
132S00093	CAP, XSR, 0.022UF, 20%, 6.3V, 01005
132S0245	CAP, CER, XSR, 0.01UF, 10%, 6.3V, 01005
132S0396	CAP, CER, XSR, 1000PF, 10%, 10V, 01005
132S0296	CAP, CER, XSR, 1000PF, 10%, 6.3V, 01005
132S0318	CAP, CER, XSR, 020PF, 10%, 10V, 01005
132S0275	CAP, CER, XSR, 470PF, 10%, 10V, 01005
131S0883	CAP, CER, NP0/C0G, 220PF, 2%, 50V, 0201
131S00170	CAP, CER, C0G, 220PF, 5%, 25V, 01005
131S00053	CAP, CER, C0G, 220PF, 5%, 10V, 01005
132S0249	CAP, CER, XTR, 220PF, 10%, 10V, 01005
131S0307	CAP, CER, NP0/C0G, 100PF, 5%, 16V, 01005

Capacitors (cont'd)

CRITICAL PART#	COMMENT
131S00003	CAP, CER, NP0/C0G, 100PF, 5%, 16V, 01005
131S00323	CAP, CER, NP0/C0G, 50PF, 5%, 25V, 01005
131S0643	CAP, CER, NP0/C0G, 50PF, 5%, 25V, 01005
131S0216	CAP, CER, NP0/C0G, 47PF, 5%, 16V, 01005
131S00004	CAP, CER, 27PF, 5%, C0G, 25V, 0201
131S0223	CAP, CER, NP0/C0G, 27PF, 5%, 10V, 01005
131S0215	CAP, CER, NP0/C0G, 22PF, 5%, 16V, 01005
131S0225	CAP, CER, NP0/C0G, 15PF, 5%, 16V, 01005
131S0220	CAP, CER, NP0/C0G, 12PF, 5%, 10V, 01005
131S00353	CAP, CER, NP0/C0G, 10PF, 5%, 16V, 01005

Ferrites

CRITICAL PART#	COMMENT
155S0576	FERR 80,10 OHM,50%,750MA,0.07 DCR,01005
155S00168	FLTR,NOISE,65 OHMZ,3.40MM,0.7-26HZ,0605

Resistors

CRITICAL PART#	COMMENT
118S00068	RES,MF,1.3 MOHM,1%,200PPM,1/20W,0201
117S0055	RES,MF,1/20W,2M OHM,5,0201,SMD

Misc.

CRITICAL PART#	COMMENT
377S0106	SUPPR,TRANS,VARISTOR,12V,33PF,01005
107S0257	THERMISTOR,NTC,10K OHM,1%,B=3435,01005

Capacitor Alternates

0.1uF, 01005

PART NUMBER	ALTERNATE FOR PART NUMBER	BOM OPTION	REF DES	COMMENTS:	CRITICAL PART#	COMMENT
132S00185	132S00316	?	ALL	CAP, CER, XSR, 0.1UF, 20%, 6.3V, 02005	132S00316	01005, 0.1uF, 6.3V

0.22uF, 01005

RefDes field intentionally left blank to allow selective single-sourcing

PART NUMBER	ALTERNATE FOR PART NUMBER	BOM OPTION	REF DES	COMMENTS:	CRITICAL PART#	COMMENT
132S00233	132S00014	?		01005, 0.22uF, 6.3V, Taiyo	132S00014	01005, 0.22uF, 6.3V
132S00304	132S00014	?		01005, 0.22uF, 6.3V, Kyocera		

0.47uF, 01005

PART NUMBER	ALTERNATE FOR PART NUMBER	BOM OPTION	REF DES	COMMENTS:	CRITICAL PART#	COMMENT
138S00133	138S00120	?	ALL	01005, 0.47uF, 6.3V, Murata	138S00120	01005, 0.47uF, 6.3V
138S00269	138S00120	?	ALL	01005, 0.47uF, 6.3V, Taiyo		

1uF, 0201

PART NUMBER	ALTERNATE FOR PART NUMBER	BOM OPTION	REF DES	COMMENTS:	CRITICAL PART#	COMMENT
138S0739	138S0706	?	ALL	CAP, CER, XSR, 1UF, 20%, 10V, 0201, SAMSUNG	138S0706	CAP, XSR, 1UF, 20%, 10V, 0201
138S0945	138S0706	?	ALL	CAP, CER, XSR, 1UF, 20%, 10V, 0201, KYOCERA		

2.2uF, 0201

RefDes field intentionally left blank to allow selective single-sourcing

PART NUMBER	ALTERNATE FOR PART NUMBER	BOM OPTION	REF DES	COMMENTS:	CRITICAL PART#	COMMENT
138S00049	138S0031	?		CAP, CER, XSR, 2.2UF, 20%, 6.3V, 0201	138S0031	CAP, CER, XSR, 2.2UF, 20%, 6.3V, 0201

3uF @ 1V, 0201

PART NUMBER	ALTERNATE FOR PART NUMBER	BOM OPTION	REF DES	COMMENTS:	CRITICAL PART#	COMMENT
138S00138	138S00139	?	ALL	0201, 3uF@1V, KYOCERA	138S00139	0201, 3uF@1V
138S00164	138S00139	?	ALL	0201, 3uF@1V, TAIYO		
138S00280	138S00139	?	ALL	0201, 3uF@1V, SAMSUNG		

4uF, 0201

RefDes field intentionally left blank to allow selective single-sourcing

PART NUMBER	ALTERNATE FOR PART NUMBER	BOM OPTION	REF DES	COMMENTS:	CRITICAL PART#	COMMENT
138S00116	138S00071	?		CAP, XSR, 4UF, 0201, 0.55MM, TAIYO	138S00071	0201, XSR, 4UF, 0201, 0.55MM, MURATA
138S00117	138S00071	?		CAP, XSR, 4UF, 0201, 0.55MM, KYOCERA		

4.7uF, 0402

PART NUMBER	ALTERNATE FOR PART NUMBER	BOM OPTION	REF DES	COMMENTS:	CRITICAL PART#	COMMENT
138S1103	138S0719	?	ALL	CAP, CER, XSR, 4.7UF, 20%, 10V, 0402	138S0719	0402, 4.7uF, 10V

15uF, 0402

All RefDes in () are single-sourced from Murata

PART NUMBER	ALTERNATE FOR PART NUMBER	BOM OPTION	REF DES	COMMENTS:	CRITICAL PART#	COMMENT
138S00048	138S00003	?		0402, 15uF, 6.3V, Kyocera	138S00003	0402, 15uF, 6.3V

(C1005, C3021, C3172, C3373, C3302)

18uF, 0402

PART NUMBER	ALTERNATE FOR PART NUMBER	BOM OPTION	REF DES	COMMENTS:	CRITICAL PART#	COMMENT
138S00221	138S00146	?	ALL	CAP, CER, XSR, 18UF, 20%, 6.3V, MUR, 0402	138S00146	CAP, CER, XSR, 18UF, 20%, 6.3V, MUR, 0402

20uF, 0402

All parts are single-sourced except for approved parts (listed below)

PART NUMBER	ALTERNATE FOR PART NUMBER	BOM OPTION	REF DES	COMMENTS:	CRITICAL PART#	COMMENT
138S00339	138S00084	?		Taiyo	138S00084	CAP, CER, XSR, 20UF, 20%, 6.3V, 0402, H=0.7MM
138S00338	138S00084	?		Kyocera		

C3002, C3302, C3303, C3306, C3004, C3003, C3402, C3403, C3025, C3035, C3403, C3007, C1006, C1014, C2000, C3002, C3302, C3303, C3306, C3004, C3003, C3402, C3403, C3025, C3035, C3403, C3007, C1006, C1014, C2000

22uF, 0402

PART NUMBER	ALTERNATE FOR PART NUMBER	BOM OPTION	REF DES	COMMENTS:	CRITICAL PART#	COMMENT
138S00144	138S00279	?	ALL	CAP, XSR, 20UF, 20%, 4V, MURATA, 0402	138S00279	CAP, XSR, 20UF, 20%, 4V, MURATA, 0402
138S00143	138S00279	?	ALL	CAP, XSR, 20UF, 20%, 4V, KYOCERA, 0402		

10uF @ 1V, 4-Term

PART NUMBER	ALTERNATE FOR PART NUMBER	BOM OPTION	REF DES	COMMENTS:	CRITICAL PART#	COMMENT
138S00148	138S00149	?	ALL	0402-3T, 10.5uF@1V, Kyocera	138S00149	0402-3T, 10.5uF@1V
138S00150	138S00149	?	ALL	0402-3T, 10.5uF@1V, SEMCO		
138S00151	138S00149	?	ALL	0402-3T, 10.5uF@1V, TY		

22uF, 0402 3T (WiFi)

PART NUMBER	ALTERNATE FOR PART NUMBER	BOM OPTION	REF DES	COMMENTS:	CRITICAL PART#	COMMENT
138S00024	138S0096	?	ALL	CAP, CER, 3-TERM, 7.5UF, 20%, 4V, 0402	138S0096	CAP, CER, 3-TERM, 7.5UF, 20%, 4V, 0402

Display Choke Alternates

PART NUMBER	ALTERNATE FOR PART NUMBER	BOM OPTION	REF DES	COMMENTS:	CRITICAL PART#	COMMENT
155S00415	155S00524	?	ALL	FLTR,NOISE,35 OHMZ,3 OHM,70HZ,50MA,0403	155S00524	FLTR,NOISE,35 OHMZ,3 OHM,70HZ,50MA,0403

Level Shifter Alternates

PART NUMBER	ALTERNATE FOR PART NUMBER	BOM OPTION	REF DES	COMMENTS:	CRITICAL PART#	COMMENT
311S00231	311S00232	?	ALL	IC,74AVC2145,XCVR,2 BIT,2 SPLY,X2S0N0	311S00232	IC,74AVC2145,XCVR,2 BIT,2 SPLY,X2S0N0
311S00230	311S00212	?	ALL	IC,74AVC2145,XCVR,1 BIT,2 SPLY,X2S0N0	311S00212	IC,74AVC2145,XCVR,1 BIT,2 SPLY,X2S0N0
311S00261	311S00233	?	ALL	IC,LSF0101,XCVR,2 BIT CF6,2 SPLY,X2S0N6	311S00233	IC,LSF0101,XCVR,2 BIT CF6,2 SPLY,X2S0N6
					311S00235	IC,WT0202,XCVR,2 BIT, 2 SPLY,BD01,DFN0

Power Inductor Alternates

PART NUMBER	ALTERNATE FOR PART NUMBER	BOM OPTION	REF DES	COMMENTS:	CRITICAL PART#	COMMENT
152S000721	152S000876	?	ALL	TAIYO, 200, ML2, 320, 2.5A, 60PM, 2010	152S000876	Cyntec, IND
152S000930	152S000897	?	ALL	CYNTEC, 200, CF10, 0.10H, 0.1A, 2700, 2010	152S000897	Taiyo, IND
152S000826	152S000821	?	ALL	TAIYO, 200, ML2, 100, 1.0A, 60PM, 2010	152S000821	Cyntec, IND
152S000831	152S000818	?	ALL	TAIYO, 200, ML2, 0.100H, 0.1A, 40PM, 1000	152S000818	Cyntec, IND
152S000991	152S000904	?	ALL	TAIYO, 200, ML2, 0.100H, 0.1A, 34PM, 1004	152S000904	Cyntec, IND
152S000992	152S000905	?	ALL	TAIYO, 200, ML2, 0.470H, 0.4A, 40PM, 2012	152S000905	Cyntec, IND
152S000909	152S000902	?	ALL	TAIYO, 200, ML2, 0.470H, 0.4A, 70PM, 2000	152S000902	Cyntec, IND
152S000824	152S000819	?	ALL	TAIYO, 200, ML2, 0.470H, 0.4A, 60PM, 1010, 05, 2012	152S000819	Cyntec, IND
152S000872	152S000918	?	ALL	TKA, 200, ML2, 0.470H, 0.4A, 30PM, 1010, 0.2010		Boost/Yeti (2117/0.8mm)
152S000847	152S000918	?	ALL	CYNTEC, 200, ML2, 0.470H, 0.4A, 30PM, 1010, 0.2010		
152S01282	152S01255	?	ALL	TAIYO, 200, ML2, 0.470H, 0.4A, 30PM, 1010, 05, 2117		Boost/Yeti (2117/0.65mm)

Misc. Alternates

PART NUMBER	ALTERNATE FOR PART NUMBER	BOM OPTION	REF DES	COMMENTS:	CRITICAL PART#	COMMENT
155S00437	155S00402	?	ALL	FERR 80,330MH,25%,1.5A,550MHM DCR,0201	155S00402	FERR 80,330MH,25%,1.5A,550MHM DCR,0201
155S00194	155S00400	?	ALL	FERR 80,1500MH,25%,200MA,0.70MM DCR,01005	155S00400	FERR 80,1500MH,25%,200MA,0.70MM DCR,01005
155S00414	155S00876	?	ALL	FERR 80,100MH,50%,1.1A,0.650MM DCR,01005	155S00876	FERR 80,100MH,50%,1.1A,0.650MM DCR,01005
155S00131	155S0755	?	ALL	FERR 80,1400MH,25L,200MA,1.0MM DCR,01005	155S0755	FERR 80,240 OHM,25%,200MA,1.0 DCR,01005
155S00583	155S00140	?	ALL	FERR 80,330MH,25%,400MA,0.200DCR,01005	155S00140	FERR 80,330MH,25%,400MA,0.200DCR,01005
377S00070	377S00001	?	ALL	TVS,BIDIR,5.8V,6PF,01005	377S00001	TVS,BIDIR,5.8V,6PF,01005
377S00140	377S00001	?	ALL	TVS,BIDIR,5.8V,6PF,01005	377S00001	SUPPRESS,TRANS,6.8V,100PF,01005
377S0168	377S00129	?	ALL	SUPPRESS,TRANS,6.8V,100PF,ANTICATA,01005		
107S0245	107S0244	?	ALL	THERMISTOR,NTC,100K OHM,1%,B=4250,01005	107S0244	THERMISTOR,NTC,100K OHM,1%,B=4250,01005

ZRB Cap ALTs

PART NUMBER	ALTERNATE FOR PART NUMBER	BOM OPTION	REF DES	COMMENTS:	CRITICAL PART#	COMMENT
138S00240	138S00187	?	ALL	CAP, XSR, 4.7UF, 20%, 25V, 0402	138S00187	CAP, XSR, 4.7UF, 20%, 25V, 0402
138S00246	138S00185	?	ALL	CAP, XSR, 2.2UF, 20%, 25V, 0402	138S00185	CAP, XSR, 2.2UF, 20%, 25V, 0402

0-ohm, 0201, 4.5A

PART NUMBER	ALTERNATE FOR PART NUMBER	BOM OPTION	REF DES	COMMENTS:	CRITICAL PART#	COMMENT
117S00040	117S00012	?	ALL	RES,MF,0 OHM,1/10W,4.5A,0201	117S00012	RES,MF,0 OHM,1/10W,4.5A,0201

2020 MLCCs

16uF, 0402, 4V

PART NUMBER	ALTERNATE FOR PART NUMBER	BOM OPTION	REF DES	COMMENTS:	
138S00316	138S00313	?	ALL	CAP, XSR, 16UF, 20%, 4V, H, 0402	Primary: Murata
138S00314	138S00313	?	ALL	CAP, XSR, 16UF, 20%, 4V, H, 0402	Taiyo
138S00315	138S00313	?	ALL	CAP, XSR, 16UF, 20%, 4V, H, 0402	Kyocera
					Samsung

11uF, 0402, 4V

PART NUMBER	ALTERNATE FOR PART NUMBER	BOM OPTION	REF DES	COMMENTS:	
138S00318	138S00317	?	ALL	CAP, XSR, 11UF, 20%, 4V, H, 0402-3T	Primary: Murata
138S00319	138S00317	?	ALL	CAP, XSR, 11UF, 20%, 4V, H, 0402-3T	Kyocera
138S00320	138S00317	?	ALL	CAP, XSR, 11UF, 20%, 4V, H, 0402-3T	Samsung
					Taiyo

2.7uF, 0201, 6.3V

PART NUMBER	ALTERNATE FOR PART NUMBER	BOM OPTION	REF DES	COMMENTS:
138S00326	138S00325	?	ALL	CAP, K58, 2, 70P, 20%, 6, 3V, 0, 0201
138S00327	138S00325	?	ALL	CAP, K58, 2, 70P, 20%, 6, 3V, 0, 0201

AP Physical

CLASS DEFINITIONS				CLASS CHECKED Y/N
CLASS NAME	...	CONSTRAINT SET	COMMA SEPARATED WITH WILDCARD SUPPORT: NET NAMES EX: DDR* DP NAMES EX: DP-DP_AA*,DP_BB* (LINE STARTS WITH FLAG DP:)	
LX/CX Nodes	PWR_SHAPE_LX	P	PWR_SHAPE	?
System-Wide	PWR_SHAPE	P	PWR_SHAPE	?
	PWR_SHAPE	P	PWR_SHAPE	?
	PWR_200UM	P	PWR_200UM	?
	PWR_100UM	P	PWR_100UM	?
	PWR_100UM	P	PWR_100UM	?
	PWR_100UM	P	PWR_100UM	?
	PWR_100UM	P	PWR_100UM	?
	PWR_100UM	P	PWR_100UM	?
	PWR_100UM	P	PWR_100UM	?
	PWR_100UM	P	PWR_100UM	?
Yeti	PWR_SHAPE	P	PWR_SHAPE	?
	PWR_SHAPE	P	PWR_SHAPE	?
	PWR_SHAPE	P	PWR_SHAPE	?
Kobo1	PWR_50UM	P	PWR_50UM	?
	PWR_SHAPE	P	PWR_SHAPE	?
AF/OIS (Shared)	PWR_SHAPE	P	PWR_SHAPE	?
	PWR_80UM	P	PWR_80UM	?

RF Physical

CLASS DEFINITIONS				CLASS CHECKED Y/N
CLASS NAME	...	CONSTRAINT SET	COMMA SEPARATED WITH WILDCARD SUPPORT: NET NAMES EX: DDR* DP NAMES EX: DP-DP_AA*,DP_BB* (LINE STARTS WITH FLAG DP:)	
SMPS 1/6	PWR_SHAPE	P	PWR_SHAPE	?
SMPS 2	PWR_SHAPE	P	PWR_SHAPE	?
SMPS 3	PWR_DEFAULT	P	PWR_DEFAULT	?
	PWR_SHAPE	P	PWR_SHAPE	?
	PWR_DEFAULT	P	PWR_DEFAULT	?
SMPS 4	PWR_SHAPE	P	PWR_SHAPE	?
	PWR_DEFAULT	P	PWR_DEFAULT	?
SMPS 5	PWR_SHAPE	P	PWR_SHAPE	?
	PWR_DEFAULT	P	PWR_DEFAULT	?
SMPS 7	PWR_SHAPE	P	PWR_SHAPE	?
	PWR_DEFAULT	P	PWR_DEFAULT	?
LD01	PWR_SHAPE	P	PWR_SHAPE	?
	PWR_500UM	P	PWR_500UM	?
	PWR_300UM	P	PWR_300UM	?
	PWR_200UM	P	PWR_200UM	?
	PWR_200UM	P	PWR_200UM	?
	PWR_100UM	P	PWR_100UM	?
	PWR_80UM	P	PWR_80UM	?
	PWR_50UM	P	PWR_50UM	?
	PWR_500UM	P	PWR_500UM	?
	PWR_SHAPE	P	PWR_SHAPE	?
LD02	PWR_SHAPE	P	PWR_SHAPE	?
	PWR_500UM	P	PWR_500UM	?
	PWR_500UM	P	PWR_500UM	?
LD03	PWR_SHAPE	P	PWR_SHAPE	?
	PWR_500UM	P	PWR_500UM	?
	PWR_500UM	P	PWR_500UM	?
LD04	PWR_SHAPE	P	PWR_SHAPE	?
	PWR_500UM	P	PWR_500UM	?
	PWR_500UM	P	PWR_500UM	?
	PWR_300UM	P	PWR_300UM	?
	PWR_200UM	P	PWR_200UM	?
	PWR_100UM	P	PWR_100UM	?
	PWR_100UM	P	PWR_100UM	?
	PWR_100UM	P	PWR_100UM	?
	PWR_100UM	P	PWR_100UM	?
	PWR_100UM	P	PWR_100UM	?
LD05	PWR_100UM	P	PWR_100UM	?
	PWR_100UM	P	PWR_100UM	?
	PWR_100UM	P	PWR_100UM	?
	PWR_100UM	P	PWR_100UM	?
	PWR_100UM	P	PWR_100UM	?
	PWR_100UM	P	PWR_100UM	?
	PWR_100UM	P	PWR_100UM	?
	PWR_100UM	P	PWR_100UM	?
	PWR_100UM	P	PWR_100UM	?
	PWR_100UM	P	PWR_100UM	?
LD06	PWR_100UM	P	PWR_100UM	?
	PWR_100UM	P	PWR_100UM	?
	PWR_100UM	P	PWR_100UM	?
	PWR_100UM	P	PWR_100UM	?
	PWR_100UM	P	PWR_100UM	?
	PWR_100UM	P	PWR_100UM	?
	PWR_100UM	P	PWR_100UM	?
	PWR_100UM	P	PWR_100UM	?
	PWR_100UM	P	PWR_100UM	?
	PWR_100UM	P	PWR_100UM	?
LD07	PWR_SHAPE	P	PWR_SHAPE	?
	PWR_SHAPE	P	PWR_SHAPE	?
	PWR_SHAPE	P	PWR_SHAPE	?
	PWR_SHAPE	P	PWR_SHAPE	?
	PWR_SHAPE	P	PWR_SHAPE	?
	PWR_SHAPE	P	PWR_SHAPE	?
	PWR_SHAPE	P	PWR_SHAPE	?
	PWR_SHAPE	P	PWR_SHAPE	?
	PWR_SHAPE	P	PWR_SHAPE	?
	PWR_SHAPE	P	PWR_SHAPE	?
LD08	PWR_SHAPE	P	PWR_SHAPE	?
	PWR_SHAPE	P	PWR_SHAPE	?
	PWR_SHAPE	P	PWR_SHAPE	?
	PWR_SHAPE	P	PWR_SHAPE	?
	PWR_SHAPE	P	PWR_SHAPE	?
	PWR_SHAPE	P	PWR_SHAPE	?
	PWR_SHAPE	P	PWR_SHAPE	?
	PWR_SHAPE	P	PWR_SHAPE	?
	PWR_SHAPE	P	PWR_SHAPE	?
	PWR_SHAPE	P	PWR_SHAPE	?
LD09	PWR_SHAPE	P	PWR_SHAPE	?
	PWR_SHAPE	P	PWR_SHAPE	?
	PWR_SHAPE	P	PWR_SHAPE	?
	PWR_SHAPE	P	PWR_SHAPE	?
	PWR_SHAPE	P	PWR_SHAPE	?
	PWR_SHAPE	P	PWR_SHAPE	?
	PWR_SHAPE	P	PWR_SHAPE	?
	PWR_SHAPE	P	PWR_SHAPE	?
	PWR_SHAPE	P	PWR_SHAPE	?
	PWR_SHAPE	P	PWR_SHAPE	?
LD010	PWR_SHAPE	P	PWR_SHAPE	?
	PWR_SHAPE	P	PWR_SHAPE	?
	PWR_SHAPE	P	PWR_SHAPE	?
	PWR_SHAPE	P	PWR_SHAPE	?
	PWR_SHAPE	P	PWR_SHAPE	?
	PWR_SHAPE	P	PWR_SHAPE	?
	PWR_SHAPE	P	PWR_SHAPE	?
	PWR_SHAPE	P	PWR_SHAPE	?
	PWR_SHAPE	P	PWR_SHAPE	?
	PWR_SHAPE	P	PWR_SHAPE	?
LD011	PWR_SHAPE	P	PWR_SHAPE	?
	PWR_SHAPE	P	PWR_SHAPE	?
	PWR_SHAPE	P	PWR_SHAPE	?
	PWR_SHAPE	P	PWR_SHAPE	?
	PWR_SHAPE	P	PWR_SHAPE	?
	PWR_SHAPE	P	PWR_SHAPE	?
	PWR_SHAPE	P	PWR_SHAPE	?
	PWR_SHAPE	P	PWR_SHAPE	?
	PWR_SHAPE	P	PWR_SHAPE	?
	PWR_SHAPE	P	PWR_SHAPE	?
LD012	PWR_SHAPE	P	PWR_SHAPE	?
	PWR_SHAPE	P	PWR_SHAPE	?
	PWR_SHAPE	P	PWR_SHAPE	?
	PWR_SHAPE	P	PWR_SHAPE	?
	PWR_SHAPE	P	PWR_SHAPE	?
	PWR_SHAPE	P	PWR_SHAPE	?
	PWR_SHAPE	P	PWR_SHAPE	?
	PWR_SHAPE	P	PWR_SHAPE	?
	PWR_SHAPE	P	PWR_SHAPE	?
	PWR_SHAPE	P	PWR_SHAPE	?
LD013	PWR_SHAPE	P	PWR_SHAPE	?
	PWR_SHAPE	P	PWR_SHAPE	?
	PWR_SHAPE	P	PWR_SHAPE	?
	PWR_SHAPE	P	PWR_SHAPE	?
	PWR_SHAPE	P	PWR_SHAPE	?
	PWR_SHAPE	P	PWR_SHAPE	?
	PWR_SHAPE	P	PWR_SHAPE	?
	PWR_SHAPE	P	PWR_SHAPE	?
	PWR_SHAPE	P	PWR_SHAPE	?
	PWR_SHAPE	P	PWR_SHAPE	?
LD014	PWR_SHAPE	P	PWR_SHAPE	?
	PWR_SHAPE	P	PWR_SHAPE	?
	PWR_SHAPE	P	PWR_SHAPE	?
	PWR_SHAPE	P	PWR_SHAPE	?
	PWR_SHAPE	P	PWR_SHAPE	?
	PWR_SHAPE	P	PWR_SHAPE	?
	PWR_SHAPE	P	PWR_SHAPE	?
	PWR_SHAPE	P	PWR_SHAPE	?
	PWR_SHAPE	P	PWR_SHAPE	?
	PWR_SHAPE	P	PWR_SHAPE	?
LD015	PWR_SHAPE	P	PWR_SHAPE	?
	PWR_SHAPE	P	PWR_SHAPE	?
	PWR_SHAPE	P	PWR_SHAPE	?
	PWR_SHAPE	P	PWR_SHAPE	?
	PWR_SHAPE	P	PWR_SHAPE	?
	PWR_SHAPE	P	PWR_SHAPE	?
	PWR_SHAPE	P	PWR_SHAPE	?
	PWR_SHAPE	P	PWR_SHAPE	?
	PWR_SHAPE	P	PWR_SHAPE	?
	PWR_SHAPE	P	PWR_SHAPE	?
LD016	PWR_SHAPE	P	PWR_SHAPE	?
	PWR_SHAPE	P	PWR_SHAPE	?
	PWR_SHAPE	P	PWR_SHAPE	?
	PWR_SHAPE	P	PWR_SHAPE	?
	PWR_SHAPE	P	PWR_SHAPE	?
	PWR_SHAPE	P	PWR_SHAPE	?
	PWR_SHAPE	P	PWR_SHAPE	?
	PWR_SHAPE	P	PWR_SHAPE	?
	PWR_SHAPE	P	PWR_SHAPE	?
	PWR_SHAPE	P	PWR_SHAPE	?

RF Physical (continued)

CLASS DEFINITIONS				CLASS CHECKED Y/N
CLASS NAME	...	CONSTRAINT SET	COMMA SEPARATED WITH WILDCARD SUPPORT: NET NAMES EX: DDR* DP NAMES EX: DP-DP_AA*,DP_BB* (LINE STARTS WITH FLAG DP:)	
QET0	PWR_500UM	P	PWR_500UM	?
	PWR_500UM	P	PWR_500UM	?
	PWR_100UM	P	PWR_100UM	?
	PWR_100UM	P	PWR_100UM	?
	PWR_100UM	P	PWR_100UM	?
	PWR_100UM	P	PWR_100UM	?
QET1	PWR_500UM	P	PWR_500UM	?
	PWR_100UM	P	PWR_100UM	?
	PWR_100UM	P	PWR_100UM	?
	PWR_100UM	P	PWR_100UM	?
	PWR_100UM	P	PWR_100UM	?
	PWR_100UM	P	PWR_100UM	?
APT (Lower)	PWR_500UM	P	PWR_500UM	?
	PWR_200UM	P	PWR_200UM	?
	PWR_200UM	P	PWR_200UM	?
	PWR_200UM	P	PWR_200UM	?
	PWR_200UM	P	PWR_200UM	?
	PWR_200UM	P	PWR_200UM	?
APT (Upper)	PWR_500UM	P	PWR_500UM	?
	PWR_200UM	P	PWR_200UM	?
	PWR_200UM	P	PWR_200UM	?
	PWR_200UM	P	PWR_200UM	?
	PWR_200UM	P	PWR_200UM	?
	PWR_200UM	P	PWR_200UM	?
PAD/DSM VDDs	PWR_50UM	P	PWR_50UM	?
	PWR_50UM	P	PWR_50UM	?
	PWR_50UM	P	PWR_50UM	?
	PWR_50UM	P	PWR_50UM	?
	PWR_50UM	P	PWR_50UM	?
	PWR_50UM	P	PWR_50UM	?
Misc Radio	PWR_100UM	P	PWR_100UM	?
	PWR_300UM	P	PWR_300UM	?
	PWR_100UM	P	PWR_100UM	?
	PWR_80UM	P	PWR_80UM	?
	PWR_80UM	P	PWR_80UM	?
	PWR_80UM	P	PWR_80UM	?
WiFi	PWR_100UM	P	PWR_100UM	?
	PWR_100UM	P	PWR_100UM	?
	PWR_100UM	P	PWR_100UM	?
	PWR_100UM	P	PWR_100UM	?
	PWR_100UM	P	PWR_100UM	?
	PWR_100UM	P	PWR_100UM	?
Fez	PWR_SHAPE	P	PWR_SHAPE	?
	PWR_100UM	P	PWR_100UM	?
	PWR_100UM	P	PWR_100UM	?
	PWR_100UM	P	PWR_100UM	?
	PWR_100UM	P	PWR_100UM	?
	PWR_100UM	P	PWR_100UM	?
GND				?
DEFAULT				?

Spacing

CLASS DEFINITIONS				CLASS CHECKED Y/N
CLASS NAME	...	CONSTRAINT SET	COMMA SEPARATED WITH WILDCARD SUPPORT: NET NAMES EX: DDR* DP NAMES EX: DP-DP_AA*,DP_BB* (LINE STARTS WITH FLAG DP:)	
GND	S	MSAP_YIELD	=	?
Power	S	DEFAULT	=	?
GND	PWR_DEFAULT	S	DEFAULT	?
	PWR_50UM	S	DEFAULT	?
	PWR_80UM	S	DEFAULT	?
	PWR_100UM	S	DEFAULT	?
	PWR_200UM	S	DEFAULT	?
	PWR_300UM	S	DEFAULT	?
	PWR_500UM	S	DEFAULT	?
	PWR_750UM	S	DEFAULT	?
	PWR_SHAPE	S	DEFAULT	?
	PWR_SHAPE_HV	S	HV_SPACING	?
RFFE	PWR_80UM_RFFE	S	DEFAULT	?
	PWR_SHAPE_LX	S	LX_SPACING	?

Class-Class Spacing

CLASS TO CLASS SPACING		
CLASS NAME	CLASS NAME	CONSTRAINT SET
PWR_100UM	GND	MSAP_YIELD_PWR
PWR_100UM	PWR_100UM	100UM-249UM_SPACING
PWR_100UM	PWR_200UM	100UM-249UM_SPACING
PWR_100UM	PWR_300UM	100UM-249UM_SPACING
PWR_100UM	PWR_500UM	100UM-249UM_SPACING
PWR_100UM	PWR_750UM	100UM-249UM_SPACING
PWR_100UM	PWR_SHAPE	100UM-249UM_SPACING
PWR_200UM	GND	MSAP_YIELD_PWR
PWR_200UM	PWR_200UM	250UM+ SPACING
PWR_200UM	PWR_300UM	250UM+ SPACING
PWR_200UM	PWR_500UM	250UM+ SPACING
PWR_200UM	PWR_750UM	250UM+ SPACING
PWR_200UM	PWR_SHAPE	250UM+ SPACING
PWR_300UM	GND	MSAP_YIELD_PWR
PWR_300UM	PWR_300UM	250UM+ SPACING
PWR_300UM	PWR_500UM	250UM+ SPACING
PWR_300UM	PWR_750UM	250UM+ SPACING
PWR_300UM	PWR_SHAPE	250UM+ SPACING
PWR_500UM	GND	MSAP_YIELD_PWR
PWR_500UM	PWR_500UM	250UM+ SPACING
PWR_500UM	PWR_750UM	250UM+ SPACING
PWR_500UM	PWR_SHAPE	250UM+ SPACING
PWR_750UM	GND	MSAP_YIELD_PWR
PWR_750UM	PWR_750UM	250UM+ SPACING
PWR_SHAPE	GND	MSAP_YIELD_PWR
PWR_SHAPE	PWR_SHAPE	250UM+ SPACING

PWR_SHAPE_LX spacing on next page

PAGE TITLE		
CONSTRAINTS: Power		
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	SHEET	

Standard 50-ohm CSets

Thin

All Things	HYBRID IMPEDANCE RULE			
	TRACE LAYER	REFERENCE LAYER(s)	REQUIRED IMPEDANCE	TRACE WIDTH (OPTIONAL)
	RF SPACING VALUES= ?			
	RULE NAME= 50_THIN		ZONE NAME=	PRIMARY
	TOP	ISL2	50	
	ISL3	ISL2, ISL4		
	ISL7	BOTTOM, ISL6	50	

Wide

		HYBRID IMPEDANCE RULE			
		TRACE LAYER	REFERENCE LAYER(S)	REQUIRED IMPEDANCE	TRACE WIDTH (OPTIONAL)
		RF SPACING VALUES= ?			
		RULE NAME= 50_WIDE		ZONE NAME= PRIMARY	
All Wides		TOP	ISL3	50	
		ISL6	BOTTOM, ISL4	50	

Hybrid 50-ohm CSets

Thin on TOP and L3, Wide on L6

		HYBRID IMPEDANCE RULE			
		TRACE LAYER	REFERENCE LAYER(S)	REQUIRED IMPEDANCE	TRACE WIDTH (OPTIONAL)
		RF SPACING VALUES: _____ ?			
		RULE NAME== 50_WIDE-L6_THIN-L1-L3		ZONE NAME== PRIMARY	
Thins	TOP	ISL2		50	
	ISL3	ISL2, ISL4		50	
Wide	ISL6	BOTTOM, ISL4		50	

Wide on TOP and L6, Thin on L3

HYBRID IMPEDANCE RULE			
TRACE LAYER	REFERENCE LAYER(S)	REQUIRED IMPEDANCE	TRACE WIDTH (OPTIONAL)
RF SPACING VALUES= ?			
RULE NAME= 58_WIDE_L1_L6_THIN-L3		ZONE NAME=	PRIMARY
TOP	ISL3	50	
ISL3	ISL2, ISL4	50	
ISL6	BOTTOM, ISL4	50	

Custom mmW Constraints

Physical

CLASS DEFINITIONS			COMMA SEPARATED WITH WILDCARD SUPPORT: NET NAMES EX: DDR*		CLASS	COMMENT
	CLASS NAME	...	CONSTRAINT SET	DP NAMES EX: DP.DP_AA.DP_BB* (LINE STARTS WITH FLAG DP.)	Y/N	Y/N
mmw IF	50_MMW_IF	P	50_OHM_MMW_IF_SE	50_MMW_*IFH*, 50_MMW_*IFV*	Y	
	50_MMW_RF	P	50_OHM_MMW_SE	50_MMW_SMR525_S1H_HP0L, 50_MMW_SMR525_S1L_HP0L	Y	
mmw RF	50_MMW_RF	P	50_OHM_MMW_SE	50_MMW_SMR525_S2H_HP0L, 50_MMW_SMR525_S2L_HP0L	Y	
	50_MMW_RF	P	50_OHM_MMW_SE	50_MMW_SMR525_S5H_HP0L, 50_MMW_SMR525_S5L_HP0L	Y	
	50_MMW_RF	P	50_OHM_MMW_SE	50_MMW_SMR525_S6H_HP0L, 50_MMW_SMR525_S6L_HP0L	Y	
	50_MMW_SMYRNA_HB_RF	P	50_OHM_MMW_SMYRNA_HB_SE	50_MMW_SMR525_S7H_HP0L	Y	
	50_MMW_SMYRNA_LB_RF	P	50_OHM_MMW_SMYRNA_LB_SE	50_MMW_SMR525_S7L_HP0L	Y	
	50_MMW_SMYRNA_HB_RF	P	50_OHM_MMW_SMYRNA_HB_SE	50_MMW_SMR525_S8H_HP0L	Y	
	50_MMW_SMYRNA_LB_RF	P	50_OHM_MMW_SMYRNA_LB_SE	50_MMW_SMR525_S8L_HP0L	Y	
	50_MMW_RF	P	50_OHM_MMW_SE	50_MMW_SMR525_S9H_VP0L, 50_MMW_SMR525_S9L_VP0L	Y	
	50_MMW_RF	P	50_OHM_MMW_SE	50_MMW_SMR525_S10H_VP0L, 50_MMW_SMR525_S10L_VP0L	Y	
	50_MMW_SMYRNA_HB_RF	P	50_OHM_MMW_SMYRNA_HB_SE	50_MMW_SMR525_S11H_VP0L	Y	
	50_MMW_SMYRNA_LB_RF	P	50_OHM_MMW_SMYRNA_LB_SE	50_MMW_SMR525_S11L_VP0L	Y	
	50_MMW_SMYRNA_HB_RF	P	50_OHM_MMW_SMYRNA_HB_SE	50_MMW_SMR525_S12H_VP0L	Y	
	50_MMW_SMYRNA_LB_RF	P	50_OHM_MMW_SMYRNA_LB_SE	50_MMW_SMR525_S12L_VP0L	Y	
	50_MMW_RF	P	50_OHM_MMW_SE	50_MMW_SMR525_S13H_VP0L, 50_MMW_SMR525_S13L_VP0L	Y	
	50_MMW_RF	P	50_OHM_MMW_SE	50_MMW_SMR525_S14H_VP0L, 50_MMW_SMR525_S14L_VP0L	Y	

Spacing

		CLASS DEFINITIONS		COMMA SEPARATED WITH WILDCARD SUPPORT: NET NAMES EX: DDR*	CLEAR OVERIDE
		CLASS NAME	CONSTRAINT SET	DP NAMES EX: DP:DP_AA*,DP_BB* (LINE STARTS WITH FLAG DP:)	Y/N
mmw	IF	50_MMW_IF	DIELECTRIC_4XV_50_MMW_IF_SE	=	Y
mmw	RF	50_MMW_RF	DIELECTRIC_4XV_50_WIDE_MMW_SE	=	Y
		50_MMW_SHYRNA_HB_RF	DIELECTRIC_4XV_50_WIDE_MMW_SE	=	Y
		50_MMW_SHYRNA_LB_RF	DIELECTRIC_4XV_50_WIDE_MMW_SE	=	Y

Class-Class Spacing

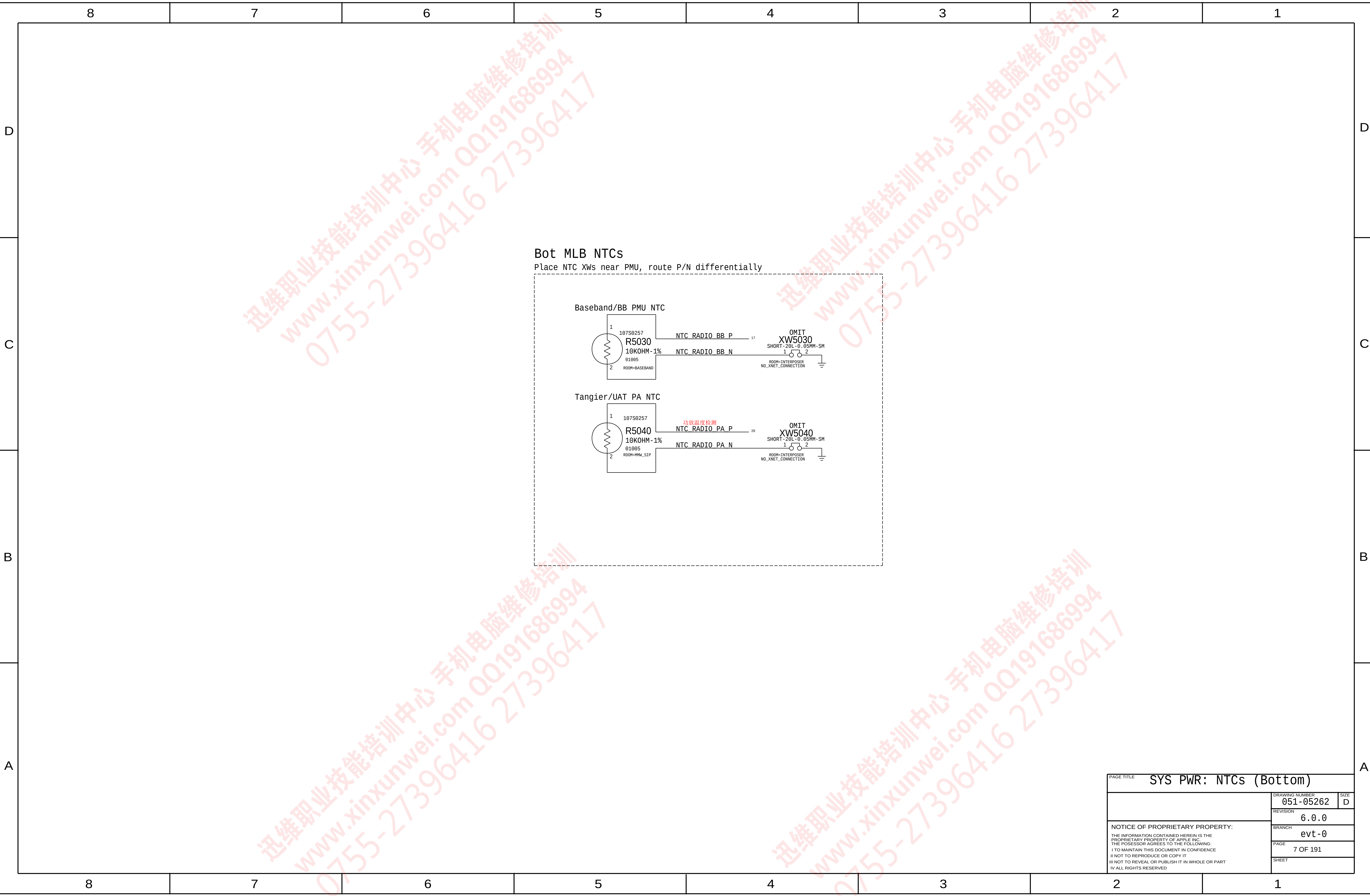
CLASS TO CLASS SPACING		
CLASS NAME	CLASS NAME	CONSTRAINT SET
50_MMW_IF	GND	DIELECTRIC_2X_50_MMW_IF_SE
50_MMW_RF	GND	DIELECTRIC_2X_50_WIDE_MMW_SE
50_MMW_SMYRNA_HB_RF	GND	DIELECTRIC_2X_50_WIDE_MMW_SMYRNA_HB_SE
50_MMW_SMYRNA_LB_RF	GND	DIELECTRIC_2X_50_WIDE_MMW_SMYRNA_LB_SE

CLASS DEFINITIONS			COMMA SEPARATED WITH WILDCARD SUPPORT: NET NAMES EX: DDR*	CLASS OVERLOAD
CLASS NAME	TYPE	CONSTRAINT SET	DP NAMES EX: DP_DP_AA*, DP_BB* (LINE STARTS WITH FLAG DP:)	Y/N
PWR_80UM	P	PWR_80UM	PP_BBPMK_VREG_RF	Y
PWR_80UM	P	PWR_80UM	PP_BBPMK_VREG_X0	Y
PWR_80UM	P	PWR_80UM	PP_PWK_REF_BYP	Y

CLASS DEFINITIONS			COMMA SEPARATED WITH WILDCARD SUPPORT: NET NAMES EX: DDR*	CLEAR OVERLAP
CLASS NAME	...	CONSTRAINT SET	DP NAMES EX: DP_DP_AA*DP_BB* (LINE STARTS WITH FLAG DP:)	Y/N
XTAL_OSC	S	NA_DIELECTRIC_3X	XO_GND	Y
XTAL_OSC	S	NA_DIELECTRIC_3X	XO_THERM	Y
XTAL_OSC	S	NA_DIELECTRIC_3X	THERMAL_GNSS_XO_THERM	Y

CLASS TO CLASS SPACING		
CLASS NAME	CLASS NAME	CONSTRAINT SET
XTAL_OSC	XTAL_OSC	DEFAULT
XTAL_OSC	GND	DEFAULT

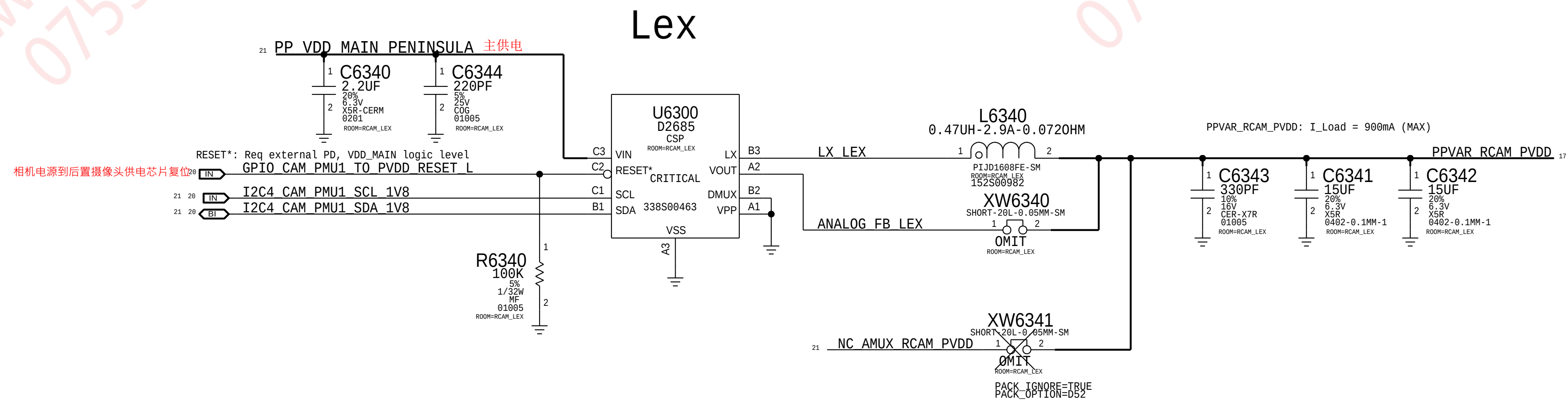
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CONSTRAINTS: Stackup/Misc. (FRW-Specific)			
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		051-05262	SIZE
		REVISION	
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		evt-0	
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PAGE TITLE SYS PWR: NTCs (Bottom)		
	DRAWING NUMBER	051-05262
	REVISION	6.0.0
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SUBSYSTEM SPECIFIC BOM TABLES

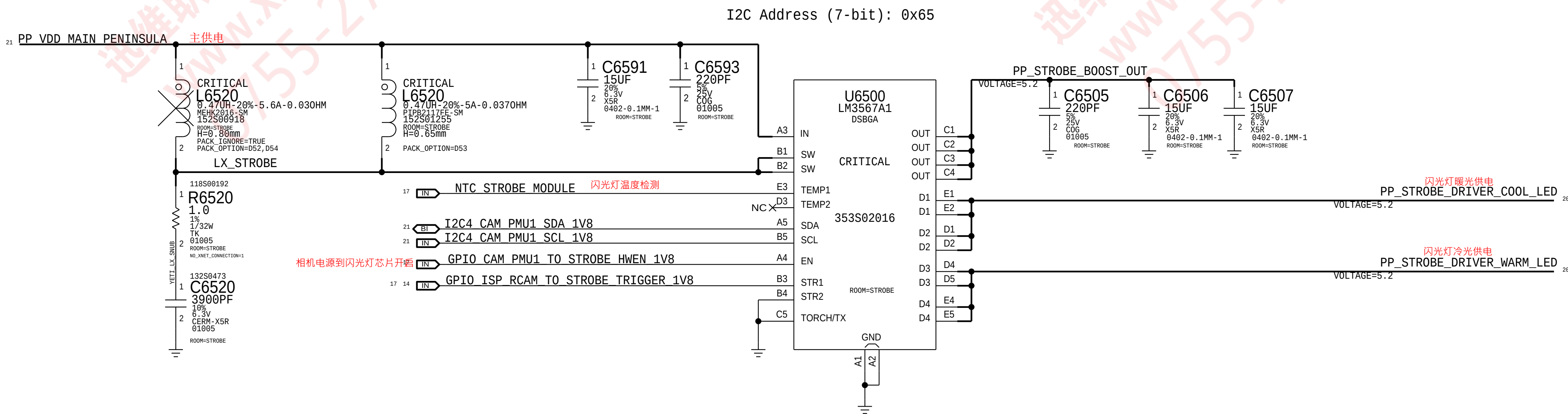
PART NUMBER	ALTERNATE FOR PART NUMBER	BOM OPTION	REF DES	COMMENTS:
138S00048	138S00003	?	(C0341,C0342)	0402, 15uF, 6.3V, Kyocera



SYNCING: D52, D53

PAGE TITLE			CAMERA: Actuator Supply (Lex)			
			DRAWING NUMBER		SIZE	
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			REVISION			
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			PAGE		8 OF 191	
			SHEET			

LED STROBE DRIVER (YETI)

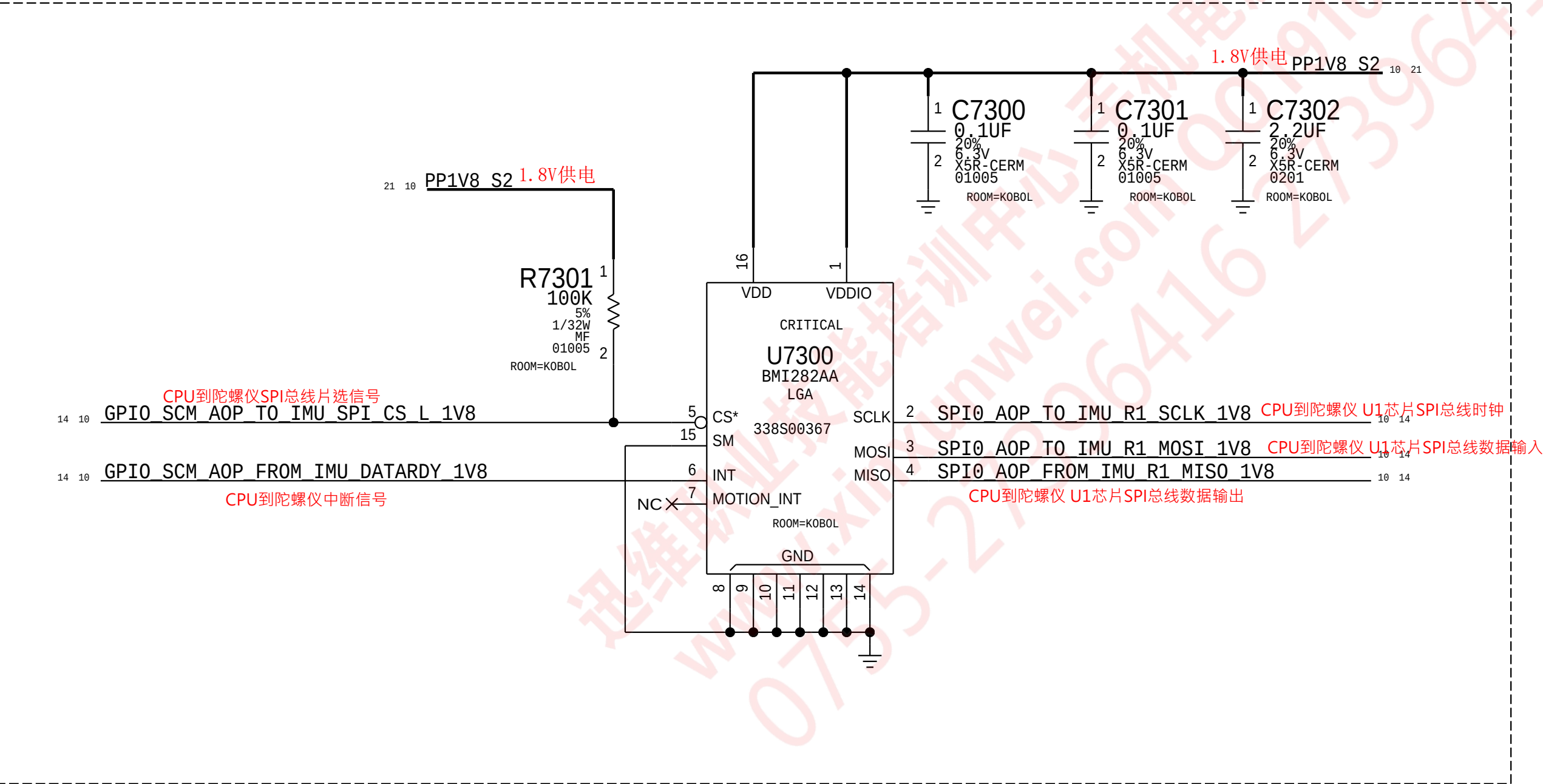


SYNCING: D52, D53, D54

PAGE TITLE		
CAMERA: Strobe Driver		
	DRAWING NUMBER	051-05262
	REVISION	6.0.0
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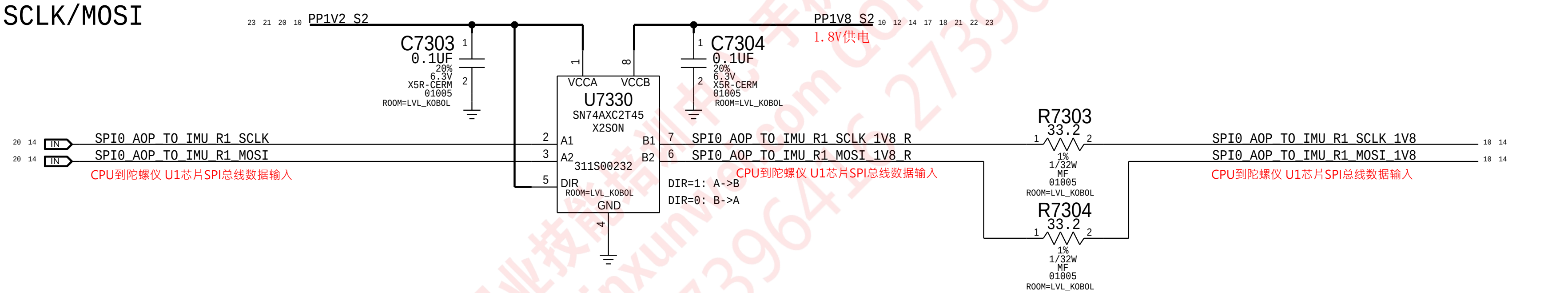
KOBOL - ACCEL & GYRO

IMU

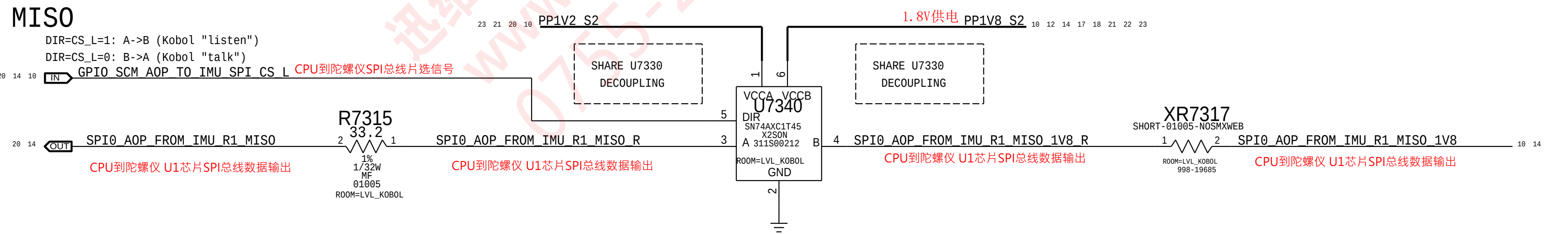


Level Translators

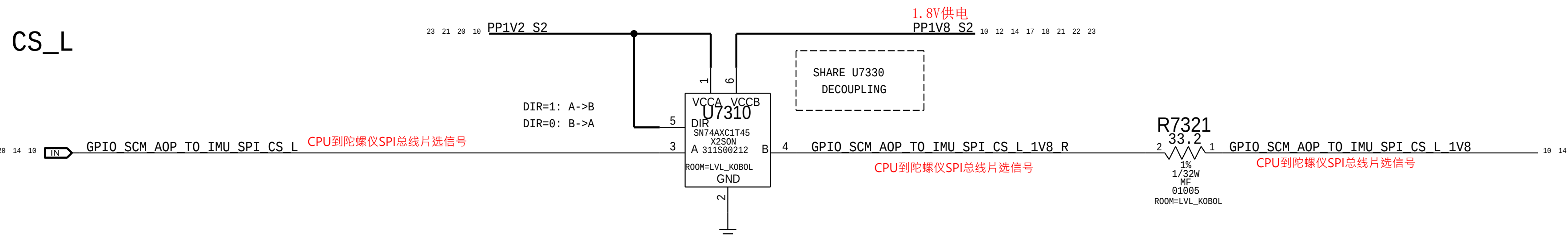
SCLK/MOSI



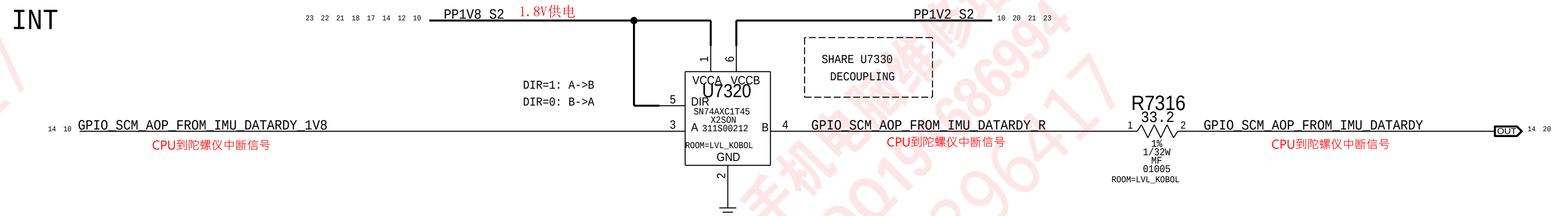
MISO



CS_L

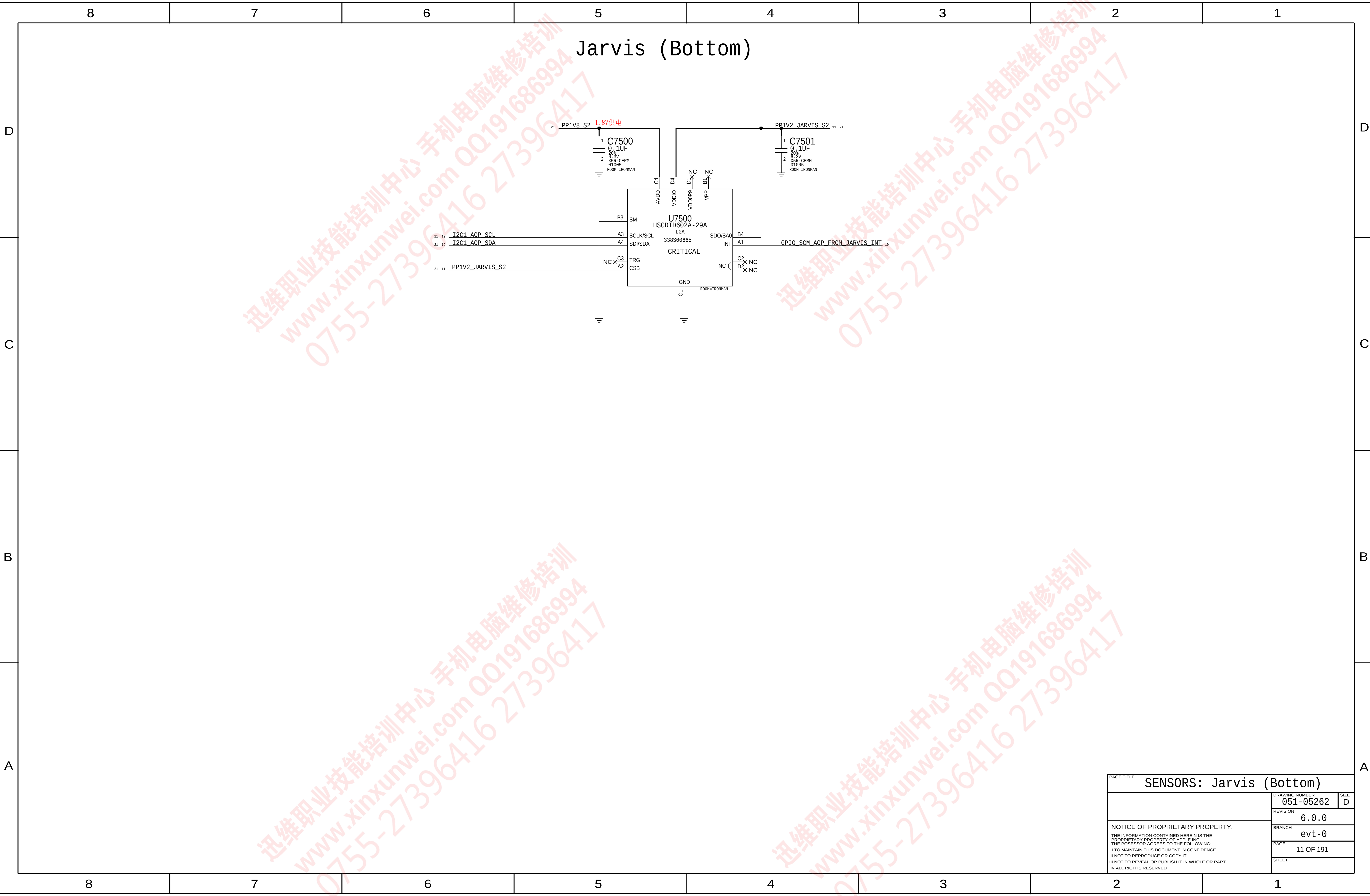


INT



SYNCING: D52, D53, D54

PAGE TITLE		
SENSORS: Accel / Gyro		
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PAGE TITLESENSORS: Jarvis (Bottom)		
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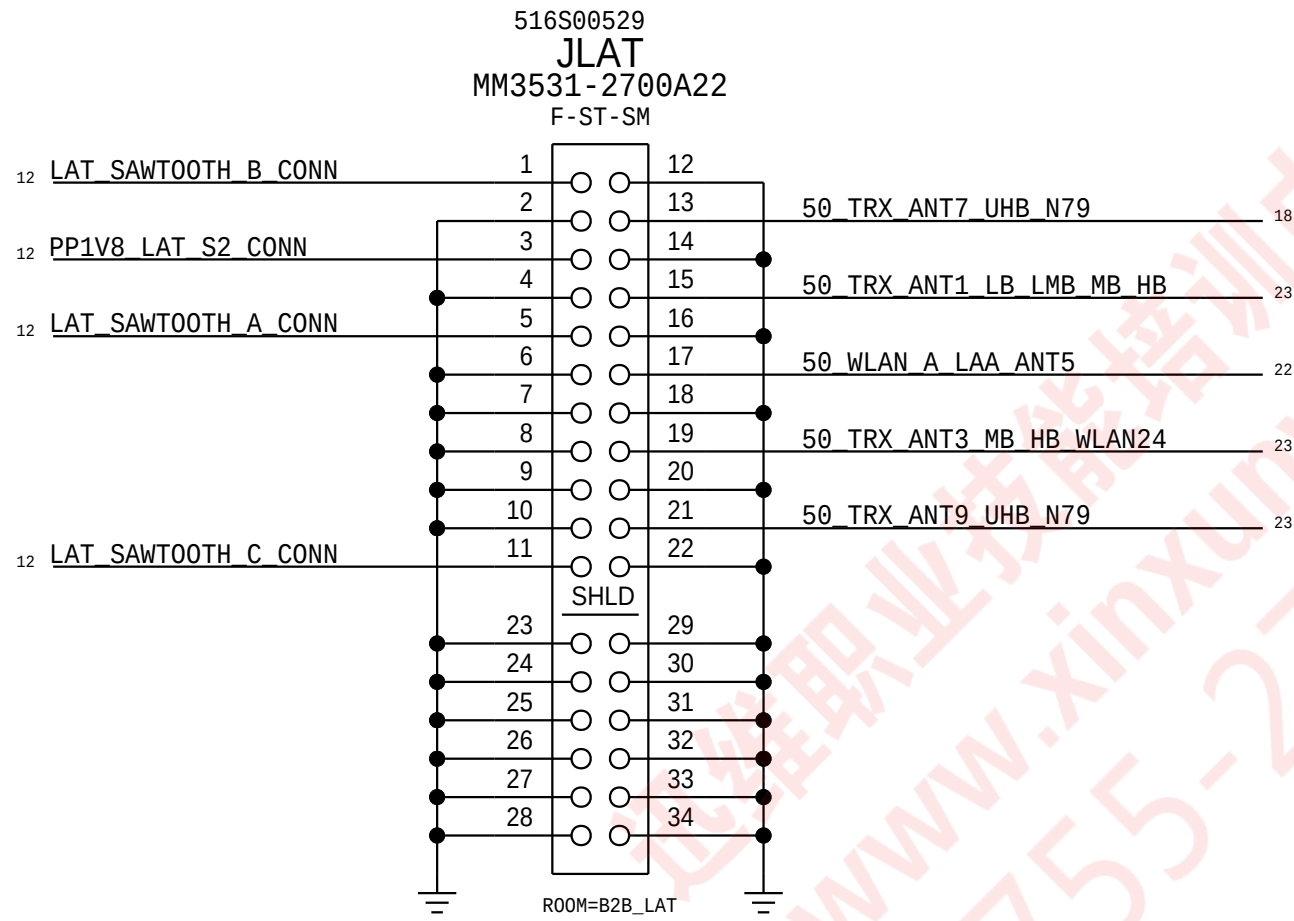
3

2

1

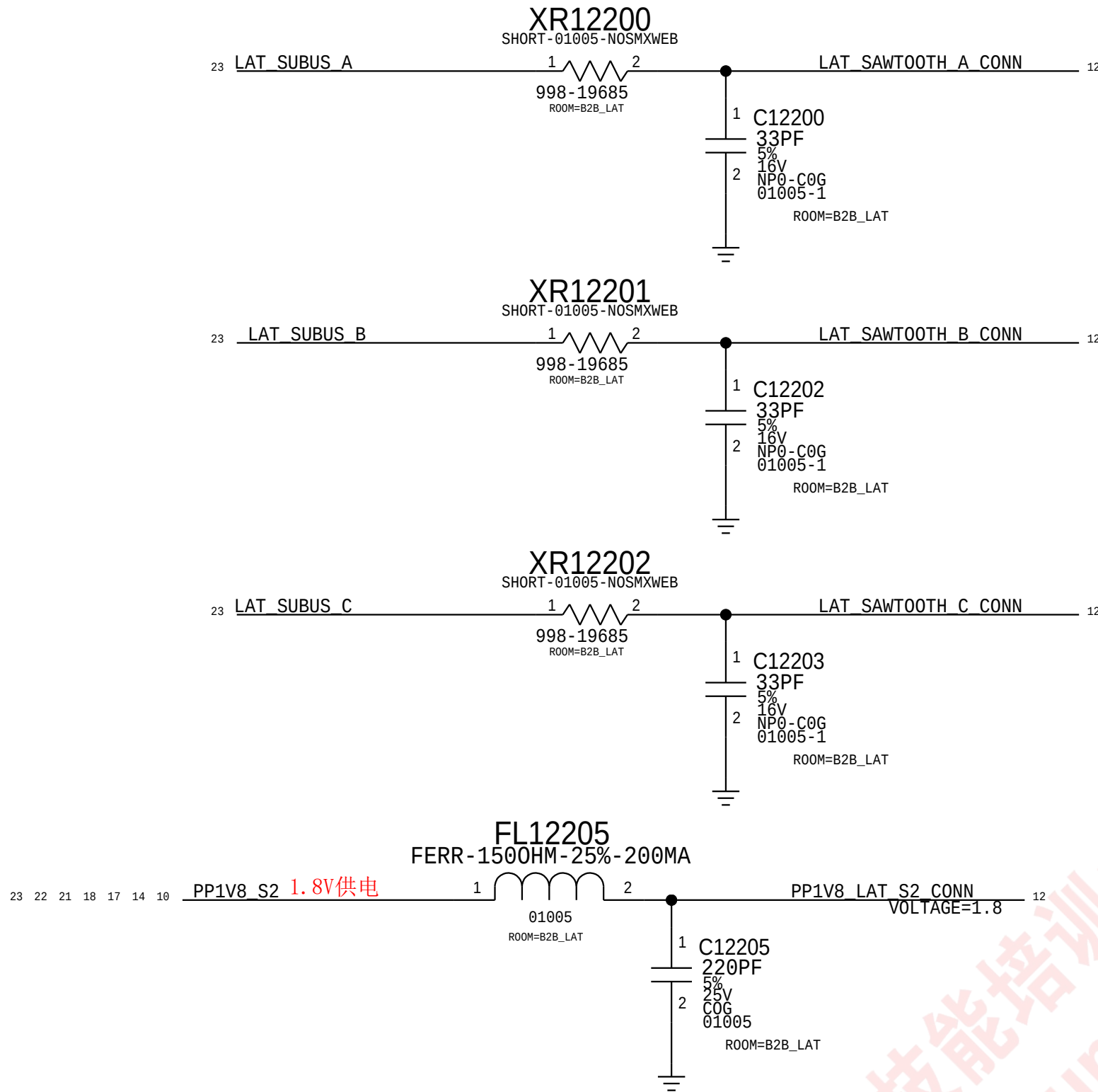
LAT B2B

Connector



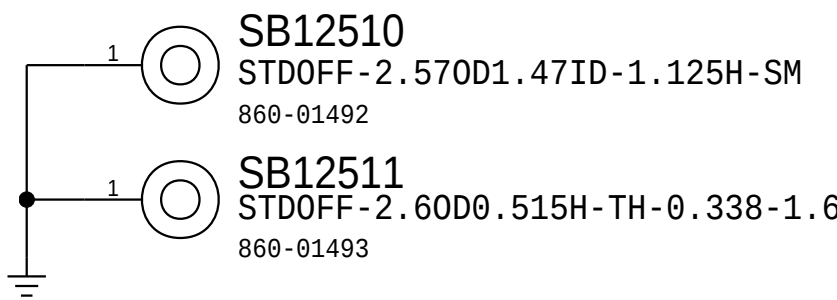
SUBUS filtering

Change SuBus to Sawtooth for flex side per antenna team request

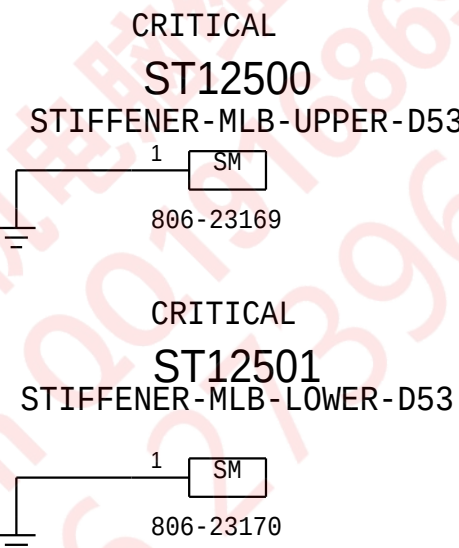


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B2B: LAT		
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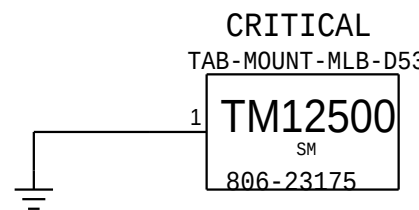
Standoffs



Stiffeners

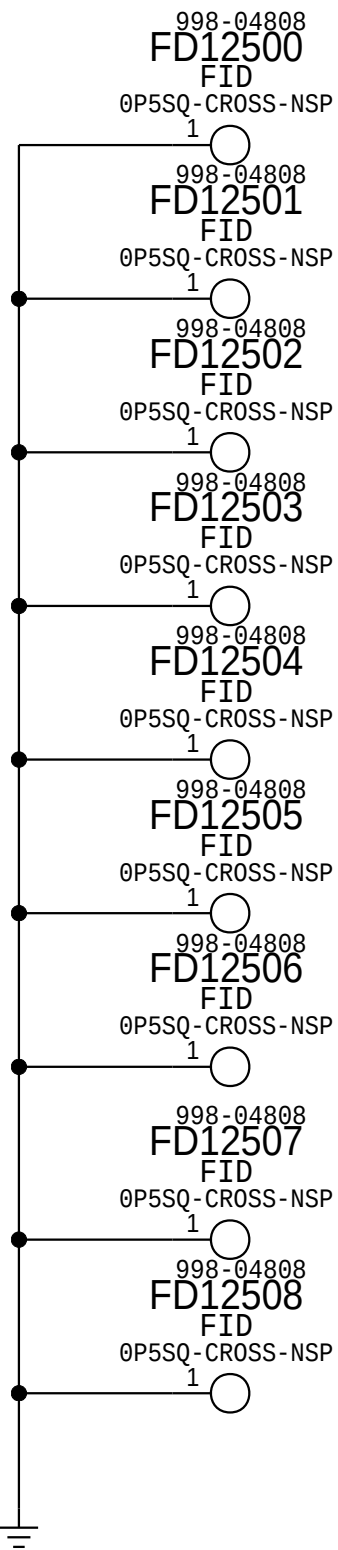


Mounts

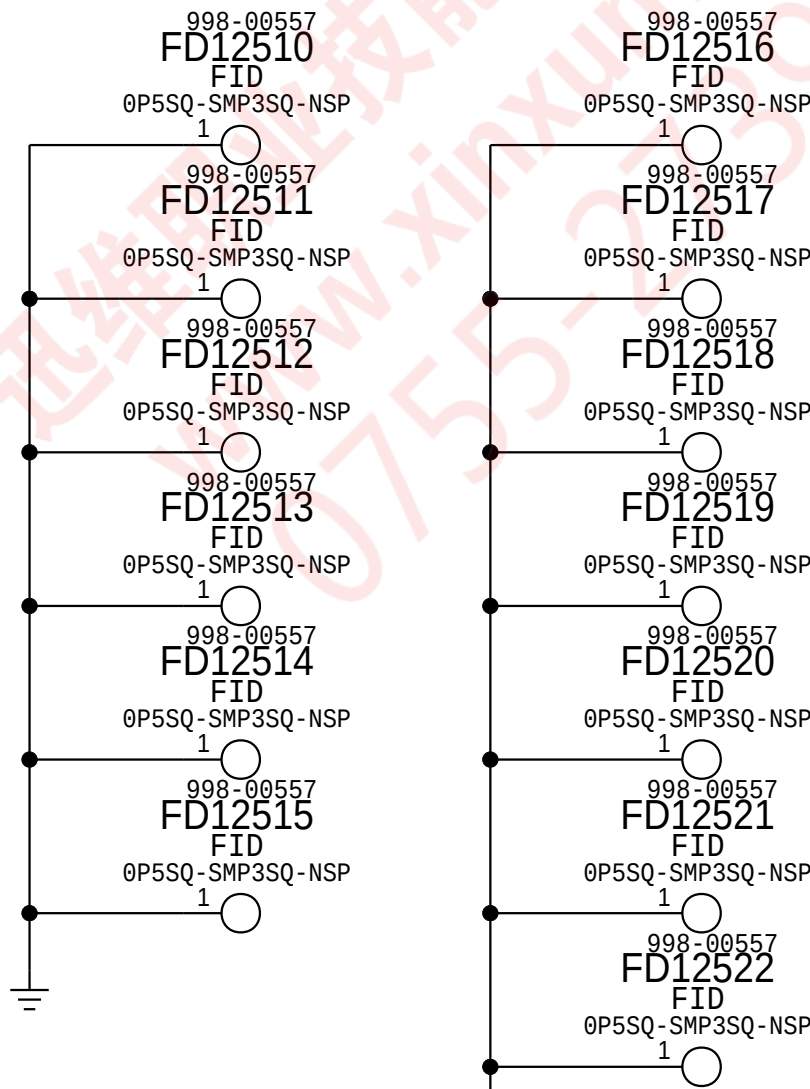


Fiducials

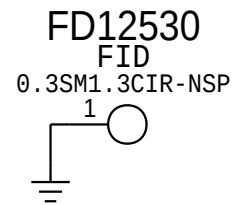
Crosses



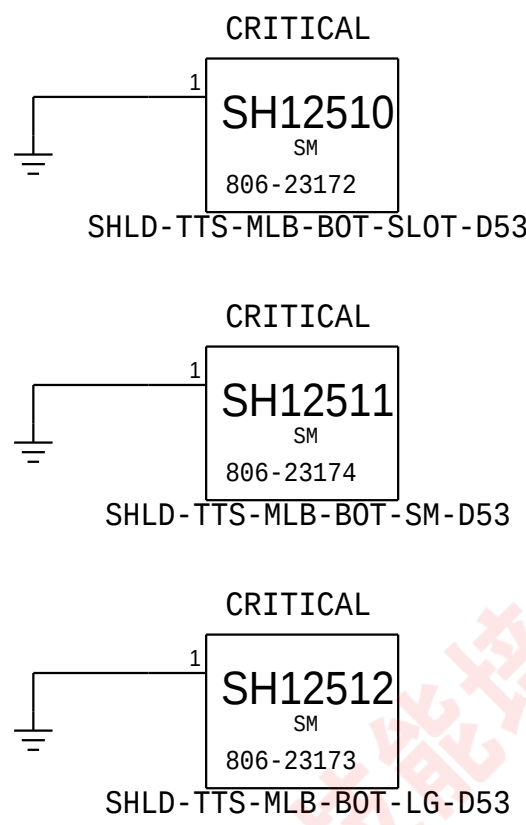
Squares



Sidefire

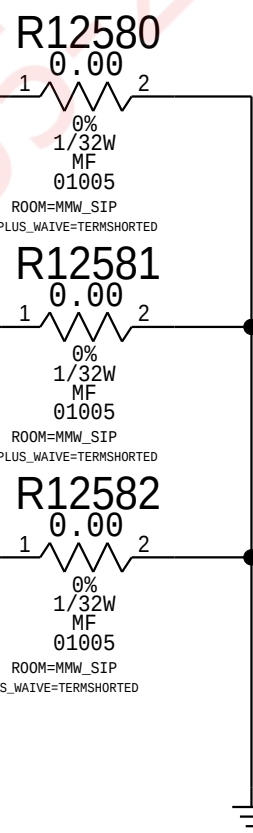
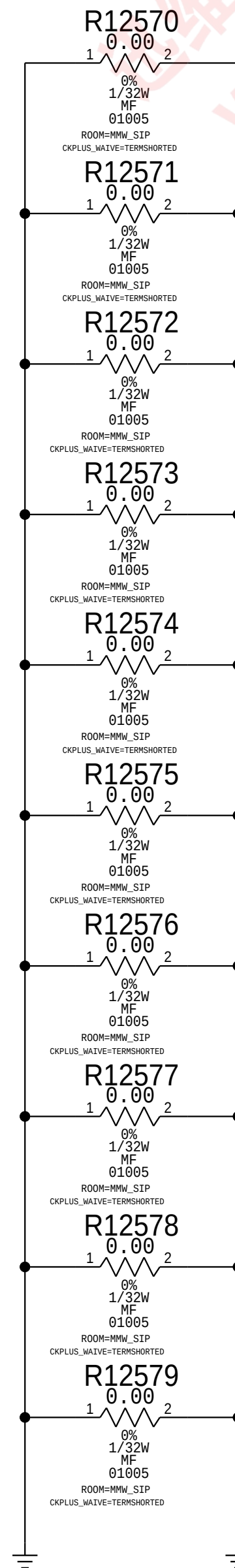
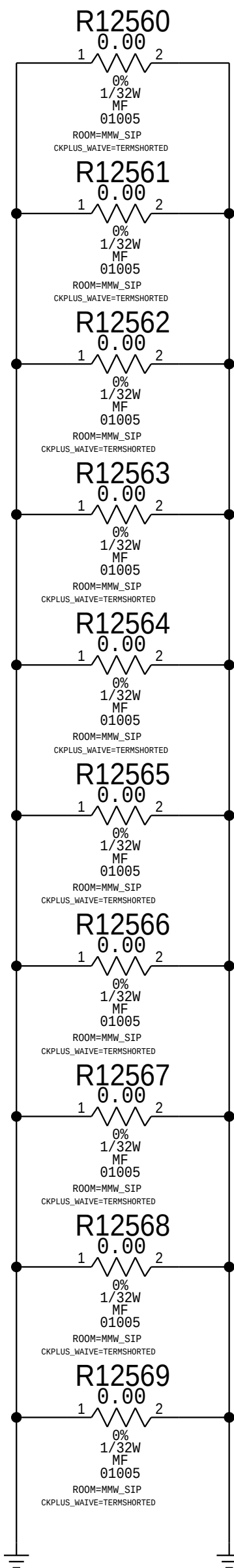
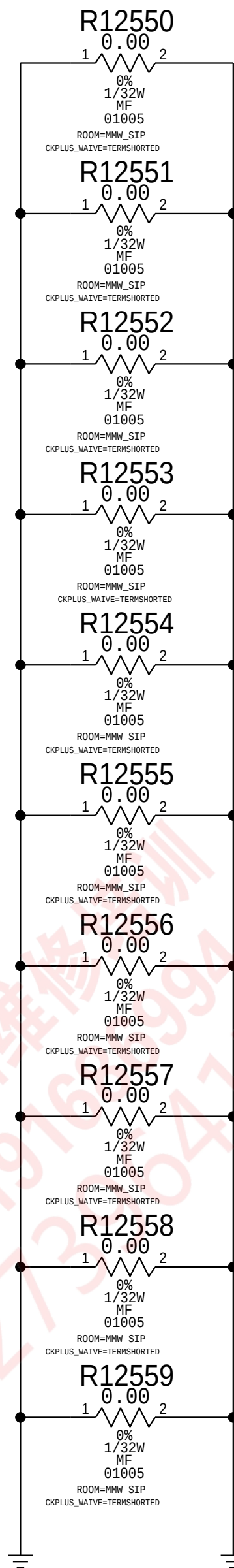


Shields

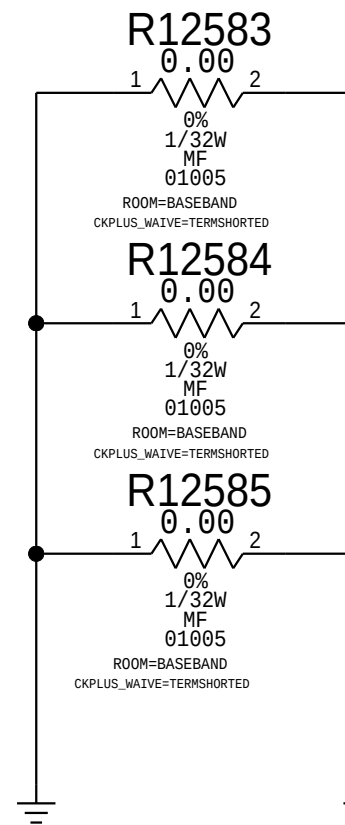


Tangier Underfill Dam

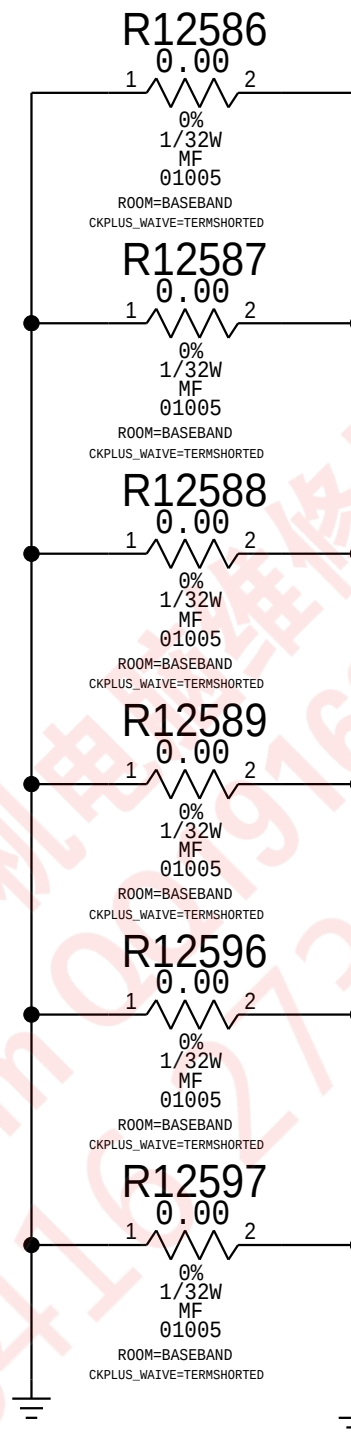
RefDes start at R12550 to avoid conflict w/ mlb_top



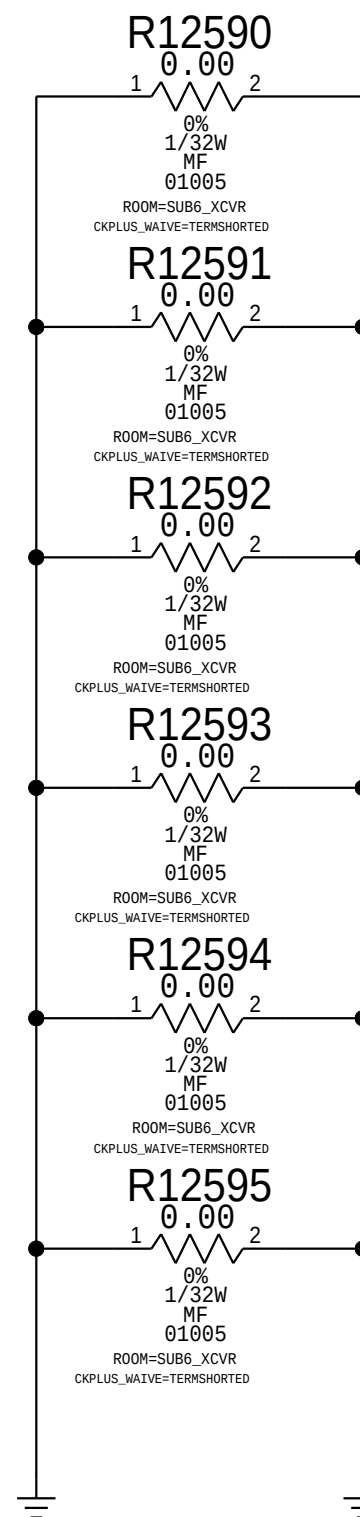
BB



WiFi



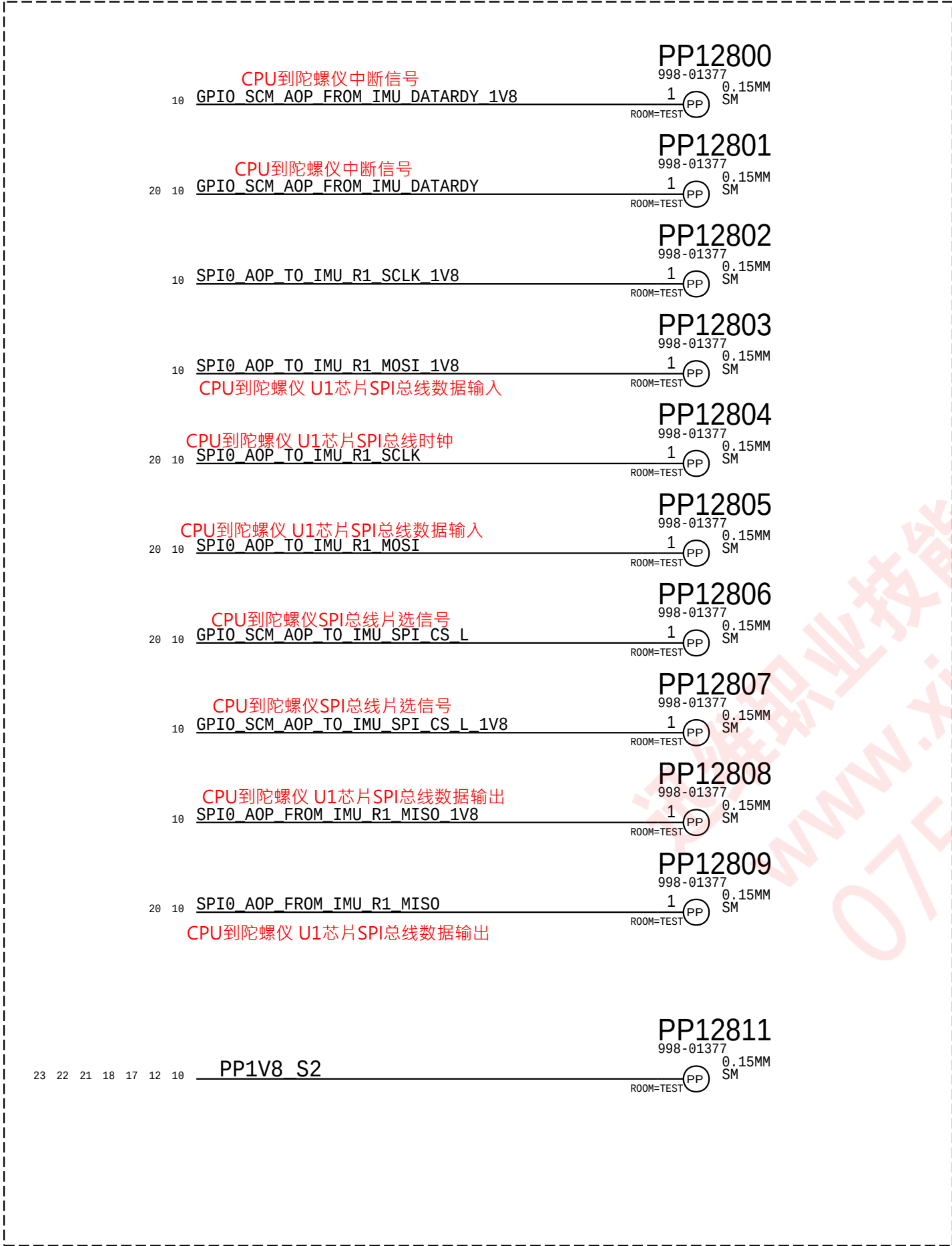
Sub6 XCVR Underfill Dam



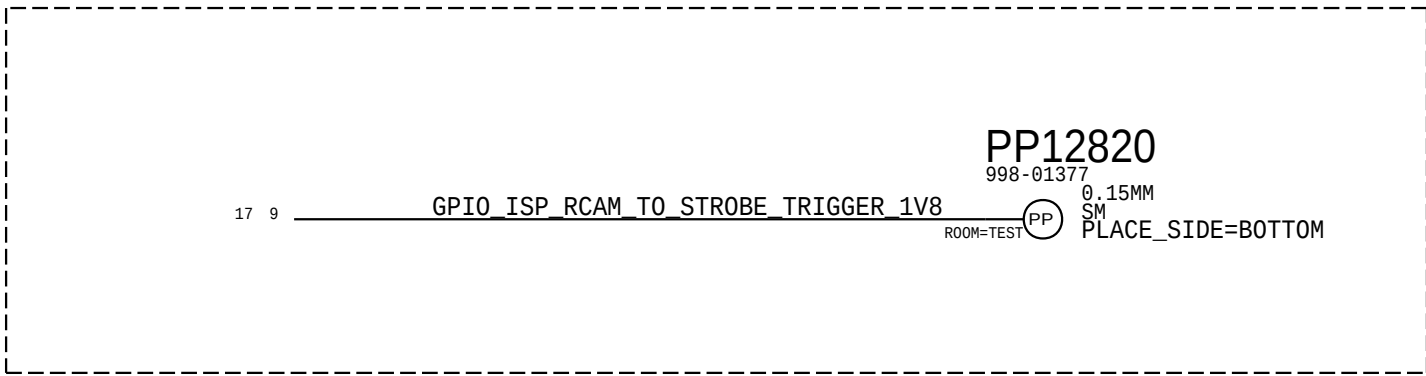
MECHANICAL: Standoffs, Shields, etc

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Kobol



Yeti



PAGE TITLE		
TESTING: Probe Points		
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Interposer Symbol (1/2)

J INT TOP
INTERPOSER-TOP-D53
SM
SYM 1 OF 4

		998-20170	
17 GND	1	IO91	91 GND
17 GND	2	IO92	92 PP_VDD_MAIN_PENINSULA
17 GND	3	IO93	93 GND
17 GND	4	IO94	94 PP_VDD_MAIN_PENINSULA
17 GND	5	IO95	95 GND
17 GND	6	IO96	96 PP_VDD_MAIN_PENINSULA
17 GPIO_AP_CANARY6_LEG2	7	IO97	97 GND
17 GND	8	IO98	98 GND
17 GND	9	IO99	99 GND
17 GPIO_AP_CANARY6_LEG2	10	IO100	100 GND
17 GND	11	IO101	101 GND
17 GND	12	IO102	102 CLK_GPIO_S4_PMU_TO_WLAN_R1_32K
17 GND	13	IO103	103 GND
17 GND	14	IO104	104 PP_VDD_BOOST_NORTH_PA
17 GPIO_AP_CANARY6_LEG1	15	IO105	105 GND
17 GPIO_AP_CANARY6_LEG1	16	IO106	106 PP1V2_S4
17 GND	17	IO107	107 GND
17 PP1V8_S2	18	IO108	108 GPIO_AOP_TO_WLAN_CONTEXT_A
17 GND	19	IO109	109 GND
17 GND	20	IO110	110 PP1V8_S4
17 GND	21	IO111	111 GND
17 GND	22	IO112	112 RFFE_WLAN_TO_UAT_2G_FEM_RST_L
17 GND	23	IO113	113 GND
17 GPIO_ISP_RCAM_TO_STROBE_TRIGGER_1V8	24	IO114	114 RFFE_WLAN_TO_UAT_5G_FEM_RST_L
17 GND	25	IO115	115 GND
17 GND	26	IO116	116 GPIO_SE_TO_BT_PKT_RDY
17 GND	27	IO117	117 GND
17 IO_R1_TO_BB_LAA_SW_CTRL	28	IO118	118 GND
17 GND	29	IO119	119 GND
17 IO_BT_TO_R1_WAKE	30	IO120	120 I2C_BT_TO_NFC_SCL
17 GND	31	IO121	121 GND
17 GND	32	IO122	122 ANALOG_UAT_5G_FEM_TO_WLAN_POTS
17 GND	33	IO123	123 GND
17 PPVAR_RCAM_PVDD	34	IO124	124 GPIO_BB_TO_ALS_GSM_BURST_IND_1V8
17 GND	35	IO125	125 GND
17 GND	36	IO126	126 GND
17 GND	37	IO127	127 GND
17 GPIO_CAM_PMU1_TO_STROBE_HWEN_1V8	38	IO128	128 I2C_BT_TO_NFC_SDA
17 GND	39	IO129	129 GND
17 GPIO_BB_TO_R1_CH9_REQ	40	IO130	130 GPIO_AP_FROM_WLAN_TIME_SYNC
17 GND	41	IO131	131 GND
17 NTC_STROBE_MODULE	42	IO132	132 GPIO_WLAN_TO_UAT_2G_FEM_LNA_BYP_EN
17 GND	43	IO133	133 GND
17 GND	44	IO134	134 GND
17 GPIO_AP_CANARY5_LEG1	45	IO135	135 GND
17 GND	46	IO136	136 GPIO_WLAN_TO_UAT_5G_FEM_LNA_BYP_EN
17 GPIO_AP_CANARY5_LEG1	47	IO137	137 GND
17 GND	48	IO138	138 GPIO_AP_TO_BB_PEAK_PWR_IND
17 GND	49	IO139	139 GND
17 GND	50	IO140	140 SWD_AP_BT_BB_SWDIO
17 GPIO_AP_CANARY5_LEG2	51	IO141	141 GND
17 GND	52	IO142	142 GND
17 GND	53	IO143	143 GND
17 GPIO_AP_CANARY5_LEG2	54	IO144	144 GPIO_AP_FROM_BB_RESET_DETECT_L
17 GND	55	IO145	145 GND
17 GND	56	IO146	146 SPMI0_EVENTS_AOP_BT_WLAN_NFC_DATA
17 GND	57	IO147	147 GND
17 GND	58	IO148	148 SPMI0_EVENTS_AOP_TO_WLAN_NFC_CLK
17 GND	59	IO149	149 GND
17 GND	60	IO150	150 GND
		IO151	151 GND
17 GND	62	IO152	152 GPIO_AP_FROM_BT_AUDIO_SYNC
17 GPIO_AP_CANARY5_LEG3	63	IO153	153 GND
17 GND	64	IO154	154 GPIO_AP_TO_BB_TIME_MARK
17 GND	65	IO155	155 GND
17 GND	66	IO156	156 NTC_RADIO_BB_P
17 GPIO_AP_CANARY5_LEG3	67	IO157	157 GND
17 GND	68	IO158	158 GND
17 GPIO_AP_CANARY5_LEG4	69	IO159	159 GND
17 GND	70	IO160	160 UART1_AOP_FROM_BB_RXD
17 GND	71	IO161	161 GND
17 GND	72	IO162	162 UART1_AOP_TO_BB_TXD
17 GND	73	IO163	163 GND
17 GND	74	IO164	164 GPIO_AOP_TO_BB_FORCE_PWM
17 GND	75	IO165	165 GND
17 GND	76	IO166	166 GND
17 GND	77	IO167	167 GND
17 GND	78	IO168	168 PP_VDD_RF_1V2
17 GND	79	IO169	169 GND
17 GND	80	IO170	170 GND
17 GND	81	IO171	171 GND
17 GND	82	IO172	172 GPIO_AOP_TO_WLAN_CONTEXT_B
17 GND	83	IO173	173 GND
17 GND	84	IO174	174 GND
17 GND	85	IO175	175 GND
17 GND	86	IO176	176 PP_VDD_BOOST_BBPMU
17 GPIO_AP_CANARY5_LEG4	87	IO177	177 GND
17 GND	88	IO178	178 GND
17 GND	89	IO179	179 GND
17 PP_VDD_MAIN_PENINSULA	90	IO180	180 PP_VDD_MAIN_QET_BBPMU_APTL

ROOM=INTERPOSER

J INT TOP
INTERPOSER-TOP-D53
SM
SYM 2 OF 4

		998-20170	
18 GND	181	IO271	271 50_TRX_LB_PAD_ANT1
18 PP_VDD_MAIN_QET_BBPMU_APTL	182	IO182	272 GND
18 GND	183	IO183	273 GND
18 PP_VDD_MAIN_QET_BBPMU_APTL	184	IO184	274 GND
18 GND	185	IO185	275 GND
18 PP_VDD_MAIN_QET_BBPMU_APTL	186	IO186	276 50_TRX_IN_LMB_PAD_2G_MB_OUT_M
18 GND	187	IO187	277 GND
18 GND	188	IO188	278 GND
18 GND	189	IO189	279 GND
18 PP3V0_S2	190	IO190	280 50_TRX_IN_LMB_PAD_LMB_ANT_M
18 GND	191	IO191	281 GND
18 GND	192	IO192	282 GND
18 GND	193	IO193	283 GND
18 GND	194	IO194	284 GND
18 GND	195	IO195	285 50_TRX_WLAN_G_BT_ANT3_NPLXR
18 GND	196	IO196	286 GND
18 GND	197	IO197	287 GND
18 GND	198	IO198	288 GND
18 GND	199	IO199	
18 GPIO_BT_TO_BT_SPKR_TRIG	200	IO200	290 GND
18 GND	201	IO201	291 50_TX_IN_XCVR_TX0_MB2G
18 GND	202	IO202	292 GND
18 GND	203	IO203	293 GND
18 GND	204	IO204	294 GND
18 GND	205	IO205	
18 GND	206	IO206	296 GND
18 GND	207	IO207	297 50_TRX_ANT7_UHB_N79
18 GND	208	IO208	298 GND
18 GND	209	IO209	299 GND
18 RFFE_WLAN_TO_LAT_2G_FEM_RST_L	210	IO210	300 GND
18 GND	211	IO211	301 50_TX_IN_XCVR_TX0_LMB
18 GPIO_WLAN_TO_LAT_2G_FEM_LNA_BYP_EN	212	IO212	302 GPIO_AOP_CANARY7_LEG3
18 GND	213	IO213	303 GND
18 ANALOG_LAT_2G_FEM_TO_WLAN_POTS	214	IO214	304 GND
18 GND	215	IO215	305 GND
18 PP_LD06_VIO_1V8	216	IO216	306 PP_VDD_MAIN_SGFEM1
18 GND	217	IO217	307 PP_VDD_MAIN_SGFEM1
18 SHIELD_RFFE_WLAN_TO_LAT_FEMS_CLK	218	IO218	308 GND
18 GND	219	IO219	309 GND
18 SHIELD_RFFE_WLAN_TO_LAT_FEMS_DATA	220	IO220	310 GPIO_AOP_CANARY7_LEG2
18 GND	221	IO221	311 GND
18 PP1V8_S2	222	IO222	312 GND
18 GND	223	IO223	313 GND
18 GND	224	IO224	314 GND
18 GND	225	IO225	315 GND
18 50_RX_LMB_PAD_LMB_PRX_OUT_M	226	IO226	316 GND
18 50_TX_IN_TX2_UHB_PAD_TX_N79_IN	227	IO227	317 GND
18 GND	228	IO228	318 GND
18 GND	229	IO229	319 GND
18 GND	230	IO230	320 GND
18 GND	231	IO231	321 GPIO_AOP_CANARY7_LEG2
18 50_TX_IN_XCVR_TX0_LB	232	IO232	322 GND
18 GND	233	IO233	323 GND
18 GND	234	IO234	324 GND
18 GND	235	IO235	325 GND
18 50_UHB_L_CPLR_OUT	236	IO236	326 GPIO_AOP_CANARY7_LEG1
18 50_RX_XCVR_PRX6_LB_A_RX0	237	IO237	327 GND
18 GND	238	IO238	328 GPIO_AOP_CANARY7_LEG1
18 GND	239	IO239	329 GND
18 GND	240	IO240	330 GND
18 GND	241	IO241	331 GND
18 PP_QET0_2G_PA	242	IO242	332 GND
18 GND	243	IO243	333 GND
18 PP_QET0_2G_PA	244	IO244	334 GND
18 GND	245	IO245	335 GND
18 PP_QET0_PA	246	IO246	336 50_TRX_PAUHL_N79_ANT2_TO_ANT9
18 GND	247	IO247	337 GND
18 PP_QET0_PA	248	IO248	338 GND
18 GND	249	IO249	339 50_TRX_UHB_PAD_PORTB_TX_M
18 GND	250	IO250	340 GND
18 GND	251	IO251	341 GND
18 50_WLAN_G_C1_TXRX_FEM_M	252	IO252	342 50_TRX_UHB_PAD_N79_PORTB_TX_M
18 GND	253	IO253	343 GND
18 GND	254	IO254	344 GND
18 GND	255	IO255	345 50_TRX_PAUHL_UHB_ANT2_TO_ANT9
18 SHIELD_RFFE1_SCLK_SDR	256	IO256	346 GND
18 GND	257	IO257	347 GND
18 GND	258	IO258	348 50_TRX_LB_PORTB_RX1_M
18 GND	259	IO259	349 GND
18 SHIELD_RFFE1_SDATA_SDR	260	IO260	350 GND
18 GND	261	IO261	
18 GND	262	IO262	352 GND
18 GPIO_AOP_CANARY7_LEG4	263	IO263	
18 GND	264	IO264	354 GND
18 GPIO_AOP_CANARY7_LEG4	265	IO265	355 50_TX_IN_XCVR_TX0_LB2G
18 GND	266	IO266	356 GND
18 GPIO_AOP_CANARY7_LEG3	267	IO267	357 50_RX_XCVR_PRX10_LB_PRX_OUT2
18 GND	268	IO268	358 50_TRX_LB_PORTB_TX_M
18 GND	269	IO269	359 GND
18 GND	270	IO270	360 GND

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19 GND	361	IO361	451 GND
19 PP_APT_L_PA	362	IO362	452 GND
19 PP_APT_L_PA	363	IO363	453 GND
19 GND	364	IO364	454 GPIO_NUB_TO_BBPMU_CLK_EN_DOTARA
19 GND	365	IO365	455 GND
19 PP_QET1_PA	366	IO366	456 I2S2_AP_FROM_BB_BCLK
19 PP_QET1_PA	367	IO367	457 GND
19 GND	368	IO368	458 I2S2_AP_FROM_BB_LRCLK
19 GND	369	IO369	459 GND
19 GND	370	IO370	460 GND
19 50_RX_XCVR_PRX15_N79_RX_OUT2	371	IO371	461 GND
19 GND	372	IO372	462 I2S2_AP_FROM_BB_DIN
19 GND	373	IO373	463 GND
19 GND	374	IO374	464 I2S2_AP_TO_BB_DOUT
19 50_RX_XCVR_PRX17_UHB_RX_OUT2	375	IO375	465 GND
19 50_RX_XCVR_PRX14_N79_RX_OUT1	376	IO376	466 GND
19 GND	377	IO377	467 GND
19 GND	378	IO378	
19 GND	379	IO379	469 GND
19 GND	380	IO380	470 GND
19 50_RX_XCVR_PRX16_UHB_RX_OUT1	381	IO381	471 GND
19 SHIELD_RFFE2_SDATA_SDR	382	IO382	472 GND
19 GND	383	IO383	473 GND
19 SHIELD_RFFE2_SCLK_SDR	384	IO384	474 GPIO_PMU_FROM_BB_PCIE_HOST_WAKE_L
19 SHIELD_RFFE8_SDATA_SDR	385	IO385	475 GND
19 SHIELD_RFFE8_SCLK_SDR	386	IO386	476 GPIO_S4_PMU_TO_WLAN_REG_ON
19 PP_VDD_BOOST_SOUTH_PA	387	IO387	477 GND
19 GND	388	IO388	478 GPIO_PMU_FROM_WLAN_HOST_WAKE
19 GND	389	IO389	479 GND
19 GND	390	IO390	480 GND
19 50_TX_IN_TX2_UHB_PAD_TX_UHB_IN_1	391	IO391	481 GND
19 GND	392	IO392	482 GPIO_SEQ_PMU_TO_BBPMU_RESET_L
19 GND	393	IO393	483 GND
19 GND	394	IO394	484 PCIE_GP1_AP_TO_WLAN_PERST_L
19 50_TX_IN_XCVR_TX1_LB_JB	395	IO395	485 GND
19 SHIELD_RFFE3_SDATA_SDR	396	IO396	486 PCIE_GP1_AP_BT_WLAN_CLKREQ_L
19 GND	397	IO397	487 GND
19 SHIELD_RFFE3_SCLK_SDR	398	IO398	488 GND
19 GND	399	IO399	489 GND
19 GND	400	IO400	490 PCIE_GP2_AP_TO_BB_PERST_L
19 GND	401	IO401	491 GND
19 PP_VDD_MAIN_QET_BBPMU_APTL	402	IO402	492 PCIE_GP2_AP_BT_BB_CLKREQ_L
19 GND	403	IO403	493 GND
19 PP_VDD_MAIN_QET_BBPMU_APTL	404	IO404	494 SWD_AP_TO_MANY_SWCLK
19 GND	405	IO405	495 GND
19 PP_VDD_MAIN_QET_BBPMU_APTL	406	IO406	496 GPIO_AP_TO_BB_COREDUMP
19 GND	407	IO407	497 GND
19 PP_VDD_MAIN_QET_BBPMU_APTL	408	IO408	
19 GND	409	IO409	499 GND
19 GND	410	IO500	500 GND
19 GND	411	IO501	501 GND
19 GND	412	IO502	502 GND
19 GND	413	IO503	503 GND
19 PP1V2_JARVIS_S2	414	IO504	504 CLK_BBPMU_TO_NFC_38M4
19 GND	415	IO505	505 GND
19 GND	416	IO506	506 GND
19 GND	417	IO507	507 GND
19 GND	418	IO508	508 90_PCIE_GP1_AP_TO_WLAN_REFCLK_N
19 GND	419	IO509	509 GND
19 GPIO_SCM_AOP_FROM_JARVIS_INT	420	IO510	510 90_PCIE_GP1_AP_TO_WLAN_REFCLK_P
19 GND	421	IO511	511 GND
19 I2C1_AOP_SCL	422	IO512	512 GND
19 GND	423	IO513	513 GND
19 I2C1_AOP_SDA	424	IO514	514 90_PCIE_GP1_AP_FROM_WLAN_RX_N
19 GND	425	IO515	515 GND
19 GND	426	IO516	516 90_PCIE_GP1_AP_FROM_WLAN_RX_P
19 GND	427	IO517	517 GND
19 90_PCIE_GP2_AP_TO_BB_TX_N	428	IO518	518 GND
19 GND	429	IO519	519 GND
19 90_PCIE_GP2_AP_TO_BB_TX_P	430	IO520	520 90_PCIE_GP1_AP_TO_WLAN_TX_P
19 GND	431	IO521	521 GND
19 GND	432	IO522	522 90_PCIE_GP1_AP_TO_WLAN_TX_N
19 GND	433	IO523	523 GND
19 90_PCIE_GP2_AP_FROM_BB_RX_P	434	IO524	524 GND
19 GND	435	IO525	525 GND
19 90_PCIE_GP2_AP_FROM_BB_RX_N	436	IO526	526 CLK_BBPMU_TO_DOTARA_38M4
19 GND	437	IO527	527 GND
19 GND	438	IO528	528 GND
19 GND	439	IO529	529 GND
19 90_PCIE_GP2_AP_TO_BB_REFCLK_N	440	IO530	530 ISP_RCAM_TO_STROBE_KRAKEN_WLAN_FLASH_TRIG
19 GND	441	IO531	531 GND
19 90_PCIE_GP2_AP_TO_BB_REFCLK_P	442	IO532	532 GND
19 GND	443	IO533	533 GND
19 GND	444	IO534	534 GND
19 GND	445	IO535	535 GND
19 PP_VDD_MAIN_QET_BBPMU_APTL	446	IO446	
19 GND	447	IO447	537 GND
19 PP_VDD_MAIN_QET_BBPMU_APTL	448	IO448	538 GND
19 GND	449	IO539	539 GND
19 PP_VDD_MAIN_QET_BBPMU_APTL	450	IO450	540 GND

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J INT TOP INTERPOSER-TOP-D53 SM SYM 4 OF 4			
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20 GND	541	IO541	IO631
20 GND	542	IO542	IO632
20 50_WLAN_A_C0_TXRX_FEM_M	543	IO543	IO633
20 GND	544	IO544	IO634
20 GND	545	IO545	IO635
20 50_WLAN_G_C0_TXRX_FEM_M	546	IO546	IO636
20 GND	547	IO547	IO637
20 GND	548	IO548	IO638
20 50_TRX_PAUHB_UHB_ANT1_TO_ANT4	549	IO549	IO639
20 GND	550	IO550	IO640
20 GND	551	IO551	IO641
20 50_TRX_PAUHB_U79_ANT1_TO_ANT4	552	IO552	IO642
20 GND	553	IO553	IO643
20 GND	554	IO554	IO644
20 GND	555	IO555	IO645
20 GND	556	IO556	IO646
20 GND	557	IO557	IO647
20 GND	558	IO558	IO648
20 GND	559	IO559	IO649
20 IO_NFC_TO_BBPMU_CLKREQ	560	IO560	IO650
20 GND	561	IO561	IO651
20 GND	562	IO562	IO652
20 GND	563	IO563	IO653
20 GND	564	IO564	IO654
20 GND	565	IO565	IO655
20 GND	566	IO566	IO656
20 GND	567	IO567	IO657
20 GND	568	IO568	IO658
20 GND	569	IO569	IO659
20 GND	570	IO570	IO660
20 GND	571	IO571	IO661
20 GND	572	IO572	IO662
20 GND	573	IO573	IO663
20 ANALOG_UAT_2G_FEM_TO_WLAN_PDTS	574	IO574	IO664
20 GND	575	IO575	IO665
20 GND	576	IO576	IO666
20 GND	577	IO577	IO667
20 GND	578	IO578	IO668
20 GND	579	IO579	IO669
20 GND	580	IO580	IO670
20 GND	581	IO581	IO671
20 50_TRX_CPLR_U_CPL2_ANT_LMB_MB_HB	582	IO582	IO672
20 GND	583	IO583	IO673
20 50_TRX_ANT8_UHB_N79	584	IO584	IO674
20 GND	585	IO585	IO675
20 GND	586	IO586	IO676
20 GND	587	IO587	IO677
20 GND	588	IO588	IO678
20 GND	589	IO589	IO679
20 GND	590	IO590	IO680
20 GPIO_AP_CANARY1_LEG1	591	IO591	IO681
20 GND	592	IO592	IO682
20 GND	593	IO593	IO683
20 GPIO_AP_CANARY1_LEG1	594	IO594	IO684
20 GND	595	IO595	IO685
20 GND	596	IO596	IO686
20 GND	597	IO597	IO687
20 GND	598	IO598	IO688
20 GND	599	IO599	IO689
20 GND	600	IO600	IO690
20 GND	601	IO601	IO691
20 GND	602	IO602	IO692
20 GND	603	IO603	IO693
20 GND	604	IO604	IO694
20 GPIO_AP_CANARY1_LEG2	605	IO605	IO695
20 GND	606	IO606	IO696
20 GND	607	IO607	IO697
20 GND	608	IO608	IO698
20 GND	609	IO609	IO699
20 GND	610	IO610	IO700
20 GND	611	IO611	IO701
20 GND	612	IO612	IO702
20 GPIO_AP_CANARY1_LEG2	613	IO613	IO703
20 PP_LD06_VIO_1V8	614	IO614	IO704
20 GND	615	IO615	IO705
20 GPIO_BB_TO_R1_COEX	616	IO616	IO706
20 GND	617	IO617	IO707
20 IO_BT_TO_R1_TIME_SYNC	618	IO618	IO708
20 GND	619	IO619	IO709
20 GPIO_WLAN_TO_R1_COEX	620	IO620	IO710
20 GND	621	IO621	IO711
20 GND	622	IO622	IO712
20 GND	623	IO623	IO713
20 GND	624	IO624	IO714
20 50_TRX_ANT2_LB_LMB_MB_HB_L1	625	IO625	IO715
20 GND	626	IO626	IO716
20 50_LAA_RX_XCVR_DRX10	627	IO627	IO717
20 GND	628	IO628	IO718
20 GND	629	IO629	IO719
20 SHIELD_RFFE5_SCLK_SDR	630	IO630	IO720
			IO721
			IO722
			IO723
			IO724
			IO725
			IO726
			IO727

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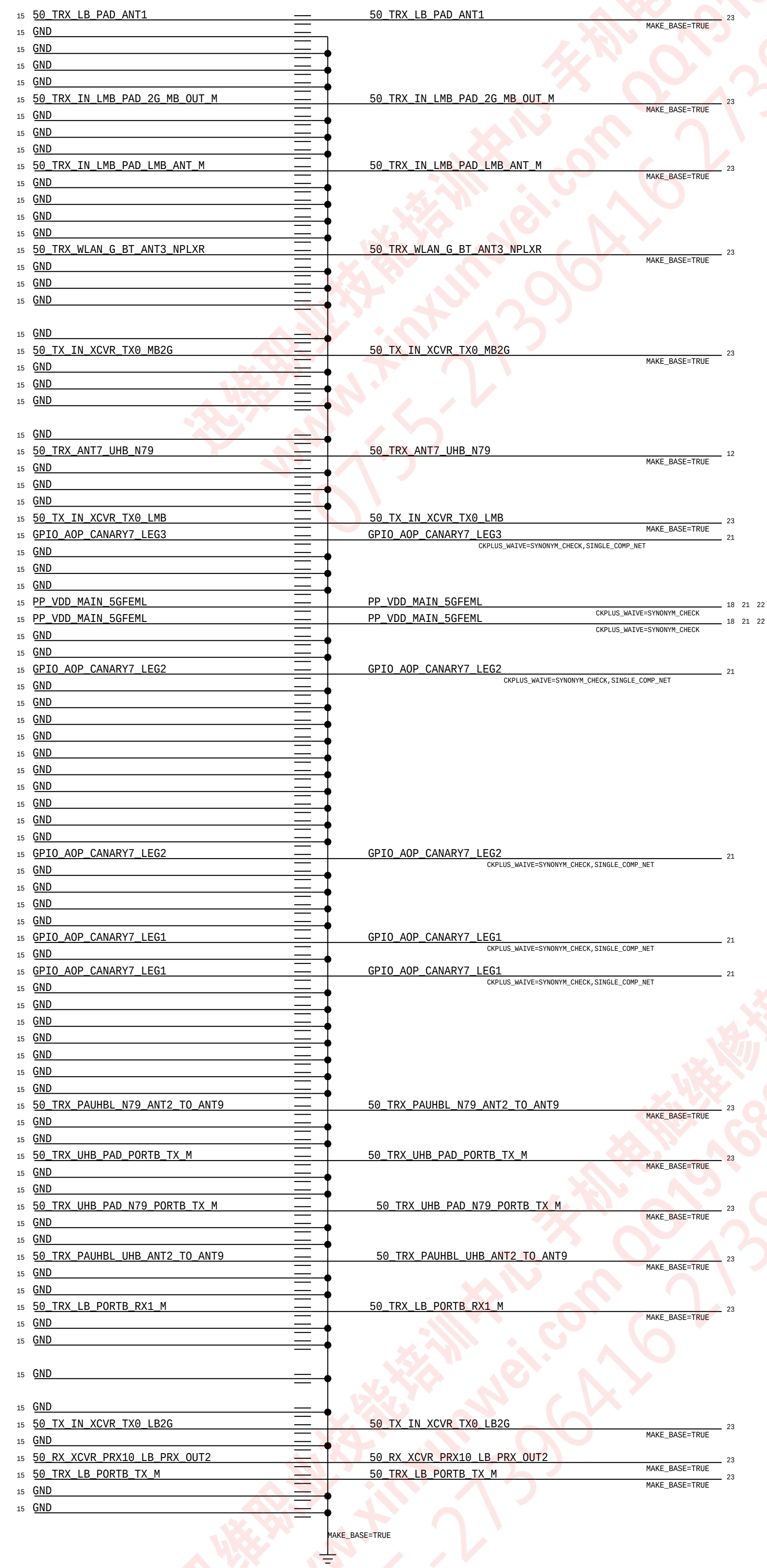
Interposer Aliases (1/4)

15	GND		
15	GND		
15	GND		
15	GND		
15	GND		
15	GND		
15	GPIO_AP_CANARY6_LEG2	GPIO_AP_CANARY6_LEG2	21
15	GND	CKPLUS_WAIVE=SYNONYM_CHECK, SINGLE_COMP_NET	
15	GND		
15	GPIO_AP_CANARY6_LEG2	GPIO_AP_CANARY6_LEG2	21
15	GND	CKPLUS_WAIVE=SYNONYM_CHECK, SINGLE_COMP_NET	
15	GND		
15	GND		
15	GND		
15	GPIO_AP_CANARY6_LEG1	GPIO_AP_CANARY6_LEG1	21
15	GPIO_AP_CANARY6_LEG1	GPIO_AP_CANARY6_LEG1	21
15	GND	CKPLUS_WAIVE=SYNONYM_CHECK, SINGLE_COMP_NET	
15	PP1V8_S2	PP1V8_S2	10 12 14 18 21 22 23
15	GND	MAKE_BASE=TRUE	
15	GND		
15	GND		
15	GND		
15	GND		
15	GPIO_ISP_RCAM_TO_STROBE_TRIGGER_1V8	GPIO_ISP_RCAM_TO_STROBE_TRIGGER_1V8	9 14
15	GND	MAKE_BASE=TRUE	
15	GND		
15	GND		
15	IO_R1_TO_BB_LAA_SW_CTRL	IO_R1_TO_BB_LAA_SW_CTRL	23
15	GND	MAKE_BASE=TRUE	
15	IO_BT_TO_R1_WAKE	IO_BT_TO_R1_WAKE	22
15	GND	MAKE_BASE=TRUE	
15	GND		
15	GND		
15	PPVAR_RCAM_PVDD	PPVAR_RCAM_PVDD	8
15	GND	MAKE_BASE=TRUE	
15	GND		
15	GND		
15	GPIO_CAM_PMU1_TO_STROBE_HWEN_1V8	GPIO_CAM_PMU1_TO_STROBE_HWEN_1V8	9
15	GND	MAKE_BASE=TRUE	
15	GPIO_BB_TO_R1_CH9_REQ	GPIO_BB_TO_R1_CH9_REQ	20
15	GND	MAKE_BASE=TRUE	
15	NTC_STROBE_MODULE	NTC_STROBE_MODULE	9
15	GND	MAKE_BASE=TRUE	
15	GND		
15	GPIO_AP_CANARY5_LEG1	GPIO_AP_CANARY5_LEG1	21
15	GND	CKPLUS_WAIVE=SYNONYM_CHECK, SINGLE_COMP_NET	
15	GPIO_AP_CANARY5_LEG1	GPIO_AP_CANARY5_LEG1	21
15	GND	CKPLUS_WAIVE=SYNONYM_CHECK, SINGLE_COMP_NET	
15	GND		
15	GND		
15	GPIO_AP_CANARY5_LEG2	GPIO_AP_CANARY5_LEG2	21
15	GND	CKPLUS_WAIVE=SYNONYM_CHECK, SINGLE_COMP_NET	
15	GND		
15	GPIO_AP_CANARY5_LEG2	GPIO_AP_CANARY5_LEG2	21
15	GND	CKPLUS_WAIVE=SYNONYM_CHECK, SINGLE_COMP_NET	
15	GND		
15	GND		
15	GND		
15	GND		
15	GND		
15	GND		
15	GND		
15	GPIO_AP_CANARY5_LEG3	GPIO_AP_CANARY5_LEG3	21
15	GND	CKPLUS_WAIVE=SYNONYM_CHECK, SINGLE_COMP_NET	
15	GND		
15	GND		
15	GPIO_AP_CANARY5_LEG3	GPIO_AP_CANARY5_LEG3	21
15	GND	CKPLUS_WAIVE=SYNONYM_CHECK, SINGLE_COMP_NET	
15	GPIO_AP_CANARY5_LEG4	GPIO_AP_CANARY5_LEG4	21
15	GND	CKPLUS_WAIVE=SYNONYM_CHECK, SINGLE_COMP_NET	
15	GND		
15	GND		
15	GND		
15	GND		
15	GND		
15	GND		
15	GND		
15	GND		
15	GND		
15	GND		
15	GND		
15	GND		
15	GPIO_AP_CANARY5_LEG4	GPIO_AP_CANARY5_LEG4	21
15	GND	CKPLUS_WAIVE=SYNONYM_CHECK, SINGLE_COMP_NET	
15	GND		
15	PP_VDD_MAIN_PENINSULA	PP_VDD_MAIN_PENINSULA	17 21 22 23
		CKPLUS_WAIVE=SYNONYM_CHECK	
		MAKE_BASE=TRUE	

15	GND		
15	PP_VDD_MAIN_PENINSULA	PP_VDD_MAIN_PENINSULA	17 21 22 23
15	GND	CKPLUS_WAIVE=SYNONYM_CHECK	
15	PP_VDD_MAIN_PENINSULA	PP_VDD_MAIN_PENINSULA	17 21 22 23
15	GND	CKPLUS_WAIVE=SYNONYM_CHECK	
15	PP_VDD_MAIN_PENINSULA	PP_VDD_MAIN_PENINSULA	17 21 22 23
15	GND	CKPLUS_WAIVE=SYNONYM_CHECK	
15	GND		
15	GND	CKPLUS_WAIVE=SYNONYM_CHECK, SINGLE_COMP_NET	21
15	GND		
15	GND		
15	GND		
15	CLK_GPIO_S4_PMU_TO_WLAN_R1_32K	CLK_GPIO_S4_PMU_TO_WLAN_R1_32K	22
15	GND	MAKE_BASE=TRUE	
15	PP_VDD_BOOST_NORTH_PA	PP_VDD_BOOST_NORTH_PA	21 23
15	GND	CKPLUS_WAIVE=SYNONYM_CHECK	
15	PP1V2_S4	PP1V2_S4	22 23
15	GND	MAKE_BASE=TRUE	
15	GPIO_AOP_TO_WLAN_CONTEXT_A	GPIO_AOP_TO_WLAN_CONTEXT_A	22
15	GND	MAKE_BASE=TRUE	
15	PP1V8_S4	PP1V8_S4	22 23
15	GND	MAKE_BASE=TRUE	
15	RFFE_WLAN_TO_UAT_2G_FEM_RST_L	RFFE_WLAN_TO_UAT_2G_FEM_RST_L	22
15	GND	MAKE_BASE=TRUE	
15	RFFE_WLAN_TO_UAT_5G_FEM_RST_L	RFFE_WLAN_TO_UAT_5G_FEM_RST_L	22
15	GND	MAKE_BASE=TRUE	
15	GPIO_SE_TO_BT_PKT_RDY	GPIO_SE_TO_BT_PKT_RDY	22
15	GND	MAKE_BASE=TRUE	
15	GND		
15	GND		
15	I2C_BT_TO_NFC_SCL	I2C_BT_TO_NFC_SCL	22
15	GND	MAKE_BASE=TRUE	
15	ANALOG_UAT_5G_FEM_TO_WLAN_PDTS	ANALOG_UAT_5G_FEM_TO_WLAN_PDTS	22
15	GND	MAKE_BASE=TRUE	
15	GPIO_BB_TO_ALS_GSM_BURST_IND_1V8	GPIO_BB_TO_ALS_GSM_BURST_IND_1V8	23
15	GND	MAKE_BASE=TRUE	
15	GND		
15	GND		
15	I2C_BT_TO_NFC_SDA	I2C_BT_TO_NFC_SDA	22
15	GND	MAKE_BASE=TRUE	
15	GPIO_AP_FROM_WLAN_TIME_SYNC	GPIO_AP_FROM_WLAN_TIME_SYNC	22
15	GND	MAKE_BASE=TRUE	
15	GPIO_WLAN_TO_UAT_2G_FEM_LNA_BYP_EN	GPIO_WLAN_TO_UAT_2G_FEM_LNA_BYP_EN	22
15	GND	MAKE_BASE=TRUE	
15	GND		
15	GND		
15	GPIO_WLAN_TO_UAT_5G_FEM_LNA_BYP_EN	GPIO_WLAN_TO_UAT_5G_FEM_LNA_BYP_EN	22
15	GND	MAKE_BASE=TRUE	
15	GPIO_AP_TO_BB_PEAK_PWR_IND	GPIO_AP_TO_BB_PEAK_PWR_IND	23
15	GND	MAKE_BASE=TRUE	
15	SWD_AP_BT_BB_SWDIO_CPU到基带的SWD总线数据	SWD_AP_BT_BB_SWDIO	23
15	GND	MAKE_BASE=TRUE	
15	GND		
15	GND		
15	GPIO_AP_FROM_BB_RESET_DETECT_L	GPIO_AP_FROM_BB_RESET_DETECT_L	23
15	GND	MAKE_BASE=TRUE	
15	SPMI0_EVENTS_AOP_BT_WLAN_NFC_DATA	SPMI0_EVENTS_AOP_BT_WLAN_NFC_DATA	22
15	GND	MAKE_BASE=TRUE	
15	SPMI0_EVENTS_AOP_TO_WLAN_NFC_CLK	SPMI0_EVENTS_AOP_TO_WLAN_NFC_CLK	22
15	GND	MAKE_BASE=TRUE	
15	GND		
15	GND		
15	GPIO_AP_FROM_BT_AUDIO_SYNC	GPIO_AP_FROM_BT_AUDIO_SYNC	22
15	GND	MAKE_BASE=TRUE	
15	GPIO_AP_TO_BB_TIME_MARK	GPIO_AP_TO_BB_TIME_MARK	23
15	GND	MAKE_BASE=TRUE	
15	NTC_RADIO_BB_P	NTC_RADIO_BB_P	7
15	GND	MAKE_BASE=TRUE	
15	GND		
15	GND		
15	UART1_AOP_FROM_BB_RXD	UART1_AOP_FROM_BB_RXD	23
15	GND	MAKE_BASE=TRUE	
15	UART1_AOP_TO_BB_TXD	UART1_AOP_TO_BB_TXD	23
15	GND	MAKE_BASE=TRUE	
15	GPIO_AOP_TO_BB_FORCE_PWM	GPIO_AOP_TO_BB_FORCE_PWM	23
15	GND	MAKE_BASE=TRUE	
15	GND		
15	GND		
15	PP_VDD_RF_1V2	PP_VDD_RF_1V2	23
15	GND	MAKE_BASE=TRUE	
15	GND		
15	GND		
15	GPIO_AOP_TO_WLAN_CONTEXT_B	GPIO_AOP_TO_WLAN_CONTEXT_B	22
15	GND	MAKE_BASE=TRUE	
15	GND		
15	GND		
15	PP_VDD_BOOST_BBPMU	PP_VDD_BOOST_BBPMU	21 23
15	GND	CKPLUS_WAIVE=SYNONYM_CHECK	
15	GND		
15	GND		
15	PP_VDD_MAIN_QET_BBPMU_APTL	PP_VDD_MAIN_QET_BBPMU_APTL	18 19 21
		CKPLUS_WAIVE=SYNONYM_CHECK	
		MAKE_BASE=TRUE	

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Interposer Aliases (2/4)



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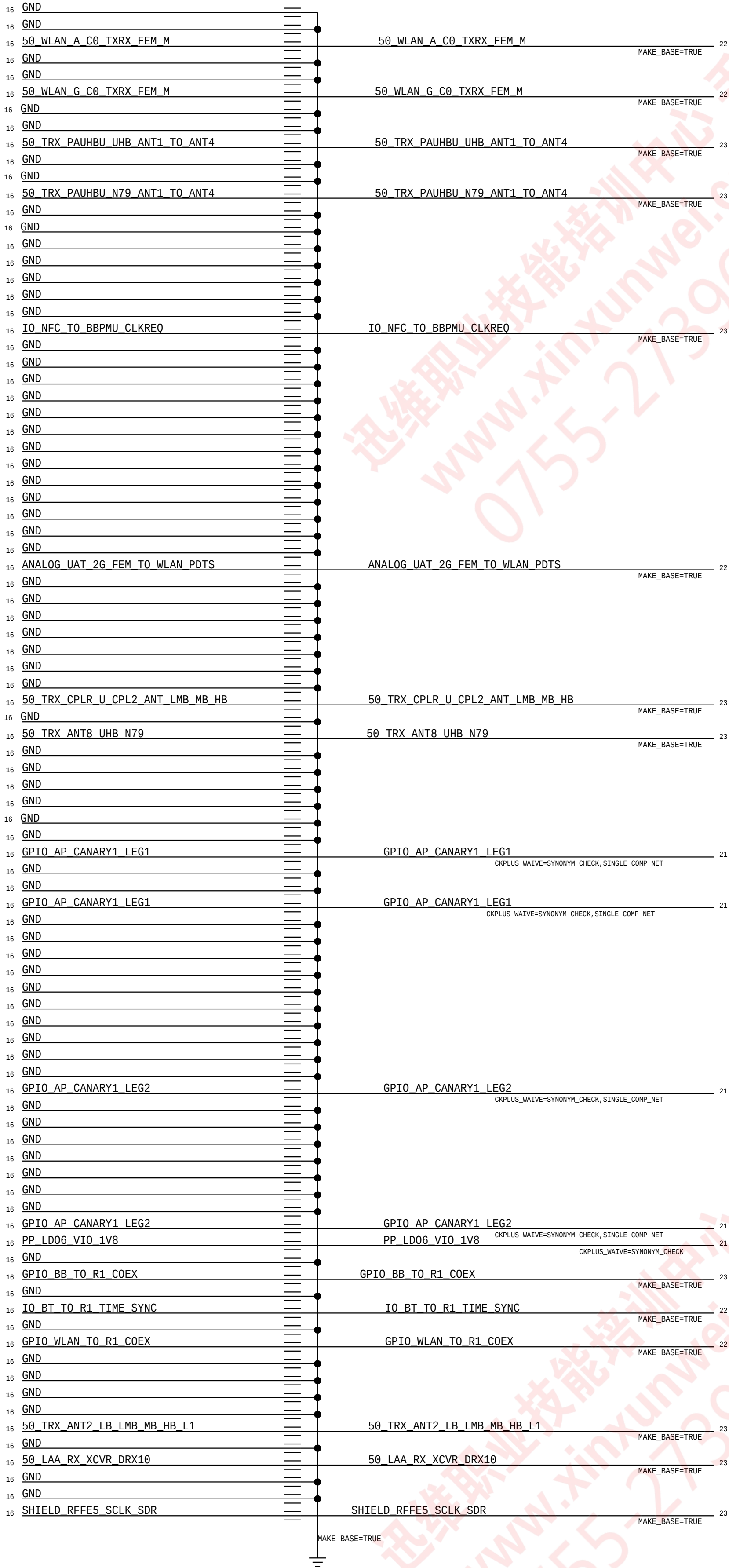
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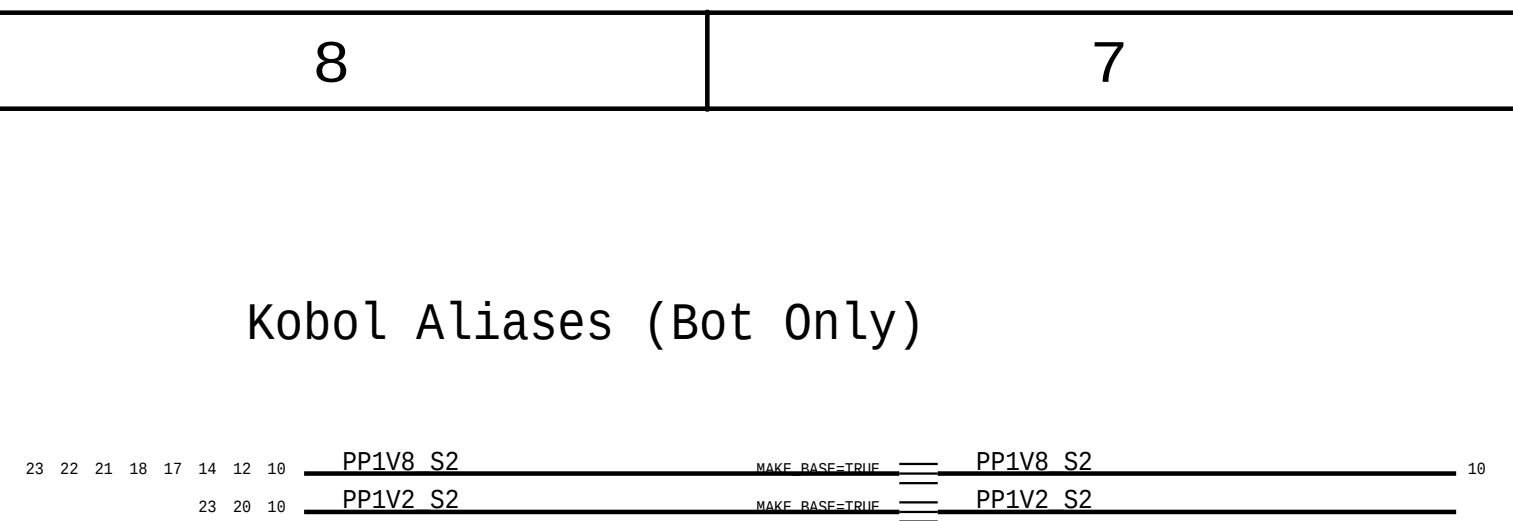
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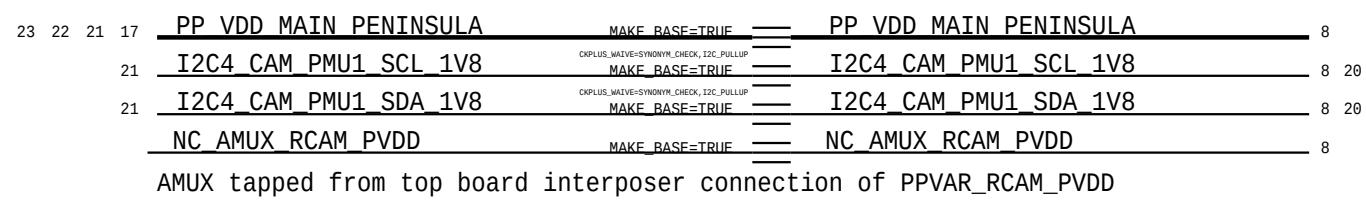
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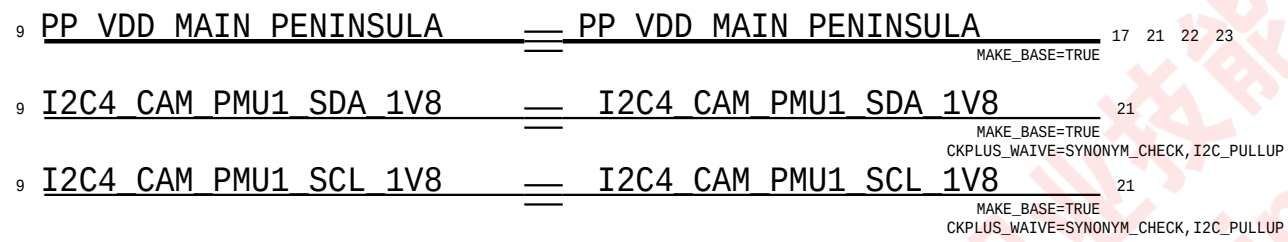
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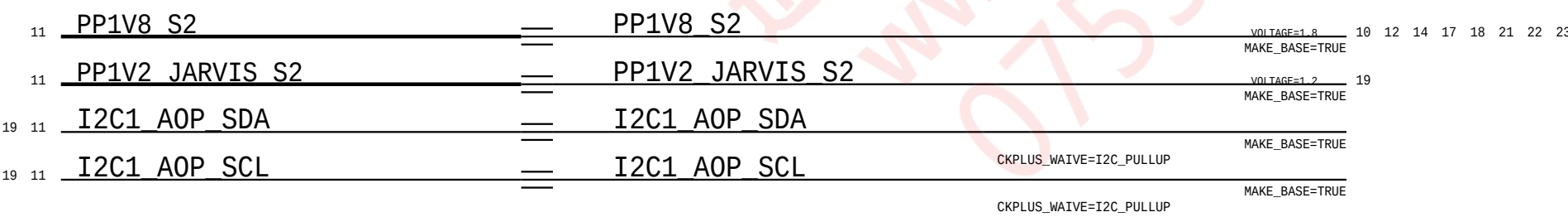
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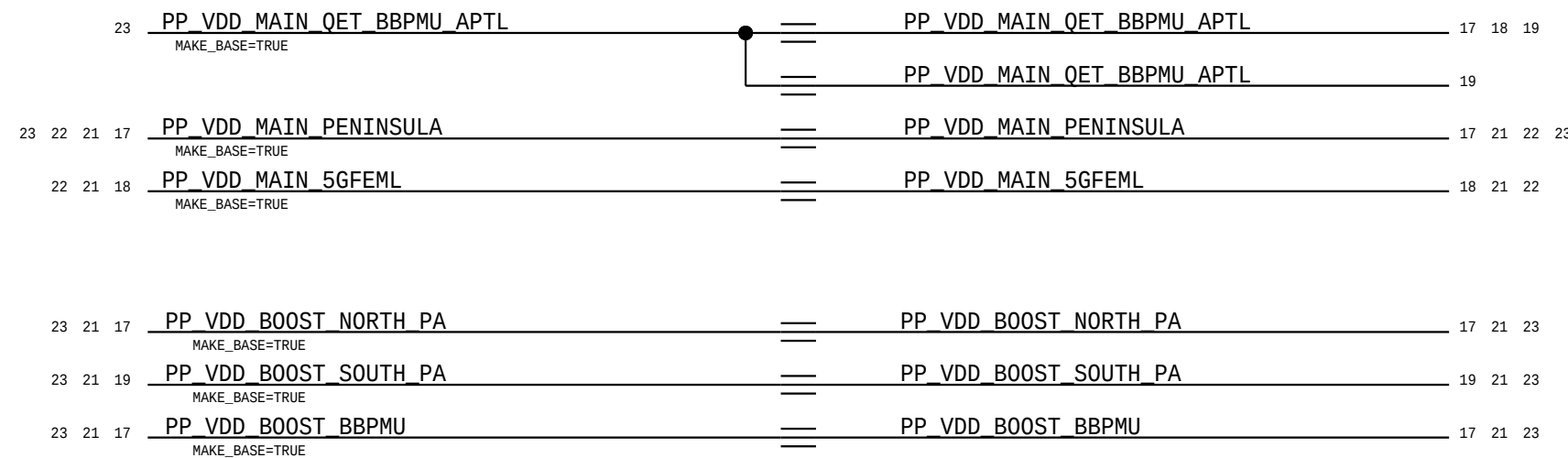
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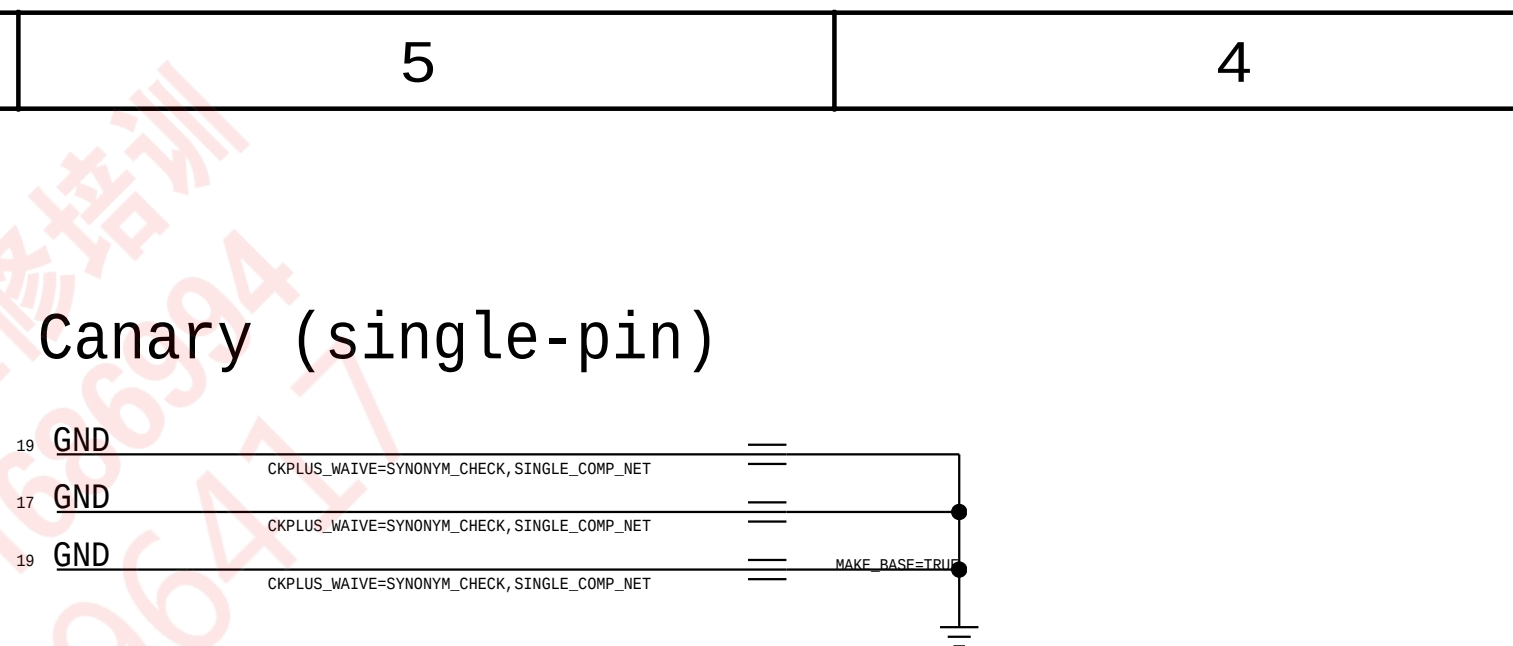
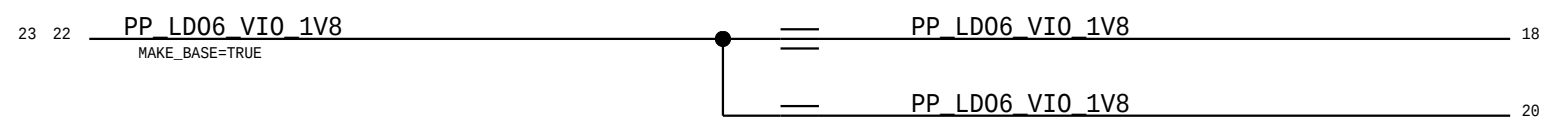
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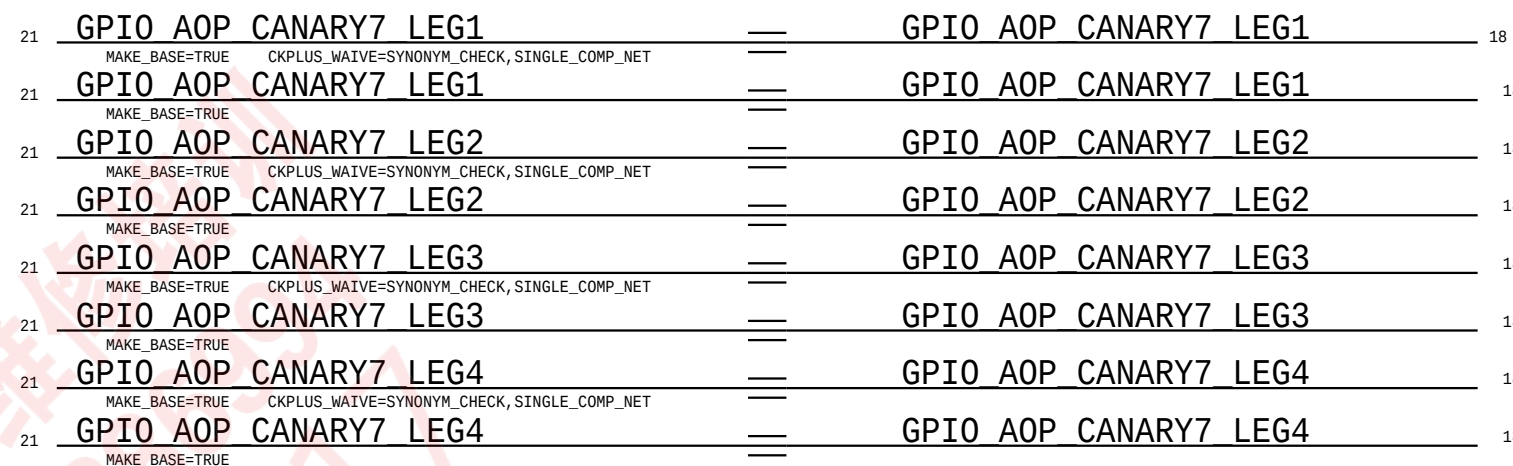
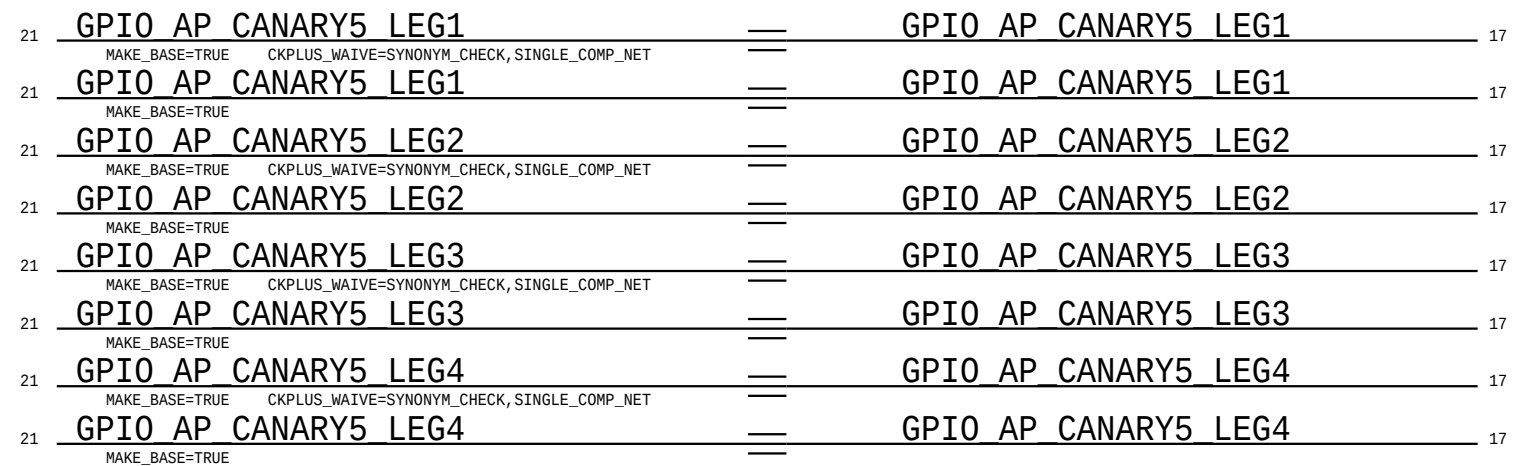
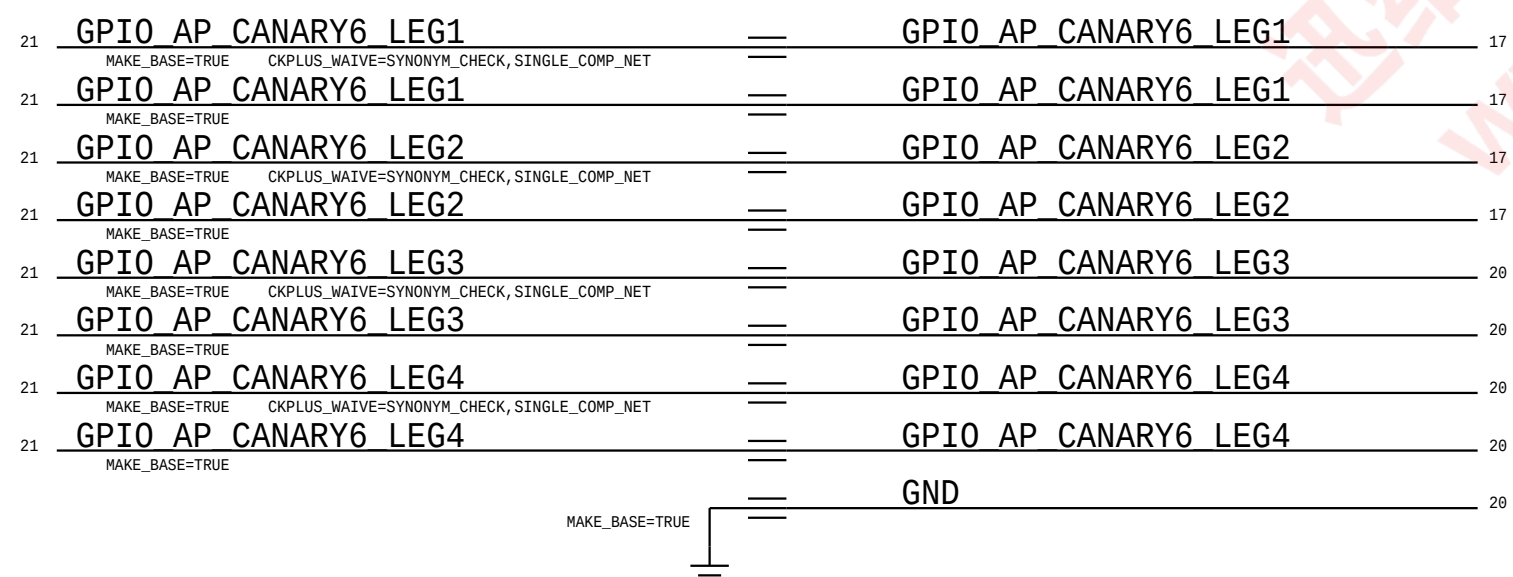
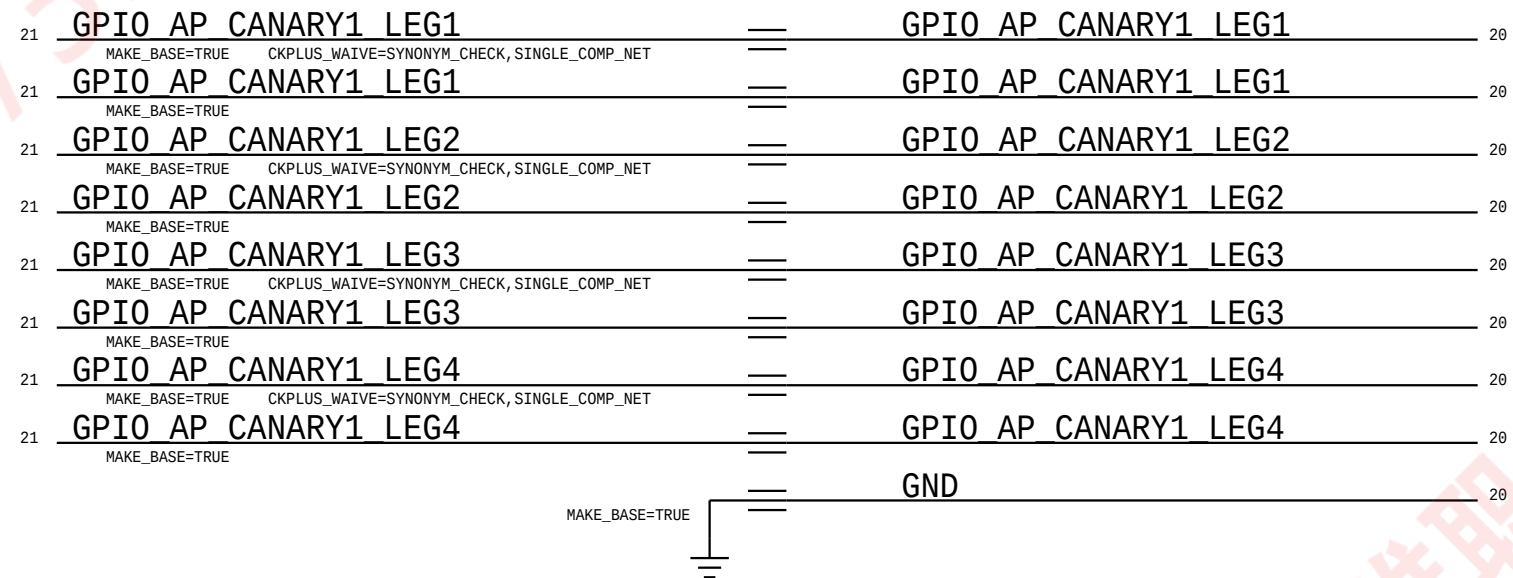
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Power Rail Splits (Top Board)

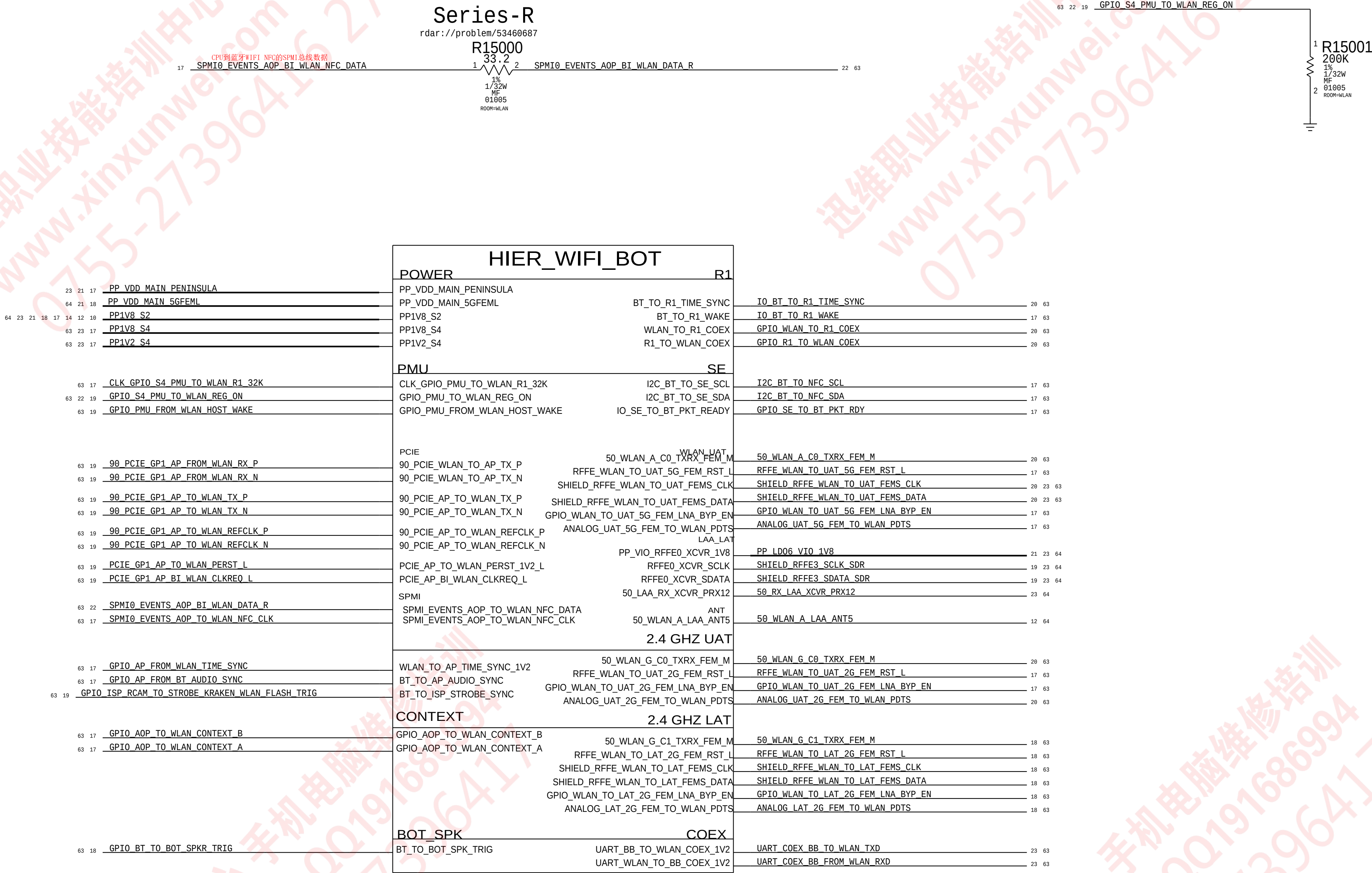


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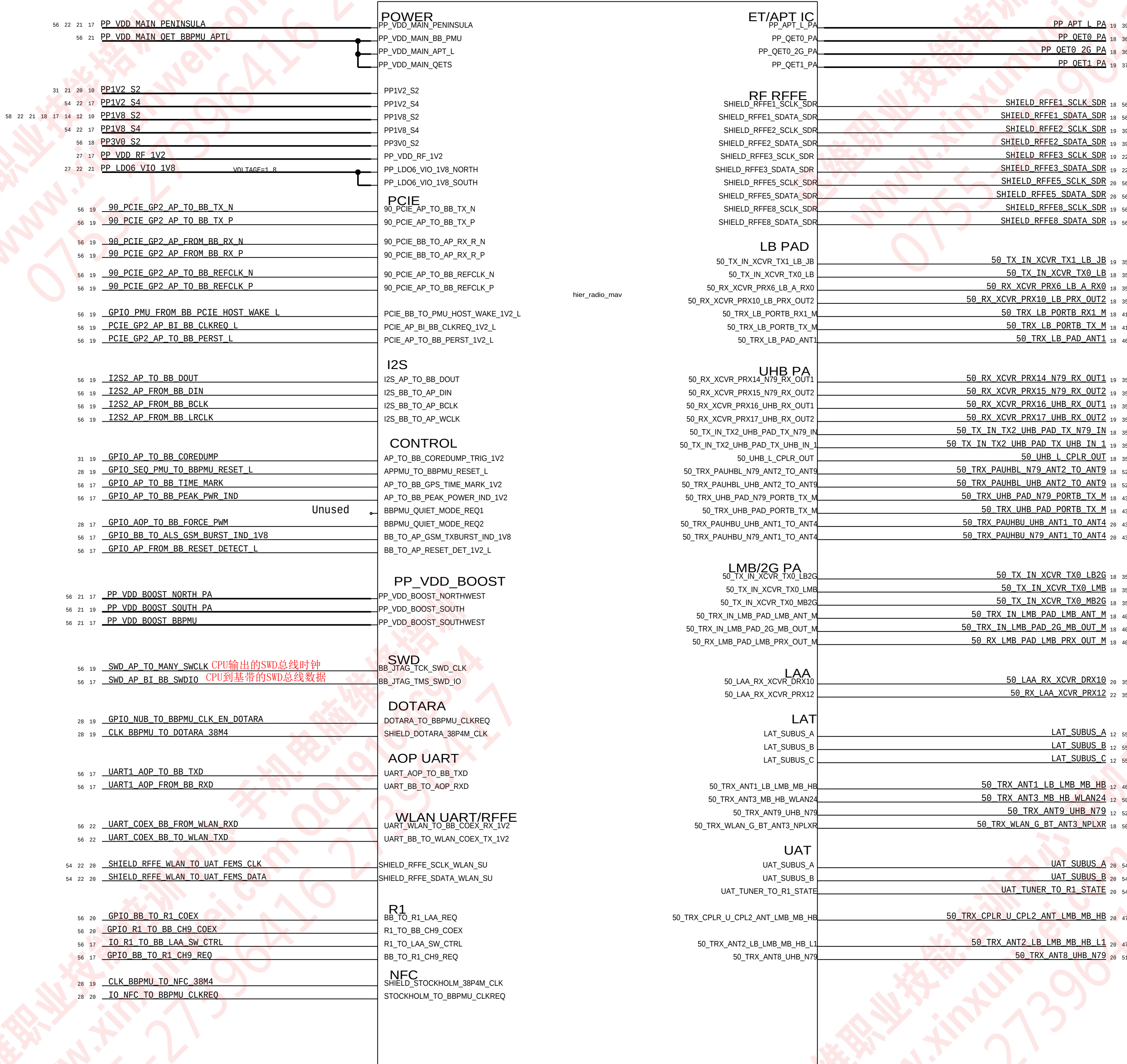
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WiFi Hierarchy



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Radio Hierarchy



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BOM TABLES AND CONSTRAINTS

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ALT TABLES

PART NUMBER	ALTERNATE FOR PART NUMBER	BOM OPTION	REF DES	COMMENTS:
197S00250	197S00249	?	Y601_E	TXC, 30.4MHZ, 8PF, 1612

PART NUMBER	ALTERNATE FOR PART NUMBER	BOM OPTION	REF DES	COMMENTS:
152S01273	152S01227	?	L401_E	TAIYO, 0.47UH, 4.8A
152S01273	152S01227	?	L402_E	TAIYO, 0.47UH, 4.8A
152S01228	152S01120	?	L405_E	SEMCO, 1.0UH, 3.4A
152S01273	152S01227	?	L406_E	TAIYO, 0.47UH, 4.8A
152S01227	152S01273	?	L407_E	SEMCO, 0.47UH, 4.9A

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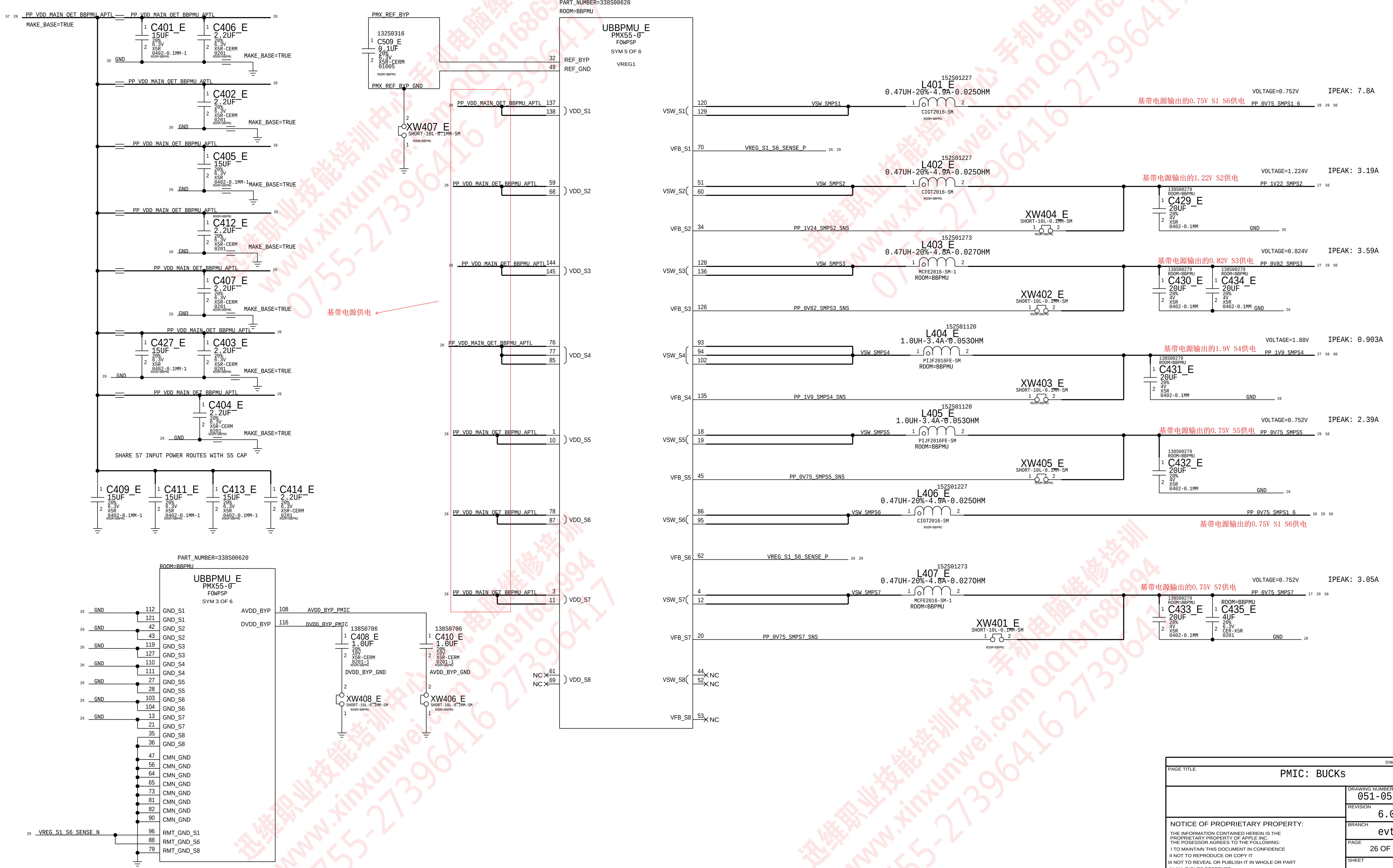
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118S0730	1	RES, MF, 12K OHM, 1%, 1/32W, 01005	R630_E	RF_SKU:WW
118S00088	1	RES, MF, 19.1K OHM, 1%, 1/32W, 01005	R630_E	RF_SKU:ROW
353S02396	1	IC, LB DSM, MW, MCD, LMRXFNJ0-K79, D5LGA20	DSM_LB_E	RF_SKU:MMRF_SKU:MM+KOLKATA
353S02397	1	IC, LB DSM, ROW, MCD, LMRXFNJ0-K83, D5LGA20	DSM_LB_E	RF_SKU:ROW
353S02391	1	IC, HB TX DSM, MW, HFQTLQFA-580, D5LGA61	DSM_HB_E	RF_SKU:MMRF_SKU:MM+KOLKATA
353S02392	1	IC, HB TX DSM, ROW, HFQTLQFA-583, D5LGA61	DSM_HB_E	RF_SKU:ROW
118S00122	1	RES, MF, 27K OHM, 1%, 1/32W, 01005	R631_E	BOARD_REV:PROTO2
118S00193	1	RES, MF, 36.5K OHM, 1%, 1/32W, 01005	R631_E	BOARD_REV:EVT
118S00060	1	RES, MF, 47.5K OHM, 1%, 1/32W, 01005	R631_E	BOARD_REV:CRB
118S00136	1	RES, MF, 60.4K OHM, 1%, 1/32W, 01005	R631_E	BOARD_REV:DVT
118S0760	1	RES, MF, 75K OHM, 1%, 1/32W, 01005	R631_E	BOARD_REV:PVT

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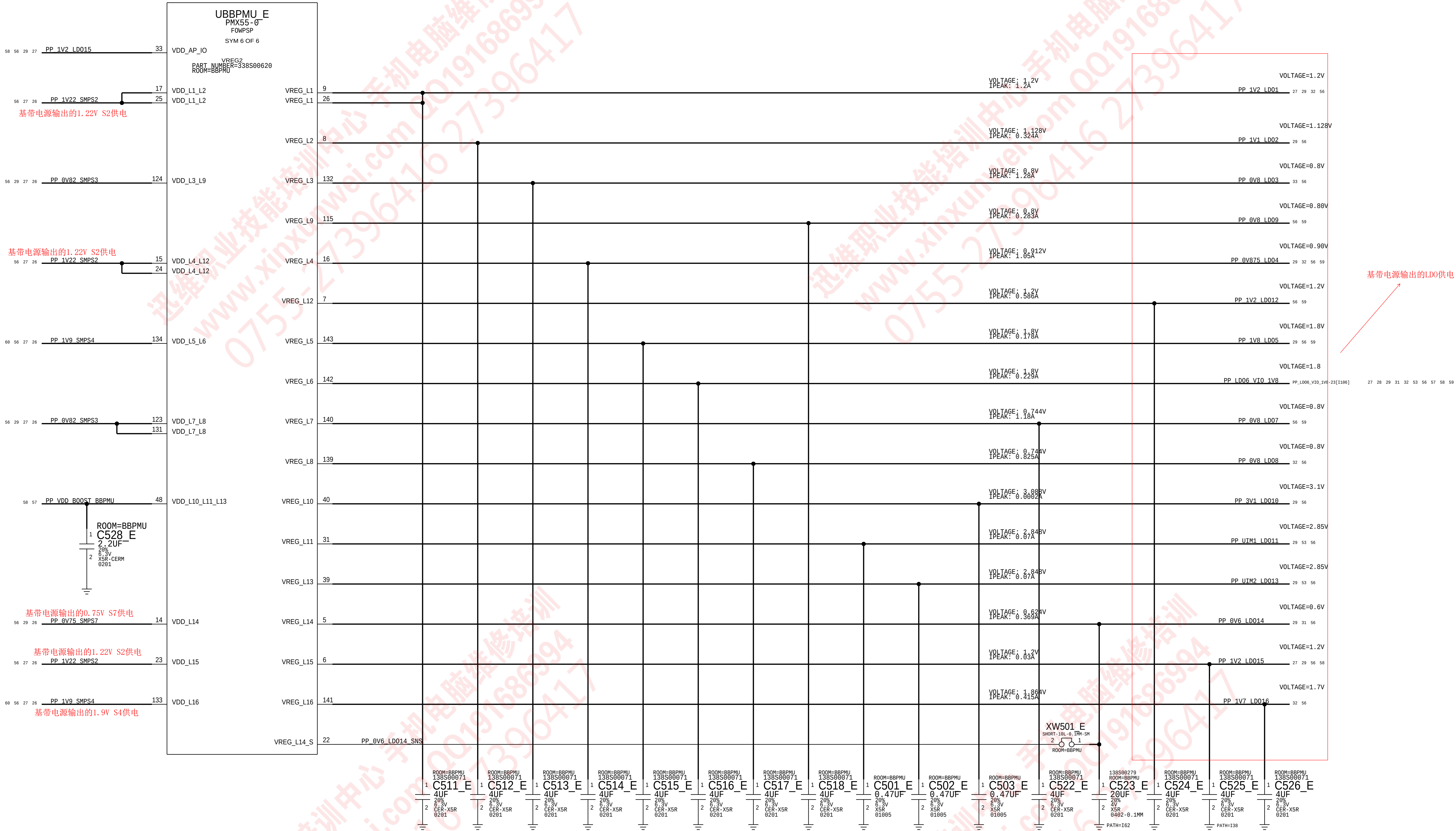
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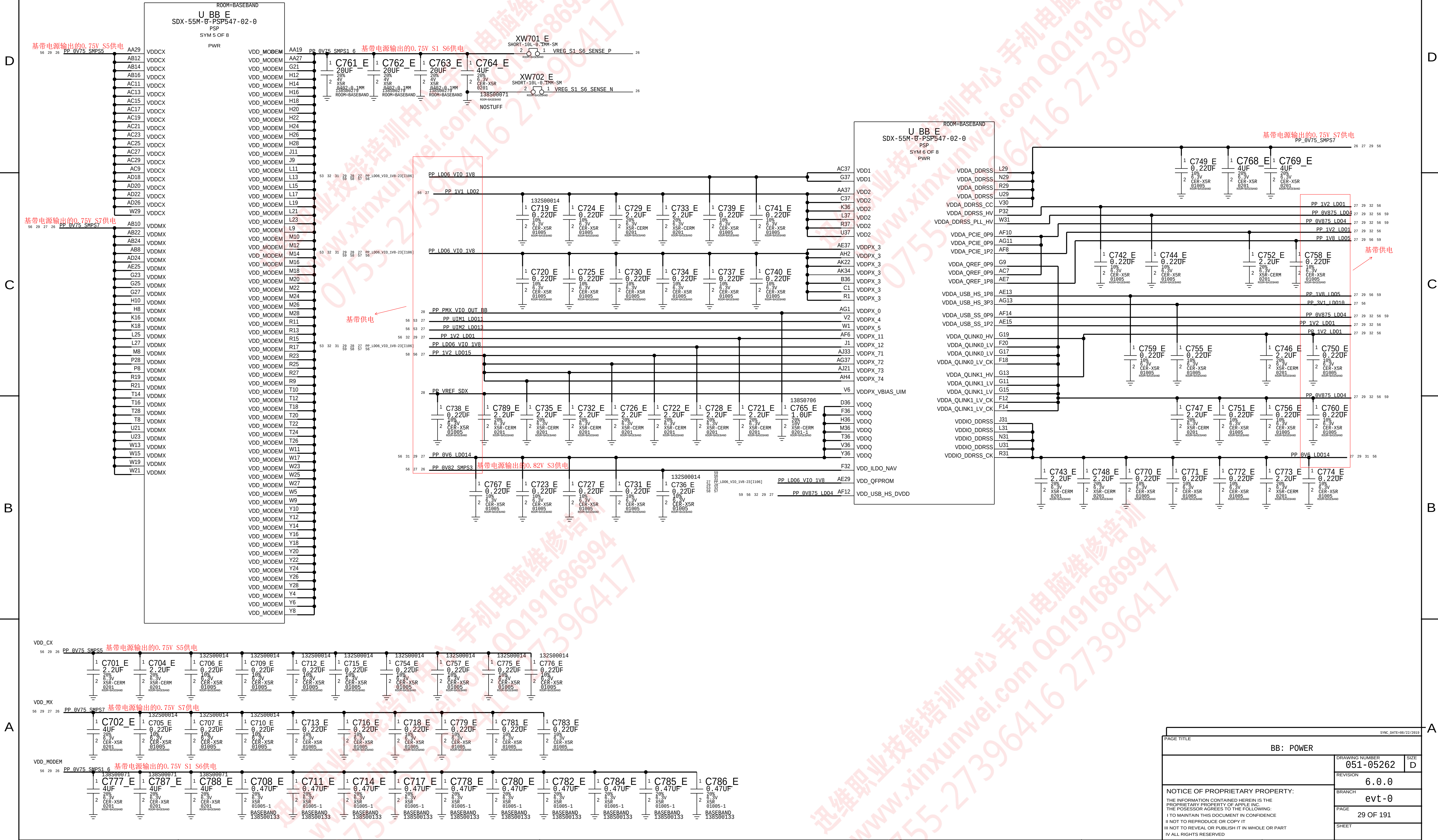
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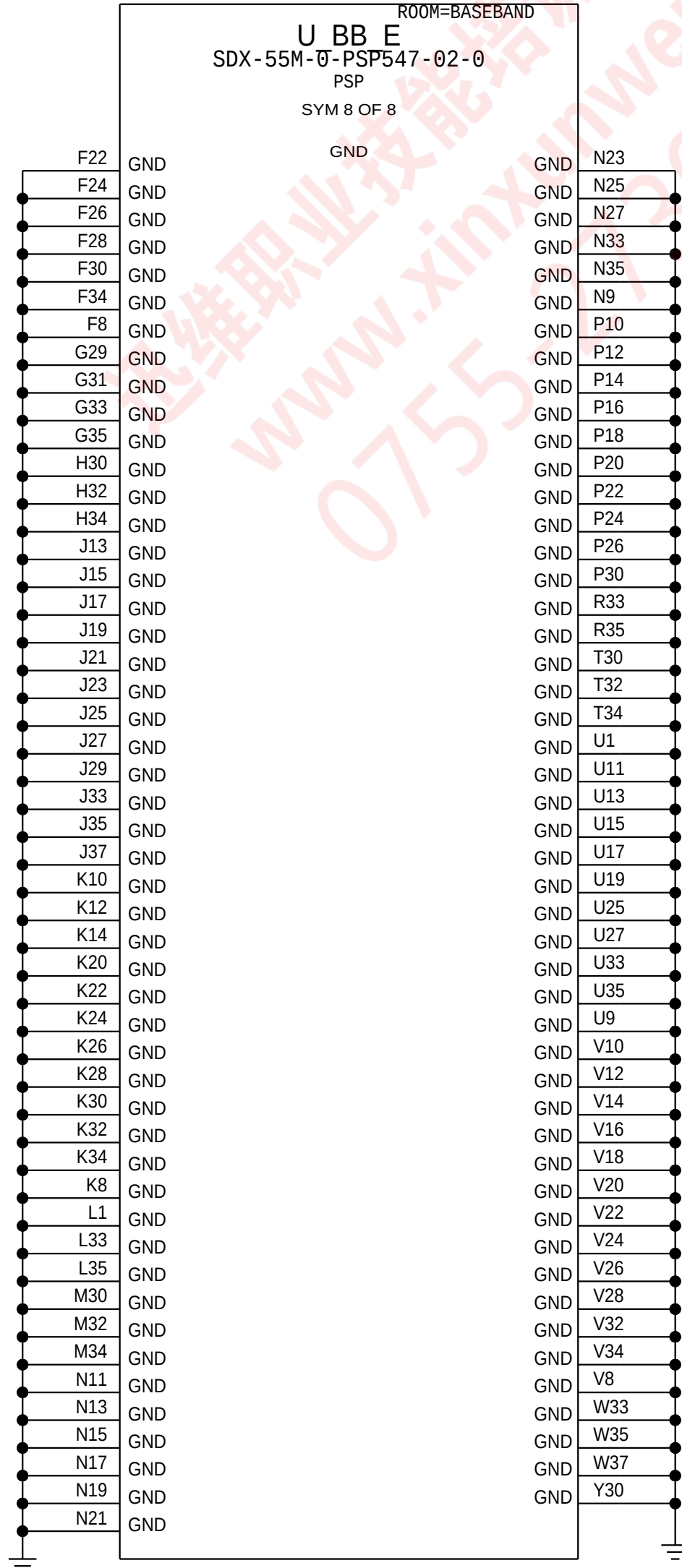
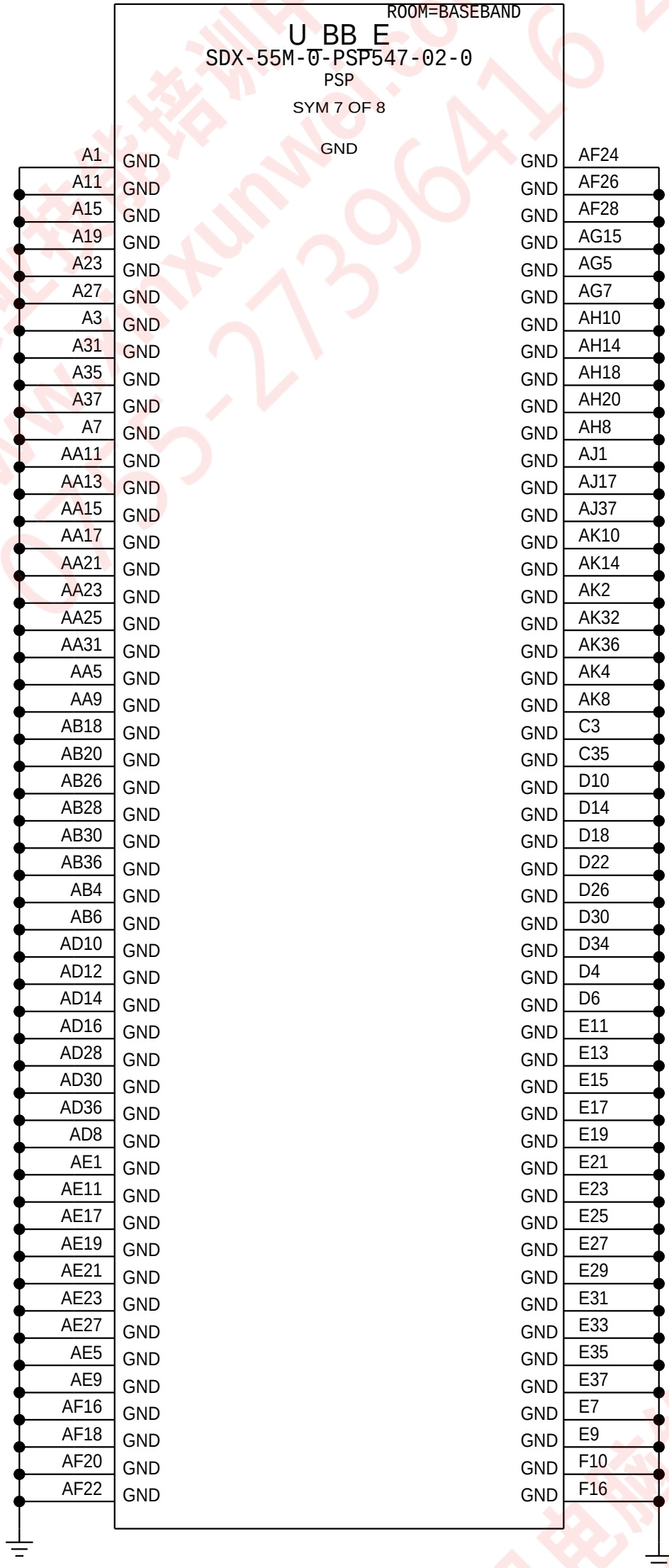
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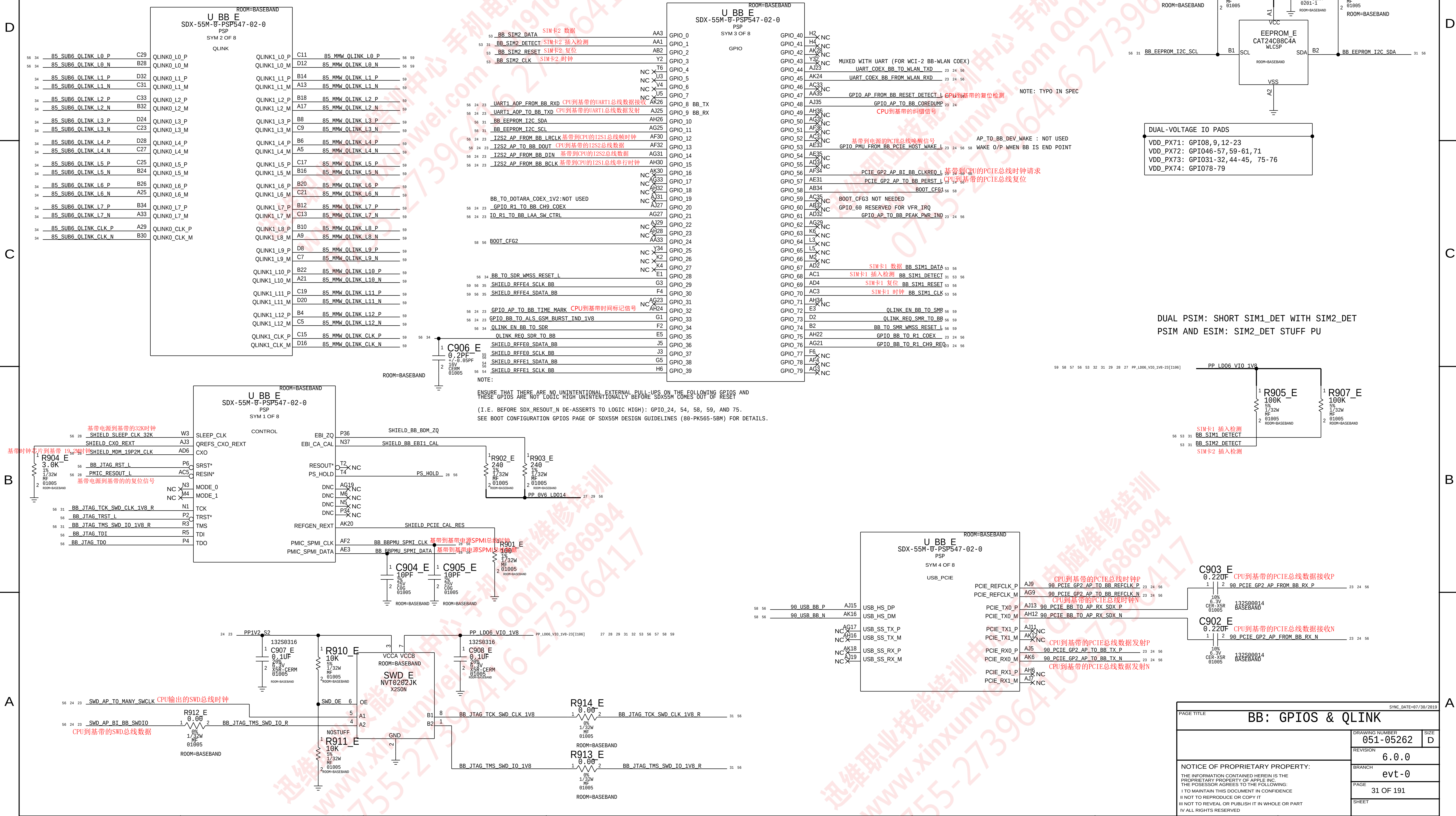
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NOTE: 4UF BULK CAPS SHOULD BE PLACED AS CLOSE AS POSSIBLE TO CORRESPONDING PIN ON SDR
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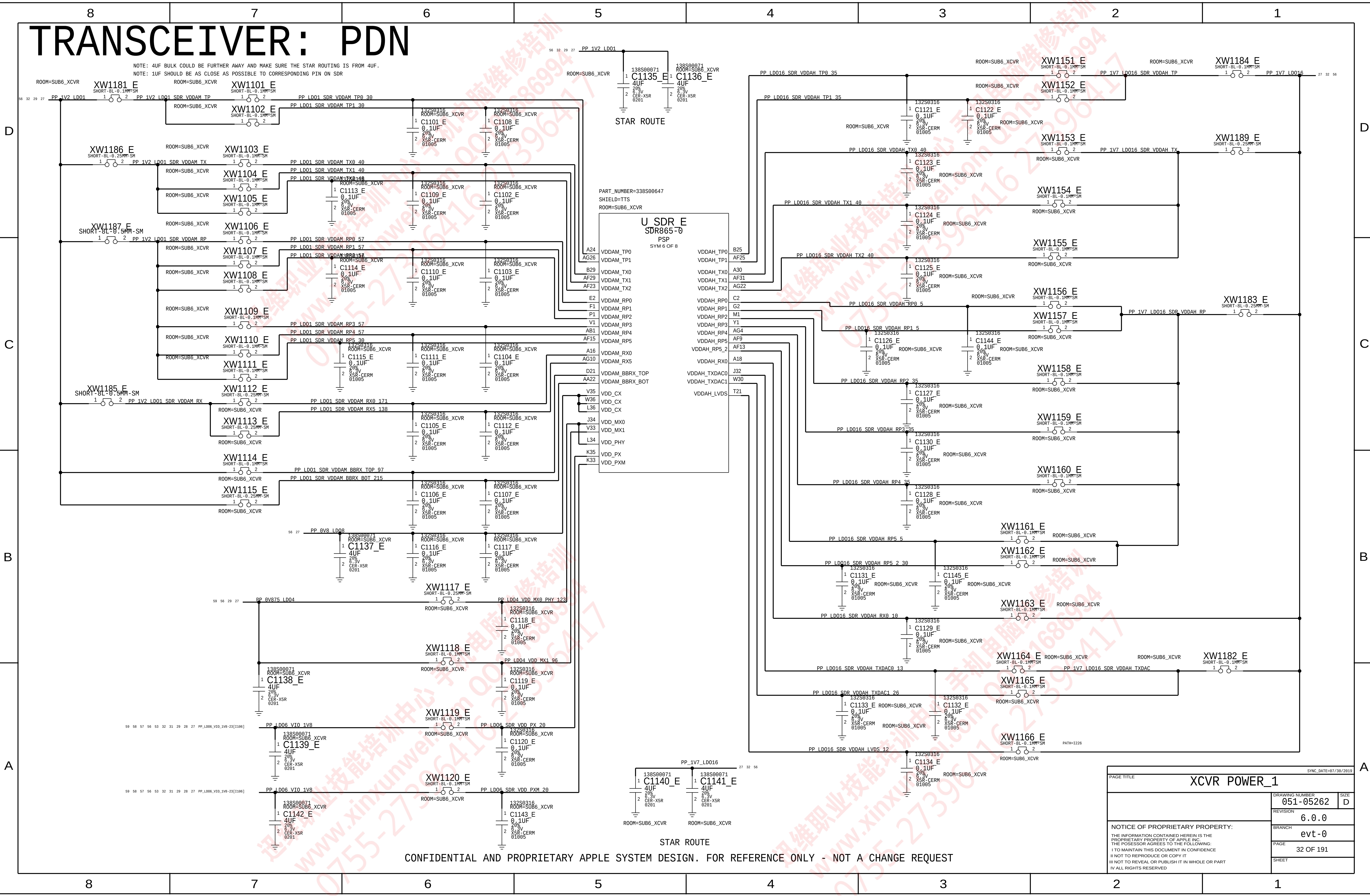
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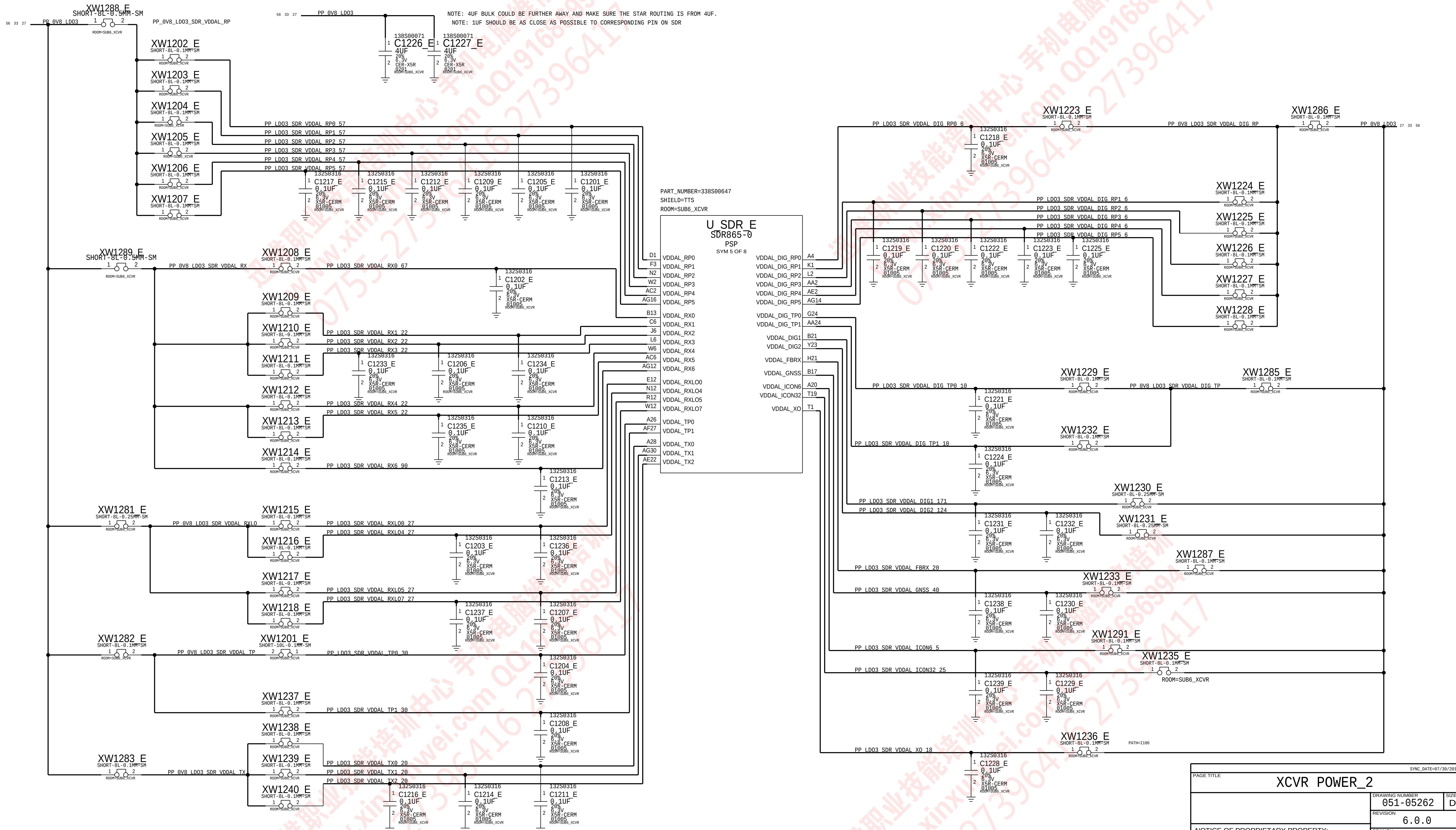
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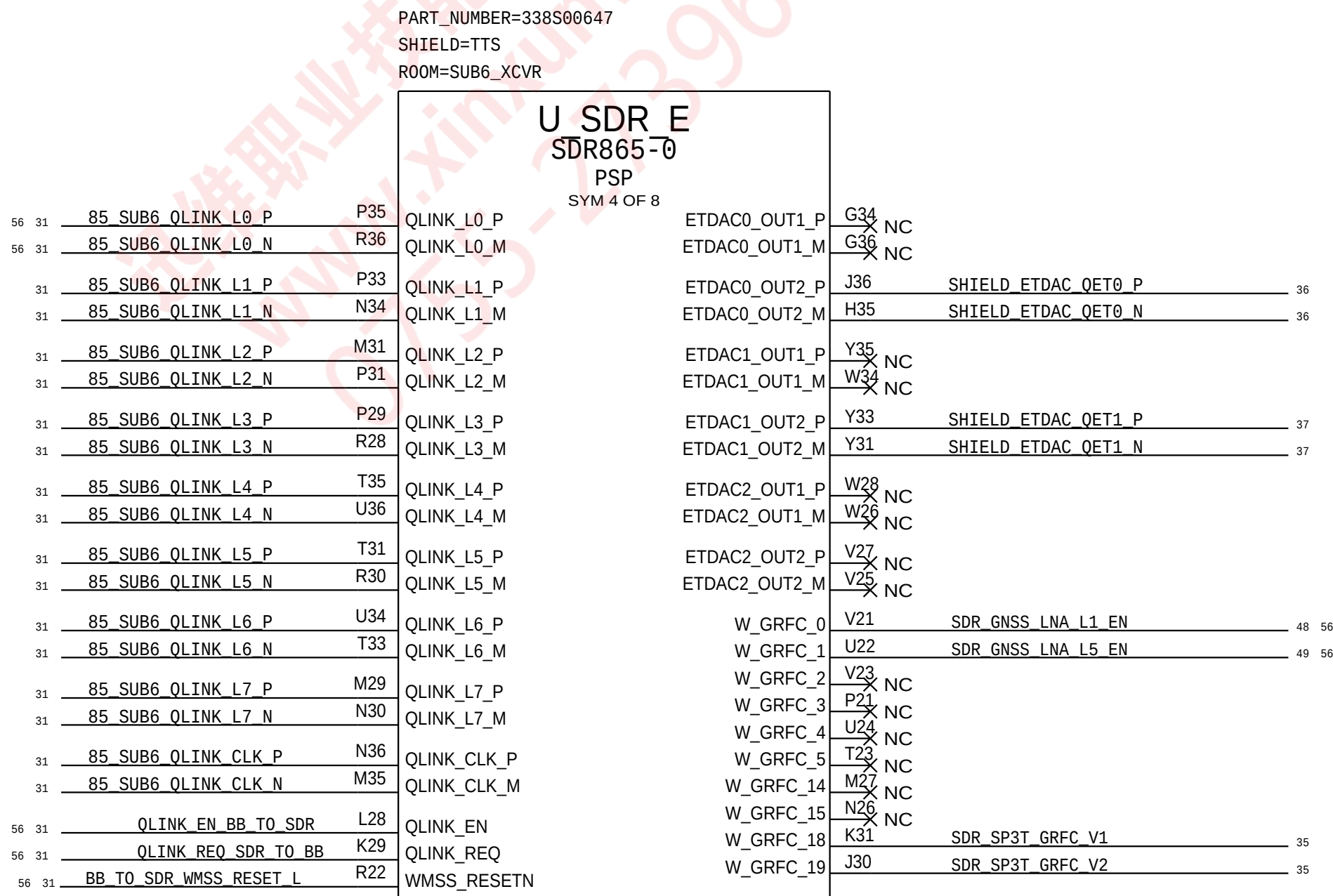
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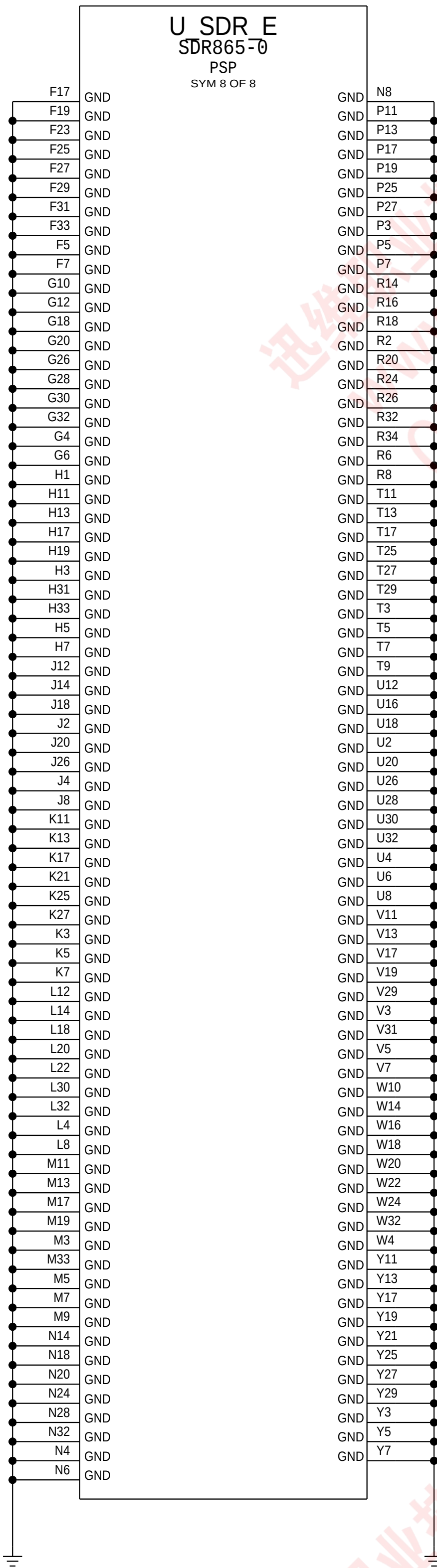
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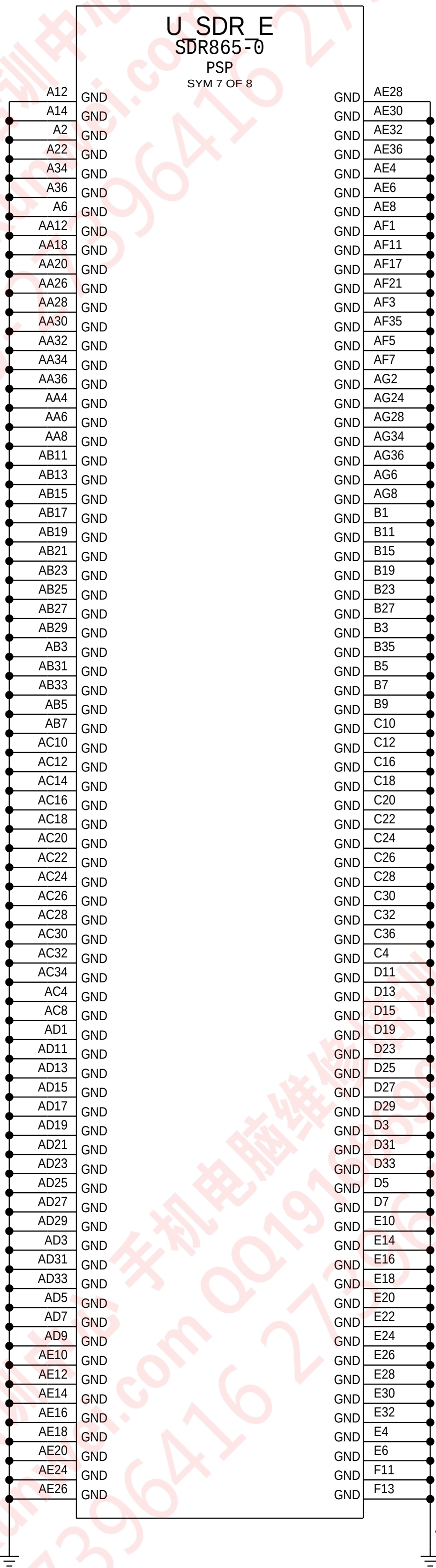
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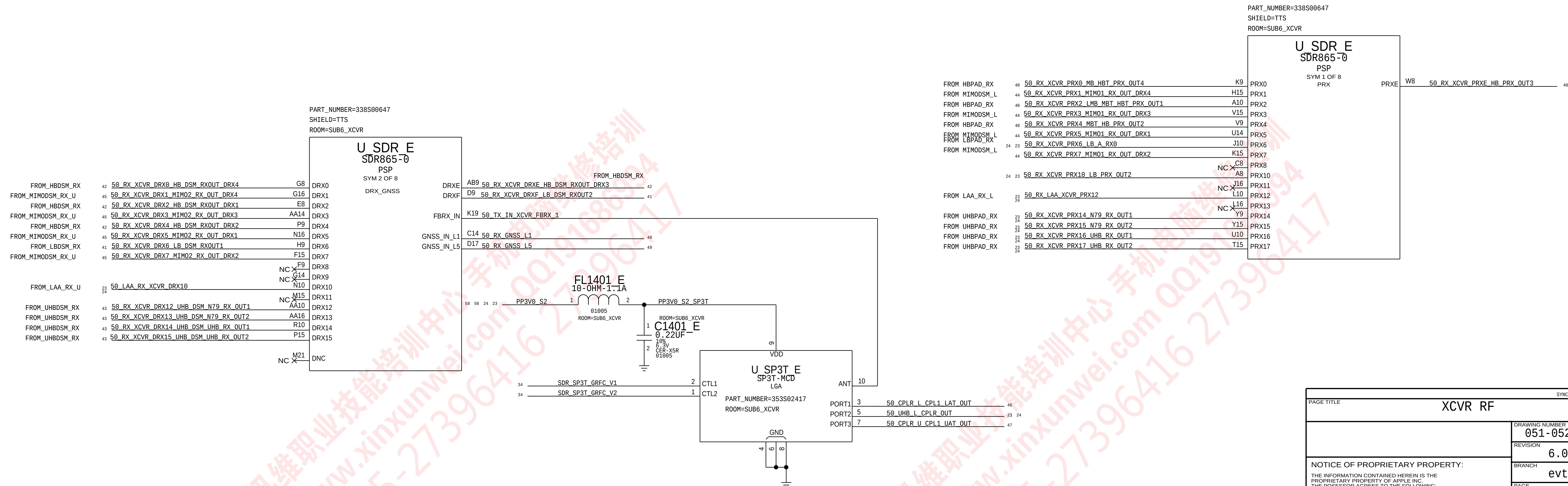
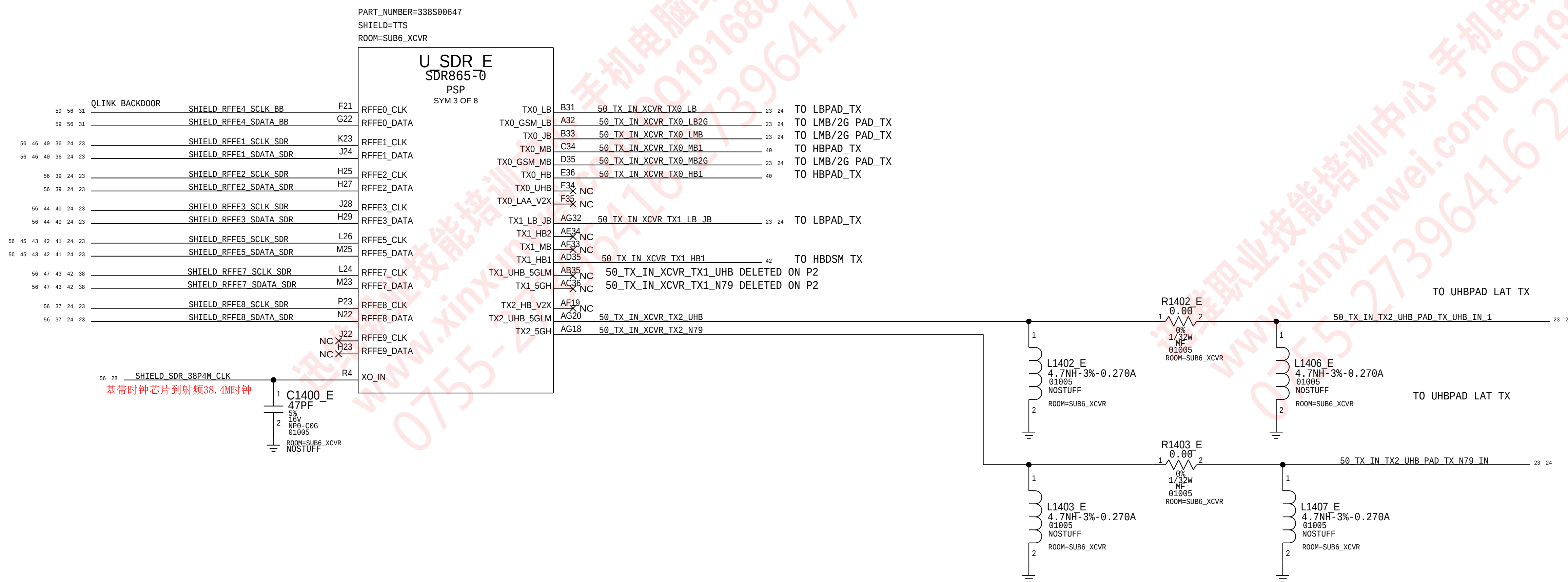
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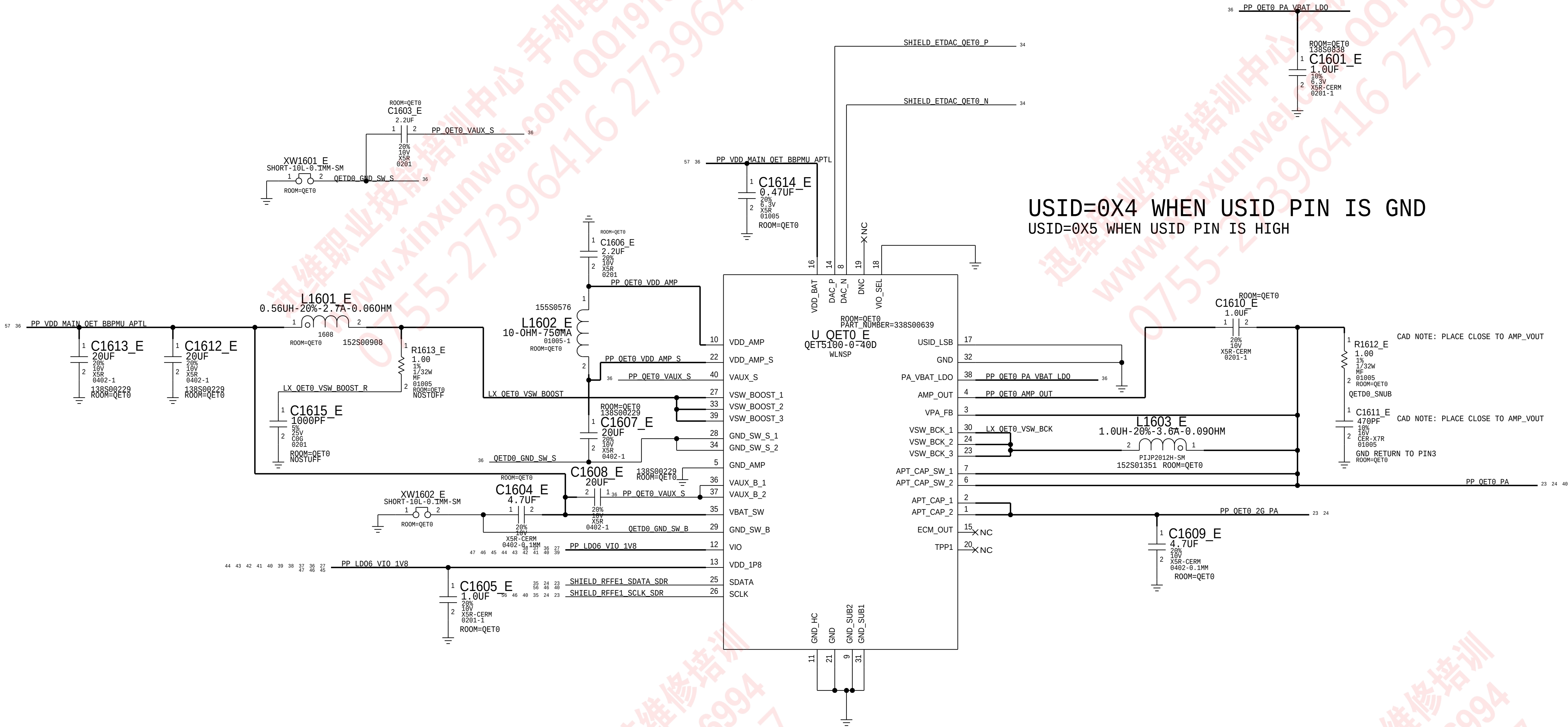


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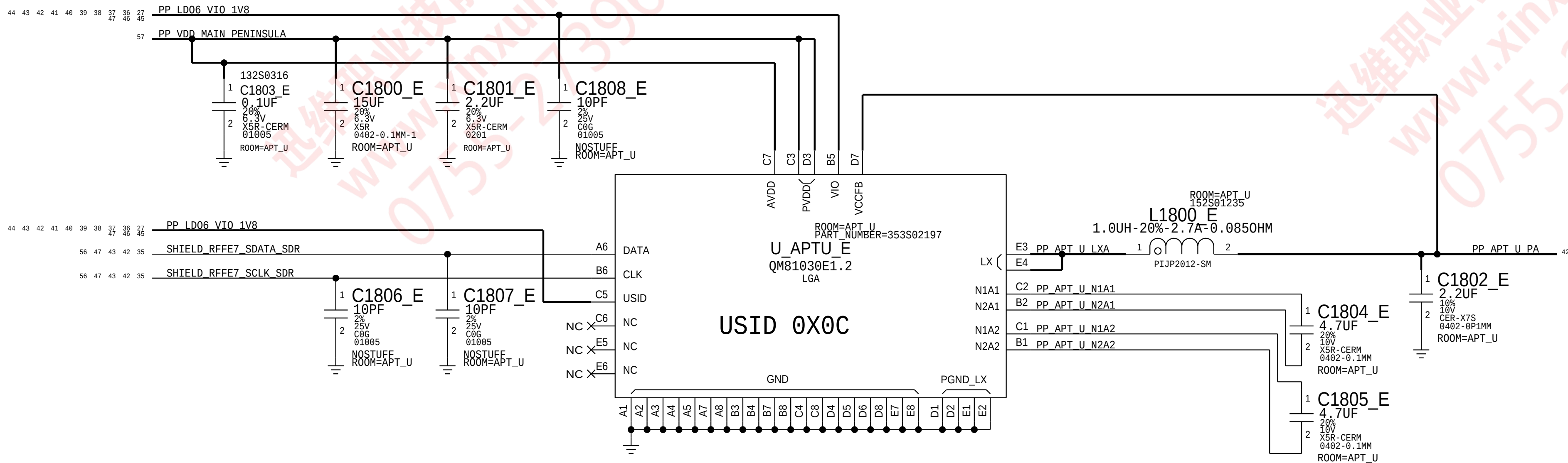
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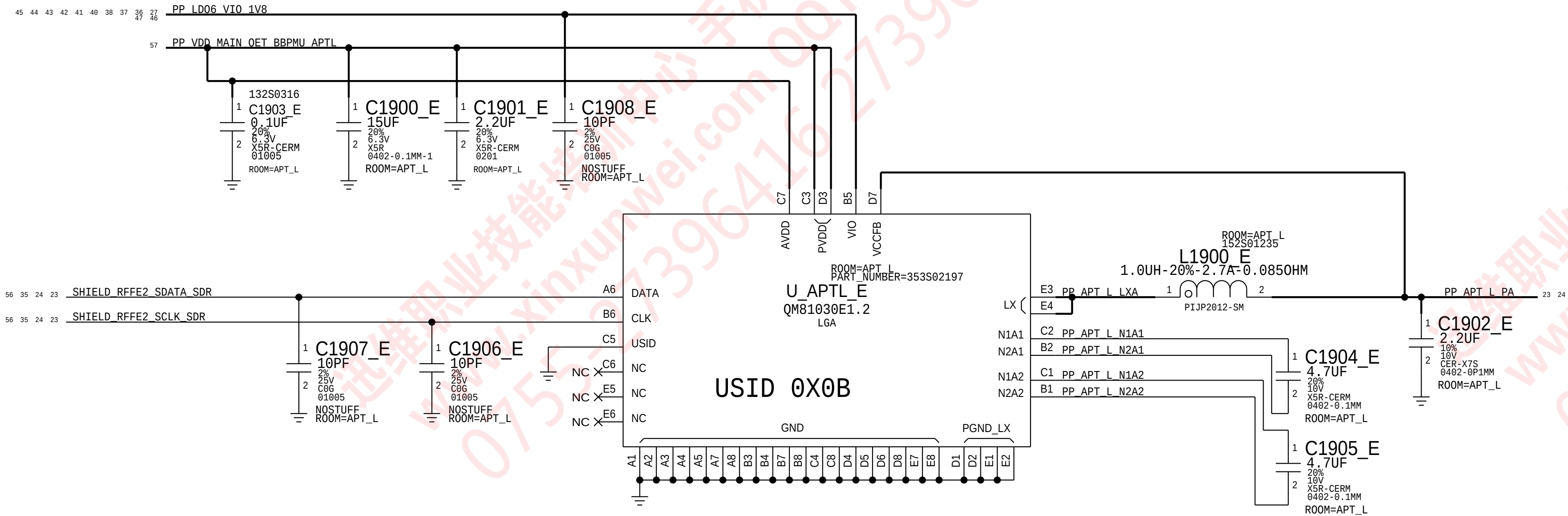
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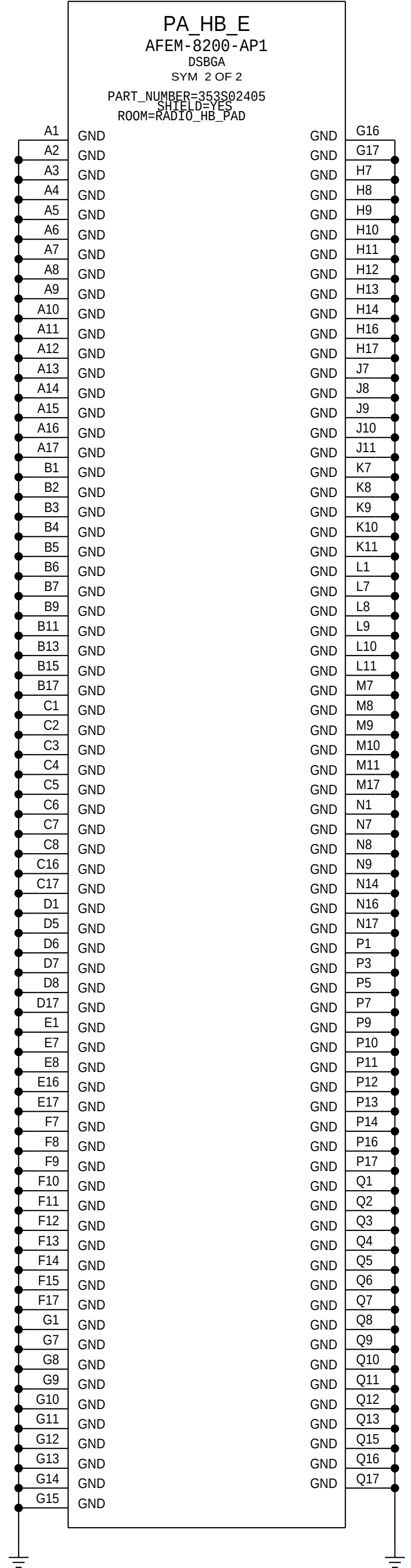
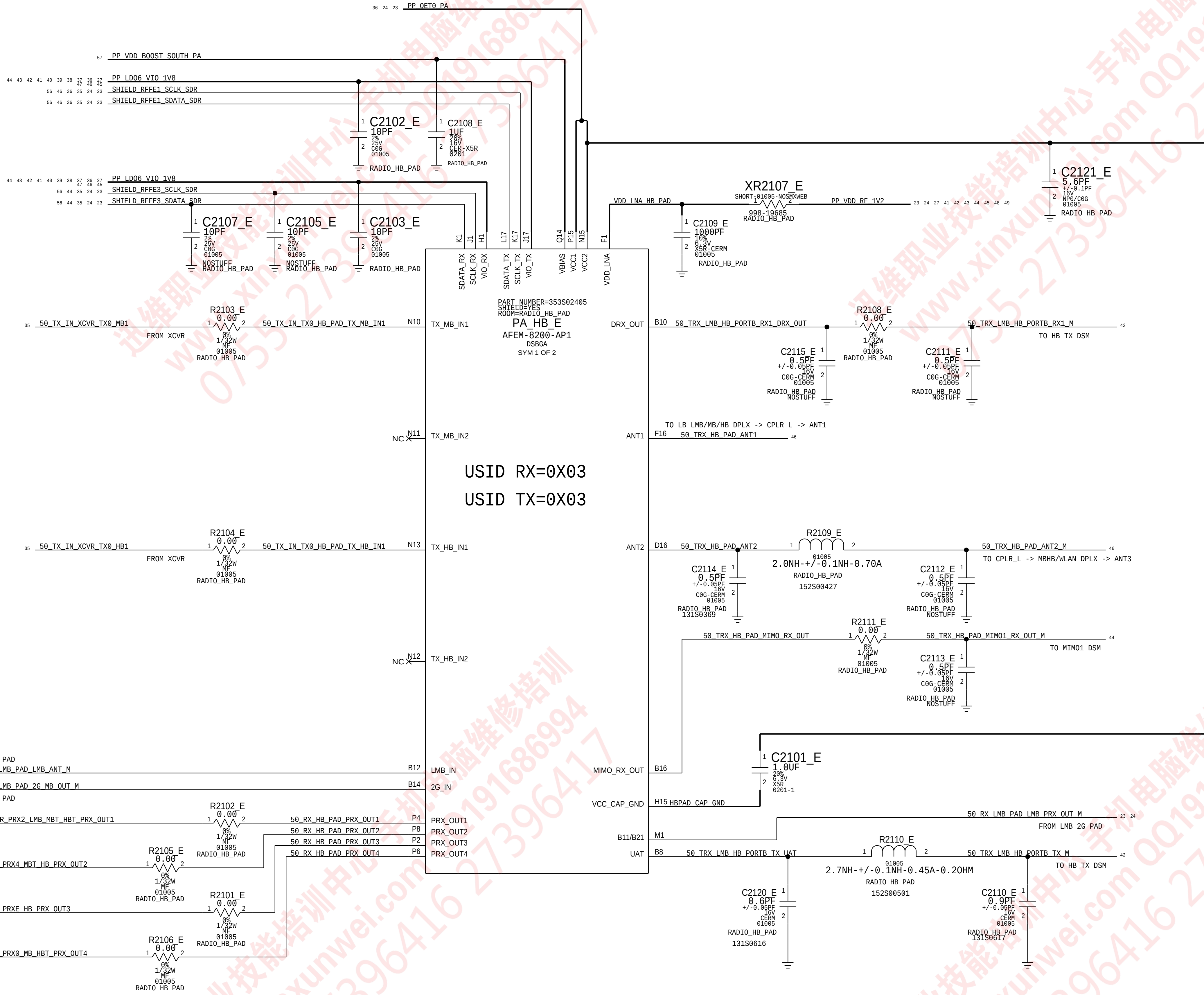
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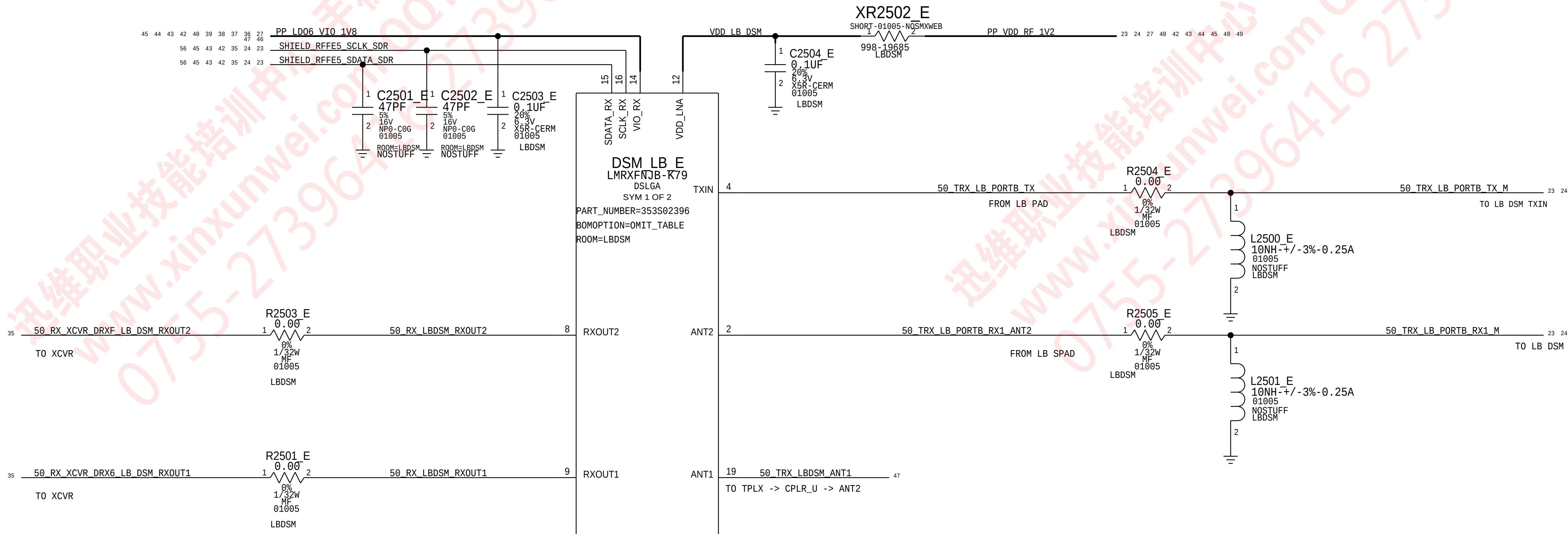
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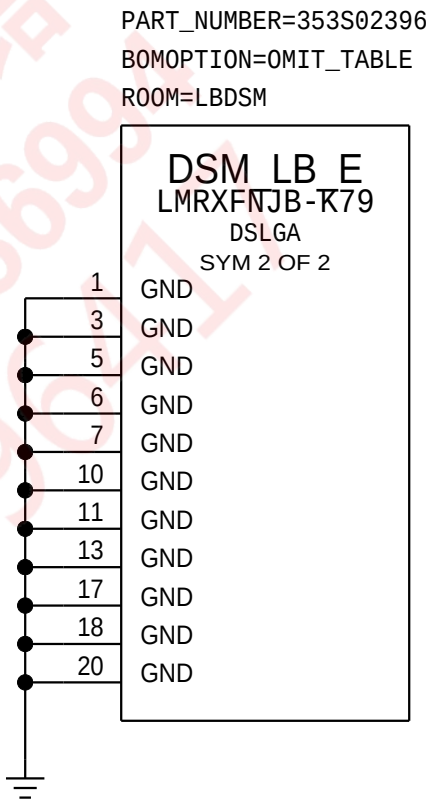


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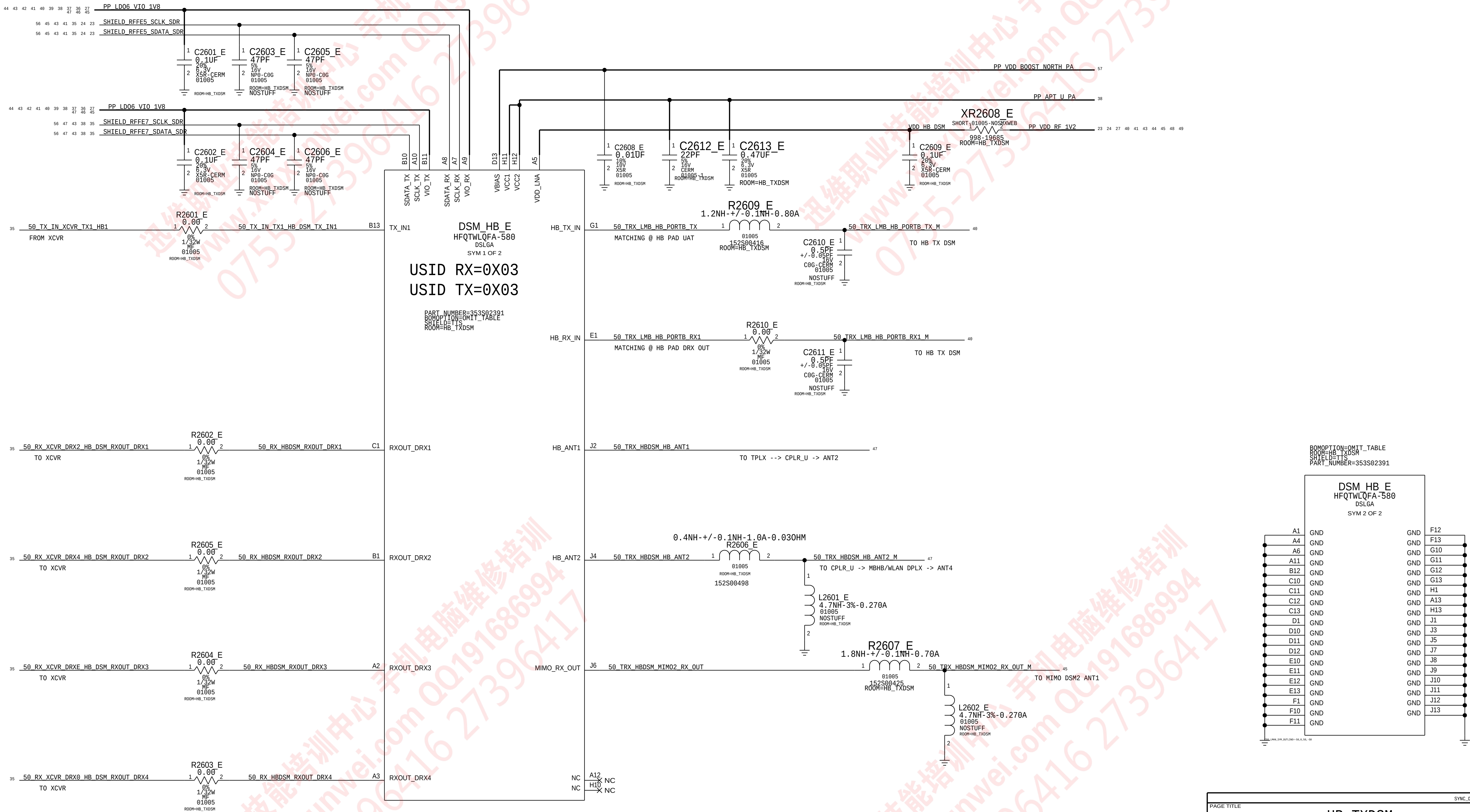


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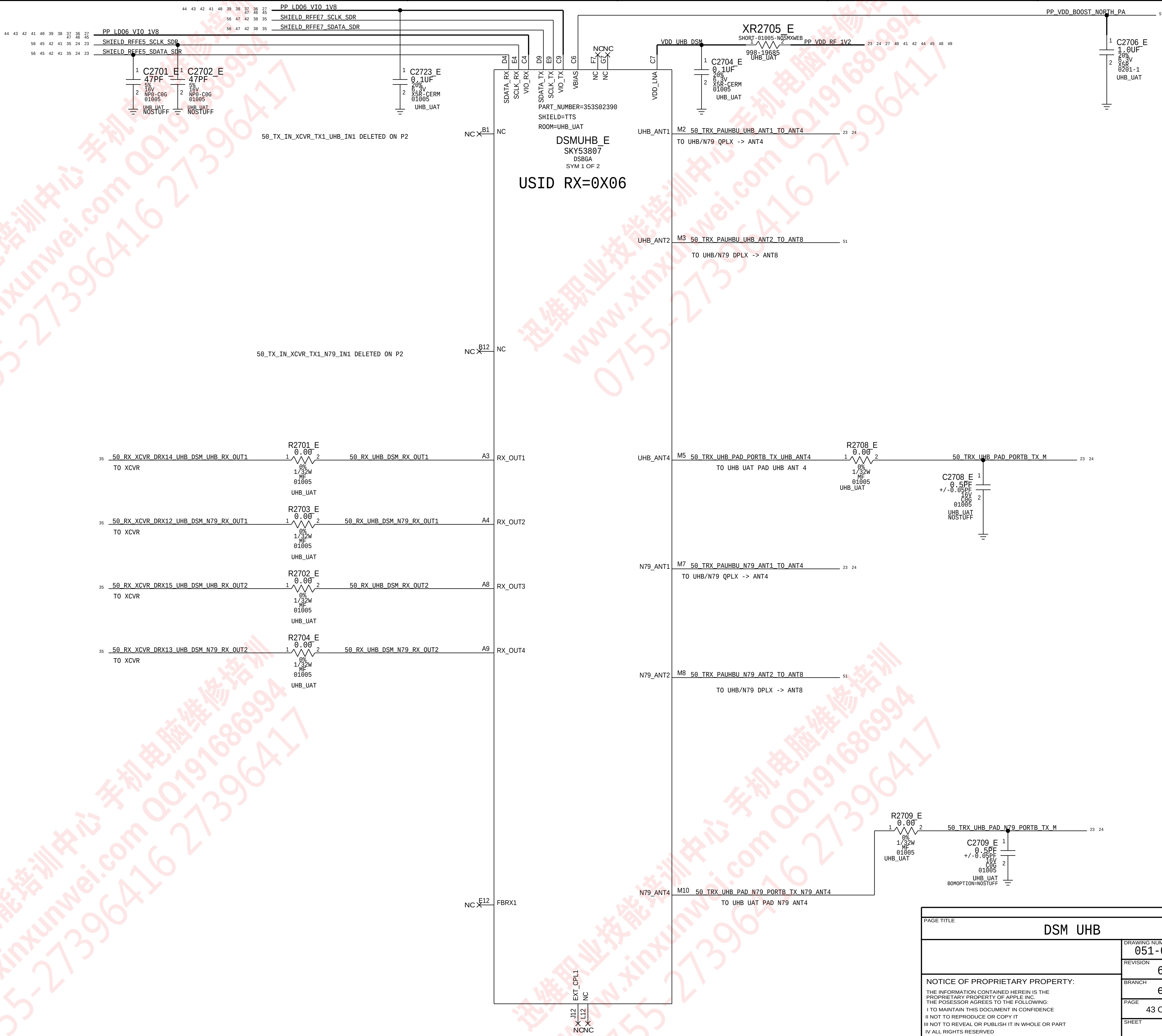
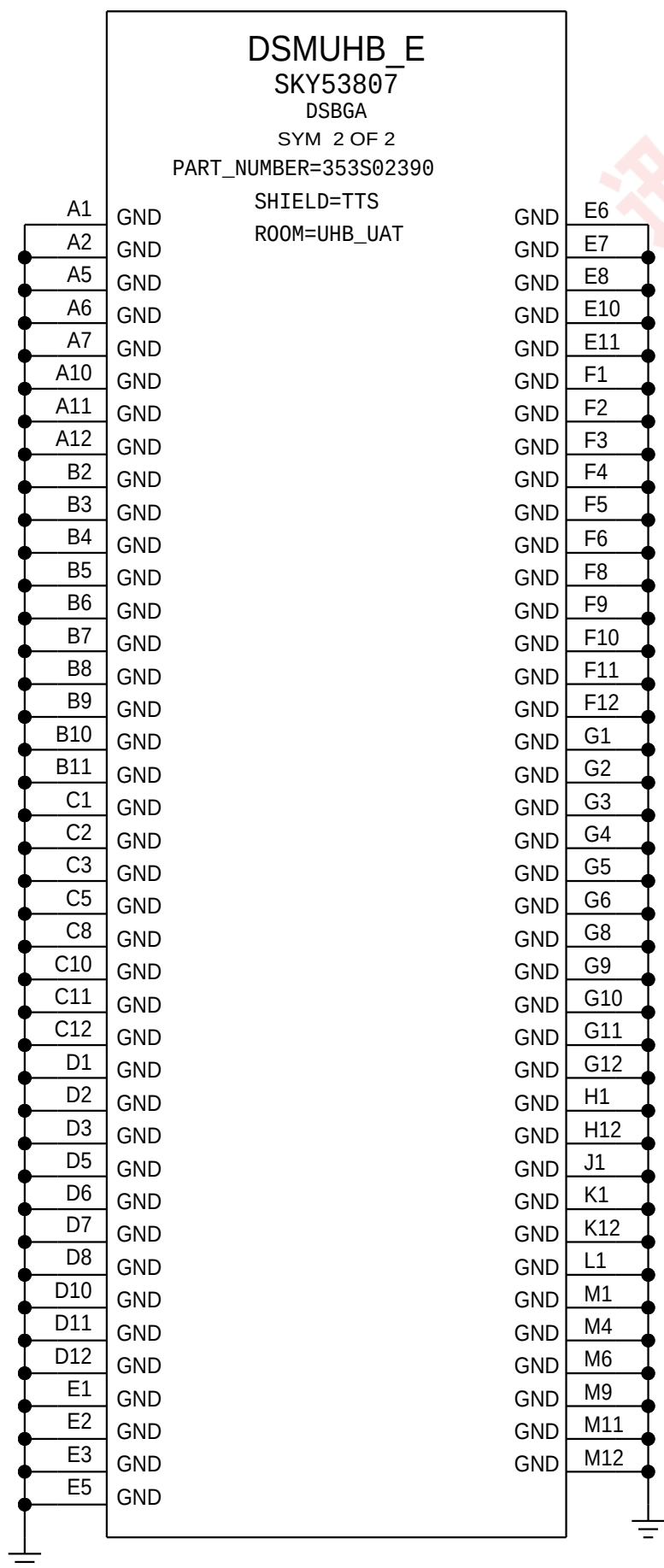
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			REVISION	6.0.0	D
NOTICE OF PROPRIETARY PROPERTY: THE INFORMATION CONTAINED HEREIN IS THE PROPRIETARY PROPERTY OF APPLE INC. THE POSSESSOR AGREES TO THE FOLLOWING: I TO MAINTAIN THIS DOCUMENT IN CONFIDENCE II NOT TO REPRODUCE OR COPY IT III NOT TO REVEAL OR PUBLISH IT IN WHOLE OR PART IV ALL RIGHTS RESERVED			BRANCH	evt-0	
			PAGE	41 OF 191	
			SHEET		

HB RECEIVE DIVERSITY & TX MODULE

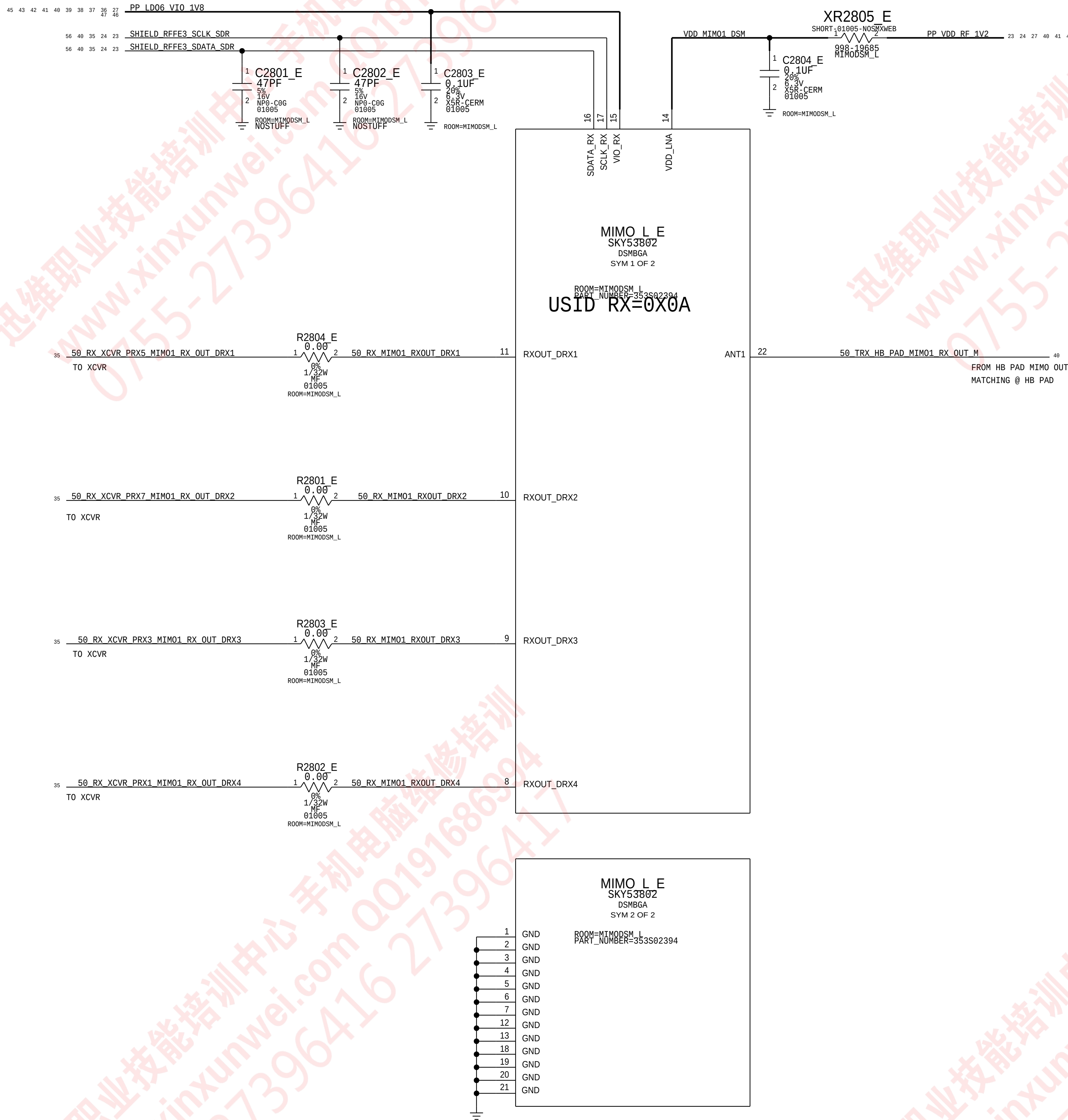


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			DRAWING NUMBER	051-05262	SIZE
			REVISION	6.0.0	D
			BRANCH	evt-0	
			PAGE	42 OF 191	
			SHEET		

UHB DIVERSITY MODULE



LOWER MIMO DIVERSITY MODULE



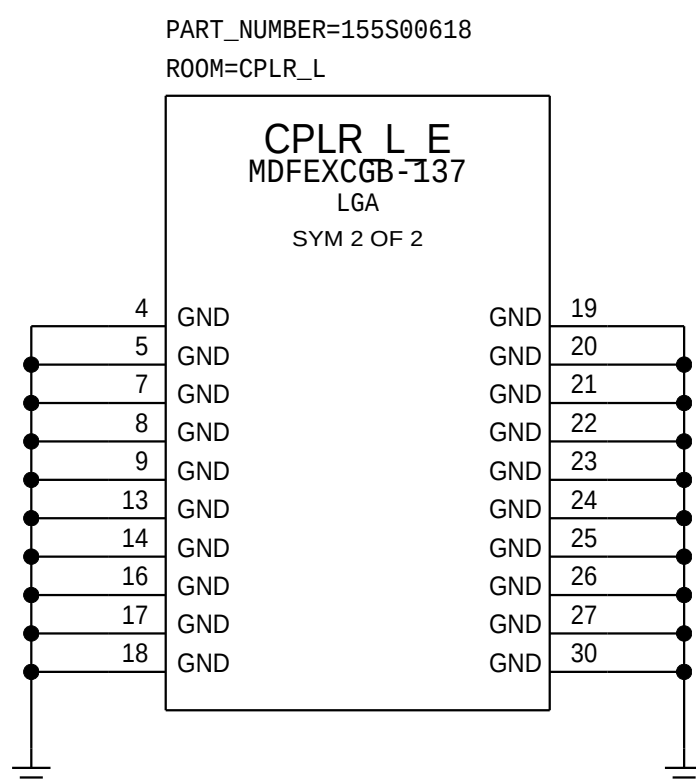
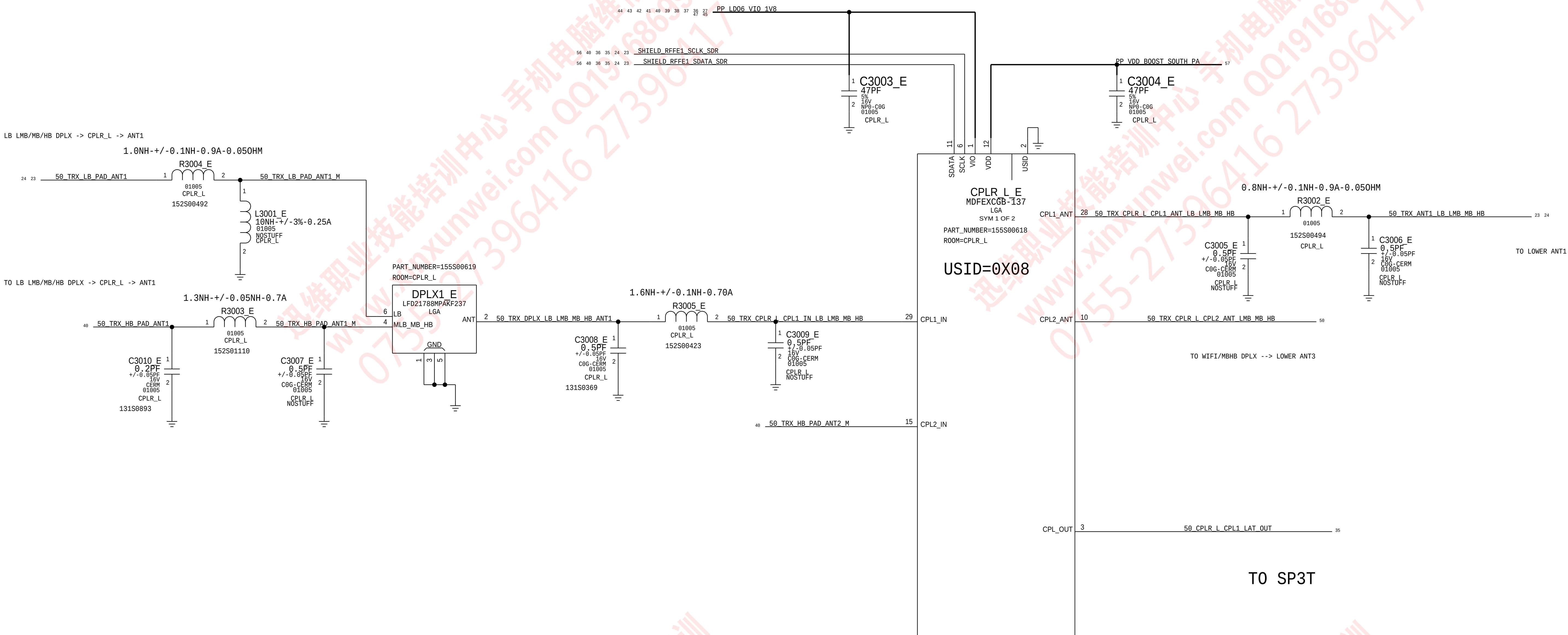
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			DRAWING NUMBER	051-05262	SIZE
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			PAGE	44 OF 191	
			SHEET		

D



A

LOWER COUPLER

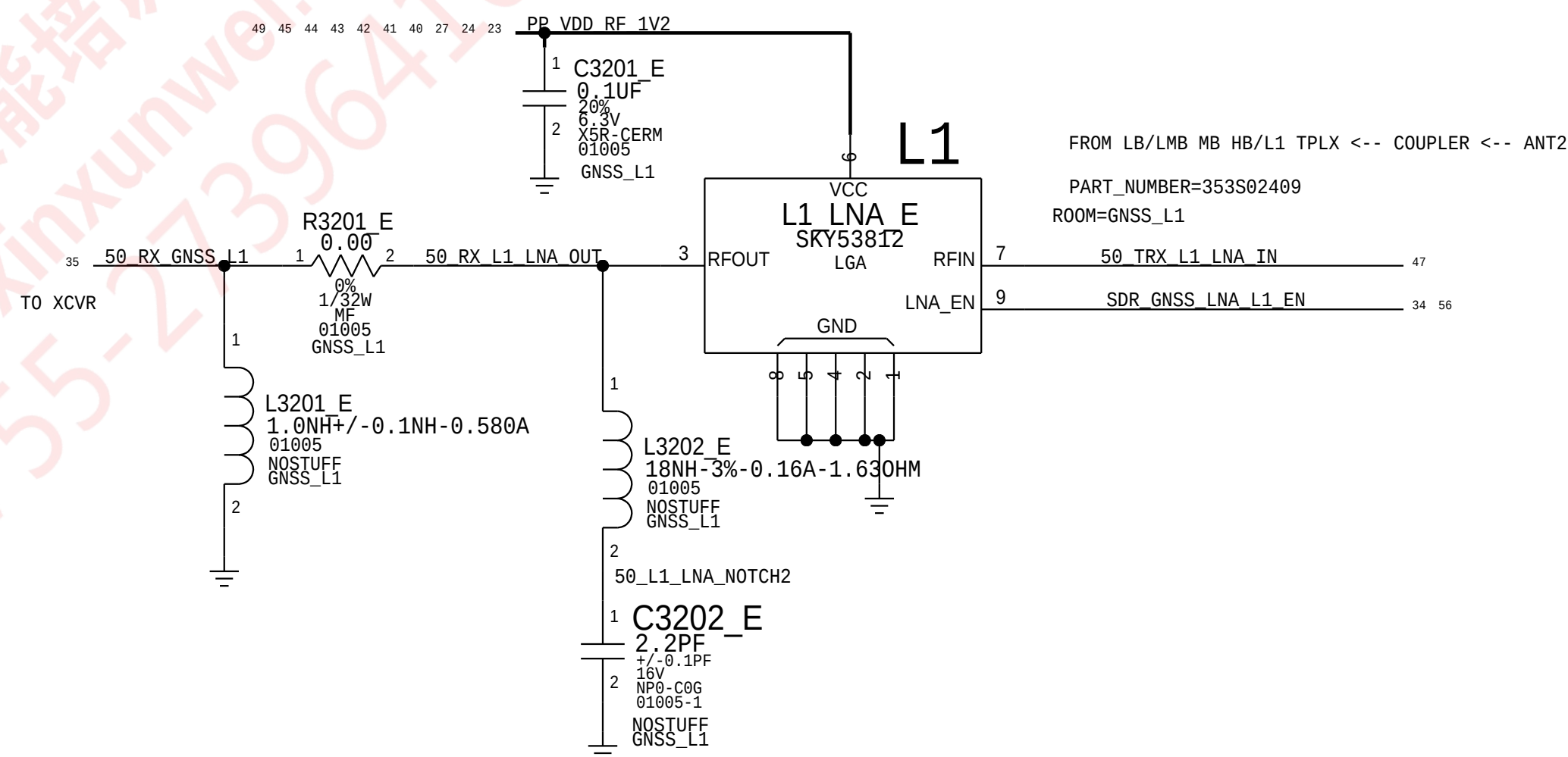


PAGE TITLE		SYNC_DATE=07/30/2019	
COUPLER LOWER			
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	REVISION	6.0.0	
	BRANCH	evt-0	
	PAGE	46 OF 191	
	SHEET		

D

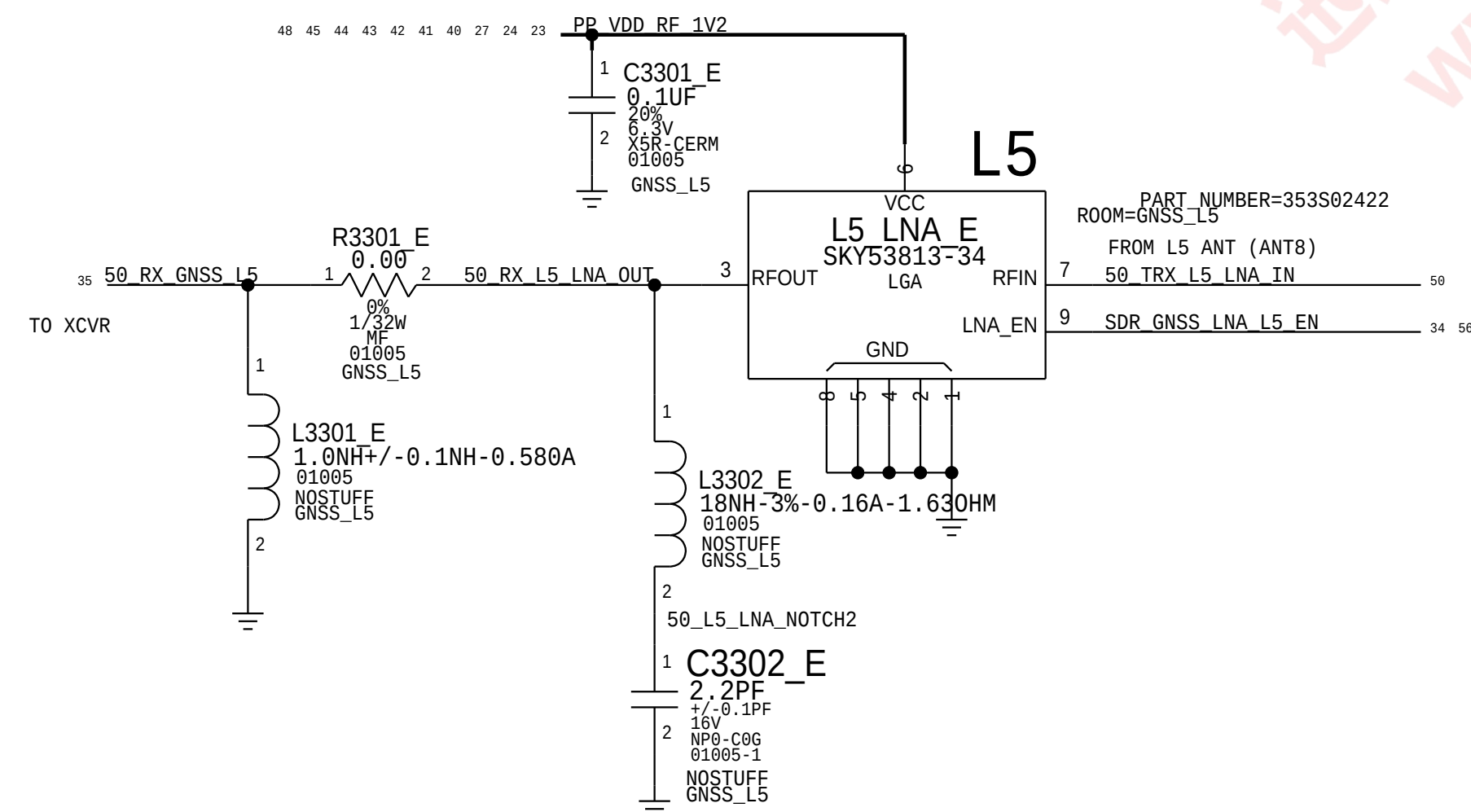
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GNSS_L1



			SYNC_DATE=07/30/2019	
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			SHEET	

GNSS_L5



PAGE TITLE		SYNC_DATE=07/30/2019	
GNSS__L5			
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	REVISION 6.0.0		
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PAGE 49 OF 191		SHEET	

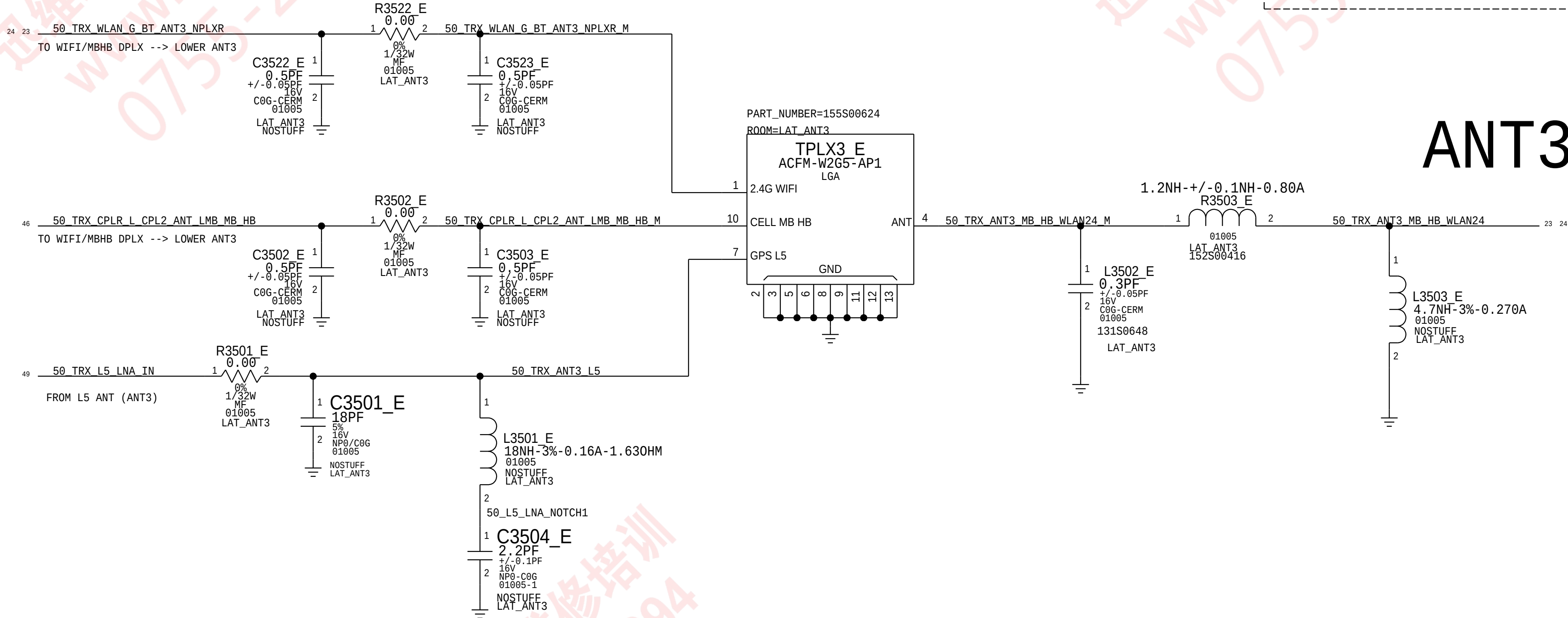
LOWER ANTENNA FEEDS: ANT3

UAT ANTENNAS

ANT2	ANT4	ANT6	ANT8
LB	MB/HB	SAM	UHB
LMB/MB/HB	2.4GHZ	5GHZ	N79
L1 GNSS	UHB		
	N79		

LAT ANTENNAS

ANT1	ANT3	ANT5	ANT7	ANT9
LB	MB/HB	5GHZ	UHB	UHB
LMB/MB/HB	2.4GHZ		N79	N79
	L5 GNSS			



5GHZ WLAN BPF IS ON LAA SCHEMATIC (ANT5)

PAGE TITLE		SYNC_DATE=07/30/2019	
LOWER ANTENNA FEEDS_ANT3			
	DRAWING NUMBER	051-05262	SIZE D
	REVISION	6.0.0	
NOTICE OF PROPRIETARY PROPERTY:	BRANCH	evt-0	
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	SHEET		

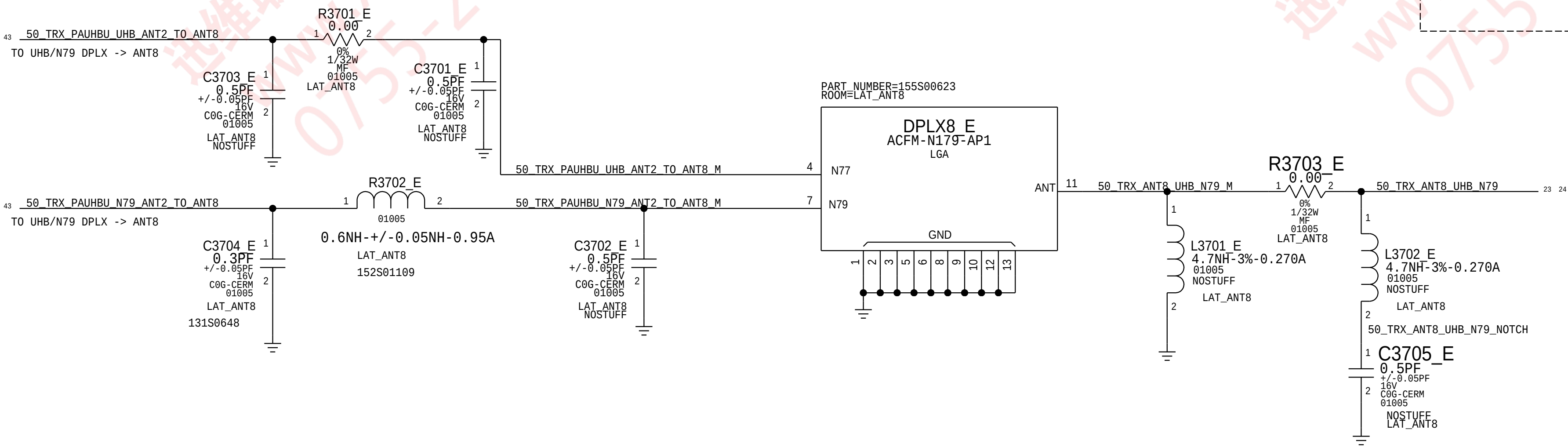
LOWER ANTENNA FEEDS: ANT8

UAT ANTENNAS

ANT2	ANT4	ANT6	ANT8
LB	MB/HB	SAM	UHB
LMB/MB/HB	2.4GHZ	5GHZ	N79
L1 GNSS	UHB		
	N79		

LAT ANTENNAS

ANT1	ANT3	ANT5	ANT7	ANT9
LB	MB/HB	5GHZ	UHB	UHB
LMB/MB/HB	2.4GHZ		N79	N79
	L5 GNSS			



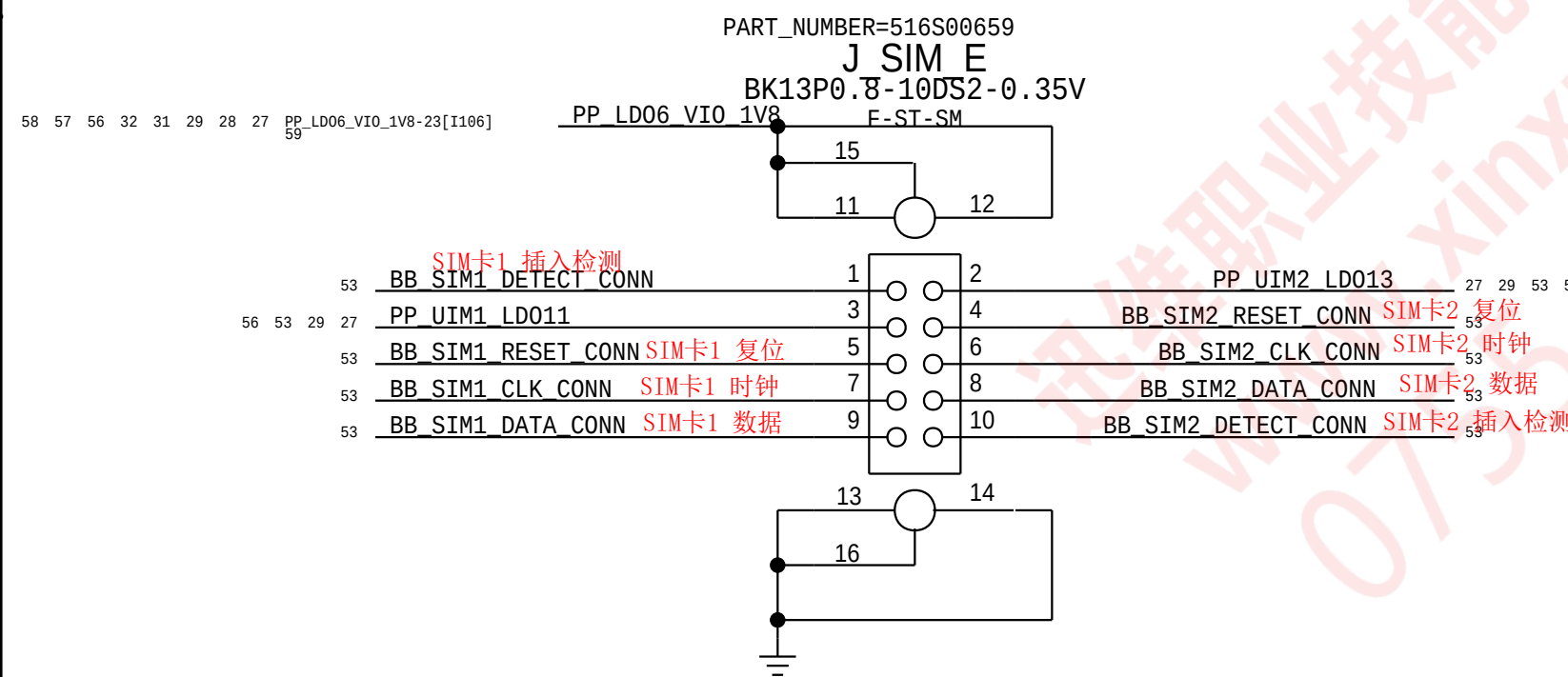
ANT8

5GHZ WLAN BPF IS ON LAA SCHEMATIC (ANT5)

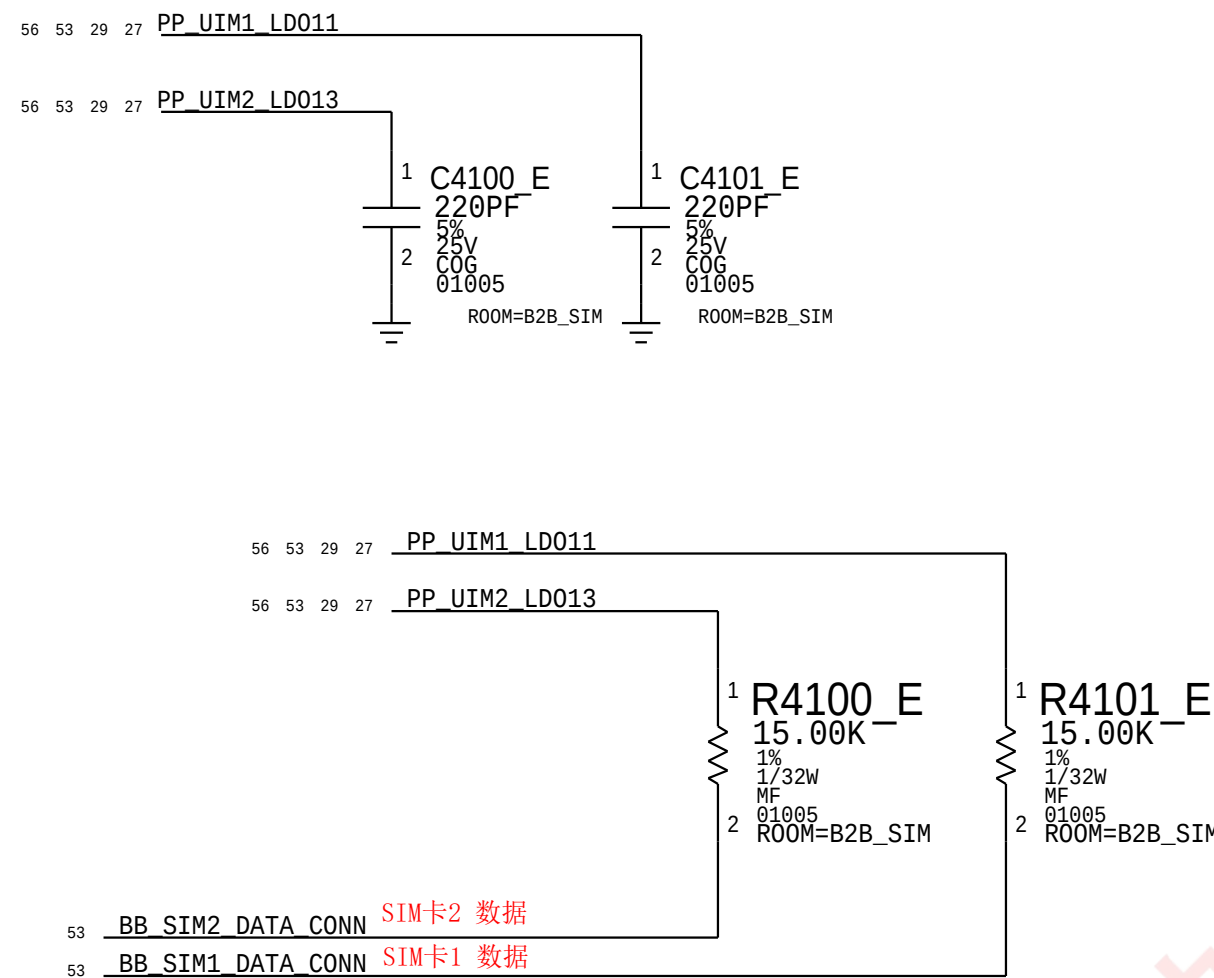
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LOWER ANTENNA FEEDS_ANT8			051-05262			D		
REVISION			6.0.0			BRANCH		
NOTICE OF PROPRIETARY PROPERTY:			evt-0			PAGE		
THE INFORMATION CONTAINED HEREIN IS THE PROPRIETARY PROPERTY OF APPLE INC. THE POSSESSOR AGREES TO THE FOLLOWING:			51 OF 191			SHEET		
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PSIM AND ESIM ARE ON FLEX

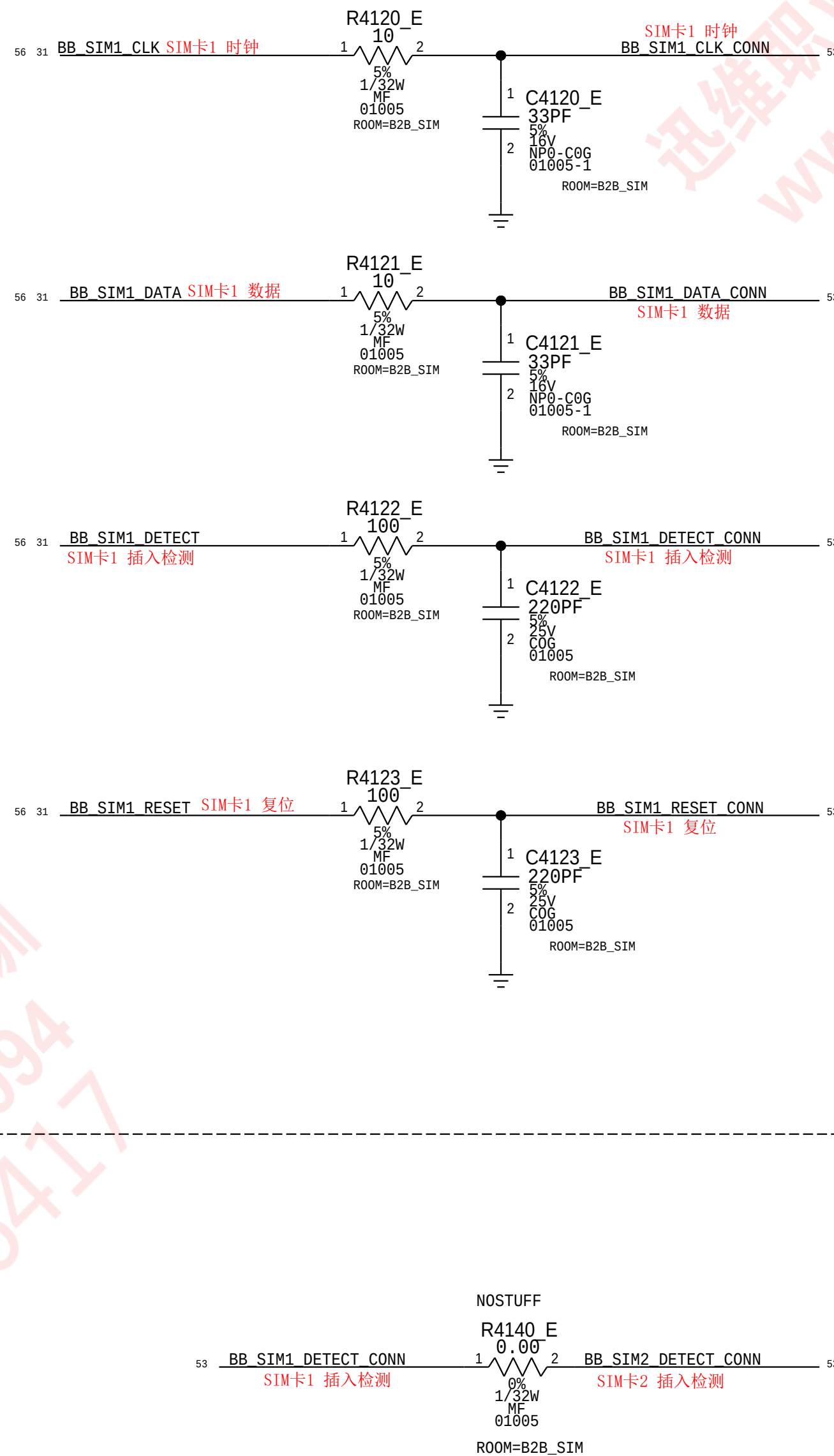
J_SIM_E (PLUG.APN:516S00659) ON MLB_BOT_MAV



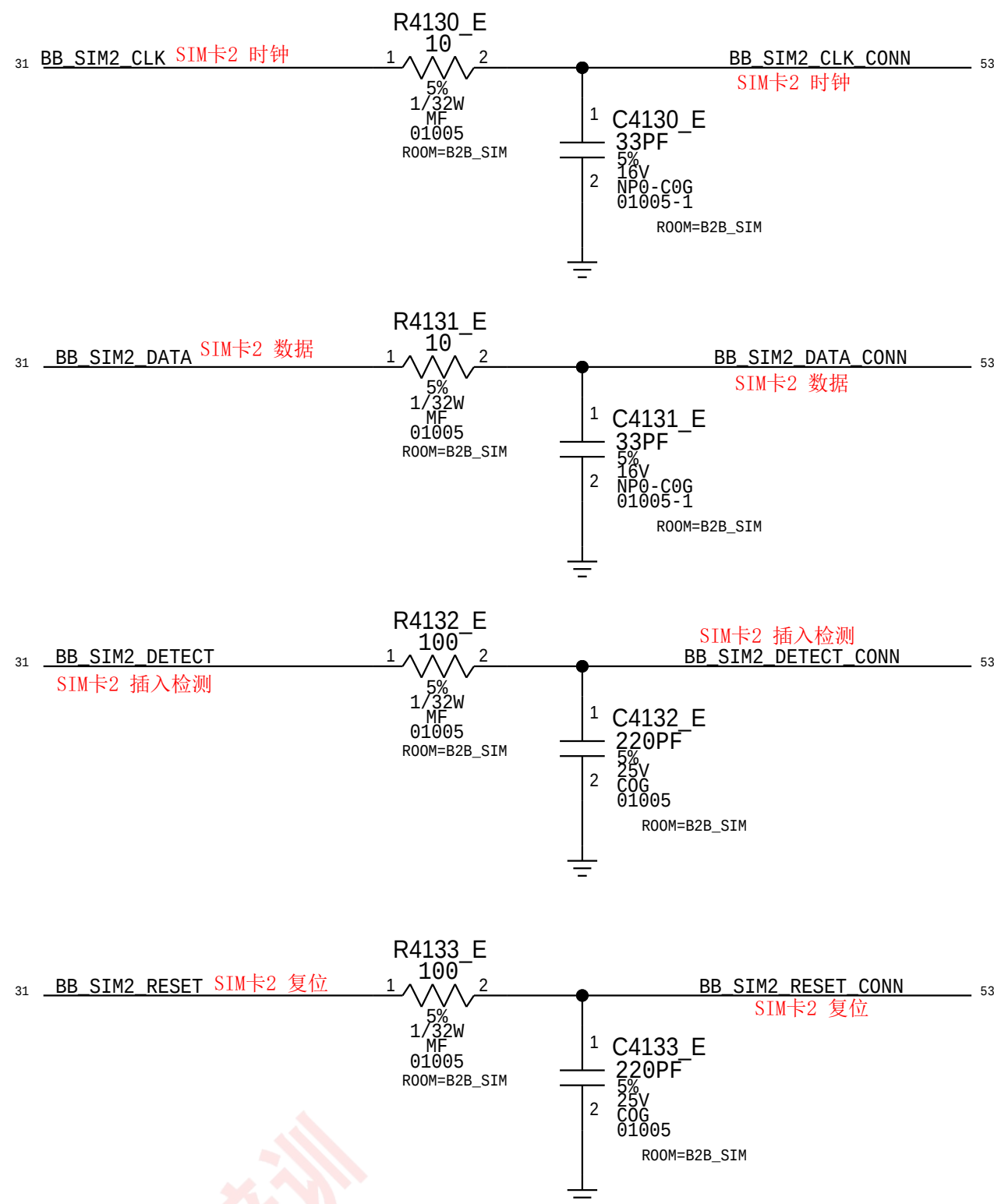
Power Filters & Pull-ups



SIM1 IO Filters



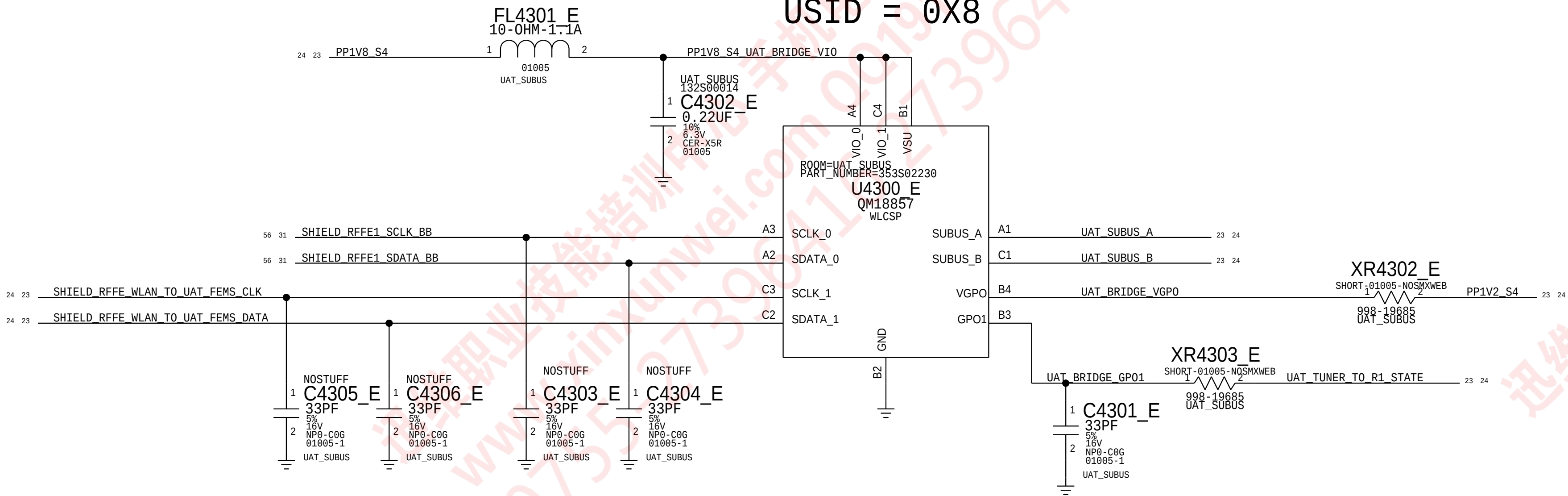
SIM2 IO Filters



PAGE TITLE			PSIM		
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			PAGE	53 OF 191	
			SHEET		

ANTENNA SYSTEM UAT

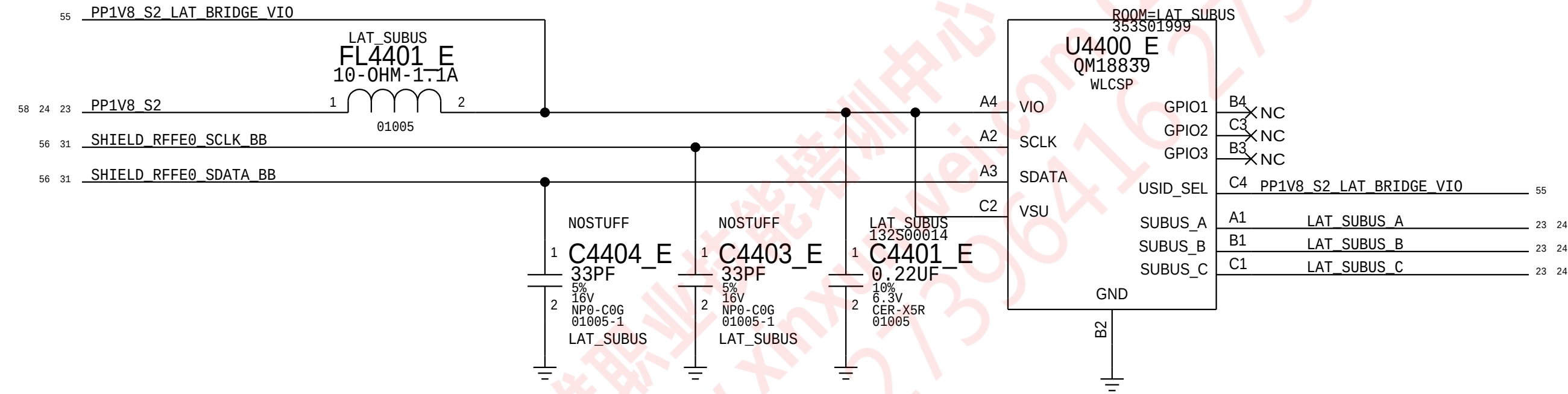
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PAGE TITLE			SYNC_DATE=07/30/2019
UAT SAWT00TH			
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	REVISION	6.0.0	
NOTICE OF PROPRIETARY PROPERTY: THE INFORMATION CONTAINED HEREIN IS THE PROPRIETARY PROPERTY OF APPLE INC. THE POSSESSOR AGREES TO THE FOLLOWING: I TO MAINTAIN THIS DOCUMENT IN CONFIDENCE II NOT TO REPRODUCE OR COPY IT III NOT TO REVEAL OR PUBLISH IT IN WHOLE OR PART IV ALL RIGHTS RESERVED	BRANCH	evt-0	
	PAGE	54 OF 191	
	SHEET		

ANTENNA SYSTEM LAT

USID = 0X8 (SEL=VIO)
USID = 0XF (SEL=GND)



PAGE TITLE		SYNC_DATE=87/38/1981	
LAT SAWTOOTH			
		DRAWING NUMBER	SIZE
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NOTICE OF PROPRIETARY PROPERTY:		BRANCH	evt-0
THE INFORMATION CONTAINED HEREIN IS THE PROPRIETARY PROPERTY OF APPLE INC. THE POSSESSOR AGREES TO THE FOLLOWING:		PAGE	55 OF 191
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IV. ALL RIGHTS RESERVED			

AP POWER

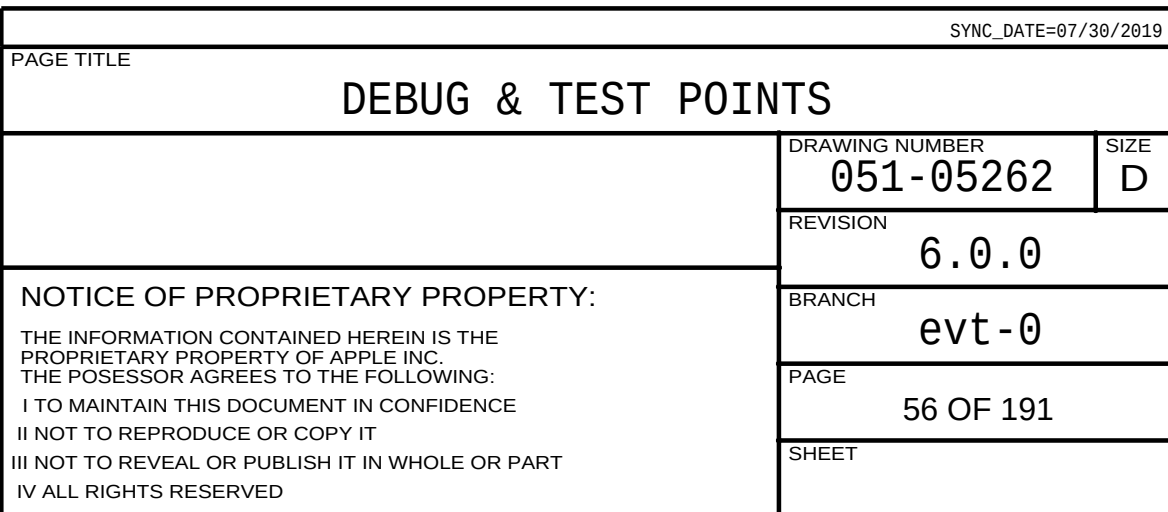
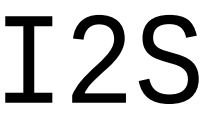
PCIE

BASEBAND

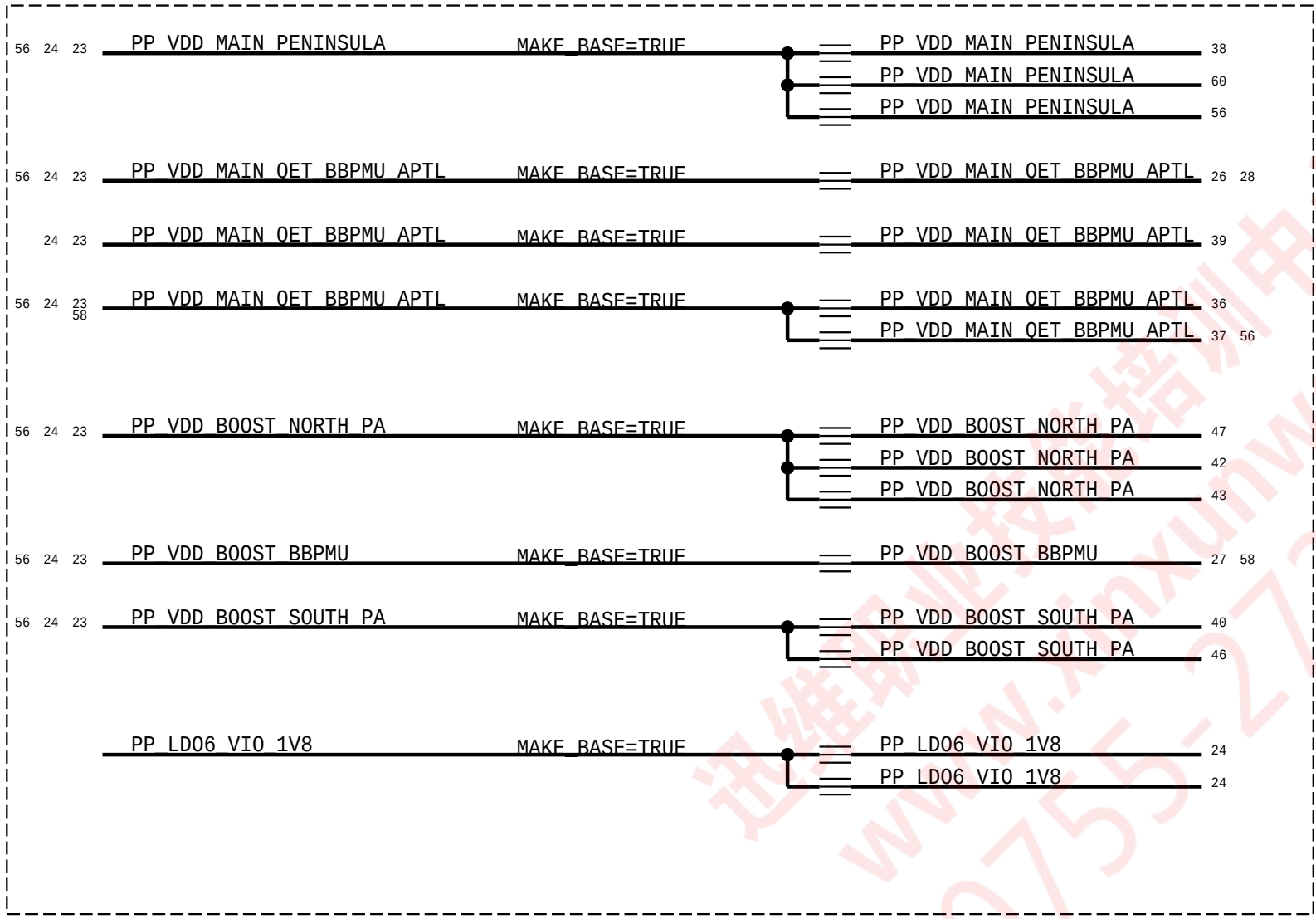
QLINK



QET DEBUG



SYNONYMS



PAGE TITLE			DRAWING NUMBER			SIZE		
SYNONYMS			051-05262			D		
REVISION			6.0.0			BRANCH		
NOTICE OF PROPRIETARY PROPERTY:			evt-0			PAGE		
THE INFORMATION CONTAINED HEREIN IS THE PROPRIETARY PROPERTY OF APPLE INC. THE POSSESSOR AGREES TO THE FOLLOWING:			57 OF 191			SHEET		
I TO MAINTAIN THIS DOCUMENT IN CONFIDENCE								
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III NOT TO REVEAL OR PUBLISH IT IN WHOLE OR PART								
IV ALL RIGHTS RESERVED								

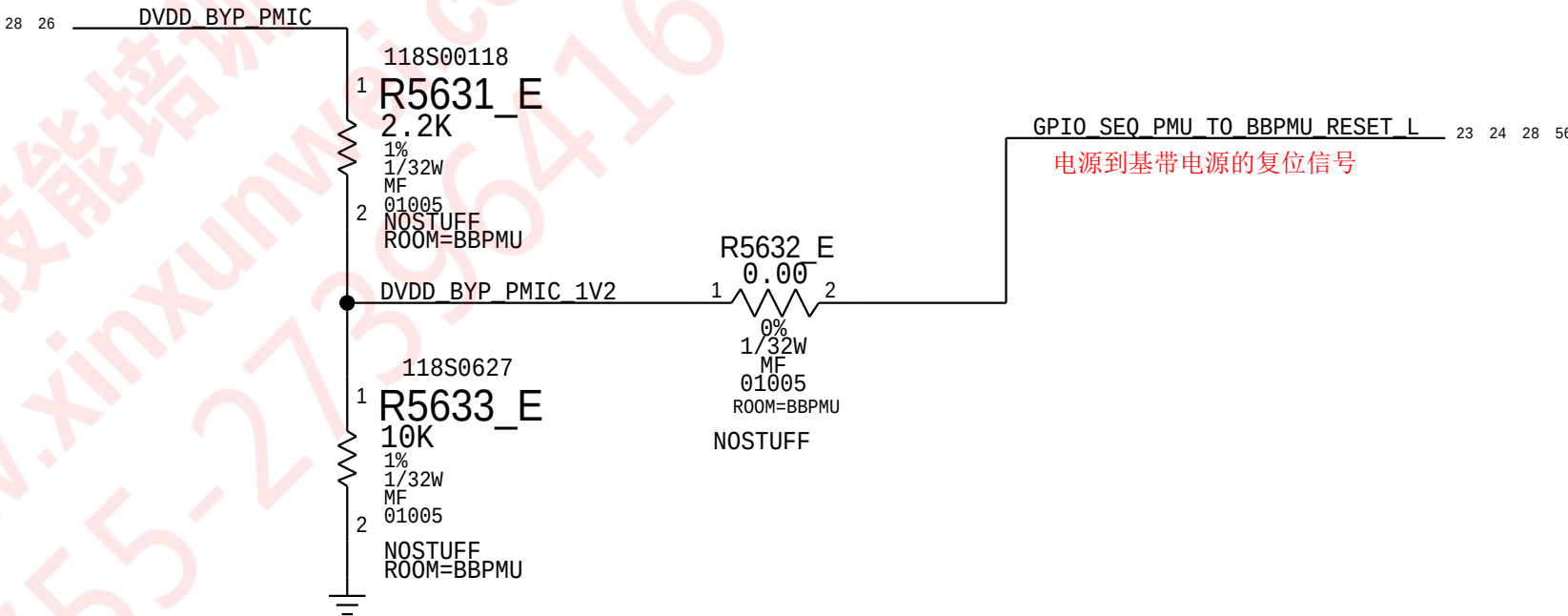
MLB ADJUSTABLES

CONFIDENTIAL AND PROPRIETARY APPLE SYSTEM DESIGN. FOR REFERENCE ONLY - NOT A CHANGE REQUEST

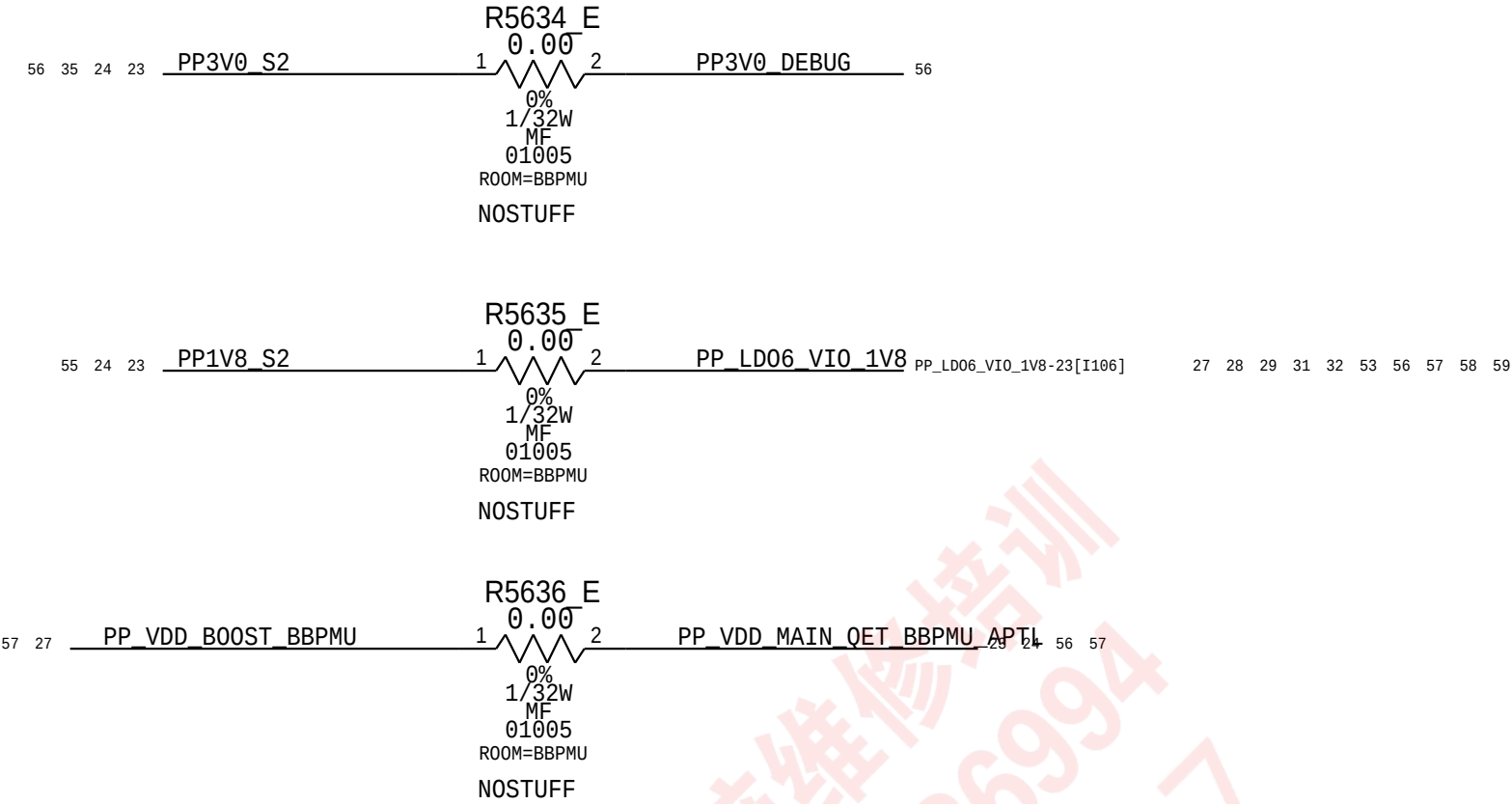
BOOT_SEQ	BOOT_CFG3	BOOT_CFG2	BOOT_CFG1	CONFIG
NAND>HSUSB	0	0	0	
PCIE	0	0	1	POR
USB	0	1	0	EUREKA
EMMC>USB	0	1	1	

POR PCIE BOOT: BOOT_CFG1 PU, BOOT_CFG2 PD
EUREKA CONFIG USB BOOT: BOOT_CFG1 PD, BOOT_CFG2 PU

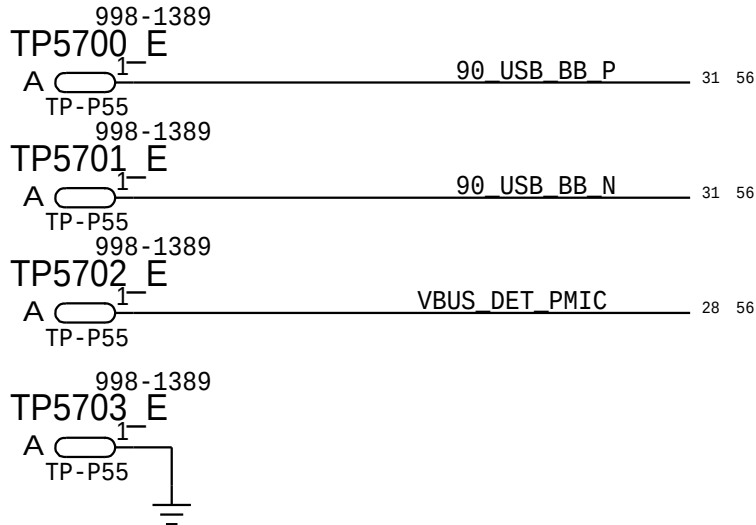
AUTO POWER ON FOR EUREKA CONFIG



STUFF ONLY FOR VENDOR CONFIG

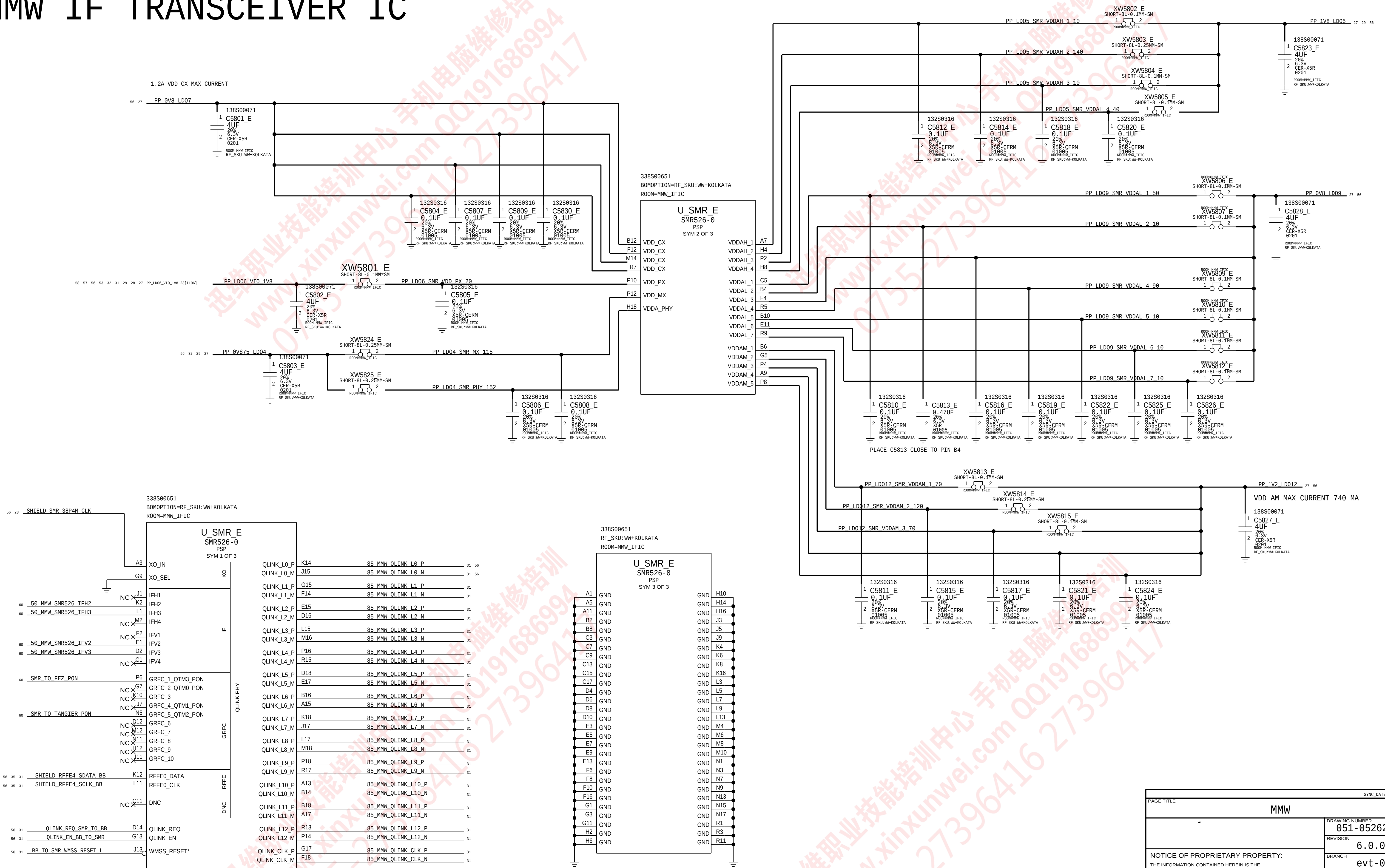


USB TP FOR TEXT FIXTURE



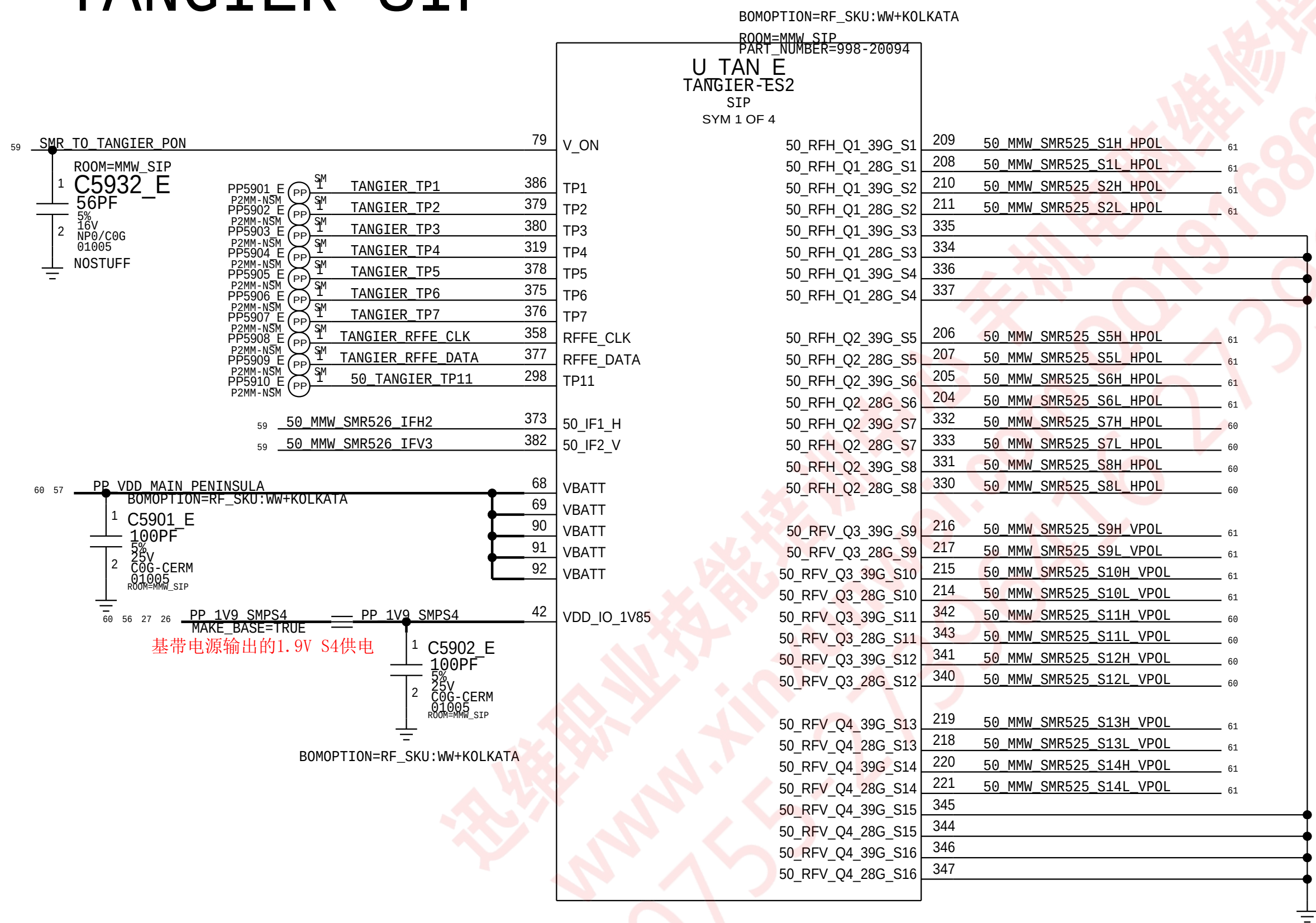
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MLB_ADJUSTABLES		
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	PAGE	58 OF 191
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MMW IF TRANSCEIVER IC

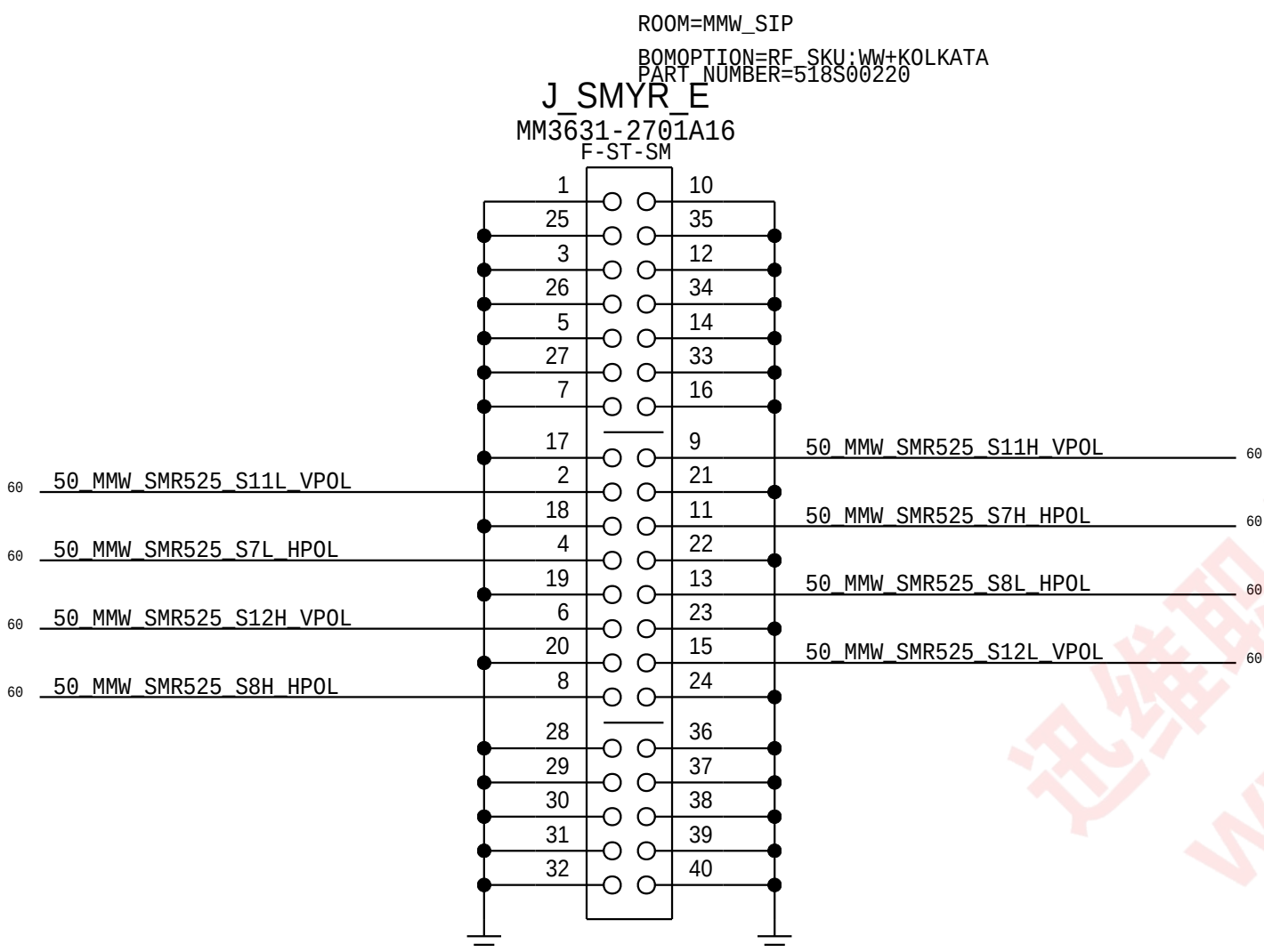


SYNC_DATE=87/38/2019			
PAGE TITLE			
MMW			
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	051-05262		D
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	6.0.0		
	BRANCH		
	evt-0		
	PAGE		
	59 OF 191		
	SHEET		

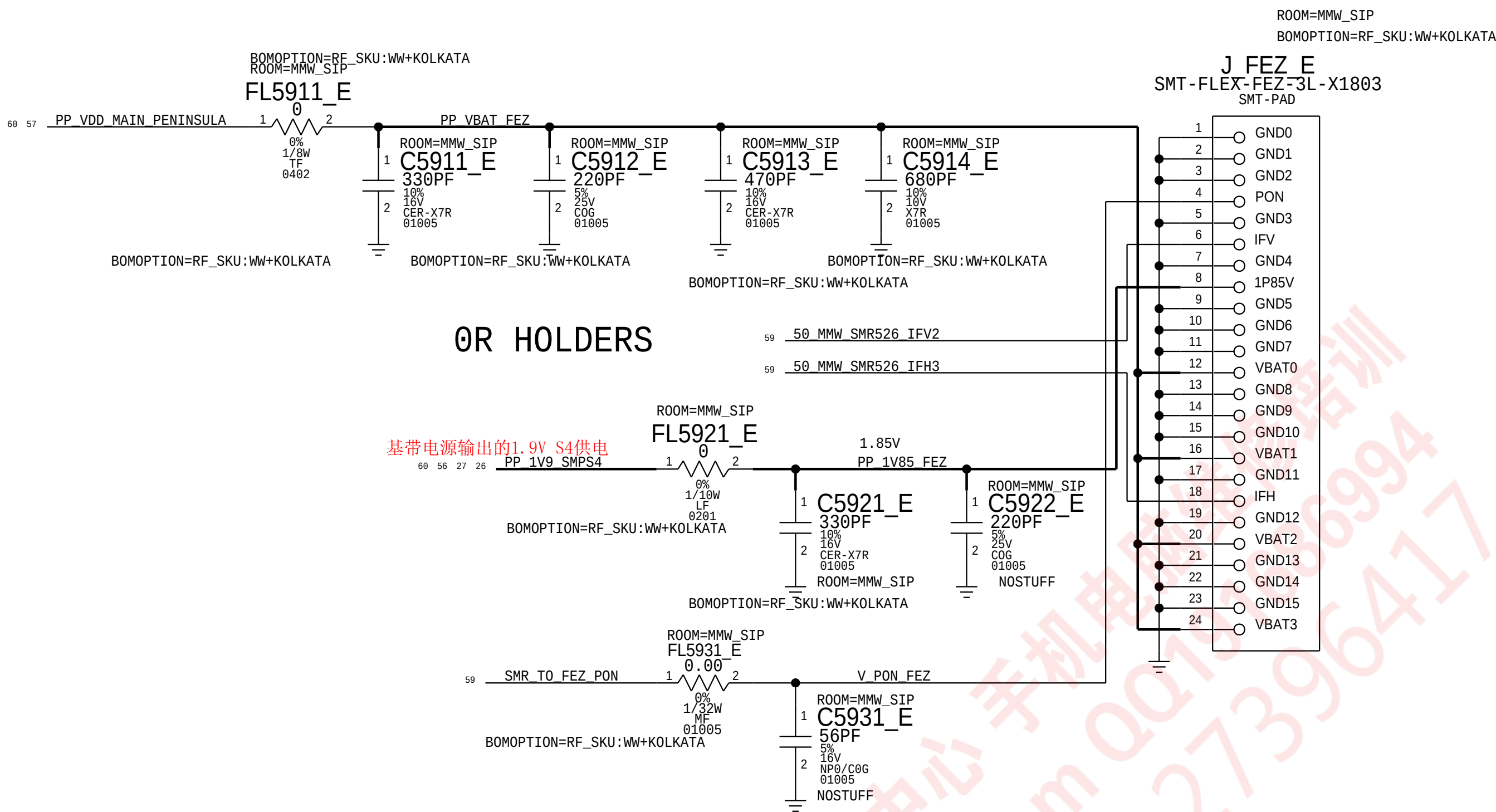
TANGIER SIP



SMYRNA



SIDE FIRE



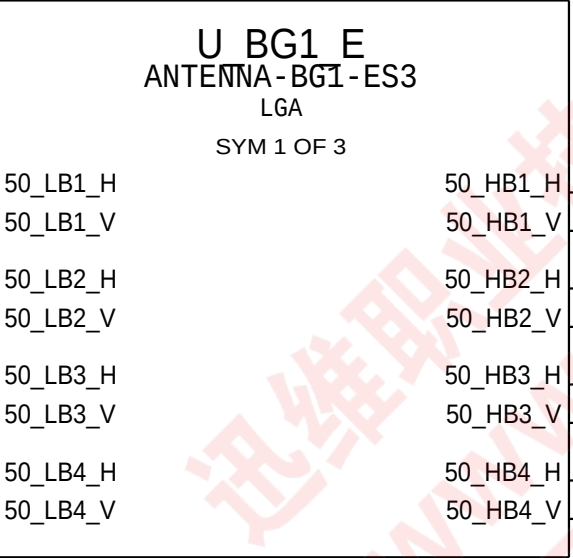
50_MMW_SMR526_IFH3 NET CONNECTS TO 50_IF1_H AND
50_MMW_SMR526_IFV2 NET CONNECTS TO 50_IF2_V IN FEZ FLEX

PAGE TITLE			SYNC_DATE=07/30/2019
TANGIER SIP		DRAWING NUMBER	051-05262
		REVISION	6.0.0
		BRANCH	evt-0
		PAGE	60 OF 191
		SHEET	
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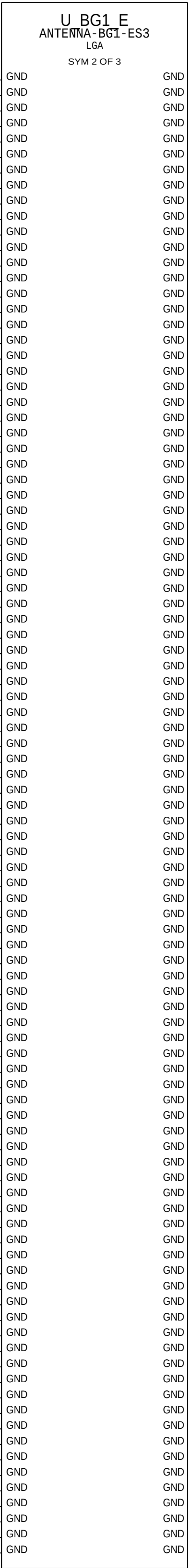
SAISS FLEX SHOULD BE ASSEMBLED TO MLB
WITH APN 632-03618

BG1 ARRAY

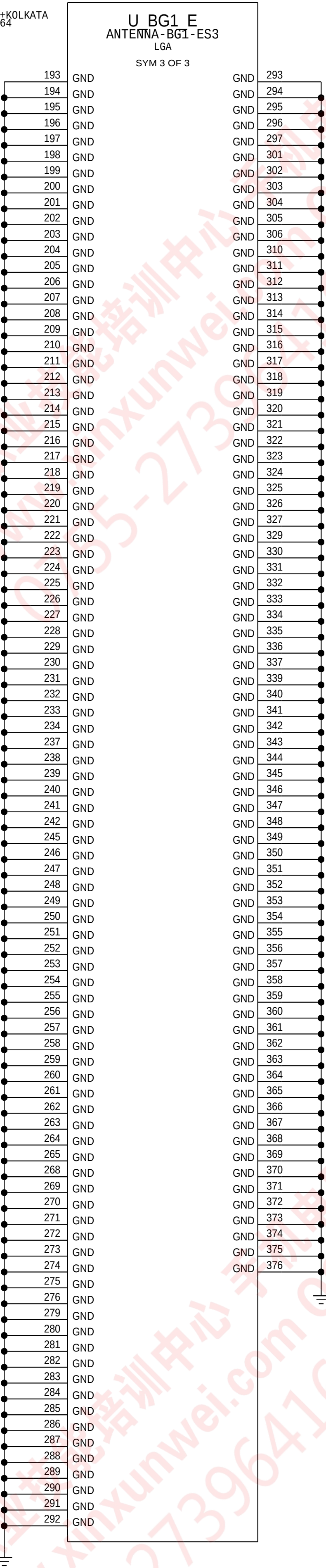
ROOM=BG1
BOMOPTION=RF_SKU:WM+KOLKATA
PART_NUMBER=998-21864



ROOM=BG1
BOMOPTION=RF_SKU:WM+KOLKATA
PART_NUMBER=998-21864



ROOM=BG1
BOMOPTION=RF_SKU:WM+KOLKATA
PART_NUMBER=998-21864



PAGE TITLE			SYNC_DATE=97/38/2019	
BG1 ARRAY			DRAWING NUMBER	SIZE
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			BRANCH	evt-0
			PAGE	61 OF 191
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1

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D



A

D



Multi-Vendor Criticals

Capacitors

In descending order of value, then package size

CRITICAL PART#	COMMENT
138S0979	CAP, CER, XSR, 10UF, 20%, 10V, 0402, H=0.65MM
138S0652	CAP, CER, XSR, 4.7UF, 20%, 6.3V, H=0.65MM, 0402
138S00070	CAP, XSR, 4.7UF, 20%, 25V, 0402
132S0663	CAP, CER, XSR, 1UF, 10%, 25V, 0402
138S0683	CAP, CER, XSR, 1UF, 10%, 25V, 0402
138S0692	CAP, CER, XSR, 1UF, 20%, 6.3V, 0201
138S00014	CAP, CER, 1UF, 20%, 16V, XSR, 0201, H=0.39MM
132S0304	CAP, CER, XSR, 0.22UF, 20%, 6.3V, 0201
132S00000	CAP, CER, 0.1UF, 10%, 50V, XTR, 0402
132S0288	CAP, CER, XSR, 0.1UF, 10%, 16V, 0201
132S0288	CAP, CER, XSR, 0.1UF, 10%, 16V, 0201
132S0534	CAP, CER, XSR, 0.1UF, 10%, 25V, 0201
132S0664	CAP, CER, 0.047UF, 10%, 25V, XSR, 0201
132S00025	CAP, CER, XSR, 0.047UF, 20%, 6.3V, 01005
132S00093	CAP, XSR, 0.022UF, 20%, 6.3V, 01005
132S0245	CAP, CER, XSR, 0.01UF, 10%, 6.3V, 01005
132S0396	CAP, CER, XSR, 1000PF, 10%, 10V, 01005
132S0296	CAP, CER, XSR, 1000PF, 10%, 6.3V, 01005
132S0318	CAP, CER, XSR, 020PF, 10%, 10V, 01005
132S0275	CAP, CER, XSR, 470PF, 10%, 10V, 01005
131S0883	CAP, CER, NP0/C0G, 220PF, 2%, 50V, 0201
131S00170	CAP, CER, C0G, 220PF, 5%, 25V, 01005
131S00053	CAP, CER, C0G, 220PF, 5%, 10V, 01005
132S0249	CAP, CER, XTR, 220PF, 10%, 10V, 01005
131S0307	CAP, CER, NP0/C0G, 100PF, 5%, 16V, 01005

Capacitors (cont'd)

CRITICAL PART#	COMMENT
131S00003	CAP, CER, NP0/C0G, 100PF, 5%, 16V, 01005
131S00323	CAP, CER, NP0/C0G, 50PF, 5%, 25V, 01005
131S0643	CAP, CER, NP0/C0G, 50PF, 5%, 25V, 01005
131S0216	CAP, CER, NP0/C0G, 47PF, 5%, 16V, 01005
131S00004	CAP, CER, 27PF, 5%, C0G, 25V, 0201
131S0223	CAP, CER, NP0/C0G, 27PF, 5%, 10V, 01005
131S0215	CAP, CER, NP0/C0G, 22PF, 5%, 16V, 01005
131S0225	CAP, CER, NP0/C0G, 15PF, 5%, 16V, 01005
131S0220	CAP, CER, NP0/C0G, 12PF, 5%, 10V, 01005
131S00353	CAP, CER, NP0/C0G, 10PF, 5%, 16V, 01005

Ferrites

CRITICAL PART#	COMMENT
155S0576	FERR 80,10 OHM,50%,750MA,0.07 DCR,01005
155S00168	FLTR,NOISE,65 OHMZ,3.40HM,0.7-26HZ,0605

Resistors

CRITICAL PART#	COMMENT
118S00068	RES,MF,1.3 MOHM,1%,200PPM,1/20W,0201
117S0055	RES,MF,1/20W,2M OHM,5,0201,SMD

Misc.

CRITICAL PART#	COMMENT
377S0106	SUPPR,TRANS,VARISTOR,12V,33PF,01005
107S0257	THERMISTOR,NTC,10K OHM,1%,B=3435,01005

Capacitor Alternates

0.1uF, 01005

PART NUMBER	ALTERNATE FOR PART NUMBER	BOM OPTION	REF DES	COMMENTS:	CRITICAL PART#	COMMENT
132S00185	132S00316	?	ALL	CAP, CER, XSR, 0.1UF, 20%, 6.3V, 02005	132S00316	01005, 0.1uF, 6.3V

0.22uF, 01005

RefDes field intentionally left blank to allow selective single-sourcing

PART NUMBER	ALTERNATE FOR PART NUMBER	BOM OPTION	REF DES	COMMENTS:	CRITICAL PART#	COMMENT
132S00233	132S00014	?		01005, 0.22uF, 6.3V, Taiyo	132S00014	01005, 0.22uF, 6.3V
132S00304	132S00014	?		01005, 0.22uF, 6.3V, Kyocera		

0.47uF, 01005

PART NUMBER	ALTERNATE FOR PART NUMBER	BOM OPTION	REF DES	COMMENTS:	CRITICAL PART#	COMMENT
138S00133	138S00120	?	ALL	01005, 0.47uF, 6.3V, Murata	138S00120	01005, 0.47uF, 6.3V
138S00269	138S00120	?	ALL	01005, 0.47uF, 6.3V, Taiyo		

1uF, 0201

PART NUMBER	ALTERNATE FOR PART NUMBER	BOM OPTION	REF DES	COMMENTS:	CRITICAL PART#	COMMENT
138S0739	138S0706	?	ALL	CAP, CER, XSR, 1UF, 20%, 10V, 0201, SAMSUNG		
138S0945	138S0706	?	ALL	CAP, CER, XSR, 1UF, 20%, 10V, 0201, KYOCERA		

2.2uF, 0201

RefDes field intentionally left blank to allow selective single-sourcing

PART NUMBER	ALTERNATE FOR PART NUMBER	BOM OPTION	REF DES	COMMENTS:	CRITICAL PART#	COMMENT
138S00049	138S0031	?		CAP, CER, XSR, 2.2UF, 20%, 6.3V, 0201	138S0031	CAP, CER, XSR, 2.2UF, 20%, 6.3V, 0201

3uF @ 1V, 0201

PART NUMBER	ALTERNATE FOR PART NUMBER	BOM OPTION	REF DES	COMMENTS:	CRITICAL PART#	COMMENT
138S00138	138S00139	?	ALL	0201, 3uF@1V, KYOCERA	138S00139	0201, 3uF@1V
138S00164	138S00139	?	ALL	0201, 3uF@1V, TAIYO		
138S00280	138S00139	?	ALL	0201, 3uF@1V, SAMSUNG		

4uF, 0201

RefDes field intentionally left blank to allow selective single-sourcing

PART NUMBER	ALTERNATE FOR PART NUMBER	BOM OPTION	REF DES	COMMENTS:	CRITICAL PART#	COMMENT
138S00116	138S00071	?		CAP, XSR, 4UF, 0201, 0.55MM, TAIYO	138S00071	0201, XSR, 4UF, 0201, 0.55MM, MURATA
138S00117	138S00071	?		CAP, XSR, 4UF, 0201, 0.55MM, KYOCERA		

4.7uF, 0402

PART NUMBER	ALTERNATE FOR PART NUMBER	BOM OPTION	REF DES	COMMENTS:	CRITICAL PART#	COMMENT
138S1103	138S0719	?	ALL	CAP, CER, XSR, 4.7UF, 20%, 10V, 0402	138S0719	0402, 4.7uF, 10V

15uF, 0402

All RefDes in () are single-sourced from Murata

PART NUMBER	ALTERNATE FOR PART NUMBER	BOM OPTION	REF DES	COMMENTS:	CRITICAL PART#	COMMENT
138S00048	138S00003	?		0402, 15uF, 6.3V, Kyocera	138S00003	0402, 15uF, 6.3V

(C1005, C3021, C3172, C3373, C3302)

18uF, 0402

PART NUMBER	ALTERNATE FOR PART NUMBER	BOM OPTION	REF DES	COMMENTS:	CRITICAL PART#	COMMENT
138S00221	138S00146	?	ALL	CAP, CER, XSR, 18UF, 20%, 6.3V, MUR, 0402	138S00146	CAP, CER, XSR, 18UF, 20%, 6.3V, MUR, 0402

20uF, 0402

All parts are single-sourced except for approved parts (listed below)

PART NUMBER	ALTERNATE FOR PART NUMBER	BOM OPTION	REF DES	COMMENTS:	CRITICAL PART#	COMMENT
138S00339	138S00084	?		Taiyo	138S00084	CAP, CER, XSR, 20UF, 20%, 6.3V, 0402, H=0.7MM
138S00338	138S00084	?		Kyocera		

C3002, C3302, C3303, C3306, C3004, C3003, C3402, C3403, C3025, C3035, C3403, C3007, C3006, C3014, C2000, C3002, C3302, C3303, C3306, C3004, C3003, C3402, C3403, C3025, C3035, C3403, C3007, C3006, C3014, C2000

22uF, 0402

PART NUMBER	ALTERNATE FOR PART NUMBER	BOM OPTION	REF DES	COMMENTS:	CRITICAL PART#	COMMENT
138S00144	138S00279	?	ALL	CAP, XSR, 20UF, 20%, 4V, MURATA, 0402	138S00279	CAP, XSR, 20UF, 20%, 4V, MURATA, 0402
138S00143	138S00279	?	ALL	CAP, XSR, 20UF, 20%, 4V, KYOCERA, 0402		

10uF @ 1V, 4-Term

PART NUMBER	ALTERNATE FOR PART NUMBER	BOM OPTION	REF DES	COMMENTS:	CRITICAL PART#	COMMENT
138S00148	138S00149	?	ALL	0402-3T, 10.5uF@1V, Kyocera	138S00149	0402-3T, 10.5uF@1V
138S00150	138S00149	?	ALL	0402-3T, 10.5uF@1V, SEMCO		
138S00151	138S00149	?	ALL	0402-3T, 10.5uF@1V, TY		

22uF, 0402 3T (WiFi)

PART NUMBER	ALTERNATE FOR PART NUMBER	BOM OPTION	REF DES	COMMENTS:	CRITICAL PART#	COMMENT
138S00024	138S0906	?	ALL	CAP, CER, 3-TERM, 7.5UF, 20%, 4V, 0402	138S0906	CAP, CER, 3-TERM, 7.5UF, 20%, 4V, 0402

Display Choke Alternates

PART NUMBER	ALTERNATE FOR PART NUMBER	BOM OPTION	REF DES	COMMENTS:	CRITICAL PART#	COMMENT
155S00415	155S00524	?	ALL	FLTR,NOISE,35 OHMZ,3 OHM,70HZ,50MA,0403	155S00524	FLTR,NOISE,35 OHMZ,3 OHM,70HZ,50MA,0403

Level Shifter Alternates

PART NUMBER	ALTERNATE FOR PART NUMBER	BOM OPTION	REF DES	COMMENTS:	CRITICAL PART#	COMMENT
311S00231	311S00232	?	ALL	IC,74AVC2145,XCVR,2 BIT,2 SPLY,X2S0N0	311S00232	IC,74AVC2145,XCVR,2 BIT,2 SPLY,X2S0N0
311S00230	311S00212	?	ALL	IC,74AVC2145,XCVR,1 BIT,2 SPLY,X2S0N0	311S00212	IC,74AVC2145,XCVR,1 BIT,2 SPLY,X2S0N0
311S00261	311S00233	?	ALL	IC,LSF0101,XCVR,2 BIT CF6,2 SPLY,X2S0N0	311S00233	IC,LSF0101,XCVR,2 BIT CF6,2 SPLY,X2S0N0
					311S00235	IC,WT0202,XCVR,2 BIT, 2 SPLY,BD01,DFN0

Power Inductor Alternates

PART NUMBER	ALTERNATE FOR PART NUMBER	BOM OPTION	REF DES	COMMENTS:	CRITICAL PART#	COMMENT
152S000721	152S000076	?	ALL	TAIYO, 200, ML2, 320, 2.5A, 40PM, 2010	152S000076	Cyntec, IND
152S000930	152S000097	?	ALL	CYNTEC, 200, CF10, 0.10M, 0.1A, 2700, 2010	152S000097	Taiyo, IND
152S000826	152S000021	?	ALL	TAIYO, 200, ML2, 100, 1.0A, 40PM, 2010	152S000021	Cyntec, IND
152S000031	152S000018	?	ALL	TAIYO, 200, ML2, 0.100M, 0.1A, 40PM, 2010	152S000018	Cyntec, IND
152S000991	152S000904	?	ALL	TAIYO, 200, ML2, 0.100M, 0.1A, 40PM, 2010	152S000904	Cyntec, IND
152S000992	152S000905	?	ALL	TAIYO, 200, ML2, 0.100M, 0.1A, 40PM, 2010	152S000905	Cyntec, IND
152S000909	152S000902	?	ALL	TAIYO, 200, ML2, 0.100M, 0.1A, 40PM, 2010	152S000902	Cyntec, IND
152S000824	152S000019	?	ALL	TAIYO, 200, ML2, 0.100M, 0.1A, 40PM, 2010	152S000019	Cyntec, IND
152S000872	152S000018	?	ALL	TKA, 200, ML2, 0.100M, 0.1A, 40PM, 2010		Boost/Yeti (2117/0.8mm)
152S000847	152S000018	?	ALL	CYNTEC, 200, ML2, 0.100M, 0.1A, 40PM, 2010		
152S01202	152S01255	?	ALL	TAIYO, 200, ML2, 0.100M, 0.1A, 40PM, 2010		Boost/Yeti (2117/0.65mm)

Misc. Alternates

PART NUMBER	ALTERNATE FOR PART NUMBER	BOM OPTION	REF DES	COMMENTS:	CRITICAL PART#	COMMENT
155S000437	155S000402	?	ALL	FERR 80,330HM,25%,1.5A,550MHM DCR,0201	155S000402	FERR 80,330HM,25%,1.5A,550MHM DCR,0201
155S000194	155S000400	?	ALL	FERR 80,1500HM,25%,200MA,0.70HM DCR,01005	155S000400	FERR 80,1500HM,25%,200MA,0.70HM DCR,01005
155S000414	155S00076	?	ALL	FERR 80,100HM,50%,1.1A,0.650HM DCR,01005	155S00076	FERR 80,100HM,50%,1.1A,0.650HM DCR,01005
155S000131	155S0755	?	ALL	FERR 80,1400HM,25%,200MA,1.0HM DCR,01005	155S0755	FERR 80,240 OHM,25%,200MA,1.0 DCR,01005
155S000583	155S000140	?	ALL	FERR 80,330HM,25%,400MA,0.200DCR,01005	155S000140	FERR 80,330HM,25%,400MA,0.200DCR,01005
377S000070	377S000001	?	ALL	TVS,BIDIR,5.0V,6PF,01005	377S000001	TVS,BIDIR,5.0V,6PF,01005
377S000140	377S000001	?	ALL	TVS,BIDIR,5.0V,6PF,01005	377S000001	SUPPRESS,TRANS,6.0V,100PF,01005
377S0168	377S000129	?	ALL	SUPPRESS,TRANS,6.0V,100PF,01005		
107S0245	107S0244	?	ALL	THERMISTOR,NTC,100K OHM,1%,B=4250,01005	107S0244	THERMISTOR,NTC,100K OHM,1%,B=4250,01005

ZRB Cap ALTs

PART NUMBER	ALTERNATE FOR PART NUMBER	BOM OPTION	REF DES	COMMENTS:	CRITICAL PART#	COMMENT
138S00240	138S00187	?	ALL	CAP, XSR, 4.7UF, 20%, 25V, 0402	138S00187	CAP, XSR, 4.7UF, 20%, 25V, 0402
138S00246	138S00185	?	ALL	CAP, XSR, 2.2UF, 20%, 25V, 0402	138S00185	CAP, XSR, 2.2UF, 20%, 25V, 0402

0-ohm, 0201, 4.5A

PART NUMBER	ALTERNATE FOR PART NUMBER	BOM OPTION	REF DES	COMMENTS:	CRITICAL PART#	COMMENT
117S00040	117S00012	?	ALL	RES,MF,0 OHM,1/10W,4.5A,0201	117S00012	RES,MF,0 OHM,1/10W,4.5A,0201

MAV20 PDN Single-source

0.22uF, 01005

All RefDes in () are single-sourced from Murata

PART NUMBER	ALTERNATE FOR PART NUMBER	BOM OPTION	REF DES	COMMENTS:
132S00233	132S00014	?	[SEE BELOW]	01005, 0.22uF, 6.3V, Taiyo
132S000304	132S00014	?	[SEE BELOW]	01005, 0.22uF, 6.3V, Kyocera

2.2uF, 0201

All RefDes in () are single-sourced from Murata

PART NUMBER	ALTERNATE FOR PART NUMBER	BOM OPTION	REF DES	COMMENTS:
138S00049	138S0031	?	[SEE BELOW]	CAP, CER, XSR, 2.2UF, 20%, 6.3V, 0201

4uF, 0201

All RefDes in () are single-sourced from Murata

PART NUMBER	ALTERNATE FOR PART NUMBER	BOM OPTION	REF DES	COMMENTS:
138S00116	138S00071	?	[SEE BELOW]	CAP, XSR, 4UF, 0201, 0.55MM, TAIYO
138S00117	138S00071	?	[SEE BELOW]	CAP, XSR, 4UF, 0201, 0.55MM, KYOCERA

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Power Constraints
Physical

CLASS DEFINITIONS	COMMA SEPARATED WITH WILDCARD SUPPORT: NET NAMES EX: DDR*		CLASS CHANGES
	CLASS NAME	CONSTRAINT SET	
LX/CX Nodes	PWR_SHAPE_LX	PWR_SHAPE	Y
	PWR_SHAPE_LX_GALENA	PWR_SHAPE	Y
	PWR_SHAPE_LX	PWR_SHAPE	Y
System-wide	PWR_SHAPE	PWR_SHAPE	Y
	PWR_SHAPE	PWR_SHAPE	Y
	PWR_50UM	PWR_50UM	Y
	PWR_SHAPE	PWR_SHAPE	Y
	PWR_SHAPE	PWR_SHAPE	Y
	PWR_200UM	PWR_200UM	Y
	PWR_200UM	PWR_200UM	Y
	PWR_100UM	PWR_100UM	Y
SoC:CPU/GPU	PWR_SHAPE	PWR_SHAPE	Y
	PWR_SHAPE	PWR_SHAPE	Y
	PWR_SHAPE	PWR_SHAPE	Y
SoC:SRAM/SOC	PWR_SHAPE	PWR_SHAPE	Y
	PWR_SHAPE	PWR_SHAPE	Y
SoC:Fixed	PWR_SHAPE	PWR_SHAPE	Y
	PWR_SHAPE	PWR_SHAPE	Y
SoC:VDD12	PWR_80UM	PWR_80UM	Y
	PWR_80UM	PWR_80UM	Y
	PWR_80UM	PWR_80UM	Y
	PWR_200UM	PWR_200UM	Y
	PWR_80UM	PWR_80UM	Y
	PWR_80UM	PWR_80UM	Y
	PWR_80UM	PWR_80UM	Y
	PWR_80UM	PWR_80UM	Y
SoC:DDR	PWR_SHAPE	PWR_SHAPE	Y
	PWR_200UM	PWR_200UM	Y
	PWR_SHAPE	PWR_SHAPE	Y
SoC:Misc.	PWR_SHAPE	PWR_SHAPE	Y
	PWR_80UM	PWR_80UM	Y
	PWR_SHAPE	PWR_SHAPE	Y
	PWR_SHAPE	PWR_SHAPE	Y
	PWR_200UM	PWR_200UM	Y
NAND	PWR_100UM	PWR_100UM	Y
	PWR_300UM	PWR_300UM	Y
	PWR_SHAPE	PWR_SHAPE	Y
	PWR_100UM	PWR_100UM	Y
	PWR_SHAPE	PWR_SHAPE	Y
PMU	PWR_SHAPE	PWR_SHAPE	Y
	PWR_SHAPE	PWR_SHAPE	Y
Yangtze	PWR_SHAPE	PWR_SHAPE	Y
	PWR_SHAPE	PWR_SHAPE	Y
Dotara	PWR_SHAPE	PWR_SHAPE	Y
	PWR_SHAPE	PWR_SHAPE	Y
	PWR_100UM	PWR_100UM	Y
	PWR_SHAPE	PWR_SHAPE	Y
	PWR_SHAPE	PWR_SHAPE	Y
	PWR_SHAPE	PWR_SHAPE	Y
	PWR_SHAPE	PWR_SHAPE	Y
	PWR_SHAPE	PWR_SHAPE	Y
	PWR_SHAPE	PWR_SHAPE	Y
	PWR_SHAPE	PWR_SHAPE	Y
Camera PMUs	PWR_SHAPE	PWR_SHAPE	Y
	PWR_100UM	PWR_100UM	Y
	PWR_100UM	PWR_100UM	Y
	PWR_SHAPE	PWR_SHAPE	Y
	PWR_SHAPE	PWR_SHAPE	Y
	PWR_SHAPE	PWR_SHAPE	Y
	PWR_SHAPE	PWR_SHAPE	Y
	PWR_SHAPE	PWR_SHAPE	Y
	PWR_100UM	PWR_100UM	Y
	PWR_100UM	PWR_100UM	Y
	PWR_100UM	PWR_100UM	Y
	PWR_100UM	PWR_100UM	Y
	PWR_100UM	PWR_100UM	Y
	PWR_100UM	PWR_100UM	Y
	PWR_100UM	PWR_100UM	Y
Will/Jasper	PWR_100UM	PWR_100UM	Y
	PWR_100UM	PWR_100UM	Y
	PWR_100UM	PWR_100UM	Y
	PWR_100UM	PWR_100UM	Y
	PWR_100UM	PWR_100UM	Y
Rigel/Pearl	PWR_100UM	PWR_100UM	Y
	PWR_SHAPE	PWR_SHAPE	Y
	PWR_SHAPE	PWR_SHAPE	Y
	PWR_SHAPE	PWR_SHAPE	Y
	PWR_SHAPE	PWR_SHAPE	Y
	PWR_SHAPE	PWR_SHAPE	Y
	PWR_SHAPE	PWR_SHAPE	Y
	PWR_SHAPE	PWR_SHAPE	Y
	PWR_SHAPE	PWR_SHAPE	Y
	PWR_SHAPE	PWR_SHAPE	Y
Audio	PWR_50UM	PWR_50UM	Y
	PWR_50UM	PWR_50UM	Y
Mic Bias	PWR_MIC	PWR_DEFAULT	Y
	PWR_MIC	RET_CODED_FROM_MIC*	Y
Codec	PWR_200UM	PWR_200UM	Y
	PWR_50UM	PWR_50UM	Y
BotSpk	PWR_SHAPE	PWR_SHAPE	Y
	PWR_SHAPE	PWR_SHAPE	Y
	PWR_80UM	PWR_80UM	Y
	PWR_50UM	PWR_50UM	Y
	PWR_80UM	PWR_80UM	Y
TopSpk	PWR_50UM	PWR_50UM	Y
	PWR_SHAPE	PWR_SHAPE	Y
	PWR_80UM	PWR_80UM	Y
	PWR_SHAPE	PWR_SHAPE	Y
	PWR_80UM	PWR_80UM	Y

Physical (continued)

CLASS DEFINITIONS	COMMA SEPARATED WITH WILDCARD SUPPORT: NET NAMES EX: DDR*		CLASS CHANGES
	CLASS NAME	CONSTRAINT SET	
Arc	PWR_50UM	PWR_50UM	Y
	PWR_50UM	PWR_50UM	Y
	PWR_SHAPE	PWR_SHAPE	Y
Touch (DITO)	PWR_SHAPE	PWR_SHAPE	Y
	PWR_80UM	PWR_80UM	Y
	PWR_100UM	PWR_100UM	Y
	PWR_80UM	PWR_80UM	Y
Touch (Both)	PWR_200UM	PWR_200UM	Y
	PWR_SHAPE	PWR_SHAPE	Y
Display PMIC	PWR_SHAPE	PWR_SHAPE	Y
	PWR_SHAPE	PWR_SHAPE	Y
Kraken	PWR_SHAPE	PWR_SHAPE	Y
	PWR_SHAPE	PWR_SHAPE	Y
	PWR_80UM	PWR_80UM	Y
	PWR_200UM	PWR_200UM	Y
Parrot	PWR_SHAPE	PWR_SHAPE	Y
	PWR_80UM	PWR_80UM	Y
	PWR_80UM	PWR_80UM	Y
AF/OIS (Shared)	PWR_SHAPE	PWR_SHAPE	Y
	PWR_SHAPE	PWR_SHAPE	Y
Wide RCAM	PWR_100UM	PWR_100UM	Y
	PWR_80UM	PWR_80UM	Y
	PWR_100UM	PWR_100UM	Y
	PWR_200UM	PWR_200UM	Y
Superflex-Cam2	PWR_100UM	PWR_100UM	Y
	PWR_100UM	PWR_100UM	Y
	PWR_200UM	PWR_200UM	Y
	PWR_100UM	PWR_100UM	Y
Singleflex-Cam	PWR_80UM	PWR_80UM	Y
	PWR_80UM	PWR_80UM	Y
	PWR_100UM	PWR_100UM	Y
	PWR_200UM	PWR_200UM	Y
FCAM	PWR_80UM	PWR_80UM	Y
	PWR_200UM	PWR_200UM	Y
	PWR_200UM	PWR_200UM	Y
	PWR_200UM	PWR_200UM	Y
Juliet IRCAM	PWR_80UM	PWR_80UM	Y
	PWR_80UM	PWR_80UM	Y
Jasper	PWR_80UM	PWR_80UM	Y
	PWR_80UM	PWR_80UM	Y
	PWR_80UM	PWR_80UM	Y
	PWR_SHAPE	PWR_SHAPE	Y
ALS	PWR_80UM	PWR_80UM	Y
	Compass	PWR_80UM	Y
Penrose	PWR_80UM	PWR_80UM	Y
	Dock	PWR_80UM	Y
Display	PWR_200UM	PWR_200UM	Y
	PWR_80UM	PWR_80UM	Y
	PWR_80UM	PWR_80UM	Y
	PWR_SHAPE	PWR_SHAPE	Y
NFC/Ironman NFC	PWR_80UM	PWR_80UM	Y
	PWR_80UM	PWR_80UM	Y
	PWR_SHAPE	PWR_SHAPE	Y
	PWR_100UM	PWR_100UM	Y
Arrow	PWR_100UM	PWR_100UM	Y
	PWR_100UM	PWR_100UM	Y
	PWR_100UM	PWR_100UM	Y
	PWR_100UM	PWR_100UM	Y
	PWR_100UM	PWR_100UM	Y
	PWR_100UM	PWR_100UM	Y
	PWR_100UM	PWR_100UM	Y
	PWR_100UM	PWR_100UM	Y
	PWR_100UM	PWR_100UM	Y
	PWR_100UM	PWR_100UM	Y
	PWR_100UM	PWR_100UM	Y
	PWR_100UM	PWR_100UM	Y
	PWR_100UM	PWR_100UM	Y
	PWR_100UM	PWR_100UM	Y
	PWR_100UM	PWR_100UM	Y
QETs/APTs	PWR_SHAPE	PWR_SHAPE	Y
	PWR_SHAPE	PWR_SHAPE	Y
	PWR_300UM	PWR_300UM	Y
	PWR_SHAPE	PWR_SHAPE	Y
	PWR_SHAPE	PWR_SHAPE	Y
Misc. Radio	PWR_100UM	PWR_100UM	Y
	PWR_100UM	PWR_100UM	Y
	PWR_100UM	PWR_100UM	Y
	PWR_100UM	PWR_100UM	Y
	GND	DEFAULT	Y

Spacing

CLASS DEFINITIONS		COMMA SEPARATED WITH WILDCARD SUPPORT: NET NAMES EX: DDR*	
CLASS NAME	CONSTRAINT SET	DP NAMES EX: DP:DP_AA*.DP_BB* (LINE STARTS WITH FLAG DP)	Y/N
GND	MSAP_YIELD	GND	Y
PWR_DEFAULT	DEFAULT	=	Y
PWR_50UM	DEFAULT	=	Y
PWR_80UM	DEFAULT	=	Y
PWR_100UM	DEFAULT	=	Y
PWR_200UM	DEFAULT	=	Y
PWR_300UM	DEFAULT	=	Y
PWR_SHAPE	DEFAULT	=	Y
PWR_100UM_HV	HV_SPACING	=	Y
PWR_MIC	DEFAULT	=	Y
PWR_NFC_ANT	DEFAULT	=	Y
PWR_SHAPE_LX	LX_SPACING	=	Y
PWR_SHAPE_LX_GALENA	LX_SPACING	=	Y

Class-Class on next page

CONSTRAINTS: Power

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Diff Pair Constraints
Electrical

CLASS DEFINITIONS			COMMA SEPARATED WITH WILDCARD SUPPORT: NET NAMES EX: DDR* DP NAMES EX: DP:DP_AA*.DP_BB* (LINE STARTS WITH FLAG DP:)	CLASS COVERED Y/N
CLASS NAME	...	CONSTRAINT SET		
E_DP_GENERIC	E	GENERIC_DP	DP:DP_CODEC_AOUT*	Y
E_DP_GENERIC	E	GENERIC_DP	DP:DP_MIC*	Y
E_DP_GENERIC	E	GENERIC_DP	DP:DP_PENROSE*	Y
E_DP_NTC	E	DIFF_PAIR_NTC	DP:DP_NTC_*	Y
E_DP_GENERIC	E	GENERIC_DP	DP:DP_*MTR*	Y
E_DP_GENERIC	E	GENERIC_DP	DP:DP_ANALOG*SENSE*	Y
E_DP_GENERIC	E	GENERIC_DP	DP:DP_ANALOG_VIN_SAKONNET_FROM_HALL	Y
E_DP_GENERIC	E	GENERIC_DP	DP:DP_PMU_VDD_MAIN_SENSE*	Y
E_DP_NC	E	GENERIC_DP	DP:DP_NC*	Y

Physical

CLASS DEFINITIONS			COMMA SEPARATED WITH WILDCARD SUPPORT: NET NAMES EX: DDR* DP NAMES EX: DP:DP_AA*.DP_BB* (LINE STARTS WITH FLAG DP:)	CLASS COVERED Y/N
CLASS NAME	...	CONSTRAINT SET		
DP_PWR	P	PWR_DP	DP:DP_CODEC_AOUT*	Y
DP_MIC	P	GENERIC_DP	DP:DP_*MIC*	Y
DP_PENROSE	P	GENERIC_DP	DP:DP_PENROSE*	Y
DP_NTC	P	GENERIC_DP	DP:DP_NTC_*	Y
DP_GENERIC	P	GENERIC_DP	DP:DP_*MTR*	Y
DP_GENERIC	P	GENERIC_DP	DP:DP_ANALOG*SENSE*	Y
DP_SAKONNET	P	GENERIC_DP	DP:DP_ANALOG_VIN_SAKONNET_FROM_HALL	Y
DP_GENERIC	P	GENERIC_DP	DP:DP_PMU_VDD_MAIN_SENSE*	Y
DP_NC	P	GENERIC_DP	DP:DP_NC*	Y

Spacing

CLASS DEFINITIONS			COMMA SEPARATED WITH WILDCARD SUPPORT: NET NAMES EX: DDR* DP NAMES EX: DP:DP_AA*.DP_BB* (LINE STARTS WITH FLAG DP:)	CLASS COVERED Y/N
CLASS NAME	...	CONSTRAINT SET		
DP_PWR	S	DEFAULT	=	Y
DP_MIC	S	A_DIELECTRIC_1.5X	=	Y
DP_PENROSE	S	A_DIELECTRIC_1.5X	=	Y
DP_NTC	S	A_DIELECTRIC_1.5X	=	Y
DP_GENERIC	S	DEFAULT	=	Y
DP_SAKONNET	S	DEFAULT	=	Y
DP_NC	S	DEFAULT	=	Y

Class-Class Spacing

CLASS TO CLASS SPACING		
CLASS NAME	CLASS NAME	CONSTRAINT SET
DP_MIC	GND	DEFAULT
DP_MIC	PWR_MIC	DEFAULT
DP_MIC	DP_SAKONNET	DEFAULT

Sensitive Analog
Physical

CLASS DEFINITIONS			COMMA SEPARATED WITH WILDCARD SUPPORT: NET NAMES EX: DDR* DP NAMES EX: DP:DP_AA*.DP_BB* (LINE STARTS WITH FLAG DP:)	CLASS COVERED Y/N
CLASS NAME	...	CONSTRAINT SET		
ANALOG_SAKONNET_BIAS	P	DEFAULT	ANALOG_BIAS_SAKONNET*, ANALOG_RET_SAKONNET*	Y
ANALOG	P	DEFAULT	ANALOG*DOTARA*	Y
ANALOG	P	DEFAULT	ANALOG*DOMBRA*	Y
ANALOG	P	DEFAULT	ANALOG*SENSE,ANALOG*SENSE_SE,ANALOG*SENSE_FILT	Y
ANALOG	P	DEFAULT	ANALOG_DDR*ZQ*	Y
ANALOG_NAND_ZQ	P	PWR_100UM	ANALOG_NAND*ZQ*	Y
ANALOG	P	DEFAULT	ANALOG*REXT*	Y
ANALOG	P	DEFAULT	ANALOG*CAL*	Y
ANALOG	P	DEFAULT	ANALOG_*REF*	Y
ANALOG	P	DEFAULT	ANALOG_FB*	Y
ANALOG	P	DEFAULT	ANALOG_KRAKEN_BYPASS,ANALOG_KRAKEN_BI_CC62B_CC	Y
ANALOG	P	DEFAULT	ANALOG_RIGEL_LSCP	Y
ANALOG_AMP_FILT	P	PWR_80UM	ANALOG_*_SPK_FILT,ANALOG_ARCAMP_FILT	Y
ANALOG_VSS_PMU_XTAL	P	DEFAULT	VSS_PMU_XTAL	Y
ANALOG_AMUX	P	DEFAULT	AMUX*	Y
ANALOG_NTC	P	DEFAULT	NTC_NFC_UAT*_POS	Y
ANALOG_NTC	P	DEFAULT	NTC_PEARL_VCSEL_TO_RIGEL	Y
ANALOG_NTC	P	DEFAULT	NTC_STROBE_MODULE*	Y
ANALOG_SPKR_SENSE	P	DEFAULT	COIL_TO_SPKRAMP_*_VSENSE_*, SOLENOID_TO_ARC_VSENSE_*	Y

Spacing

CLASS DEFINITIONS			COMMA SEPARATED WITH WILDCARD SUPPORT: NET NAMES EX: DDR* DP NAMES EX: DP:DP_AA*.DP_BB* (LINE STARTS WITH FLAG DP:)	CLASS COVERED Y/N
CLASS NAME	...	CONSTRAINT SET		
ANALOG	S	A_DIELECTRIC_1.5X	=	Y
ANALOG_SAKONNET_BIAS	S	A_DIELECTRIC_1.5X	=	Y
ANALOG_AMP_FILT	S	A_DIELECTRIC_1.5X	=	Y
ANALOG_VSS_PMU_XTAL	S	A_DIELECTRIC_1.5X	=	Y
ANALOG_AMUX	S	A_DIELECTRIC_1.5X	=	Y
ANALOG_NTC	S	A_DIELECTRIC_1.5X	=	Y
ANALOG_SPKR_SENSE	S	A_DIELECTRIC_1.5X	=	Y
ANALOG_NAND_ZQ	S	A_DIELECTRIC_1.5X	=	Y

Class-Class Spacing

CLASS TO CLASS SPACING		
CLASS NAME	CLASS NAME	CONSTRAINT SET
ANALOG	GND	DEFAULT
ANALOG_SAKONNET_BIAS	GND	DEFAULT
ANALOG_AMP_FILT	GND	DEFAULT
ANALOG_VSS_PMU_XTAL	GND	DEFAULT
ANALOG_AMUX	GND	DEFAULT
ANALOG_NTC	GND	DEFAULT
ANALOG_SPKR_SENSE	GND	DEFAULT
ANALOG_SAKONNET_BIAS	ANALOG_SAKONNET_BIAS	DEFAULT
ANALOG_SAKONNET_BIAS	DP_SAKONNET	DEFAULT
ANALOG_NAND_ZQ	GND	DEFAULT

Clocks
Physical

CLASS DEFINITIONS			COMMA SEPARATED WITH WILDCARD SUPPORT: NET NAMES EX: DDR* DP NAMES EX: DP:DP_AA*.DP_BB* (LINE STARTS WITH FLAG DP:)	CLASS COVERED Y/N
CLASS NAME	...	CONSTRAINT SET		
CLK	P	DEFAULT	CLK_*	Y
CLK_SPMI	P	DEFAULT	SPMI*CLK*	Y
CLK_I2S	P	DEFAULT	I2S*MCLK*, I2S*BCLK*	Y
CLK_SPI	P	DEFAULT	SPI*SCLK*	Y

Spacing

CLASS DEFINITIONS			COMMA SEPARATED WITH WILDCARD SUPPORT: NET NAMES EX: DDR* DP NAMES EX: DP:DP_AA*.DP_BB* (LINE STARTS WITH FLAG DP:)	CLASS COVERED Y/N
CLASS NAME	...	CONSTRAINT SET		
CLK	S	A_DIELECTRIC_1.5X	=	Y
CLK_SPMI	S	A_DIELECTRIC_1.5X	=	Y
CLK_I2S	S	A_DIELECTRIC_1.5X	=	Y
CLK_SPI	S	A_DIELECTRIC_1.5X	=	Y

Class-Class Spacing

CLASS TO CLASS SPACING		
CLASS NAME	CLASS NAME	CONSTRAINT SET
CLK	GND	DEFAULT
CLK_SPMI	GND	DEFAULT
CLK_I2S	GND	DEFAULT
CLK_SPI	GND	DEFAULT

Grouping Constraints
Physical

CLASS DEFINITIONS			COMMA SEPARATED WITH WILDCARD SUPPORT: NET NAMES EX: DDR* DP NAMES EX: DP:DP_AA*.DP_BB* (LINE STARTS WITH FLAG DP:)	CLASS COVERED Y/N
CLASS NAME	...	CONSTRAINT SET		
GRP_GPIO	P	DEFAULT	GPIO*, IO_*	Y
GRP_NC	P	DEFAULT	NC_*	Y
GRP_I2C	P	DEFAULT	I2C*	Y
GRP_I2S	P	DEFAULT	I2S*LRCLK*, I2S*DOUT*, I2S*DIN*	Y
GRP_SPI	P	DEFAULT	SPI*MOSI*, SPI*MISO*, SPI*CS*	Y
GRP_UART	P	DEFAULT	UART*	Y
GRP_SWD	P	DEFAULT	SWD*	Y
GRP_SPMI_DATA	P	DEFAULT	SPMI*DATA*	Y
GRP_PCIE_SIDE	P	DEFAULT	PCIE*CLKREQ*, PCIE*PERST*	Y
GRP_LPDP_AUX	P	DEFAULT	LPDP*AUX*	Y
GRP_CODEC_FILT	P	PWR_200UM	CODEC_*FILT*, CODEC_*FILT*	Y
RFFE	P	DEFAULT	SHIELD_RFFE*	Y

Spacing

CLASS DEFINITIONS			COMMA SEPARATED WITH WILDCARD SUPPORT: NET NAMES EX: DDR* DP NAMES EX: DP:DP_AA*.DP_BB* (LINE STARTS WITH FLAG DP:)	CLASS COVERED Y/N
CLASS NAME	...	CONSTRAINT SET		
GRP_GPIO	S	DEFAULT	=	Y
GRP_NC	S	DEFAULT	=	Y
GRP_I2C	S	DEFAULT	=	Y
GRP_I2S	S	DEFAULT	=	Y
GRP_SPI	S	DEFAULT	=	Y
GRP_UART	S	DEFAULT	=	Y
GRP_SWD	S	DEFAULT	=	Y
GRP_SPMI_DATA	S	DEFAULT	=	Y
GRP_PCIE_SIDE	S	DEFAULT	=	Y
GRP_LPDP_AUX	S	DEFAULT	=	Y
GRP_CODEC_FILT	S	DEFAULT	=	Y
RFFE	S	A_DIELECTRIC_1.5X	=	Y

Class-Class Spacing

CLASS TO CLASS SPACING		
CLASS NAME	CLASS NAME	CONSTRAINT SET
RFFE	50_TX0_THIN	A_DIELECTRIC_3X
RFFE	50_TX1_THIN	A_DIELECTRIC_3X
RFFE	50_TX2_THIN	A_DIELECTRIC_3X
RFFE	50_RX_THIN	A_DIELECTRIC_3X
RFFE	RFFE	DEFAULT
RFFE	GND	DEFAULT

Power (cont'd from prev. page)
Class-Class Spacing

CLASS TO CLASS SPACING		
CLASS NAME	CLASS NAME	CONSTRAINT SET
PWR_100UM	GND	MSAP_YIELD_PWR
PWR_100UM	PWR_100UM	100UM-249UM_SPACING
PWR_100UM	PWR_200UM	100UM-249UM_SPACING
PWR_100UM	PWR_300UM	100UM-249UM_SPACING
PWR_100UM	PWR_SHAPE	100UM-249UM_SPACING
PWR_200UM	GND	MSAP_YIELD_PWR
PWR_200UM	PWR_200UM	100UM-249UM_SPACING
PWR_200UM	PWR_300UM	100UM-249UM_SPACING
PWR_200UM	PWR_SHAPE	100UM-249UM_SPACING
PWR_300UM	GND	MSAP_YIELD_PWR
PWR_300UM	PWR_300UM	250UM+ SPACING
PWR_300UM	PWR_SHAPE	250UM+ SPACING
PWR_SHAPE	GND	MSAP_YIELD_PWR
PWR_SHAPE	PWR_SHAPE	250UM+ SPACING
PWR_SHAPE_LX	GND	DEFAULT
PWR_SHAPE_LX	PWR_SHAPE_LX	DEFAULT
PWR_SHAPE_LX	ANALOG	A_DIELECTRIC_3X
PWR_SHAPE_LX	ANALOG_SAKONNET_BIAS	A_DIELECTRIC_3X
PWR_SHAPE_LX	ANALOG_AMP_FILT	A_DIELECTRIC_3X
PWR_SHAPE_LX	ANALOG_VSS_PMU_XTAL	A_DIELECTRIC_3X
PWR_SHAPE_LX	ANALOG_NTC	A_DIELECTRIC_3X

CLASS TO CLASS SPACING		
CLASS NAME	CLASS NAME	CONSTRAINT SET
PWR_SHAPE_LX	ANALOG_AMUX	A_DIELECTRIC_3X
PWR_SHAPE_LX	ANALOG_SPKR_SENSE	A_DIELECTRIC_3X
PWR_SHAPE_LX	CLK	A_DIELECTRIC_3X
PWR_SHAPE_LX	CLK_SPMI	A_DIELECTRIC_3X
PWR_SHAPE_LX	CLK_I2S	A_DIELECTRIC_3X
PWR_SHAPE_LX	CLK_SPI	A_DIELECTRIC_3X
PWR_SHAPE_LX_GALENA	GND	A_DIELECTRIC_3X
PWR_100UM_HV	GND	DEFAULT

PAGE TITLE		
CONSTRAINTS: Misc.		
		DRAWING NUMBER 051-05215
		REVISION 4.6.0
		BRANCH
		PAGE 69 OF 191
		SHEET
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Hybrid RF CSets

50-Thin				50-Wide			
HYBRID IMPEDANCE RULE				HYBRID IMPEDANCE RULE			
TRACE LAYER	REFERENCE LAYER(S)	REQUIRED IMPEDANCE	TRACE WIDTH (OPTIONAL)	TRACE LAYER	REFERENCE LAYER(S)	REQUIRED IMPEDANCE	TRACE WIDTH (OPTIONAL)
RF SPACING VALUES= 2				RF SPACING VALUES= 2			
RULE NAME= 50_THIN		ZONE NAME= PRIMARY		RULE NAME= 50_WIDE		ZONE NAME= PRIMARY	
ISL3	ISL2, ISL4	50		ISL3	TOP, ISL5	50	
ISL6	ISL7, ISL5	50		ISL6	ISL8, ISL4	50	
ISL8	ISL9, ISL7	50		ISL7	ISL9, ISL5	50	
ISL9	BOTTOM, ISL8	50		BOTTOM	ISL8	50	
BOTTOM	ISL9	50					
50-Wide w/ Thin on L1/L10							
HYBRID IMPEDANCE RULE							
TRACE LAYER	REFERENCE LAYER(S)	REQUIRED IMPEDANCE	TRACE WIDTH (OPTIONAL)				
RF SPACING VALUES= 2							
RULE NAME= 50_WIDE_SURFACE_THIN		ZONE NAME= PRIMARY					
ISL3	TOP, ISL5	50					
ISL6	ISL8, ISL4	50					
ISL7	ISL9, ISL5	50					
BOTTOM	ISL9	50					

50-ohm (Thin) Constraints Physical

CLASS DEFINITIONS				COMMA SEPARATED WITH WILDCARD SUPPORT: NET NAMES EX: DDR* DP NAMES EX: DP:DP_AA*.DP_BB* (LINE STARTS WITH FLAG DP:)		CLASS OCCURS Y/N
CLASS NAME	---	CONSTRAINT SET				
Cellular	50_TX0_THIN	P	A_50_THIN_SE	50_TX_IN.*TX0_*		Y
	50_TX1_THIN	P	A_50_THIN_SE	50_TX_IN.*TX1_*		Y
	50_TX2_THIN	P	A_50_THIN_SE	50_TX_IN.*TX2_*		Y
	50_RX_THIN	P	A_50_THIN_SE	50_RX_*		Y
	50_RX_THIN	P	A_50_THIN_SE	50_LAA_RX_*		Y
	50_RX_THIN	P	A_50_THIN_SE	50_RX_LMB_PAD_LMB_PRX_OUT_M		Y
	50_RX_THIN	P	A_50_THIN_SE	50_RX_LMB_PAD_LMB_PRX_OUT		Y
	50_TRX_THIN	P	A_50_THIN_SE	50_UHB_L_CPLR_OUT		Y
	50_TRX_THIN	P	A_50_THIN_SE	50_TRX_WLAN_A_C0*		Y
	50_TRX_THIN	P	A_50_THIN_SE	50_WLAN_A_UAT*		Y
WiFi	50_TRX_THIN	P	A_50_THIN_SE	50_TRX_WLAN_G_C0*		Y
	50_TRX_THIN	P	A_50_THIN_SE	50_WLAN_G_UAT_TXRX_FEM		Y
	50_TRX_THIN	P	A_50_THIN_SE	50_TRX_WLAN_G_C1*		Y
	50_TRX_THIN	P	A_50_THIN_SE	50_WLAN_G_LAT_TXRX_FEM		Y
	50_TRX_THIN	P	A_50_THIN_SE	50_TRX_LB_PORTB_RX1, 50_TRX_LB_PORTB_TX_ANT2		Y
	50_TRX_THIN	P	A_50_THIN_SE	50_TRX_PAUHL_N79_ANT1_TO_ANT7		Y
	50_TRX_THIN	P	A_50_THIN_SE	50_TRX_ANT4_MB_HB_WLAN24		Y
	50_TRX_THIN	P	A_50_THIN_SE	50_UMB_ANT2_6G_FL_OUT		Y
	50_TRX_THIN	P	A_50_THIN_SE	50_UMB_ANT2_6G_T_MATCHING		Y
	50_TRX_THIN	P	A_50_THIN_SE			Y

Spacing

CLASS DEFINITIONS				COMMA SEPARATED WITH WILDCARD SUPPORT: NET NAMES EX: DDR* DP NAMES EX: DP:DP_AA*.DP_BB* (LINE STARTS WITH FLAG DP:)		CLASS OCCURS Y/N
CLASS NAME	---	CONSTRAINT SET				
Everything	50_TX_IN_THIN	S	NA_DIELECTRIC_2XDS_50_THIN_SE	=		Y
	50_TX0_THIN	S	NA_DIELECTRIC_2XDS_50_THIN_SE	=		Y
	50_TX1_THIN	S	NA_DIELECTRIC_2XDS_50_THIN_SE	=		Y
	50_TX2_THIN	S	NA_DIELECTRIC_2XDS_50_THIN_SE	=		Y
	50_RX_THIN	S	NA_DIELECTRIC_2XDS_50_THIN_SE	=		Y
	50_TRX_THIN	S	NA_DIELECTRIC_2XDS_50_THIN_SE	=		Y
	50_TRX_THIN	S	NA_DIELECTRIC_2XDS_50_THIN_SE	=		Y

Class-Class Spacing

CLASS TO CLASS SPACING			
	CLASS NAME	CLASS NAME	CONSTRAINT SET
Same Class	50_TX0_THIN	50_TX0_THIN	NA_DIELECTRIC_2X_50_THIN_SE
	50_TX1_THIN	50_TX1_THIN	NA_DIELECTRIC_2X_50_THIN_SE
	50_TX2_THIN	50_TX2_THIN	NA_DIELECTRIC_2X_50_THIN_SE
	50_RX_THIN	50_RX_THIN	NA_DIELECTRIC_2X_50_THIN_SE
	50_TRX_THIN	50_TRX_THIN	NA_DIELECTRIC_2X_50_THIN_SE
	50_TRX_THIN	50_TRX_THIN	NA_DIELECTRIC_2X_50_THIN_SE
GND	50_TX0_THIN	GND	NA_DIELECTRIC_2X_50_THIN_SE
	50_TX1_THIN	GND	NA_DIELECTRIC_2X_50_THIN_SE
	50_TX2_THIN	GND	NA_DIELECTRIC_2X_50_THIN_SE
	50_RX_THIN	GND	NA_DIELECTRIC_2X_50_THIN_SE
	50_TRX_THIN	GND	NA_DIELECTRIC_2X_50_THIN_SE
	50_TRX_THIN	GND	NA_DIELECTRIC_2X_50_THIN_SE
Thin-Thin	50_TX0_THIN	50_TX1_THIN	NA_DIELECTRIC_4XD_50_THIN_SE
	50_TX0_THIN	50_TX2_THIN	NA_DIELECTRIC_4XD_50_THIN_SE
	50_TX0_THIN	50_RX_THIN	NA_DIELECTRIC_4XD_50_THIN_SE
	50_TX0_THIN	50_TRX_THIN	NA_DIELECTRIC_4XD_50_THIN_SE
	50_TX1_THIN	50_TX2_THIN	NA_DIELECTRIC_4XD_50_THIN_SE
	50_TX1_THIN	50_RX_THIN	NA_DIELECTRIC_4XD_50_THIN_SE
	50_TX1_THIN	50_TRX_THIN	NA_DIELECTRIC_4XD_50_THIN_SE
	50_TX2_THIN	50_RX_THIN	NA_DIELECTRIC_4XD_50_THIN_SE
	50_TX2_THIN	50_TRX_THIN	NA_DIELECTRIC_4XD_50_THIN_SE
	50_RX_THIN	50_TRX_THIN	NA_DIELECTRIC_4XD_50_THIN_SE
	50_TRX_THIN	50_TRX_THIN	NA_DIELECTRIC_4XD_50_THIN_SE
	50_TRX_THIN	50_TRX_THIN	NA_DIELECTRIC_4XD_50_THIN_SE
	50_TRX_THIN	50_TRX_THIN	NA_DIELECTRIC_4XD_50_THIN_SE
	50_TRX_THIN	50_TRX_THIN	NA_DIELECTRIC_4XD_50_THIN_SE
Thin-Wide	50_TX0_THIN	50_TRX_WIDE	NA_DIELECTRIC_4XD_50_THIN_SE
	50_TX1_THIN	50_TRX_WIDE	NA_DIELECTRIC_4XD_50_THIN_SE
	50_TX2_THIN	50_TRX_WIDE	NA_DIELECTRIC_4XD_50_THIN_SE
	50_RX_THIN	50_TRX_WIDE	NA_DIELECTRIC_4XD_50_THIN_SE
	50_TRX_THIN	50_TRX_WIDE	NA_DIELECTRIC_4XD_50_THIN_SE
	50_TRX_THIN	50_TRX_WIDE	NA_DIELECTRIC_4XD_50_THIN_SE
	50_TRX_THIN	50_TRX_WIDE	NA_DIELECTRIC_4XD_50_THIN_SE
	50_TRX_THIN	50_TRX_WIDE	NA_DIELECTRIC_4XD_50_THIN_SE
	50_TRX_THIN	50_TRX_WIDE	NA_DIELECTRIC_4XD_50_THIN_SE
	50_TRX_THIN	50_TRX_WIDE	NA_DIELECTRIC_4XD_50_THIN_SE

50-ohm (Wide) Constraints Physical

CLASS DEFINITIONS				COMMA SEPARATED WITH WILDCARD SUPPORT: NET NAMES EX: DDR* DP NAMES EX: DP:DP_AA*.DP_BB* (LINE STARTS WITH FLAG DP:)		CLASS OCCURS Y/N
CLASS NAME	---	CONSTRAINT SET				
Cellular	50_TRX_WIDE	P	A_50_WIDE_SE	50_TRX_ANT2_LB_LMB_MB_HB_L1		Y
	50_TRX_WIDE	P	A_50_WIDE_SE	50_TRX_ANT4_MB_HB_WLAN24_M		Y
	50_TRX_WIDE	P	A_50_WIDE_SE	50_TRX_ANT7_UHB_N79		Y
	50_TRX_WIDE	P	A_50_WIDE_SE	50_TRX_ANT7_UHB_N79_M		Y
	50_TRX_WIDE	P	A_50_WIDE_SE	50_TRX_ANT8*		Y
	50_TRX_WIDE	P	A_50_WIDE_SE	50_TRX_CPLR_U_*		Y
	50_TRX_WIDE	P	A_50_WIDE_SE	50_TRX_LB_PAD_*		Y
	50_TRX_WIDE_SURFACE_THIN	P	A_50_WIDE_SURFACE_THIN_SE	50_TRX_LB_PORTB_TX_M		Y
	50_TRX_WIDE_SURFACE_THIN	P	A_50_WIDE_SURFACE_THIN_SE	50_TRX_LB_PORTB_RX1_M		Y
	50_TRX_WIDE_SURFACE_THIN	P	A_50_WIDE_SURFACE_THIN_SE	50_TRX_PAUHL_UHB_ANT1_*		Y
WiFi	50_TRX_WIDE	P	A_50_WIDE_SE	50_TRX_PAUHL_UHB_ANT2_*		Y
	50_TRX_WIDE	P	A_50_WIDE_SE	50_TRX_PAUHL_UHB_ANT2_*		Y
	50_TRX_WIDE	P	A_50_WIDE_SE	50_TRX_PAUHL_UHB_ANT2_*		Y
	50_TRX_WIDE	P	A_50_WIDE_SE	50_TRX_PAUHL_UHB_ANT2_*		Y
	50_TRX_WIDE	P	A_50_WIDE_SE	50_TRX_PAUHL_UHB_ANT2_*		Y
	50_TRX_WIDE	P	A_50_WIDE_SE	50_TRX_PAUHL_UHB_ANT2_*		Y
	50_TRX_WIDE	P	A_50_WIDE_SE	50_TRX_PAUHL_UHB_ANT2_*		Y
	50_TRX_WIDE	P	A_50_WIDE_SE	50_TRX_PAUHL_UHB_ANT2_*		Y
	50_TRX_WIDE	P	A_50_WIDE_SE	50_TRX_PAUHL_UHB_ANT2_*		Y
	50_TRX_WIDE	P	A_50_WIDE_SE	50_TRX_PAUHL_UHB_ANT2_*		Y
Arrow	50_TRX_WIDE_SURFACE_THIN	P	A_50_WIDE_SURFACE_THIN_SE	50_TRX_PAUHL_UHB_ANT2_*		Y
	50_TRX_WIDE_SURFACE_THIN	P	A_50_WIDE_SURFACE_THIN_SE	50_TRX_PAUHL_UHB_ANT2_*		Y
	50_TRX_WIDE_SURFACE_THIN	P	A_50_WIDE_SURFACE_THIN_SE	50_TRX_PAUHL_UHB_ANT2_*		Y
	50_TRX_WIDE_SURFACE_THIN	P	A_50_WIDE_SURFACE_THIN_SE	50_TRX_PAUHL_UHB_ANT2_*		Y
	50_TRX_WIDE_SURFACE_THIN	P	A_50_WIDE_SURFACE_THIN_SE	50_TRX_PAUHL_UHB_ANT2_*		Y
	50_TRX_WIDE_SURFACE_THIN	P	A_50_WIDE_SURFACE_THIN_SE	50_TRX_PAUHL_UHB_ANT2_*		Y
	50_TRX_WIDE_SURFACE_THIN	P	A_50_WIDE_SURFACE_THIN_SE	50_TRX_PAUHL_UHB_ANT2_*		Y
	50_TRX_WIDE_SURFACE_THIN	P	A_50_WIDE_SURFACE_THIN_SE	50_TRX_PAUHL_UHB_ANT2_*		Y
	50_TRX_WIDE_SURFACE_THIN	P	A_50_WIDE_SURFACE_THIN_SE	50_TRX_PAUHL_UHB_ANT2_*		Y
	50_TRX_WIDE_SURFACE_THIN	P	A_50_WIDE_SURFACE_THIN_SE	50_TRX_PAUHL_UHB_ANT2_*		Y

Spacing

CLASS DEFINITIONS				COMMA SEPARATED WITH WILDCARD SUPPORT: NET NAMES EX: DDR* DP NAMES EX: DP:DP_AA*.DP_BB* (LINE STARTS WITH FLAG DP:)		CLASS OCCURS Y/N
CLASS NAME	---	CONSTRAINT SET				
50_TRX_WIDE	S	NA_DIELECTRIC_2XDS_50_WIDE_SE	=			Y
50_TRX_WIDE_SURFACE_THIN	S	NA_DIELECTRIC_2XDS_50_WIDE_SURFACE_THIN_SE	=			Y

Class-Class Spacing

CLASS TO CLASS SPACING			
	CLASS NAME	CLASS NAME	CONSTRAINT SET
GND	50_TRX_WIDE	GND	NA_DIELECTRIC_2X_50_WIDE_SE
	50_TRX_WIDE_SURFACE_THIN	GND	NA_DIELECTRIC_2X_50_WIDE_SURFACE_THIN_SE
Wide-Wide	50_TRX_WIDE	50_TRX_WIDE	NA_DIELECTRIC_4XD_50_WIDE_SE
	50_TRX_WIDE_SURFACE_THIN	50_TRX_WIDE	NA_DIELECTRIC_4XD_50_WIDE_SURFACE_THIN_SE
	50_TRX_WIDE_SURFACE_THIN	50_TRX_WIDE_SURFACE_THIN	NA_DIELECTRIC_4XD_50_WIDE_SURFACE_THIN_SE

Pin Delay Check

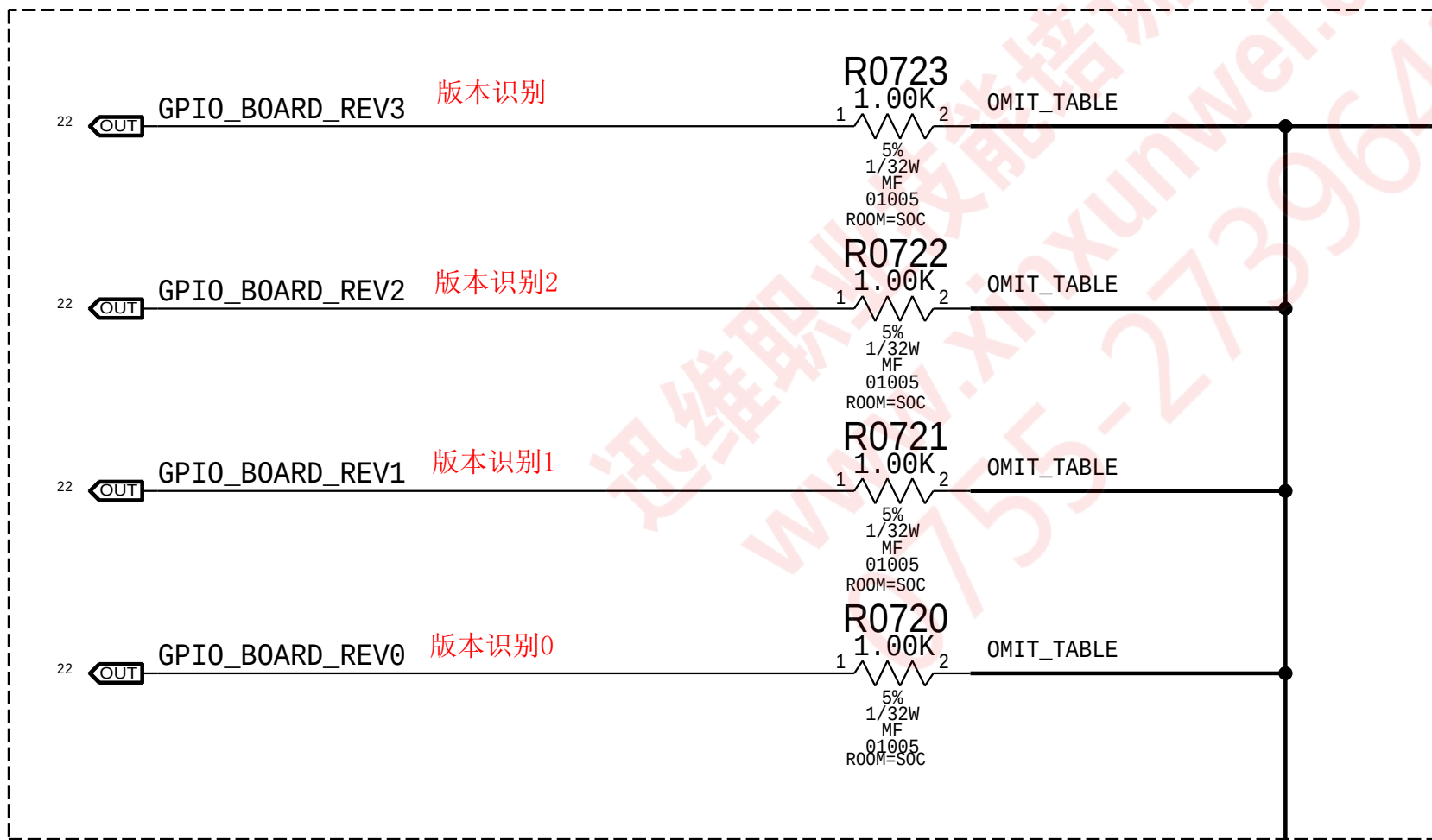
PIN DELAY MAPPING FILE	
REFERENCE DESIGNATOR	PIN DELAY CSV FILE NAME
U1000	SicilyPinDelay.csv
J10800	D53-JulietPinDelay.csv

*Location: /physical/role/pin delays

PAGE TITLE		
CONSTRAINTS: Stackup/Misc. (FF-Specific)		
DRAWING NUMBER		051-05215
REVISION		4.6.0
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BOOTSTRAPPING

BOARD REV + BOARD ID + BOOT CONFIG



Board Rev [3:0]

* Float = 0 | PU = 1

Note: iBoot uses the inverse of BOARD_REV[3:0], so that it counts up (Pre-Proto = 0x0, PVT = 0xF)

	BOARD_REV[3:0]	[3]	[2]	[1]	[0]
Pre-Proto	4'b1111	1	1	1	1
Proto 1	4'b1110	1	1	1	0
Proto 1.5	4'b1101	1	1	0	1
Proto 2	4'b1100	1	1	0	0
Proto 2.5	4'b1011	1	0	1	1
Pre-EVT	4'b1010	1	0	1	0
EVT	4'b1001	1	0	0	1
Carrier	4'b1000	1	0	0	0

(Allocate more as necessary in descending order)

DVT	4'b0001	0	0	0	1
PVT	4'b0000	0	0	0	0

PART#	QTY	DESCRIPTION	REFERENCE DESIGNATOR(S)	BOM OPTION
117S0156	1	RES, MF, 1K OHM, 5%, 1/32W, 01005	R0723, R0722, R0721, R0720	BOARD_REV:PROT00
117S0156	1	RES, MF, 1K OHM, 5%, 1/32W, 01005	R0723, R0722, R0721	BOARD_REV:PROT01
117S0156	1	RES, MF, 1K OHM, 5%, 1/32W, 01005	R0723, R0722, R0720	BOARD_REV:PROT01.5
117S0156	1	RES, MF, 1K OHM, 5%, 1/32W, 01005	R0723, R0722	BOARD_REV:PROT02
117S0156	1	RES, MF, 1K OHM, 5%, 1/32W, 01005	R0723, R0721, R0720	BOARD_REV:PROT02.5
117S0156	1	RES, MF, 1K OHM, 5%, 1/32W, 01005	R0723, R0721	BOARD_REV:PRE-EVT
117S0156	1	RES, MF, 1K OHM, 5%, 1/32W, 01005	R0723, R0720	BOARD_REV:EVT
117S0156	1	RES, MF, 1K OHM, 5%, 1/32W, 01005	R0723	BOARD_REV:CRB
117S0156	1	RES, MF, 1K OHM, 5%, 1/32W, 01005	R0720	BOARD_REV:DVT

(PVT NOSTUFF ALL)

Board ID [4:0]

* Float = 0 | PU = 1

	BOARD_ID[4:0]	[4] Unused	[3] 1=MAV20	[2]	[1]	[0] 0=MLB, 1=DEV
MLB D52G	5'b01010	0	1	0	1	0
DEV D52G	5'b01011	0	1	0	1	1
MLB D53G	5'b01100	0	1	1	0	0
DEV D53G	5'b01101	0	1	1	0	1
MLB D53P	5'b01110	0	1	1	1	0
DEV D53P	5'b01111	0	1	1	1	1
MLB D54P	5'b01000	0	1	0	0	0
DEV D54P	5'b01001	0	1	0	0	1

PART#	QTY	DESCRIPTION	REFERENCE DESIGNATOR(S)	BOM OPTION
117S0156	1	RES, MF, 1K OHM, 5%, 1/32W, 01005	R0712	BOARD_ID:D52
117S0156	1	RES, MF, 1K OHM, 5%, 1/32W, 01005	R0713	BOARD_ID:D53G
117S0156	1	RES, MF, 1K OHM, 5%, 1/32W, 01005	R0713, R0712	BOARD_ID:D53P
(D54 both NOSTUFF)				

D52 = 01
D53G = 10
D53P = 11
D54 = 00

Boot Config [2:0]

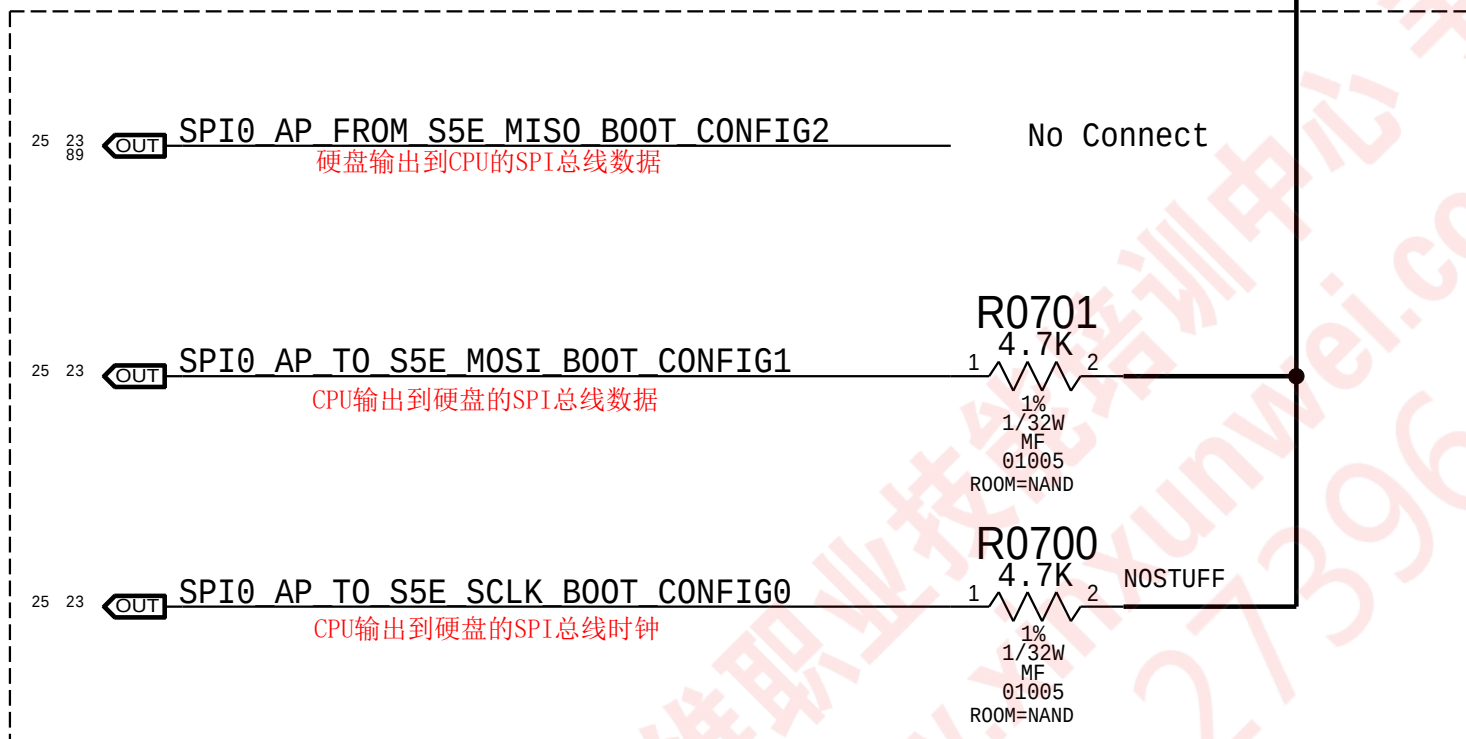
* Float = 0 | PU = 1

	USAGE	SPEED	TEST	[2]	[1]	[0]
SPI1 NOR	--	12MHz	--	0	0	0
SPI1 NOR	--	12MHz	Test	0	0	1
SPI0 NAND	POR	12MHz	--	0	1	0
SPI0 NAND	Proto	12MHz	Test	0	1	1
SPI1 NOR	--	24MHz	--	1	0	0
SPI1 NOR	--	24MHz	Test	1	0	1
SPI1 NOR	--	6MHz	--	1	1	0
SPI1 NOR	--	6MHz	Test	1	1	1

<-- POR

<-- Proto Builds

<--- Remove at EVT



PAGE TITLE		
SYSTEM: Bootstrapping		
	DRAWING NUMBER	051-05215
	REVISION	4.6.0
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SOC

4GB DRAM

PART#	QTY	DESCRIPTION	REFERENCE DESIGNATOR(S)	CRITICAL	BOM OPTION
998-19672	1	Sicily,A0,4GB,M	U1000	CRITICAL	BOARD_ID:052AB0ARD_ID:0505

PART NUMBER	ALTERNATE FOR PART NUMBER	BOM OPTION	REF DES	COMMENTS:
998-19673	998-19672	BOARD_ID:052AB0ARD_ID:0506	U1000	Sicily,A0,4GB,H
998-19674	998-19672	BOARD_ID:052AB0ARD_ID:0508	U1000	Sicily,A0,4GB,S

6GB DRAM

PART#	QTY	DESCRIPTION	REFERENCE DESIGNATOR(S)	CRITICAL	BOM OPTION
998-19675	1	Sicily,A0,6GB,M	U1000	CRITICAL	BOARD_ID:052PB0ARD_ID:0504

PART NUMBER	ALTERNATE FOR PART NUMBER	BOM OPTION	REF DES	COMMENTS:
998-19676	998-19675	BOARD_ID:052PB0ARD_ID:0504	U1000	Sicily,A0,6GB,H
998-19677	998-19675	BOARD_ID:052PB0ARD_ID:0504	U1000	Sicily,A0,6GB,S

24M XTAL Alternates

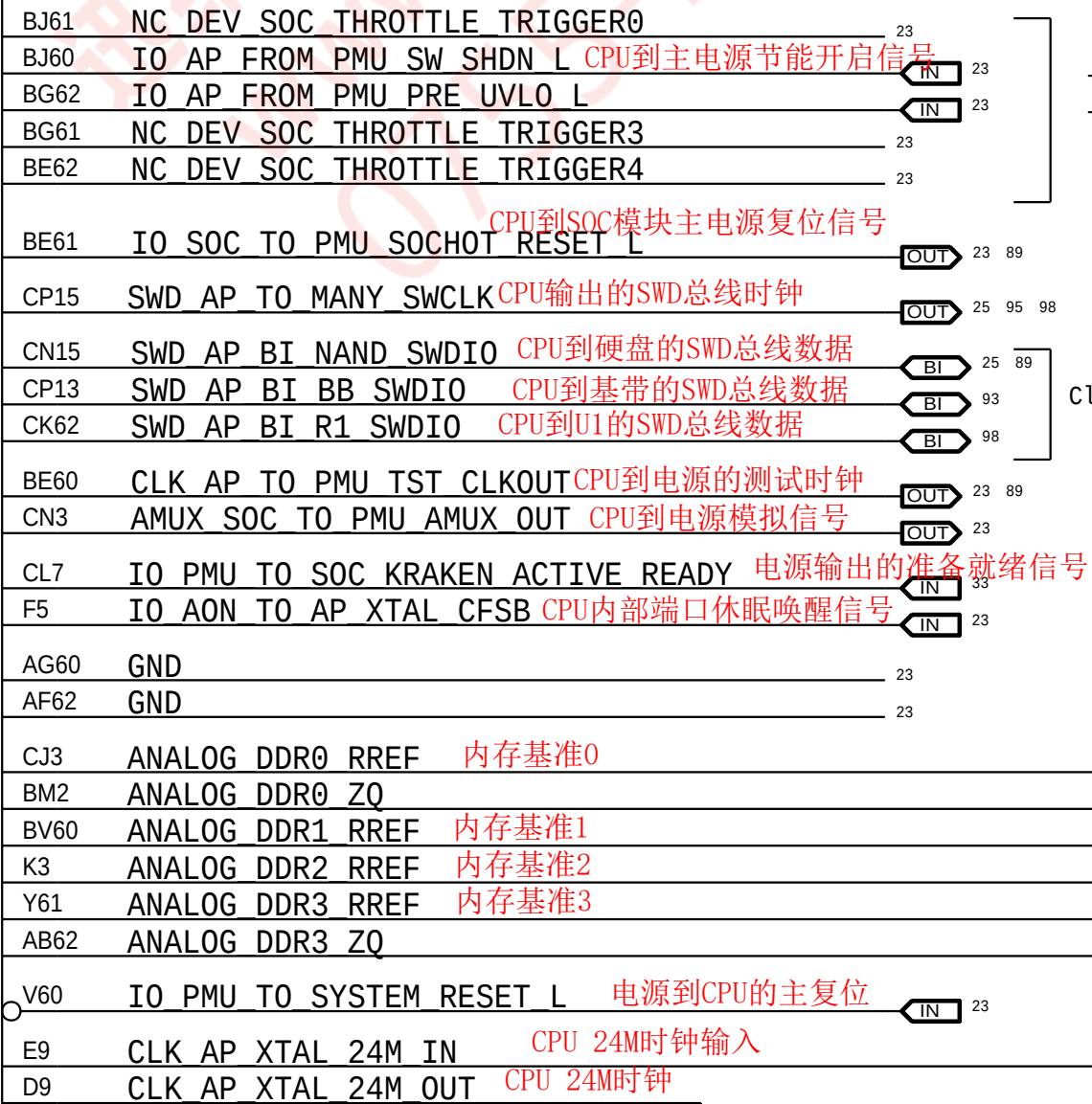
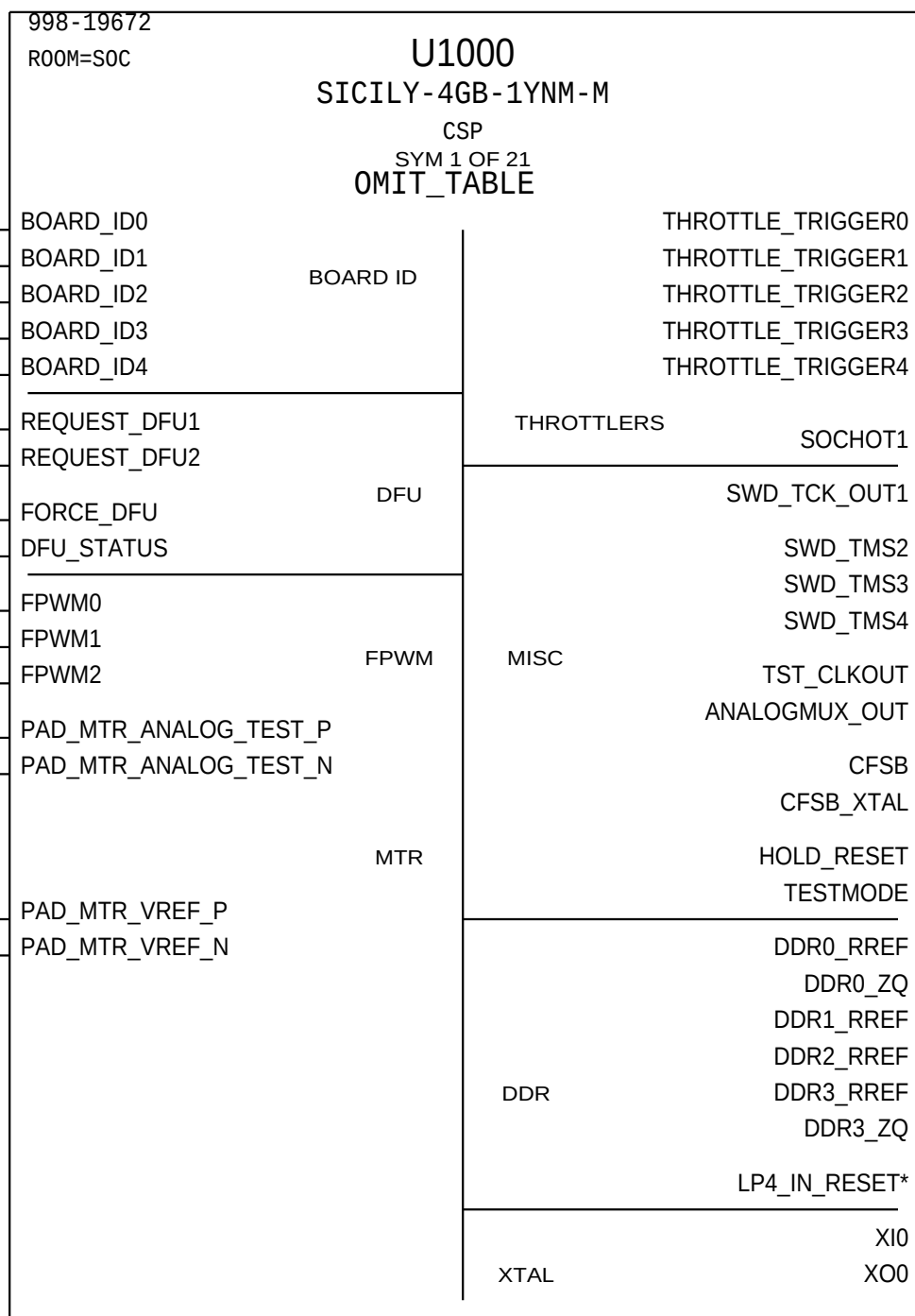
PART NUMBER	ALTERNATE FOR PART NUMBER	BOM OPTION	REF DES	COMMENTS:
197S0012	197S00118	?	Y1000	XTAL, 24M, 1612
197S00120	197S00118	?	Y1000	XTAL, 24M, 1612

SOC: Misc

REQUEST_DFUx: Legacy button detection
MLB: Hard tie to PP1V2_IO
DEV: Wire to PMU BUTTON0

PP (dev board only)

PP (dev board only)



THROTTLE_TRIGGER[0:4]:
These need an internal pull-up enabled on Soc

Clocked by SWD_TCK_OUT1

PP0V6_VDDQL_S1

R1021

499

1/32W 01005 2 ROOM=SOC_XTAL

CLK_AP_XTAL_24M_OUT_R

1

2

197S00118 CRITICAL ROOM=SOC_XTAL

Y1000 1.60X1.20MM-SM 24MHZ-30PPM-9.5PF-600HM

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NC

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GND

C1020 12PF 25V NP0-C0G 01005 1 ROOM=SOC_XTAL

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SOC_XTAL_GND

C1021 12PF 25V NP0-C0G 01005 1 ROOM=SOC_XTAL

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XW1000 SHORT-20L-0.05MM-SM NO_XNET_CONNECTION ROOM=SOC_XTAL

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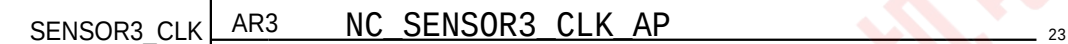
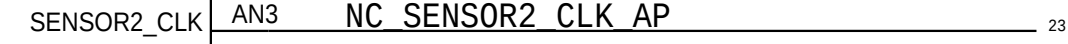
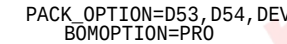
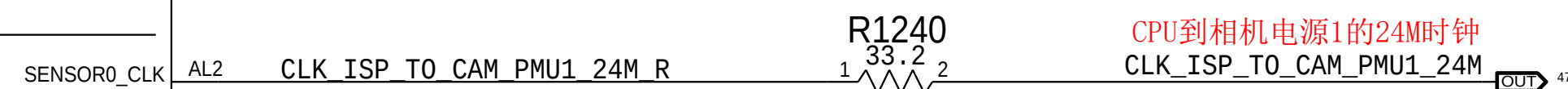
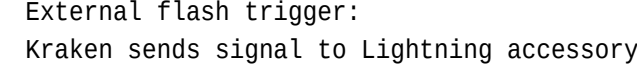
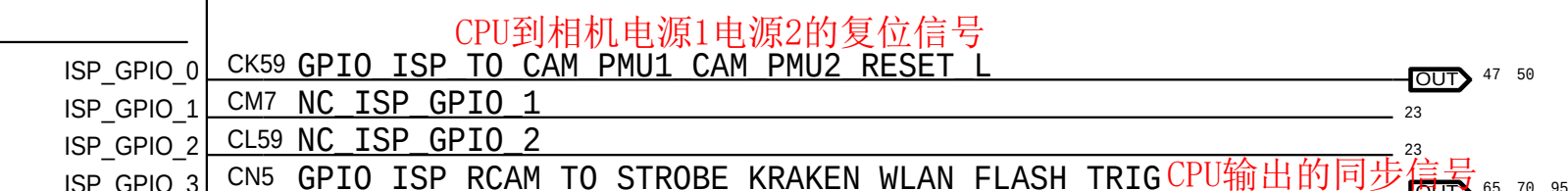
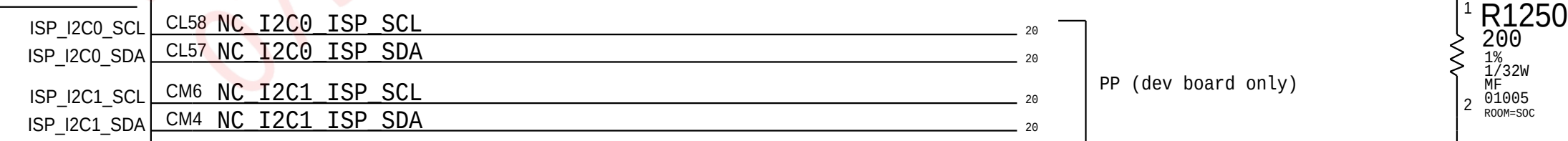
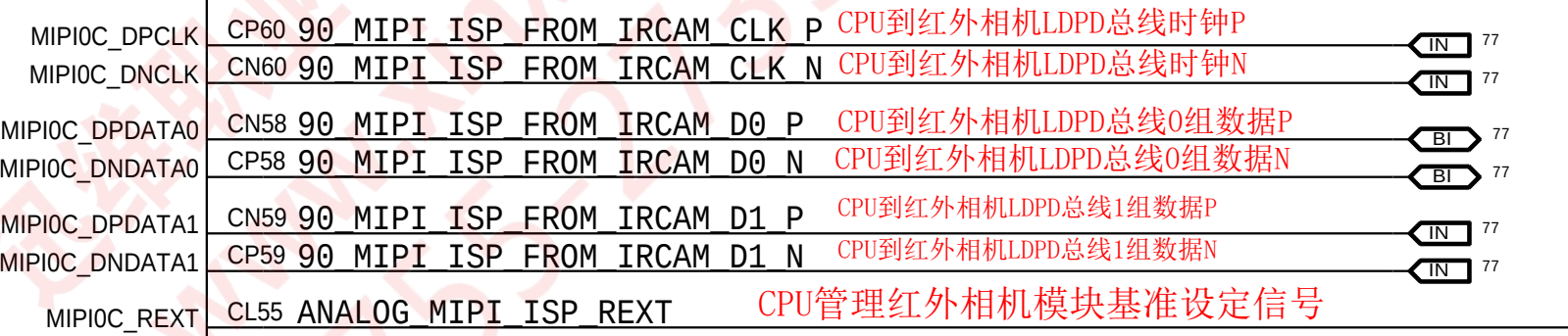
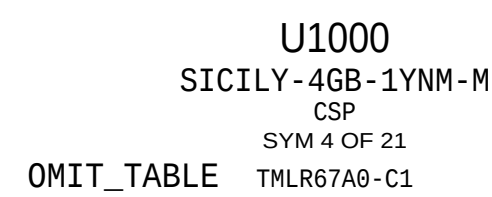
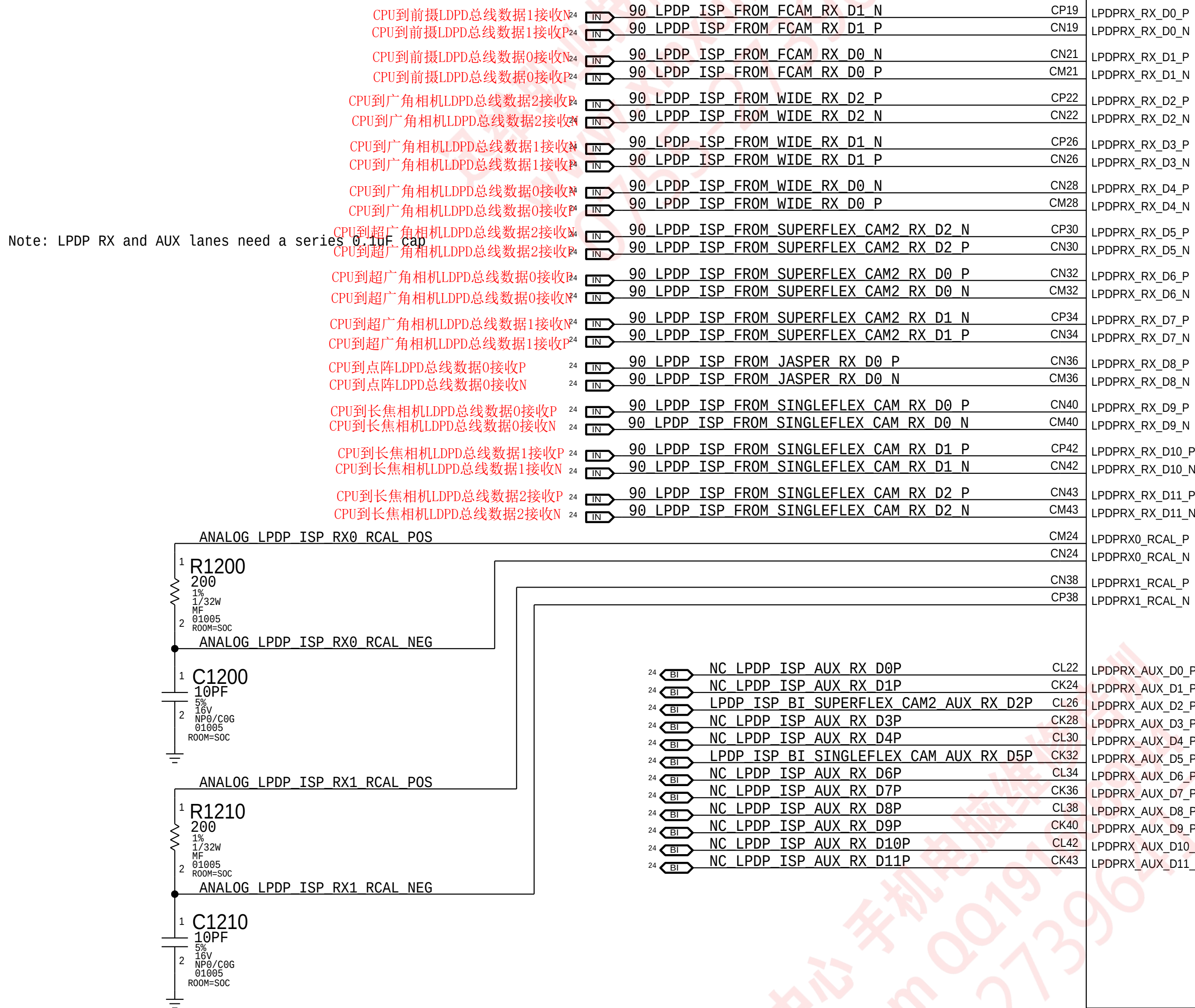
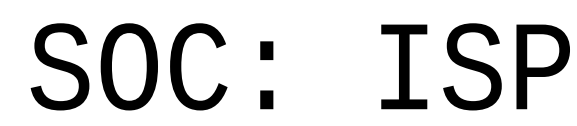
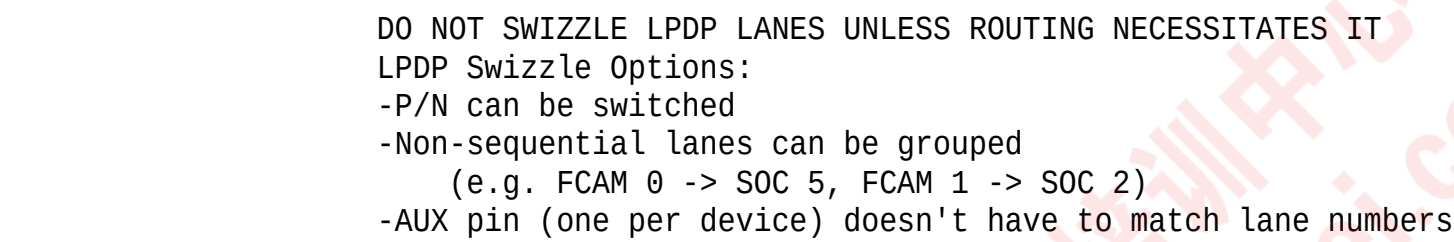
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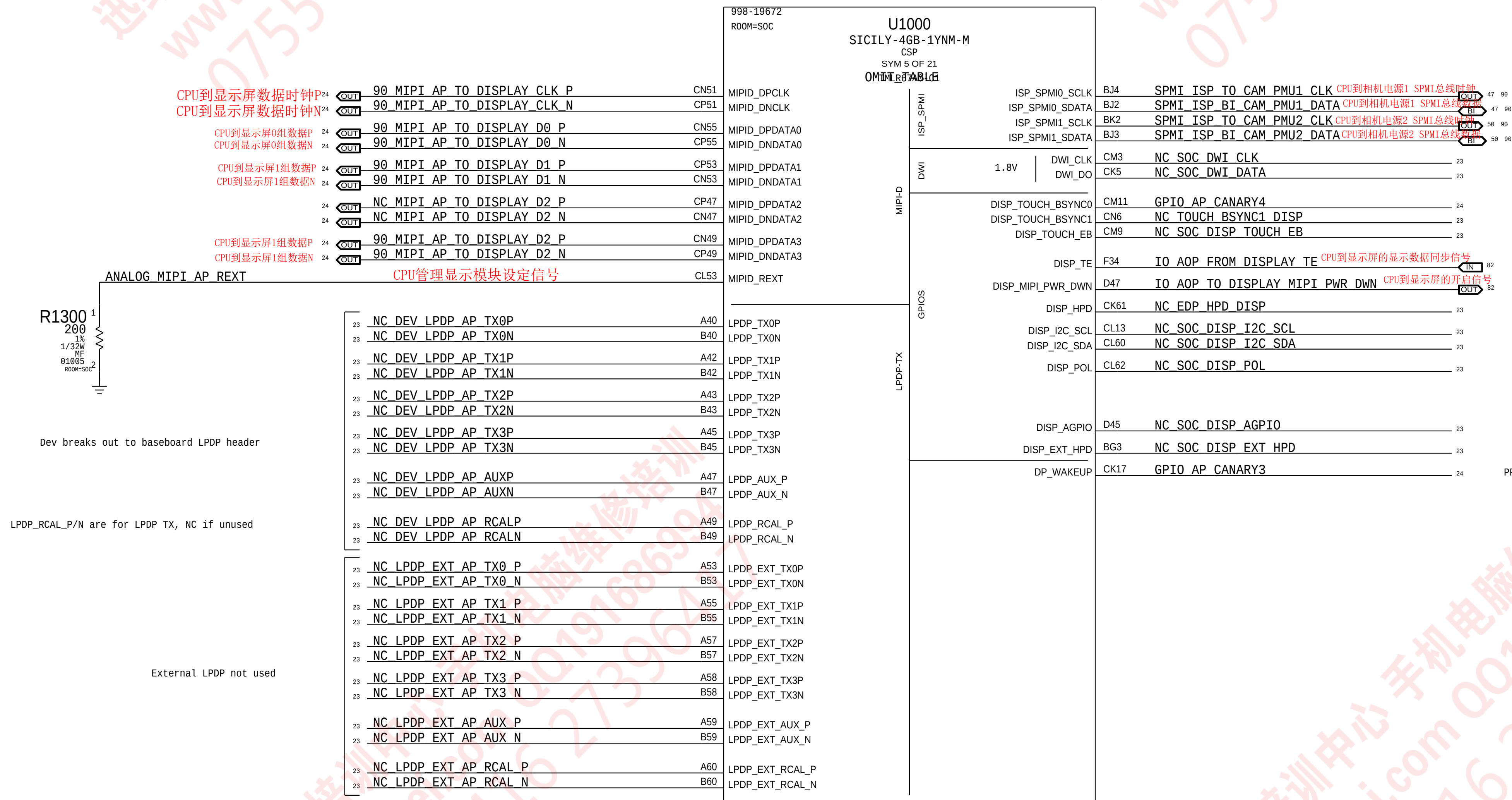
2



SYNCING: D52, D53, D54, DEV

PAGE TITLE		SQC: ISP	
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SOC: DISPLAY / ISP



SPMI Address Map

BUS	DEVICE	ADDR	LOCATION
SPMI0 ISP	Adams1	0x09	Top MLB
SPMI1 ISP	Adams2	0x09	Top MLB

NOTE: SPMI1 ISP is for D53P/D54 only

DISPLAY_TE also provides WDG functionality

PP (dev board only)

SYNCING: D52, D53, D54, DEV

PAGE TITLE		S0C: Display	
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SOC: AP Serial

D

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A

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C

B

A

CPU到USB芯片UART总线调试数据接收
CPU到USB芯片UART总线调试数据发射

IN	UART0 AP FROM KRAKEN DEBUG RXD	CH62
OUT	UART0 AP TO KRAKEN DEBUG TXD	CP5
IN	NC UART1 AP CTS L	BP2
IN	NC UART1 AP RTS L	BM3
IN	NC UART1 AP RXD	BK4
IN	NC UART1 AP TXD	BK3
IN	NC UART2 AP CTS L	BT4
IN	NC UART2 AP RTS L	BT2
IN	NC UART2 AP RXD	BP4
IN	NC UART2 AP TXD	BP3
IN	NC UART3 AP CTS L	CJ61
IN	GPIO AP CANARY6	CM13
IN	NC UART3 AP RXD	CP6
IN	GPIO AP CANARY5	CL15

CPU到USB芯片UART总线数据接收
CPU到USB芯片UART总线数据发射

IN	UART4 AP FROM KRAKEN ACC RXD	AL61
OUT	UART4 AP TO KRAKEN ACC TXD	AL62
IN	NC UART6 AP RXD 1V8	CK6
OUT	GPIO AP TO TOUCH RESET L 1V8 CPU到触摸的复位信号	CL2

Use UART6_TXD as 1.8V GPIO

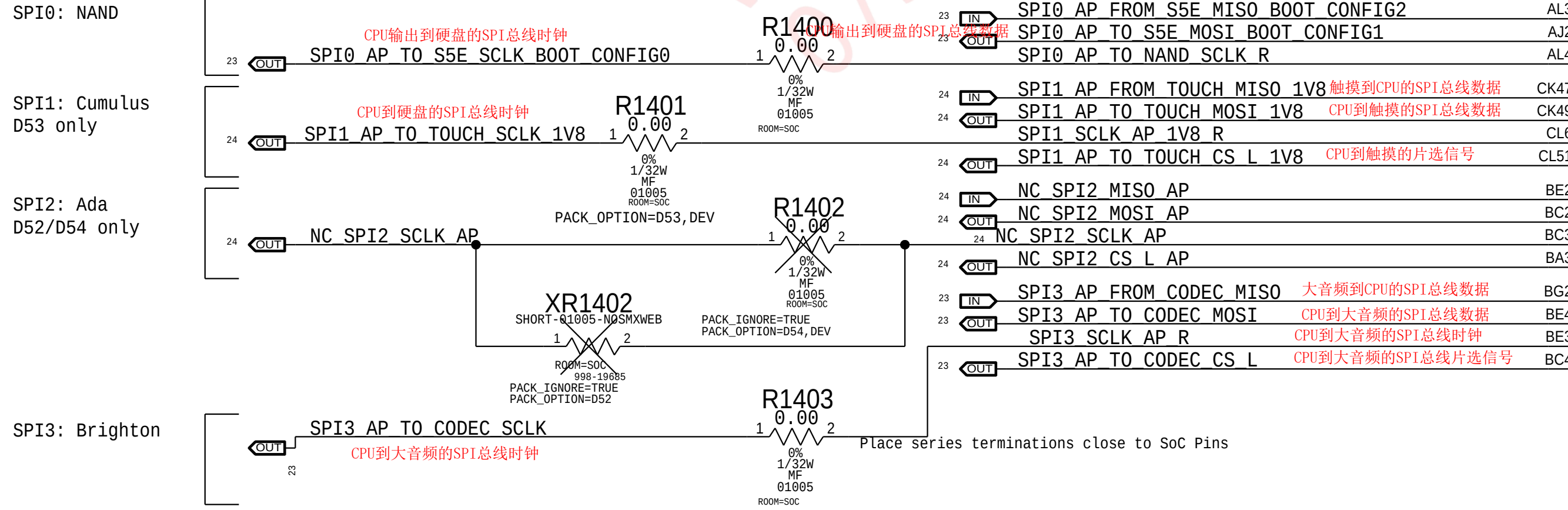
SPI0: NAND

SPI1: Cumulus
D53 only

SPI2: Ada
D52/D54 only

SPI3: Brighton

PP (dev board only)



IN	GPIO PMU TO SOC DOUBLE CLICK DET L	AW3
IN	NC SOC S GPIO1	AW4
OUT	I2C0 S SCL	AU1
BI	I2C0 S SDA	AU2

998-19672 ROOM=SOC	U1000 SICILY-4GB-1YNM-M CSP SYM 7 OF 21 OMIT_TABLE	I2C0_SCL I2C0_SDA I2C1_SCL I2C1_SDA I2C2_SCL I2C2_SDA I2C3_SCL I2C3_SDA I2C4_SCL I2C4_SDA	AG62 AJ61 AG4 AF2 AF3 AD2 CL5 CL49 CL4 CL47	NC I2C0 AP SCL NC I2C0 AP SDA NC I2C1 AP SCL NC I2C1 AP SDA I2C2 AP SCL I2C2 AP SDA I2C3 AP SCL 1V8 I2C3 AP SDA 1V8 I2C DISPLAY SCL 1V8 I2C DISPLAY SDA 1V8	24 24 24 24 20 20 20 20 20 20
-----------------------	--	--	--	--	--

I2S0_DIN I2S0_DOUT I2S0_BCLK I2S0_LRCK I2S0_MCK I2S1_DIN I2S1_DOUT I2S1_BCLK I2S1_LRCK I2S1_MCK I2S2_DIN I2S2_DOUT I2S2_BCLK I2S2_LRCK I2S2_MCK	P2 T4 P4 M2 P3 AB4 AD3 Y3 Y2 AB3 V2 Y4 T2 T3 V3	I2S0 AP FROM CODEC ASP3 DIN 音频到CPU的I2S0总线数据 I2S0 AP TO CODEC ASP3 DOUT CPU到音频的I2S0总线数据 I2S0 AP FROM CODEC ASP3 BCLK 音频到CPU的I2S0总线时钟 I2S0 AP FROM CODEC ASP3 LRCLK 音频到CPU的I2S0总线片选信号 I2S0 AP TO SPKRAMP TOP MCLK R CPU到音频的I2S0总线主时钟 I2S1 AP FROM CODEC ASP4 DIN 音频到CPU的I2S1总线数据 I2S1 AP TO CODEC ASP4 DOUT CPU到音频的I2S1总线数据 I2S1 AP FROM CODEC ASP4 BCLK 音频到CPU的I2S1总线串行时钟 I2S1 AP FROM CODEC ASP4 LRCLK 音频到CPU的I2S1总线片选信号 NC I2S1 AP MCLK I2S2 AP FROM BB DIN 基带到CPU的I2S2总线数据 I2S2 AP TO BB DOUT CPU到基带的I2S2总线数据 I2S2 AP FROM BB BCLK 基带到CPU的I2S1总线串行时钟 I2S2 AP FROM BB LRCLK 基带到CPU的I2S1总线片选信号 NC I2S2 AP MCLK	58 89 58 89 58 89 58 89 58 89 23 95 95 95 23
---	---	--	--

SPI0_MISO SPI0_MOSI SPI0_SCLK SPI1_MISO SPI1_MOSI SPI1_SCLK SPI1_SSIN SPI2_MISO SPI2_MOSI SPI2_SCLK SPI2_SSIN SPI3_MISO SPI3_MOSI SPI3_SCLK SPI3_SSIN	AL3 AJ2 AL4 CK47 CK49 CL6 CL51 BE2 BC2 BC3 BA3 BG2 BE4 BE3 BC4	1.8V
---	--	------

998-19672 ROOM=SOC	U1000 SICILY-4GB-1YNM-M CSP SYM 8 OF 21 OMIT_TABLE	SSPI0_MISO SSPI0_MOSI SSPI0_SCLK	AW2 AU4 AU3	NC SSPI0 MISO AP NC SSPI0 MOSI AP NC SSPI0 SCLK AP	23 23 23
-----------------------	--	--	-------------------	--	----------------

NOTE:
I2S DIN/DOUT are SOC-centric,
SOC DIN routes to load DOUT and vice-versa

R1410
CPU到音频的I2S0总线主时钟
I2S0 AP TO SPKRAMP TOP MCLK

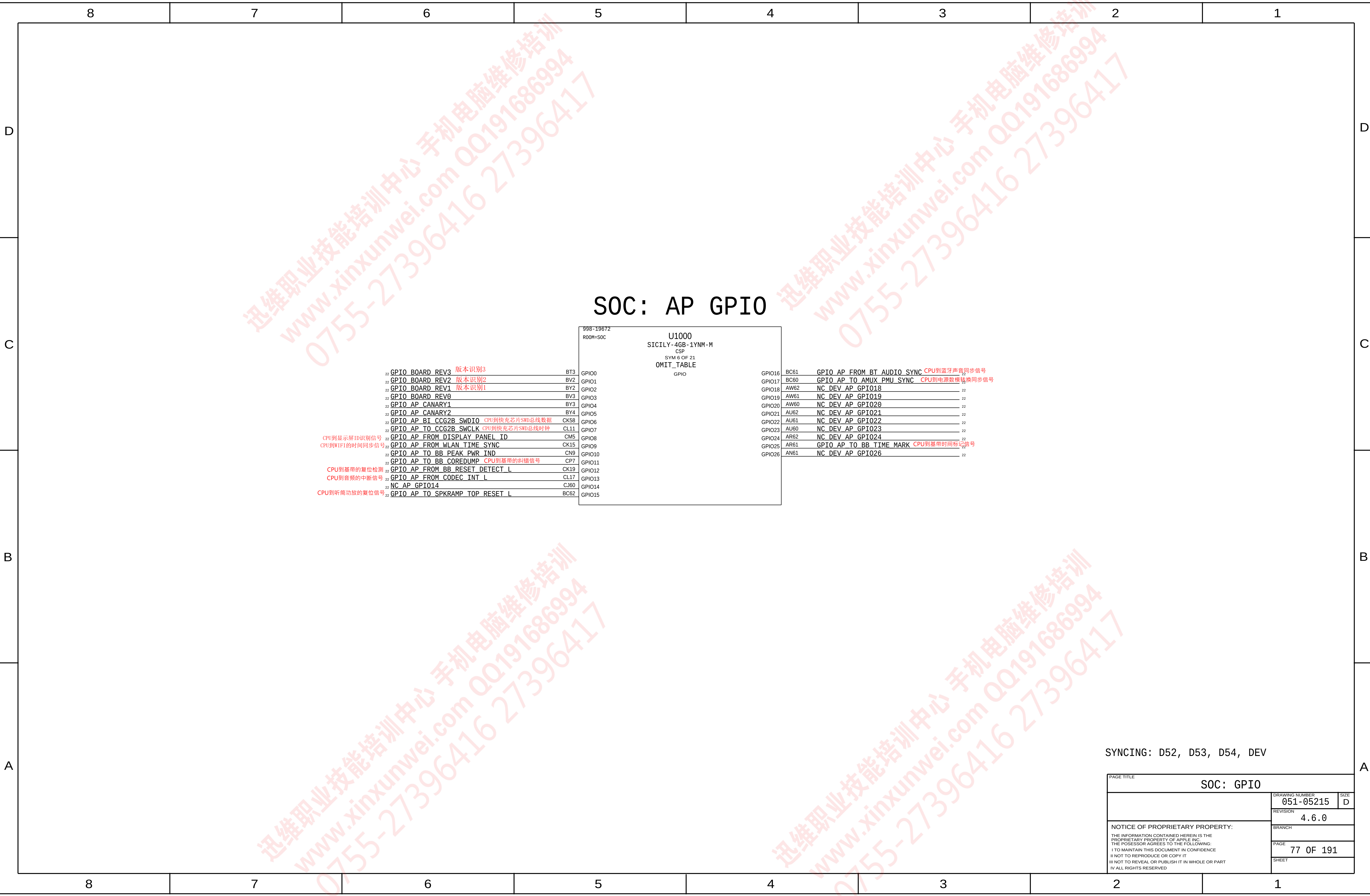
AP I2S0
Used for MikeyBus
(MCLK to Top Spk)

AP I2S1
Used for Mics

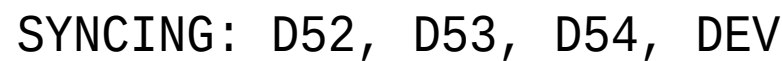
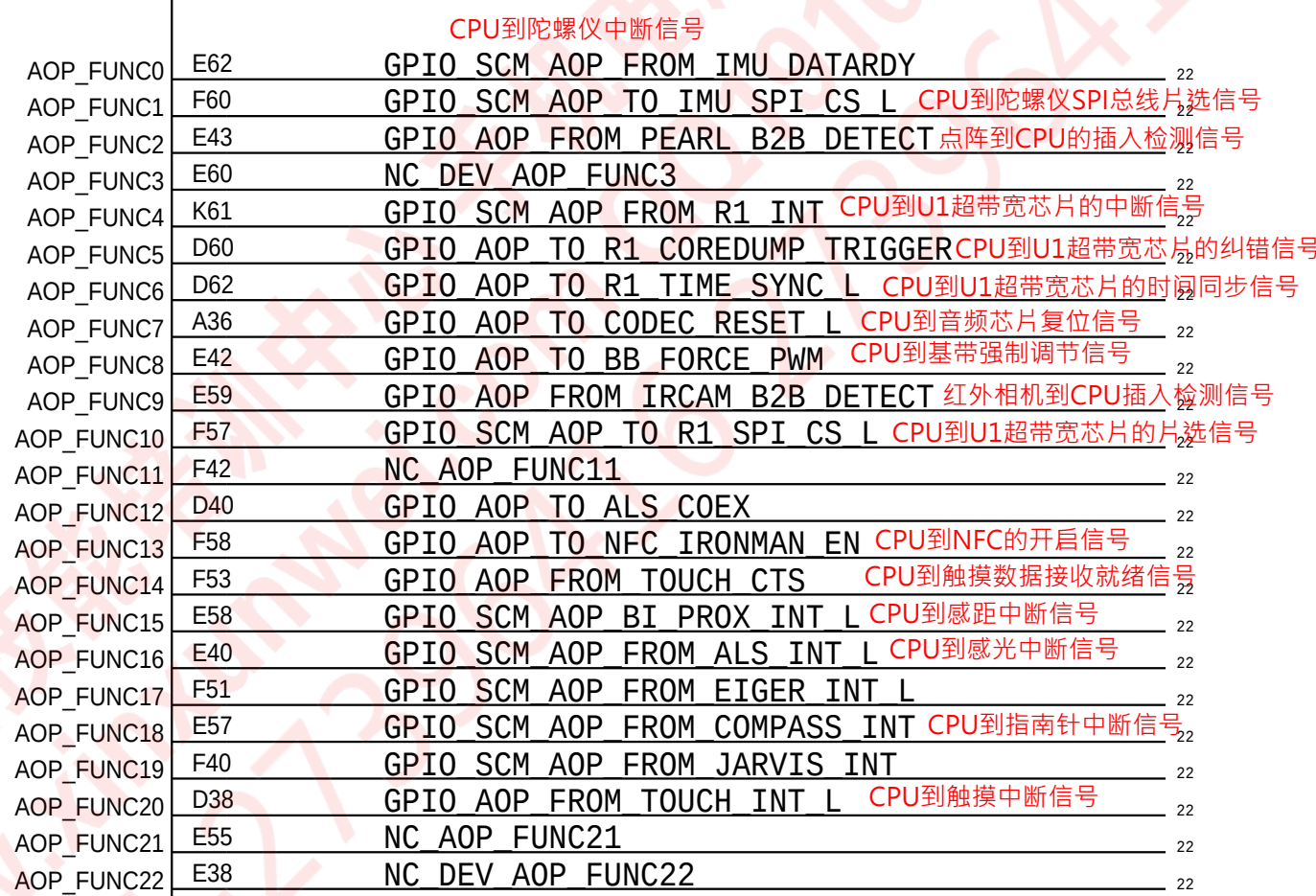
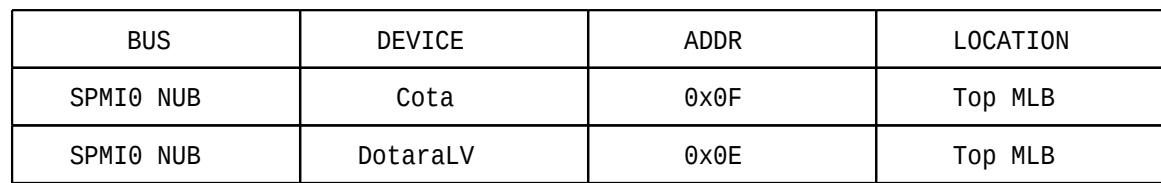
I2S0总线是指音频总线，主要用来传输声音信号

SYNCING: D52, D53, D54, DEV

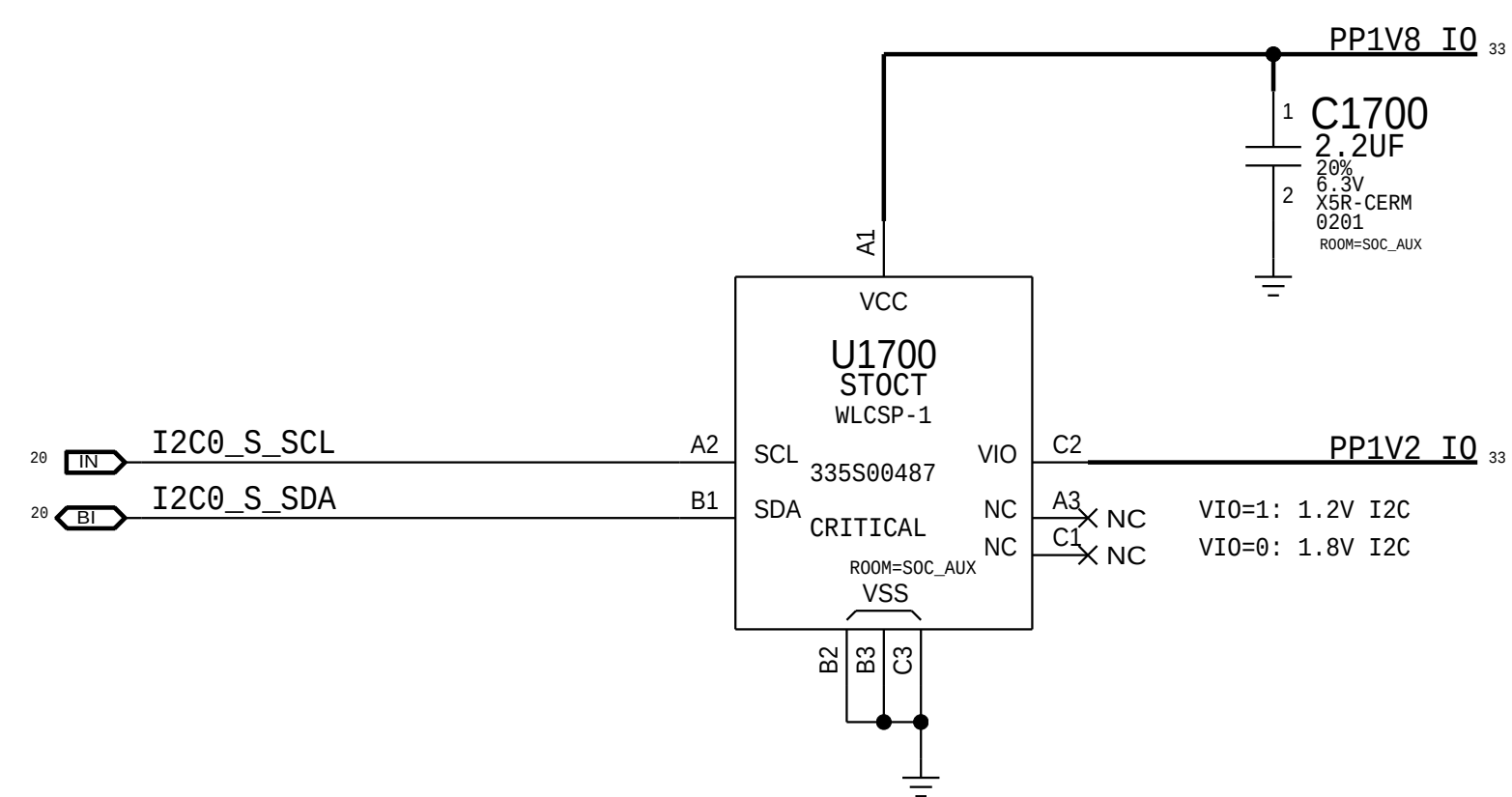
PAGE TITLE SOC: Serial		
	DRAWING NUMBER 051-05215	SIZE D
	REVISION 4.6.0	BRANCH
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SOC: SMC

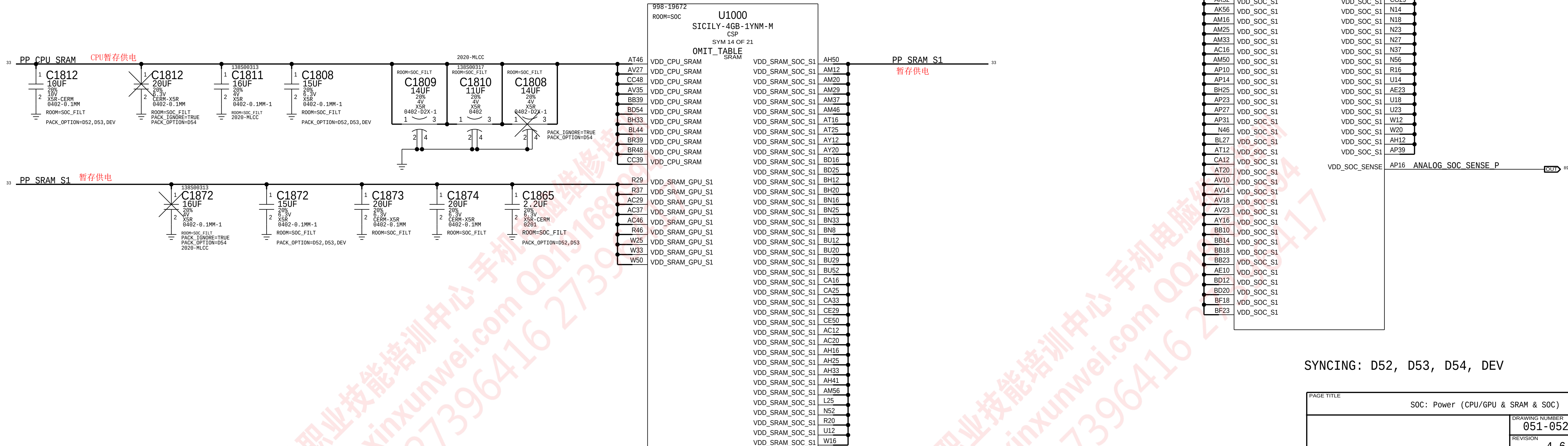
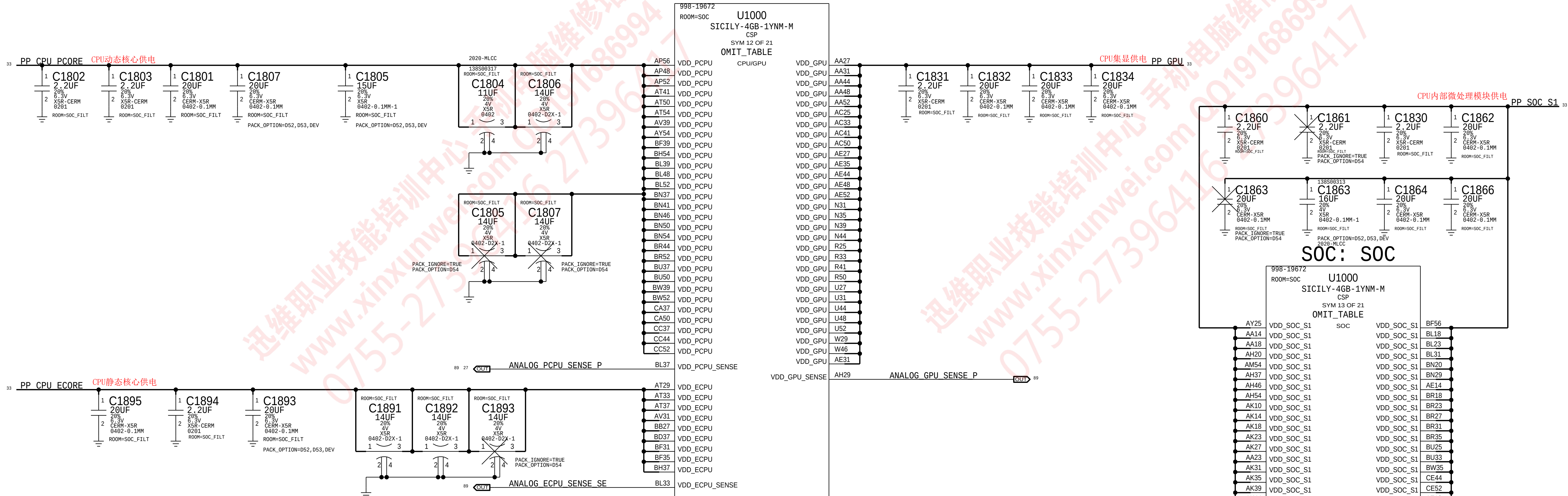


PAGE TITLE		
SOC: AOP & SMC & NUB		
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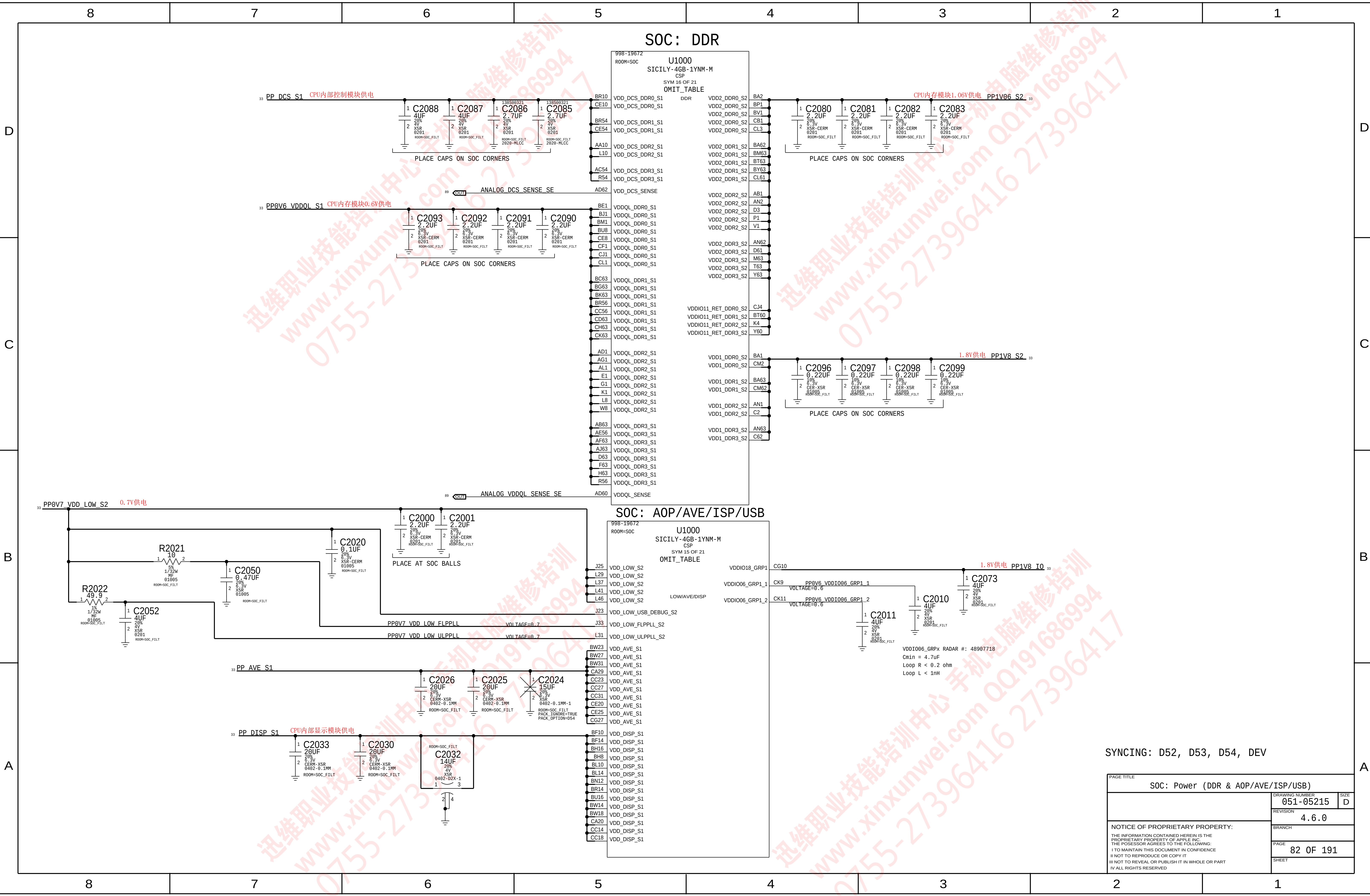
SYNCING: D52, D53, D54, DEV

PAGE TITLE		
SOC: Ocelot		
DRAWING NUMBER 051-05215		SIZE D
REVISION 4.6.0		
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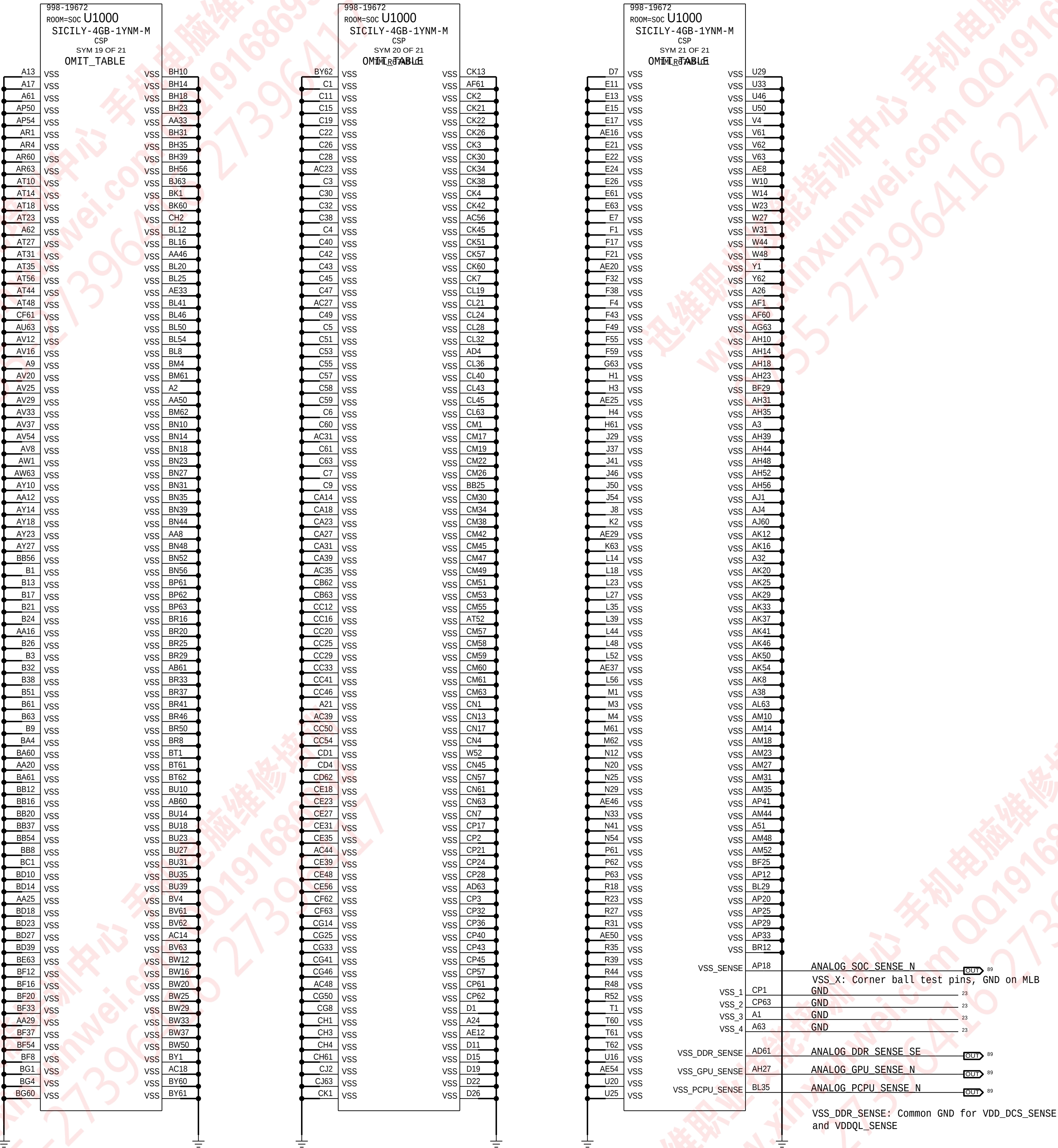


SYNCING: D52, D53, D54, DEV

PAGE TITLE			SOC: Power (CPU/GPU & SRAM & SOC)		
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SOC: GND



SYNCING: D52, D53, D54, DEV

PAGE TITLE		
SOC: Power (GND)		
	DRAWING NUMBER	051-05215
	REVISION	4.6.0
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AP/ISP I2C

AP I2C0 (Unused)

AP I2C1 (Unused)

AP I2C2



MASTER	AP	NUMBER	I2C2	DIAGS NUMBER	2	SPEED	1MHz
DEVICE	VOLTAGE	7-BIT ADDR	8-BIT ADDR	MAX SPEED	LOCATION		
Top Spk Amp	1.2V	0x40	0x80, 0x81	1MHz	MLB		

PP1V8_I0

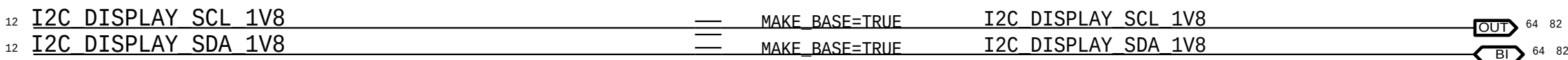


MASTER	AP	NUMBER	I2C3	DIAGS NUMBER	3	SPEED	400kHz
DEVICE		VOLTAGE	7-BIT ADDR	8-BIT ADDR		MAX SPEED	LOCATION
Touch EEPROM		1.8V	0x51	0xA2, 0xA3		400kHz	Touch Flex
Babbage (TFE)		1.8V	0x4B	0x96, 0x97		1MHz	Touch Flex
NOTE: For D52/D54, bus can either be mastered by SoC or Ada							
Roswell		1.8V	0x10	0x20, 0x21		400kHz	Touch Flex
NOTE: Roswell is I2C for D53-only (AID for D52/D54)							

D52/D54 only

D53 only

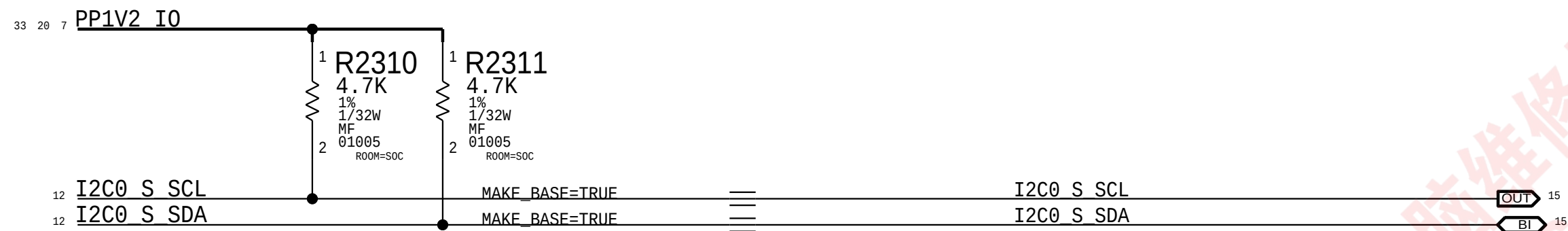
AP I2C4 (Legacy 1.8V)



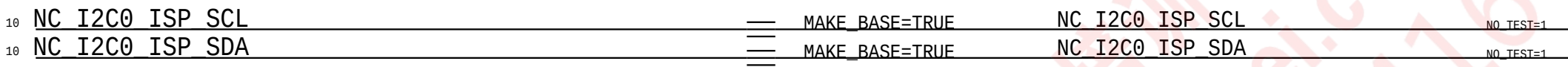
MAKE_BASE for Display PMIC intentional, test feature only

MASTER	AP	NUMBER	I2C4	DIAGS NUMBER	4	SPEED	400kHz
DEVICE		VOLTAGE		7-BIT ADDR	8-BIT ADDR	MAX SPEED	LOCATION
Display PMIC		1.8V		0x50	0xA0, 0xA1	400kHz	MLB
NOTE: SoC is master for FCT *ONLY*, DDIC is master for normal operation							

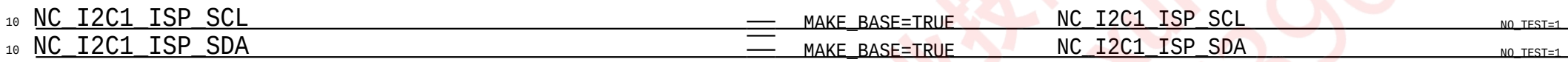
SI2C0



ISP I2C0



ISP I2C1 (Unused)

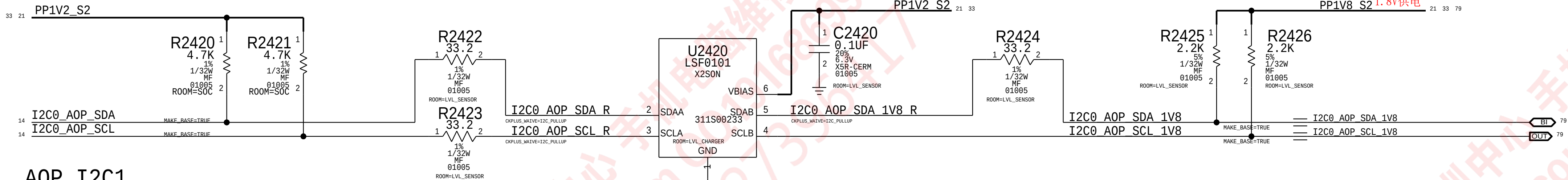


SYNCING: D52, D53, D54

PAGE TITLE		
SOC: Aliases: I2C AP/ISP		
	DRAWING NUMBER	SIZE
	051-05215	D
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	PAGE	84 OF 191
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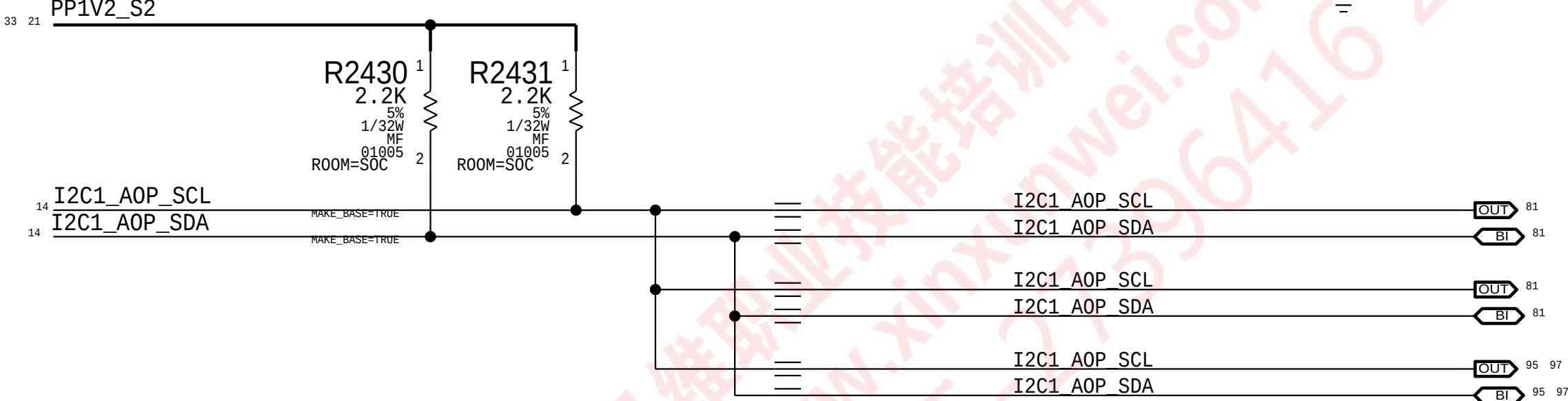
AOP/SMC I2C

AOP I2C0



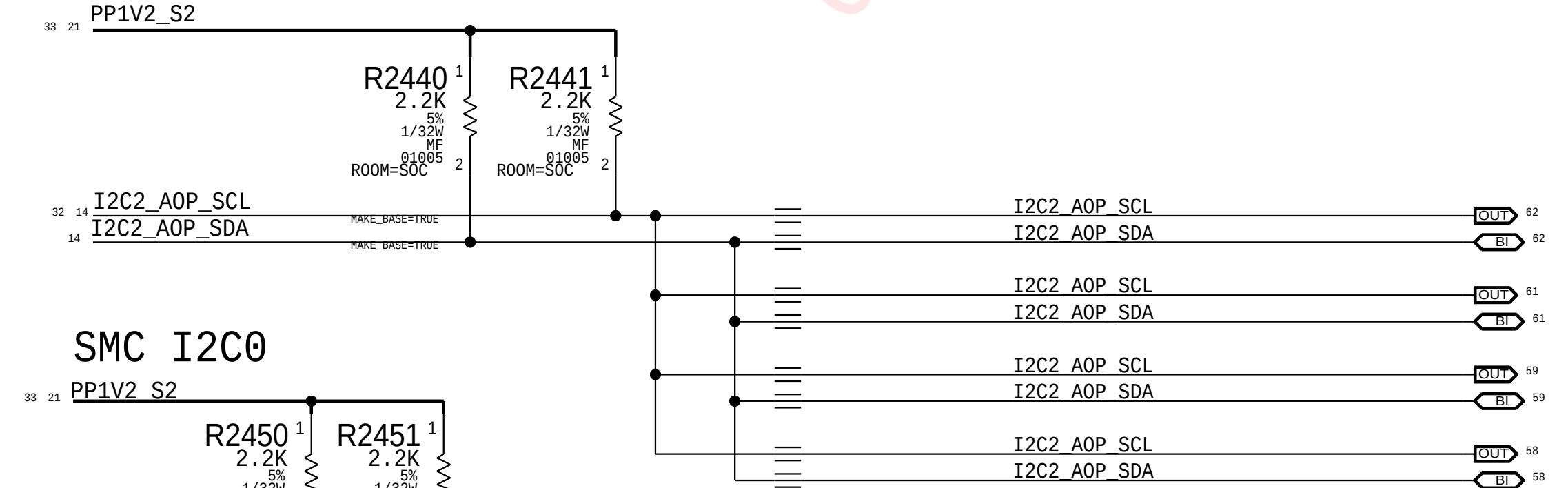
MASTER	AOP	NUMBER	I2C0	DIAGS NUMBER	5	SPEED	750kHz
DEVICE	VOLTAGE	7-BIT ADDR	8-BIT ADDR	MAX SPEED	LOCATION		
Prox	1.8V	0x58	0xB0, 0xB1	1MHz	Sensor Flex		
ALS	1.8V	0x29	0x52, 0x53	1MHz	Sensor Flex		
Grievous	1.8V	0x33	0x66, 0x67	1MHz	Sensor Flex		
Compass	1.8V	0x0E	0x1C, 0x1D	1MHz	Sensor Flex		

AOP I2C1



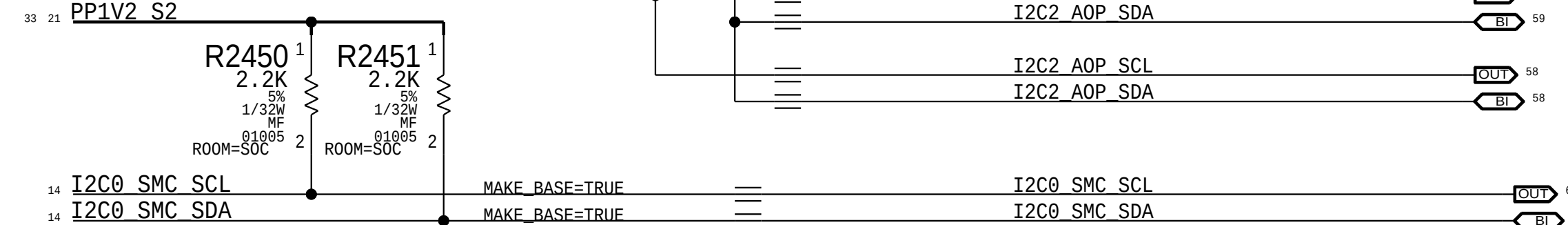
MASTER	AOP	NUMBER	I2C1	DIAGS NUMBER	6	SPEED	400kHz
DEVICE	VOLTAGE	7-BIT ADDR	8-BIT ADDR	MAX SPEED	LOCATION		
Eiger	1.2V	0x76	0xEC, 0xED	1MHz	DocK		
Arc EEPROM	1.2V	0x50	0xA0, 0xA1	1MHz	Arc Flex		
Jarvis	1.2V	0x0F	0x1E, 0x1F	1MHz	MLB		

AOP I2C2



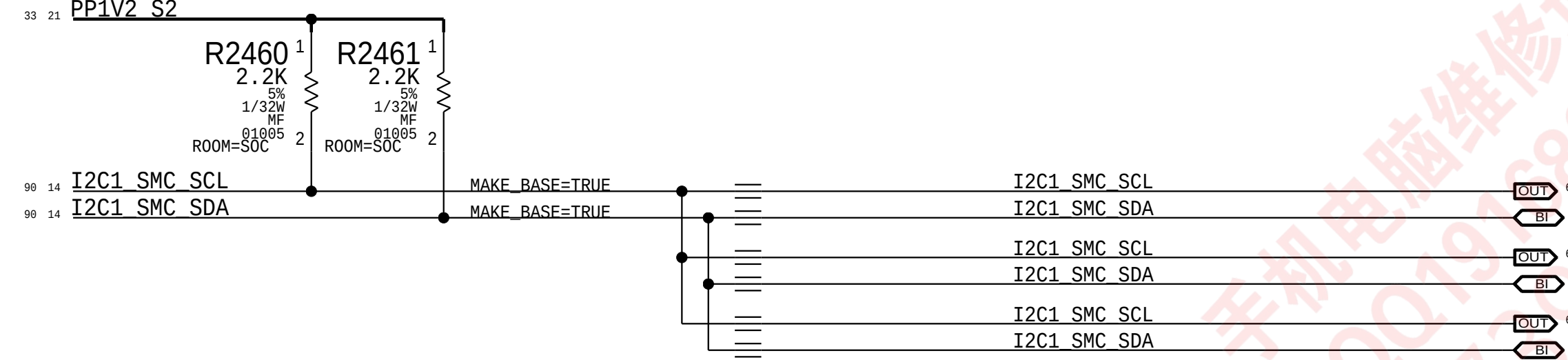
MASTER	AOP	NUMBER	I2C2	DIAGS NUMBER	7	SPEED	1MHz
DEVICE	VOLTAGE	7-BIT ADDR	8-BIT ADDR	MAX SPEED	LOCATION		
Sakonnet	1.2V	0x08	0x10, 0x11	1MHz	MLB		
Codec	1.2V	0x4A	0x94, 0x95	1MHz	MLB		
Arc Amp	1.2V	0x42	0x84, 0x85	1MHz	MLB		
Bot Spk Amp	1.2V	0x40	0x80, 0x81	1MHz	MLB		

SMC I2C0



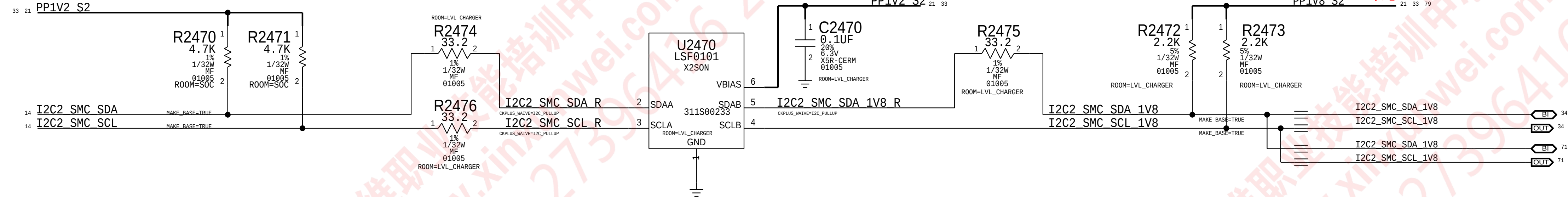
MASTER	SMC	NUMBER	I2C0	DIAGS NUMBER	8	SPEED	400kHz
DEVICE	VOLTAGE	7-BIT ADDR	8-BIT ADDR	MAX SPEED	LOCATION		
CC62B	1.2V	0x12	0x24, 0x25	1MHz	MLB		

SMC I2C1



MASTER	SMC	NUMBER	I2C1	DIAGS NUMBER	9	SPEED	400kHz
DEVICE	VOLTAGE	7-BIT ADDR	8-BIT ADDR	MAX SPEED	LOCATION		
Kraken	1.2V	0x1A	0x34, 0x35	1MHz	MLB		
Parrot	1.2V	0x21	0x42, 0x43	1MHz	MLB		
Gecko2	1.2V	0x52	0xA4, 0xA5	1MHz	MLB		

SMC I2C2



MASTER	SMC	NUMBER	I2C2	DIAGS NUMBER	10	SPEED	400kHz
DEVICE	VOLTAGE	7-BIT ADDR	8-BIT ADDR	MAX SPEED	LOCATION		
Yangtze	1.8V	0x71	0xE2, 0xE3	400kHz	MLB		
Veridian	1.8V	0x0B	0x16, 0x17	400kHz	Battery Flex		

SYNCING: D52, D53, D54

PAGE TITLE		SOC: Aliases: I2C AOP/SMC			
		DRAWING NUMBER	051-05215	SIZE	D
		REVISION	4.6.0		
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		SHEET			

AOP GPIOs

* All AOP GPIOs tie into SCM block

SPMI0	SPI	AOP_FUNC0
SPMI0	SPI	AOP_FUNC1
SPMI0	SPI	AOP_FUNC2
SPMI0	SPI	AOP_FUNC3
SPMI1	SPI	AOP_FUNC4
SPMI1	SPI	AOP_FUNC5
SPMI1	SPI	AOP_FUNC6
SPMI1	SPI	AOP_FUNC7
	SPI	AOP_FUNC8
	SPI	AOP_FUNC9
	SPI	AOP_FUNC10
	SPI	AOP_FUNC11
I2C0	SPI	AOP_FUNC12
I2C0	SPI	AOP_FUNC13
I2C0	SPI	AOP_FUNC14
I2C1, I2C0	SPI	AOP_FUNC15
I2C1	I2C0	AOP_FUNC16
I2C1	I2C0	AOP_FUNC17
I2C1	I2C0	AOP_FUNC18
	I2C1	AOP_FUNC19
	I2C1	AOP_FUNC20
	I2C1	AOP_FUNC21
	I2C1	AOP_FUNC22

14	GPIO SCM AOP FROM IMU DATARDY	==	GPIO SCM AOP FROM IMU DATARDY	MAKE_BASE=TRUE	IN	96
14	GPIO SCM AOP TO IMU SPI CS L	==	GPIO SCM AOP TO IMU SPI CS L	MAKE_BASE=TRUE	OUT	96
14	GPIO AOP FROM PEARL B2B DETECT	==	GPIO AOP FROM PEARL B2B DETECT	MAKE_BASE=TRUE	IN	78
14	NC DEV AOP_FUNC3	==	NC_DEV AOP_FUNC3			24
14	GPIO SCM AOP FROM R1 INT	==	GPIO SCM AOP FROM R1 INT	MAKE_BASE=TRUE	IN	98
14	GPIO AOP TO R1 COREDUMP TRIGGER	==	GPIO AOP TO R1 COREDUMP TRIGGER	MAKE_BASE=TRUE	OUT	98
14	GPIO AOP TO R1 TIME SYNC L	==	GPIO AOP TO R1 TIME SYNC L	MAKE_BASE=TRUE	OUT	98
14	GPIO AOP TO CODEC RESET L	==	GPIO AOP TO CODEC RESET L	MAKE_BASE=TRUE	OUT	58 89
14	GPIO AOP TO BB FORCE PWM	==	GPIO AOP TO BB FORCE PWM	MAKE_BASE=TRUE	OUT	93
14	GPIO AOP FROM IRCAM B2B DETECT	==	GPIO AOP FROM IRCAM B2B DETECT	MAKE_BASE=TRUE	IN	77
14	GPIO SCM AOP TO R1 SPI CS L	==	GPIO SCM AOP TO R1 SPI CS L	MAKE_BASE=TRUE	OUT	98
14	NC AOP_FUNC11	==	NC_AOP_FUNC11			23
14	GPIO AOP TO ALS COEX	==	GPIO AOP TO ALS COEX	MAKE_BASE=TRUE	OUT	79
14	GPIO AOP TO NFC IRONMAN EN	==	GPIO AOP TO NFC IRONMAN EN	MAKE_BASE=TRUE	OUT	99
14	GPIO AOP FROM TOUCH CTS	==	GPIO AOP FROM TOUCH CTS		IN	24 83
14	GPIO SCM AOP BI PROX INT L	==	GPIO SCM AOP BI PROX INT L	MAKE_BASE=TRUE	IN	78
14	GPIO SCM AOP FROM ALS INT L	==	GPIO SCM AOP FROM ALS INT L	MAKE_BASE=TRUE	IN	79
14	GPIO SCM AOP FROM EIGER INT L	==	GPIO SCM AOP FROM EIGER INT L	MAKE_BASE=TRUE	IN	81
14	GPIO SCM AOP FROM COMPASS INT	==	GPIO_SCM_AOP_FROM_COMPASS_INT	MAKE_BASE=TRUE	IN	78
14	GPIO SCM AOP FROM JARVIS INT	==	GPIO_SCM_AOP_FROM_JARVIS_INT	MAKE_BASE=TRUE	IN	89 95
14	GPIO AOP FROM TOUCH INT L	==	GPIO AOP FROM TOUCH INT L	MAKE_BASE=TRUE	IN	83
14	NC AOP_FUNC21	==	NC_AOP_FUNC21			23
14	NC DEV AOP_FUNC22	==	NC_DEV AOP_FUNC22			24

AP GPIOs

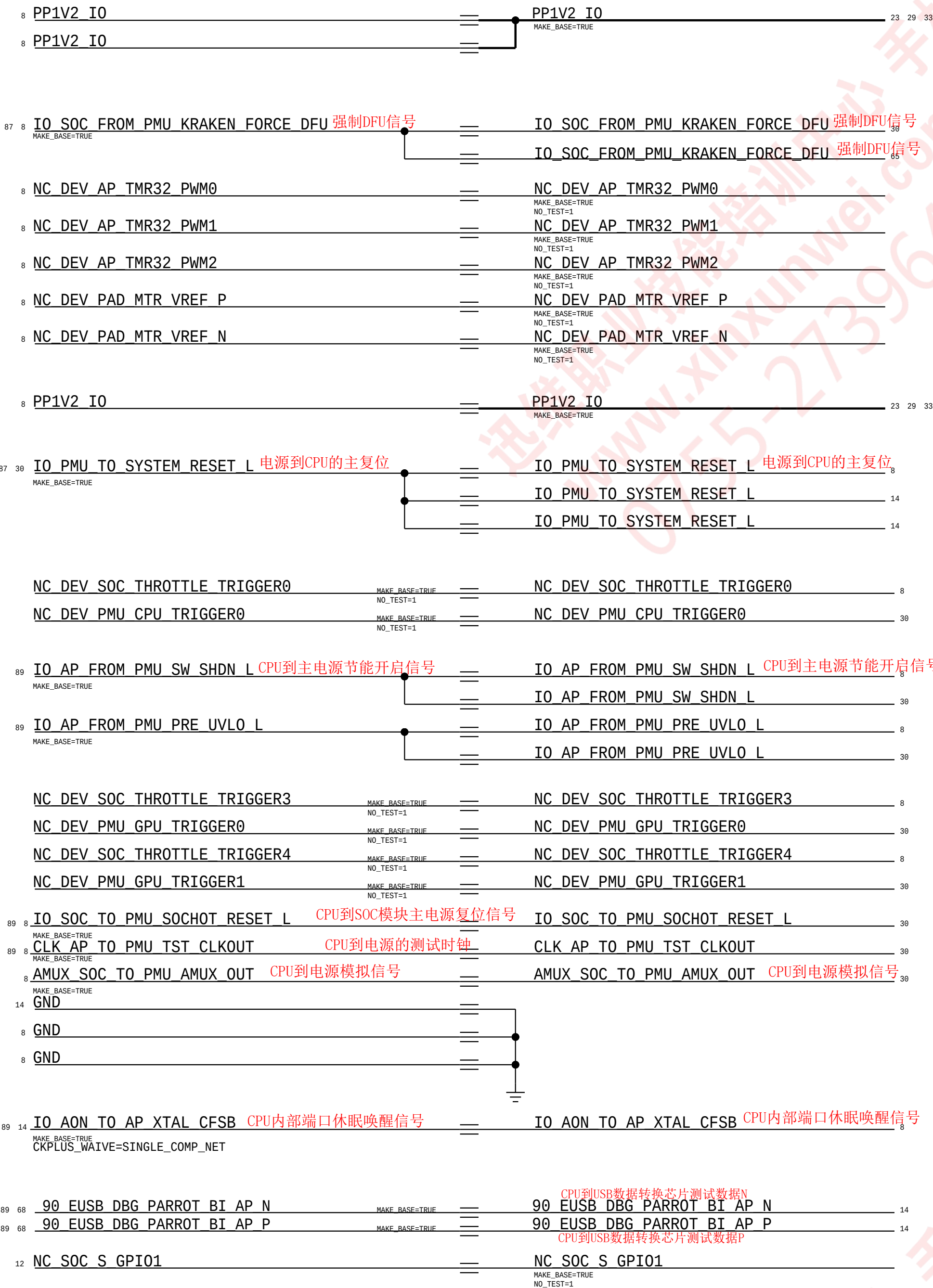
AP_GPIO0	13	GPIO_BOARD_REV3	版本识别3	==	GPIO_BOARD_REV3	版本识别3	MAKE_BASE=TRUE	IN	7
AP_GPIO1	13	GPIO_BOARD_REV2	版本识别2	==	GPIO_BOARD_REV2	版本识别2	MAKE_BASE=TRUE	IN	7
AP_GPIO2	13	GPIO_BOARD_REV1	版本识别1	==	GPIO_BOARD_REV1	版本识别1	MAKE_BASE=TRUE	IN	7
AP_GPIO3	13	GPIO_BOARD_REV0	版本识别0	==	GPIO_BOARD_REV0	版本识别0	MAKE_BASE=TRUE	IN	7
AP_GPIO4	13	GPIO_AP_CANARY1		==	GPIO_AP_CANARY1		MAKE_BASE=TRUE	IN	97
AP_GPIO5	13	GPIO_AP_CANARY2		==	GPIO_AP_CANARY2		MAKE_BASE=TRUE	IN	95 97
AP_GPIO6	13	GPIO_AP_BI_CCG2B_SWDIO	CPU到快充芯片SWD总线数据	==	GPIO_AP_BI_CCG2B_SWDIO		MAKE_BASE=TRUE	OUT	66 89
AP_GPIO7	13	GPIO_AP_TO_CCG2B_SWCLK	CPU到快充芯片SWD总线时钟	==	GPIO_AP_TO_CCG2B_SWCLK		MAKE_BASE=TRUE	OUT	66 89
AP_GPIO8	13	GPIO_AP_FROM_DISPLAY_PANEL_ID	CPU到显示屏ID识别信号	==	GPIO_AP_FROM_DISPLAY_PANEL_ID		MAKE_BASE=TRUE	IN	82
AP_GPIO9	13	GPIO_AP_FROM_WLAN_TIME_SYNC		==	GPIO_AP_FROM_WLAN_TIME_SYNC		MAKE_BASE=TRUE	IN	93
AP_GPIO10	13	GPIO_AP_TO_BB_PEAK_PWR_IND		==	GPIO_AP_TO_BB_PEAK_PWR_IND		MAKE_BASE=TRUE	OUT	93
AP_GPIO11	13	GPIO_AP_TO_BB_COREDUMP		==	GPIO_AP_TO_BB_COREDUMP		MAKE_BASE=TRUE	OUT	95
AP_GPIO12	13	GPIO_AP_FROM_BB_RESET_DETECT_L		==	GPIO_AP_FROM_BB_RESET_DETECT_L		MAKE_BASE=TRUE	IN	93
AP_GPIO13	13	GPIO_AP_FROM_CODEC_INT_L		==	GPIO_AP_FROM_CODEC_INT_L		MAKE_BASE=TRUE	IN	58 89
AP_GPIO14	13	NC_AP_GPIO14		==	NC_AP_GPIO14				23
AP_GPIO15	13	GPIO_AP_TO_SPKRAMP_TOP_RESET_L		==	GPIO_AP_TO_SPKRAMP_TOP_RESET_L		MAKE_BASE=TRUE	OUT	60
AP_GPIO16	13	GPIO_AP_FROM_BT_AUDIO_SYNC		==	GPIO_AP_FROM_BT_AUDIO_SYNC		MAKE_BASE=TRUE	IN	93
AP_GPIO17	13	GPIO_AP_TO_AMUX_PMU_SYNC		==	GPIO_AP_TO_AMUX_PMU_SYNC			OUT	24
AP_GPIO18	13	NC_DEV_AP_GPIO18		==	NC_DEV_AP_GPIO18				24
AP_GPIO19	13	NC_DEV_AP_GPIO19		==	NC_DEV_AP_GPIO19				24
AP_GPIO20	13	NC_DEV_AP_GPIO20		==	NC_DEV_AP_GPIO20				24
AP_GPIO21	13	NC_DEV_AP_GPIO21		==	NC_DEV_AP_GPIO21				24
AP_GPIO22	13	NC_DEV_AP_GPIO22		==	NC_DEV_AP_GPIO22				24
AP_GPIO23	13	NC_DEV_AP_GPIO23		==	NC_DEV_AP_GPIO23				24
AP_GPIO24	13	NC_DEV_AP_GPIO24		==	NC_DEV_AP_GPIO24				24
AP_GPIO25	13	GPIO_AP_TO_BB_TIME_MARK		==	GPIO_AP_TO_BB_TIME_MARK		MAKE_BASE=TRUE	OUT	93
AP_GPIO26	13	NC_DEV_AP_GPIO26		==	NC_DEV_AP_GPIO26				24

SYNCING: D52, D53, D54, DEV

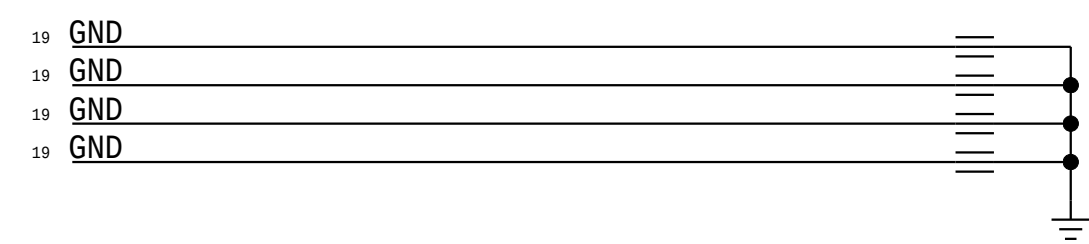
PAGE TITLE			SOC: Aliases: GPIOs		
			DRAWING NUMBER	051-05215	SIZE
			REVISION	4.6.0	
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Misc. SoC Aliases

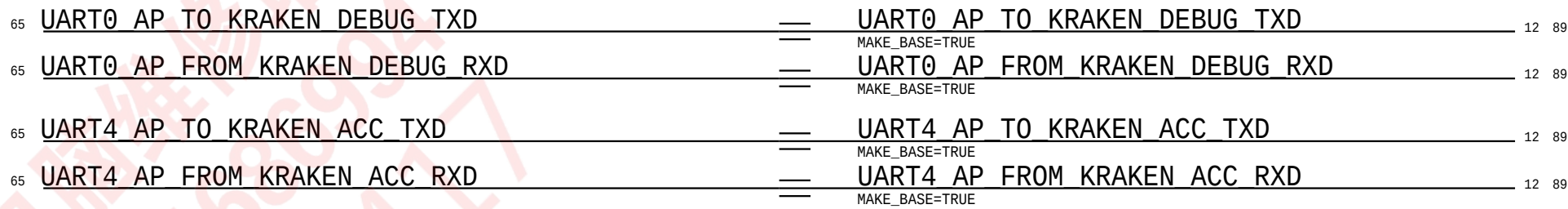
NAND + USB & MISC



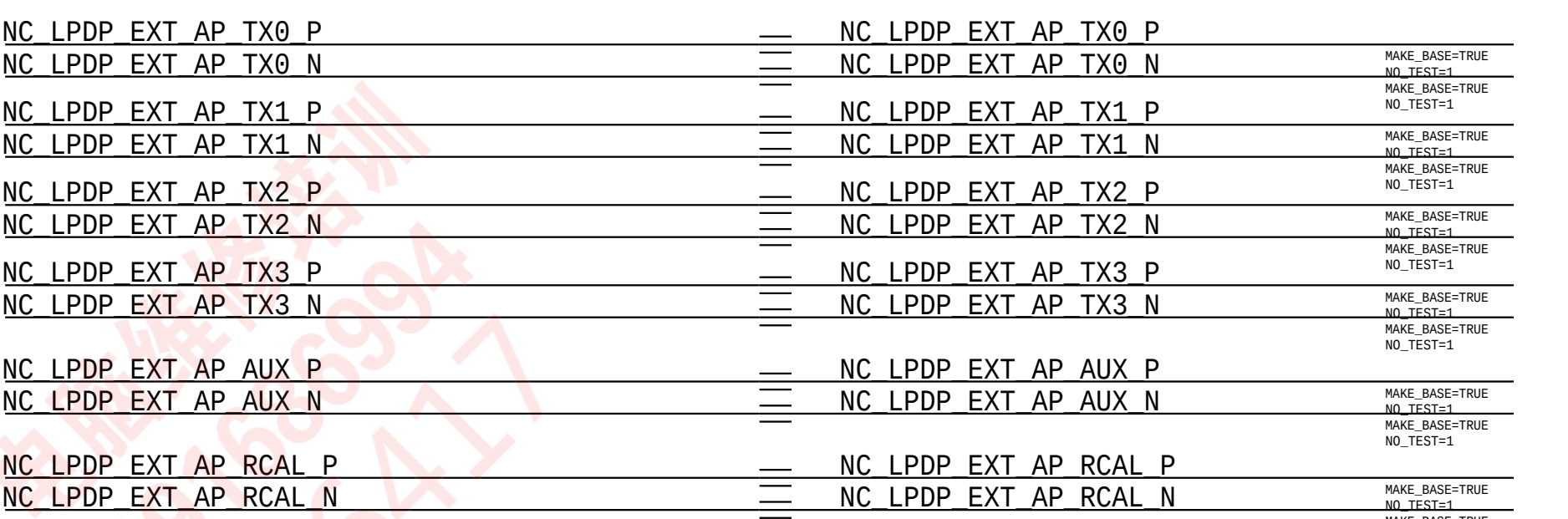
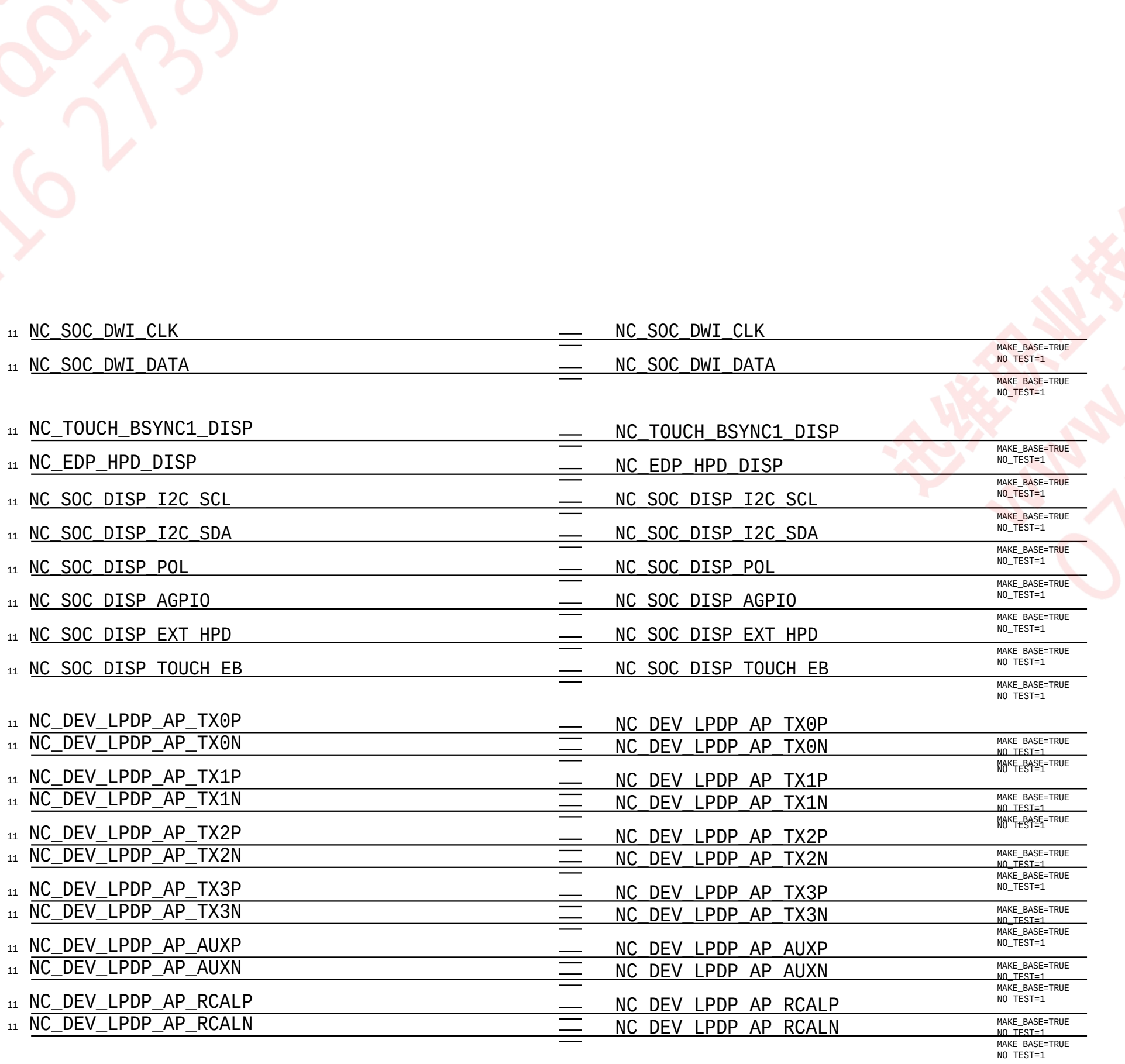
VSS CORNER BALLS



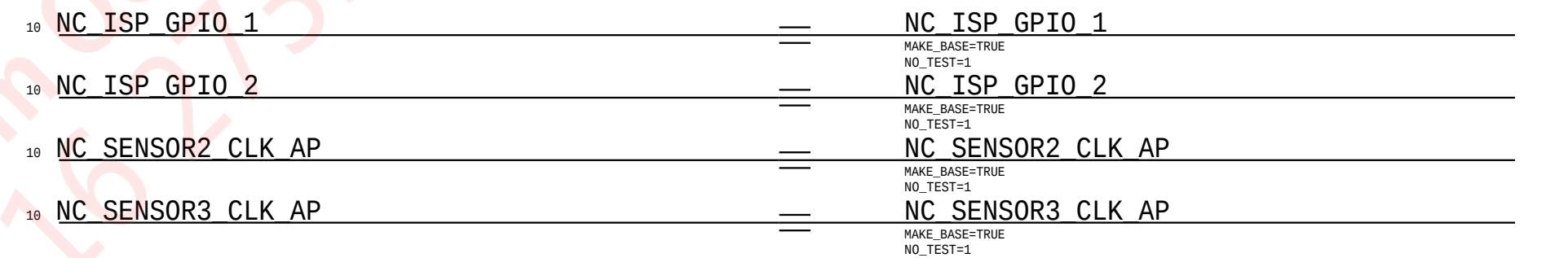
UART: KRAKEN



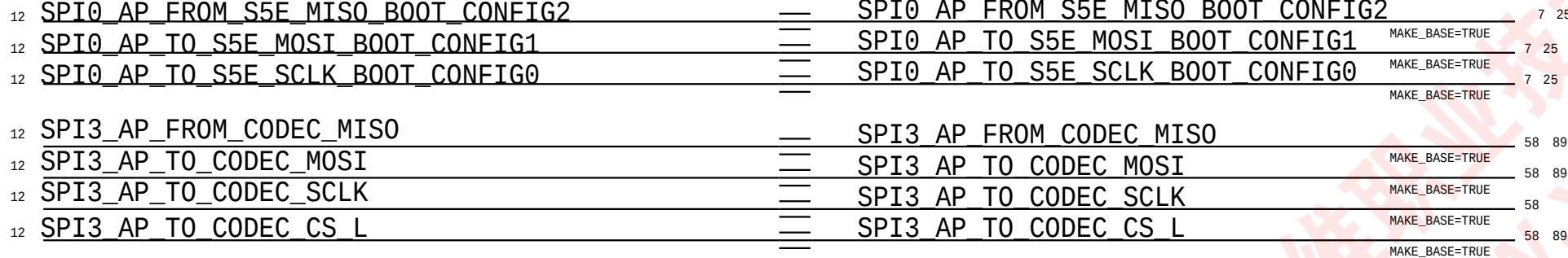
DISPLAY



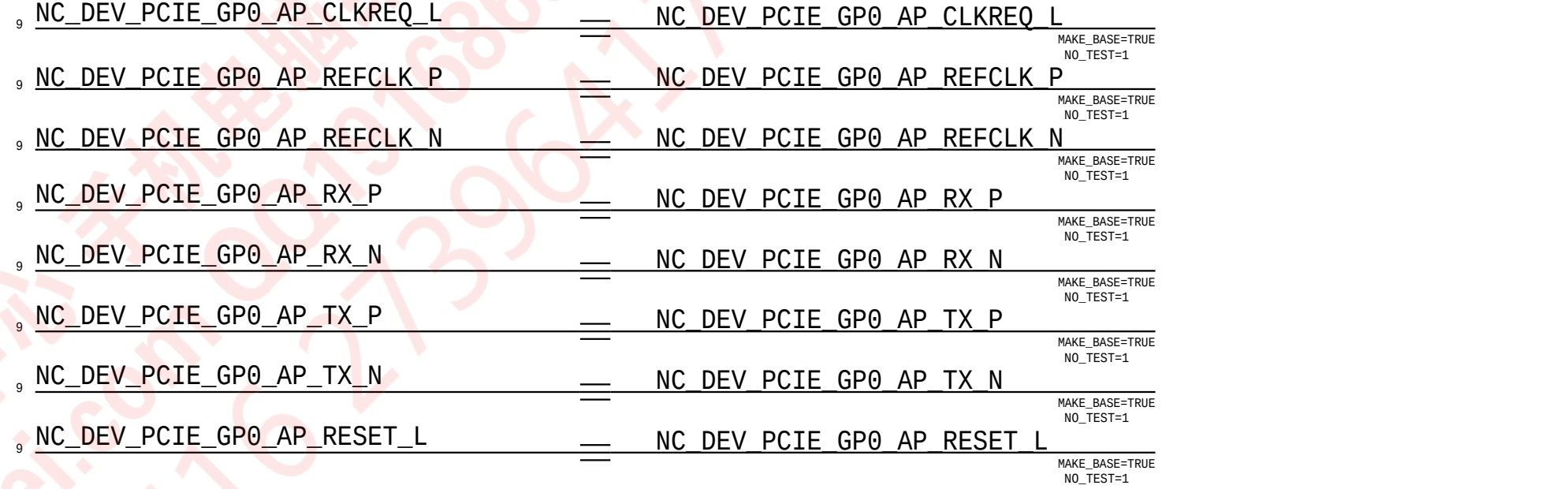
ISP



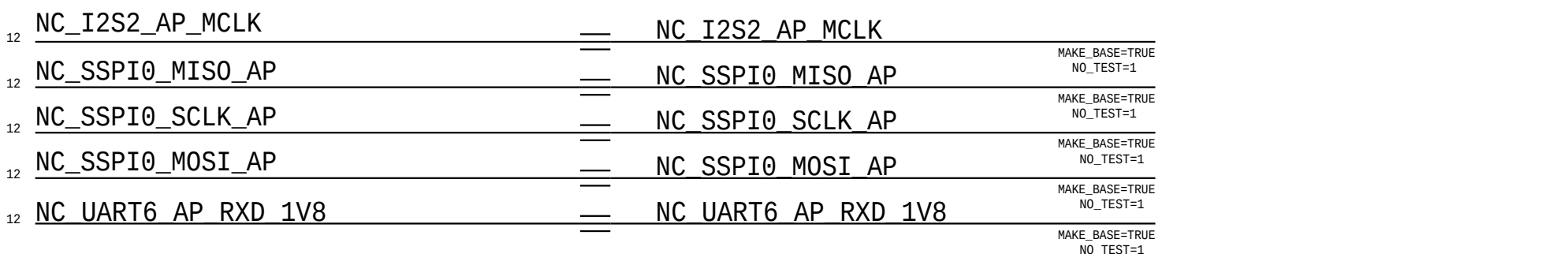
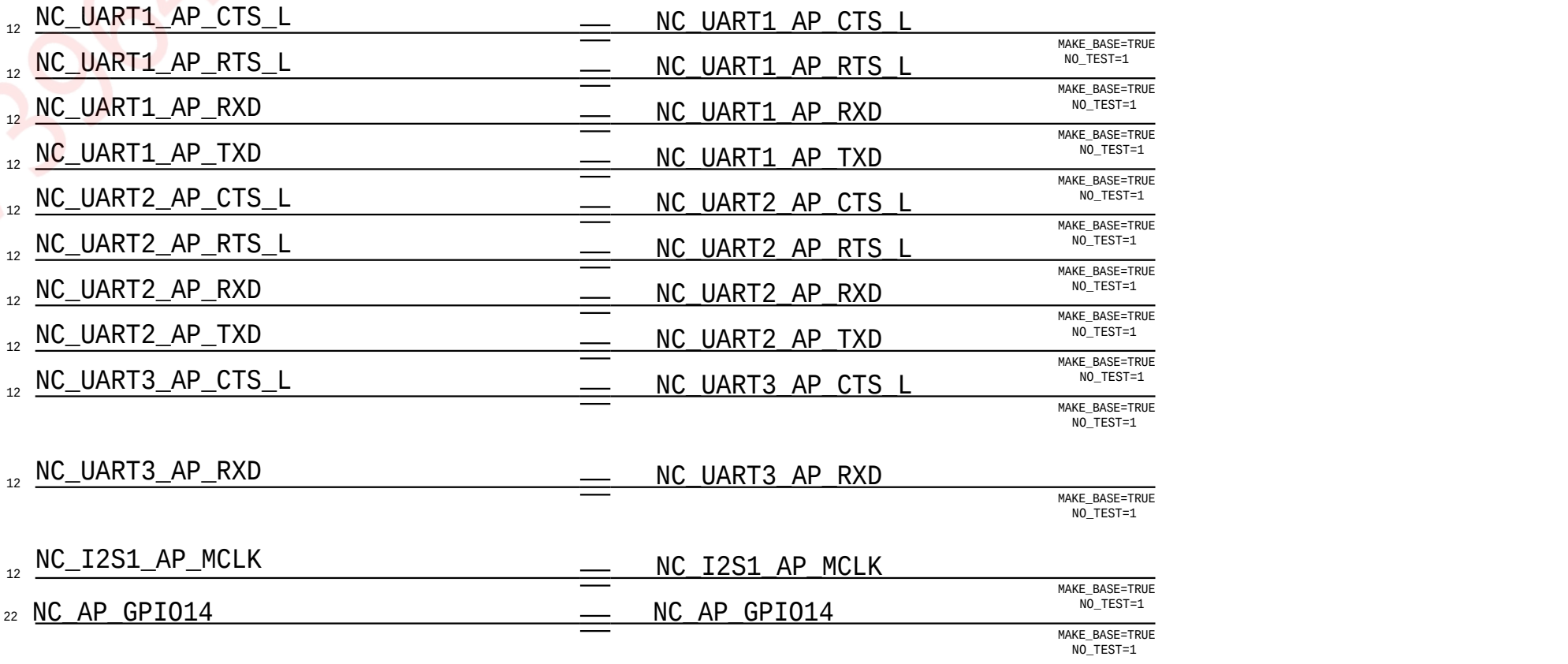
SPI (AP)



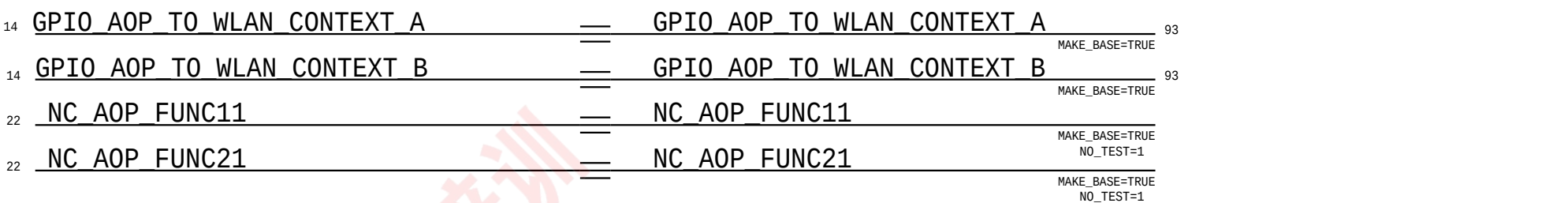
SOC: PCIe



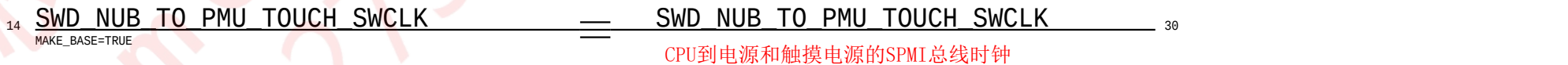
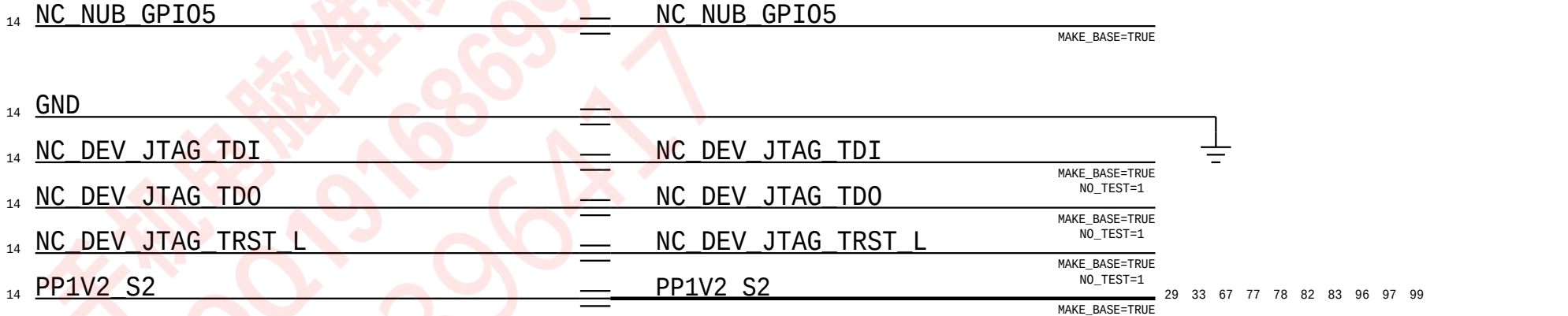
SOC: Serial



SOC: AOP

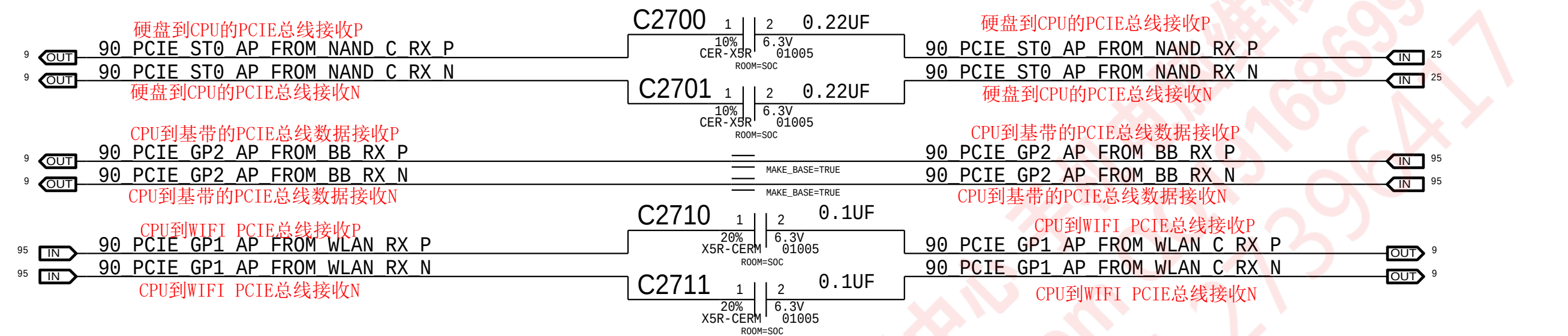


NUB



SYNCING: D52, D53, D54

PAGE TITLE		
SOC: Aliases: Misc		
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DISPLAY

90	MIPI AP TO DISPLAY CLK P	==	90	MIPI AP TO DISPLAY CLK P	80
11	90 MIPI AP TO DISPLAY CLK N	==	90	MIPI AP TO DISPLAY CLK N	80
				MAKE_BASE=TRUE	
90	MIPI AP TO DISPLAY D0 P	==	90	MIPI AP TO DISPLAY D0 P	80
11	90 MIPI AP TO DISPLAY D0 N	==	90	MIPI AP TO DISPLAY D0 N	80
				MAKE_BASE=TRUE	
90	MIPI AP TO DISPLAY D1 P	==	90	MIPI AP TO DISPLAY D1 P	80
11	90 MIPI AP TO DISPLAY D1 N	==	90	MIPI AP TO DISPLAY D1 N	80
				MAKE_BASE=TRUE	
90	MIPI AP TO DISPLAY D2 P	==	90	MIPI AP TO DISPLAY D2 P	80
11	90 MIPI AP TO DISPLAY D2 N	==	90	MIPI AP TO DISPLAY D2 N	80
				MAKE_BASE=TRUE	
90	MIPI AP TO DISPLAY D2 P	==	90	MIPI AP TO DISPLAY D2 P	80
11	90 MIPI AP TO DISPLAY D2 N	==	90	MIPI AP TO DISPLAY D2 N	80
				MAKE_BASE=TRUE	

LPDP

ISP: LPDP Lanes

90	LDPD	ISP	FROM	FCAM	RX	D1	N	==	MAKE_BASE=TRUE	90	LDPD	ISP	FROM	FCAM	RX	D1	N	76		
90	LDPD	ISP	FROM	FCAM	RX	D1	P	==	MAKE_BASE=TRUE	90	LDPD	ISP	FROM	FCAM	RX	D1	P	76		
								====	KCPUS_VALVE-DIFFERATR_BAOTERN											
90	LDPD	ISP	FROM	FCAM	RX	D0	N	==	MAKE_BASE=TRUE	90	LDPD	ISP	FROM	FCAM	RX	D0	N	76		
90	LDPD	ISP	FROM	FCAM	RX	D0	P	==	MAKE_BASE=TRUE	90	LDPD	ISP	FROM	FCAM	RX	D0	P	76		
								====	KCPUS_VALVE-DIFFERATR_BAOTERN											
90	LDPD	ISP	FROM	WIDE	RX	D2	P	==	MAKE_BASE=TRUE	90	LDPD	ISP	FROM	WIDE	RX	D2	P	76		
90	LDPD	ISP	FROM	WIDE	RX	D2	N	==	MAKE_BASE=TRUE	90	LDPD	ISP	FROM	WIDE	RX	D2	N	74		
								====	KCPUS_VALVE-DIFFERATR_BAOTERN											
90	LDPD	ISP	FROM	WIDE	RX	D1	N	==	MAKE_BASE=TRUE	90	LDPD	ISP	FROM	WIDE	RX	D1	N	74		
90	LDPD	ISP	FROM	WIDE	RX	D1	P	==	MAKE_BASE=TRUE	90	LDPD	ISP	FROM	WIDE	RX	D1	P	74		
								====	KCPUS_VALVE-DIFFERATR_BAOTERN											
90	LDPD	ISP	FROM	WIDE	RX	D0	N	==	MAKE_BASE=TRUE	90	LDPD	ISP	FROM	WIDE	RX	D0	N	74		
90	LDPD	ISP	FROM	WIDE	RX	D0	P	==	MAKE_BASE=TRUE	90	LDPD	ISP	FROM	WIDE	RX	D0	P	74		
								====	KCPUS_VALVE-DIFFERATR_BAOTERN											
90	LDPD	ISP	FROM	SUPERFLEX	CAM2	RX	D2	N	==	MAKE_BASE=TRUE	90	LDPD	ISP	FROM	SUPERFLEX	CAM2	RX	D2	N	74
90	LDPD	ISP	FROM	SUPERFLEX	CAM2	RX	D2	P	==	MAKE_BASE=TRUE	90	LDPD	ISP	FROM	SUPERFLEX	CAM2	RX	D2	P	74
								====	KCPUS_VALVE-DIFFERATR_BAOTERN											
90	LDPD	ISP	FROM	SUPERFLEX	CAM2	RX	D0	P	==	MAKE_BASE=TRUE	90	LDPD	ISP	FROM	SUPERFLEX	CAM2	RX	D0	P	74
90	LDPD	ISP	FROM	SUPERFLEX	CAM2	RX	D0	N	==	MAKE_BASE=TRUE	90	LDPD	ISP	FROM	SUPERFLEX	CAM2	RX	D0	N	74
								====	KCPUS_VALVE-DIFFERATR_BAOTERN											
90	LDPD	ISP	FROM	SUPERFLEX	CAM2	RX	D1	N	==	MAKE_BASE=TRUE	90	LDPD	ISP	FROM	SUPERFLEX	CAM2	RX	D1	N	74
90	LDPD	ISP	FROM	SUPERFLEX	CAM2	RX	D1	P	==	MAKE_BASE=TRUE	90	LDPD	ISP	FROM	SUPERFLEX	CAM2	RX	D1	P	74
								====	KCPUS_VALVE-DIFFERATR_BAOTERN											
90	LDPD	ISP	FROM	JASPER	RX	D0	P	==	MAKE_BASE=TRUE	90	LDPD	ISP	FROM	JASPER	RX	D0	P	75		
90	LDPD	ISP	FROM	JASPER	RX	D0	N	==	MAKE_BASE=TRUE	90	LDPD	ISP	FROM	JASPER	RX	D0	N	75		
								====	KCPUS_VALVE-DIFFERATR_BAOTERN											
90	LDPD	ISP	FROM	SINGLEFLEX	CAM	RX	D0	P	==	MAKE_BASE=TRUE	90	LDPD	ISP	FROM	SINGLEFLEX	CAM	RX	D0	P	73
90	LDPD	ISP	FROM	SINGLEFLEX	CAM	RX	D0	N	==	MAKE_BASE=TRUE	90	LDPD	ISP	FROM	SINGLEFLEX	CAM	RX	D0	N	73
								====	KCPUS_VALVE-DIFFERATR_BAOTERN											
90	LDPD	ISP	FROM	SINGLEFLEX	CAM	RX	D1	P	==	MAKE_BASE=TRUE	90	LDPD	ISP	FROM	SINGLEFLEX	CAM	RX	D1	P	73
90	LDPD	ISP	FROM	SINGLEFLEX	CAM	RX	D1	N	==	MAKE_BASE=TRUE	90	LDPD	ISP	FROM	SINGLEFLEX	CAM	RX	D1	N	73
								====	KCPUS_VALVE-DIFFERATR_BAOTERN											
90	LDPD	ISP	FROM	SINGLEFLEX	CAM	RX	D2	P	==	MAKE_BASE=TRUE	90	LDPD	ISP	FROM	SINGLEFLEX	CAM	RX	D2	P	73
90	LDPD	ISP	FROM	SINGLEFLEX	CAM	RX	D2	N	==	MAKE_BASE=TRUE	90	LDPD	ISP	FROM	SINGLEFLEX	CAM	RX	D2	N	73
								====	KCPUS_VALVE-DIFFERATR_BAOTERN											

ISP: LPDP Aux

10	NC LPDP ISP AUX RX D0P	==	NC LPDP ISP AUX RX D0P	74
			MAKE_BASE=TRUE NO_TEST=1	
10	NC LPDP ISP AUX RX D1P	==	NC LPDP ISP AUX RX D1P	
			MAKE_BASE=TRUE NO_TEST=1	
10	LPDP ISP BI SUPERFLEX CAM2 AUX RX D2P	==	LPDP ISP BI SUPERFLEX CAM2 AUX RX D2P	74
			MAKE_BASE=TRUE	
10	NC LPDP ISP AUX RX D3P	==	NC LPDP ISP AUX RX D3P	
			MAKE_BASE=TRUE NO_TEST=1	
10	NC LPDP ISP AUX RX D4P	==	NC LPDP ISP AUX RX D4P	
			MAKE_BASE=TRUE NO_TEST=1	
10	LPDP ISP BI SINGLEFLEX CAM AUX RX D5P	==	LPDP ISP BI SINGLEFLEX CAM AUX RX D5P	73
			MAKE_BASE=TRUE	
10	NC LPDP ISP AUX RX D6P	==	NC LPDP ISP AUX RX D6P	
			MAKE_BASE=TRUE NO_TEST=1	
10	NC LPDP ISP AUX RX D7P	==	NC LPDP ISP AUX RX D7P	
			MAKE_BASE=TRUE NO_TEST=1	
10	NC LPDP ISP AUX RX D8P	==	NC LPDP ISP AUX RX D8P	
			MAKE_BASE=TRUE NO_TEST=1	
10	NC LPDP ISP AUX RX D9P	==	NC LPDP ISP AUX RX D9P	
			MAKE_BASE=TRUE NO_TEST=1	
10	NC LPDP ISP AUX RX D10P	==	NC LPDP ISP AUX RX D10P	
			MAKE_BASE=TRUE NO_TEST=1	
10	NC LPDP ISP AUX RX D11P	==	NC LPDP ISP AUX RX D11P	
			MAKE_BASE=TRUE	

12	NC AP SPMI2 SCLK	==	NC AP SPMI2 SCLK	MAKE_BASE=TRUE NO_TEST=1
12	NC AP SPMI2 SDATA	==	NC AP SPMI2 SDATA	MAKE_BASE=TRUE NO_TEST=1
12	NC I2C0 AP SCL	==	NC I2C0 AP SCL	MAKE_BASE=TRUE NO_TEST=1
12	NC I2C0 AP SDA	==	NC I2C0 AP SDA	MAKE_BASE=TRUE NO_TEST=1
12	NC I2C1 AP SCL	==	NC I2C1 AP SCL	MAKE_BASE=TRUE NO_TEST=1
12	NC I2C1 AP SDA	==	NC I2C1 AP SDA	MAKE_BASE=TRUE NO_TEST=1
22	NC_DEV_AOP_FUNC22	==	NC_DEV_AOP_FUNC22	MAKE_BASE=TRUE
22	NC_DEV_AOP_FUNC3	==	NC_DEV_AOP_FUNC3	MAKE_BASE=TRUE NO_TEST=1
22	GPIO_AOP_FROM_TOUCH_CTS	==	GPIO_AOP_FROM_TOUCH_CTS	MAKE_BASE=TRUE
22	GPIO_AP_TO_AMUX_PMU_SYNC	==	GPIO_AP_TO_AMUX_PMU_SYNC	MAKE_BASE=TRUE
22	NC_DEV_AP_GPIO18	==	NC_DEV_AP_GPIO18	MAKE_BASE=TRUE
22	NC_DEV_AP_GPIO19	==	NC_DEV_AP_GPIO19	MAKE_BASE=TRUE
22	NC_DEV_AP_GPIO20	==	NC_DEV_AP_GPIO20	MAKE_BASE=TRUE
22	NC_DEV_AP_GPIO21	==	NC_DEV_AP_GPIO21	MAKE_BASE=TRUE NO_TEST=1
22	NC_DEV_AP_GPIO22	==	NC_DEV_AP_GPIO22	MAKE_BASE=TRUE NO_TEST=1
22	NC_DEV_AP_GPIO23	==	NC_DEV_AP_GPIO23	MAKE_BASE=TRUE NO_TEST=1
22	NC_DEV_AP_GPIO24	==	NC_DEV_AP_GPIO24	MAKE_BASE=TRUE NO_TEST=1
22	NC_DEV_AP_GPIO26	==	NC_DEV_AP_GPIO26	MAKE_BASE=TRUE NO_TEST=1

SPI (AP)

11	SPI1 AP FROM TOUCH MISO 1V8	=====	SPI1 AP FROM TOUCH MISO 1V8		83
12	SPI1 AP TO TOUCH MOSI 1V8	=====	SPI1 AP TO TOUCH MOSI 1V8	MAKE_BASE=TRUE	
11	SPI1 AP TO TOUCH SCLK 1V8	=====	SPI1 AP TO TOUCH SCLK 1V8	MAKE_BASE=TRUE	83
12	SPI1 AP TO TOUCH CS L 1V8	=====	SPI1 AP TO TOUCH CS L 1V8	MAKE_BASE=TRUE	83
		=====		MAKE_BASE=TRUE	
12	NC SPI2 MISO AP	=====	NC SPI2 MISO AP	NO_TEST=1	
12	NC SPI2 MOSI AP	=====	NC SPI2 MOSI AP	MAKE_BASE=TRUE	NO_TEST=1
12	NC SPI2 SCLK AP	=====	NC SPI2 SCLK AP	MAKE_BASE=TRUE	NO_TEST=1
12	NC SPI2 CS L AP	=====	NC SPI2 CS L AP	MAKE_BASE=TRUE	NO_TEST=1
		=====		MAKE_BASE=TRUE	

Ada 24MHz clock (not used for DIT0)

Hacky NO_TEST fix

```
51 NC_CAM_PMU2_BUCK0                                     NC_CAM_PMU2_BUCK0                                     NO_TEST=1                                             51
```

Extra Canary

12	GPIO_AP_CANARY6	==	GPIO_AP_CANARY6	93
12	GPIO_AP_CANARY5	==	GPIO_AP_CANARY5	93
12	GPIO_AP_CANARY3	==	GPIO_AP_CANARY3	93
14	GPIO_A0P_CANARY7	==	GPIO_A0P_CANARY7	94
11	GPIO_AP_CANARY4	==	GPIO_AP_CANARY4	95

SYNCING: D52, D53, D54

PAGE TITLE		SOC: Aliases: FF-Specific	
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S5E NAND

SUBSYSTEM SPECIFIC BOM TABLES

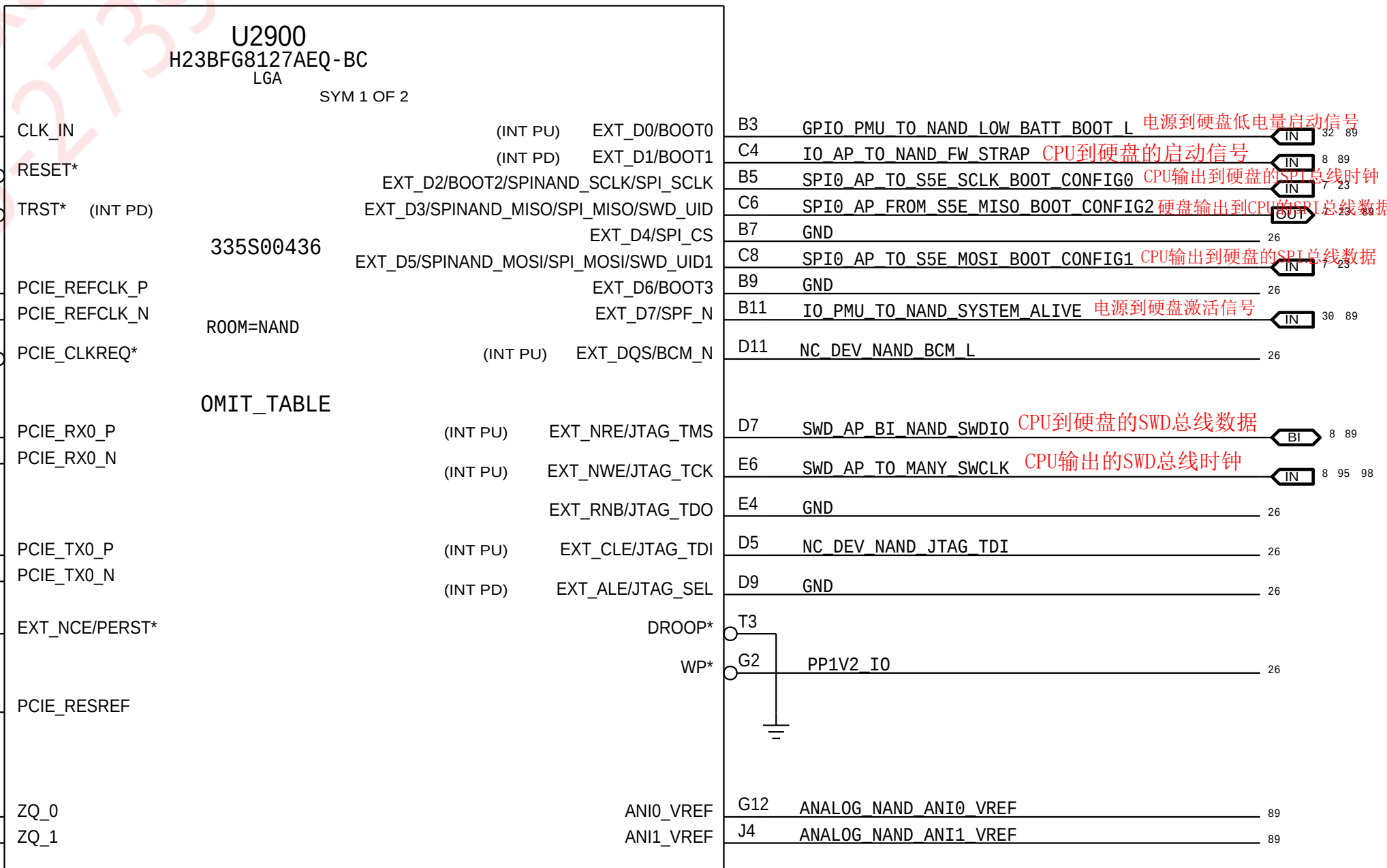
ZQ Resistor

PART#	QTY	DESCRIPTION	REFERENCE DESIGNATOR(S)	CRITICAL	BOM OPTION
118S0784	1	RES, 3800Hm, 1%, 1/32W, 01005	R2901	CRITICAL	S5E

Note: Dev Board adds an S4E option

NOTES:

INT PU = internal pull up to VDDIO_1
INT PD = internal pull down to VSS
Internal pulls are 40kOhm (min), 80kOhm (typ), 165kOhm (max)



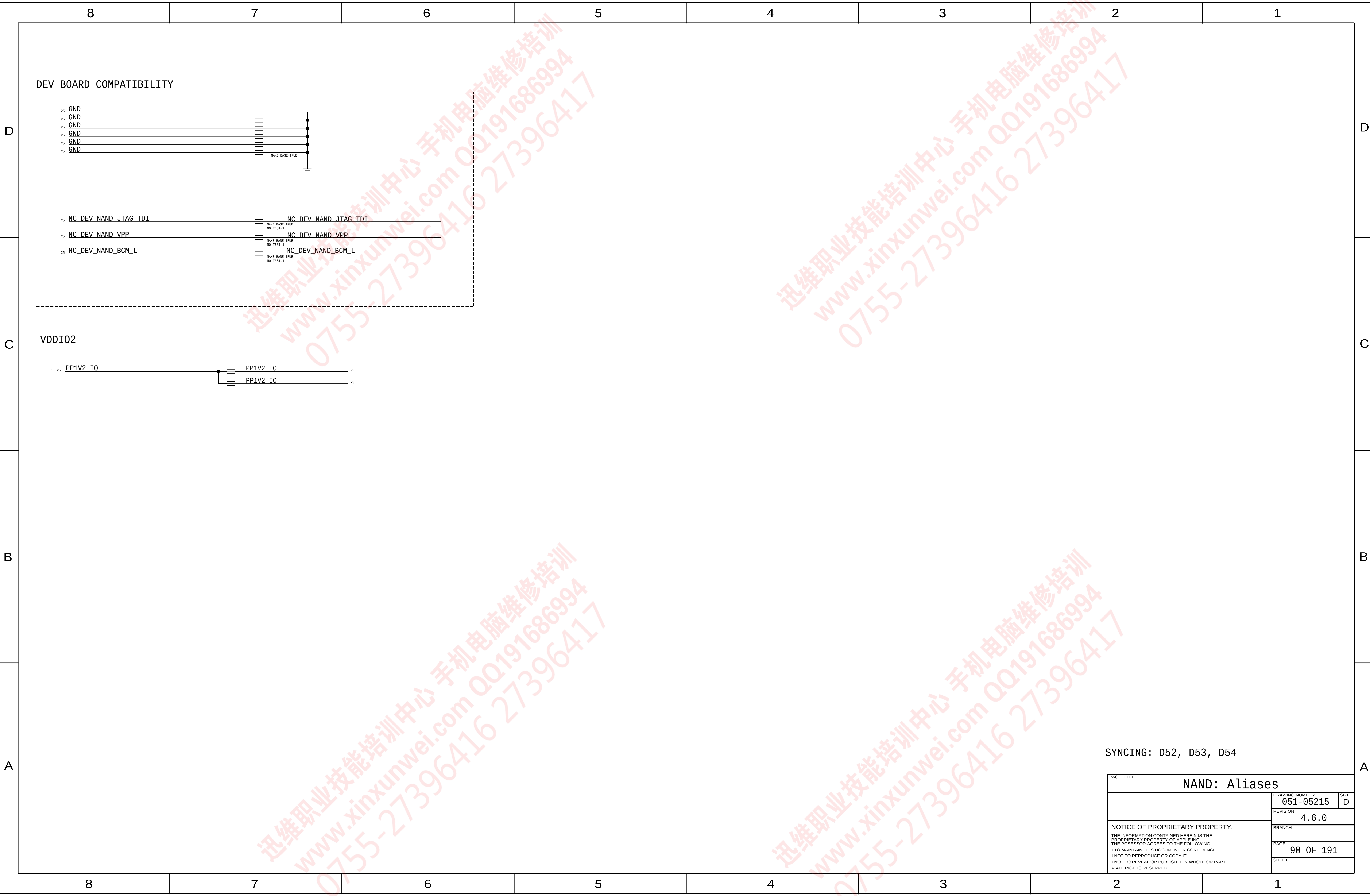
NAND Capacities

PART#	QTY	DESCRIPTION	REFERENCE DESIGNATOR(S)	CRITICAL	BOM OPTION
335S00436	1	HYNIX, 30v5, 64Gb, Ultimate	U2900	CRITICAL	NAND:ULTIMATE
335S00437	1	HYNIX, 30v5, 128Gb, Supreme	U2900	CRITICAL	NAND:SUPREME
335S00438	1	HYNIX, 30v5, 256Gb, Extreme	U2900	CRITICAL	NAND:EXTREME
335S00439	1	HYNIX, 30v5, 512Gb, Max	U2900	CRITICAL	NAND:MAX

PART NUMBER	ALTERNATE FOR PART NUMBER	BOM OPTION	REF DES	COMMENTS:
335S00449	335S00436	NAND:ULTIMATE	U2900	TOSHIBA, B1C54, 64Gb, Ultimate
335S00446	335S00437	NAND:SUPREME	U2900	SAMSUNG, 30v5, 128Gb, Supreme
335S00447	335S00438	NAND:EXTREME	U2900	SAMSUNG, 30v5, 256Gb, Extreme
335S00450	335S00438	NAND:EXTREME	U2900	TOSHIBA, B1C54, 256Gb, Extreme
335S00448	335S00439	NAND:MAX	U2900	SAMSUNG, 30v5, 512Gb, Max

SYNCING: D52, D53, D54

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NAND			
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VDDIO2

53

25

PP1V2_I0

PP1V2_I0

PP1V2_I0

25

25

SYNCING: D52, D53, D54

PAGE TITLE

NAND: Aliases

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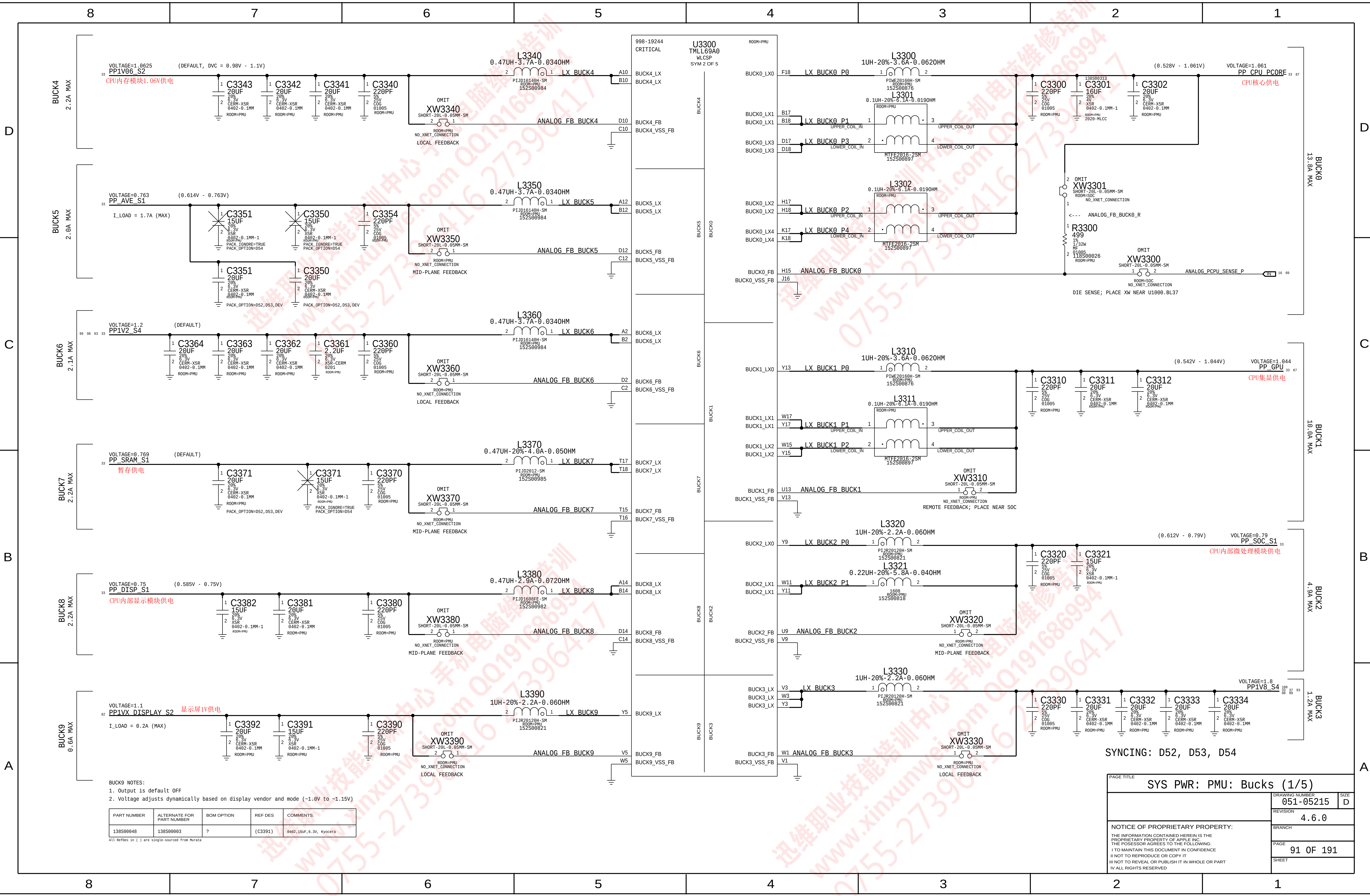
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PAGE TITLE				
SYS PWR: PMU: Bucks (1/5)				
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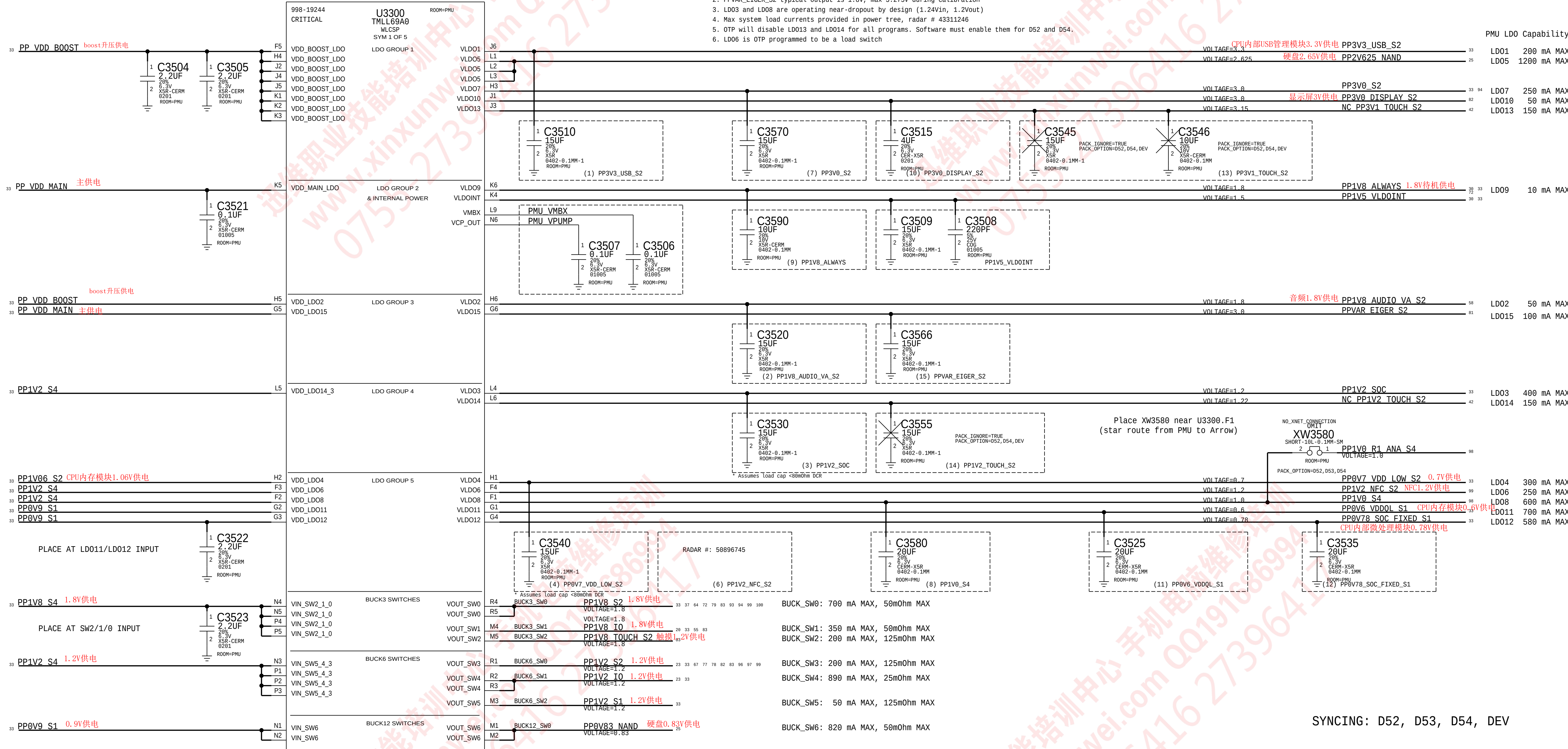
SUBSYSTEM SPECIFIC BOM TABLES

4uF 0201 Capacitors (single-source Murata)				
PART NUMBER	ALTERNATE FOR PART NUMBER	BOM OPTION	REF DES	COMMENTS:
138S00116	138S00071	?	(C3515)	CAP, XSR, 4uF, 0201, 0.559H, TAIYO
138S00117	138S00071	?	(C3515)	CAP, XSR, 4uF, 0201, 0.559H, KYOCERA
15uF 0402 Capacitors (single-source Murata)				
PART NUMBER	ALTERNATE FOR PART NUMBER	BOM OPTION	REF DES	COMMENTS:
138S00048	138S00063	?	[SEE BELOW]	0402, 15uF, 6.3V, Kyocera
All Refdes in () are single-sourced from Murata				
(C3509, C3510, C3520, C3530, C3540, C3545, C3555, C3566, C3570)				

PMU : LD0s

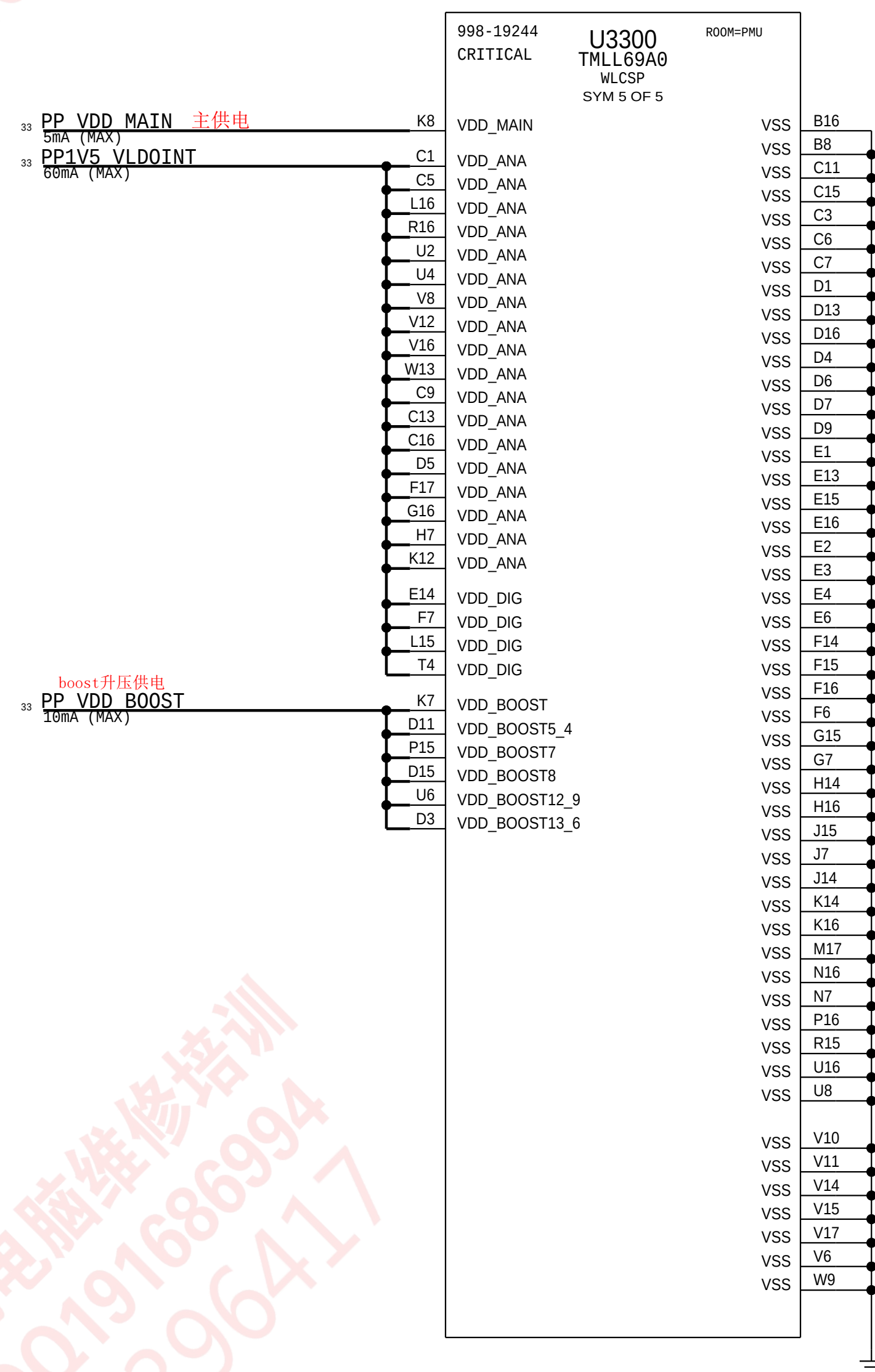
NOTES:

1. PP1V2_S0C is low noise SOC rail, PP1V2_I0 is its noisy cousin
2. PPVAR_EIGER_S2 typical output is 1.8V, max 3.275V during calibration
3. LD03 and LD08 are operating near-dropout by design (1.24Vin, 1.2Vout)
4. Max system load currents provided in power tree, radar # 43311246
5. OTP will disable LD013 and LD014 for all programs. Software must enable them for D52 and D54.
6. LD06 is OTP programmed to be a load switch



SYNCING: D52, D53, D54, DEV

PAGE TITLE		
SYS PWR: PMU: LD0s (3/5)		
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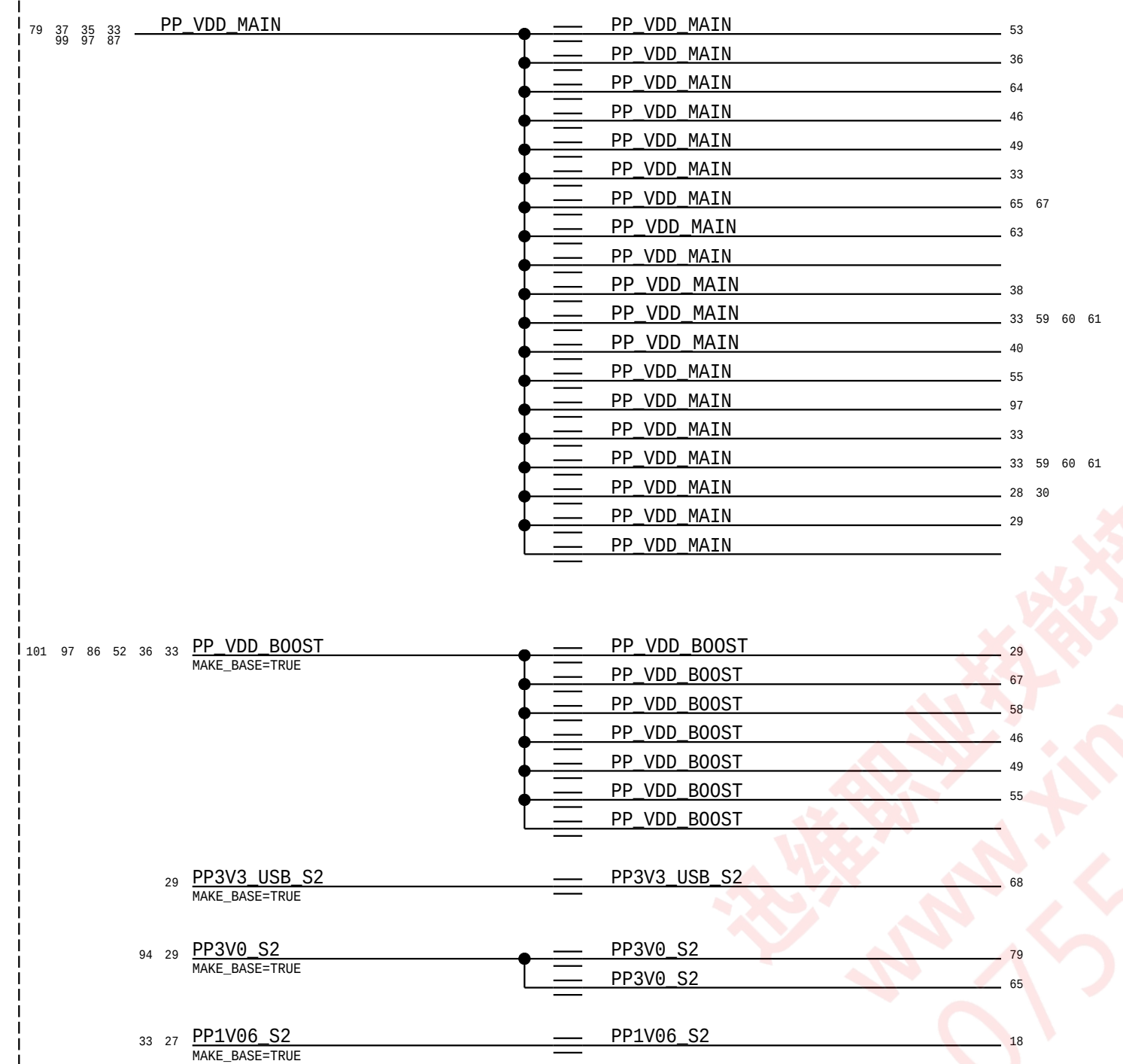


PMU GPIOs

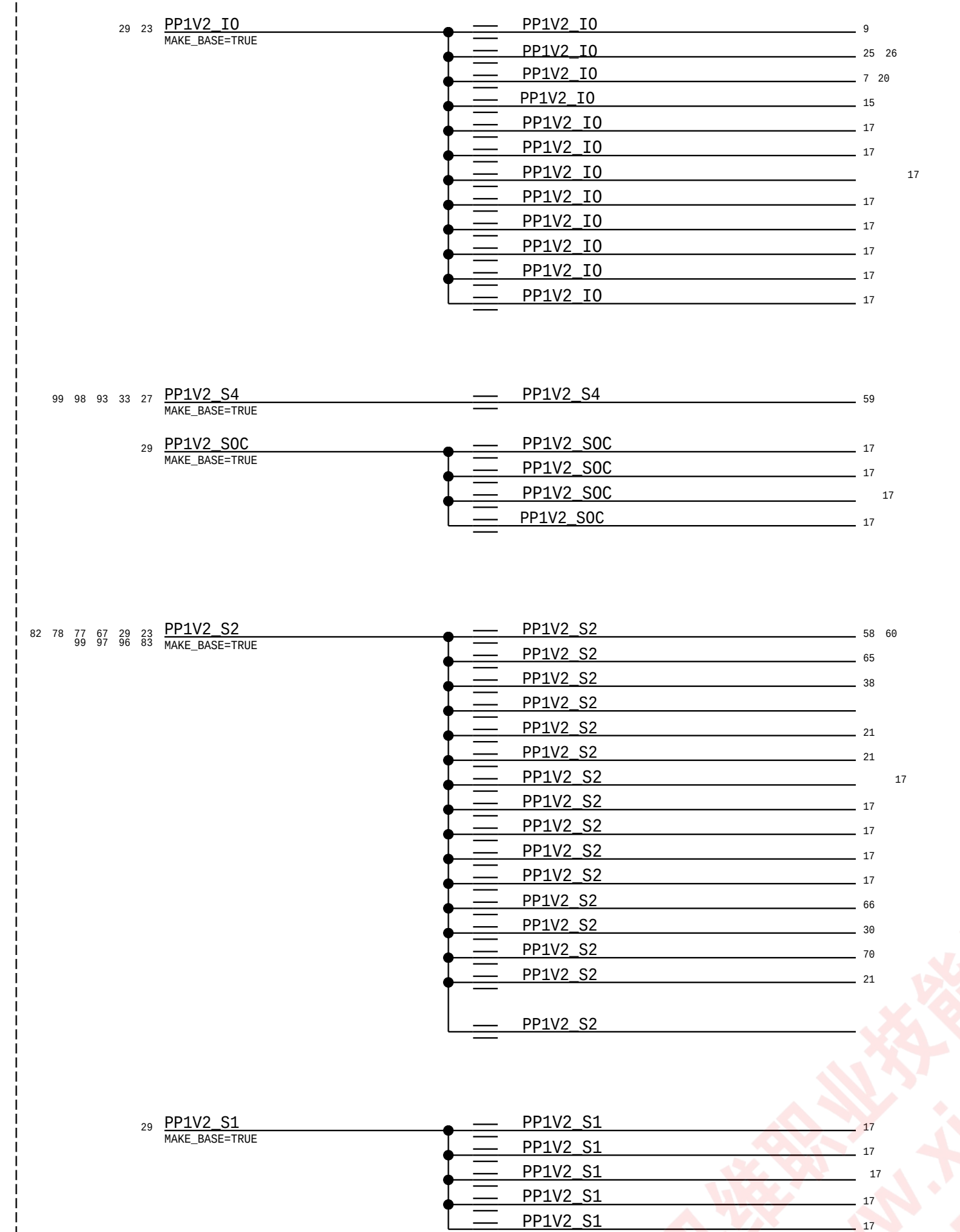
PMU_GPIO1	30	NC_DEV_PMU_GPIO_1	==	NC_DEV_PMU_GPIO_1	33
PMU_GPIO2	30	GPIO_S4_PMU_TO_SPKRAMP_BOT_RESET_L MAKE_BASE=TRUE	==	GPIO_S4_PMU_TO_SPKRAMP_BOT_RESET_L	59 89
PMU_GPIO3	30	SWD_NUB_B1_PMU_SWDIO	==	SWD_NUB_B1_PMU_SWDIO	33
PMU_GPIO4	30	GPIO_S4_PMU_TO_R1_RESET_L MAKE_BASE=TRUE	==	GPIO_S4_PMU_TO_R1_RESET_L	98
PMU_GPIO5	30	GPIO_S4_PMU_TO_WLAN_REG_ON MAKE_BASE=TRUE	==	GPIO_S4_PMU_TO_WLAN_REG_ON	95
PMU_GPIO6	30	CLK_GPIO_S4_PMU_TO_WLAN_R1_32K OPLUS_WALVE-SINGLE_COMP_NET MAKE_BASE=TRUE	==	CLK_GPIO_S4_PMU_TO_WLAN_R1_32K	30 89 93 98
PMU_GPIO7	30	NC_DEV_PMU_GPIO_7	==	NC_DEV_PMU_GPIO_7	33
PMU_GPIO8	30	GPIO_PMU_FROM_WLAN_HOST_WAKE MAKE_BASE=TRUE	==	GPIO_PMU_FROM_WLAN_HOST_WAKE	95
PMU_GPIO9	30	GPIO_PMU_FROM_BB_PCIE_HOST_WAKE_L MAKE_BASE=TRUE	==	GPIO_PMU_FROM_BB_PCIE_HOST_WAKE_L	95
PMU_GPIO10	30	GPIO_PMU_TO_CCG2B_RESET_L MAKE_BASE=TRUE	==	GPIO_PMU_TO_CCG2B_RESET_L	66 89
PMU_GPIO11	30	GPIO_PMU_FROM_CHARGER_VBUS_DET_INT_L_1V8 MAKE_BASE=TRUE	==	GPIO_PMU_FROM_CHARGER_VBUS_DET_INT_L_1V8	30 33 34 89
PMU_GPIO12	30	GPIO_PMU_FROM_CODEC_WAKE_L MAKE_BASE=TRUE	==	GPIO_PMU_FROM_CODEC_WAKE_L	58 89
PMU_GPIO13	30	GPIO_PMU_TO_AMUX_PMU_SUPER_TP OPLUS_WALVE-SINGLE_COMP_NET MAKE_BASE=TRUE	==	GPIO_PMU_TO_AMUX_PMU_SUPER_TP	30
PMU_GPIO14	30	GPIO_PMU_NFC_TO_ARCAMP_RESET_L MAKE_BASE=TRUE	==	GPIO_PMU_NFC_TO_ARCAMP_RESET_L	61
PMU_GPIO15	30	GPIO_PMU_TO_NAND_LOW_BATT_BOOT_L MAKE_BASE=TRUE	==	GPIO_PMU_TO_NAND_LOW_BATT_BOOT_L	25 89
PMU_GPIO16	30	GPIO_PMU_NFC_TO_ARCAMP_TRIG MAKE_BASE=TRUE	==	GPIO_PMU_NFC_TO_ARCAMP_TRIG	61
PMU_GPIO17	30	I2C2_AOP_SCL MAKE_BASE=TRUE	==	I2C2_AOP_SCL	14 21
PMU_GPIO18	30	CLK_GPIO_SEQ_PMU_TO_DISPLAY_32K_1V8 MAKE_BASE=TRUE	==	CLK_GPIO_SEQ_PMU_TO_DISPLAY_32K_1V8	82
PMU_GPIO19	30	GPIO_SEQ_PMU_TO_DISPLAY_RESET_L_1V8 MAKE_BASE=TRUE	==	GPIO_SEQ_PMU_TO_DISPLAY_RESET_L_1V8	64 82
PMU_GPIO20	30	GPIO_SEQ_PMU_TO_DOTARA_EN_EXT_1V8 MAKE_BASE=TRUE	==	GPIO_SEQ_PMU_TO_DOTARA_EN_EXT_1V8	30 89
PMU_GPIO21	30	GPIO_SEQ_PMU_TO_BBPMU_RESET_L MAKE_BASE=TRUE	==	GPIO_SEQ_PMU_TO_BBPMU_RESET_L	95
PMU_GPIO22	30	GPIO_SEQ_PMU_TO_NFC_EN MAKE_BASE=TRUE	==	GPIO_SEQ_PMU_TO_NFC_EN	99
PMU_GPIO23	30	GPIO_SEQ_PMU_TO_BOOST_EN_1V8 MAKE_BASE=TRUE	==	GPIO_SEQ_PMU_TO_BOOST_EN_1V8	36 89
PMU_GPIO24	30	GPIO_SEQ_PMU_TO_DISPLAY_PANICB_L_1V8 MAKE_BASE=TRUE	==	GPIO_SEQ_PMU_TO_DISPLAY_PANICB_L_1V8	64 89
PMU_GPIO25	30	GPIO_SEQ_PMU_TO_BOOST_BYPASS_L_1V8 MAKE_BASE=TRUE	==	GPIO_SEQ_PMU_TO_BOOST_BYPASS_L_1V8	30 36

SYNCING: D52, D53, D54, DEV

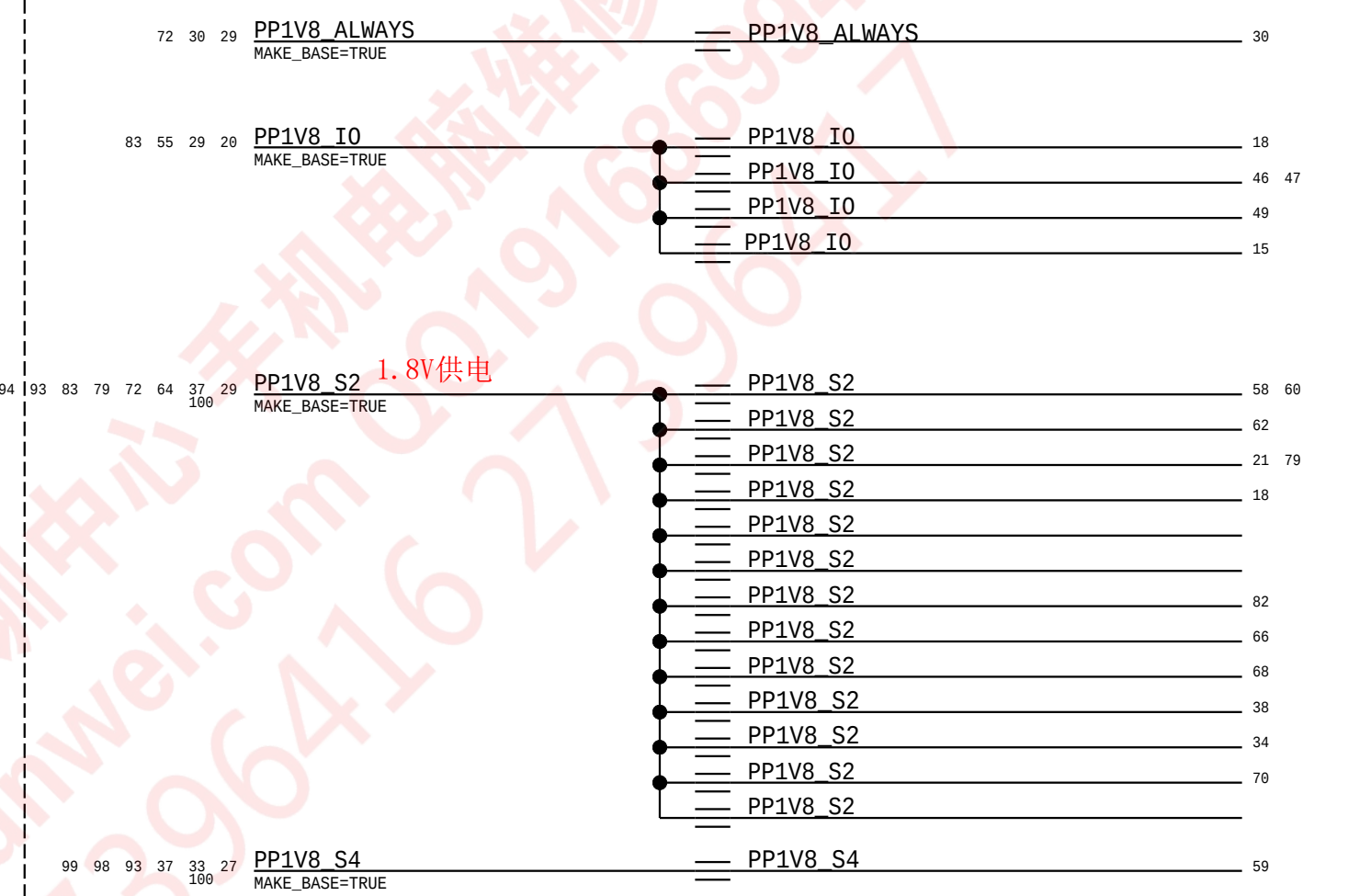
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SYS PWR: PMU: Aliases: GPIO		
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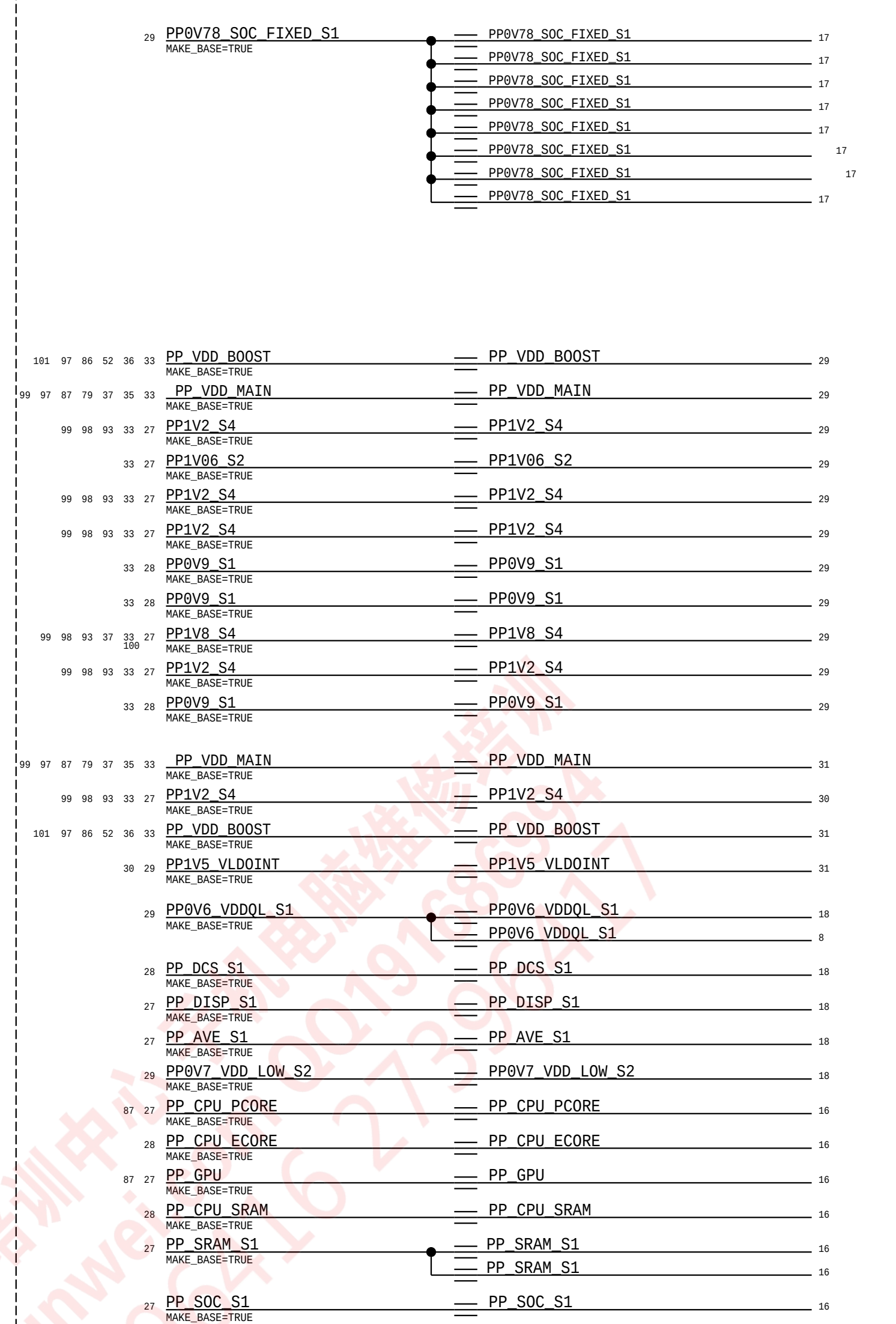
PP1V2 Domains



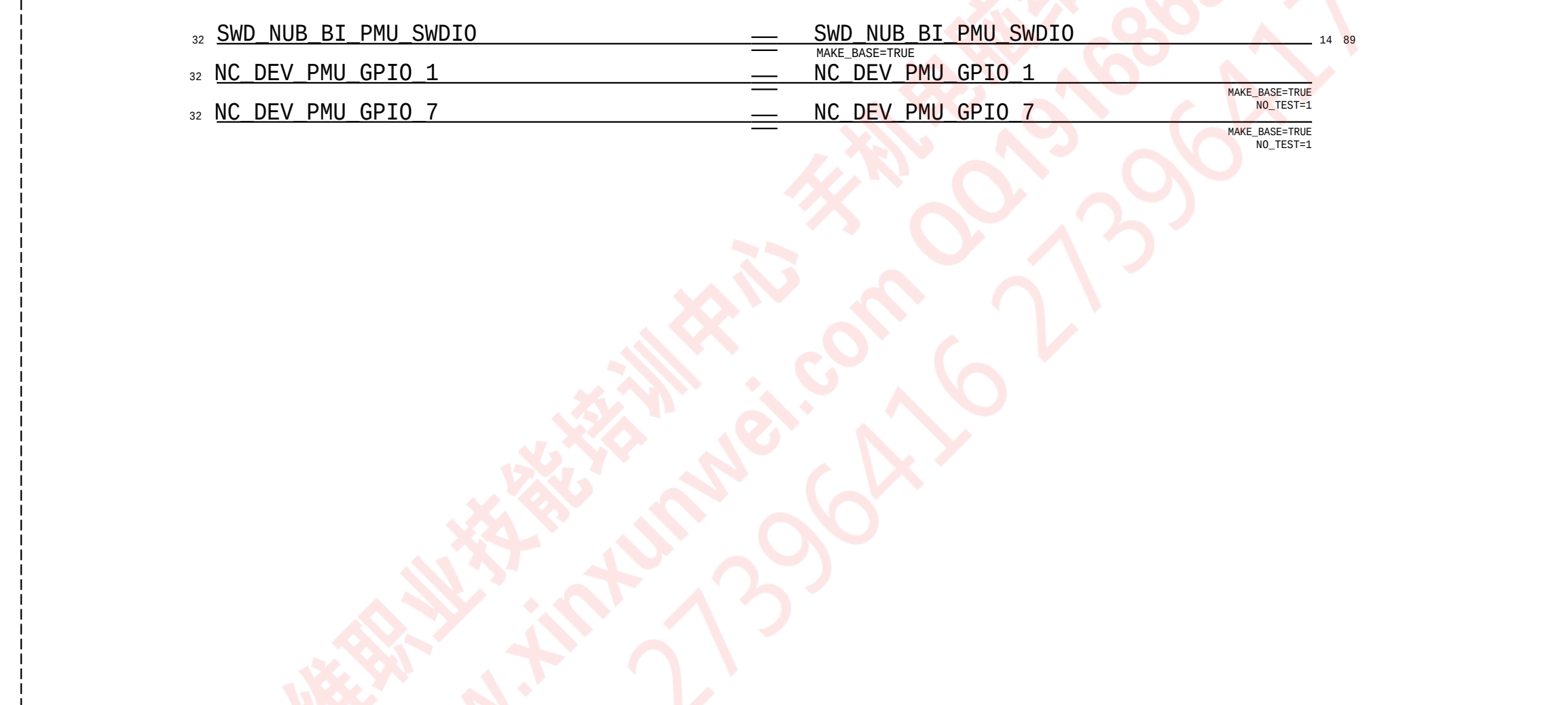
PP1V8 Domains



SOC + PMU Power Assignments



DEV Board Compatability



PMU Dedicated IO Assignments

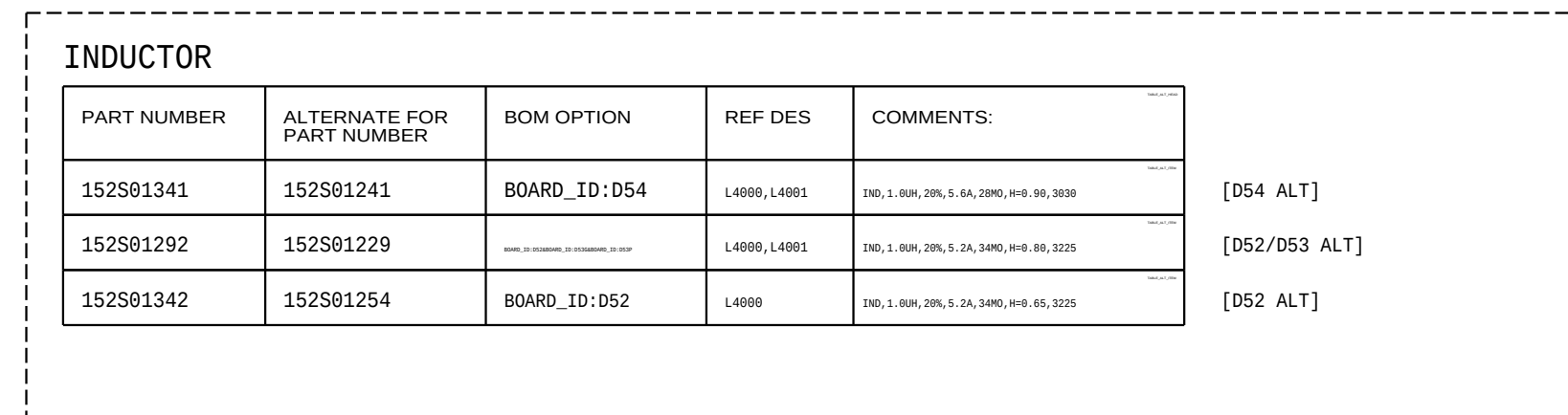


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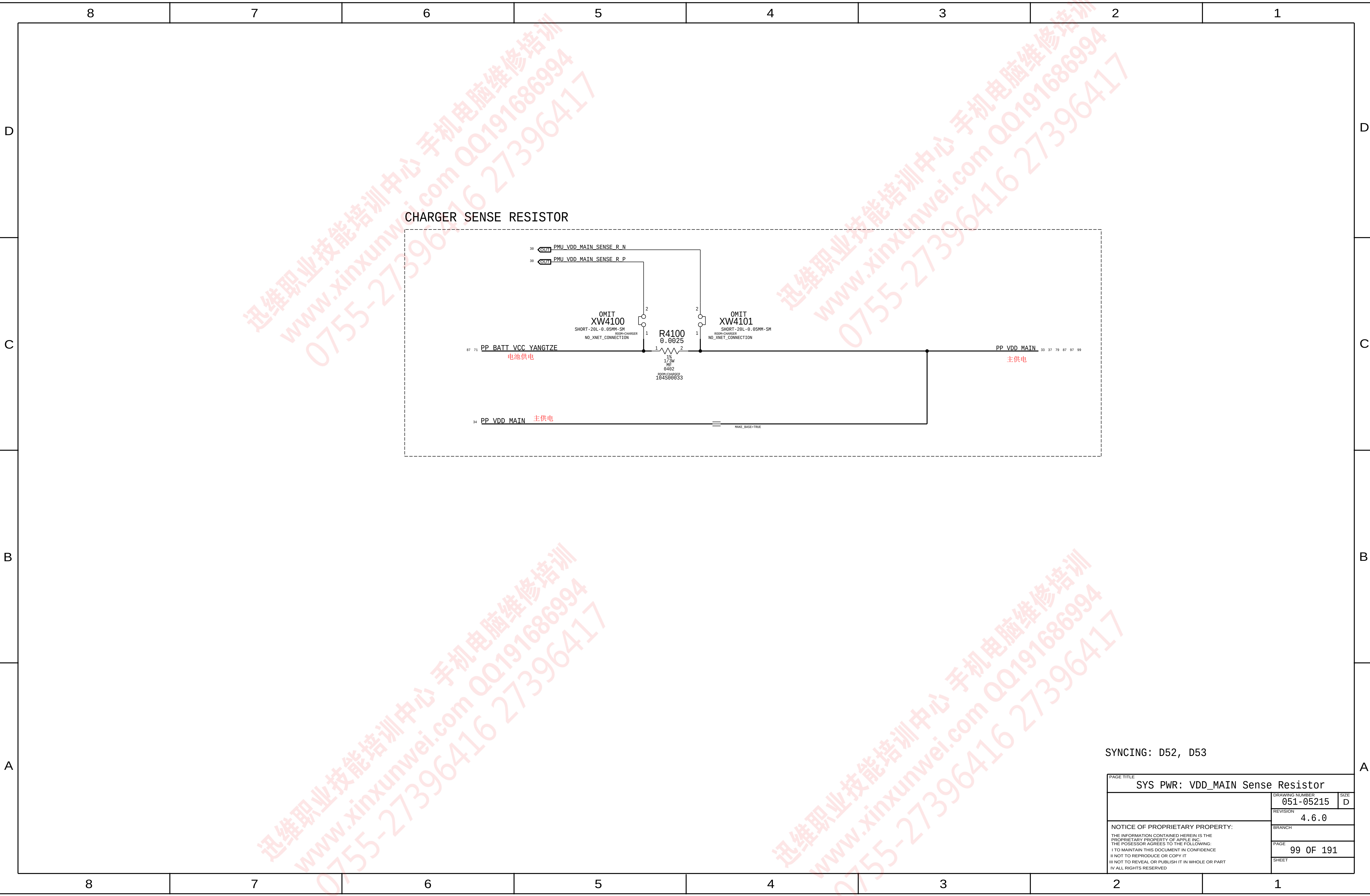
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SYS PWR: PMU: Aliases: Misc.		
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INDUCTOR					
PART NUMBER	ALTERNATE FOR PART NUMBER	BOM OPTION	REF DES	COMMENTS:	
152S01341	152S01241	BOARD_ID:D54	L4000, L4001	100, 1.00K, 20K, 5.6A, 20M, 40K, 90, 100	[D54 ALT]
152S01292	152S01229	same as 152S01241, 20 100K	L4000, L4001	100, 1.00K, 20K, 5.6A, 20M, 40K, 90, 100, 2225	[D52/D53 ALT]
152S01342	152S01254	BOARD_ID:D52	L4000	100, 1.00K, 20K, 5.6A, 20M, 40K, 90, 100, 2225	[D52 ALT]

INDUCTOR					
PART NUMBER	ALTERNATE FOR PART NUMBER	BOM OPTION	REF DES	COMMENTS:	
152S01341	152S01241	BOARD_ID:D54	L4000, L4001	100, 1.00K, 20K, 5.6A, 20M, 40K, 90, 100	[D54 ALT]
152S01292	152S01229	same as 152S01241, 20 100K	L4000, L4001	100, 1.00K, 20K, 5.6A, 20M, 40K, 90, 100, 2225	[D52/D53 ALT]
152S01342	152S01254	BOARD_ID:D52	L4000	100, 1.00K, 20K, 5.6A, 20M, 40K, 90, 100, 2225	[D52 ALT]



PAGE TITLE		SYS PWR: Charger	
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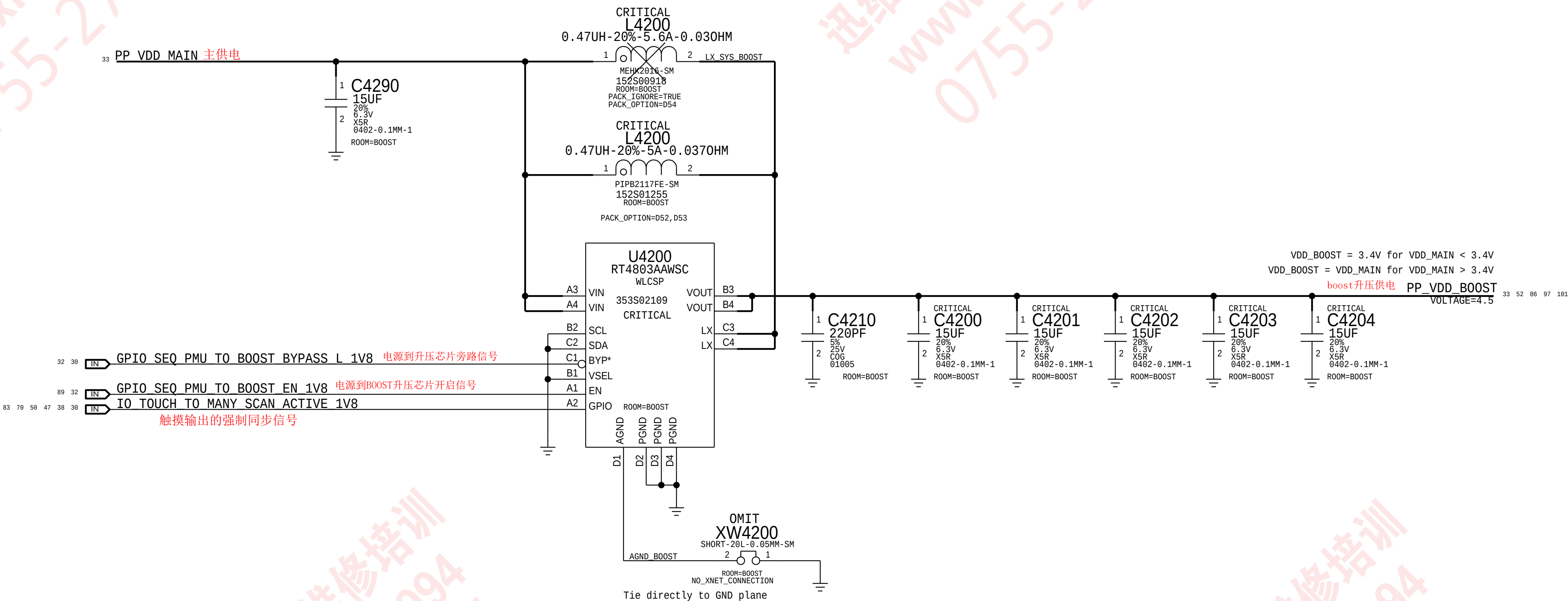
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SYS PWR: VDD_MAIN Sense Resistor		
	DRAWING NUMBER	051-05215
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SUBSYSTEM SPECIFIC BOM TABLES

Boost

PART NUMBER	ALTERNATE FOR PART NUMBER	BOM OPTION	REF DES	COMMENTS:
353S01124	353S02109	?	U4200	IC, 3W62200E, BOOST, 3.0V, 4.5A, CSP16

BOOST



SYNCING: D52, D53, D54

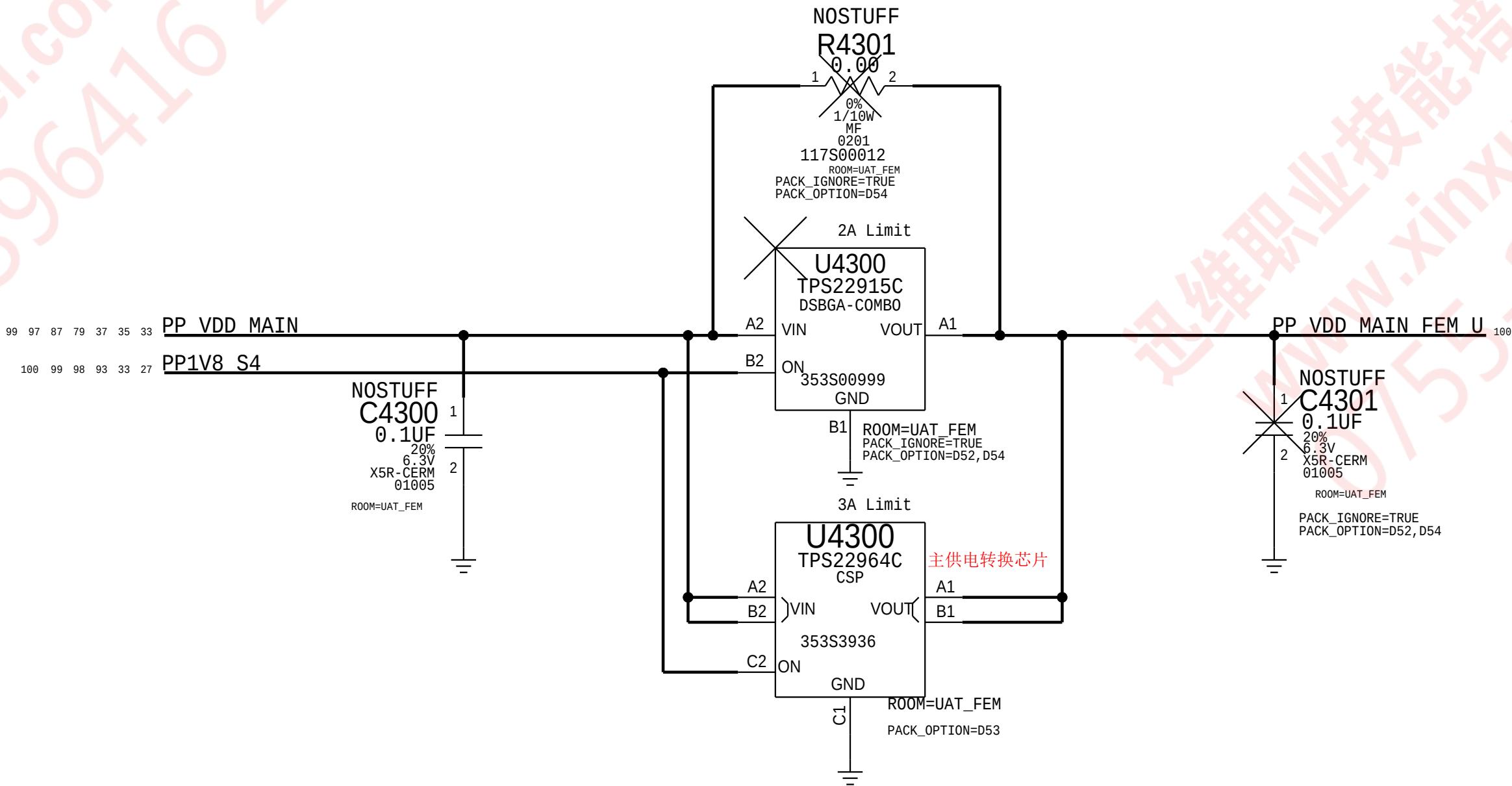
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WiFi FEM VDD_MAIN Load Switches

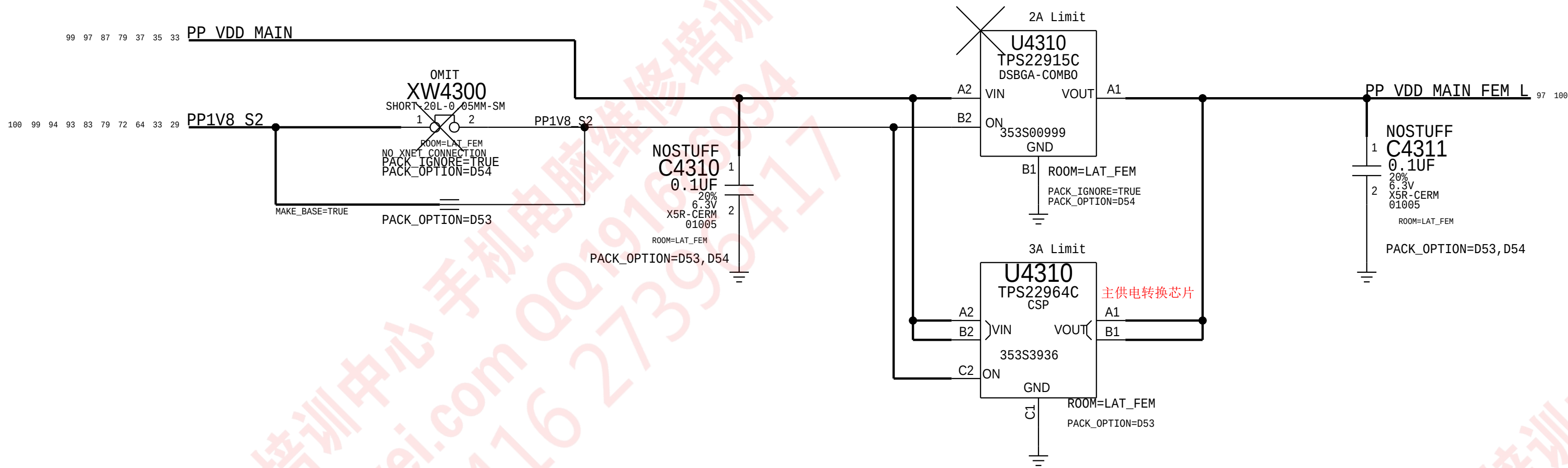
LOAD SWITCH

PART NUMBER	ALTERNATE FOR PART NUMBER	BOM OPTION	REF DES	COMMENTS:	CRITICAL PART#	COMMENT
353S01007	353S00999	?	ALL	IC, LOAD SWITCH, 1.5A, ON-SER	353S00999	IC, LOAD_SWITCH, 2A, T1

UAT FEM Switch

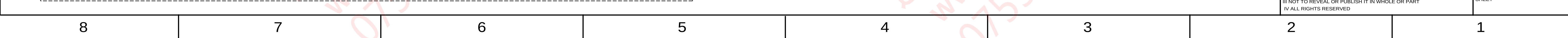


LAT FEM Switch



PAGE TITLE		
SYS PWR: WiFi FEM Load Switches		
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REVISION		4.6.0
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Dotara FET				
PART NUMBER	ALTERNATE FOR PART NUMBER	BOM OPTION	REF DES	COMMENTS:
376S00362	376S00295	?	Q4400	NFET, 120V, onSem1



	8	7	6	5	4	3	2	1
D								
C								
B								
A								
	8	7	6	5	4	3	2	1

DOTARA ALIASES

SYNCING: D52, D53, D54, DEV

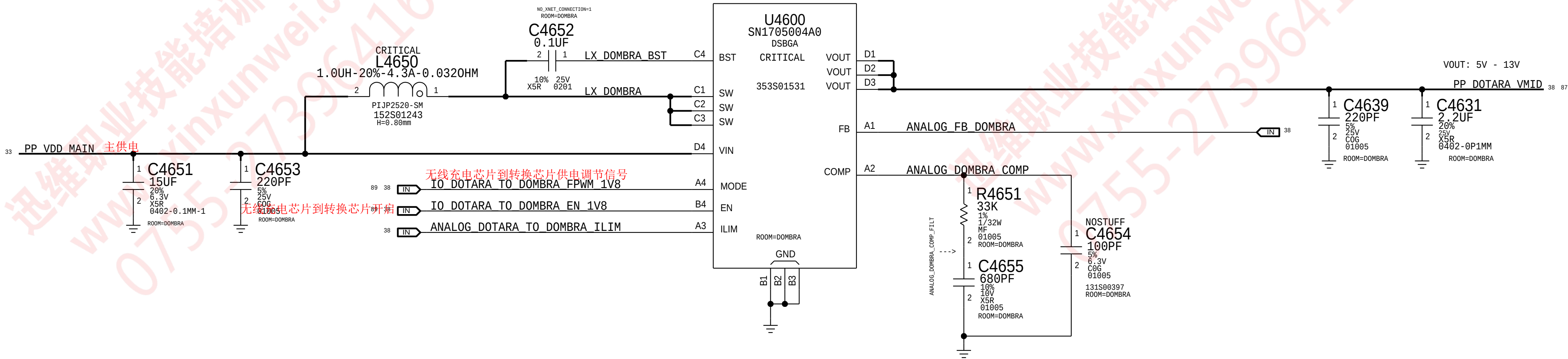
PAGE TITLE SYS PWR: Wireless Charger: Level Shifters		
	DRAWING NUMBER 051-05215	SIZE D
	REVISION 4.6.0	
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SUBSYSTEM SPECIFIC BOM TABLES

INDUCTOR

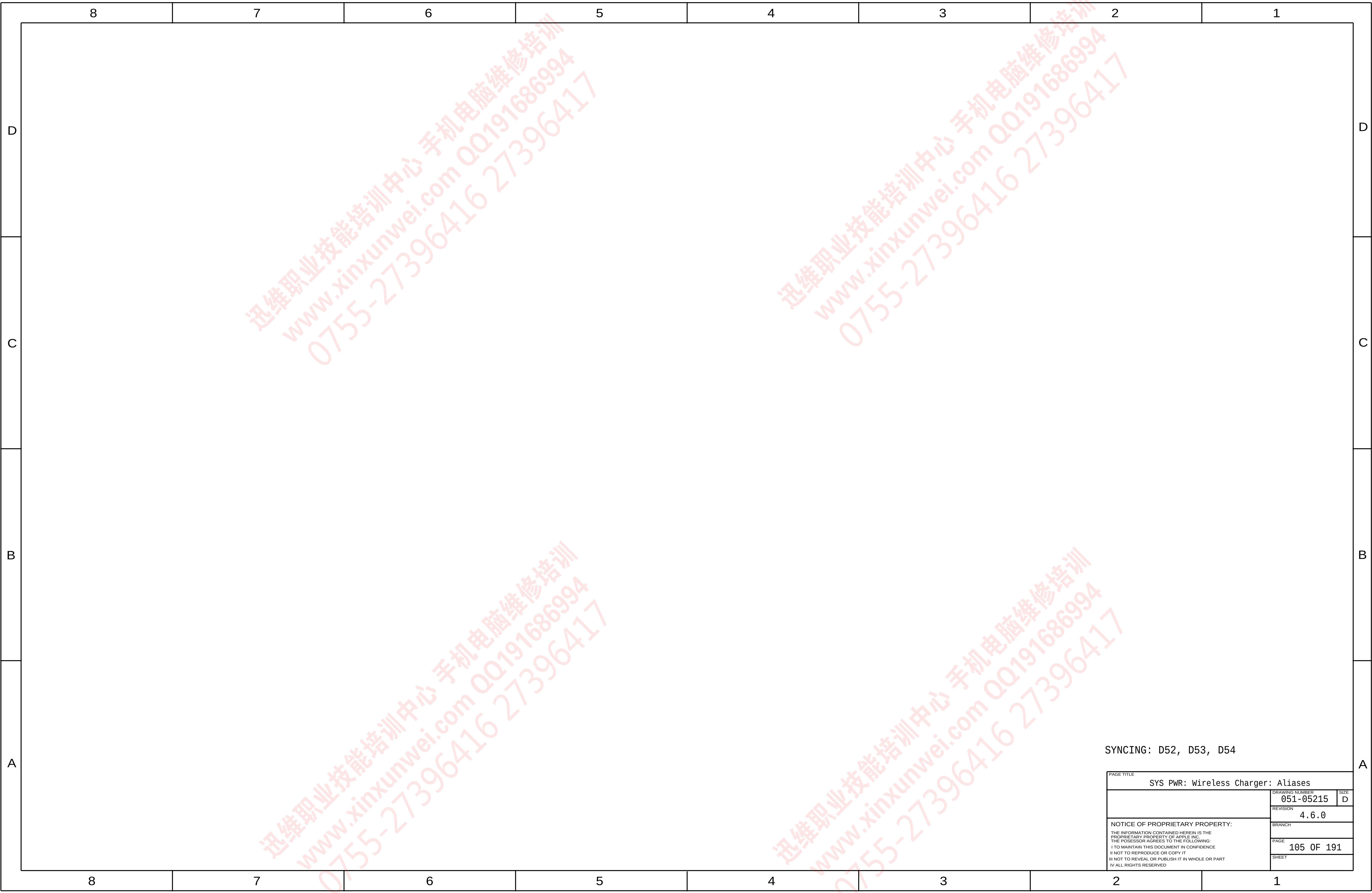
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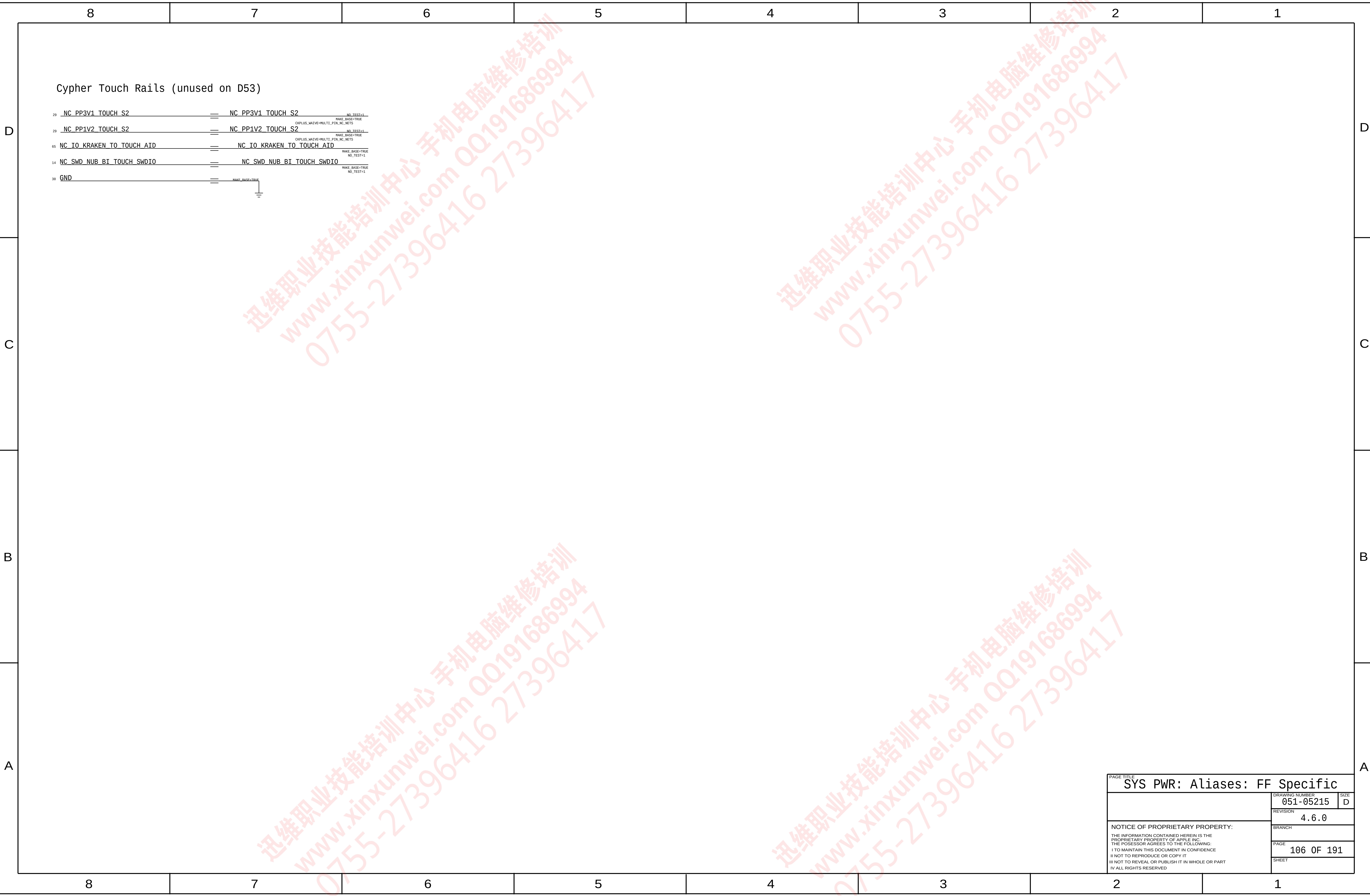
Dombra



SYNCING: D52, D53, D54

PAGE TITLE		
SYS PWR: Wireless Charger: Boost		
	DRAWING NUMBER	051-05215
	REVISION	4.6.0
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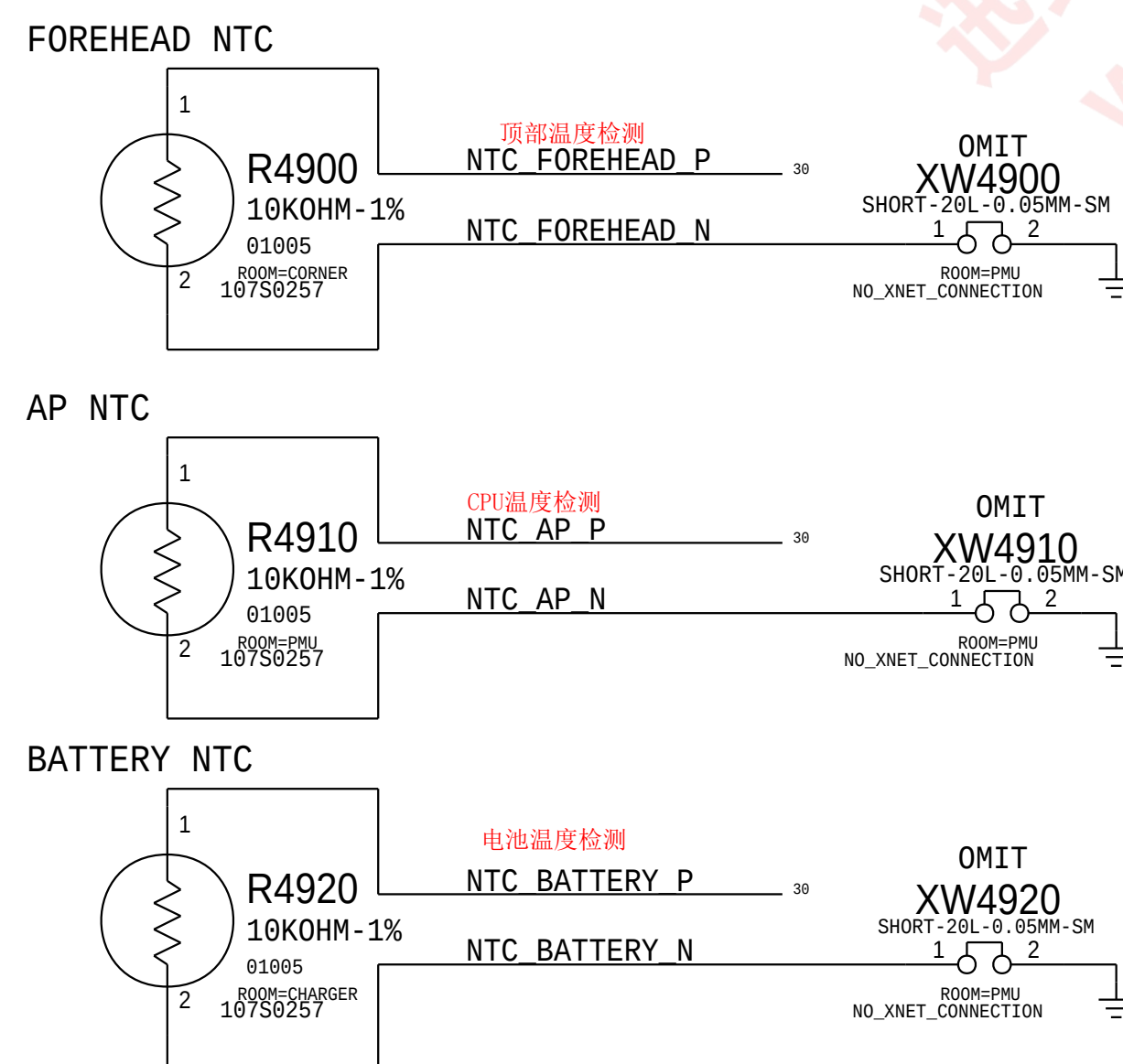
2

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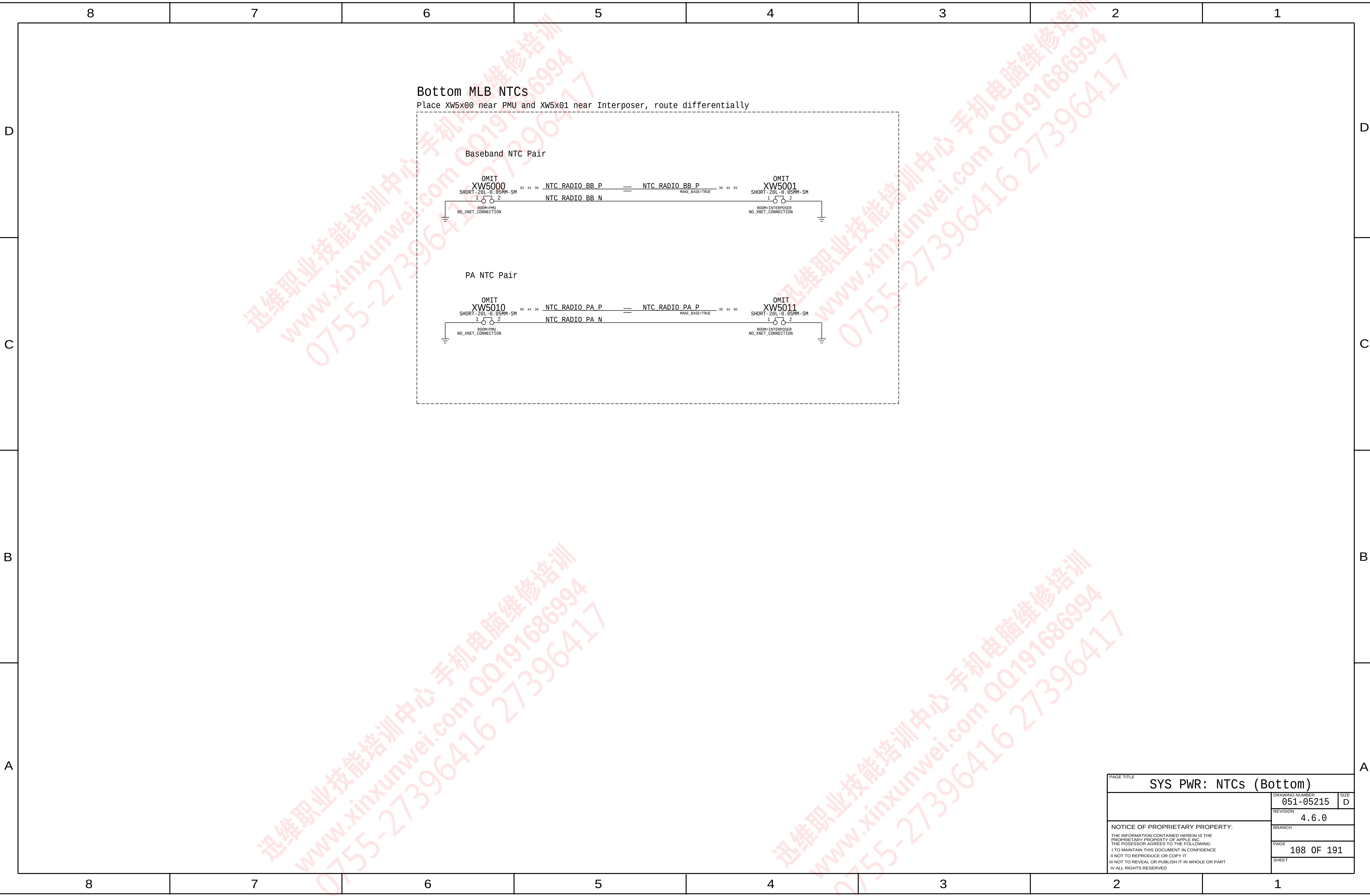
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SYS PWR: Aliases: FF Specific		
	DRAWING NUMBER	SIZE
	051-05215	D
	REVISION	
	4.6.0	
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Top MLB NTCs

Place NTC Xws near PMU, route P/N differentially



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SYS PWR: NTCs (Top)		
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PAGE TITLE			SYS PWR: NTCs (Bottom)		
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1

D53G/P Variant Decoder

	D53G	D53P
SUPERFLEX-CAM2	None	Swide
SINGLEFLEX-CAM	Swide	Tele
BOMOPTION PRO	NOSTUFF	STUFF

PAGE TITLE

CAMERA: P/G Variant

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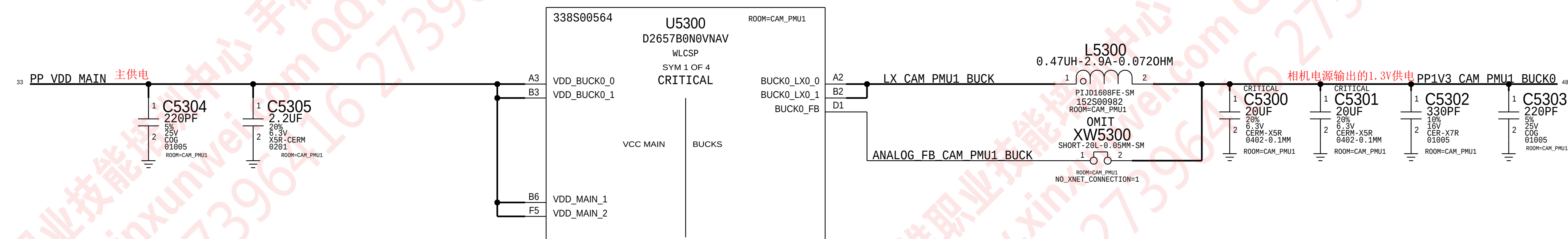
SUBSYSTEM SPECIFIC BOM TABLES

4uF 0201 Capacitors (single-source Murata)

PART NUMBER	ALTERNATE FOR PART NUMBER	BOM OPTION	REF DES	COMMENTS:
138S00116	138S00071	?	[SEE BELOW]	CAP_XSR_4UF_0201_0.55MM_TAIYO
138S00117	138S00071	?	[SEE BELOW]	CAP_XSR_4UF_0201_0.55MM_KYOCERA

(C5310, C5330, C5340, C5350, C5360, C5370, C5380, C5390)
(C5310, C5330, C5340, C5350, C5360, C5370, C5380, C5390)

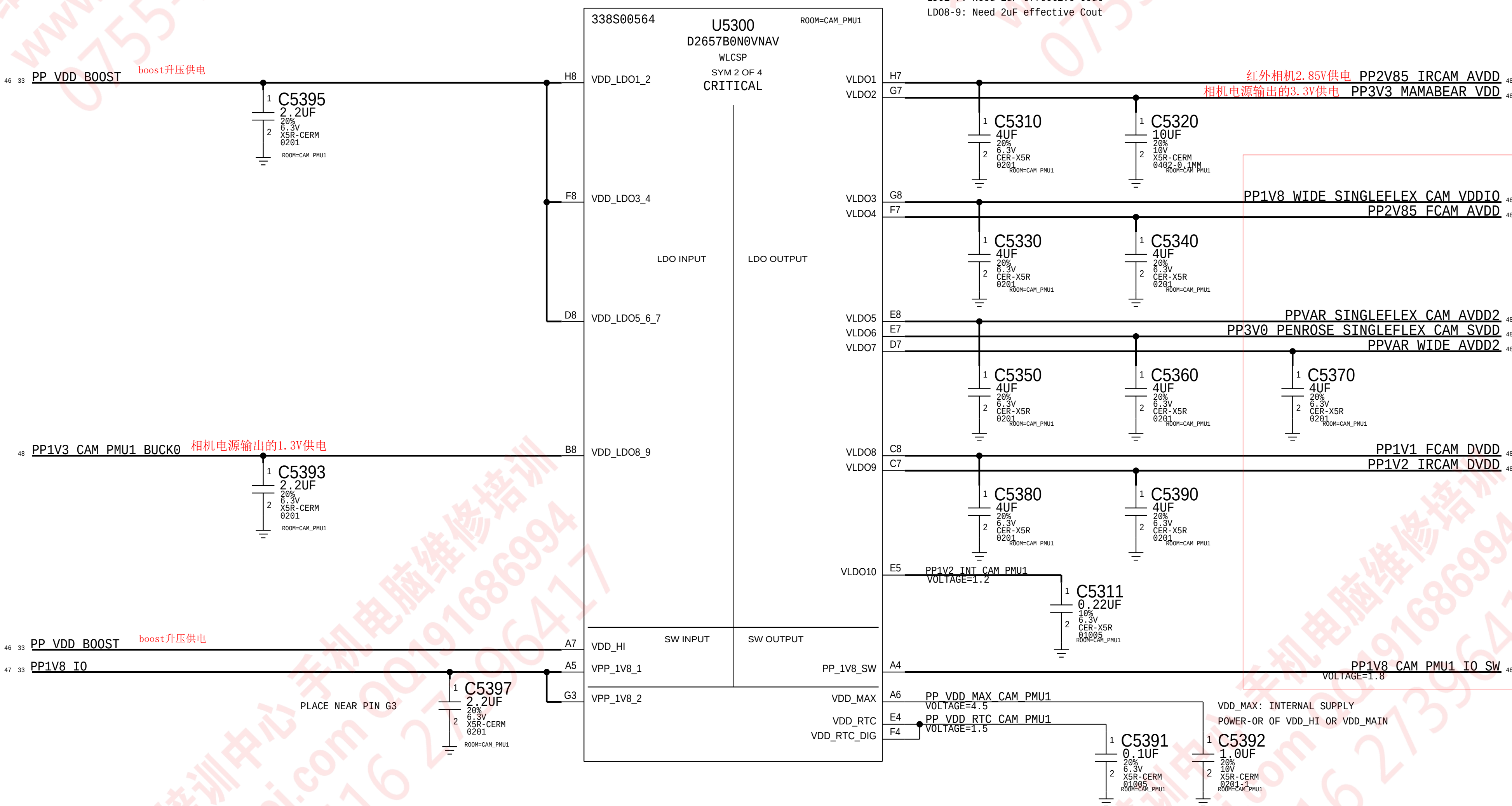
CAMERA PMU1



Adams LDO Design Guidelines

LD01-7: Need 1uF effective Cout

LD08-9: Need 2uF effective Cout

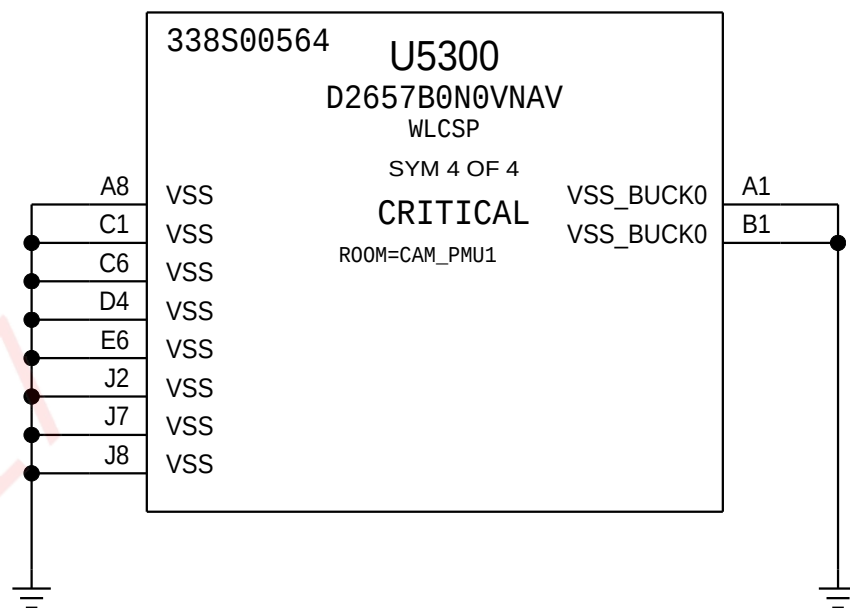
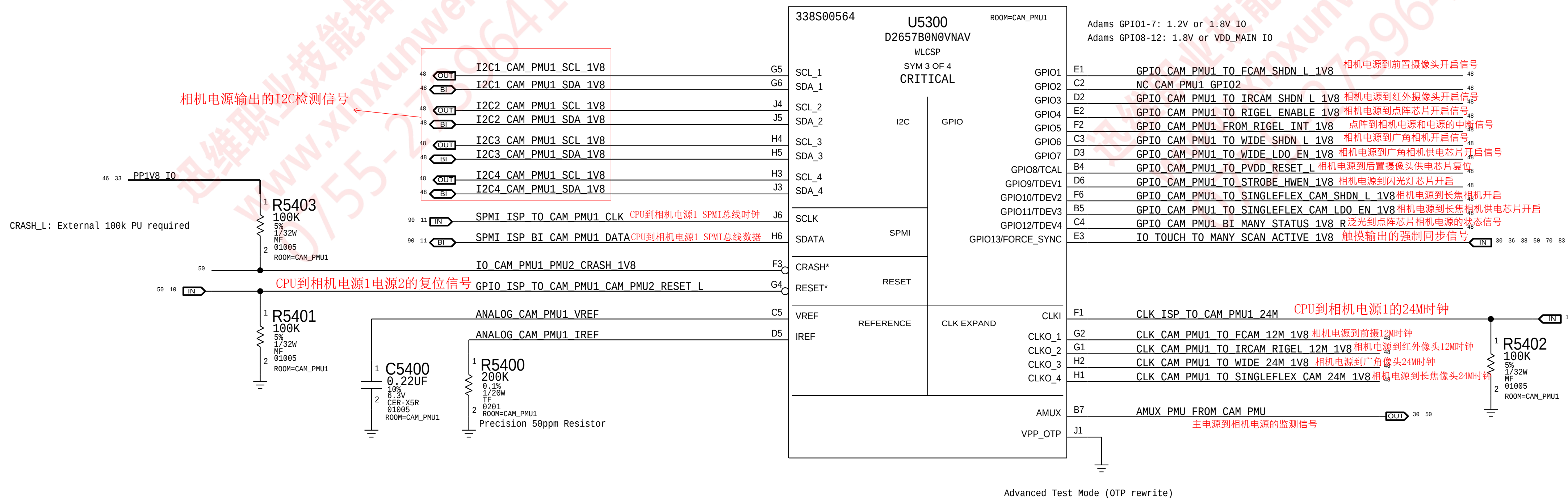


相机电源输出的供电

SYNCING: D52, D53, D54

PAGE TITLE	CAMERA: PMU 1: Power (1/2)		
	DRAWING NUMBER	051-05215	SIZE D
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CAMERA PMU1

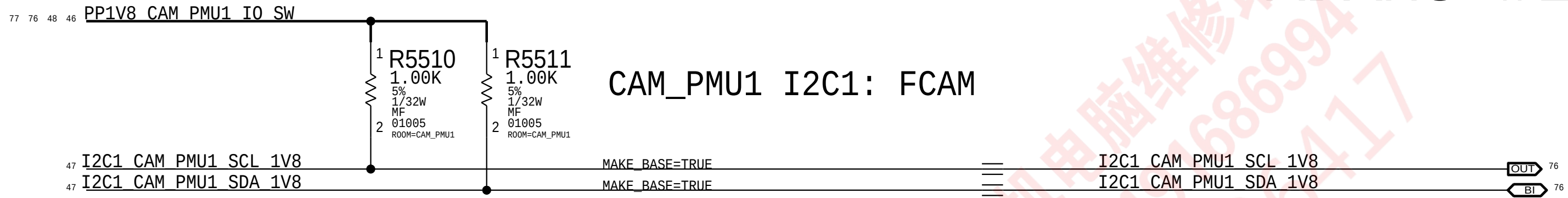


SYNCING: D52, D53, D54

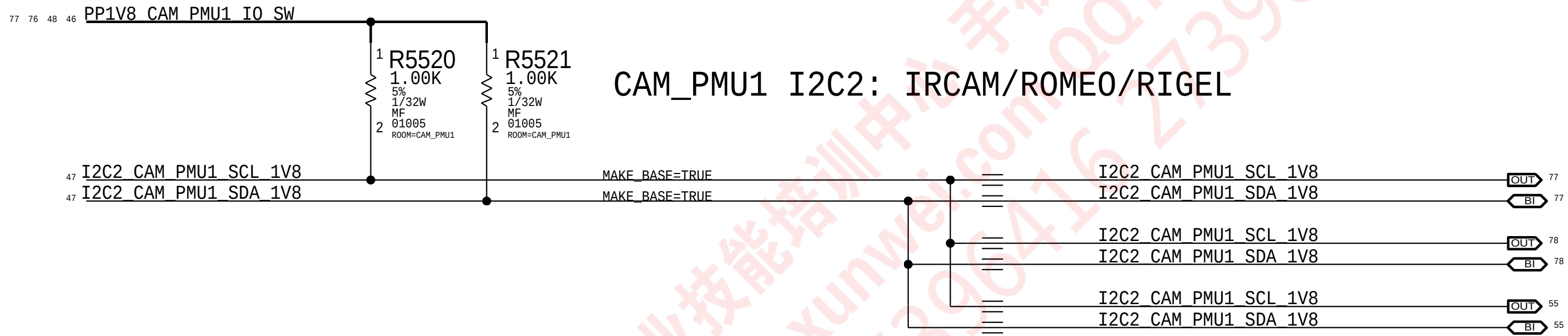
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CAMERA: PMU 1: I/O (2/2)		
DRAWING NUMBER	051-05215	SIZE
REVISION	4.6.0	D
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ADAMS #1 ALIASES

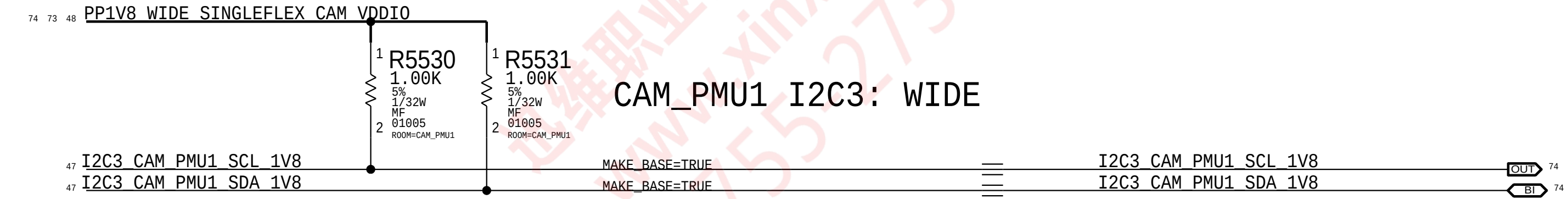
CAM_PMU1 I2C1: FCAM



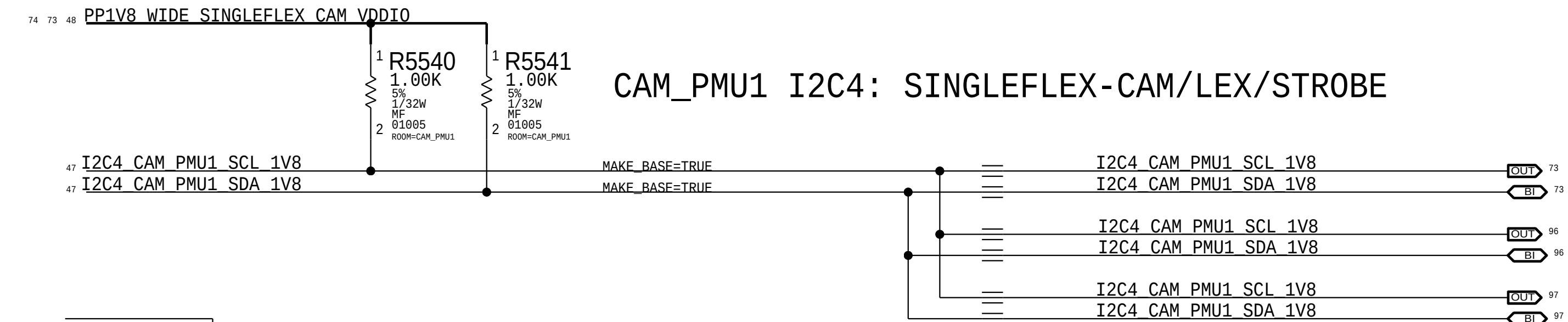
CAM_PMU1 I2C2: IRCAM/ROMEO/RIGEL



CAM_PMU1 I2C3: WIDE



CAM_PMU1 I2C4: SINGLEFLEX-CAM/LEX/STROBE



SIGNAL

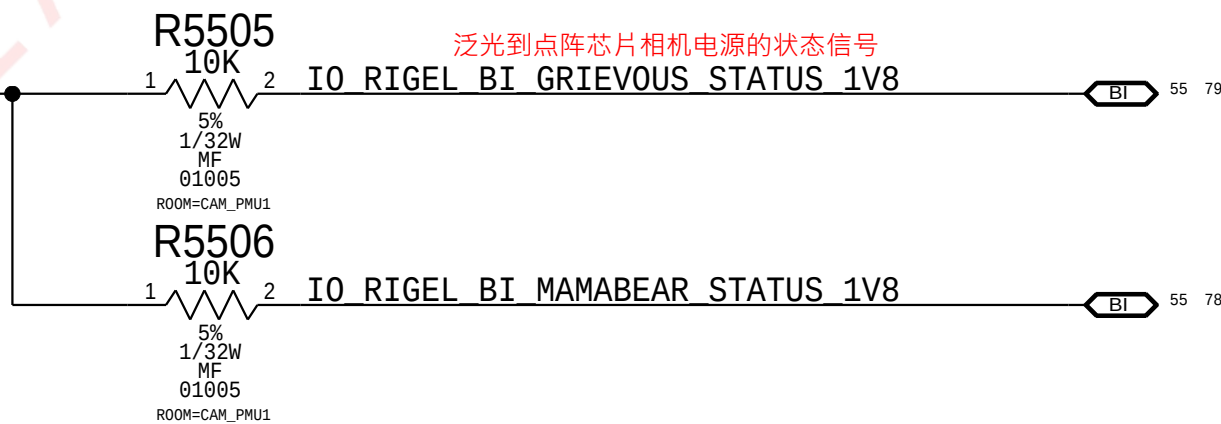
GPIO_1	GPIO CAM PMU1 TO FCAM SHDN L 1V8	GPIO CAM PMU1 TO FCAM SHDN L 1V8	MAKE_BASE=TRUE	OUT	76
GPIO_2	NC CAM PMU1 GPIO2	NC CAM PMU1 GPIO2	MAKE_BASE=TRUE	NO_TEST=1	
GPIO_3	GPIO CAM PMU1 TO IRCAM SHDN L 1V8	GPIO CAM PMU1 TO IRCAM SHDN L 1V8	MAKE_BASE=TRUE	OUT	77
GPIO_4	GPIO CAM PMU1 TO RIGEL ENABLE 1V8	GPIO CAM PMU1 TO RIGEL ENABLE 1V8	MAKE_BASE=TRUE	OUT	55 89
GPIO_5	GPIO CAM PMU1 FROM RIGEL INT 1V8	GPIO CAM PMU1 FROM RIGEL INT 1V8	MAKE_BASE=TRUE	IN	59 55 89
GPIO_6	GPIO CAM PMU1 TO WIDE SHDN L 1V8	GPIO CAM PMU1 TO WIDE SHDN L 1V8	MAKE_BASE=TRUE	OUT	74
GPIO_7	GPIO CAM PMU1 TO WIDE LDO EN 1V8	GPIO CAM PMU1 TO WIDE LDO EN 1V8	MAKE_BASE=TRUE	OUT	52 89
GPIO_8	GPIO CAM PMU1 TO PVDD RESET L	GPIO CAM PMU1 TO PVDD RESET L	MAKE_BASE=TRUE	OUT	96
GPIO_9	GPIO CAM PMU1 TO STROBE HWEN 1V8	GPIO CAM PMU1 TO STROBE HWEN 1V8	MAKE_BASE=TRUE	OUT	89 93
GPIO_10	GPIO CAM PMU1 TO SINGLEFLEX CAM SHDN L 1V8	GPIO CAM PMU1 TO SINGLEFLEX CAM SHDN L 1V8	MAKE_BASE=TRUE	OUT	73
GPIO_11	GPIO CAM PMU1 TO SINGLEFLEX CAM LDO EN 1V8	GPIO CAM PMU1 TO SINGLEFLEX CAM LDO EN 1V8	MAKE_BASE=TRUE	OUT	52 89
GPIO_12	GPIO CAM PMU1 BI MANY STATUS 1V8 R	GPIO CAM PMU1 BI MANY STATUS 1V8 R	MAKE_BASE=TRUE	OUT	

Camera PMU1 GPIO13 intentionally not aliased

CLK0_1	CLK CAM PMU1 TO FCAM 12M 1V8	CLK CAM PMU1 TO FCAM 12M 1V8	MAKE_BASE=TRUE	OUT	76
CLK0_2	CLK CAM PMU1 TO IRCAM RIGEL 12M 1V8	CLK CAM PMU1 TO IRCAM RIGEL 12M 1V8	MAKE_BASE=TRUE	OUT	55 77
CLK0_3	CLK CAM PMU1 TO WIDE 24M 1V8	CLK CAM PMU1 TO WIDE 24M 1V8	MAKE_BASE=TRUE	OUT	74
CLK0_4	CLK CAM PMU1 TO SINGLEFLEX CAM 24M 1V8	CLK CAM PMU1 TO SINGLEFLEX CAM 24M 1V8	MAKE_BASE=TRUE	OUT	73

POWER

VLD01	PP2V85 IRCAM AVDD	PP2V85 IRCAM AVDD	VOLTAGE=2.9	77
VLD02	PP3V3 MAMABEAR VDD	PP3V3 MAMABEAR VDD	VOLTAGE=3.3	78
VLD03	PP1V8 WIDE SINGLEFLEX CAM VDDIO	PP1V8 WIDE SINGLEFLEX CAM VDDIO	VOLTAGE=1.8	48 73 74
VLD04	PP2V85 FCAM AVDD	PP2V85 FCAM AVDD	VOLTAGE=2.9	76
VLD05	PPVAR SINGLEFLEX CAM AVDD2	PPVAR SINGLEFLEX CAM AVDD2	VOLTAGE=3.0	73
VLD06	PP3V0 PENROSE SINGLEFLEX CAM SVDD	PP3V0 PENROSE SINGLEFLEX CAM SVDD	VOLTAGE=3.0	52 73 89
VLD07	PPVAR WIDE AVDD2	PPVAR WIDE AVDD2	VOLTAGE=3.0	74
VLD08	PP1V1 FCAM DVDD	PP1V1 FCAM DVDD	VOLTAGE=1.1	76
VLD09	PP1V2 IRCAM DVDD	PP1V2 IRCAM DVDD	VOLTAGE=1.2	77
VLD010	Camera PMU1 LD010 unused			
BUCK0	PP1V3 CAM PMU1 BUCK0	PP1V3 CAM PMU1 BUCK0	VOLTAGE=1.4	46
		PP1V3 CAM PMU1 BUCK0		49



MASTER	CAM PMU1	NUMBER	I2C1	DIAGS NUMBER	4	SPEED	1MHz
DEVICE	VOLTAGE	7-BIT ADDR	8-BIT ADDR	MAX SPEED	LOCATION		
FCAM	1.8V	0x10	0x20, 0x21	1MHz	FCAM MD Flex		

MASTER	CAM PMU1	NUMBER	I2C2	DIAGS NUMBER	5	SPEED	1MHz
DEVICE	VOLTAGE	7-BIT ADDR	8-BIT ADDR	MAX SPEED	LOCATION		
IRCAM	1.8V	0x18	0x30, 0x31	1MHz	Juliet Flex		
Romeo	1.8V	0x66	0xCC, 0xCD	1MHz	Romeo Flex		
Rigel	1.8V	0x55	0xAA, 0xAB	1MHz	Top MLB		

MASTER	CAM PMU1	NUMBER	I2C3	DIAGS NUMBER	6	SPEED	1MHz
DEVICE	VOLTAGE	7-BIT ADDR	8-BIT ADDR	MAX SPEED	LOCATION		
Wide	1.8V	0x10	0x20, 0x21	1MHz	RCAM MD Flex		
Lorentz	1.8V	0x74	0xE8, 0xE9	1MHz	RCAM MD Flex		

MASTER	CAM PMU1	NUMBER	I2C4	DIAGS NUMBER	7	SPEED	1MHz
DEVICE	VOLTAGE	7-BIT ADDR	8-BIT ADDR	MAX SPEED	LOCATION		
Tele (D53P)	1.8V	0x20	0x40, 0x41	1MHz	RCAM ED Flex		
Swide (D53G)	1.8V	0x20	0x40, 0x41	1MHz	RCAM BR Flex		
Raman (D53P)	1.8V	0x1C	0x38, 0x39	1MHz	RCAM ED Flex		
NOTE: Tele vs Swide depends on P vs G variant (see page 52)							
Lex	1.8V	0x75	0xEA, 0xEB	1MHz	Bottom MLB		
Strobe	1.8V	0x65	0xCA, 0xCB	1MHz	Bottom MLB		

SYNCING: NONE

PAGE TITLE		
CAMERA: PMU 1: Aliases		
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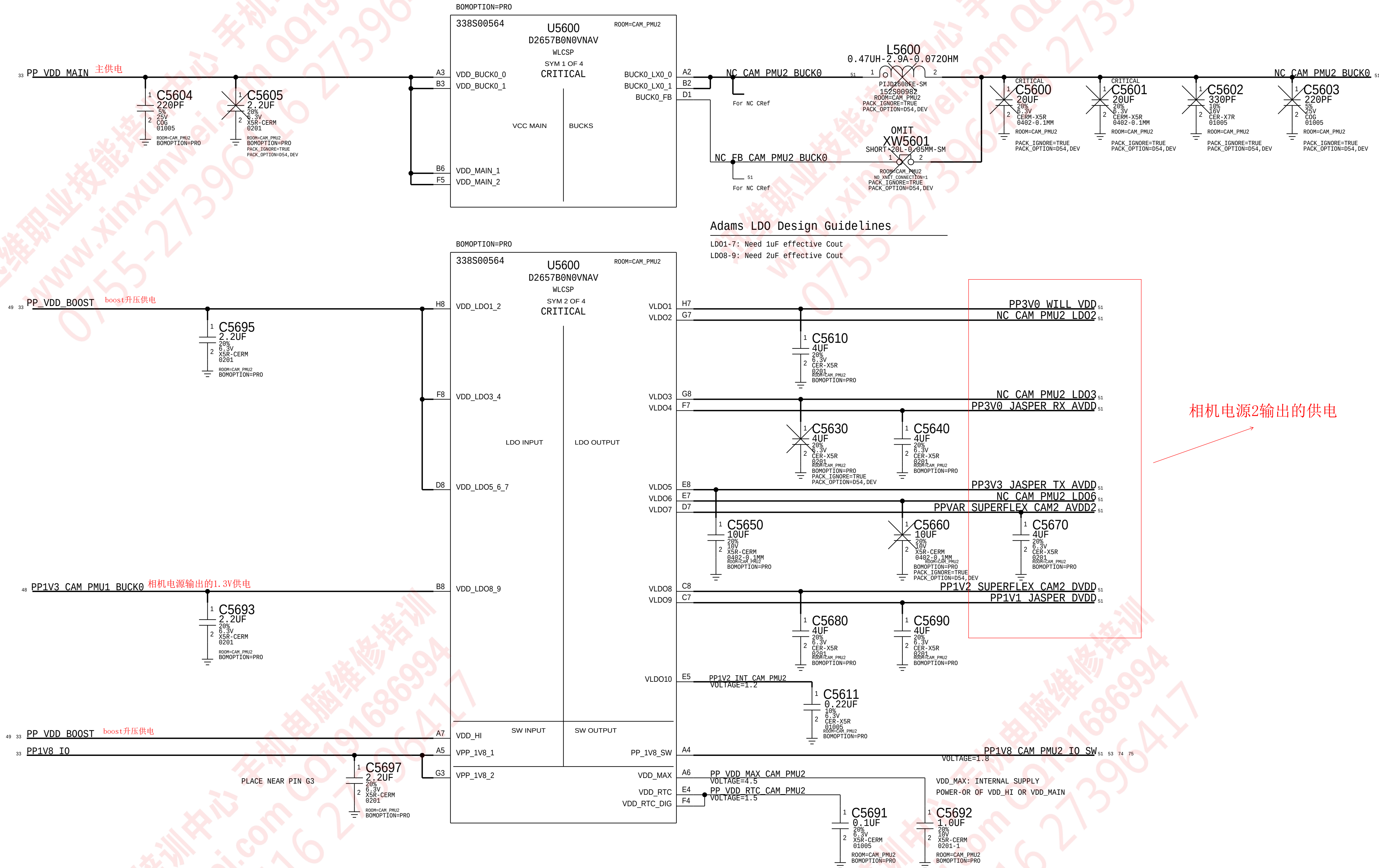
SUBSYSTEM SPECIFIC BOM TABLES

4uF 0201 Capacitors (single-source Murata)

PART NUMBER	ALTERNATE FOR PART NUMBER	BOM OPTION	REF DES	COMMENTS:
138S00116	138S00071	?	[SEE BELOW]	CAP_XSR_4UF_R201_0_50PM_KYOCERA
138S00117	138S00071	?	[SEE BELOW]	CAP_XSR_4UF_R201_0_50PM_KYOCERA

(C5618, C5638, C5648, C5668, C5678, C5688, C5698)
(C5618, C5638, C5648, C5668, C5678, C5688, C5698)

CAMERA PMU2



Adams LDO Design Guidelines

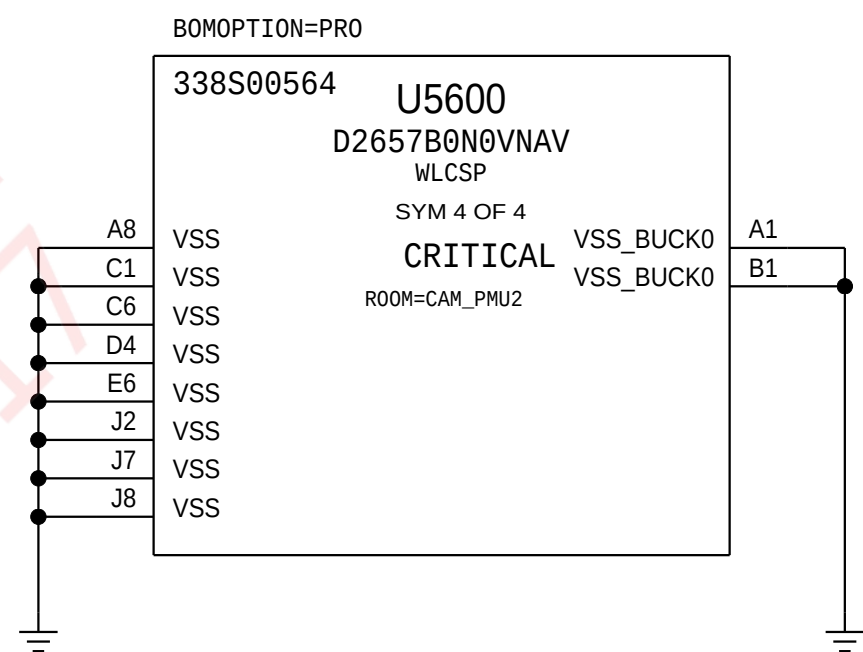
LD01-7: Need 1uF effective Cout
LD08-9: Need 2uF effective Cout

相机电源2输出的供电

SYNCING: D53, D54

PAGE TITLE		
CAMERA: PMU 2: Power (1/2)		
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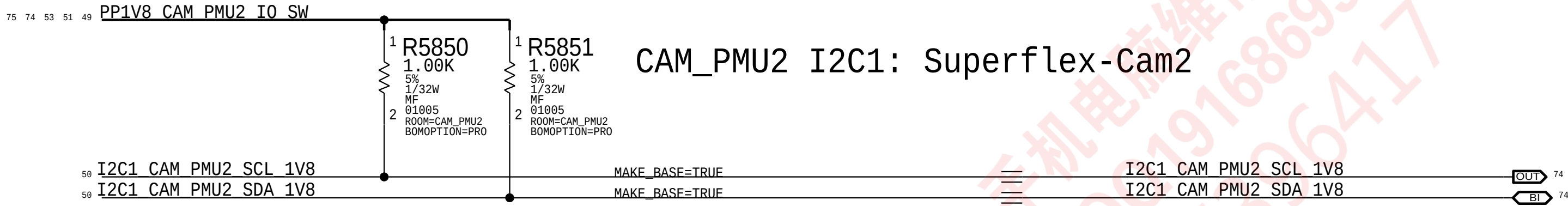
相机电源2输出的I2C检测信号



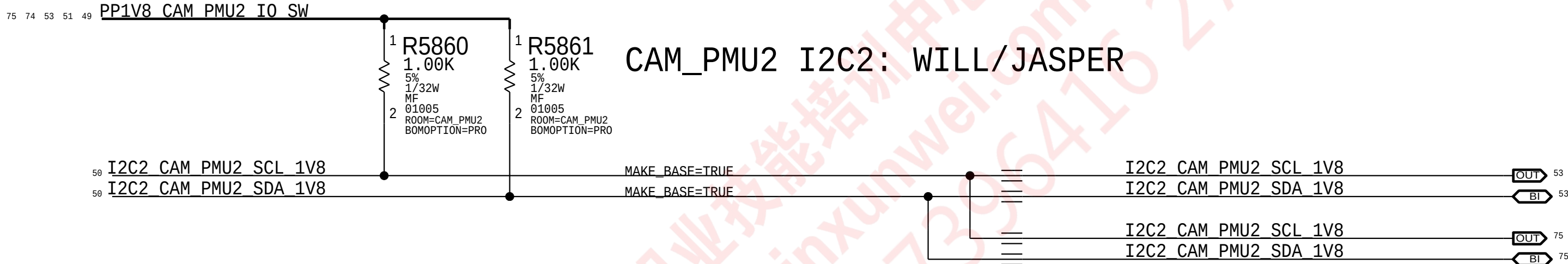
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CAMERA: PMU 2: I/O (2/2)		
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ADAMS #2 ALIASES

CAM_PMU2 I2C1: Superflex-Cam2



CAM_PMU2 I2C2: WILL/JASPER



MASTER	CAM PMU2	NUMBER	I2C1	DIAGS NUMBER	8	SPEED	1MHZ
DEVICE	VOLTAGE	7-BIT ADDR	8-BIT ADDR	MAX SPEED	LOCATION		
Swipe (D53P)	1.8V	0x20	0x40,0x41	1MHZ	RCAM M0 Flex		

MASTER	CAM PMU2	NUMBER	I2C2	DIAGS NUMBER	9	SPEED	1MHZ
DEVICE	VOLTAGE	7-BIT ADDR	8-BIT ADDR	MAX SPEED	LOCATION		
Will	1.8V	0x40	0x00,0x81	1MHZ	Top MLB		
Periscope	1.8V	0x10	0x20,0x21	1MHZ	Jasper Flex		
Riker (Main Core)	1.8V	0x33	0x66,0x67	1MHZ	Jasper Flex		
Riker (OVP Core)	1.8V	0x34	0x68,0x69	1MHZ	Jasper Flex		
Jasper (EEPROM)	1.8V	0x51	0xA2,0xA3	1MHZ	Jasper Flex		

SIGNAL

GPI0_1	GP10 CAM PMU2 TO SUPERFLEX CAM2 SHDN L 1V8	GPIO CAM PMU2 TO SUPERFLEX CAM2 SHDN L 1V8	MAKE_BASE=TRUE	OUT	74
GPI0_2	GP10 CAM PMU2 TO SUPERFLEX CAM2 LD0 EN 1V8	GPIO CAM PMU2 TO SUPERFLEX CAM2 LD0 EN 1V8	MAKE_BASE=TRUE	OUT	52 99
GPI0_3	GP10 CAM PMU2 TO JASPER SHDN L 1V8	GPIO CAM PMU2 TO JASPER SHDN L 1V8	MAKE_BASE=TRUE	OUT	75
GPI0_4	NC CAM PMU2 GPI04	NC CAM PMU2 GPI04	NO_TEST=1	MAKE_BASE=TRUE	
GPI0_5	GP10 CAM PMU2 FROM WILL JASPER INT L 1V8	GPIO CAM PMU2 FROM WILL JASPER INT L 1V8	MAKE_BASE=TRUE	IN	53 75
GPI0_6	GP10 CAM PMU2 TO WILL EN 1V8	GPIO CAM PMU2 TO WILL EN 1V8	MAKE_BASE=TRUE	OUT	53 99
GPI0_7	NC CAM PMU2 GPI07	NC CAM PMU2 GPI07	MAKE_BASE=TRUE		
GPI0_8	NC CAM PMU2 GPI08	NC CAM PMU2 GPI08	MAKE_BASE=TRUE		
GPI0_9	NC CAM PMU2 GPI09	NC CAM PMU2 GPI09	MAKE_BASE=TRUE		
GPI0_10	NC CAM PMU2 GPI010	NC CAM PMU2 GPI010	MAKE_BASE=TRUE		
GPI0_11	NC CAM PMU2 GPI011	NC CAM PMU2 GPI011	MAKE_BASE=TRUE		
GPI0_12	NC CAM PMU2 GPI012	NC CAM PMU2 GPI012	MAKE_BASE=TRUE		
GPI0_13	Camera PMU2 GPI013 intentionally not aliased				

POWER

CLK0_1	GP10 CAM PMU2 TO SUPERFLEX CAM2 24M 1V8	CLK CAM PMU2 TO SUPERFLEX CAM2 24M 1V8	MAKE_BASE=TRUE	OUT	74
CLK0_2	CLK CAM PMU2 TO JASPER 12M 1V8	CLK CAM PMU2 TO JASPER 12M 1V8	MAKE_BASE=TRUE	OUT	75
CLK0_3	NC CAM PMU2 CLK03	NC CAM PMU2 CLK03	NO_TEST=1	MAKE_BASE=TRUE	
CLK0_4	NC CAM PMU2 CLK04	NC CAM PMU2 CLK04	NO_TEST=1	MAKE_BASE=TRUE	
VLD01	PP3V0 WILL VDD	PP3V0 WILL VDD	VOLTAGE=3.0		53
VLD02	NC CAM PMU2 LD02	NC CAM PMU2 LD02	MAKE_BASE=TRUE		
VLD03	NC CAM PMU2 LD03	NC CAM PMU2 LD03	MAKE_BASE=TRUE		
VLD04	PP3V0 JASPER RX AVDD	PP3V0 JASPER RX AVDD	VOLTAGE=3.0		75
VLD05	PP3V3 JASPER TX AVDD	PP3V3 JASPER TX AVDD	VOLTAGE=3.3		75
VLD06	NC CAM PMU2 LD06	NC CAM PMU2 LD06	MAKE_BASE=TRUE		
VLD07	PPVAR SUPERFLEX CAM2 AVDD2	PPVAR SUPERFLEX CAM2 AVDD2	VOLTAGE=3.0		74
VLD08	PP1V2 SUPERFLEX CAM2 DVDD	PP1V2 SUPERFLEX CAM2 DVDD	VOLTAGE=1.2		74
VLD09	PP1V1 JASPER DVDD	PP1V1 JASPER DVDD	VOLTAGE=1.1		75
VLD010	Camera PMU2 LD010 unused				
BUCK0	NC CAM PMU2 BUCK0	NC CAM PMU2 BUCK0	MAKE_BASE=TRUE		24
	NC FB CAM PMU2 BUCK0	NC FB CAM PMU2 BUCK0	MAKE_BASE=TRUE		
	NC CAM PMU2 BUCK0	NC CAM PMU2 BUCK0	MAKE_BASE=TRUE		24

SYNCING: NONE

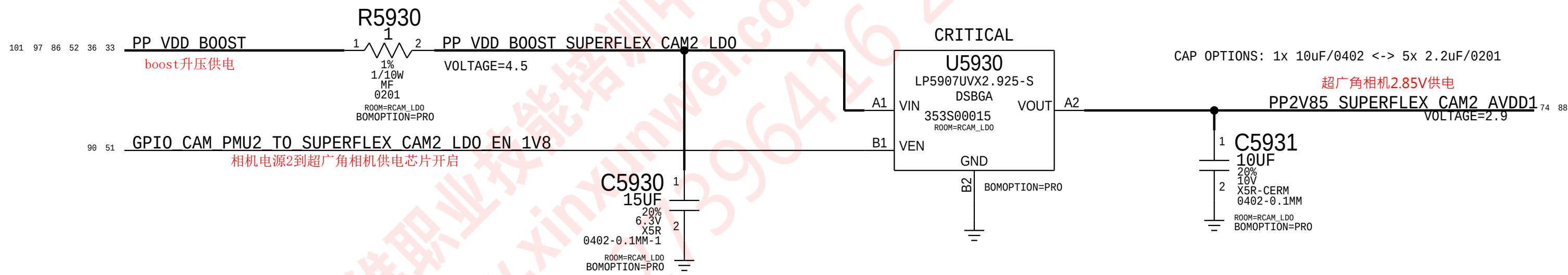
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CAMERA: PMU 2: Aliases		
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BRANCH		
PAGE	115 OF 191	
SHEET		

SUBSYSTEM SPECIFIC BOM TABLES

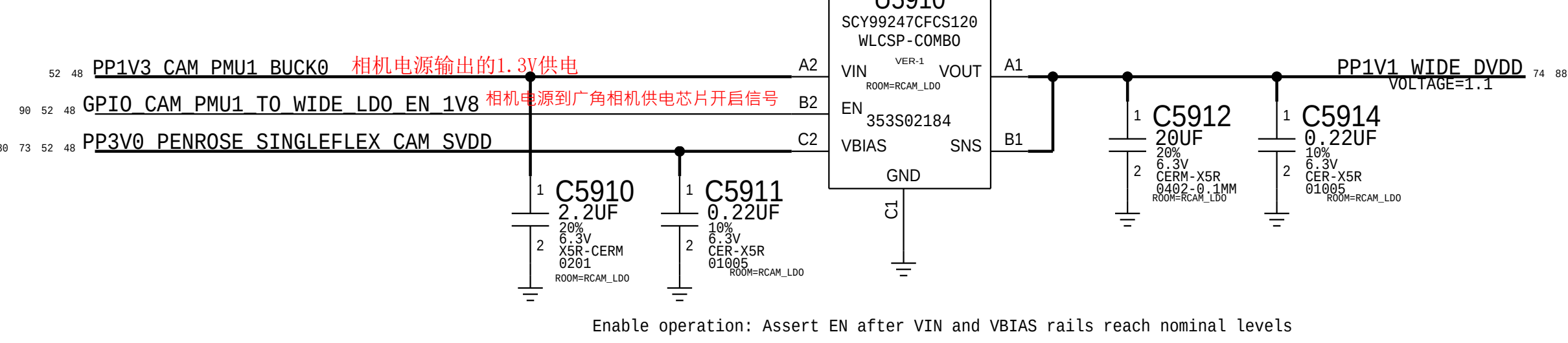
2.925V LDO				
PART NUMBER	ALTERNATE FOR PART NUMBER	BOM OPTION	REF DES	COMMENTS:
353S02185	353S00015	?	ALL	IC, LDO, REG, 2.925V, 250MA, ONSEMI

1.2V LDO				
PART NUMBER	ALTERNATE FOR PART NUMBER	BOM OPTION	REF DES	COMMENTS:
353S02127	353S02184	?	ALL	IC, LDO, REG, 1.20V, 1000MA, RICHTEK

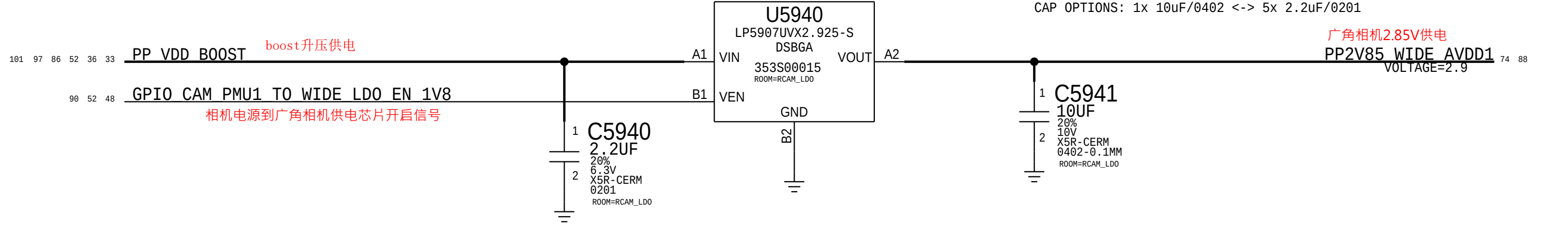
SUPERFLEX-CAM2 VDDH1 LDO



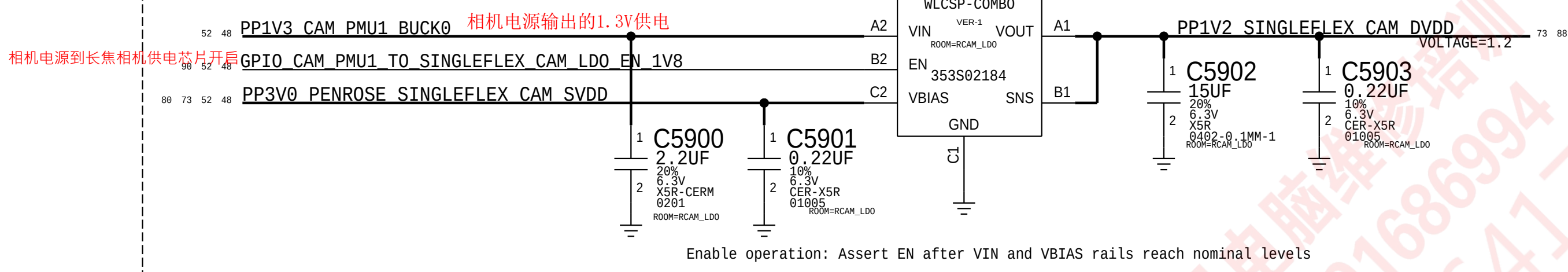
WIDE VDDL LDO



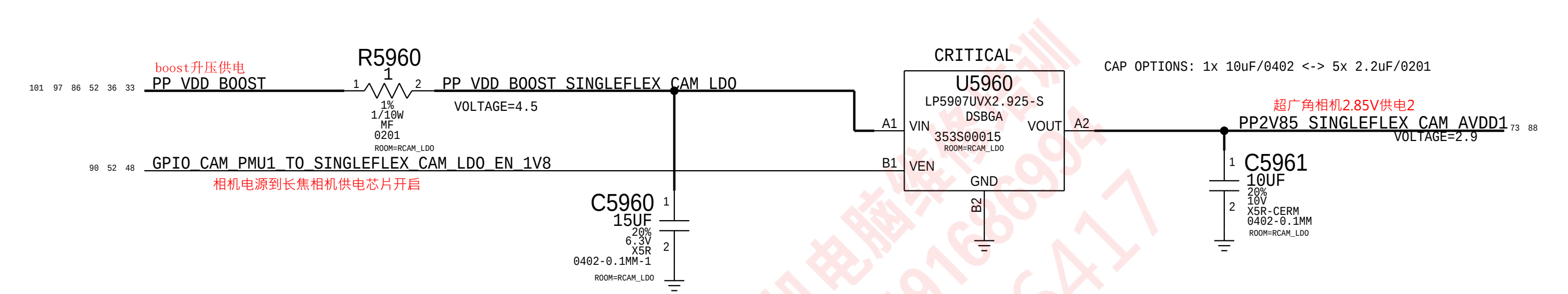
WIDE VDDH1 LDO



SINGLEFLEX-CAM VDDL LDO



SINGLEFLEX-CAM VDDH1 LDO

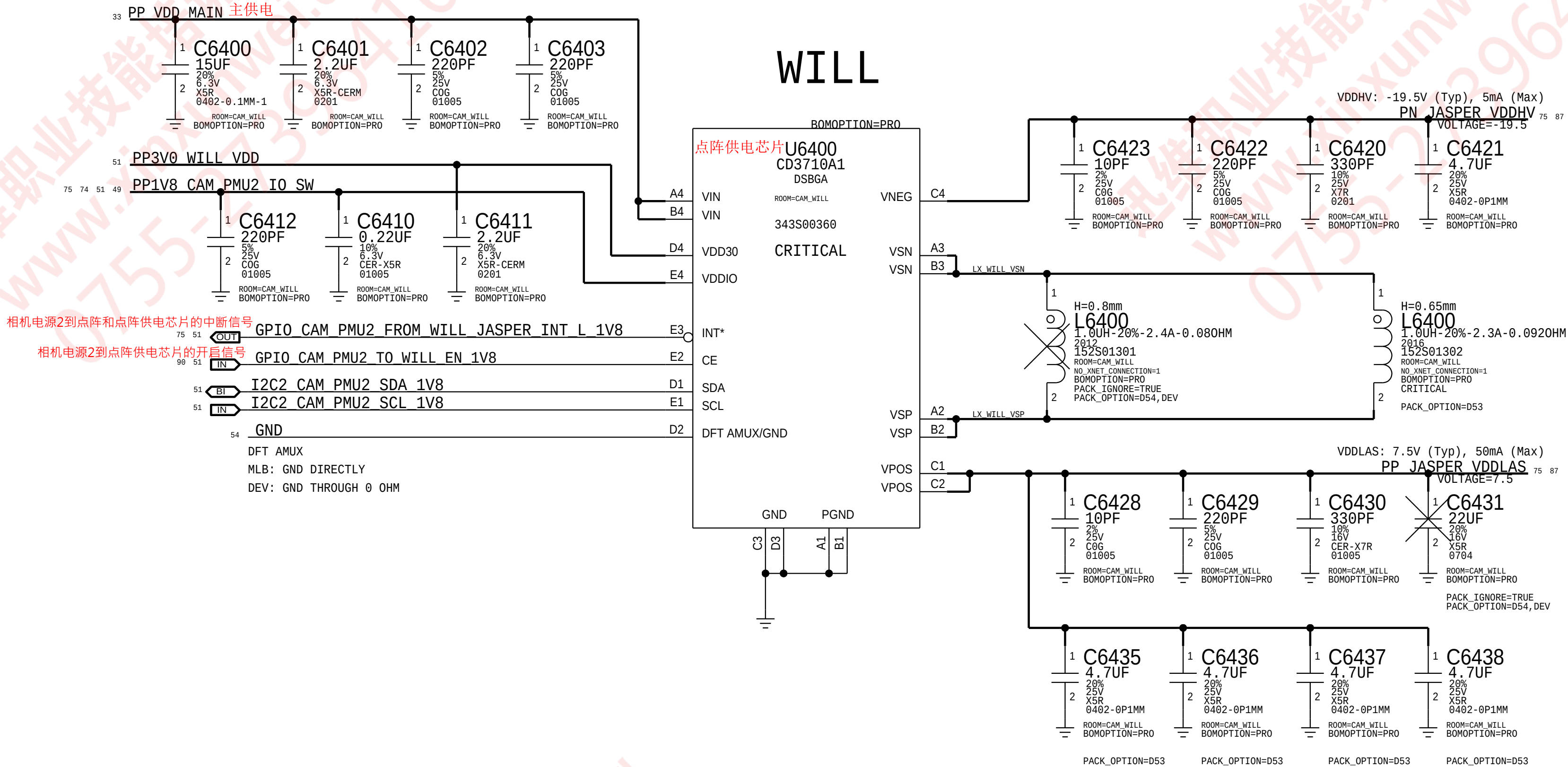


SYNCING: NONE

PAGE TITLE		
CAMERA: Discrete LDOs		
DRAWING NUMBER	051-05215	SIZE
REVISION	4.6.0	D
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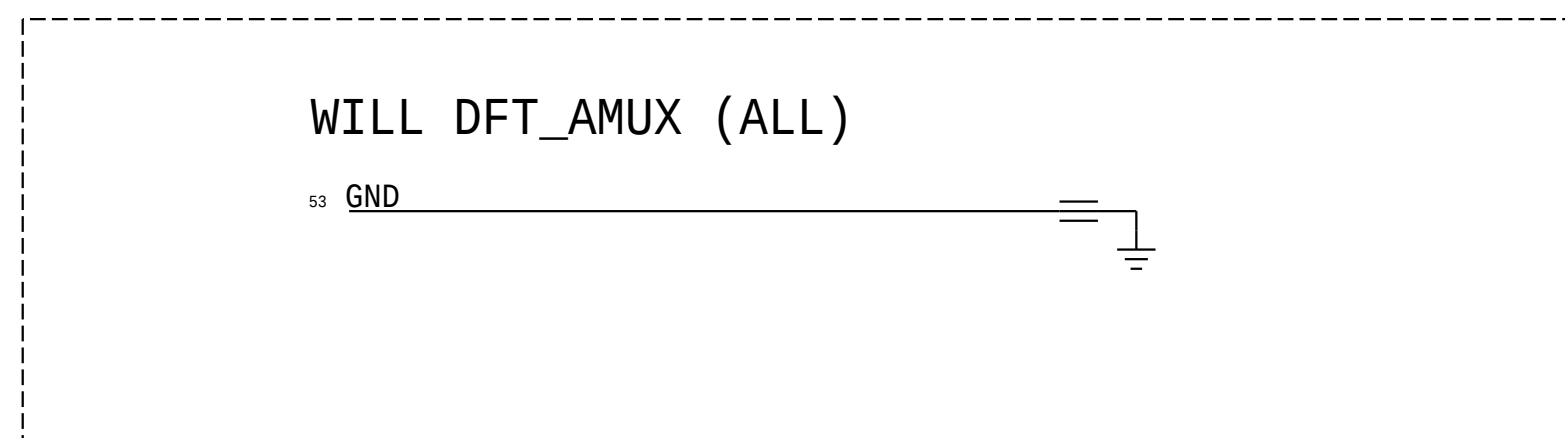
SUBSYSTEM SPECIFIC BOM TABLES

INDUCTOR						
PART NUMBER	ALTERNATE FOR PART NUMBER	BOM OPTION	REF DES	COMMENTS:	CRITICAL PART#	COMMENT
152S01343	152S01301	?	ALL	IND, ML0, 1.00H, 20%, 2.5A, 5200W, H=0.8, 2012	152S01301	IND, ML0, 1.00H, 20%, 2.5A, 5200W, H=0.8, 2012



SYNCING: D53, D54

PAGE TITLE		
CAMERA: Will		
	DRAWING NUMBER	051-05215
	REVISION	4.6.0
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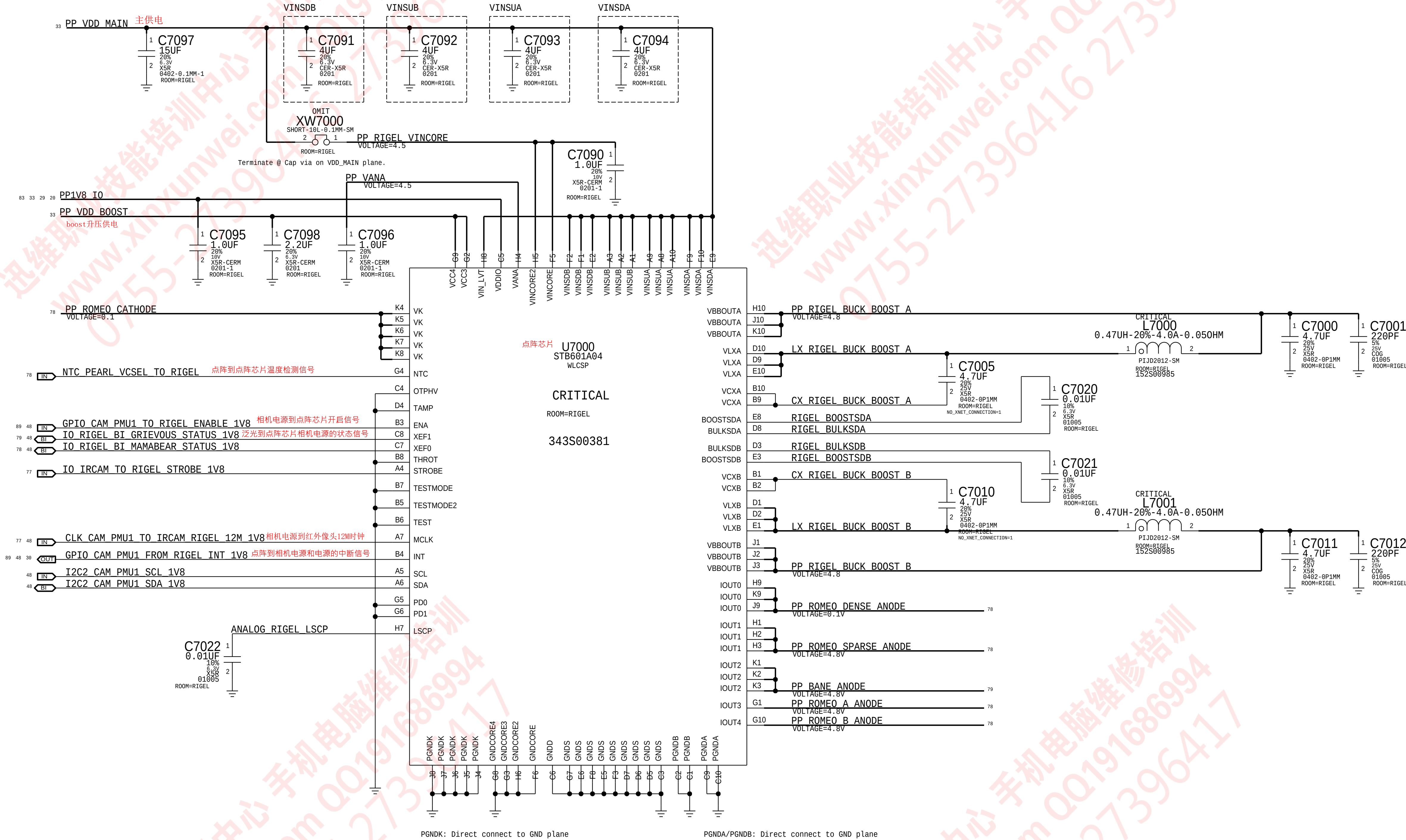


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	DRAWING NUMBER 051-05215	SIZE D
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SUBSYSTEM SPECIFIC BOM TABLES

INDUCTOR: Alt covered with PMU inductor BOM table

VCSEL DRIVER
RIGEL



SYNCING: D52, D53, D54

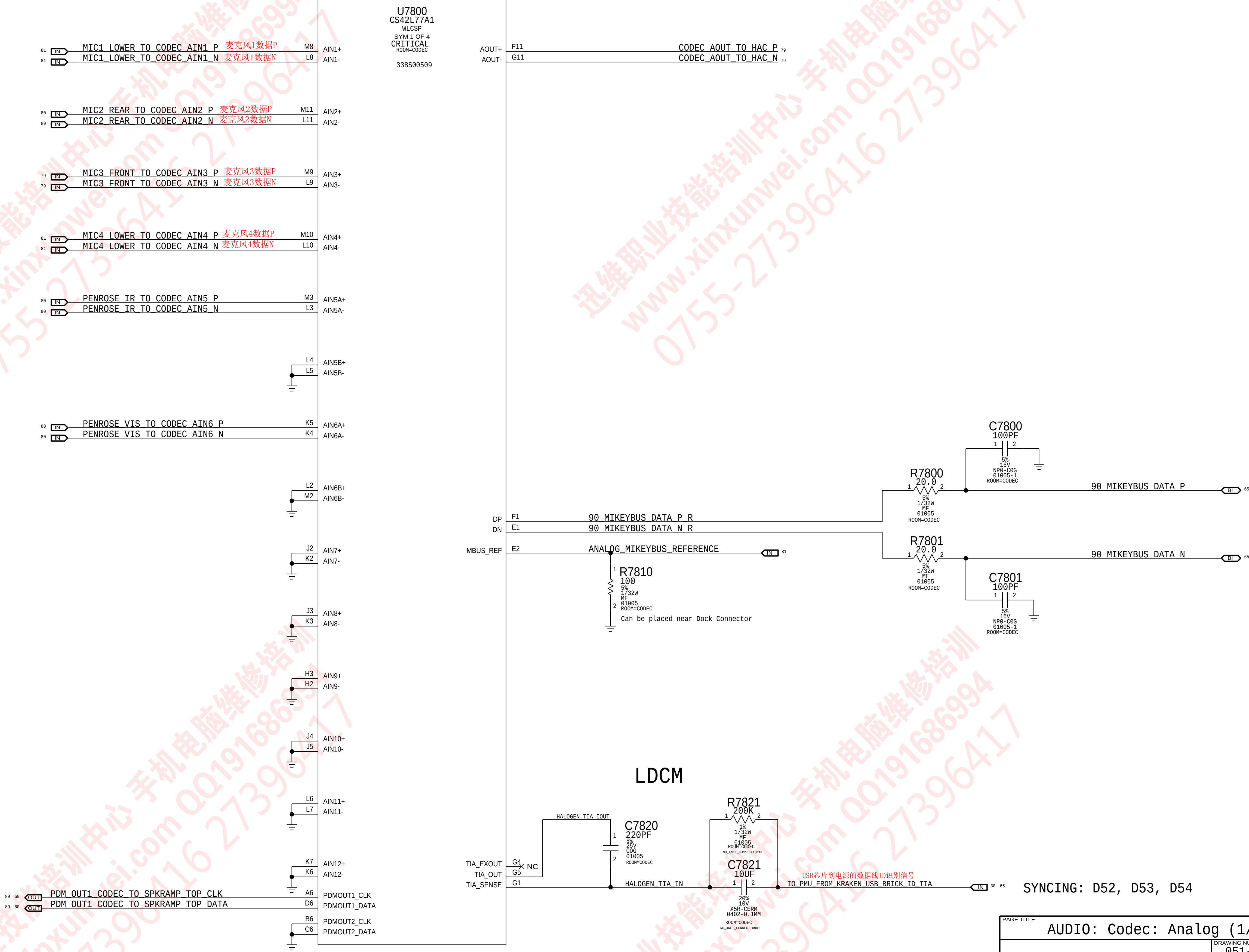
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PEARL: VCSEL Driver		
	DRAWING NUMBER	SIZE
	051-05215	D
	REVISION	
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	8	7	6	5	4	3	2	1
D								
C								
B								
A								
	8	7	6	5	4	3	2	1

SYNCING: D52, D53, D54

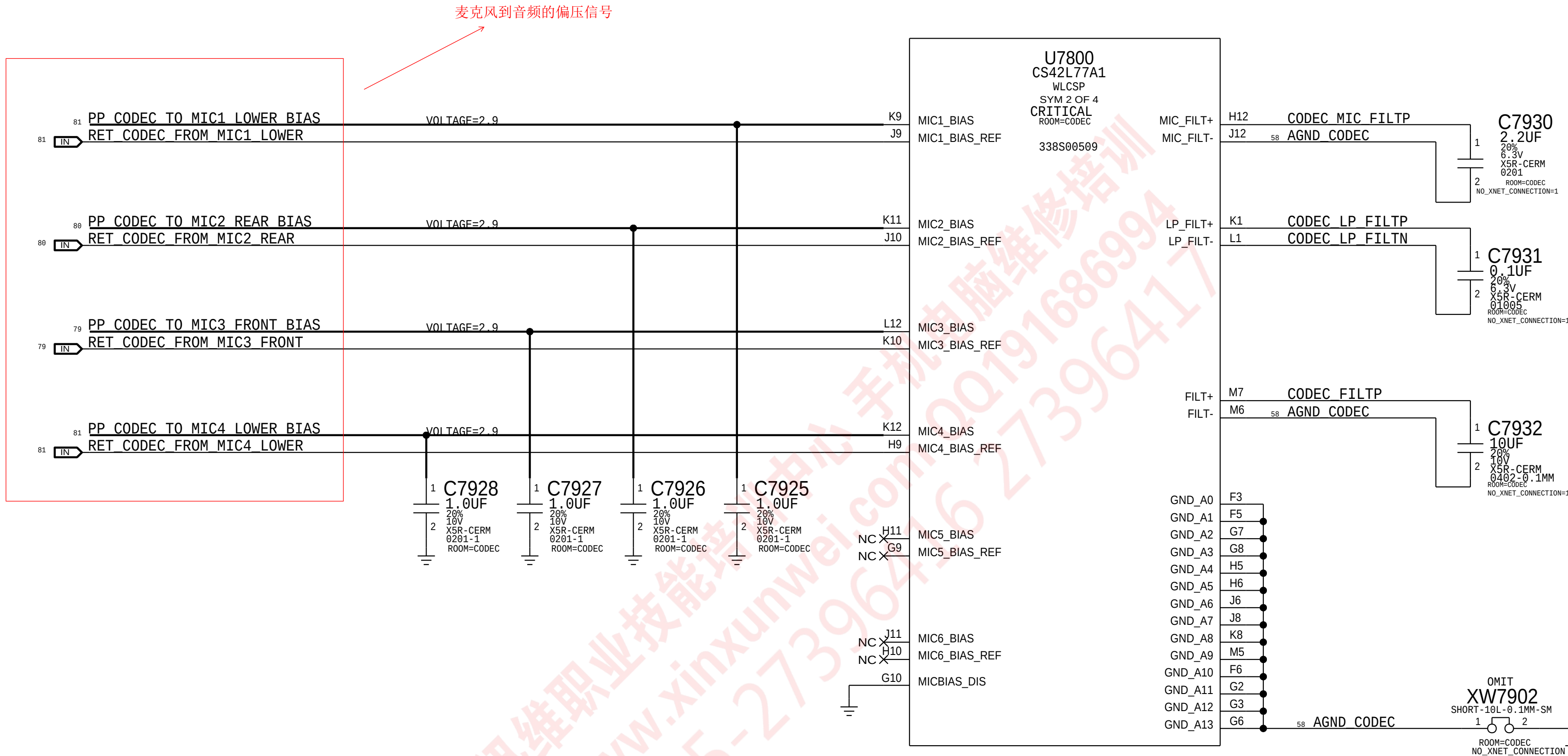
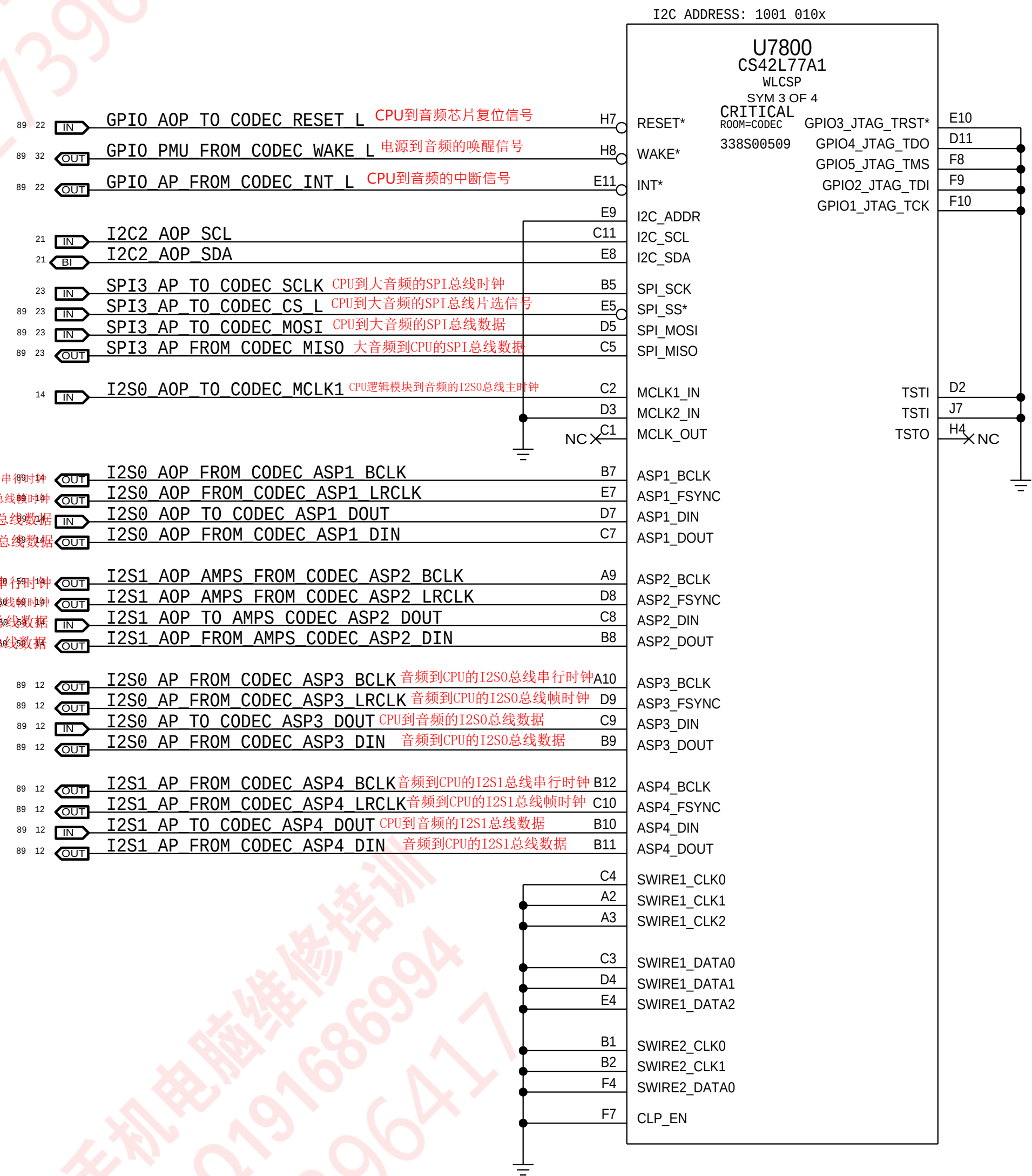
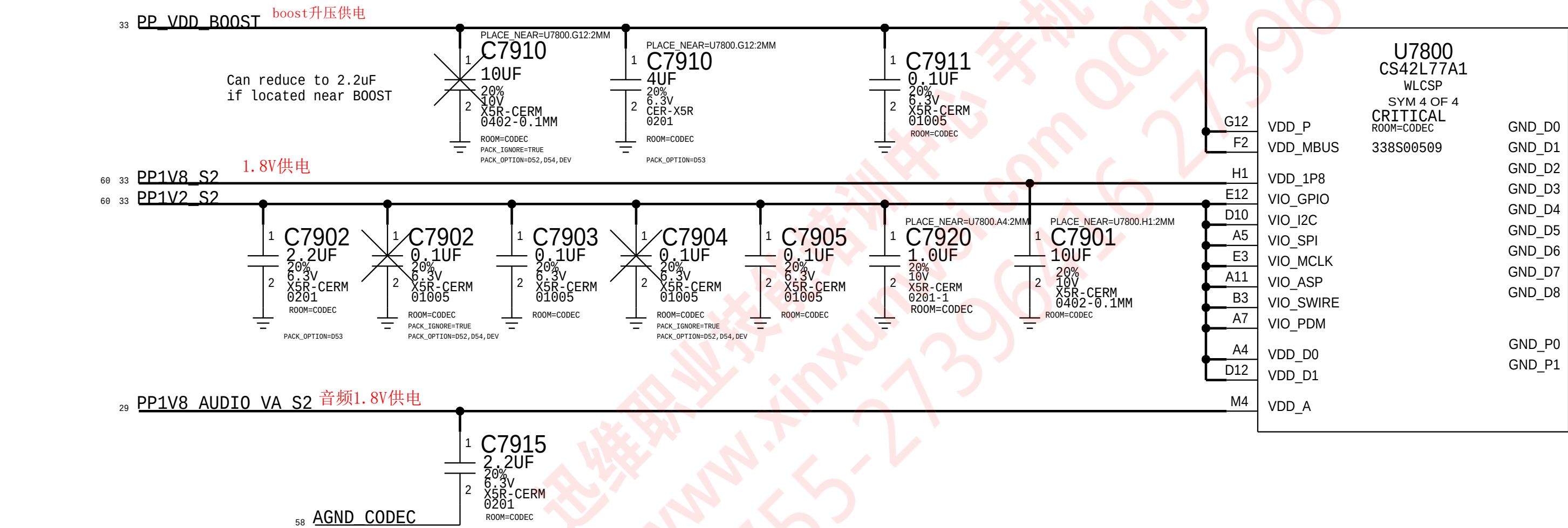
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BRIGHTON AUDIO CODEC (ANALOG INPUTS & OUTPUTS)



PAGE TITLE		
AUDIO: Codec: AnaLog (1/2)		
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REVISION	4.6.0	D
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BRIGHTON AUDIO CODEC (POWER & I/O)



SYNCING: D52, D53, D54

PAGE TITLE				
AUDIO: Codec: Power & I/O (2/2)				
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	REVISION			4.6.0
	BRANCH			
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D

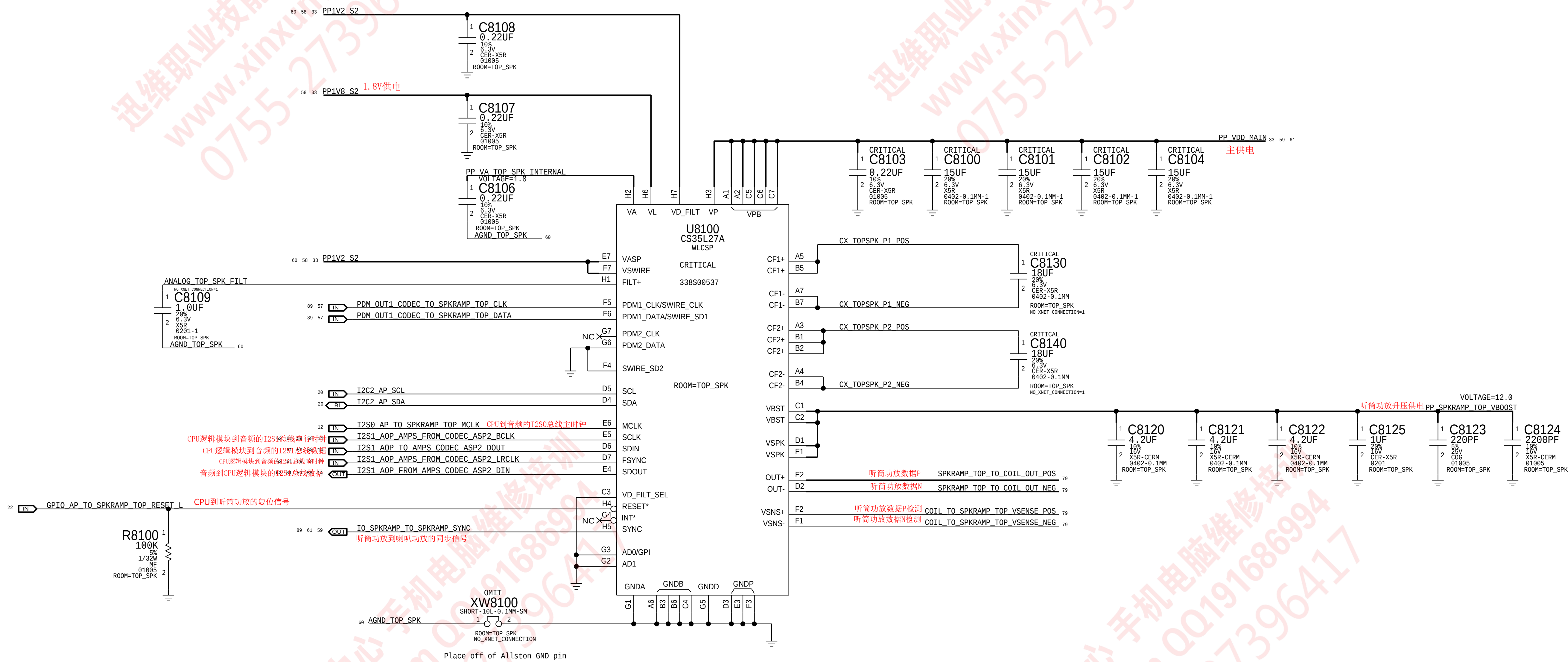


B

A

TOP SPEAKER AMP

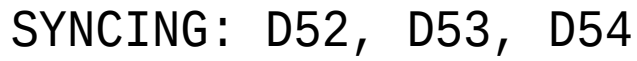
ALLSTON



SYNCING: D52, D53, D54

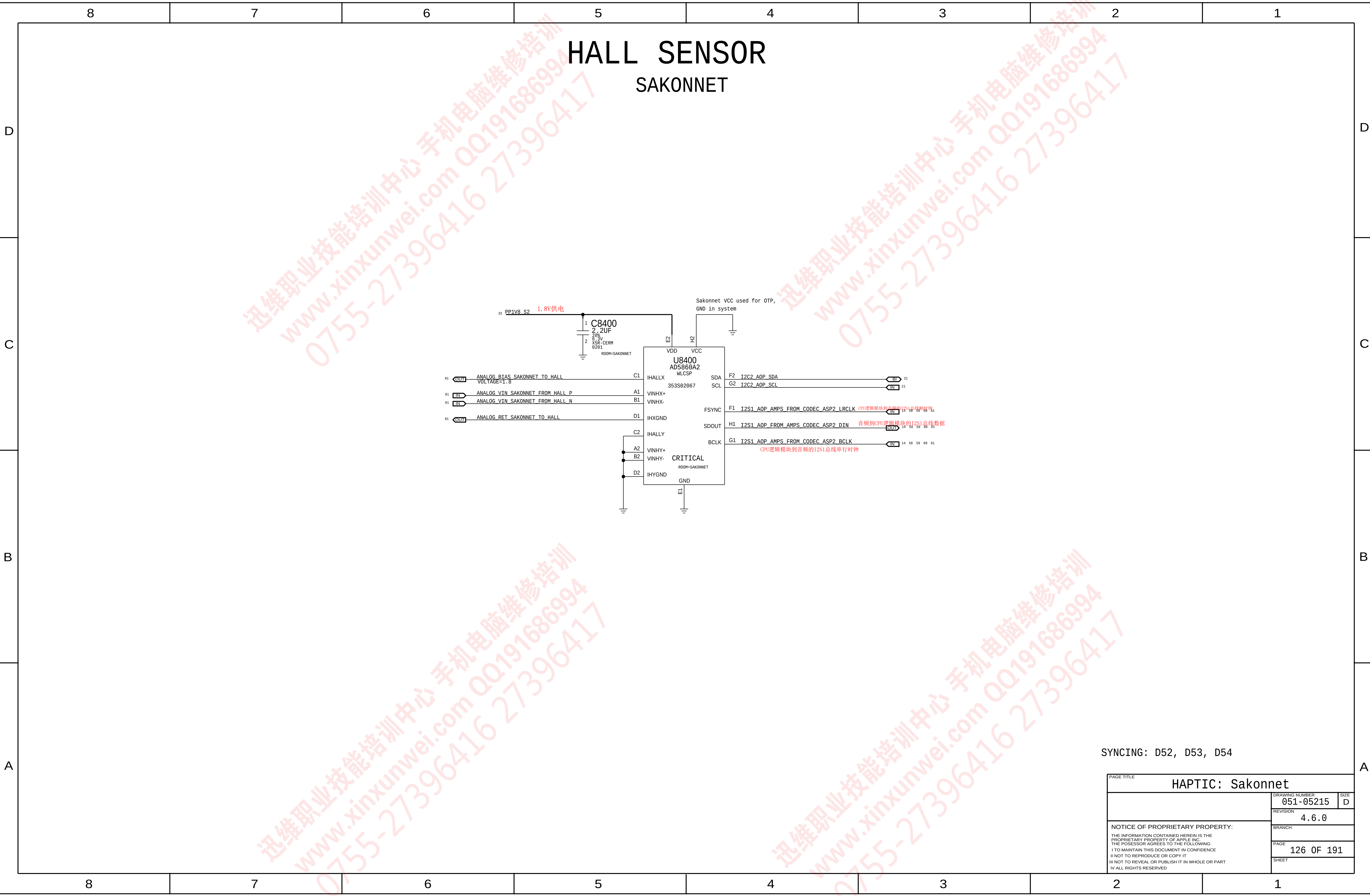
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PLACEMENT NOTES:
A1,A2: 15uF (1x), 2.2uF(1x)
C5,C6,C7: 15uF (3x)



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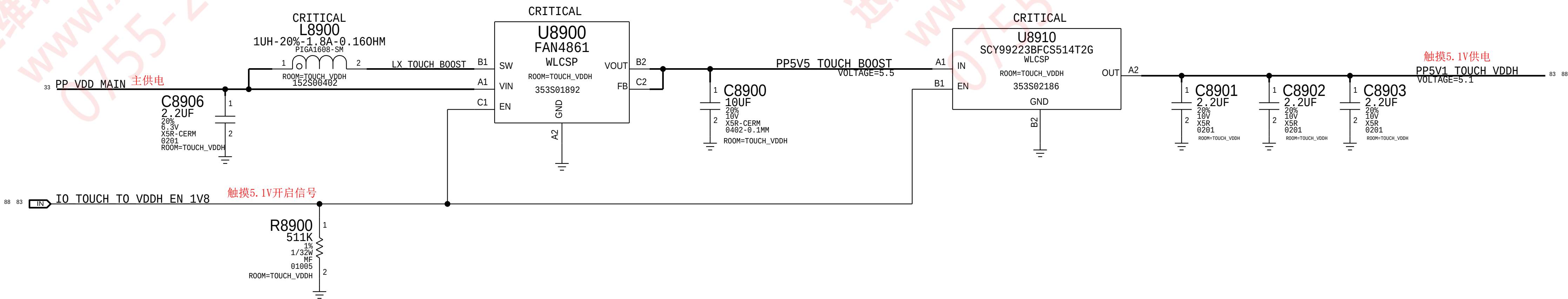


PAGE TITLE		
HAPTIC: Sakonnet		
	DRAWING NUMBER	051-05215
	REVISION	4.6.0
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SUBSYSTEM SPECIFIC BOM TABLES

INDUCTOR				
PART NUMBER	ALTERNATE FOR PART NUMBER	BOM OPTION	REF DES	COMMENTS:
152S00366	152S00402	?	ALL	150, PWR, 1, 80A, 20%, 1, 5A, 2500000, 15270, 1500

TOUCH
TOUCH POWER



SYNCING: D53

PAGE TITLE		
TOP MODULE: Touch Power		
	DRAWING NUMBER	051-05215
	REVISION	4.6.0
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PART#	QTY	DESCRIPTION	REFERENCE DESIGNATOR(S)	CRITICAL	BOM OPTION
338S00655	1	Galena A2 OTP-1	U9100	CRITICAL	BOARD_ID:0536&BOARD_ID:054
338S00578	1	Chelan	U9100	CRITICAL	BOARD_ID:052&BOARD_ID:0539

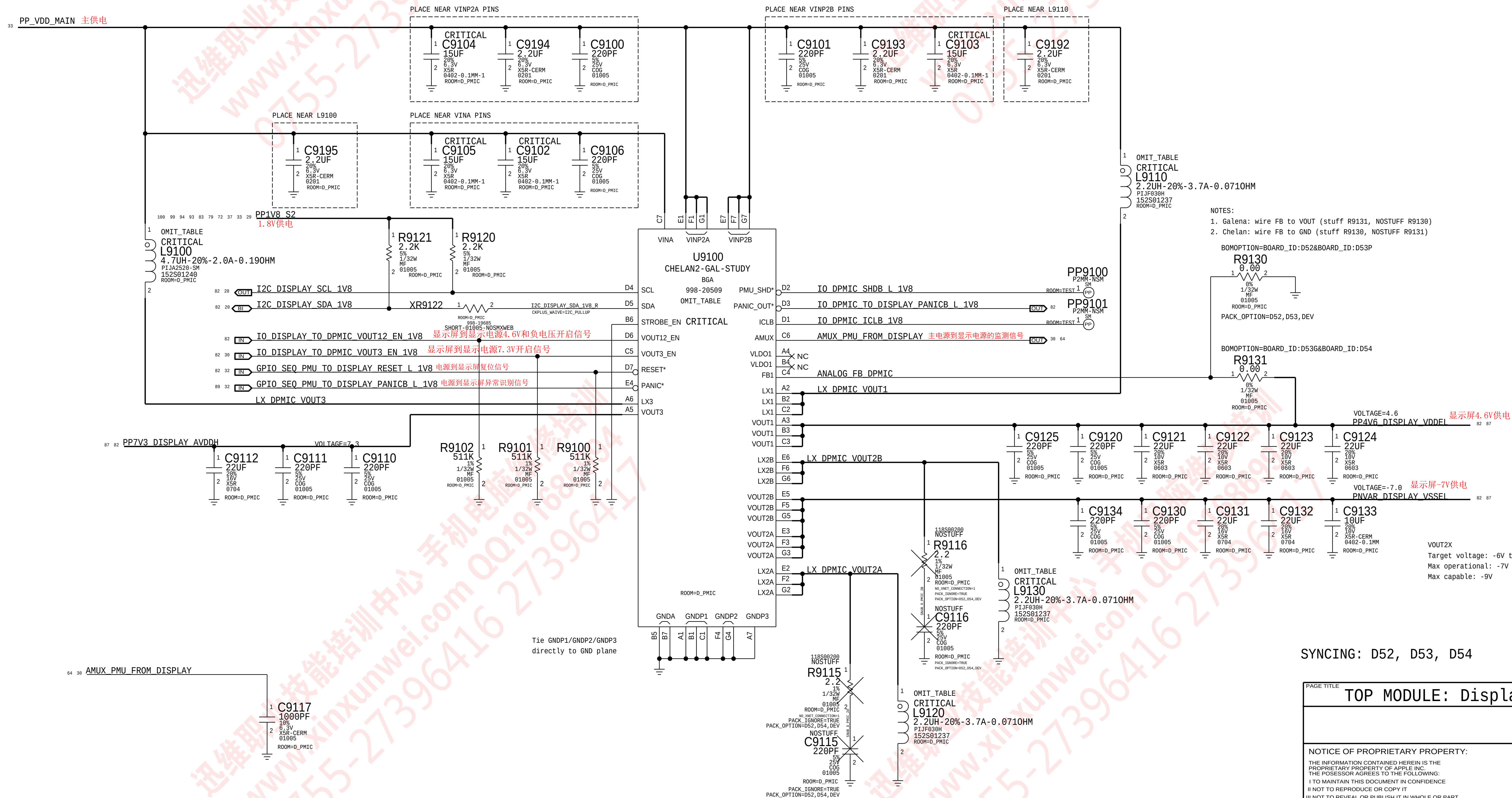
DISPLAY PMIC

GALENA/CHELAN

Inductors

PART#	QTY	DESCRIPTION	REFERENCE DESIGNATOR(S)	CRITICAL	BOM OPTION
152501355	1	Cytech V0073 Inductor (6.8uH 2520 0.050m)	L9100	CRITICAL	BOARD_20_052
152501347	1	Cytech V0073 Inductor (6.8uH 2520 1.00m)	L9100	CRITICAL	BOARD_20_053P
152501313	1	Cytech V0073 Inductor (4.7uH 2520 1.00m)	L9100	CRITICAL	BOARD_20_053G&BOARD_20_054
152501237	3	Cytech V0071/2 Inductor (2.2uH 3030 0.05m)	L9110	CRITICAL	BOARD_20_052
152501357	3	Cytech V0071/2 Inductor (2.2uH 3030 0.050m)	L9120, L9130	CRITICAL	BOARD_20_052
152501314	3	Cytech V0071/2 Inductor (2.2uH 3030 1.00m)	L9110, L9120, L9130	CRITICAL	BOARD_20_053G&BOARD_20_053P&BOARD_20_054

PART NUMBER	ALTERNATE FOR PART NUMBER	BOM OPTION	REF DES	COMMENTS
152S01356	152S01355	8040, 11-052	L9100	Taiyo 6.8uH 2520 0.95mm
152S01346	152S01347	8040, 11-050P	L9100	Taiyo 6.8uH 2520 1.0mm
152S01339	152S01313	8040, 11-051680A00, 10-354	L9100	Taiyo 4.7uH 2520 1.0mm
152S01290	152S01237	8040, 11-052	L9110	Taiyo 2.2uH 3030 0.95mm
152S01358	152S01357	8040, 11-053	L9120, L9130	Taiyo 2.2uH 3030 0.8mm
152S01340	152S01314	8040, 11-050600, 10-354	L9110, L9120, L9130	Taiyo 2.2uH 3030 1.0mm

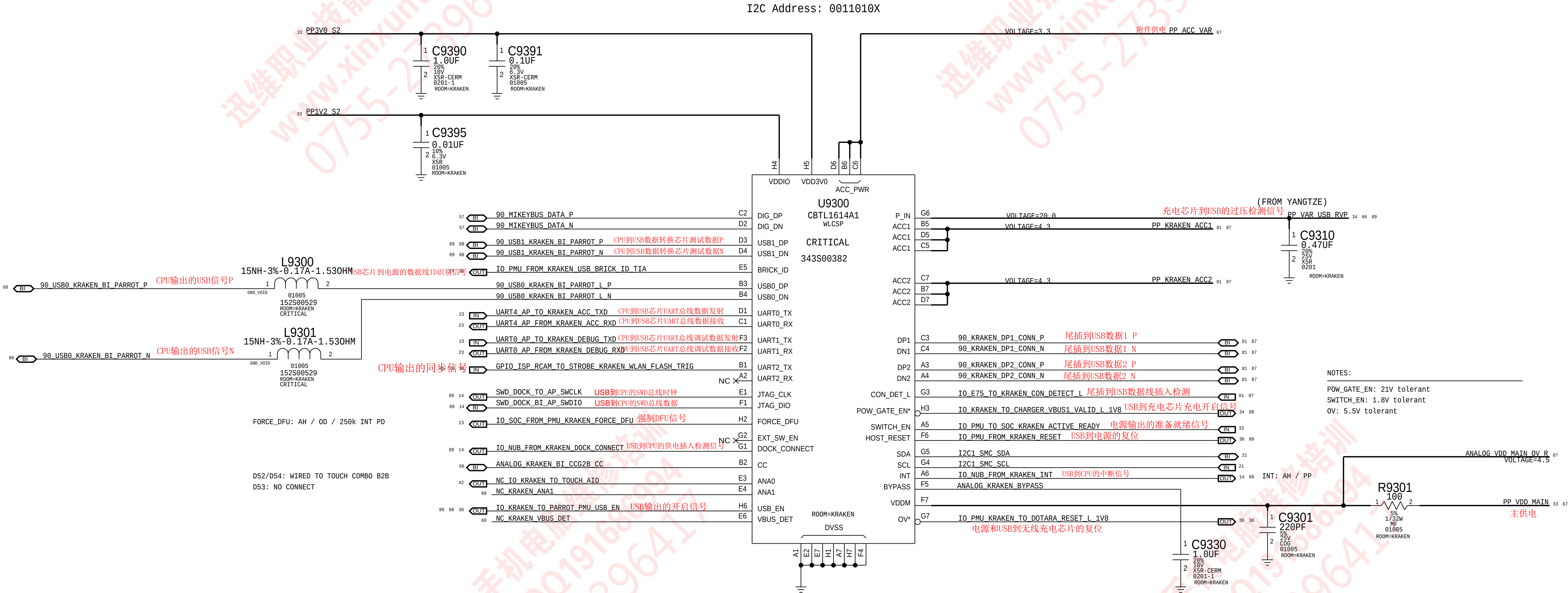


PAGE TITLE		TOP MODULE: Display Power	
		DRAWING NUMBER	SIZE
		051-05215	D
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		4.6.0	
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LIGHTNING CONTROLLER

KRAKEN

SUBSYSTEM SPECIFIC BOM TABLES

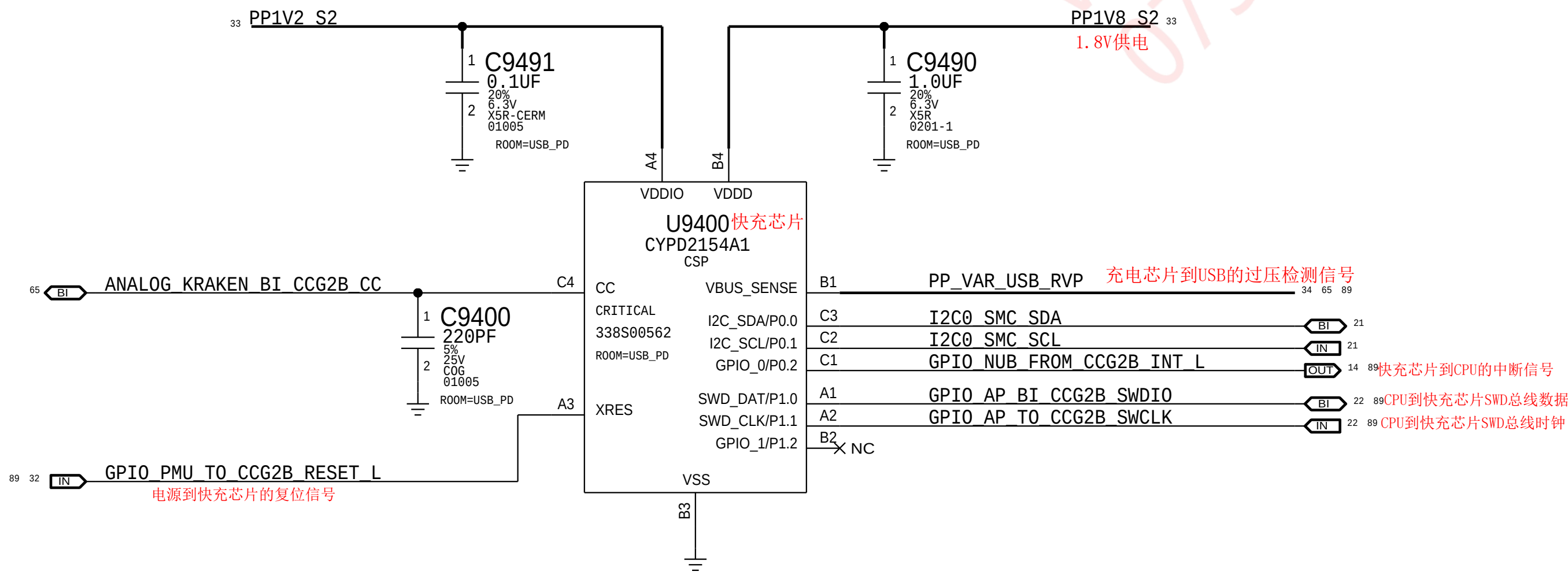


SYNCING: D52, D53, D54

PAGE TITLE		
LIGHTNING: Lightning Controller		
DRAWING NUMBER	051-05215	SIZE
REVISION	4.6.0	D
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USB - PD

CCG2B



SYNCING: D52, D53, D54

PAGE TITLE		
LIGHTNING: USB-PD		
	DRAWING NUMBER	051-05215
	REVISION	4.6.0
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SUBSYSTEM SPECIFIC BOM TABLES

Inductor

PART NUMBER	ALTERNATE FOR PART NUMBER	BOM OPTION	REF DES	COMMENTS:
152S01279	152S00854	?	L9550	1M0, PAB, 0.470uH, 20%, 2.7A, TY

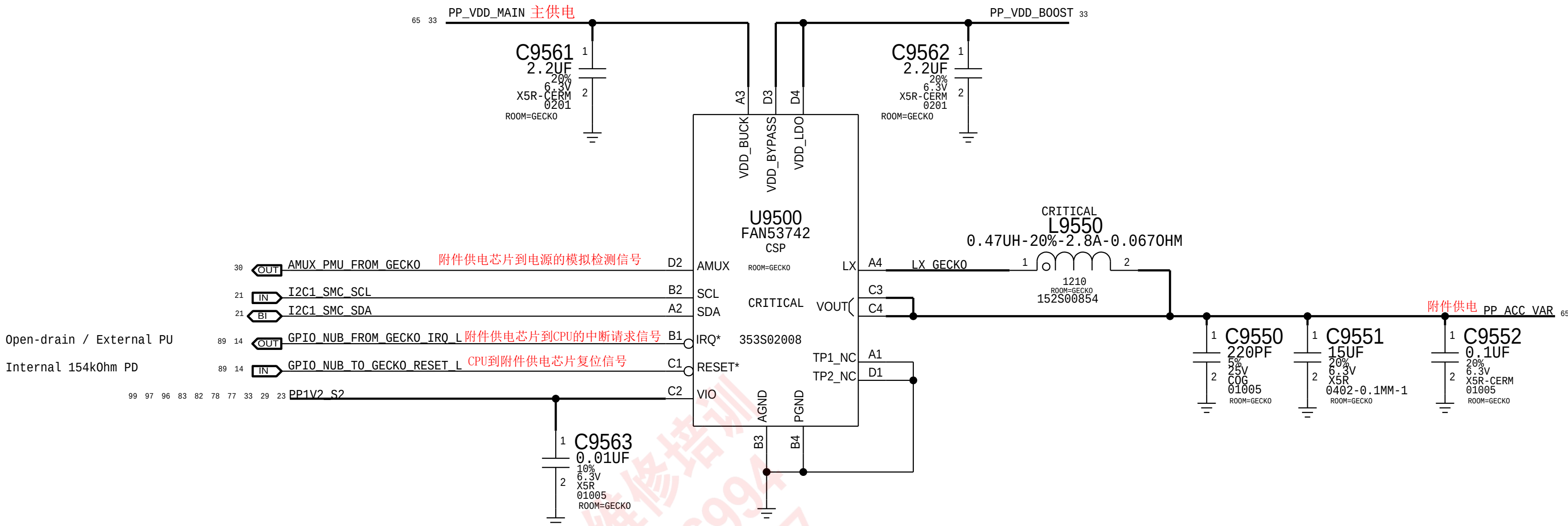
Output Capacitor (single-source Murata)

PART NUMBER	ALTERNATE FOR PART NUMBER	BOM OPTION	REF DES	COMMENTS:
138S00040	138S00003	?	(C9551)	0402, 15uF, 6.3V, Kyocera

All RefDes in () are single-sourced from Murata

ACCESSORY POWER
GECK02

I2C ADDRESS: 0X52



SYNCING: D52, D53, D54

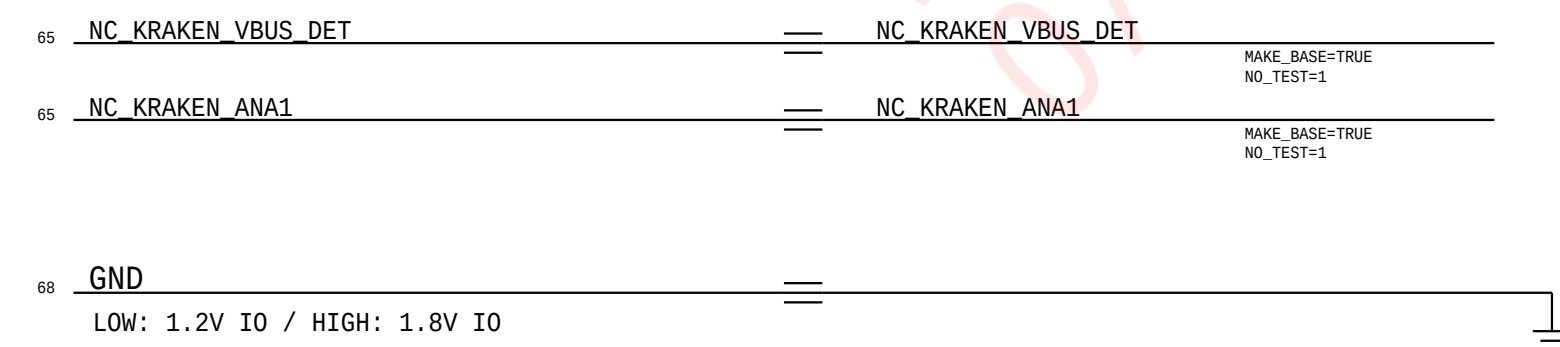
PAGE TITLE LIGHTNING: Accessory Buck		
	DRAWING NUMBER 051-05215	SIZE D
	REVISION 4.6.0	BRANCH
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D



B

A

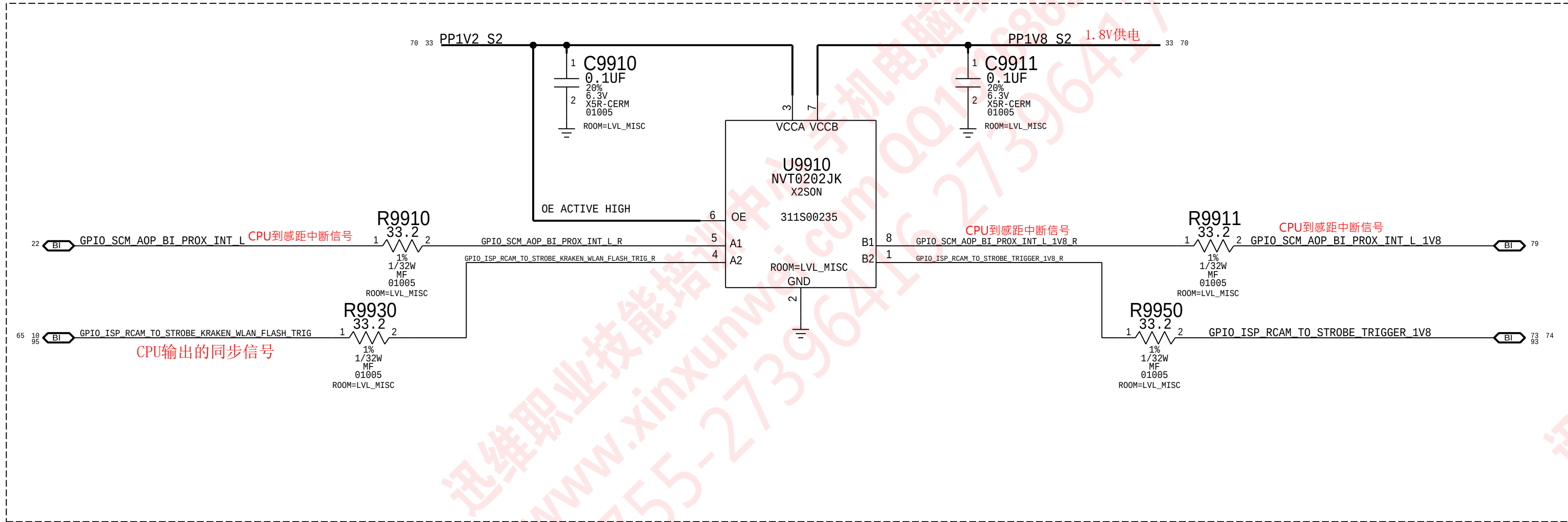


SYNCING: D52, D53, D54

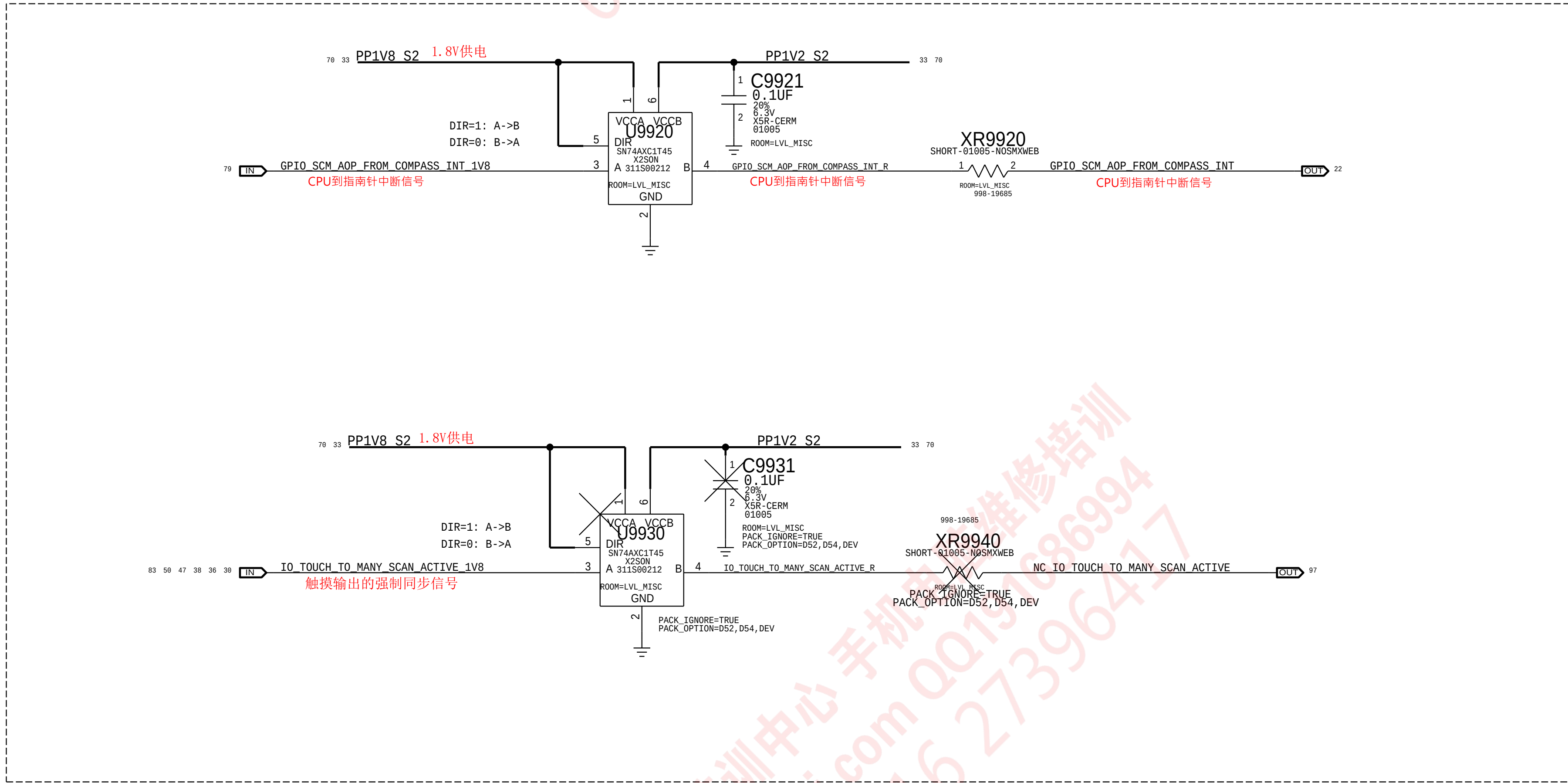
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MISCELLANEOUS LEVEL TRANSLATORS

PROX INTERRUPT + STROBE TRIGGER

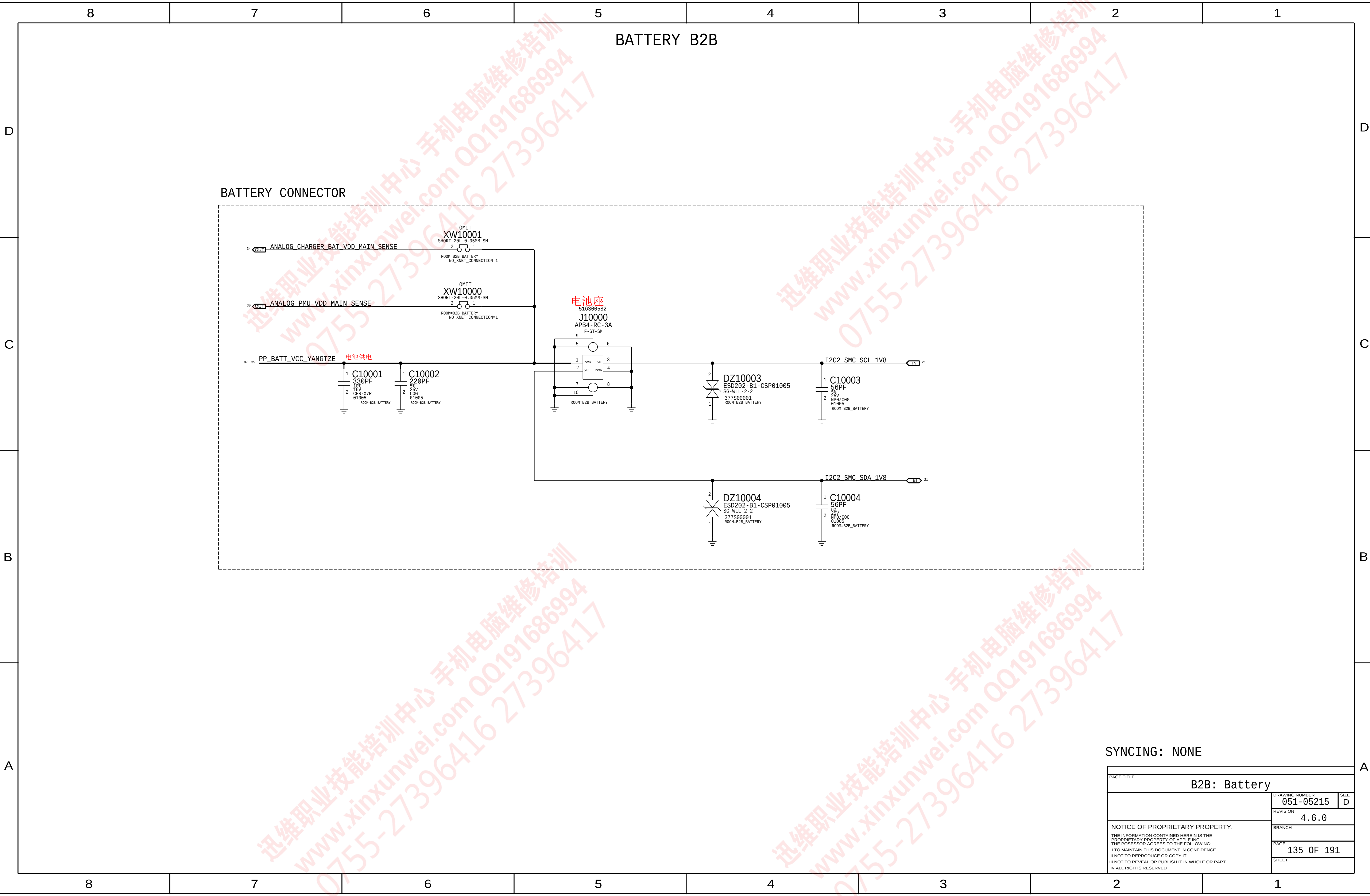


TOUCH SCAN ACTIVE + COMPASS INTERRUPT



SYNCING: D52, D53, D54, DEV

PAGE TITLE		
LVL SHIFT: Misc Nets		
	DRAWING NUMBER	051-05215
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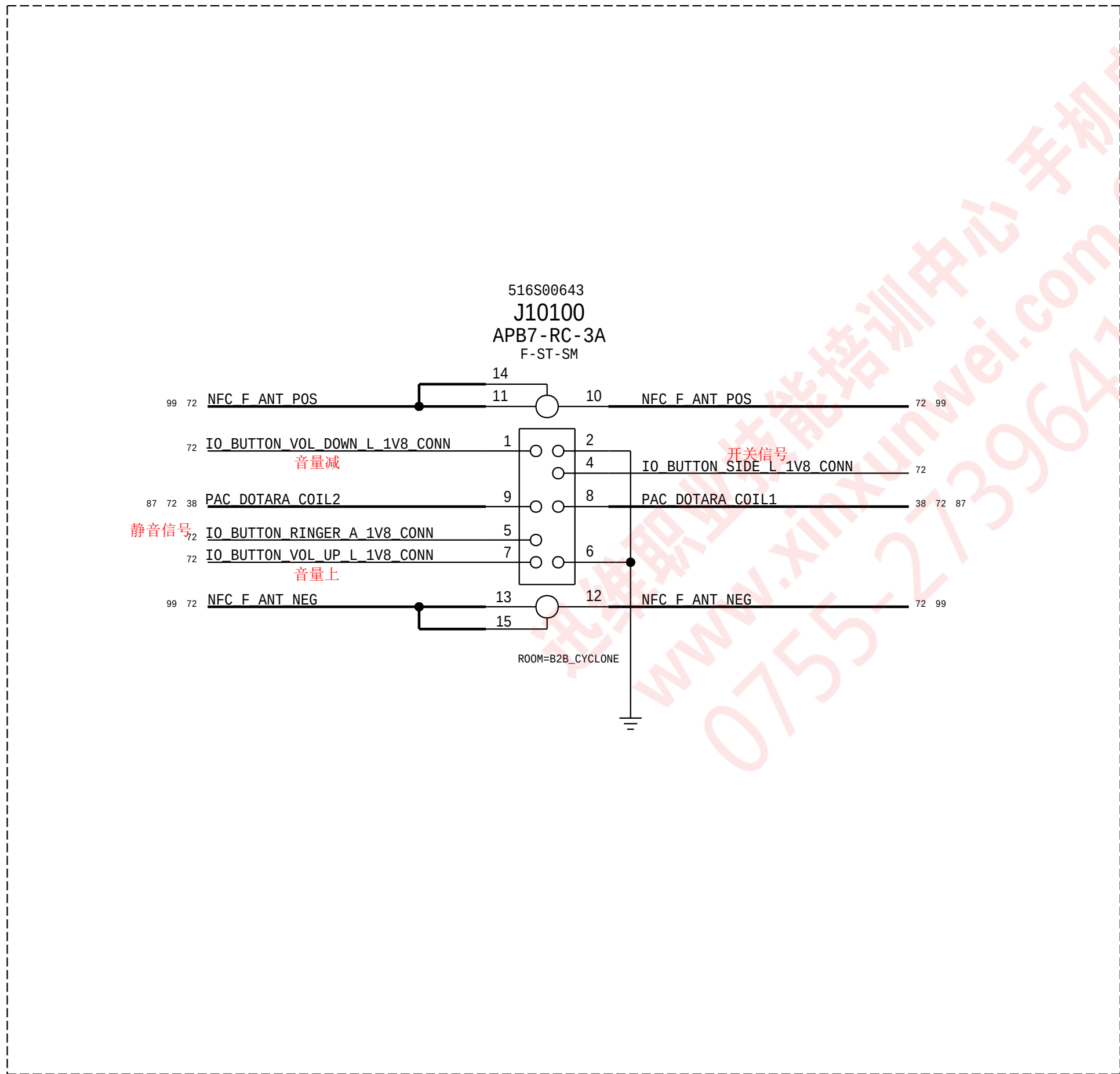


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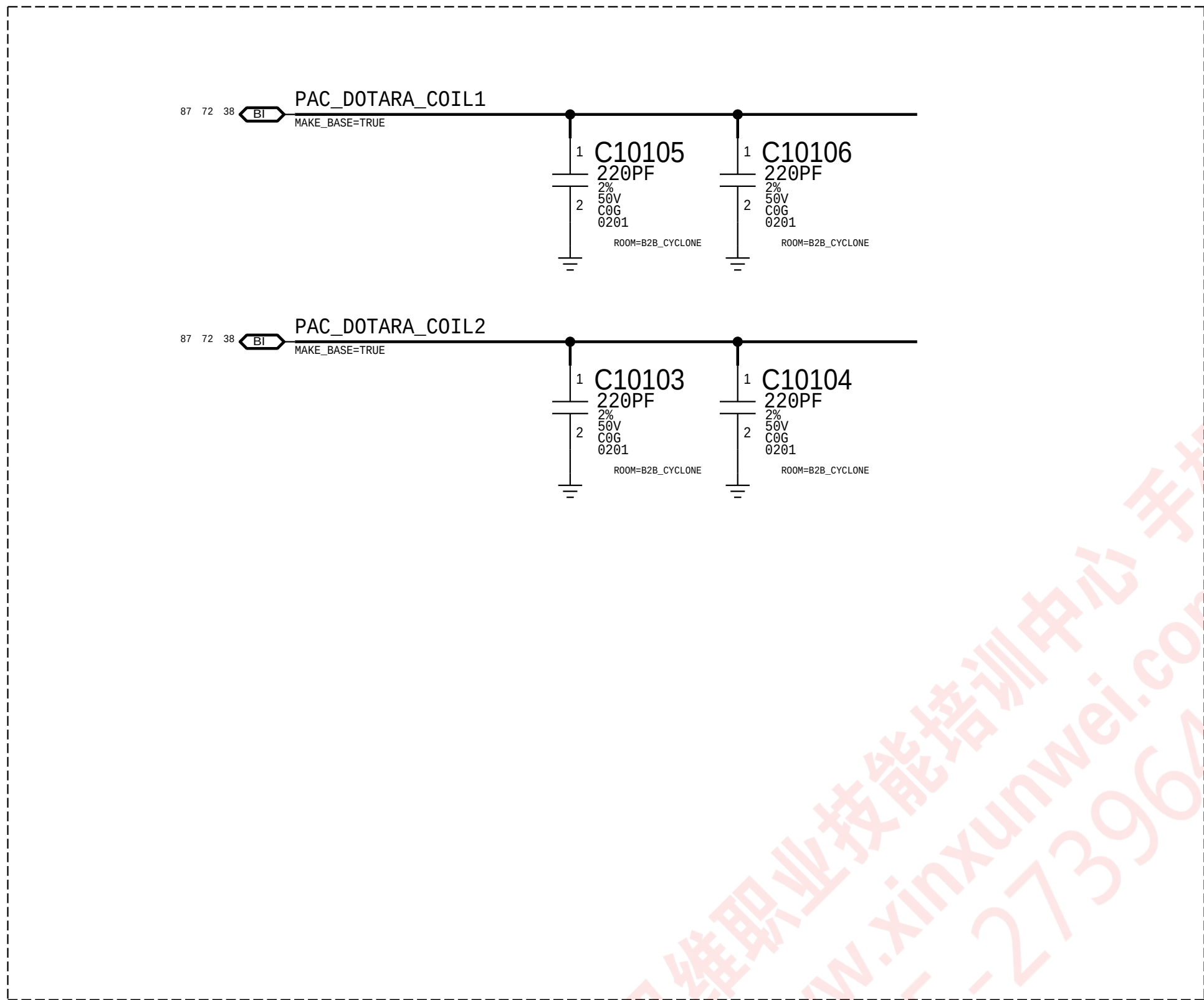
PAGE TITLE		
B2B: Battery		
	DRAWING NUMBER	051-05215
	REVISION	4.6.0
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Cyclone B2B

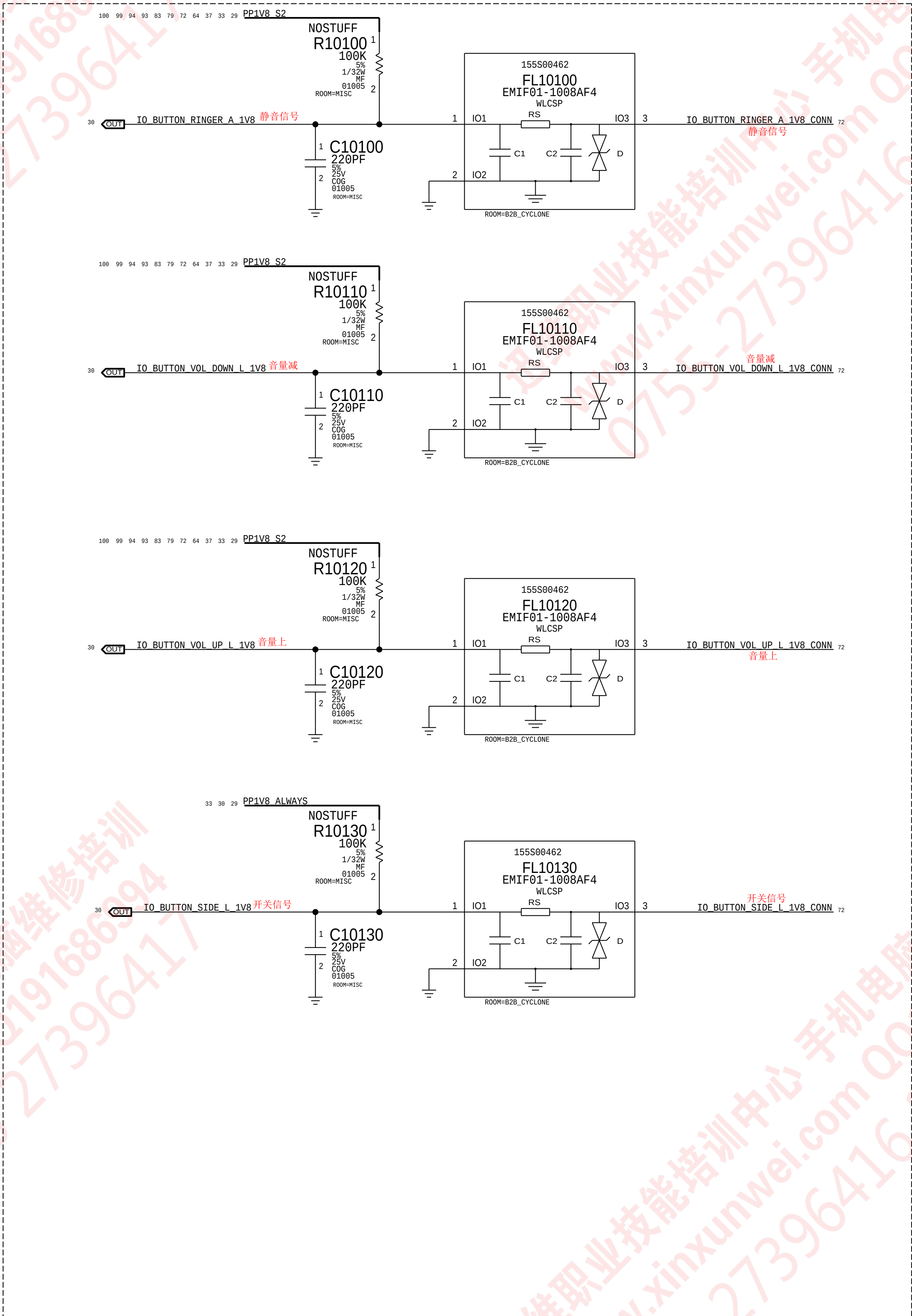
CONNECTOR



DOTARA FILTERS



BUTTON FILTERS



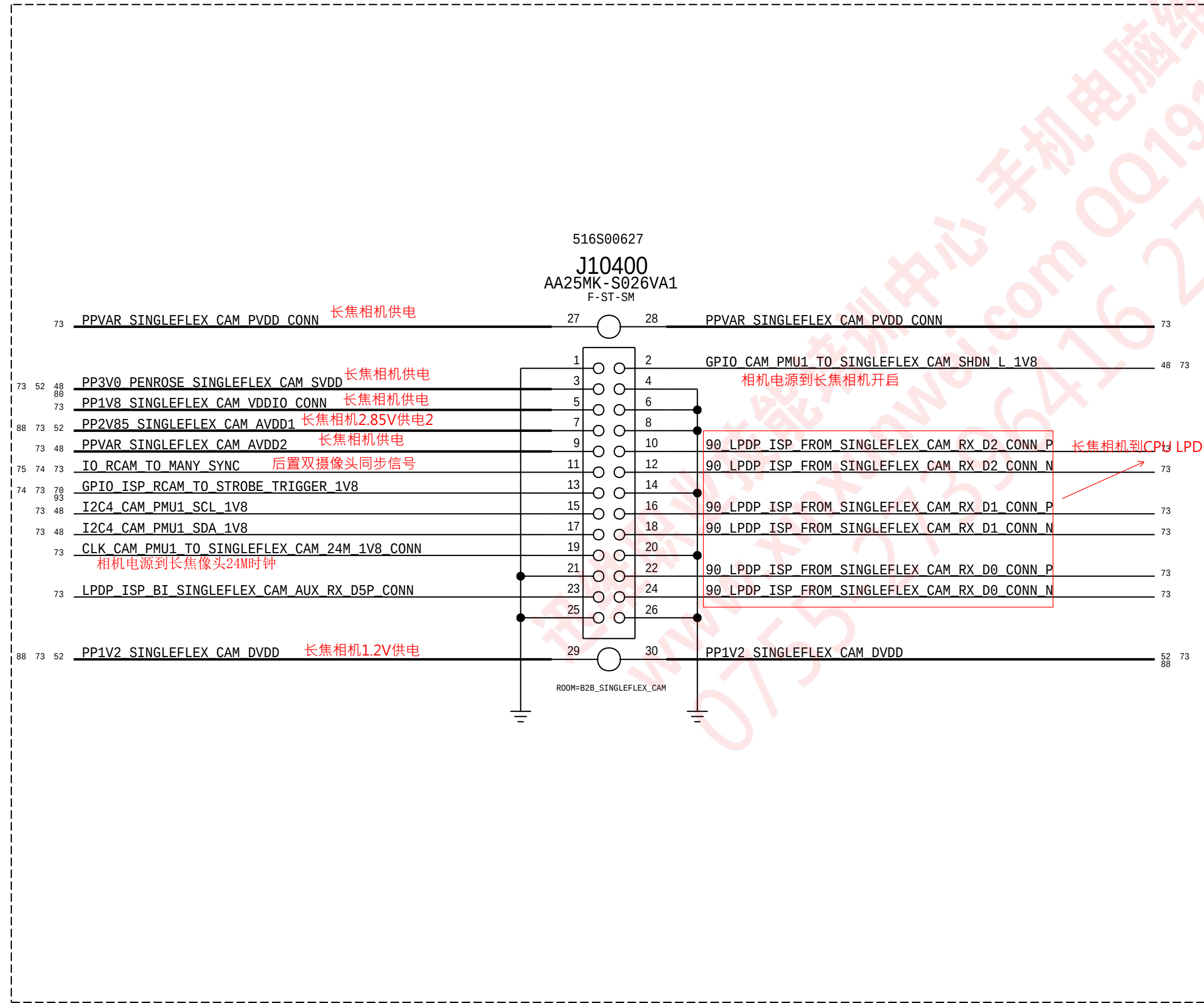
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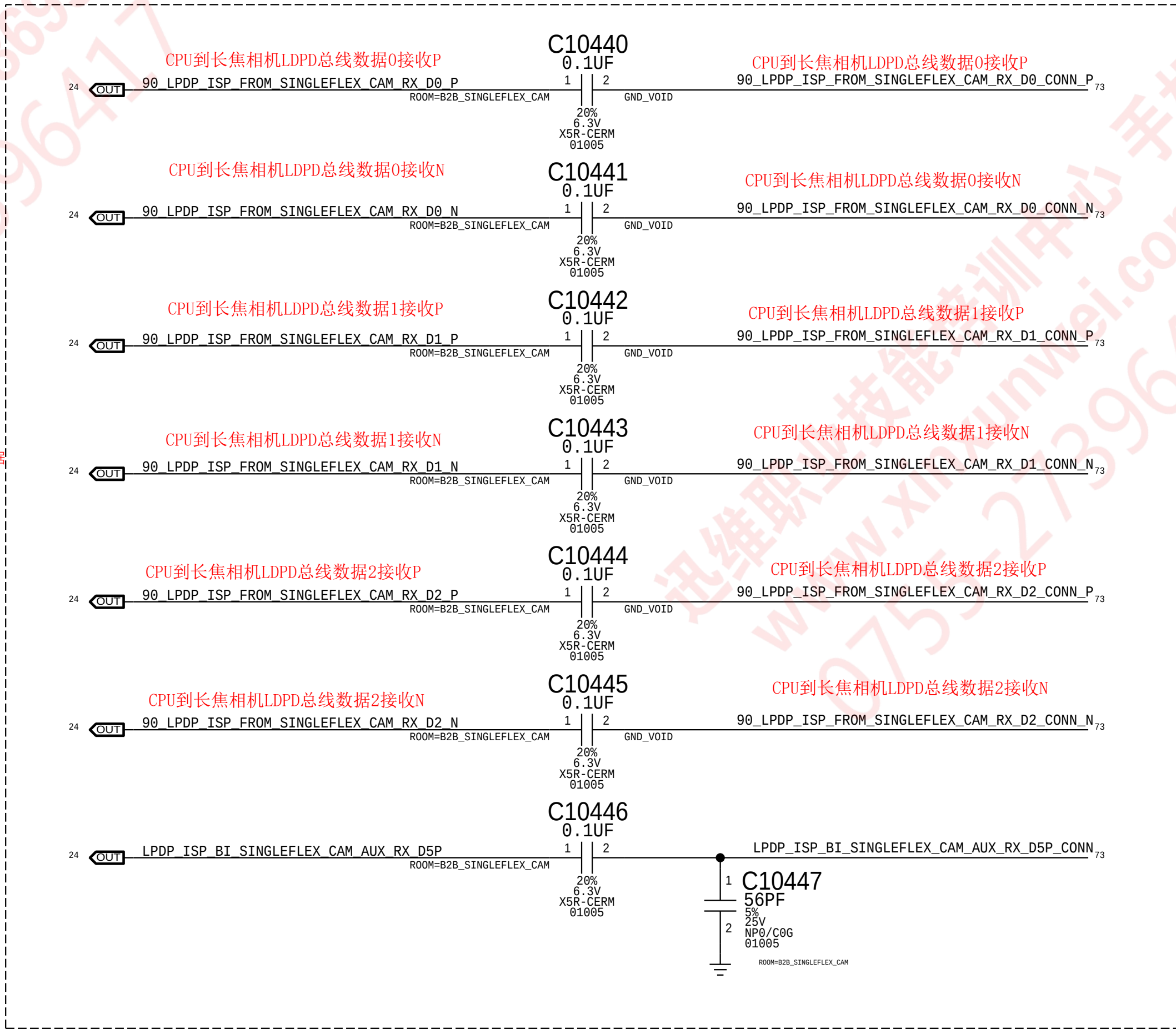
SINGLE-FLEX CAMERA B2B

See page 52 for details

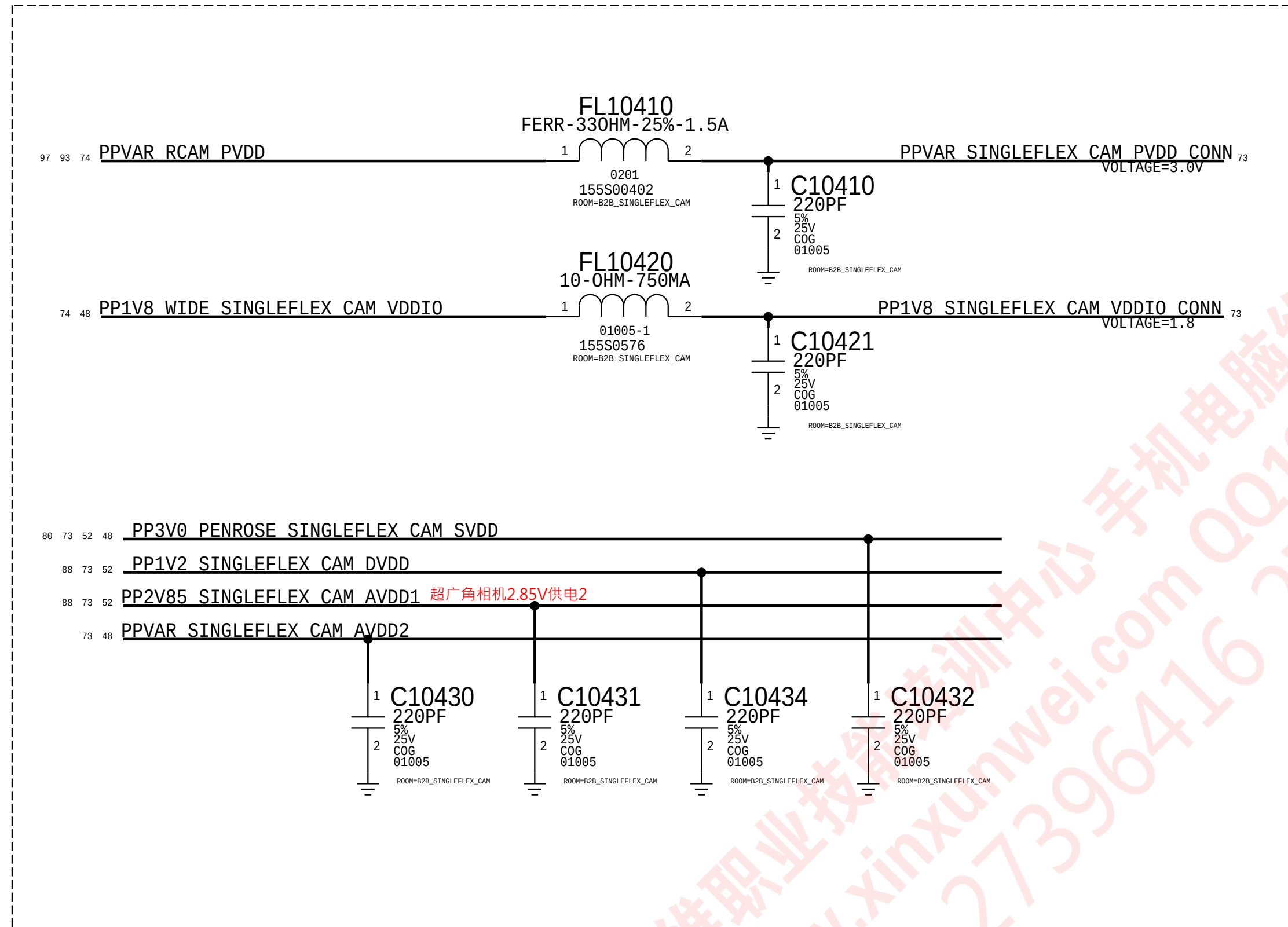
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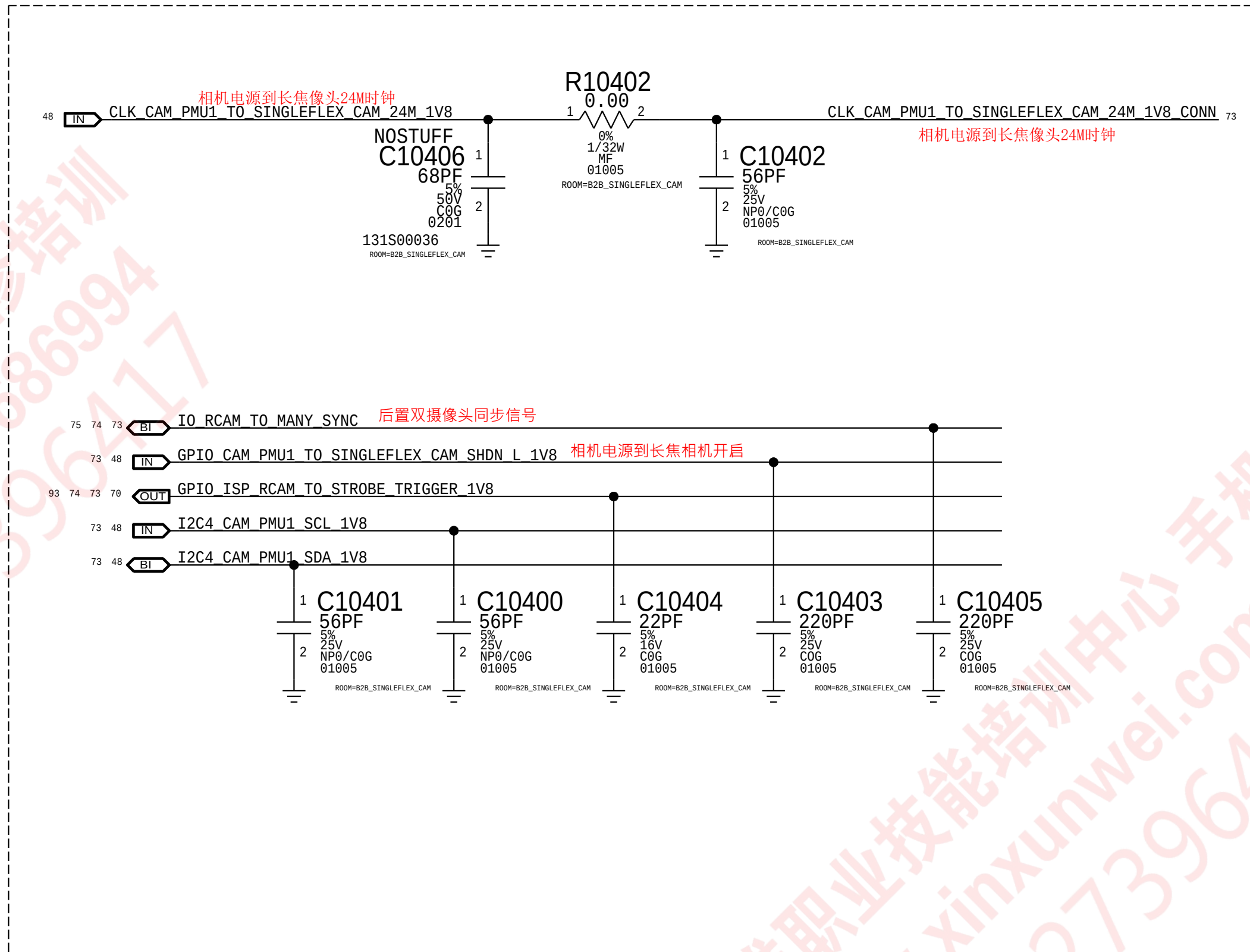
LPDP FILTERS



POWER FILTERS



I/O FILTERS



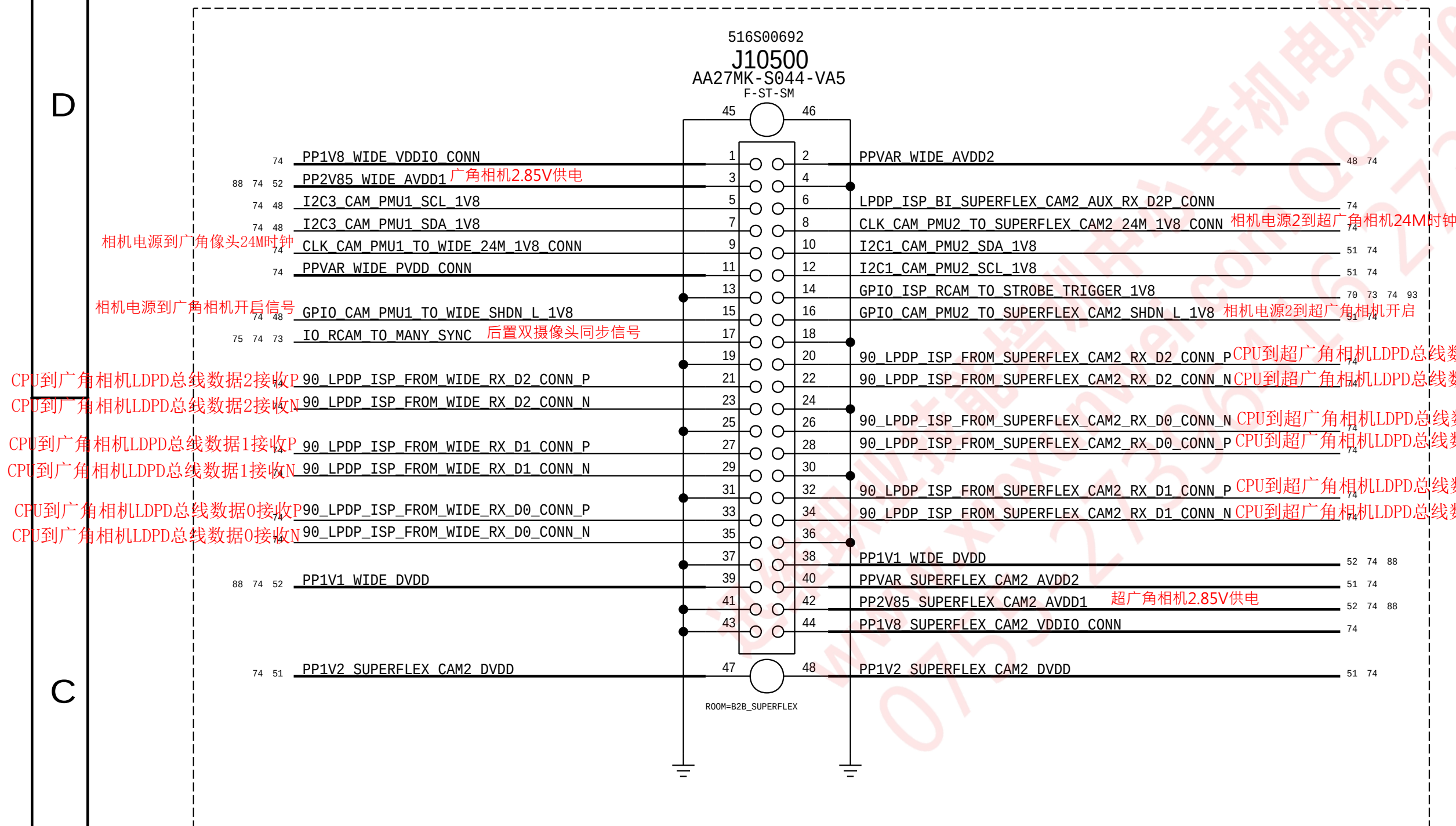
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B2B: RCAM: Single-flex		
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REVISION		4.6.0
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SUPER-FLEX B2B

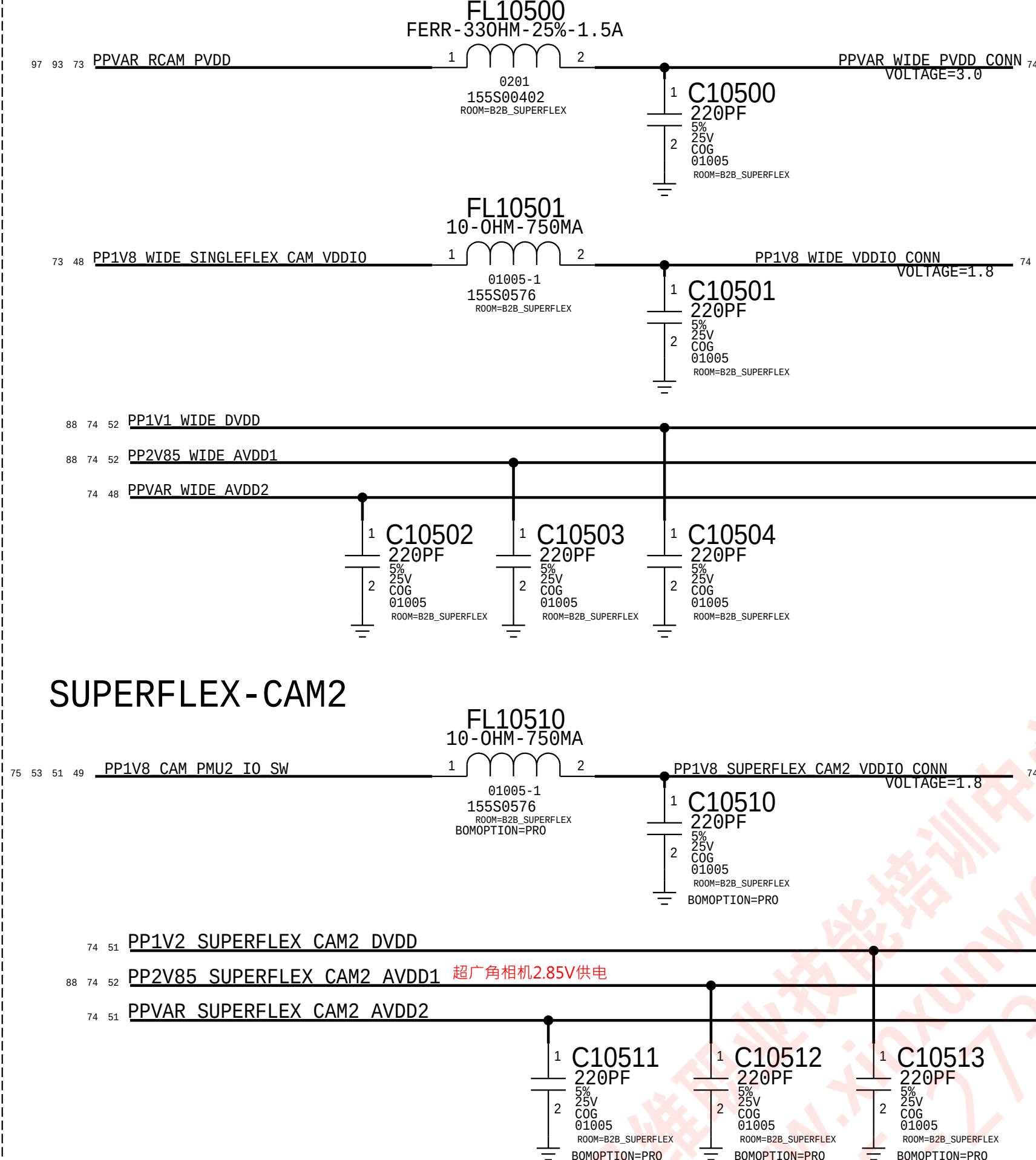
See page 52 for details

CONNECTOR



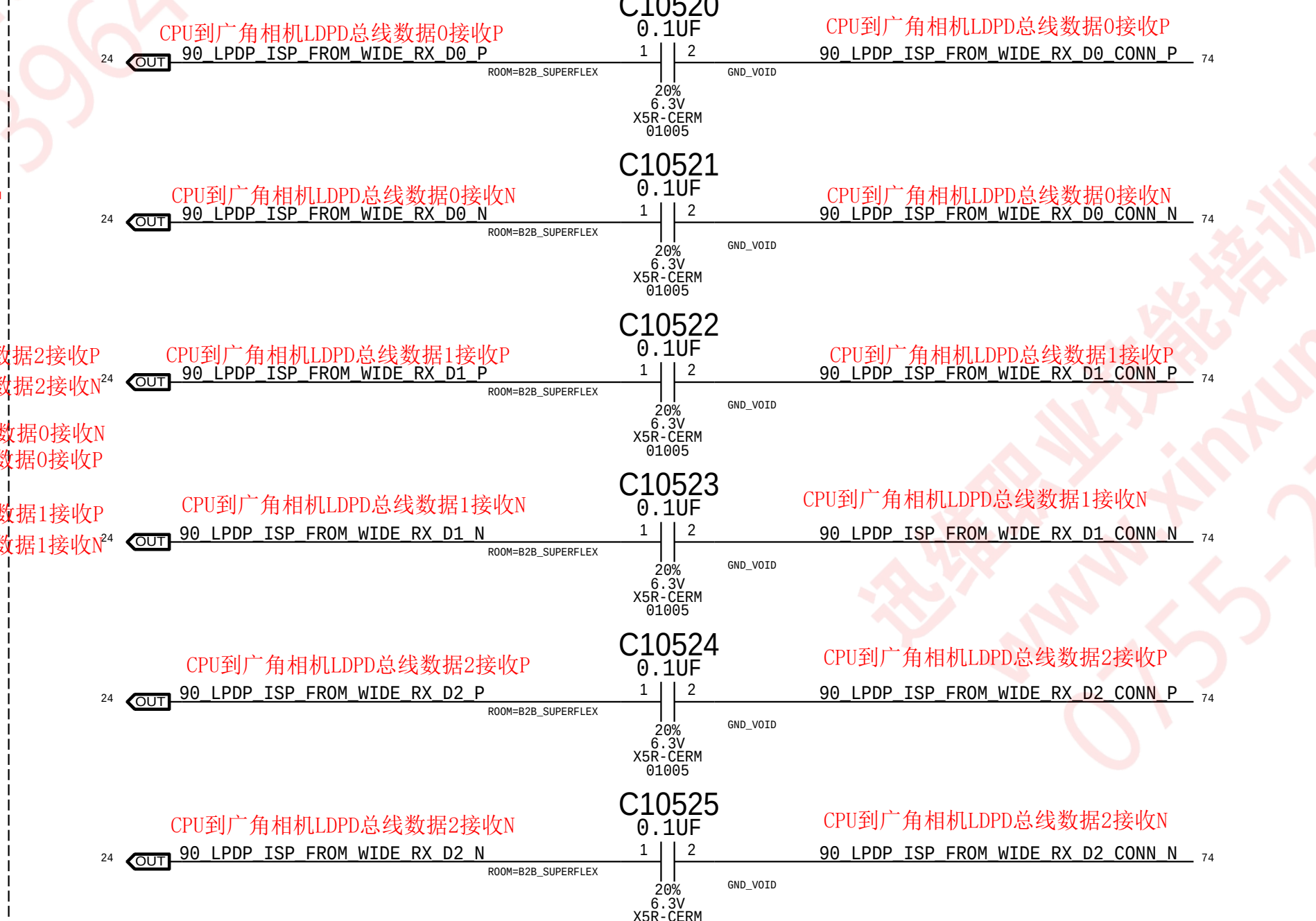
POWER FILTERS

WIDE

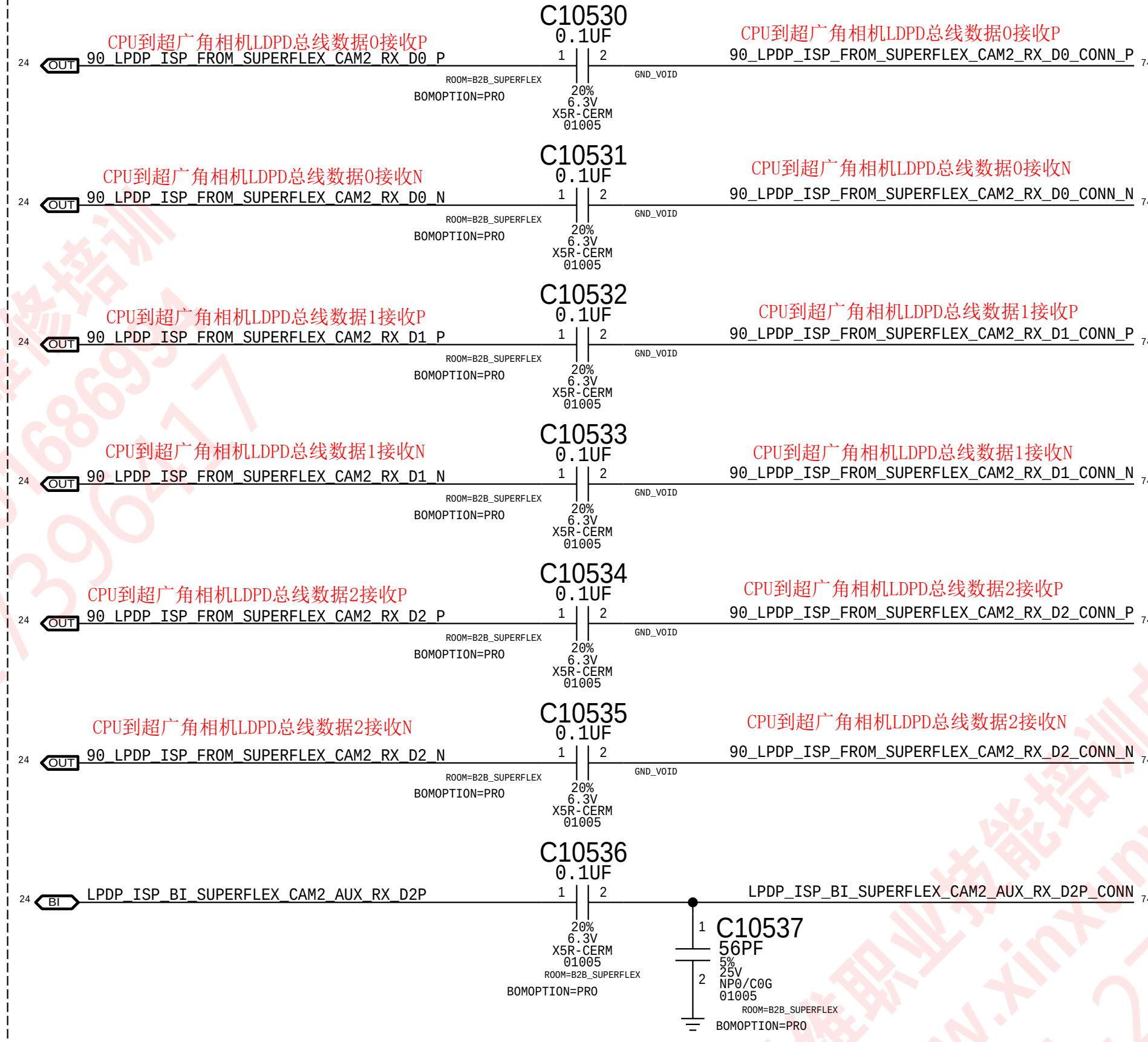


LPDP FILTERS

WIDE

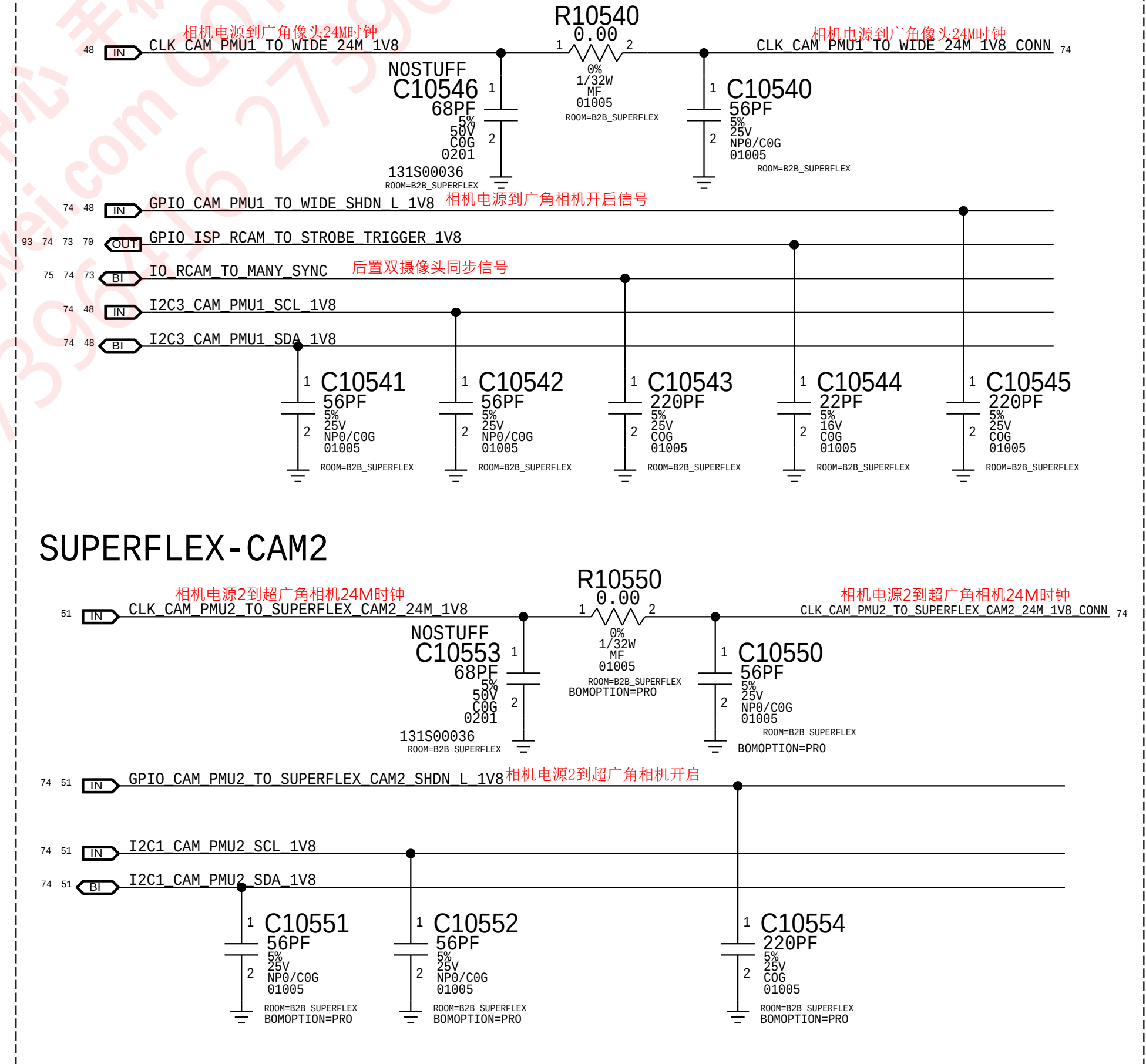


SUPERFLEX-CAM2



IO FILTERS

WIDE



PAGE TITLE			
B2B: RCAM: Super-flex			
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	REVISION 4.6.0		
	BRANCH		
	PAGE 138 OF 191	SHEET	

JASPER CAMERA B2B

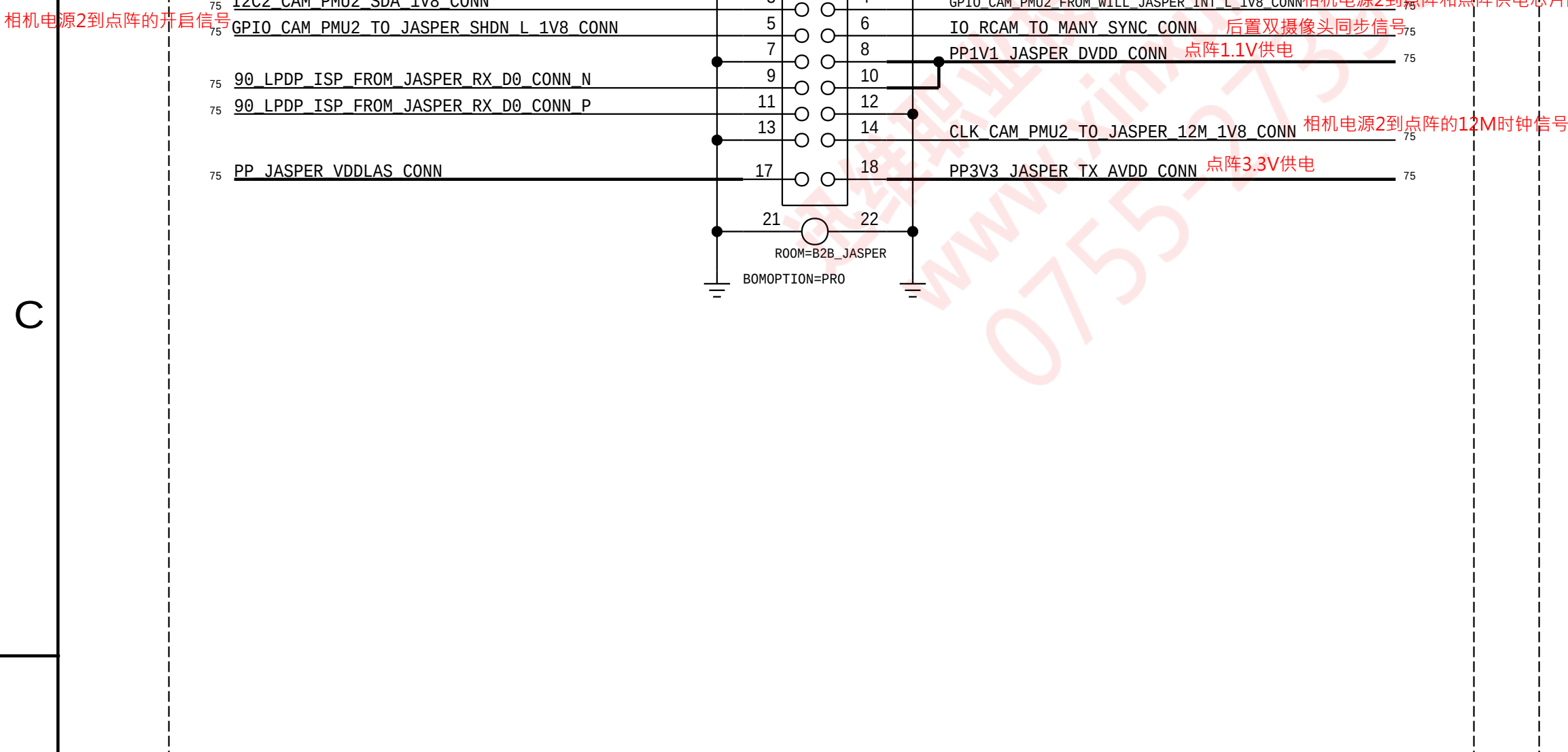
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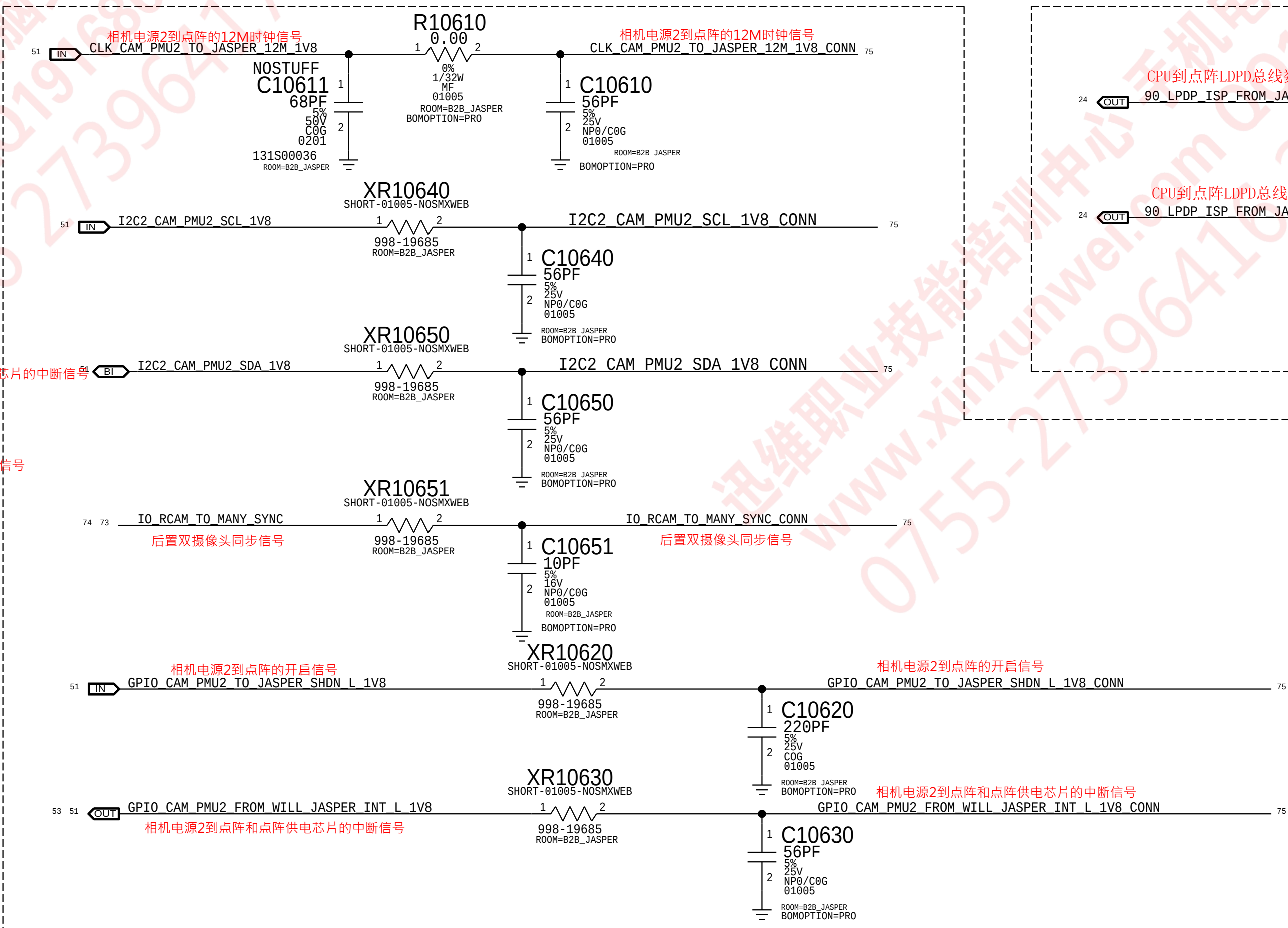
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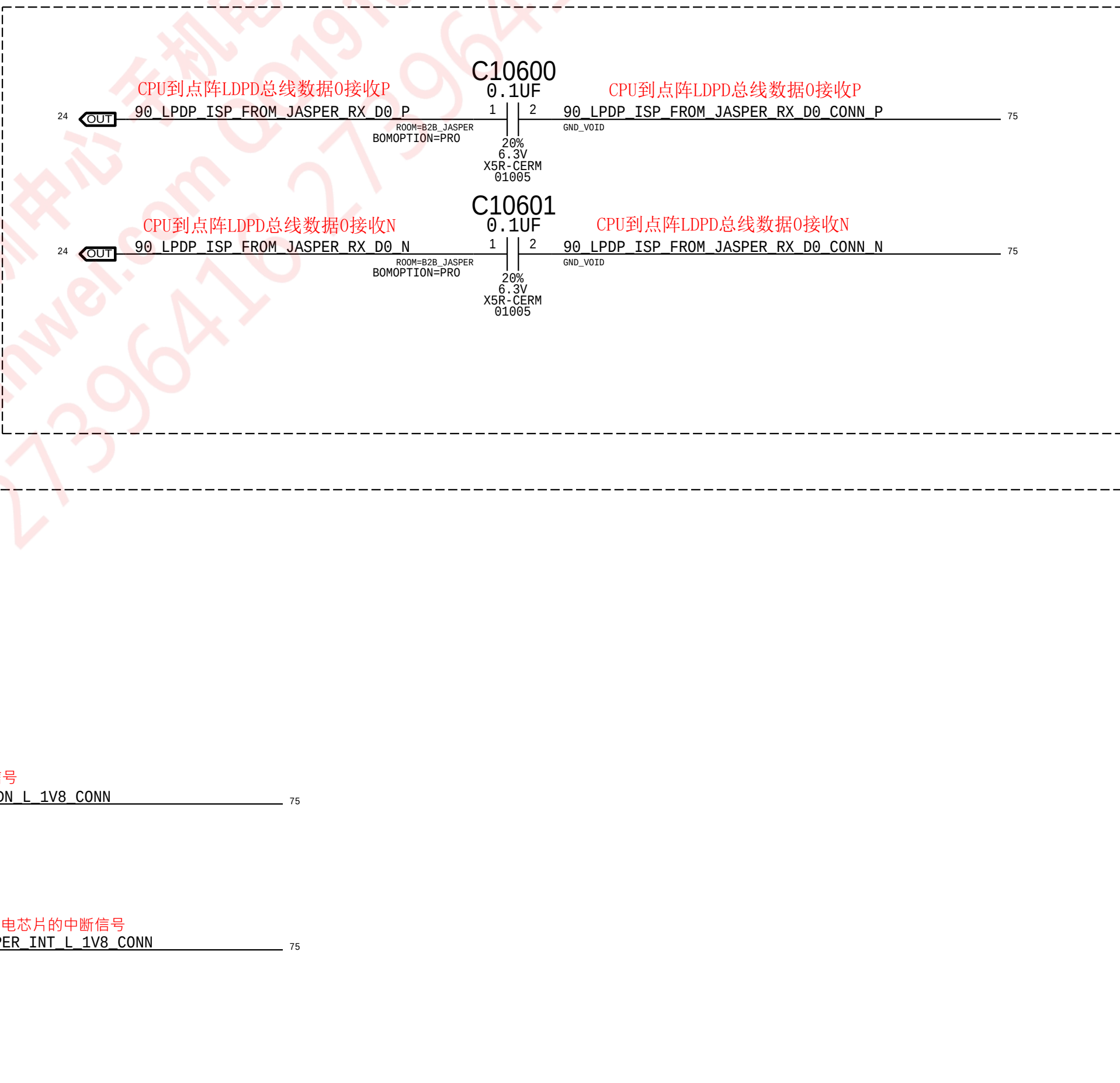
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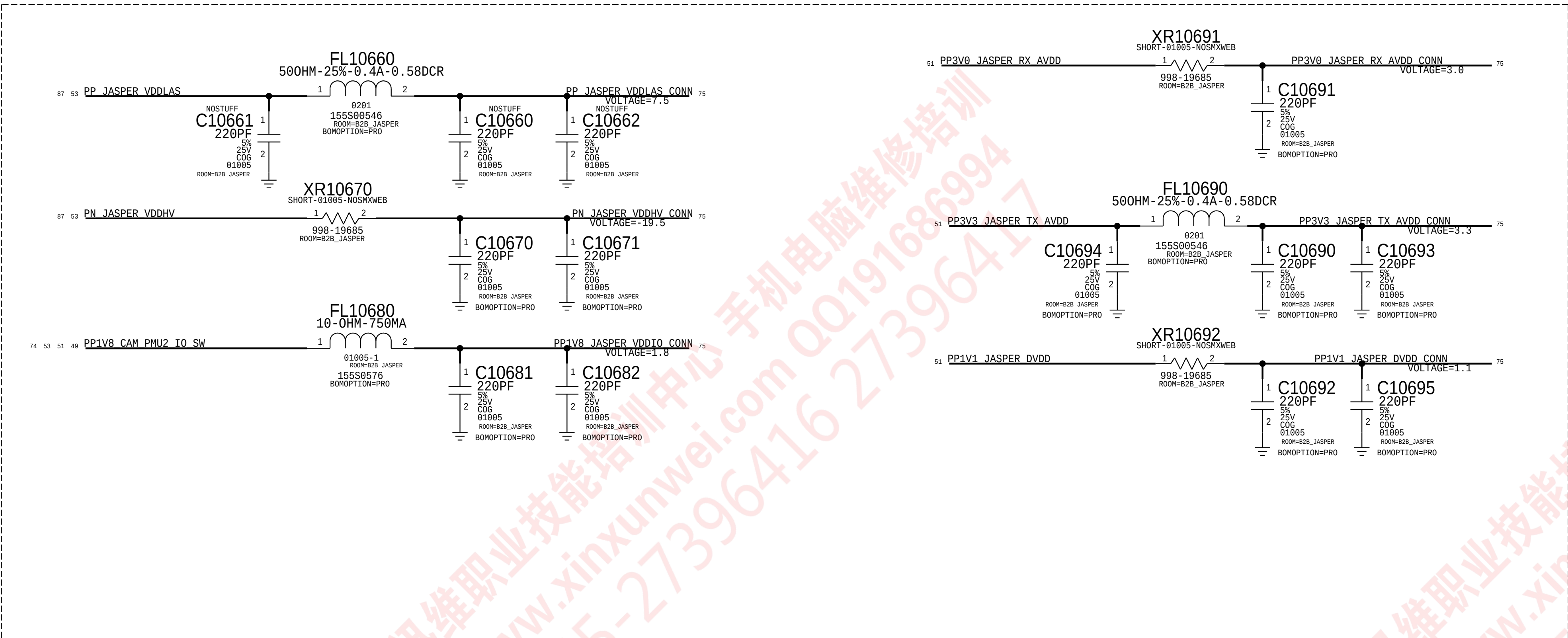
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LPDP FILTERS



POWER FILTERS



SYNCING: NONE

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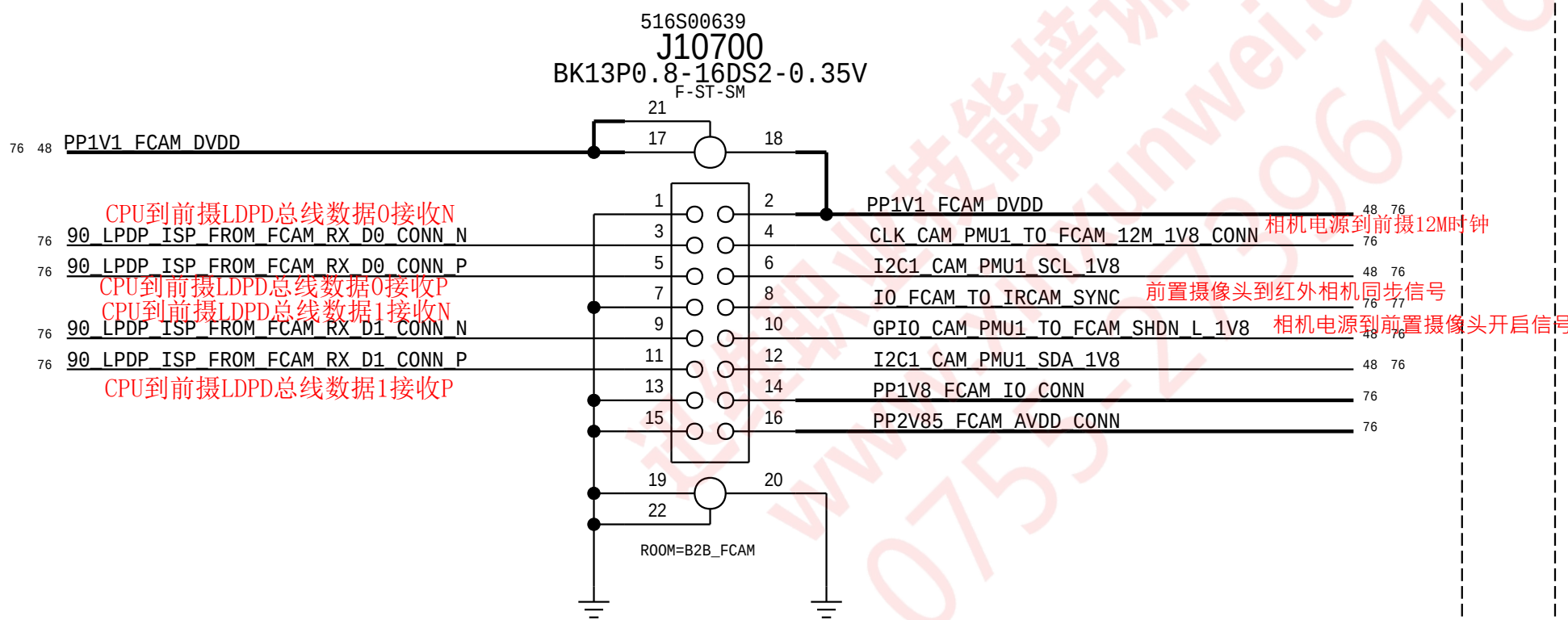
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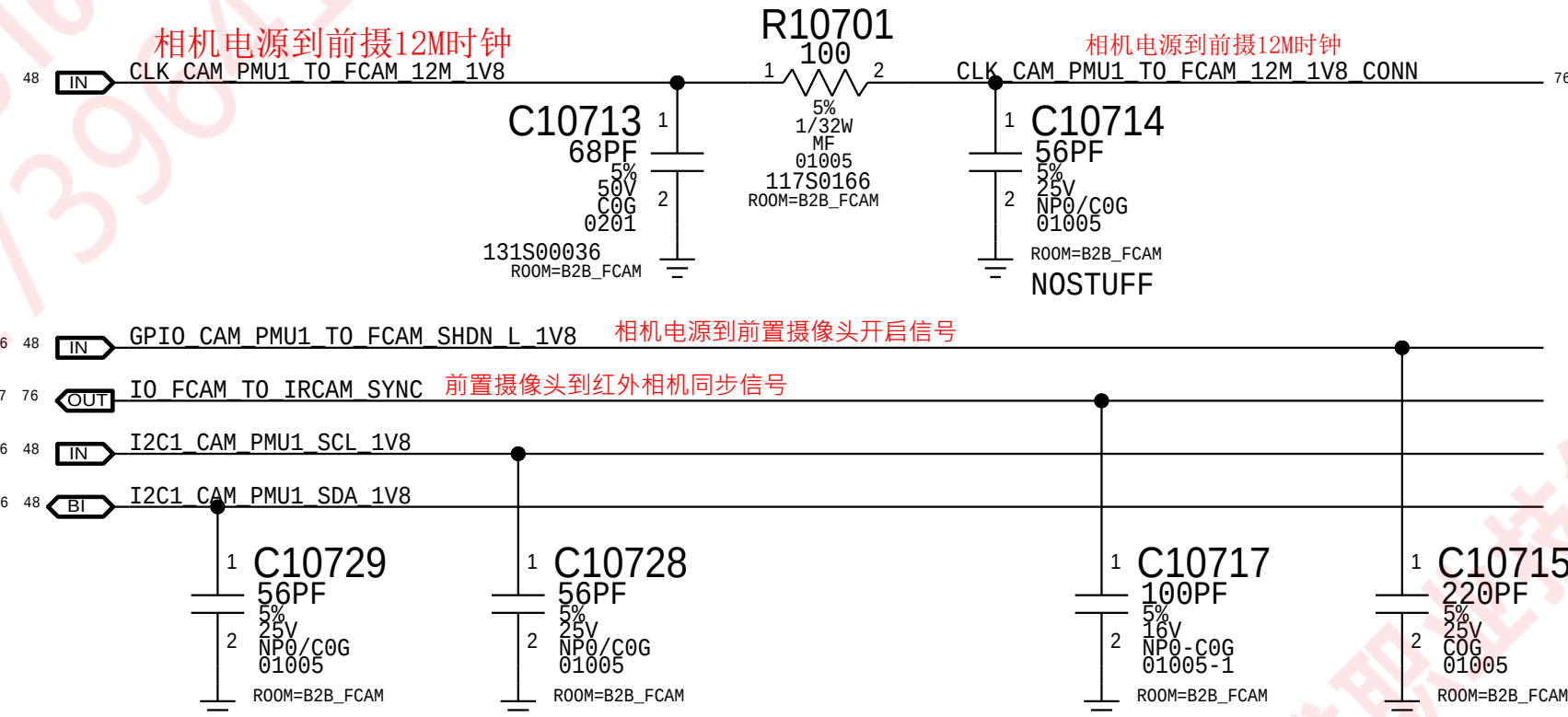
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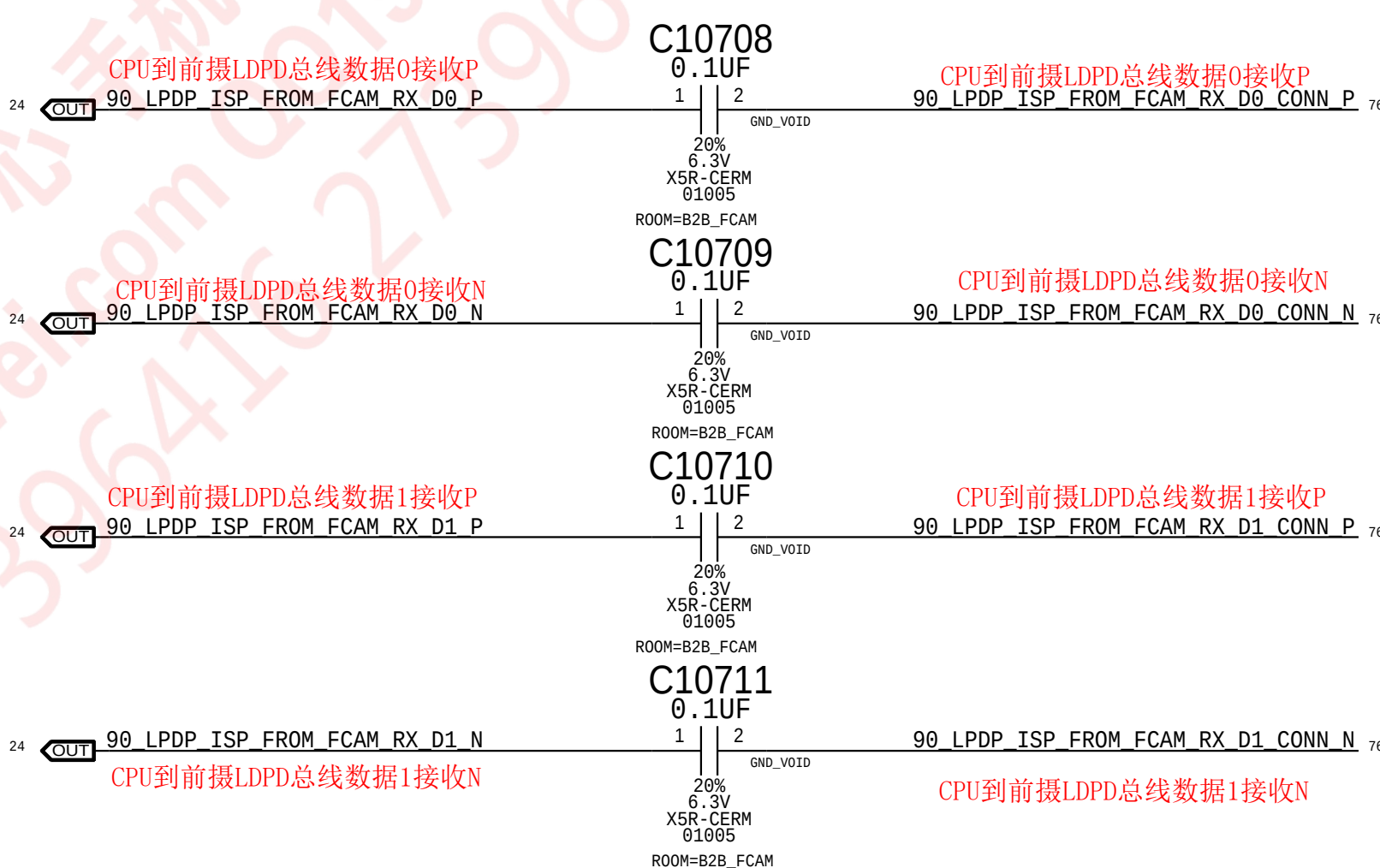
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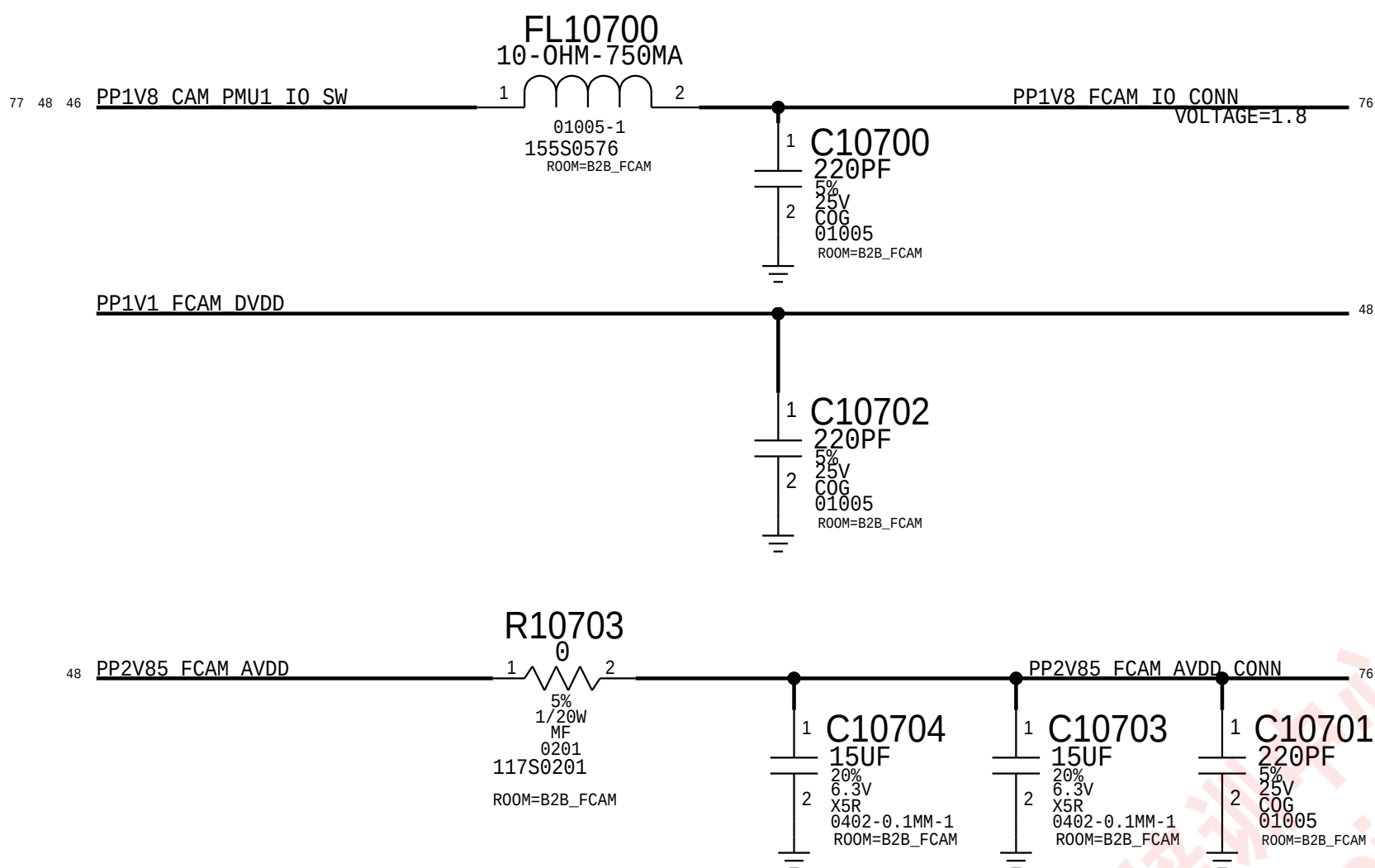
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LPDP FILTERS



POWER FILTERS

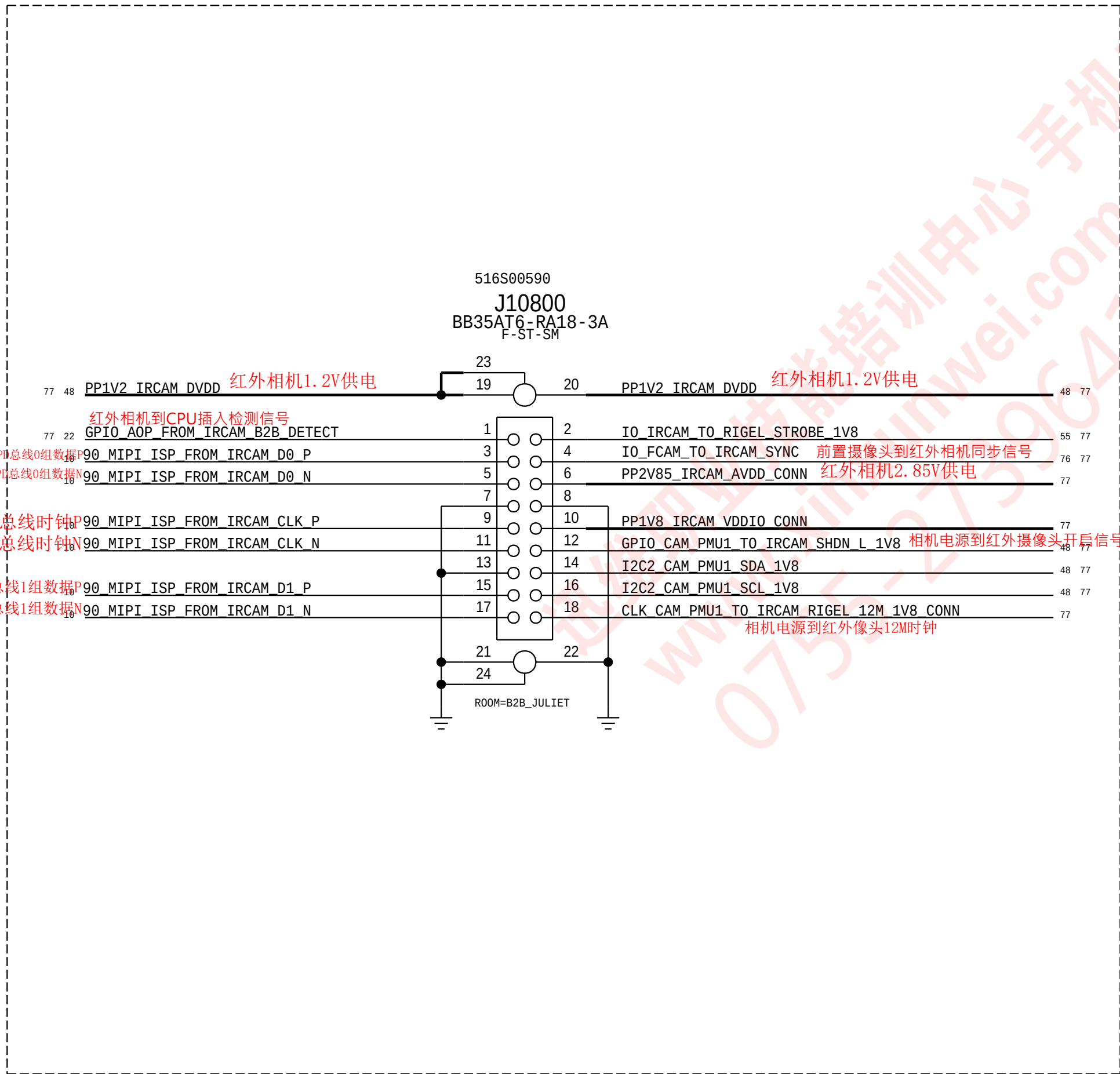


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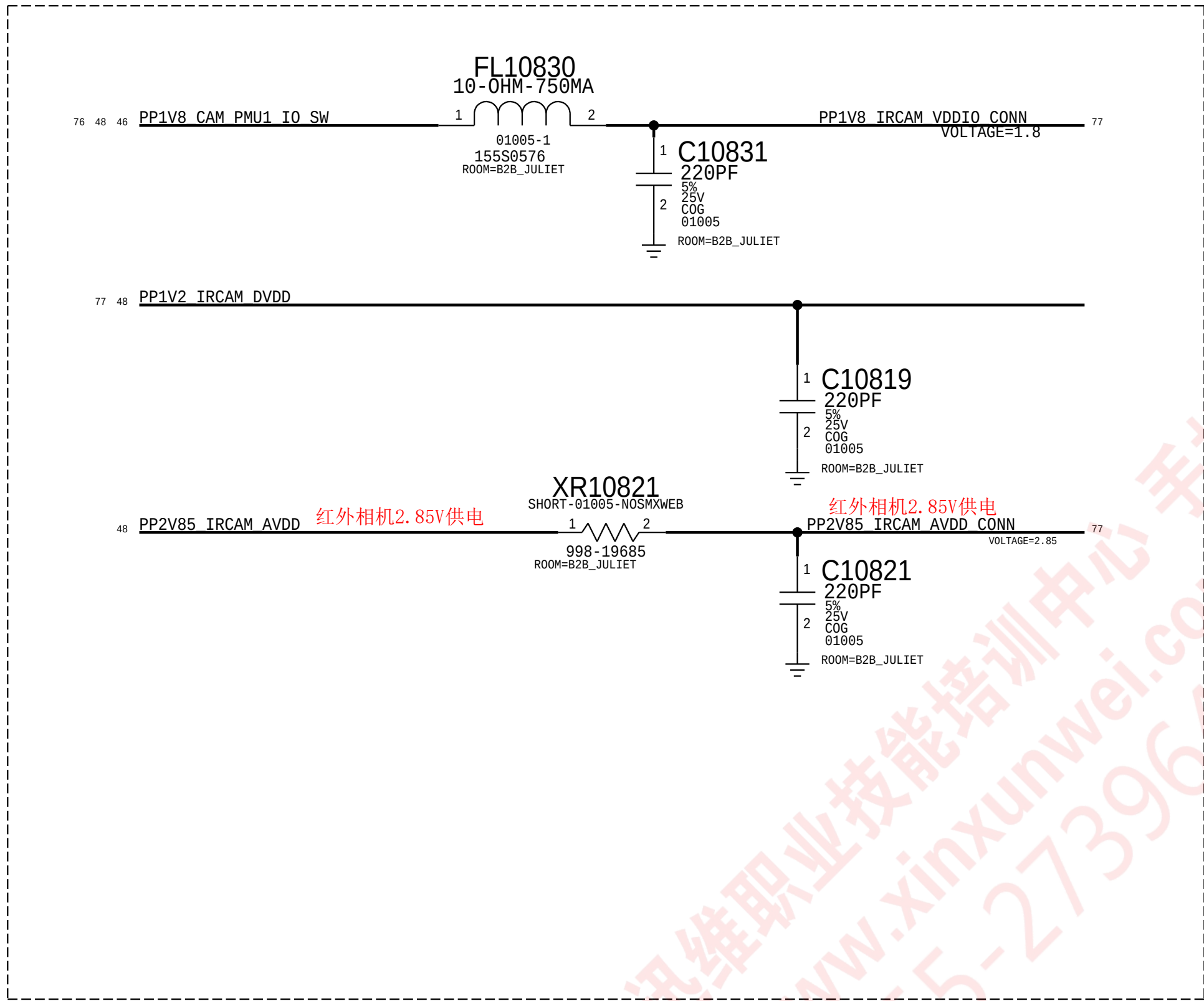
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JULIET B2B

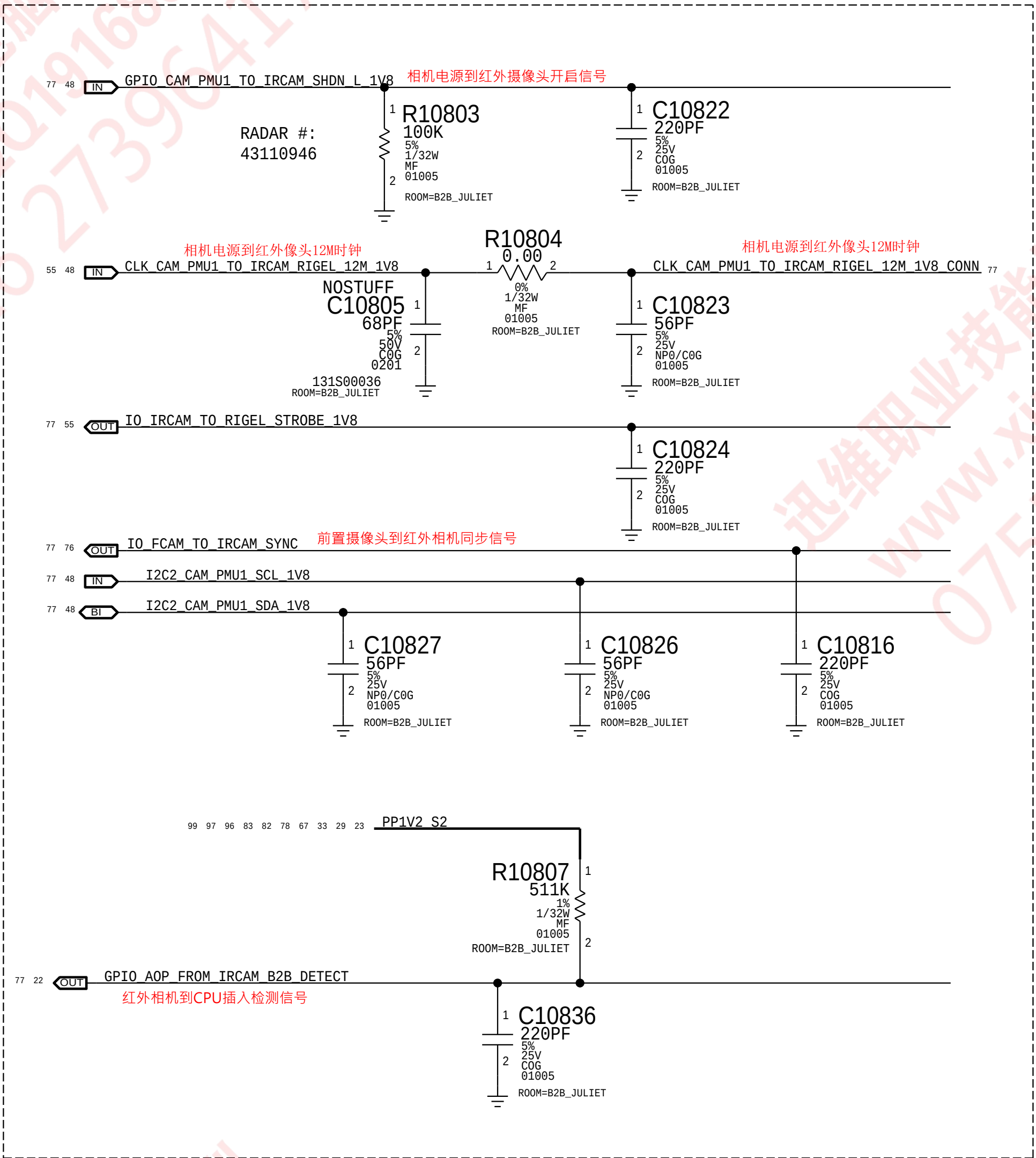
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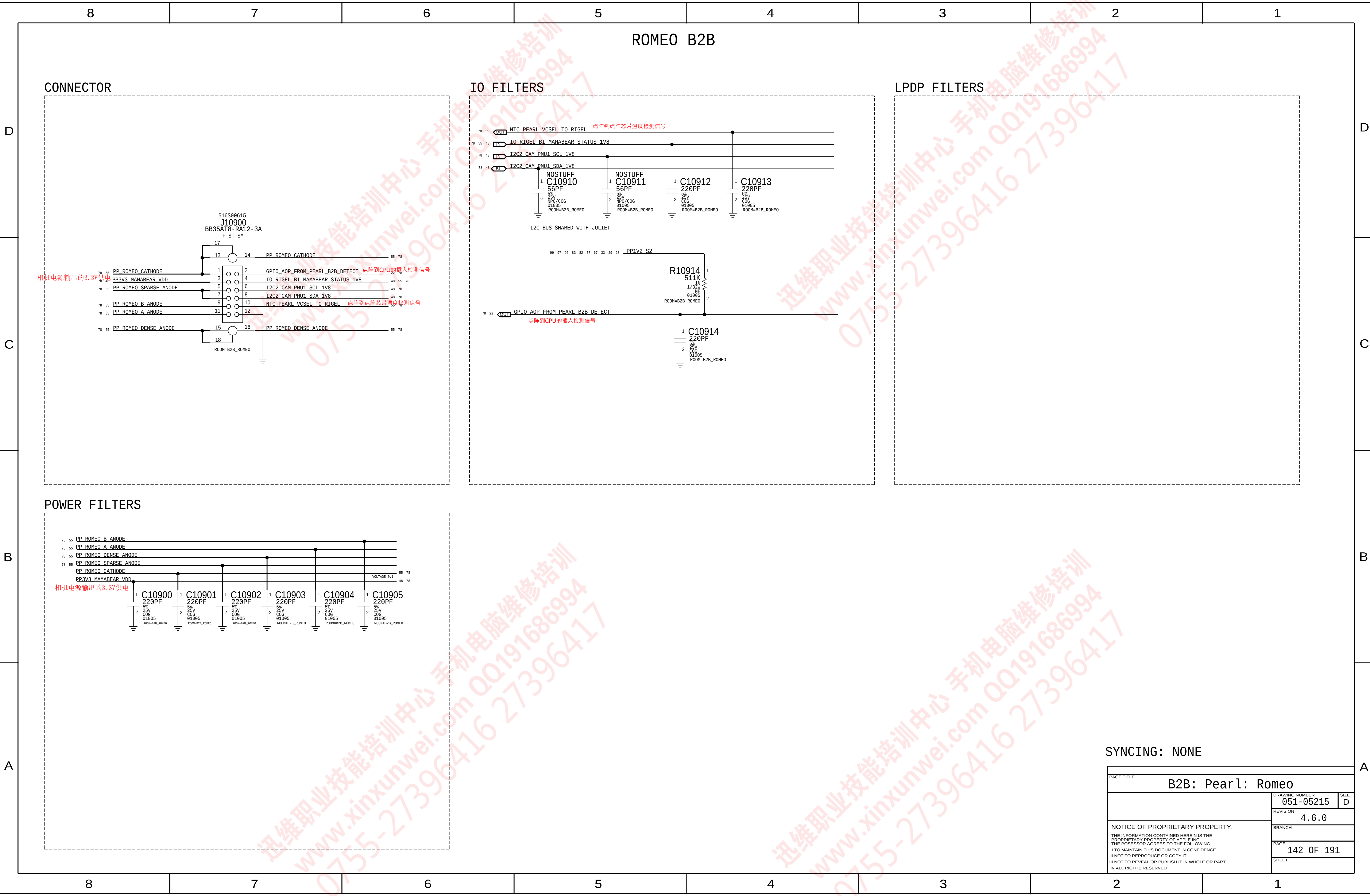
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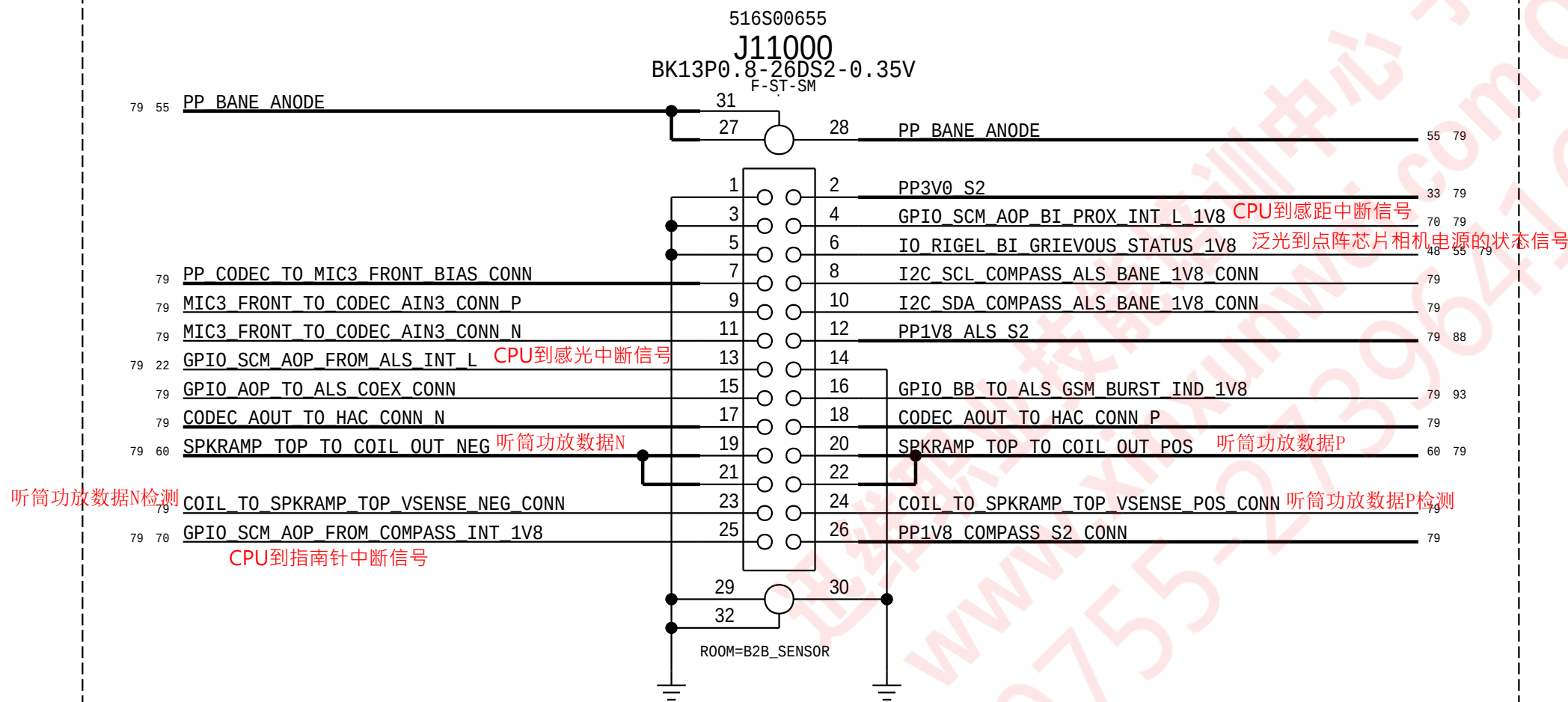


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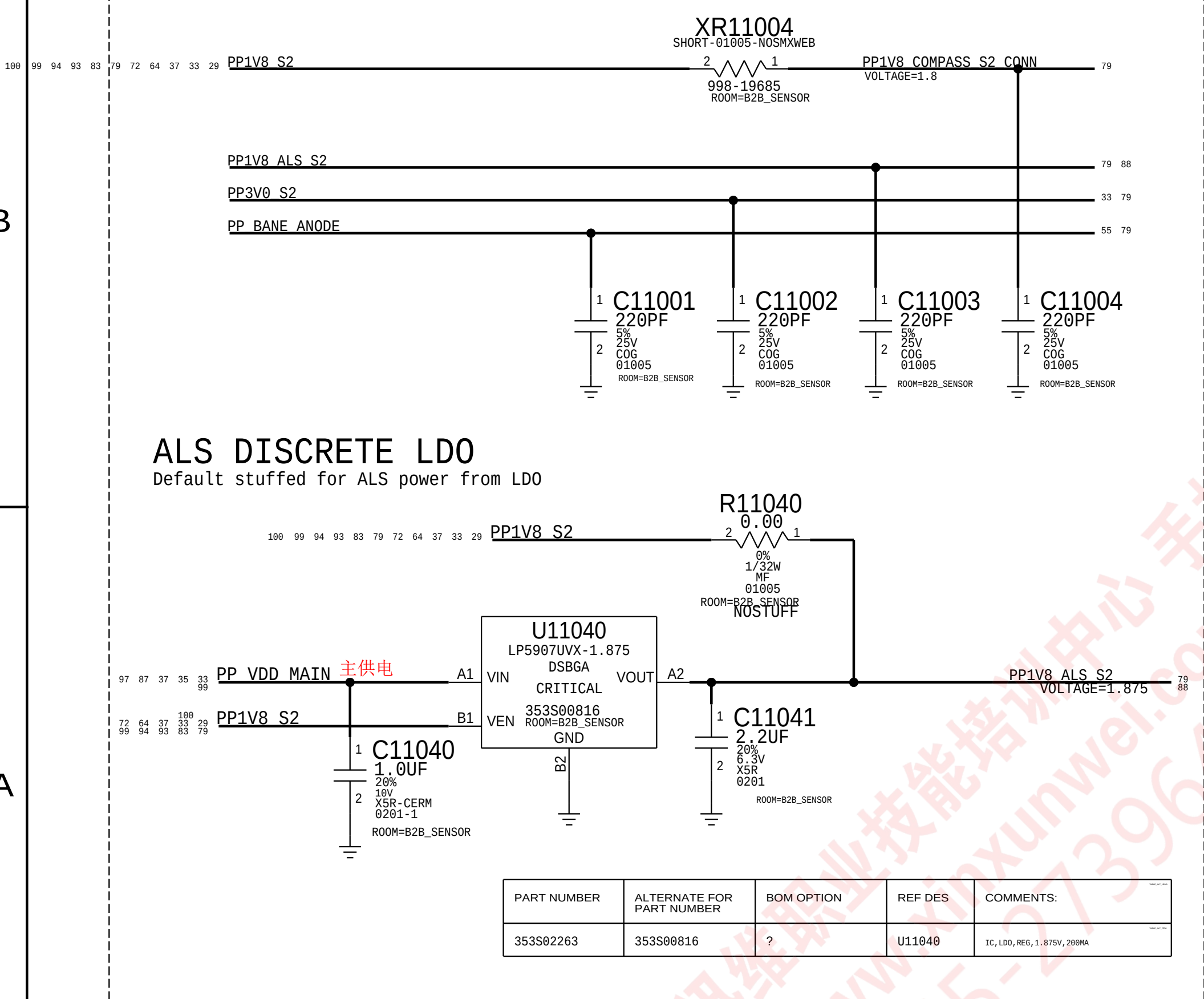


SENSOR B2B

CONNECTOR



POWER FILTERS

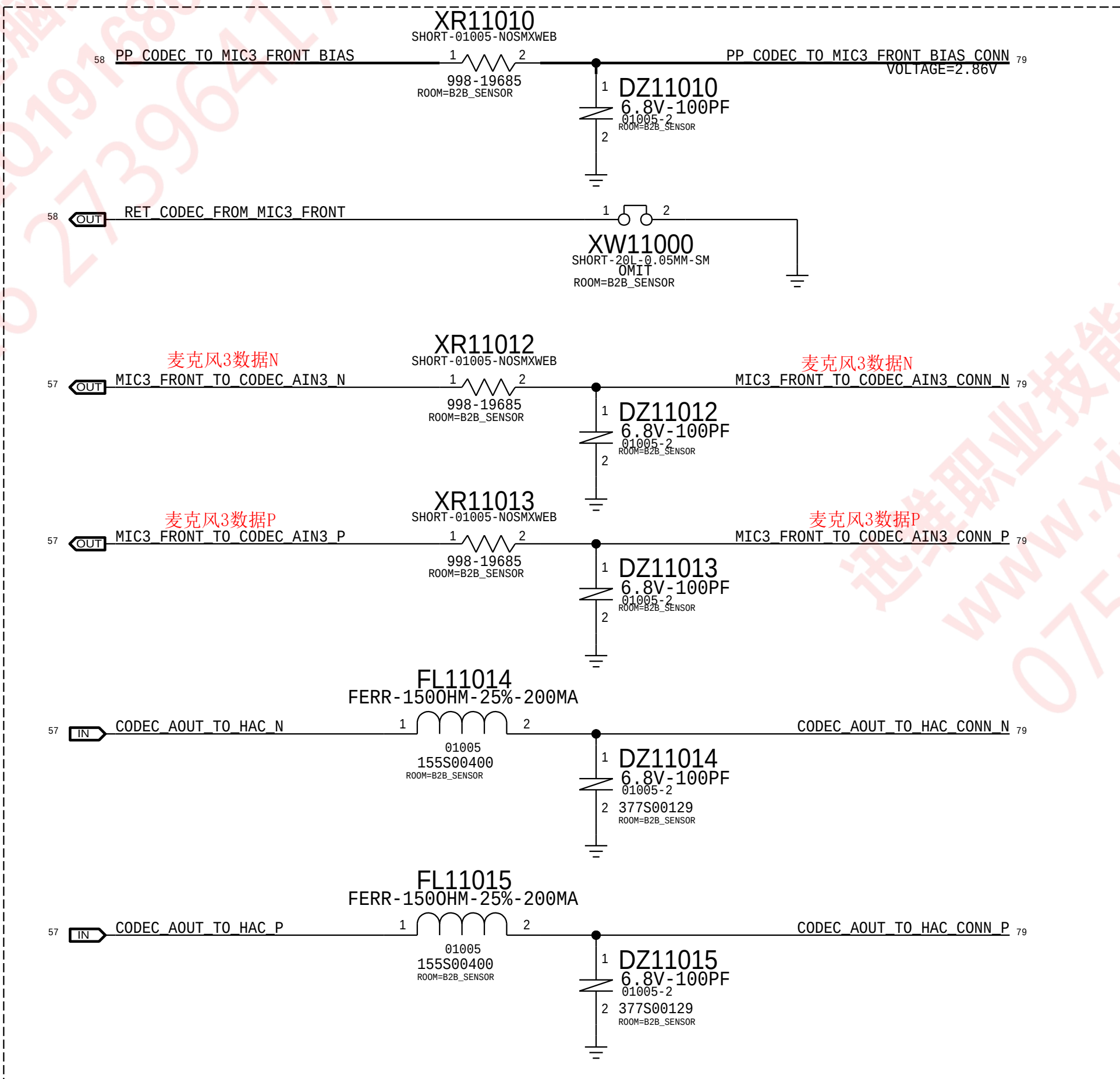


ALS DISCRETE LDO

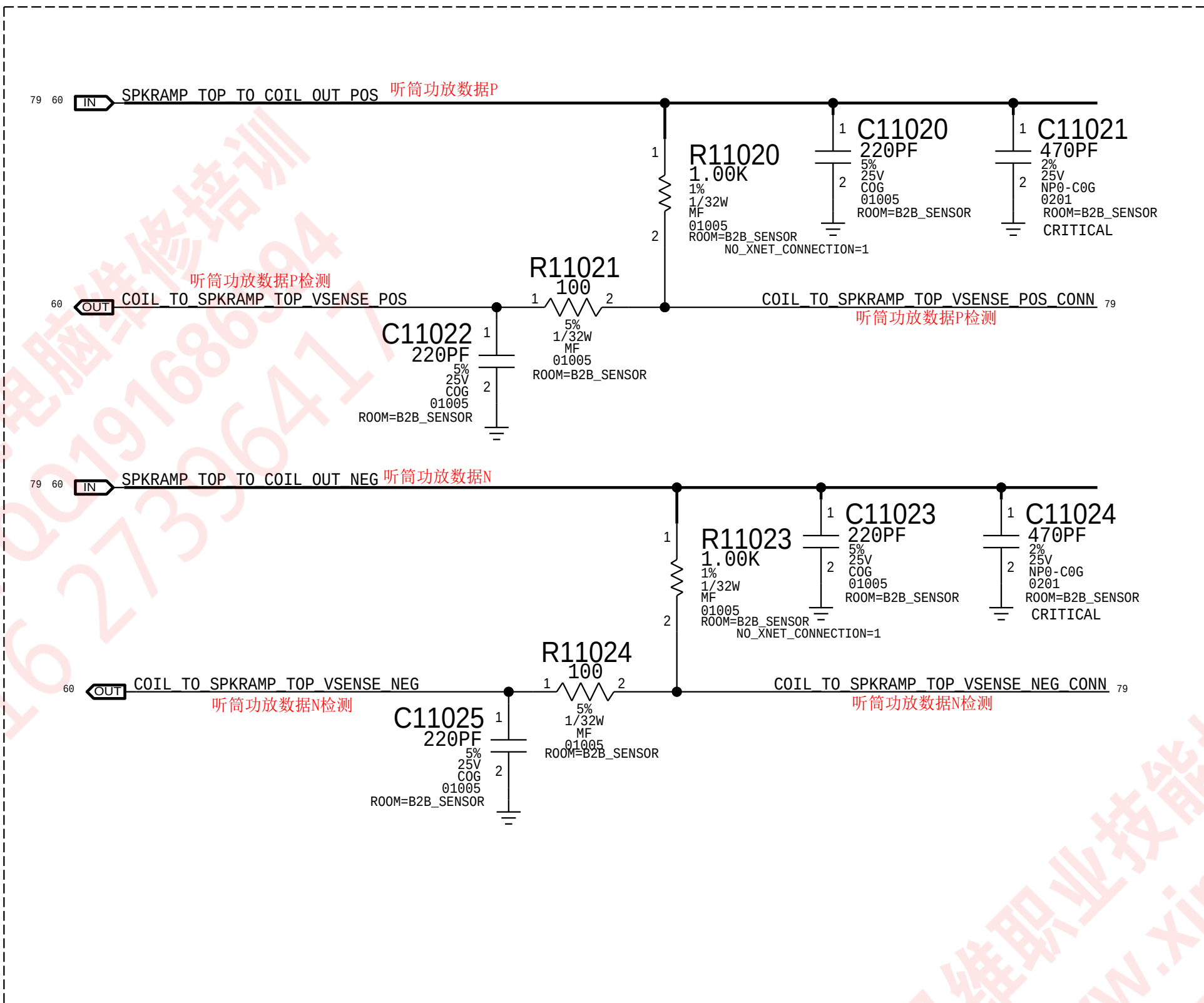
Default stuffed for ALS power from LDO

PART NUMBER	ALTERNATE FOR PART NUMBER	BOM OPTION	REF DES	COMMENTS:
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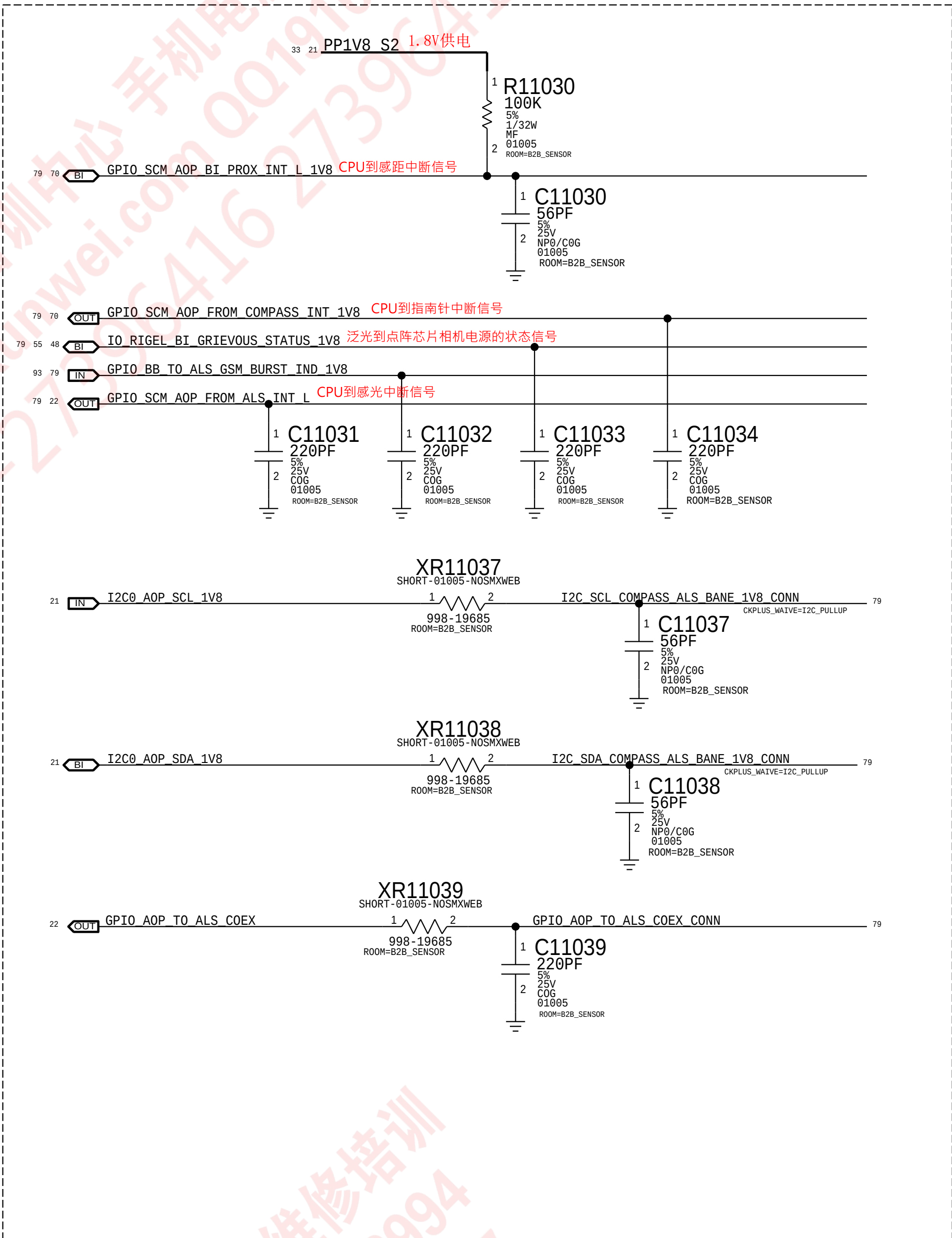
MIC3



SPEAKER2



IO FILTERS

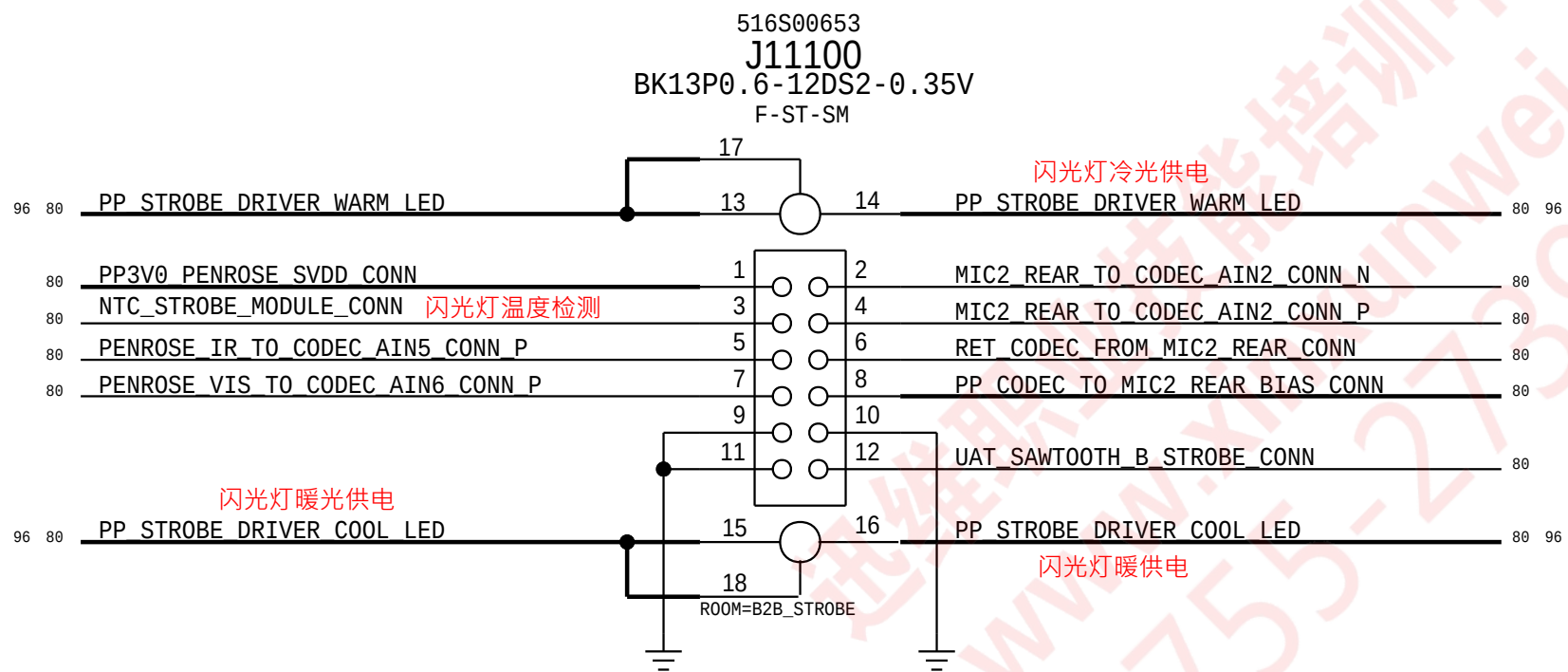


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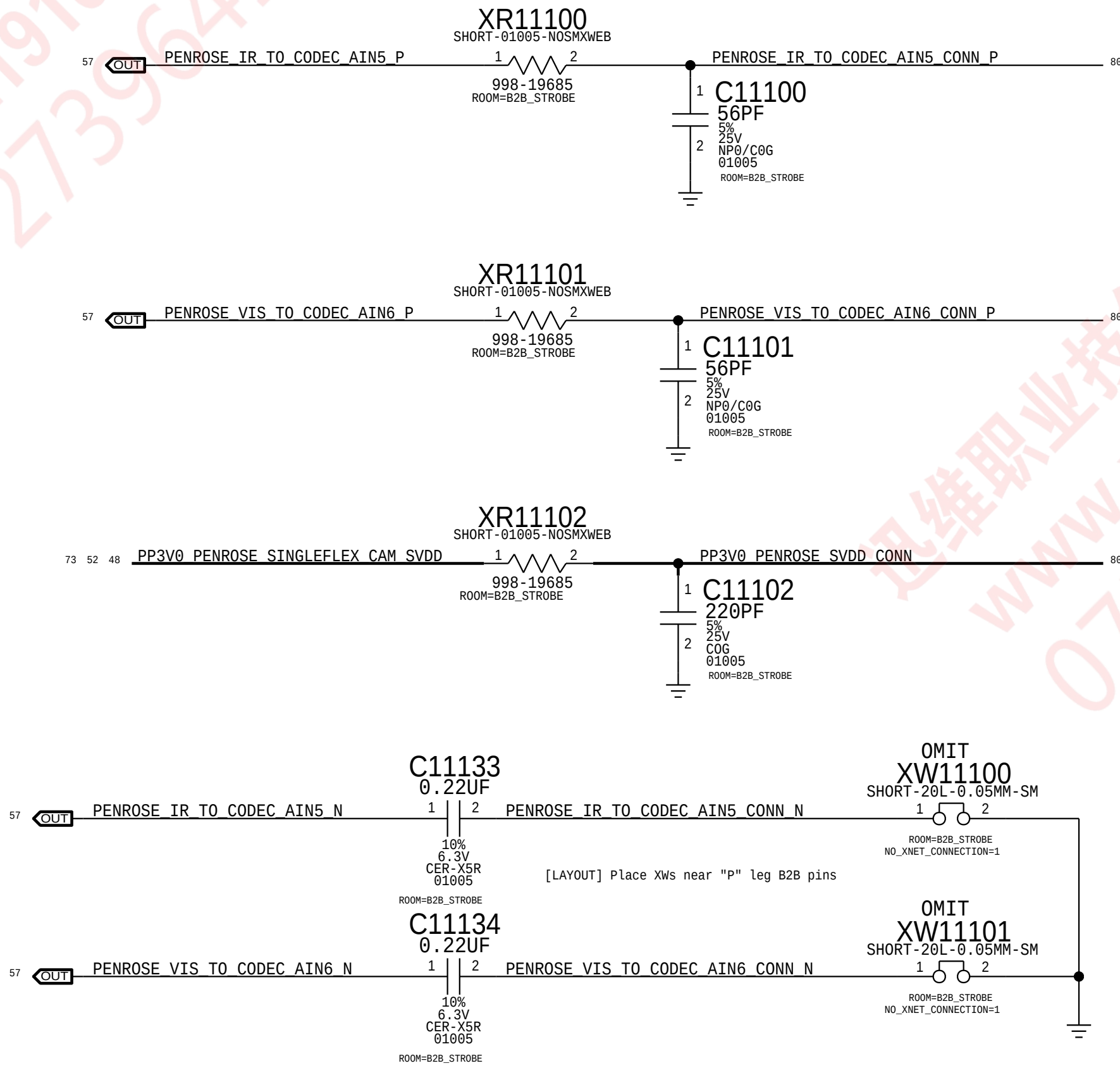
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STROBE B2B

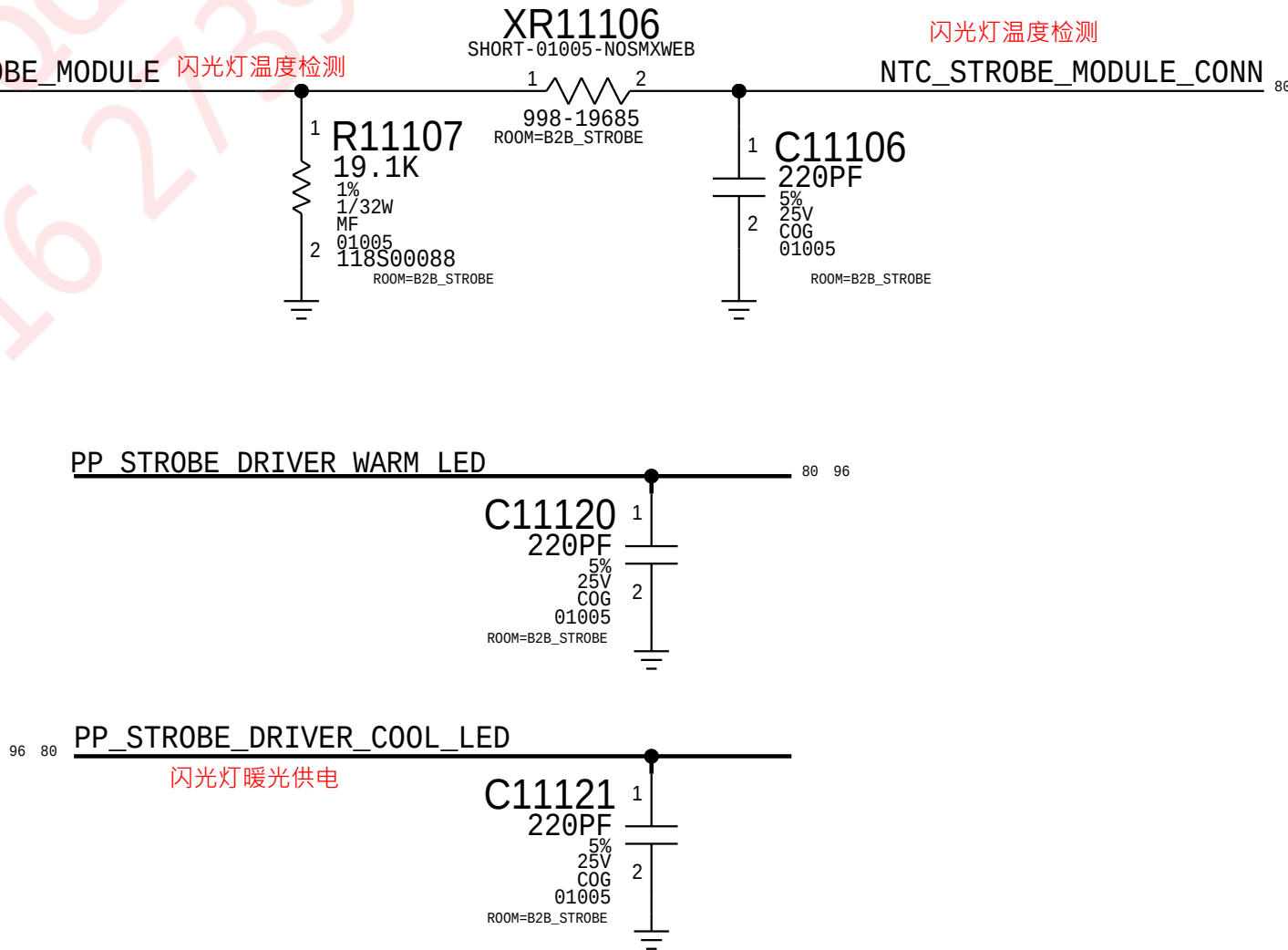
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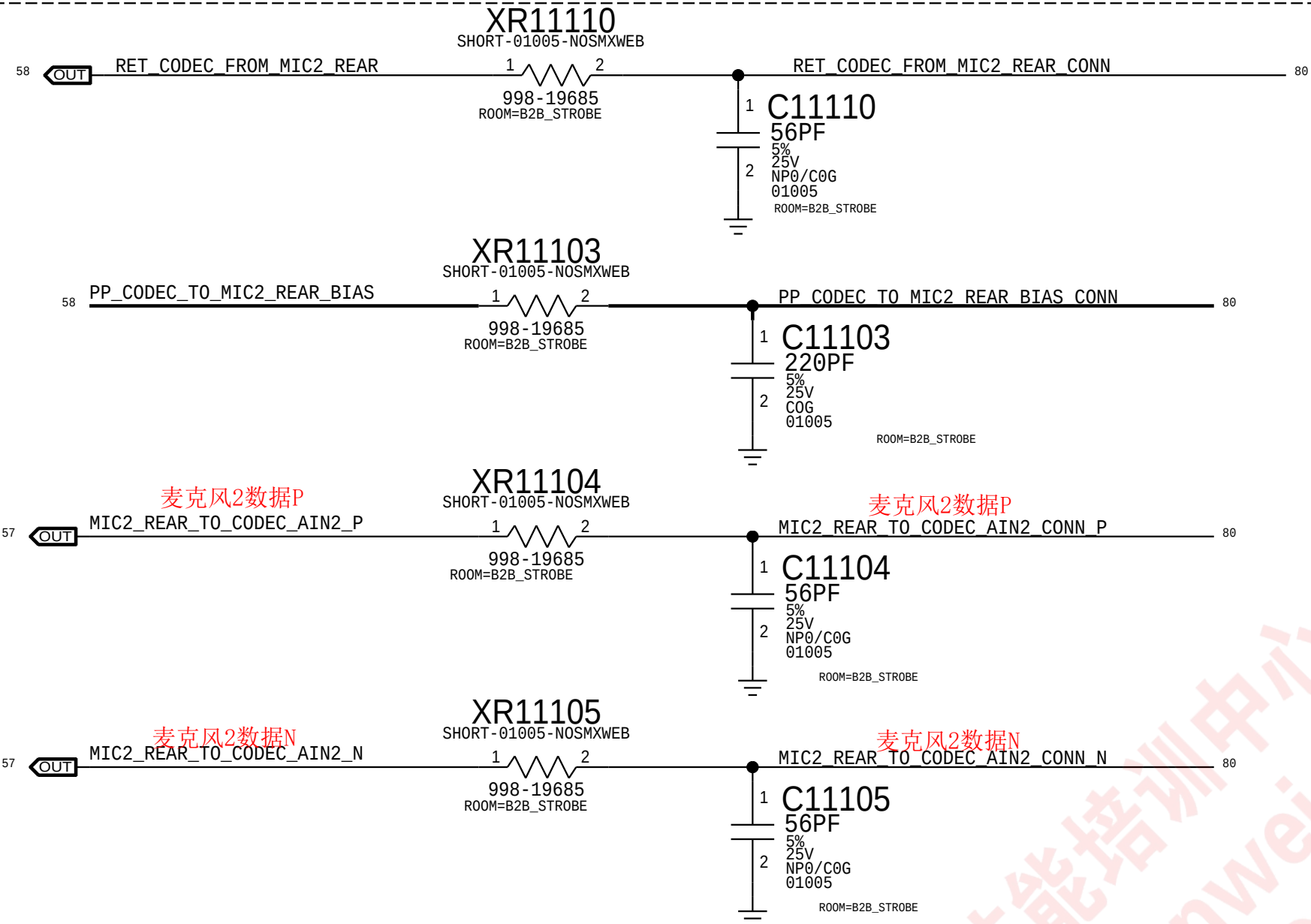
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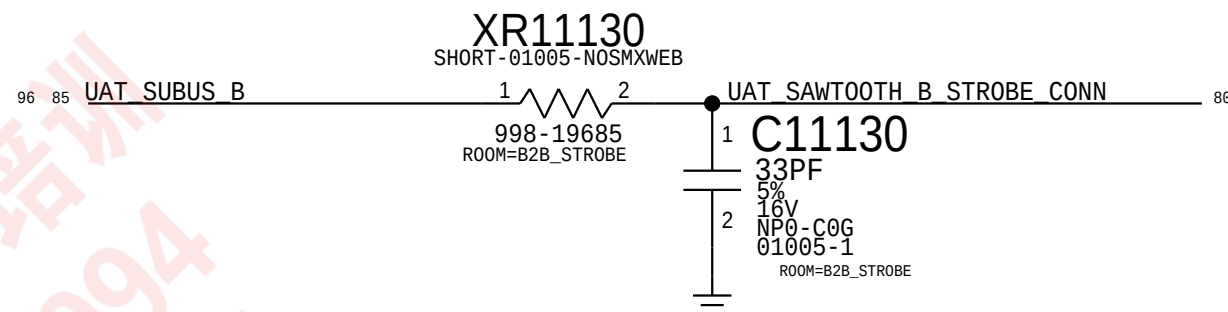
Strobe



MIC2



Change SuBus to Sawtooth for flex side per antenna team request

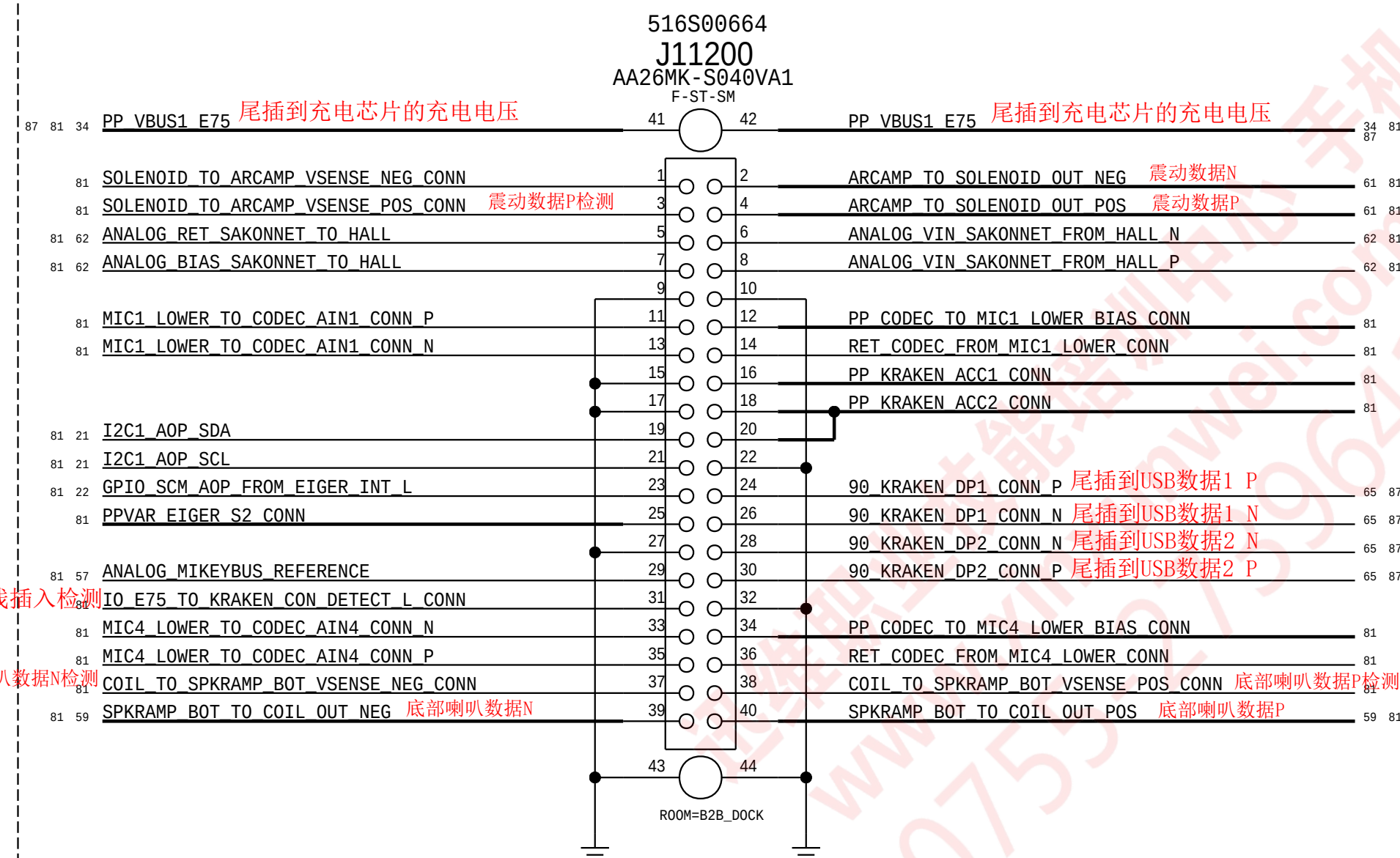


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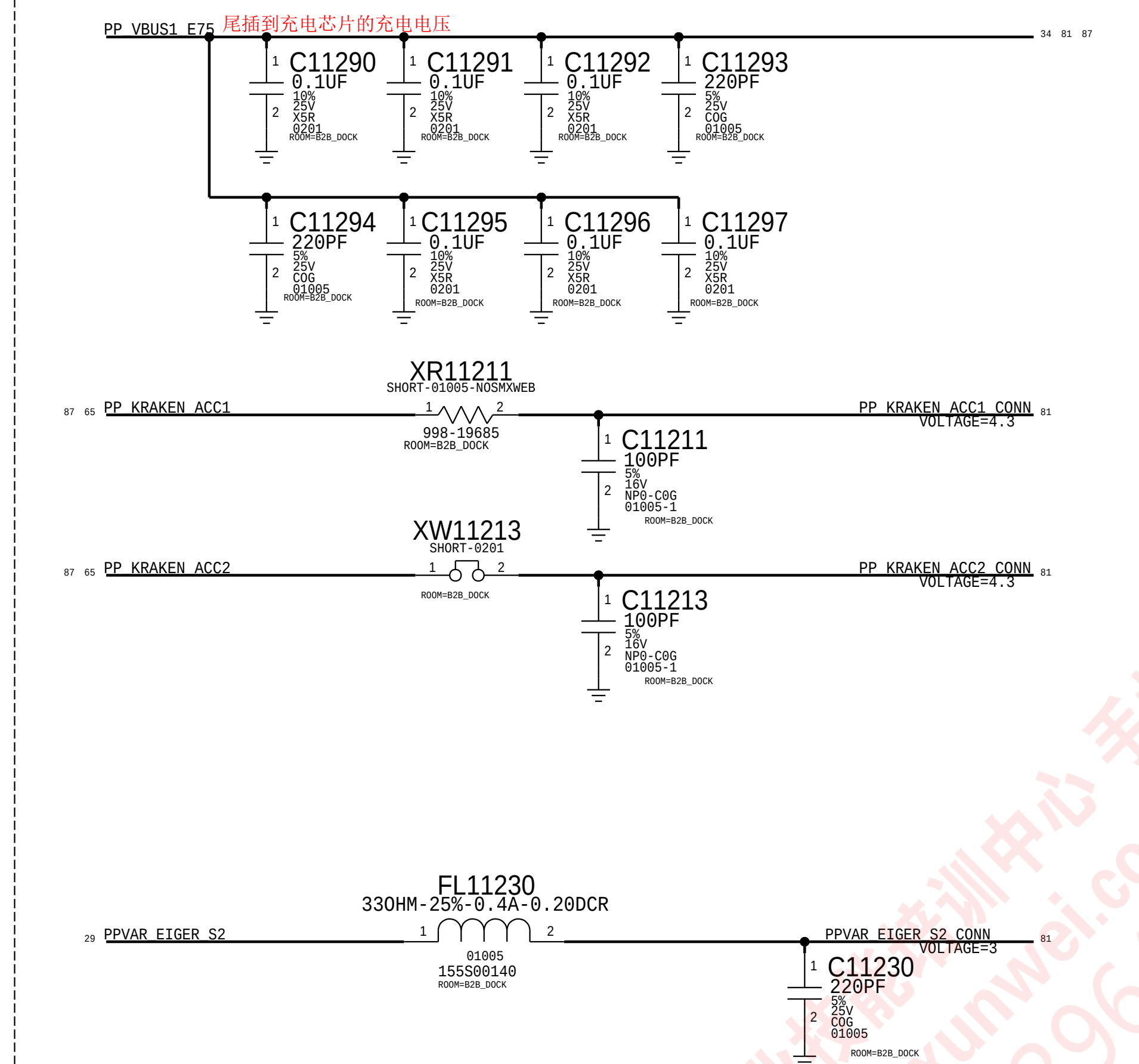
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DOCK B2B

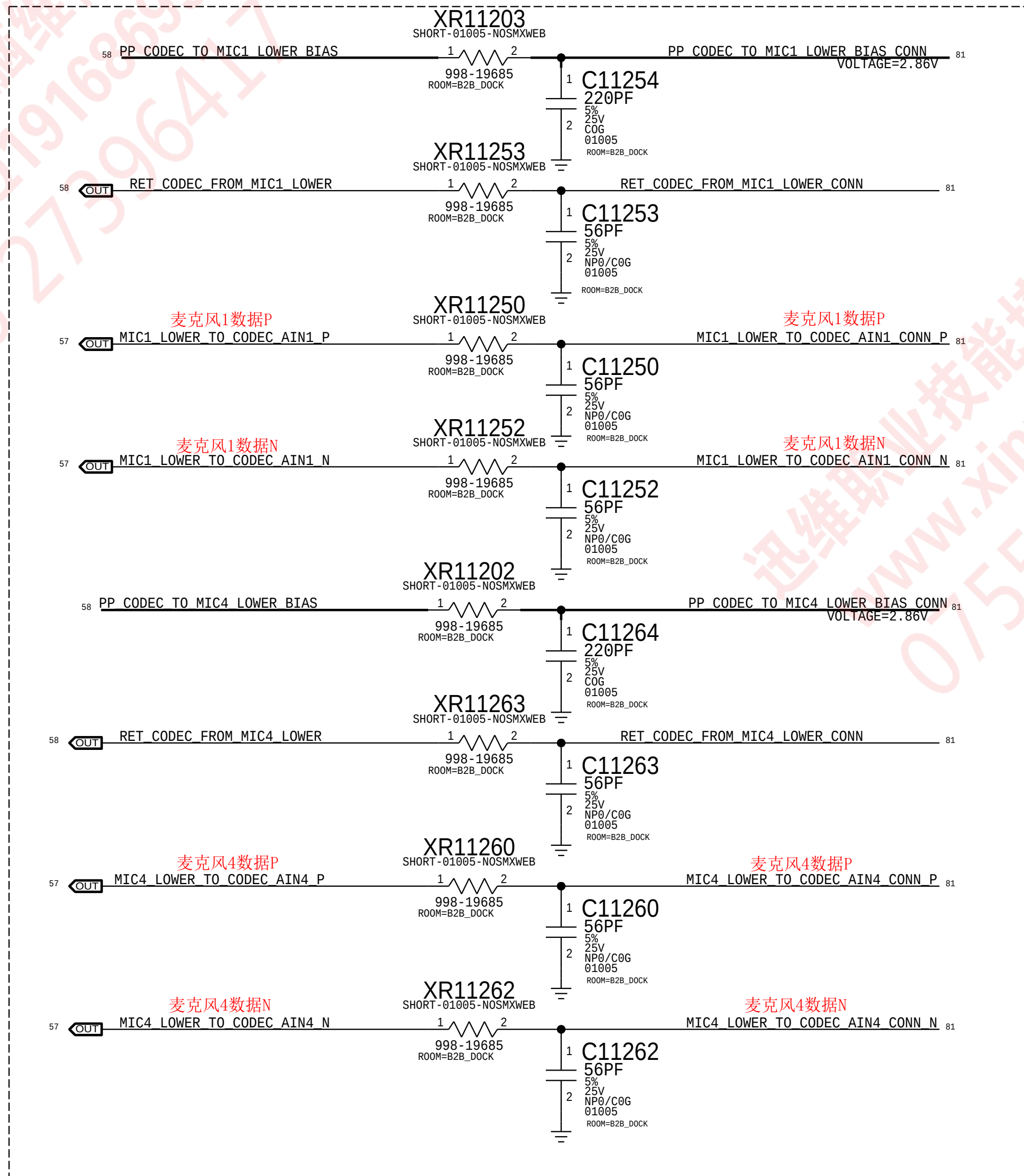
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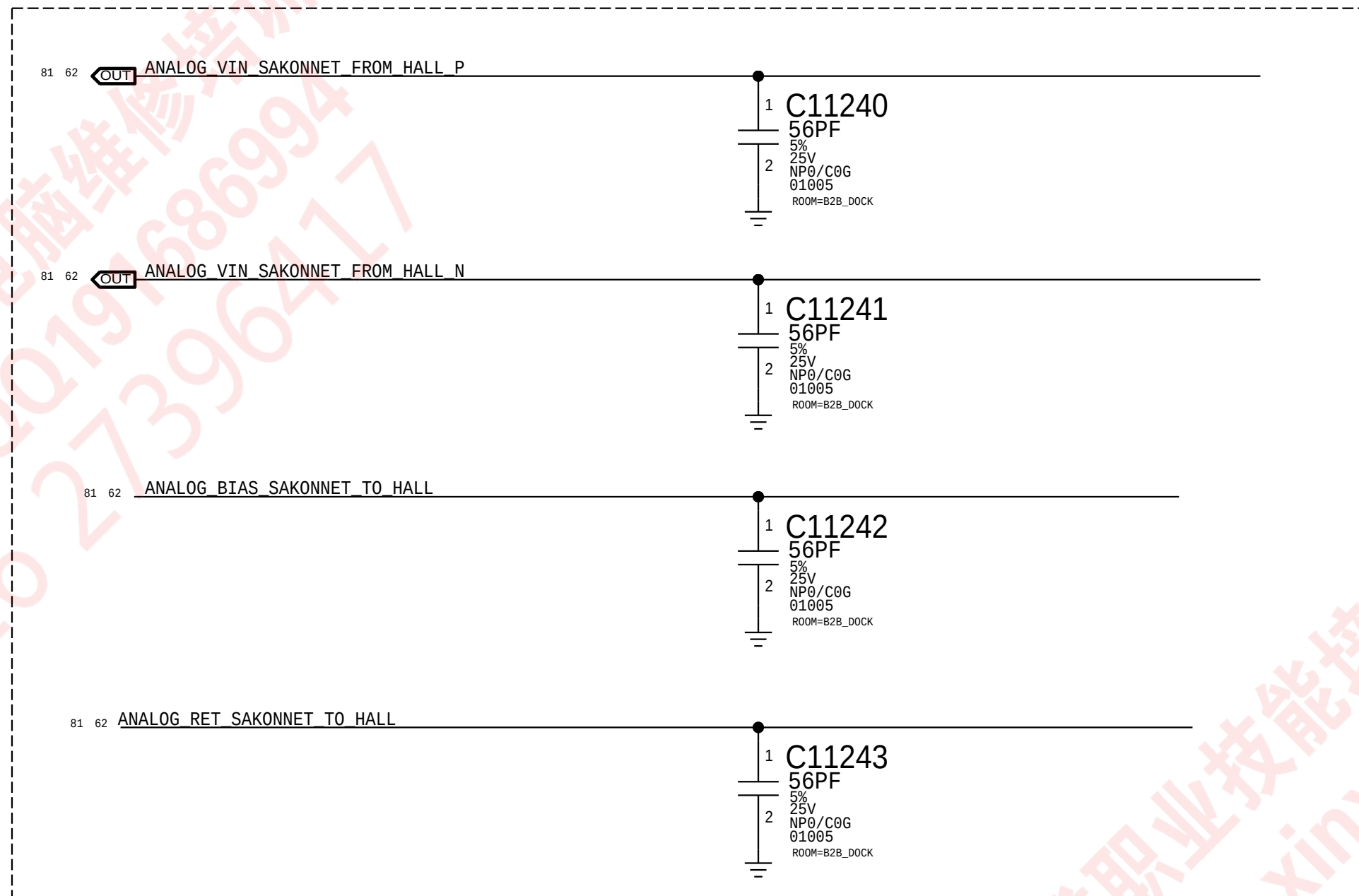
POWER FILTERS



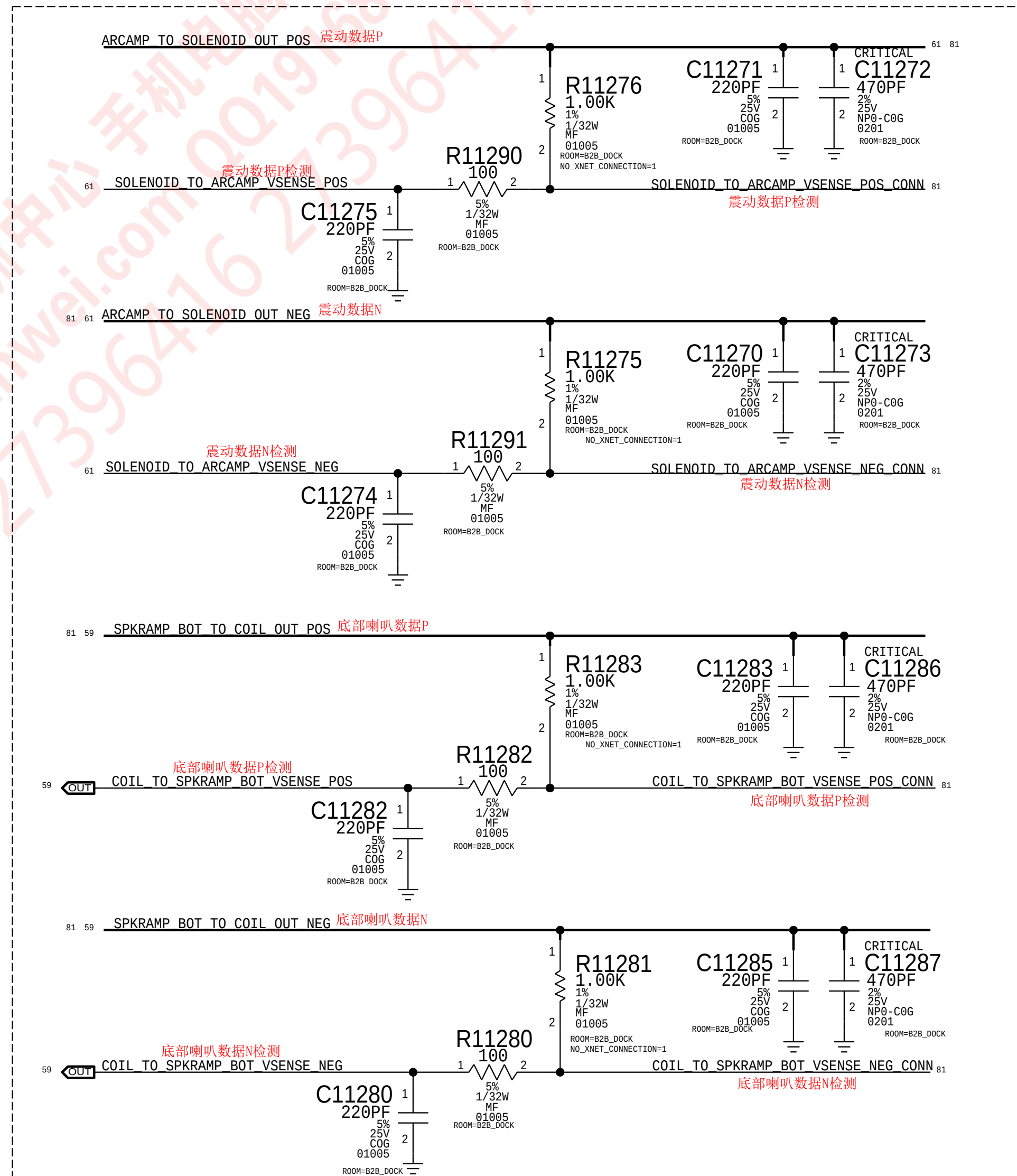
MIC1 + MIC4



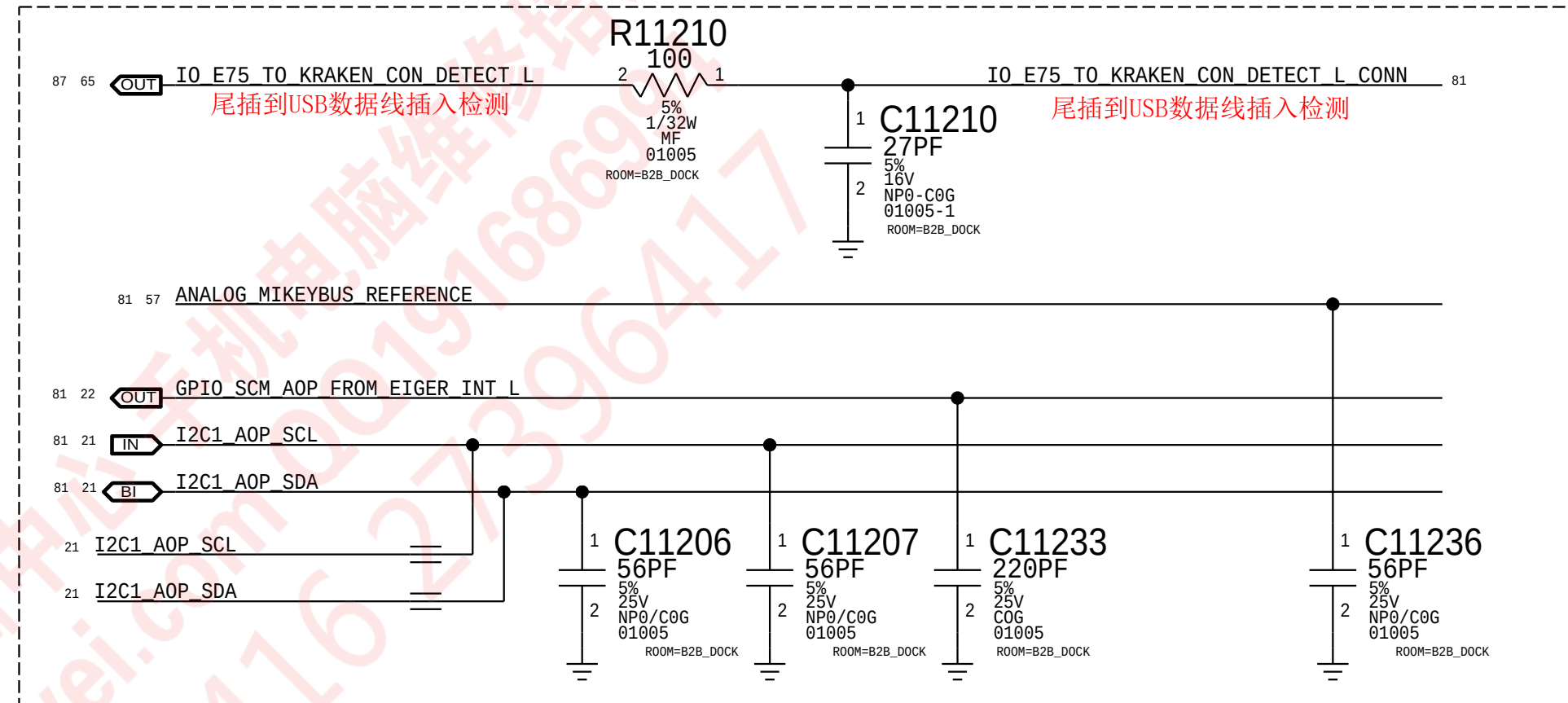
SAKONNET



ARC + BOT SPEAKER



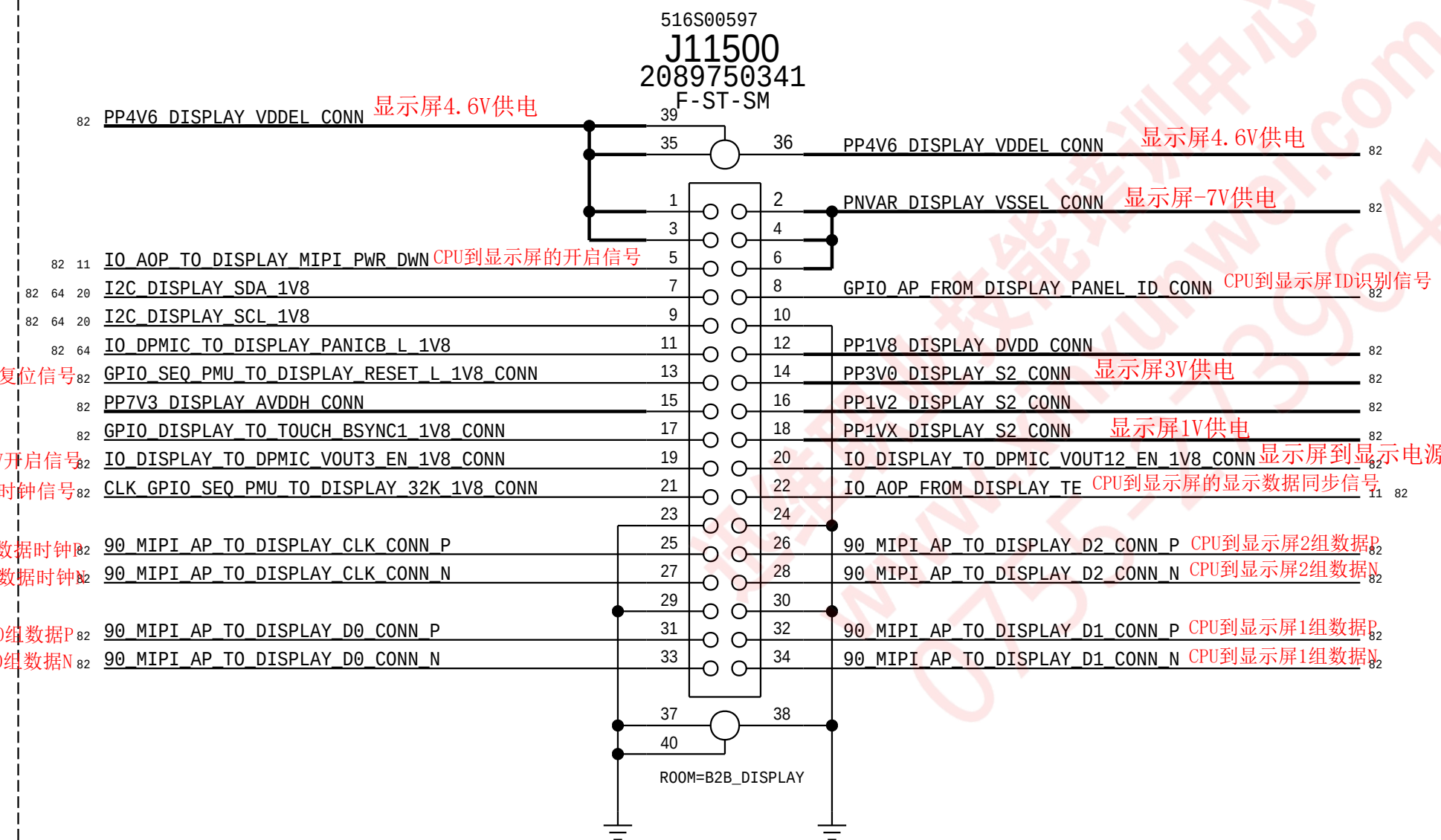
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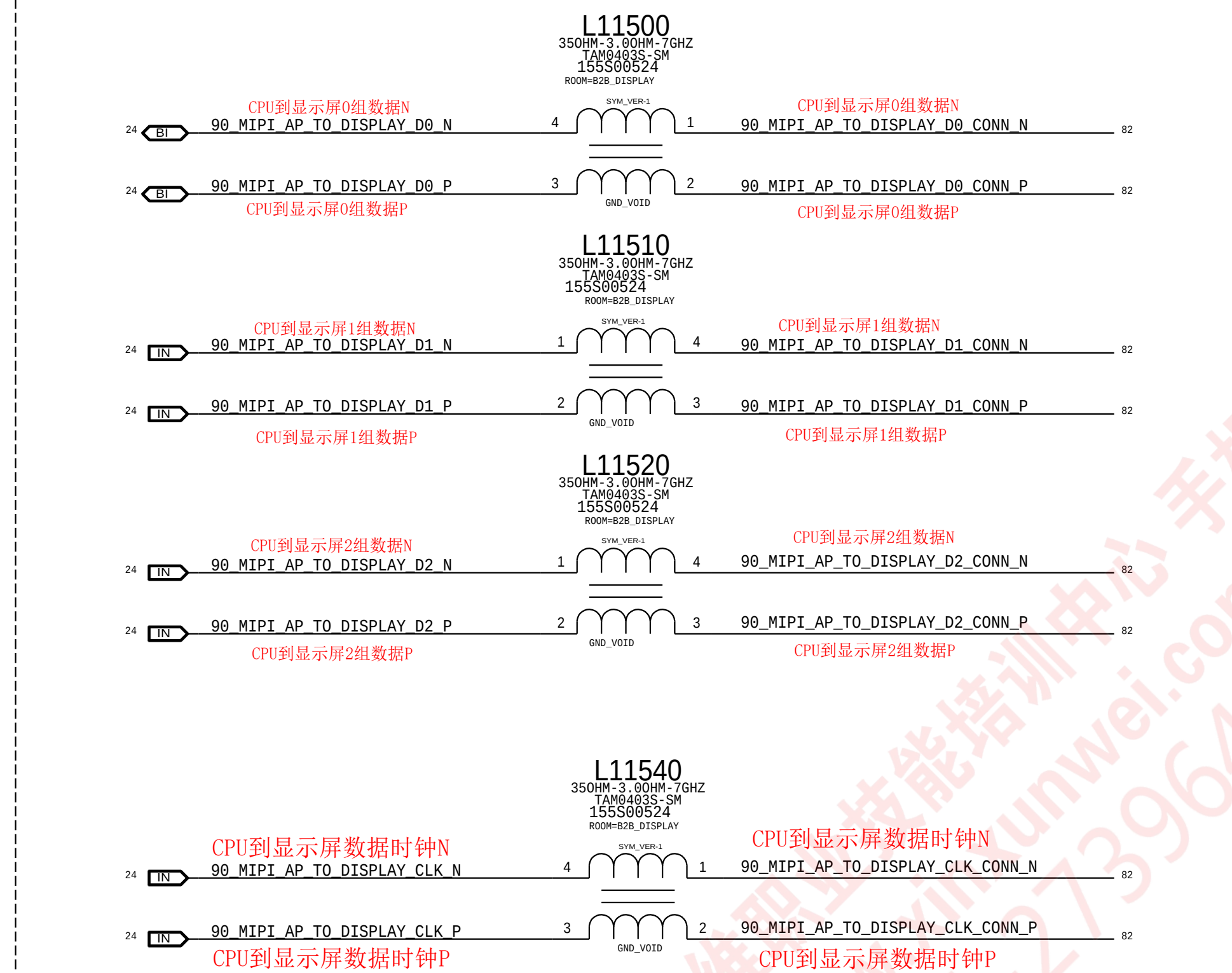
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DISPLAY B2B

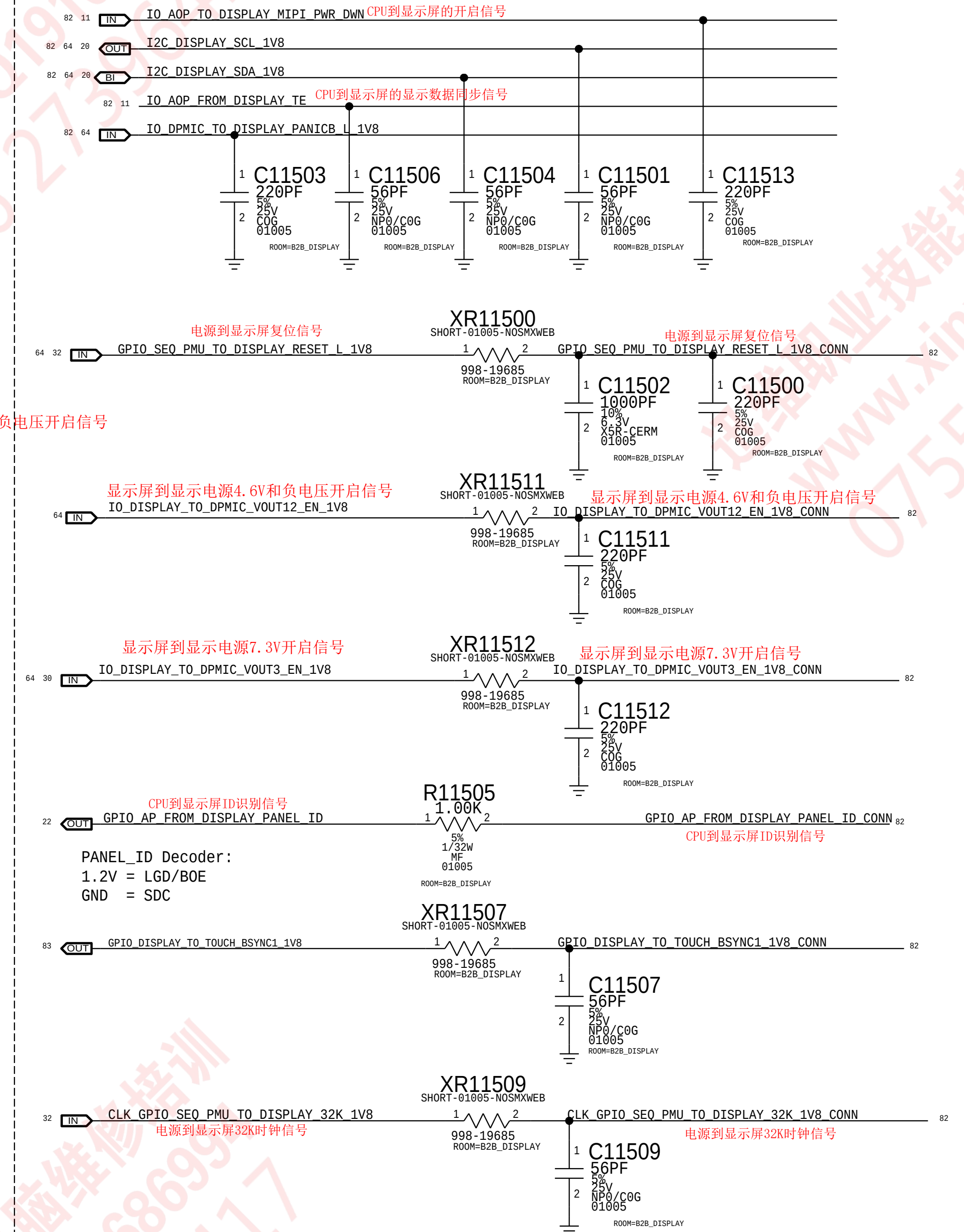
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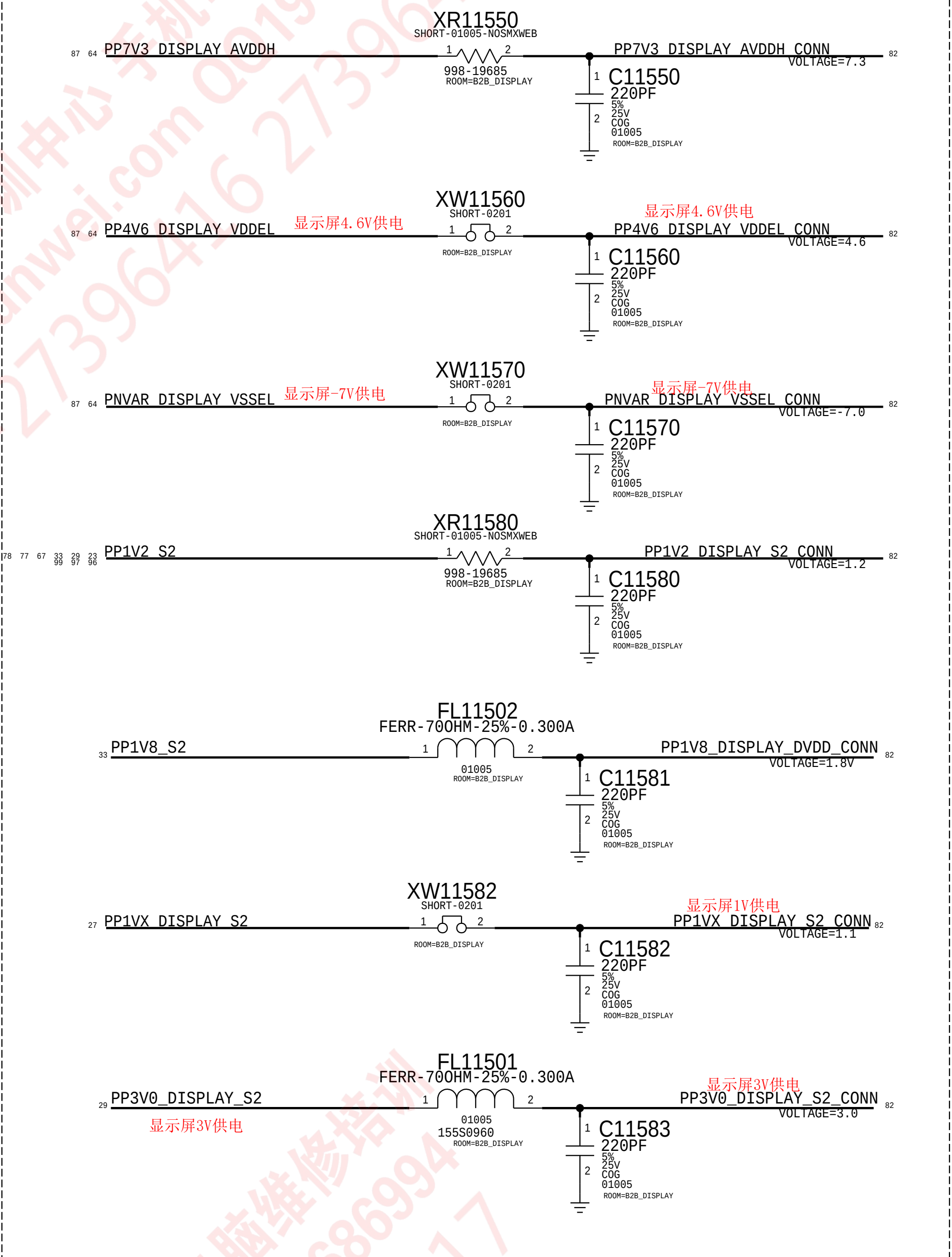
MIPI FILTERS



IO FILTERS



POWER FILTERS



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TOUCH B2B

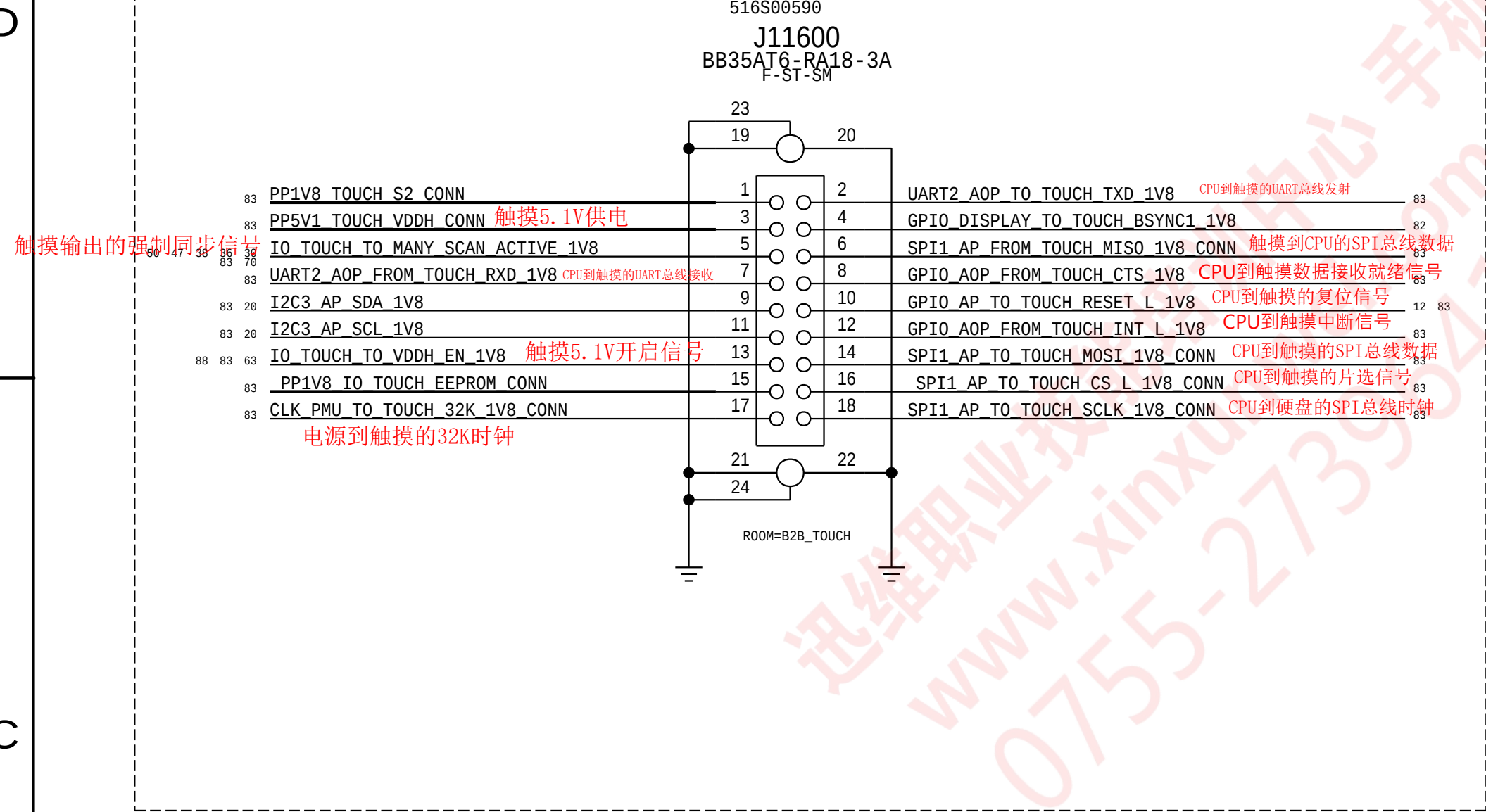
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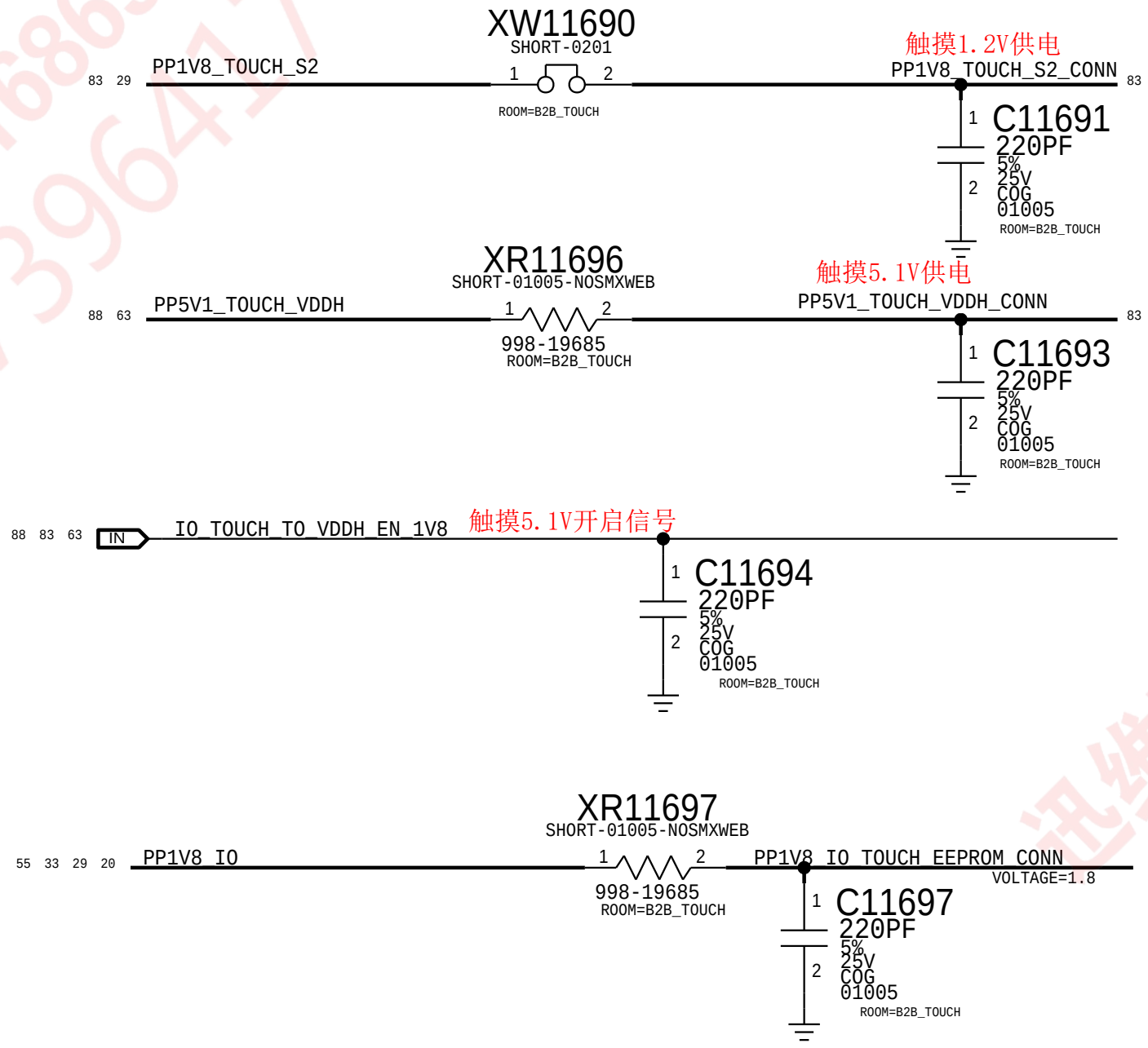
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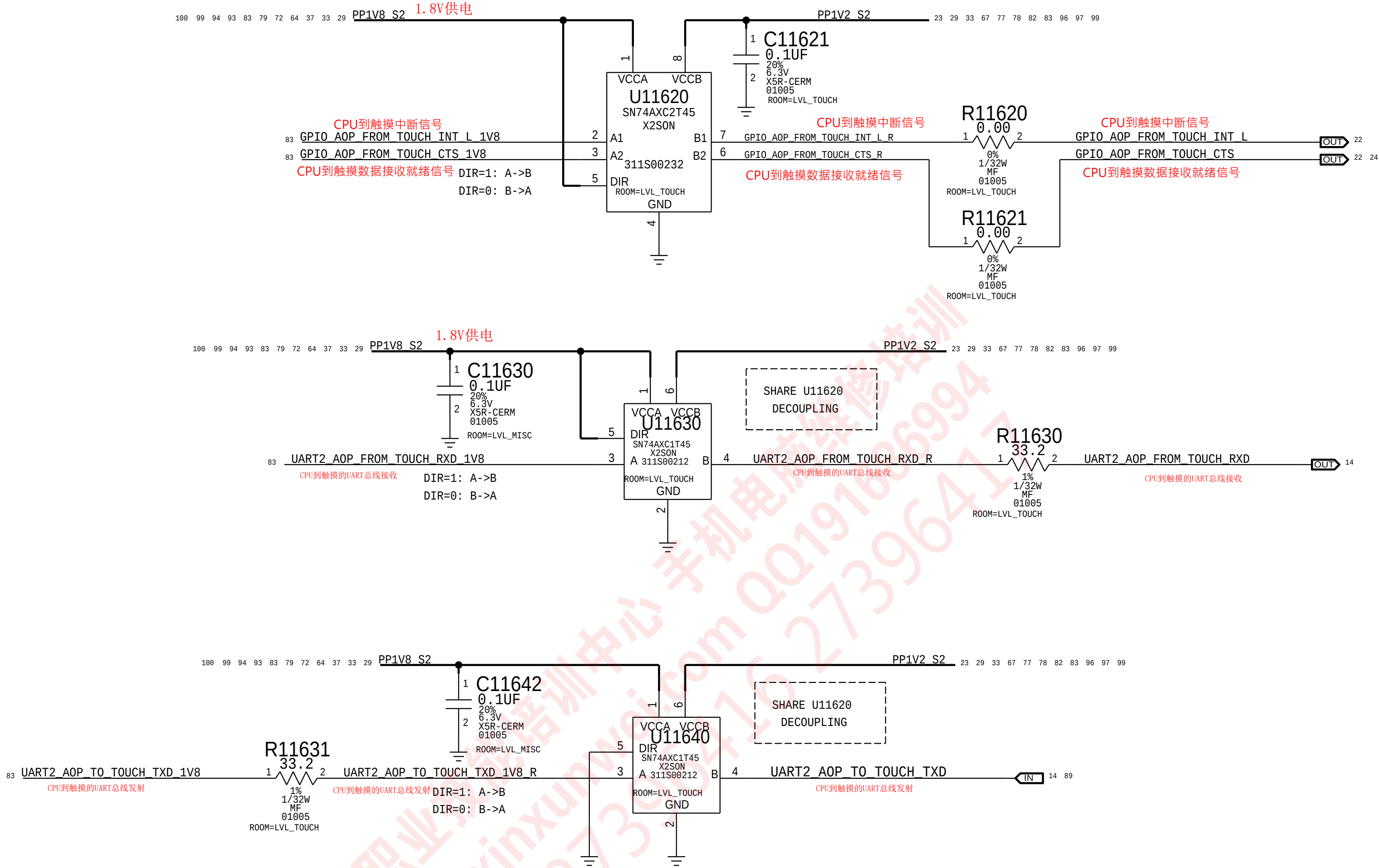
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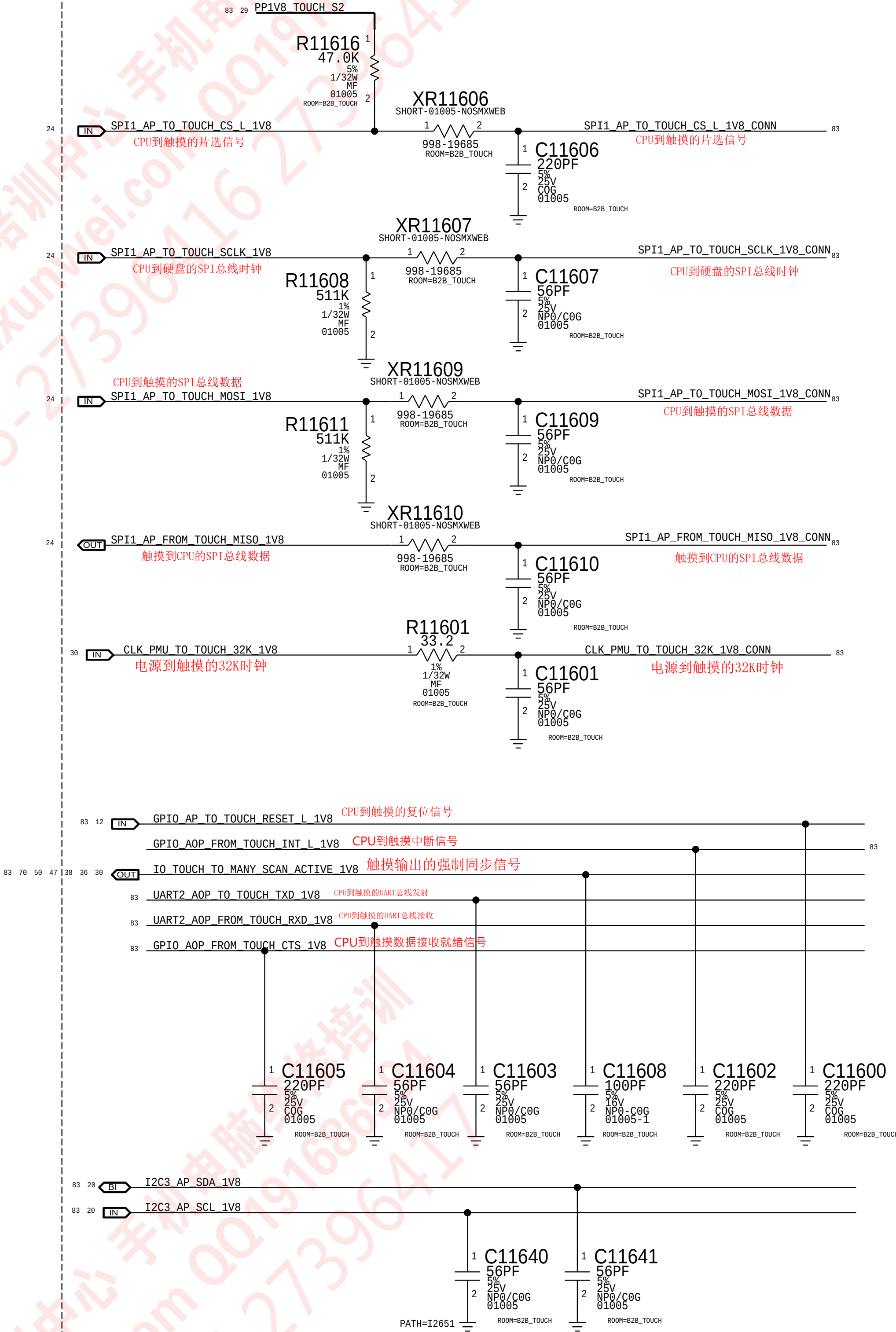
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Level Translators

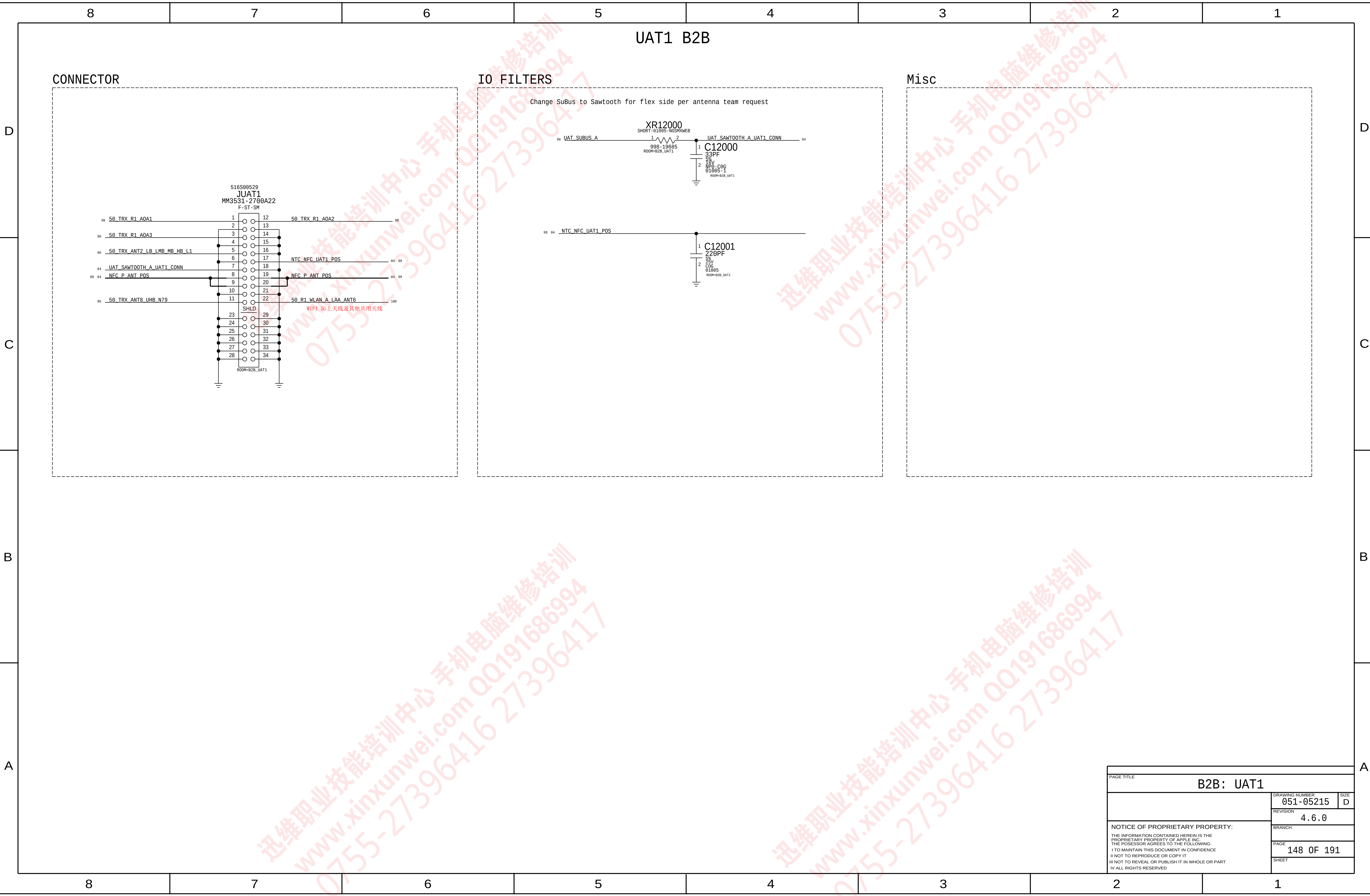


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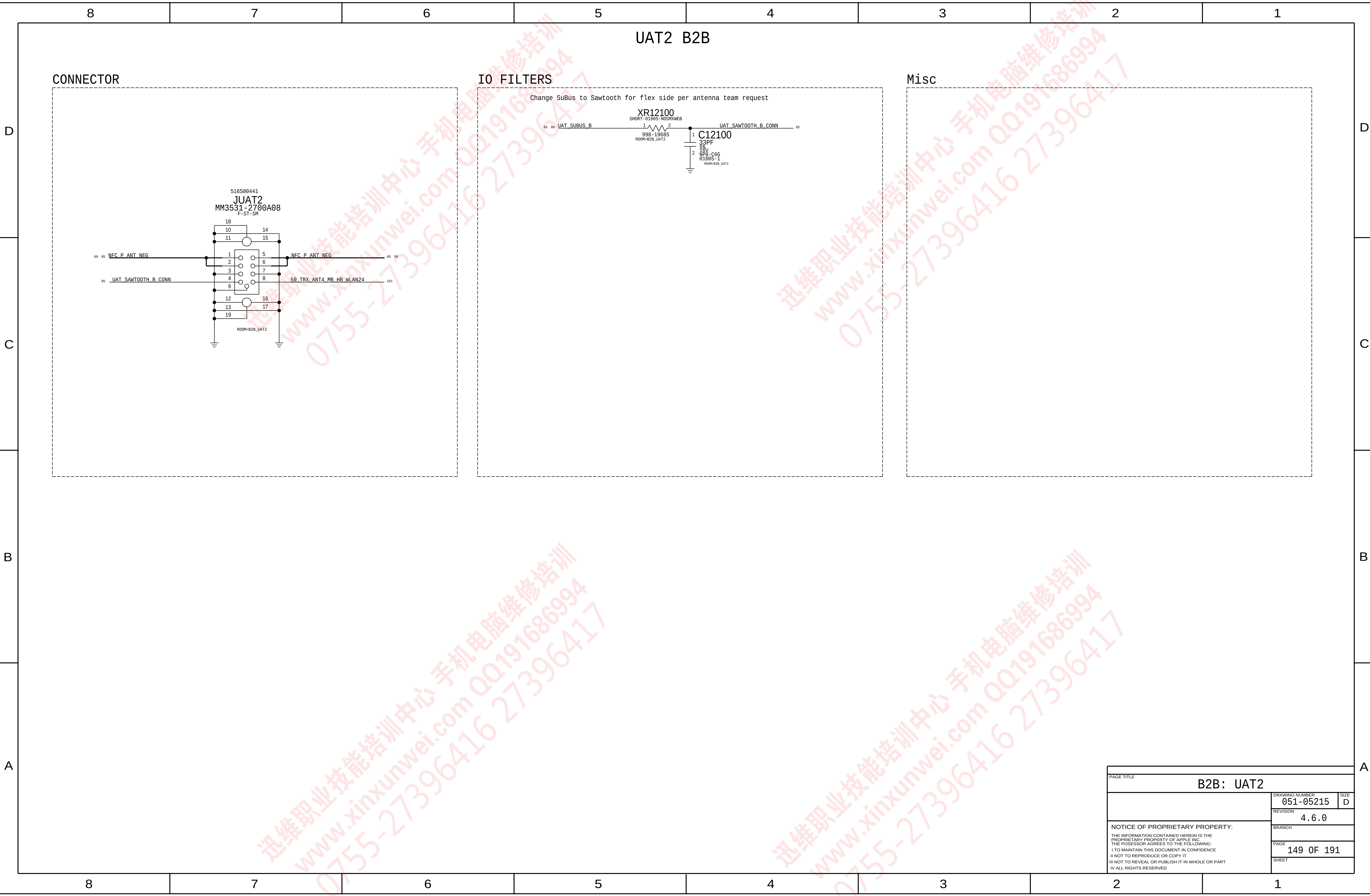


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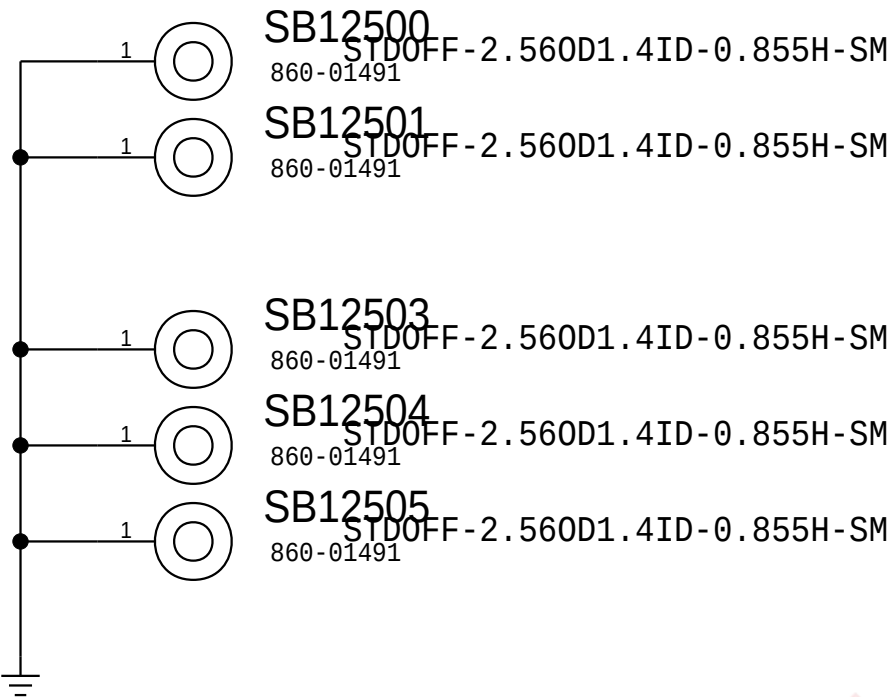
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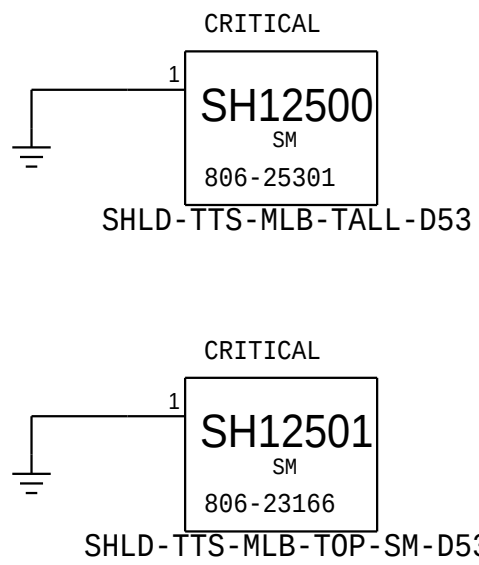
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Mechanical Parts

Standoffs

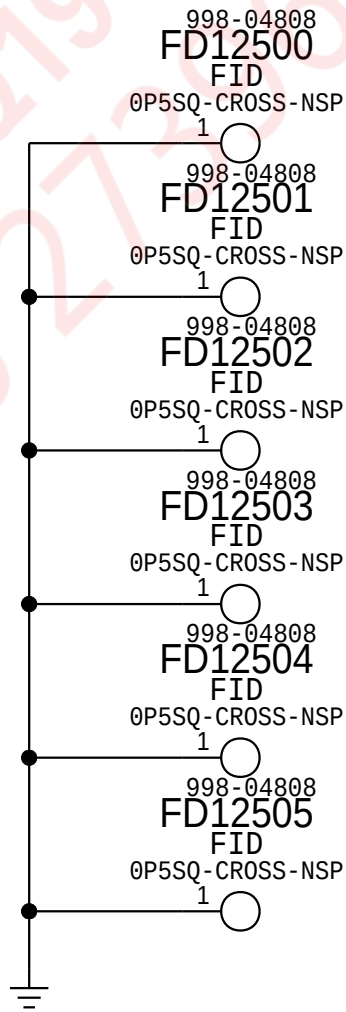


Shields

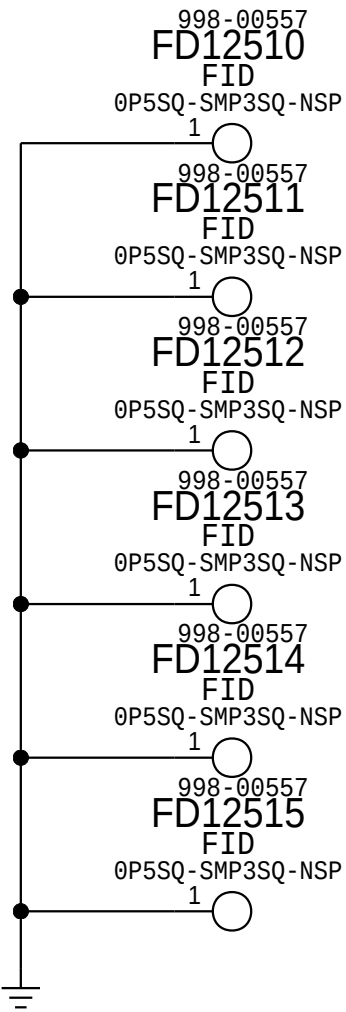


Fiducials

Crosses



Squares



Interposer Spacers

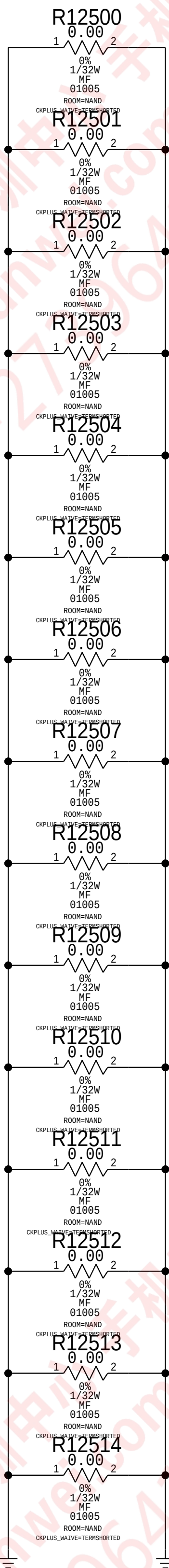
RefDes called out below

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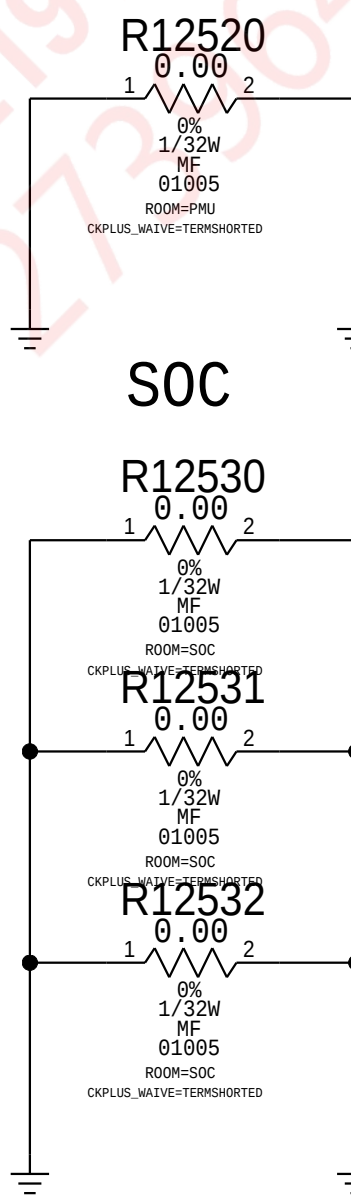
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Underfill Dam

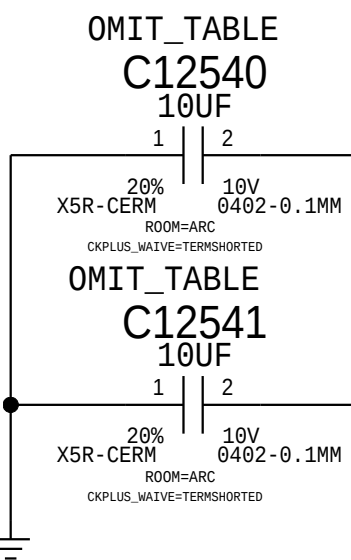
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PMU

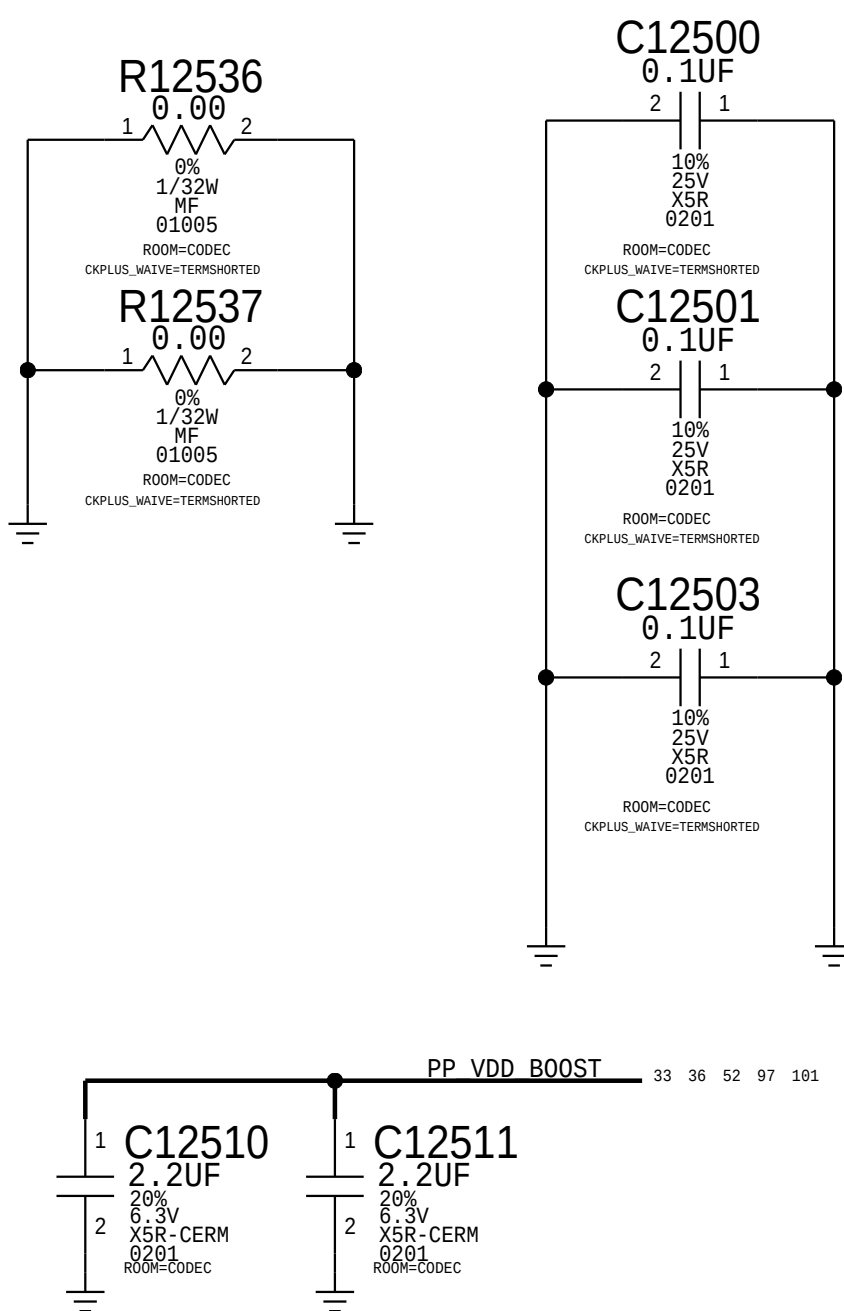


Arc Amp



PART#	QTY	DESCRIPTION	REFERENCE DESIGNATOR(S)	CRITICAL	BOM OPTION
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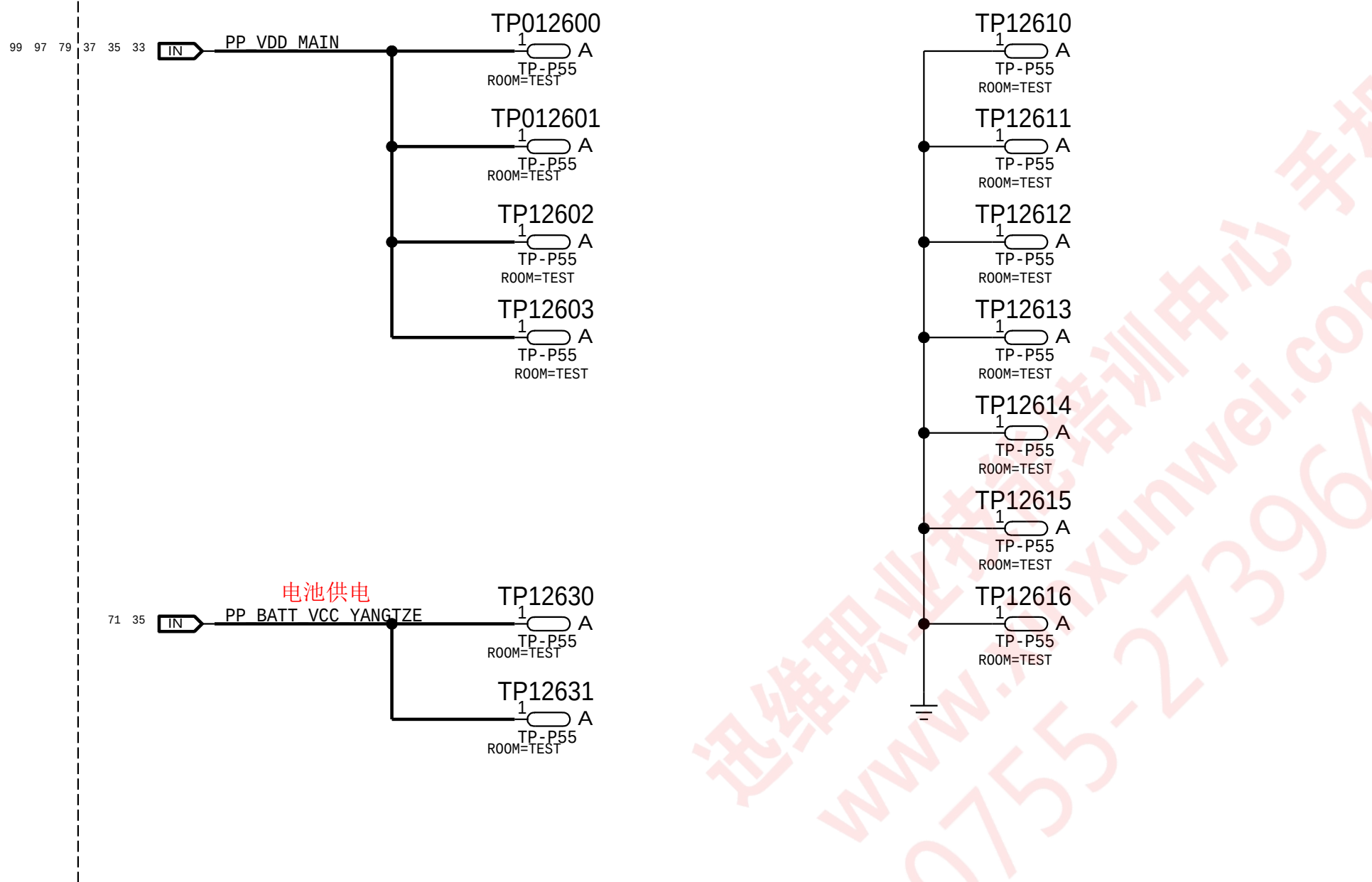
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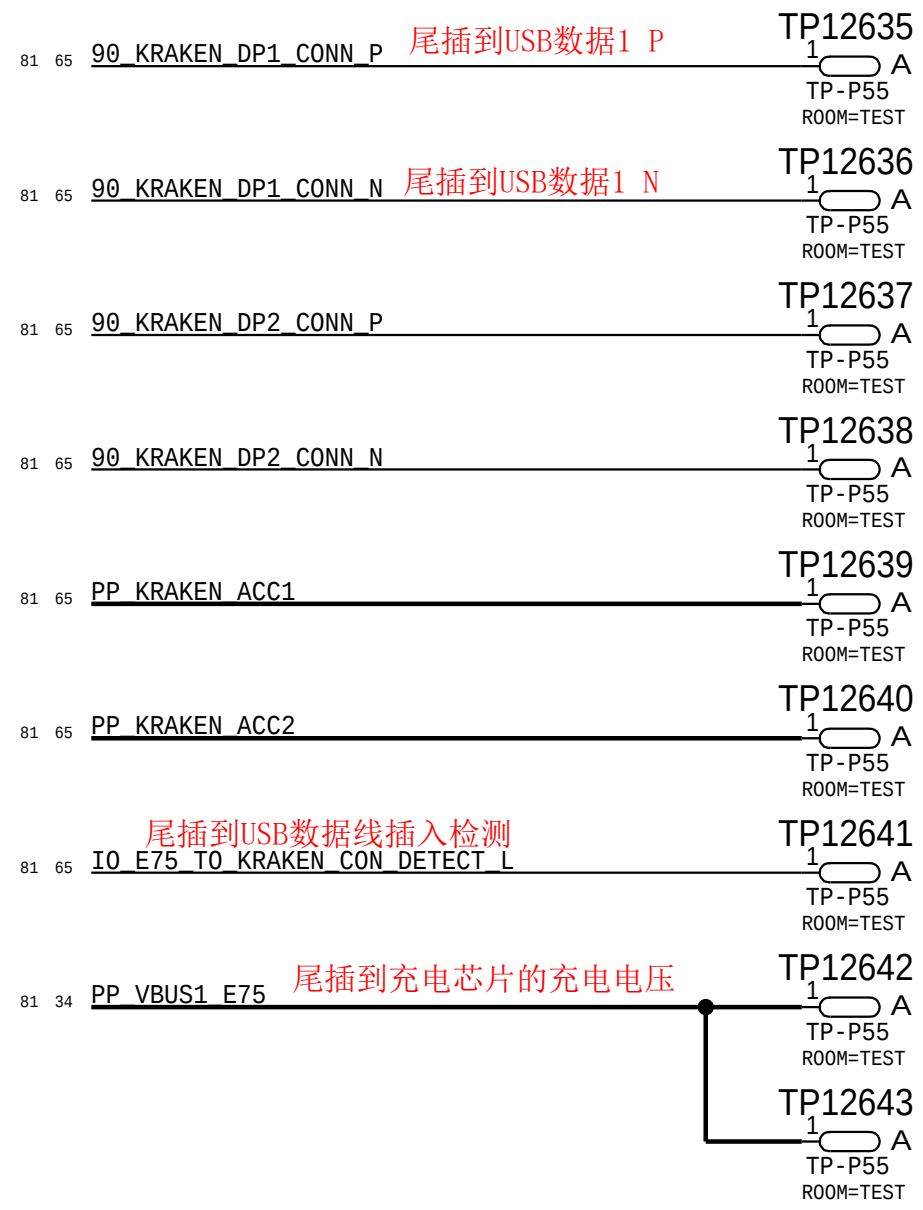
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Test Points

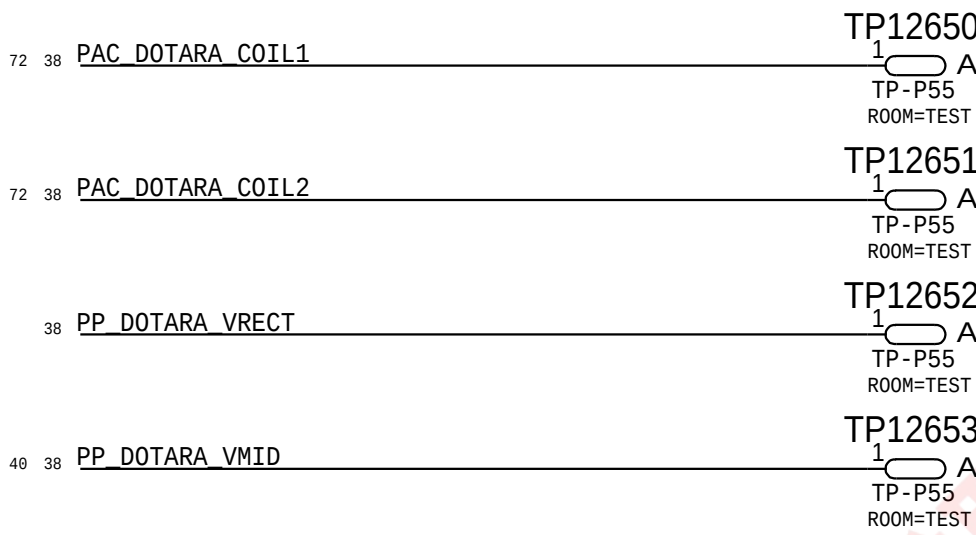
Power + GND



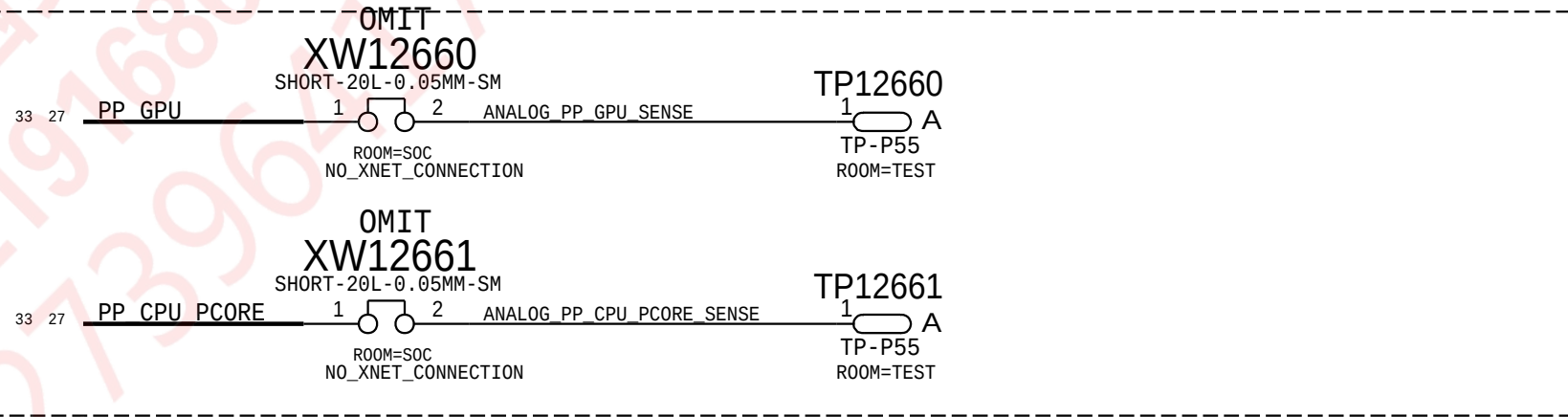
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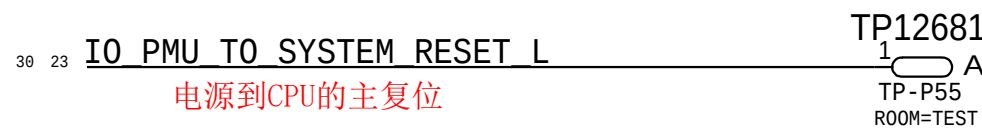
Dotara Power



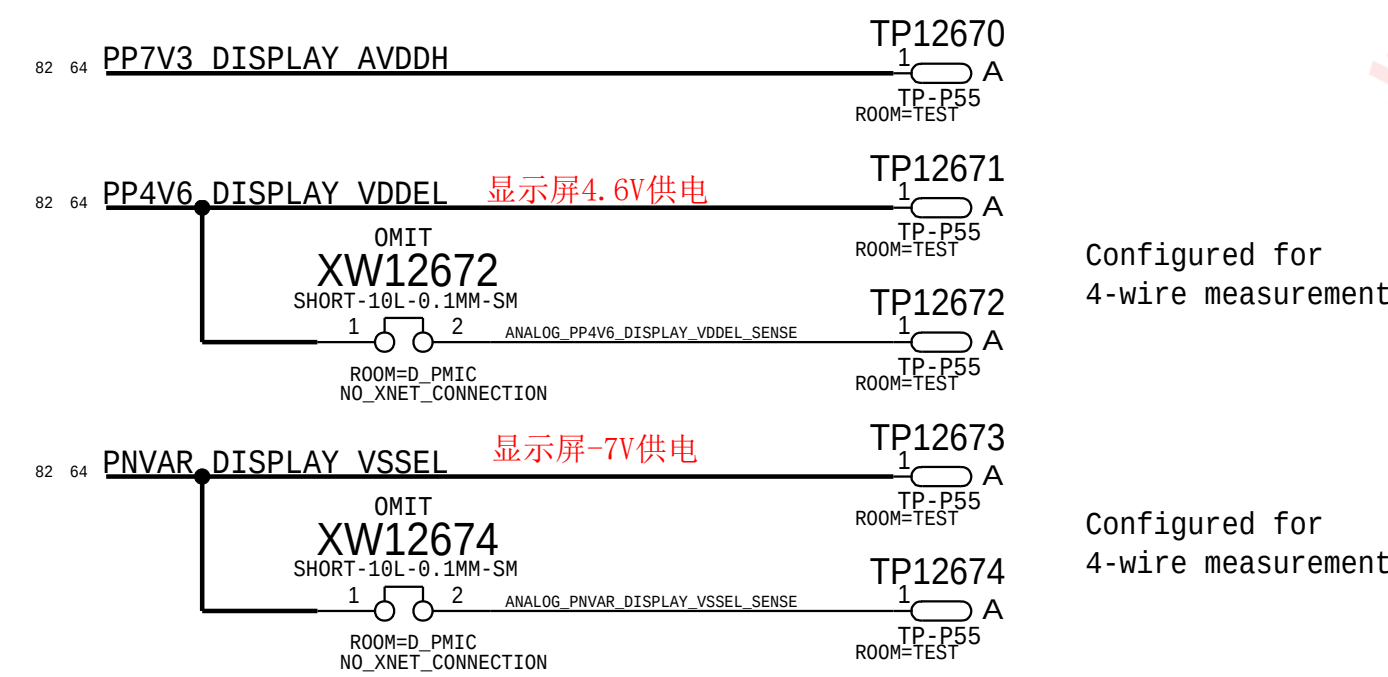
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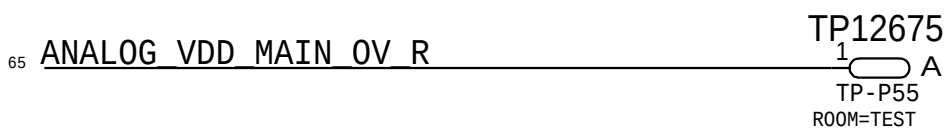
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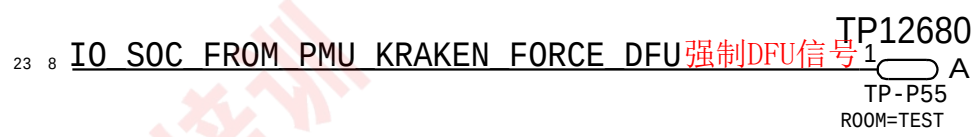
Galena



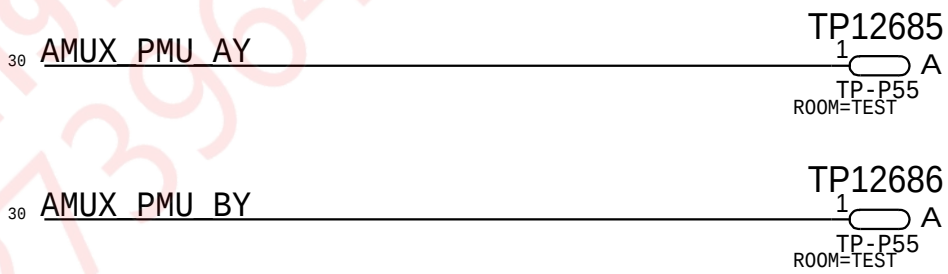
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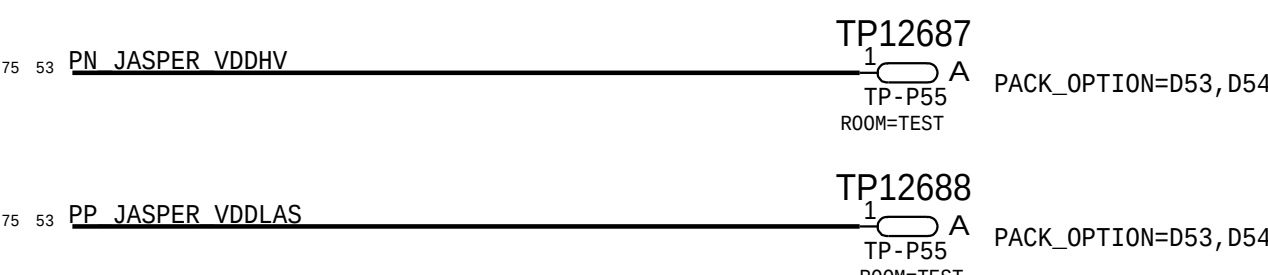
DFU



AMUX



Will



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D

C

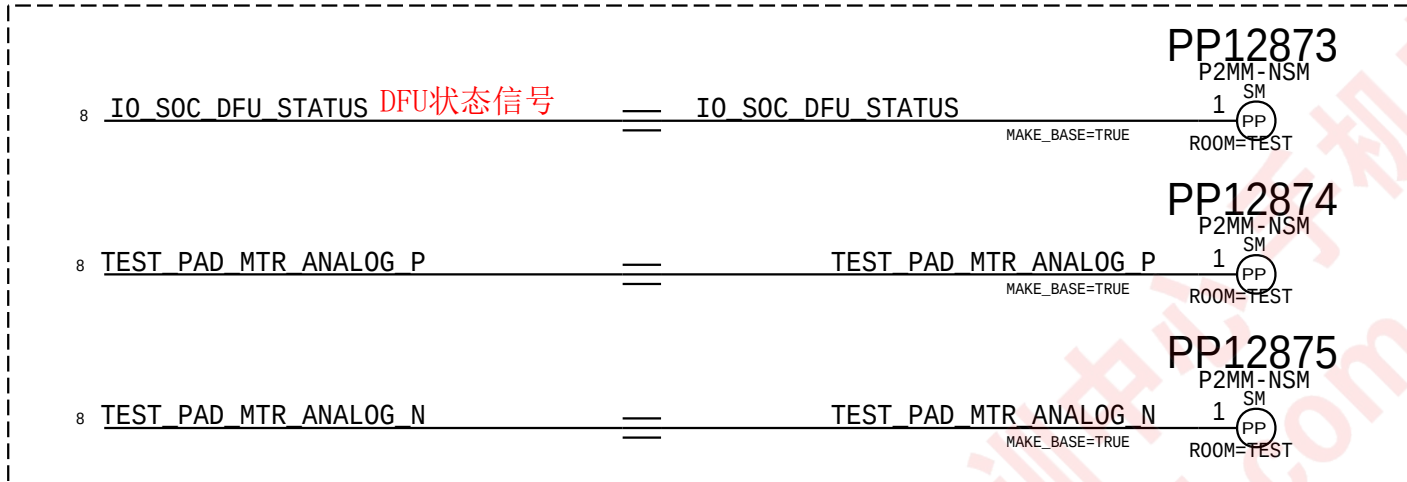
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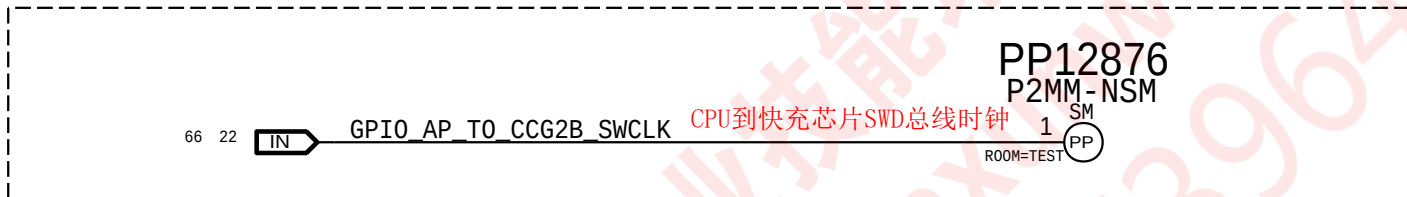
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Probe Points

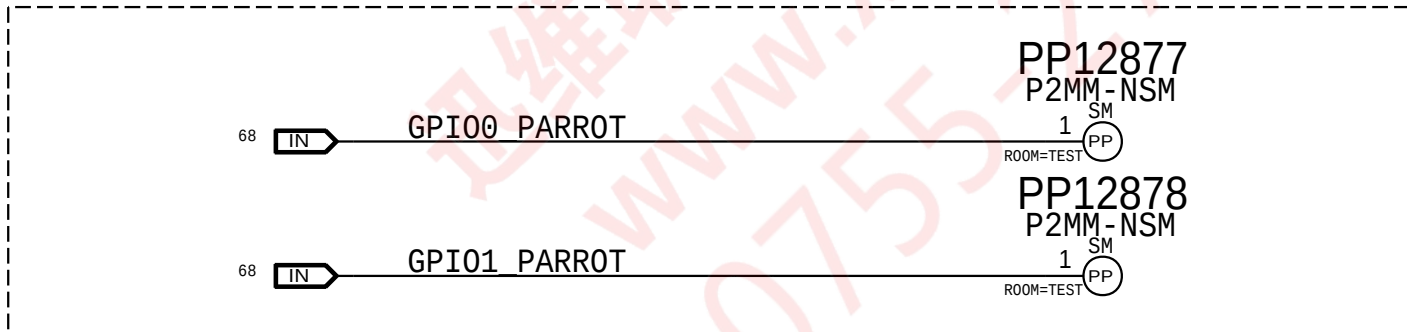
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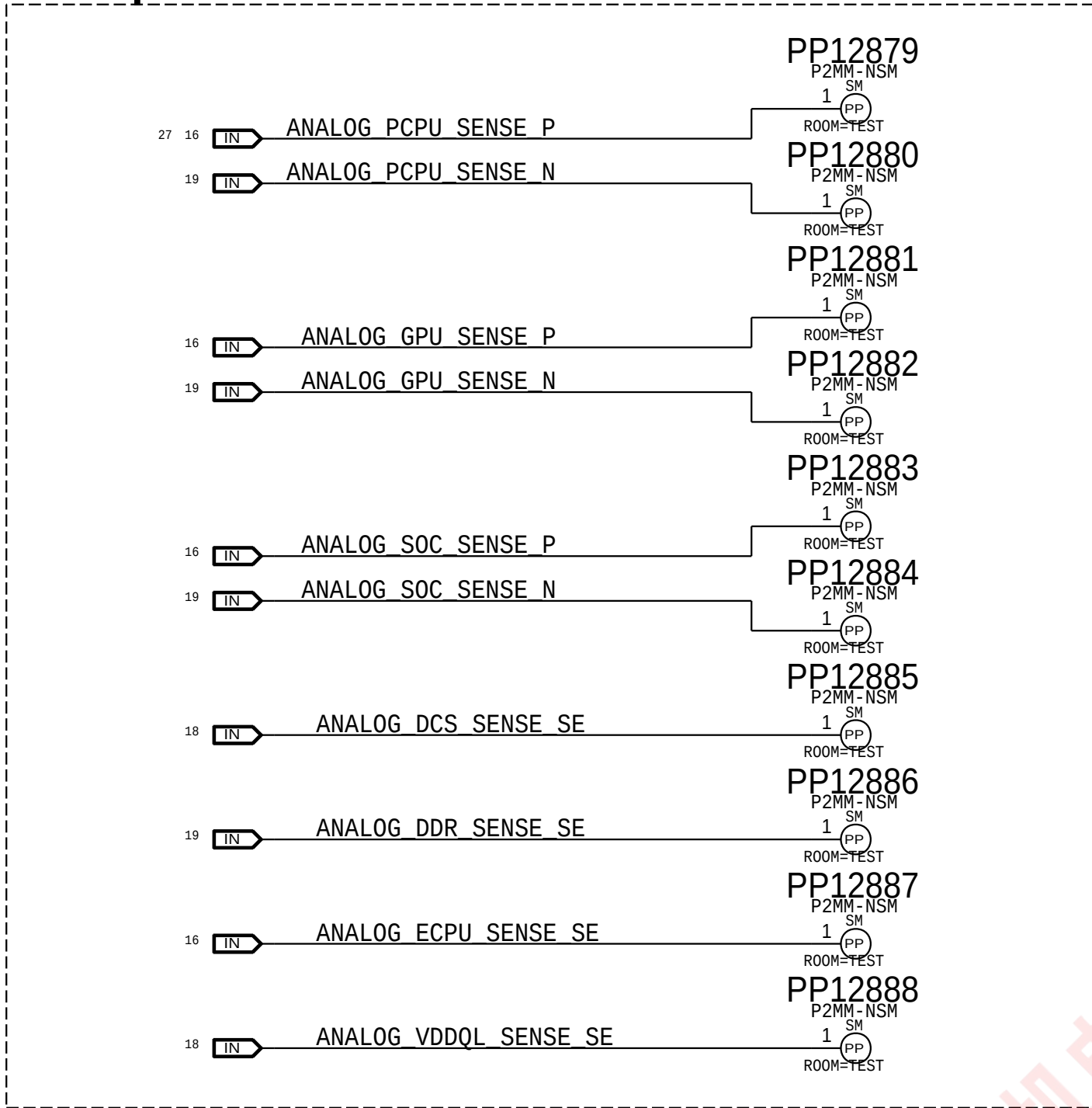
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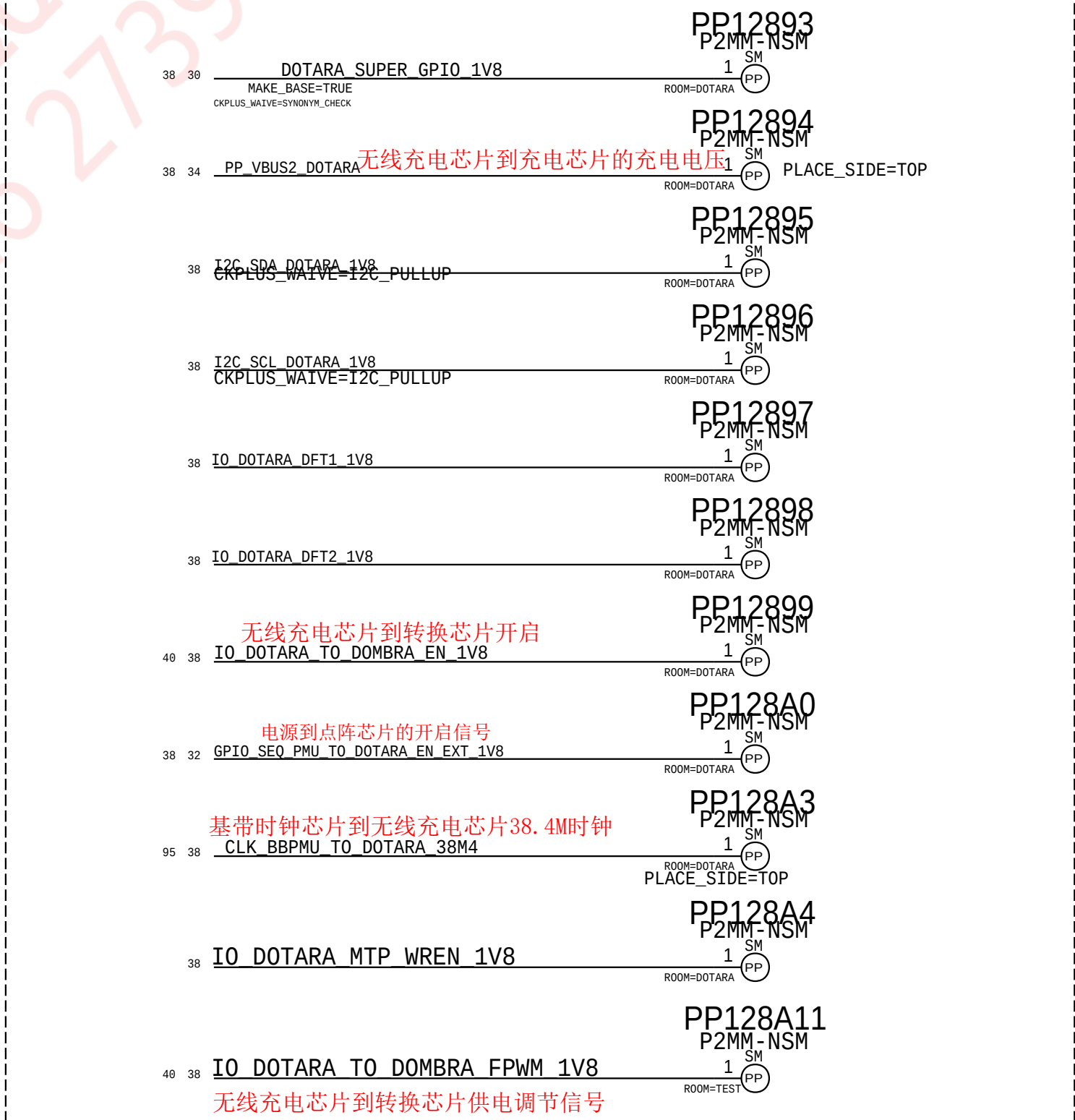
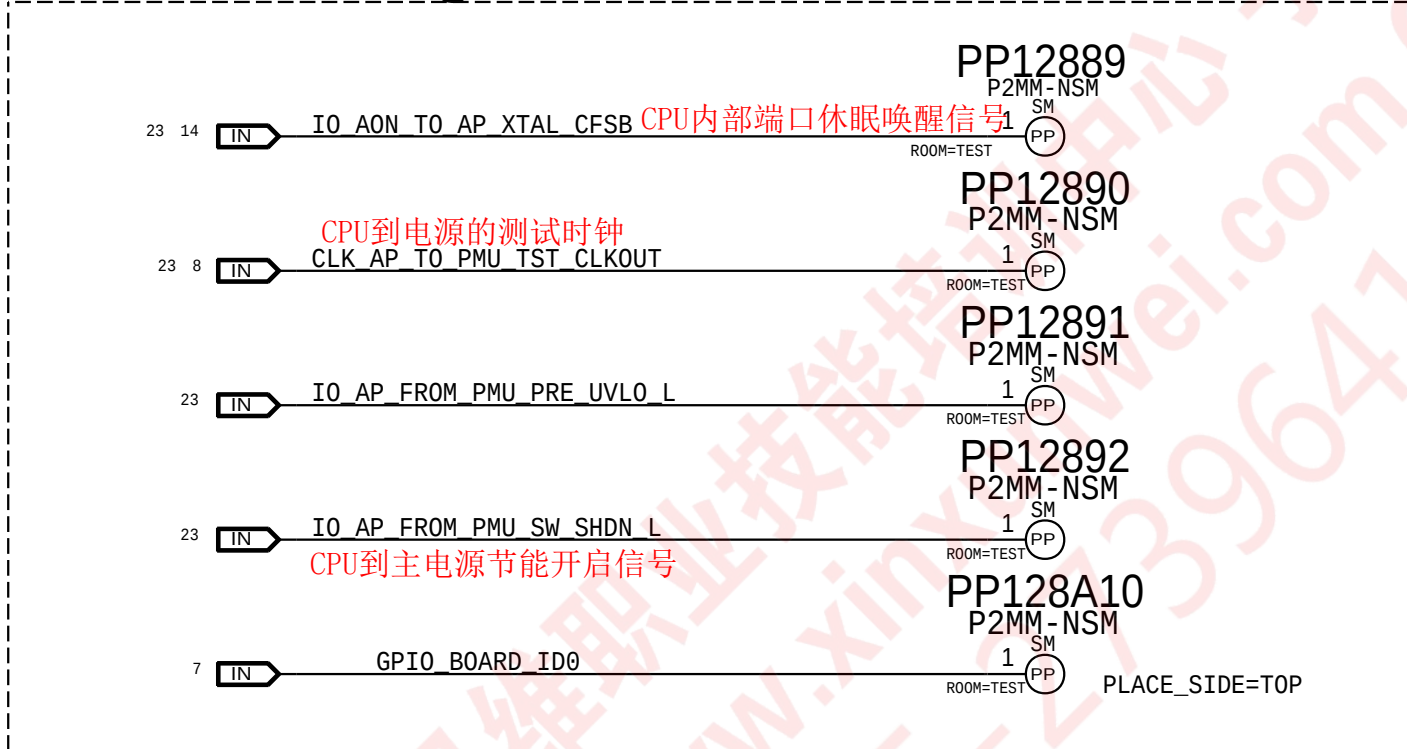
Parrot



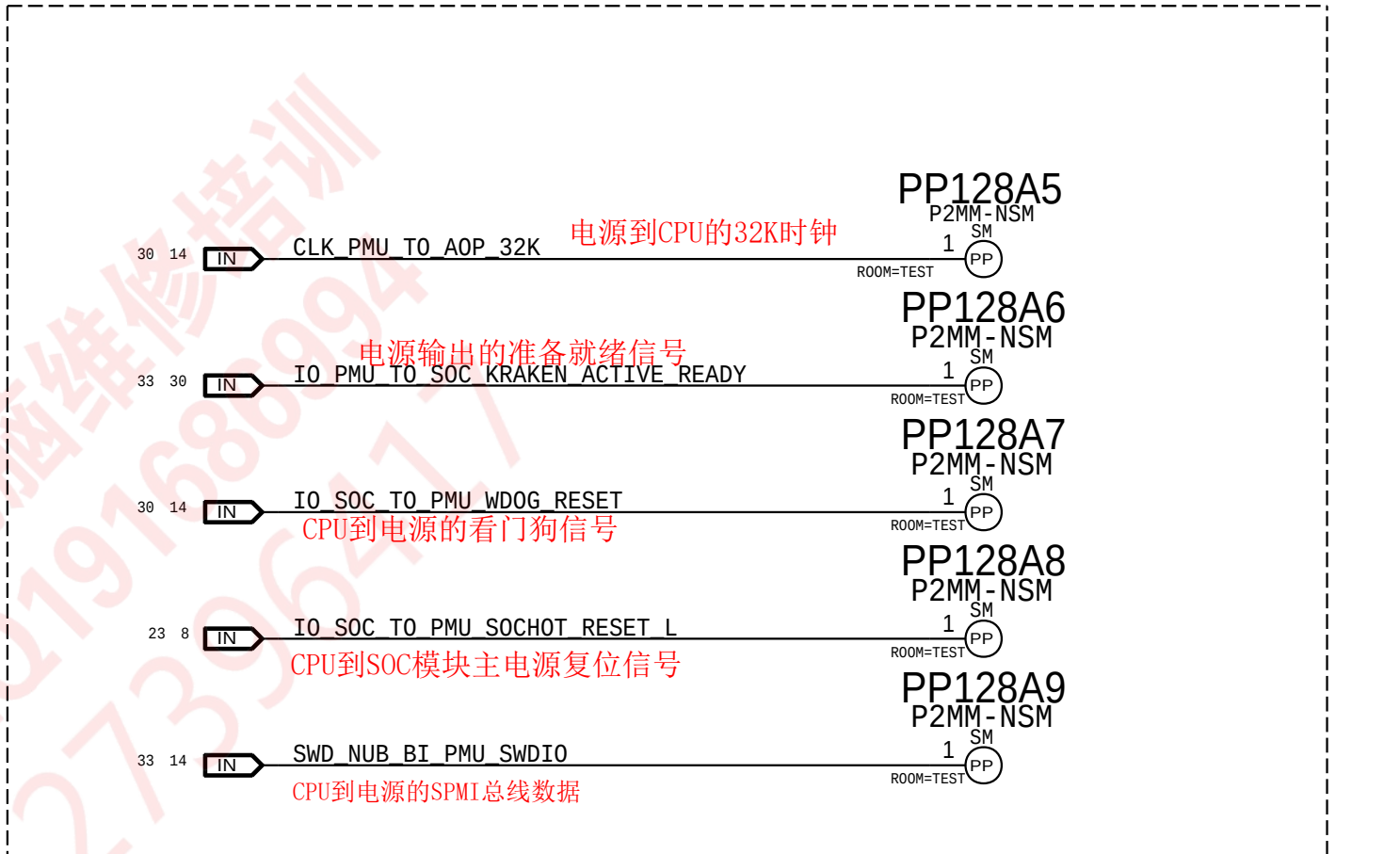
Bump Sense



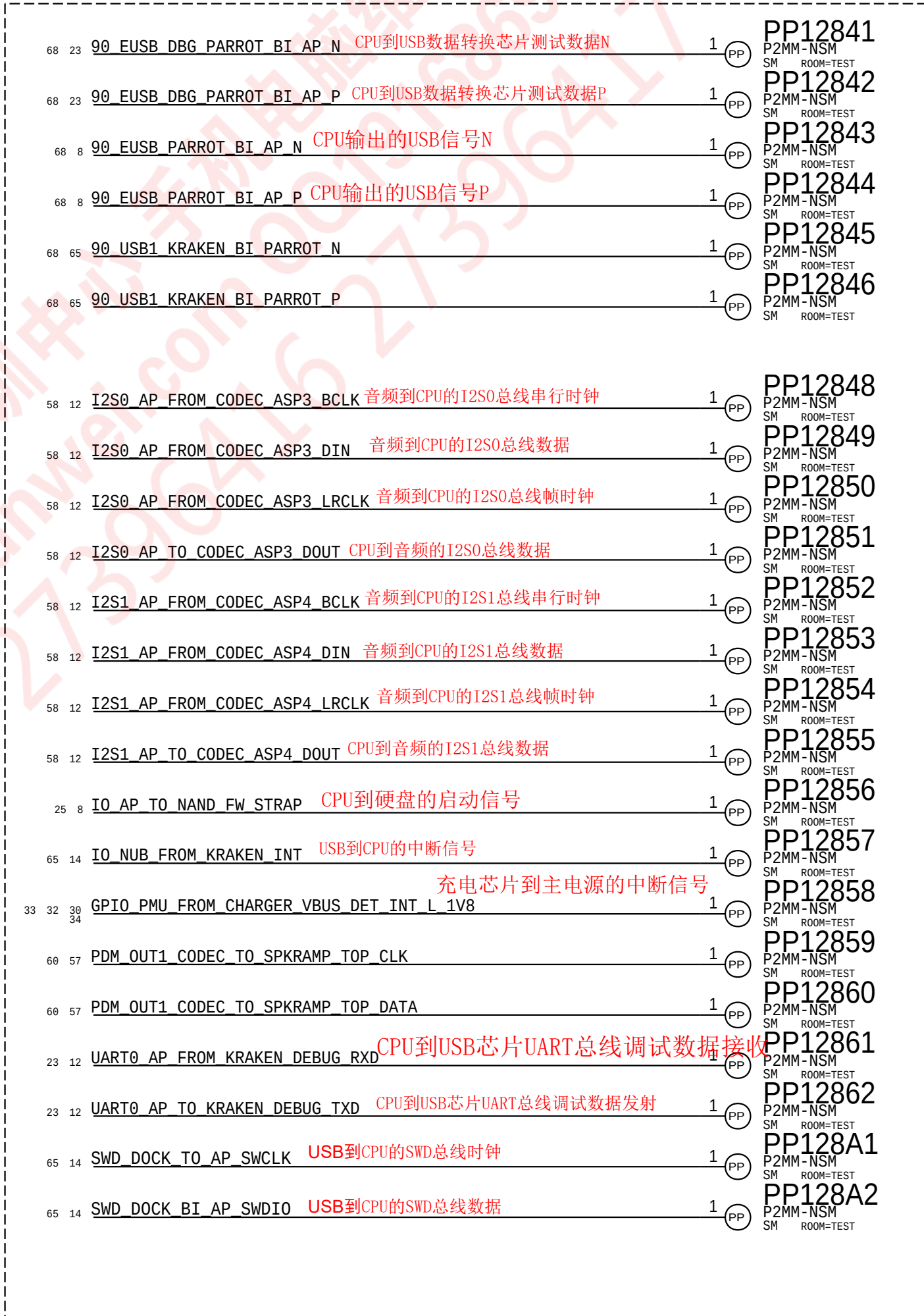
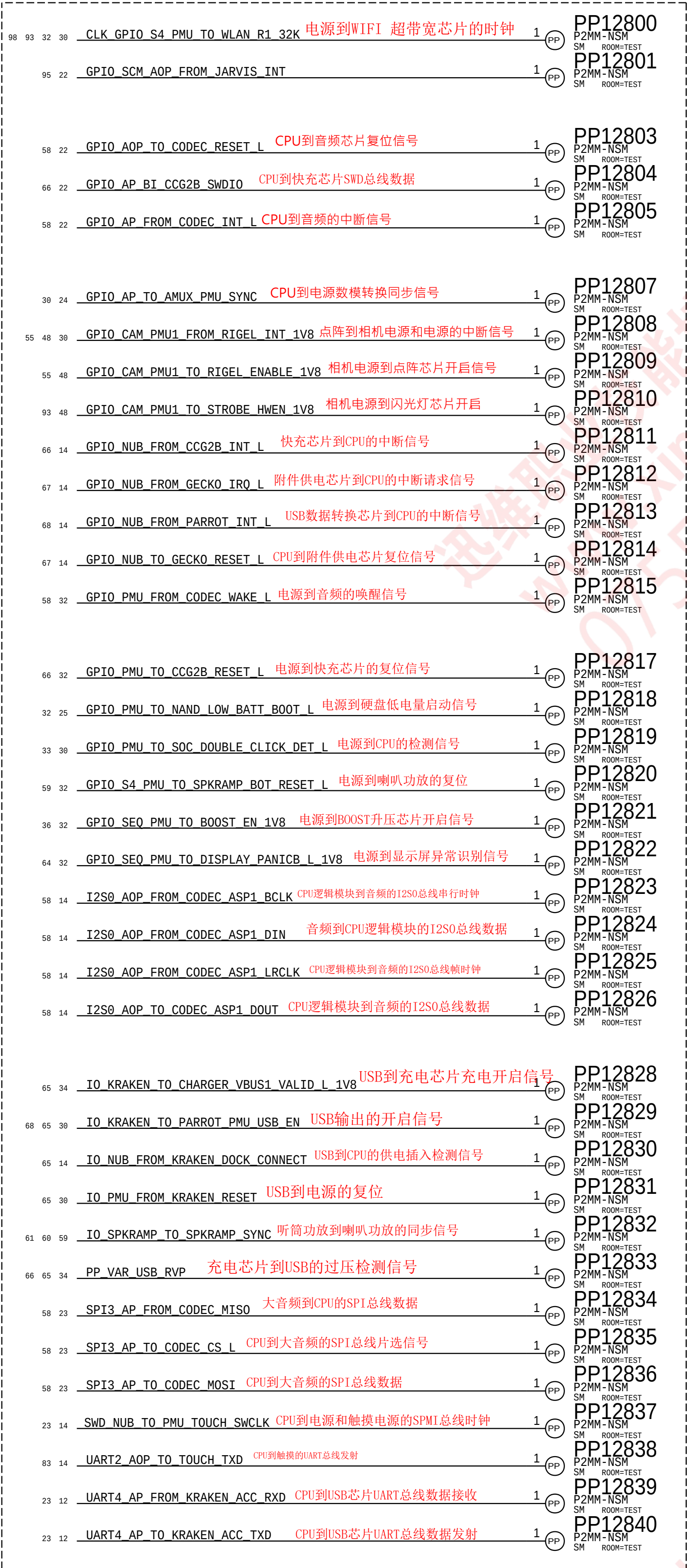
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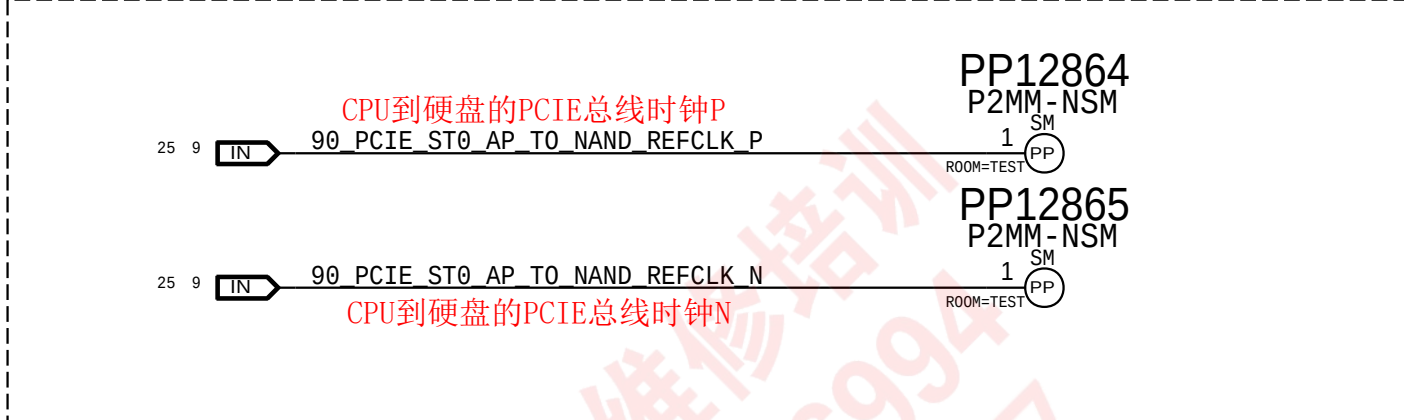
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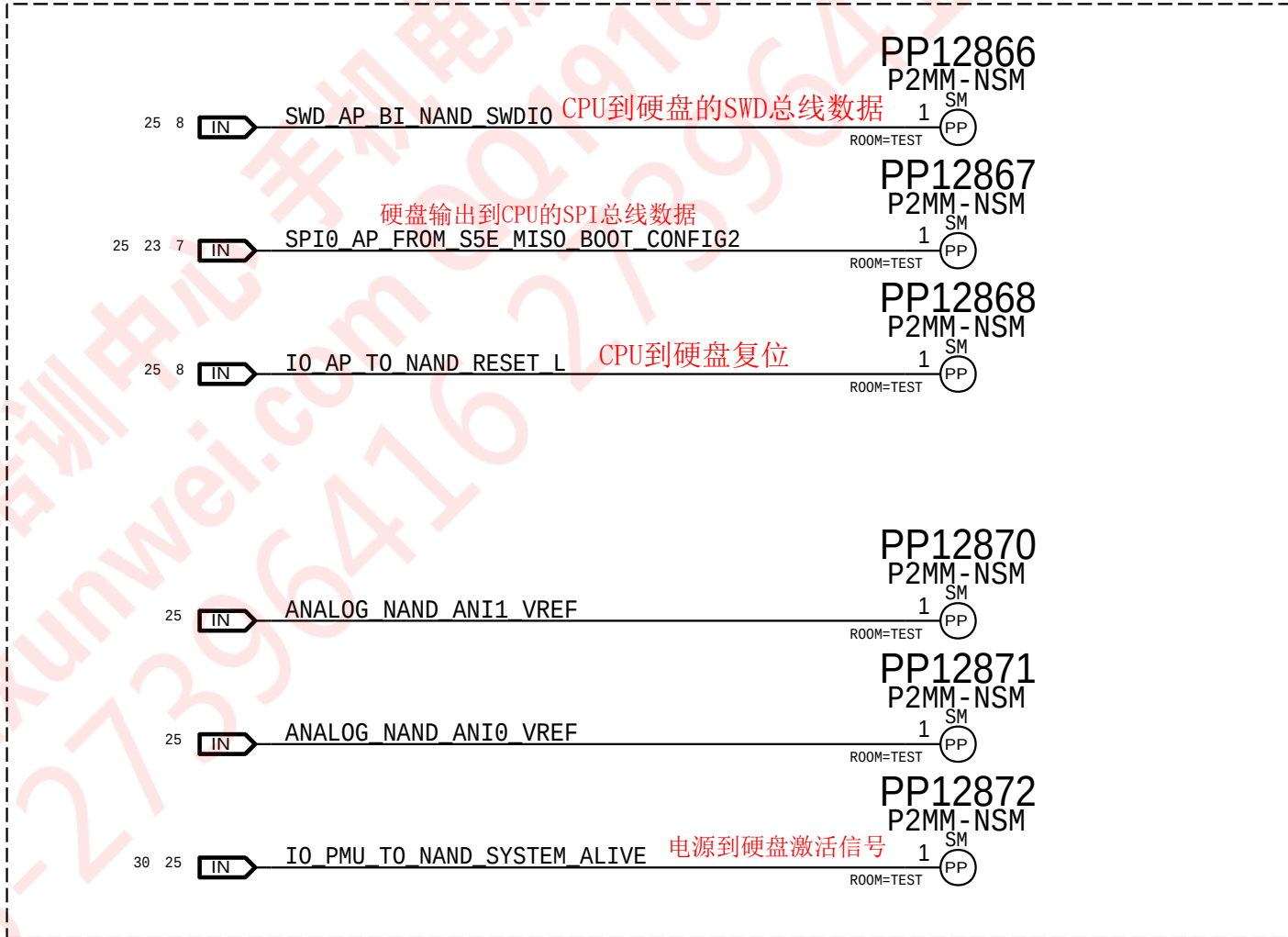
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	DRAWING NUMBER	051-05215
	REVISION	4.6.0
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PCIe

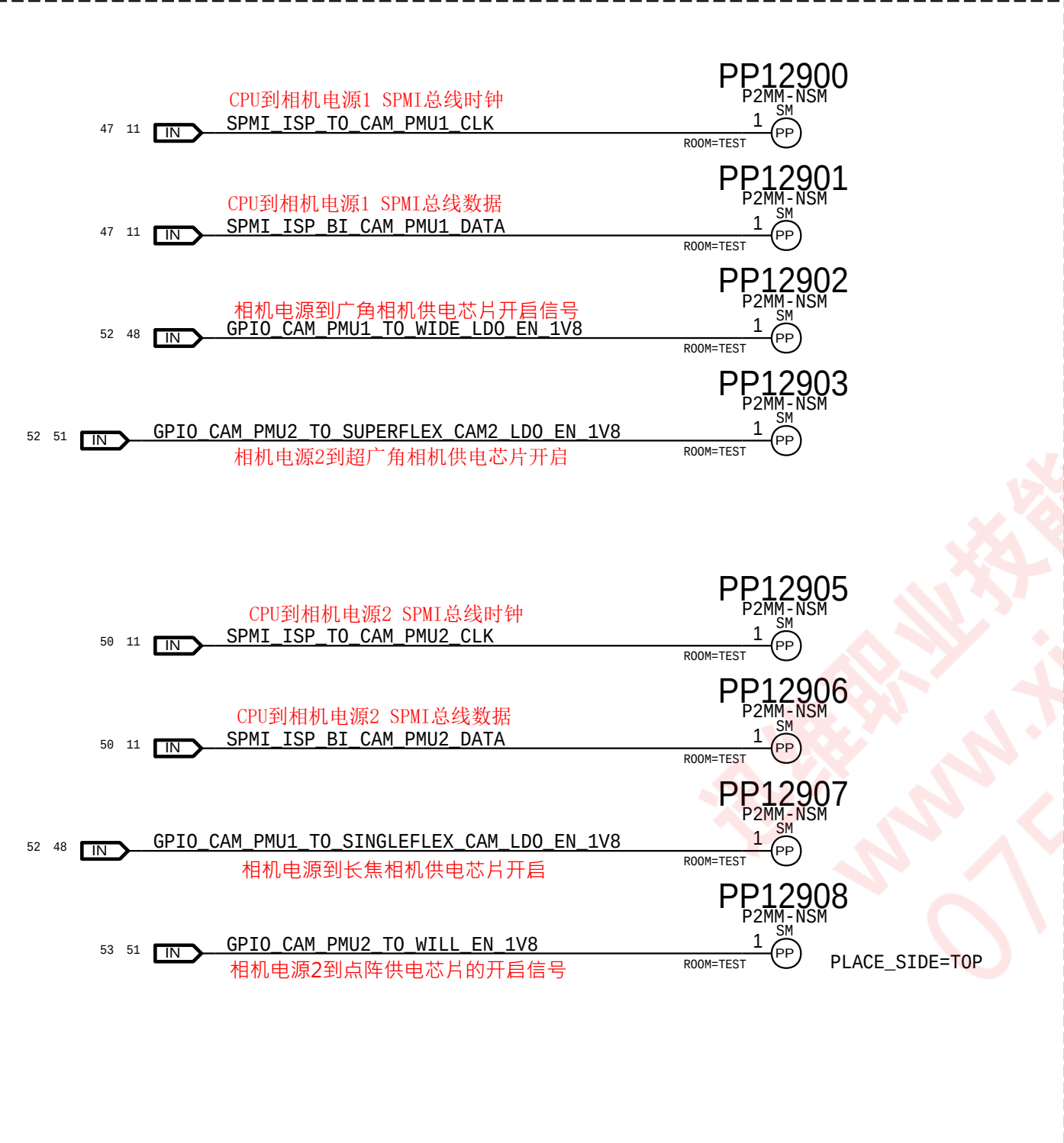


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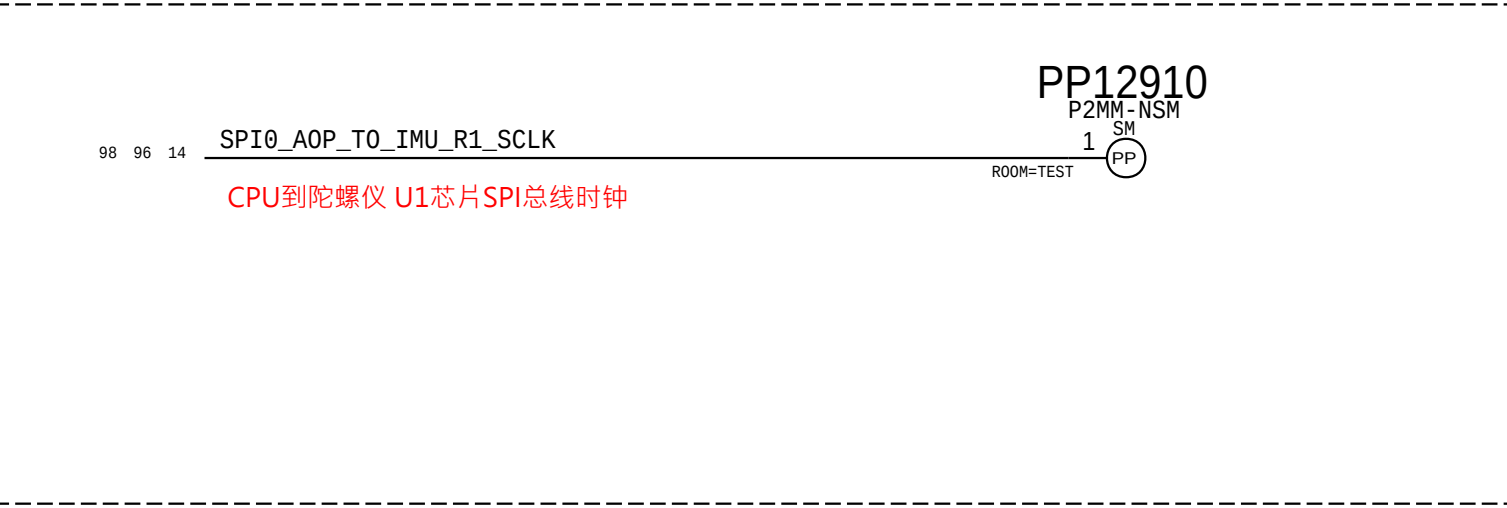


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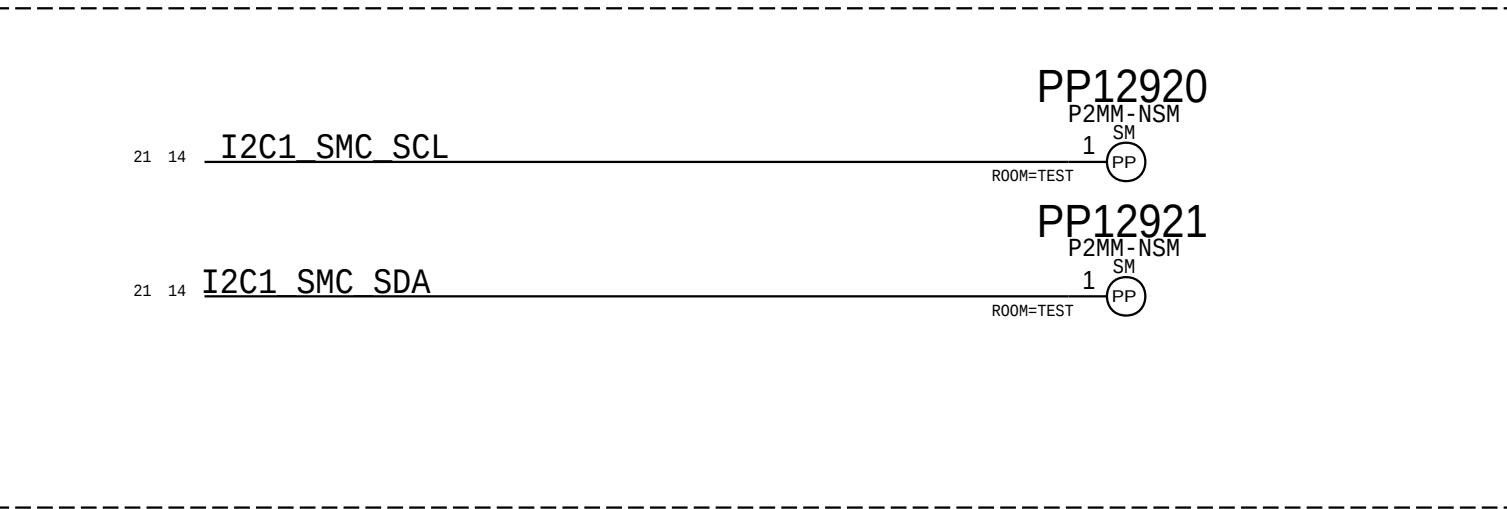
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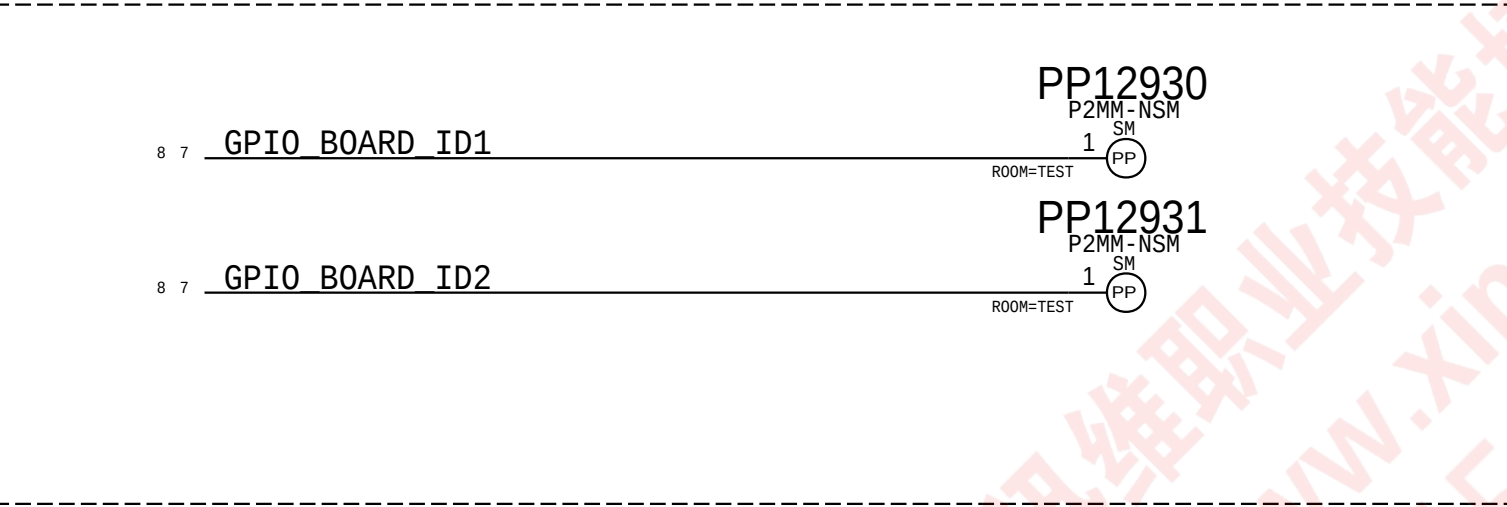
SPI0_A0P (IMU/R1)



I2C



BOARD_ID



PAGE TITLE		
TESTING: Probe Points (FF-Specific)		
	DRAWING NUMBER	051-05215
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Interposer Symbol (1/2)

J INT BOT
INTERPOSER-BOT-D53
SM
SYM 1 OF 4

998-20169		
GND	1	IO1
GND	2	IO2
GND	3	IO3
GND	4	IO4
GND	5	IO5
GND	6	IO6
GPIO AP CANARY6_LEG2	7	IO7
GND	8	IO8
GND	9	IO9
GPIO AP CANARY6_LEG1	10	IO10
GND	11	IO11
GND	12	IO12
GND	13	IO13
GND	14	IO14
GPIO AP CANARY6_LEG1	15	IO15
GPIO AP CANARY6	16	IO16
GND	17	IO17
PP1V8_S2	18	IO18
GND	19	IO19
GND	20	IO20
GND	21	IO21
GND	22	IO22
GND	23	IO23
GPIO ISP RCAM TO STROBE_TRIGGER_1V8	24	IO24
GND	25	IO25
GND	26	IO26
GND	27	IO27
IO_R1_TO_BB_LAA_SW_CTRL	28	IO28
GND	29	IO29
IO_BT_TO_R1_WAKE	30	IO30
GND	31	IO31
GND	32	IO32
GND	33	IO33
PPVAR_RCAMPVDD	34	IO34
GND	35	IO35
GND	36	IO36
GND	37	IO37
GPIO_CAM_PMU1_TO_STROBE_HWMEN_1V8	38	IO38
GND	39	IO39
GPIO_BB_TO_R1_CH9_REQ	40	IO40
GND	41	IO41
NTC_STROBE_MODULE	42	IO42
GND	43	IO43
GND	44	IO44
GPIO AP CANARY5	45	IO45
GND	46	IO46
GPIO AP CANARY5_LEG1	47	IO47
GND	48	IO48
GND	49	IO49
GND	50	IO50
GPIO AP CANARY5_LEG1	51	IO51
GND	52	IO52
GND	53	IO53
GPIO AP CANARY5_LEG2	54	IO54
GND	55	IO55
GND	56	IO56
GND	57	IO57
GND	58	IO58
GND	59	IO59
GND	60	IO60
GND	62	IO62
GPIO AP CANARY5_LEG2	63	IO63
GND	64	IO64
GND	65	IO65
GND	66	IO66
GPIO AP CANARY5_LEG3	67	IO67
GND	68	IO68
GPIO AP CANARY5_LEG3	69	IO69
GND	70	IO70
GND	71	IO71
GND	72	IO72
GND	73	IO73
GND	74	IO74
GND	75	IO75
GND	76	IO76
GND	77	IO77
GND	78	IO78
GND	79	IO79
GND	80	IO80
GND	81	IO81
GND	82	IO82
GND	83	IO83
GND	84	IO84
GND	85	IO85
GND	86	IO86
GND	87	IO87
GND	88	IO88
GND	89	IO89
PP_VDD_MAIN	90	IO90

ROOM=INTERPOSER

J INT BOT
INTERPOSER-BOT-D53
SM
SYM 2 OF 4

998-20169		
GND	181	IO181
PP_VDD_MAIN	182	IO182
GND	183	IO183
PP_VDD_MAIN	184	IO184
GND	185	IO185
PP_VDD_MAIN	186	IO186
GND	187	IO187
GND	188	IO188
GND	189	IO189
PP3V0_S2	190	IO190
GND	191	IO191
GND	192	IO192
GND	193	IO193
GND	194	IO194
GND	195	IO195
GND	196	IO196
GND	197	IO197
GND	198	IO198
GND	199	IO199
GPIO_BT_TO_BT_SPKR_TRIG	200	IO200
GND	201	IO201
GND	202	IO202
GND	203	IO203
GND	204	IO204
GND	205	IO205
GND	206	IO206
GND	207	IO207
GND	208	IO208
GND	209	IO209
RFFE_WLAN_TO_LAT_2G_FEM_RST_L	210	IO210
GND	211	IO211
GPIO_WLAN_TO_LAT_2G_FEM_LNA_BYP_EN	212	IO212
GND	213	IO213
ANALOG_LAT_2G_FEM_TO_WLAN_PDS	214	IO214
GND	215	IO215
PP_LD06_VIO_1V8_SOUTH	216	IO216
GND	217	IO217
SHIELD_RFFE_WLAN_TO_LAT_FEMS_CLK	218	IO218
GND	219	IO219
SHIELD_RFFE_WLAN_TO_LAT_FEMS_DATA	220	IO220
GND	221	IO221
PP1V8_S2	222	IO222
GND	223	IO223
GND	224	IO224
GND	225	IO225
50_RX_LMB_PAD_LMB_PRX_OUT_M	226	IO226
50_TX_IN_TX2_UHB_PAD_TX_N79_IN	227	IO227
GND	228	IO228
GND	229	IO229
GND	230	IO230
GND	231	IO231
50_TX_IN_XCVR_TX0_LB	232	IO232
GND	233	IO233
GND	234	IO234
GND	235	IO235
50_UHB_L_CPLR_OUT	236	IO236
50_RX_XCVR_PRX6_LB_A_RX0	237	IO237
GND	238	IO238
GND	239	IO239
GND	240	IO240
GND	241	IO241
PP_QET0_2G_PA	242	IO242
GND	243	IO243
PP_QET0_2G_PA	244	IO244
GND	245	IO245
PP_QET0_PA	246	IO246
GND	247	IO247
PP_QET0_PA	248	IO248
GND	249	IO249
GND	250	IO250
GND	251	IO251
50_WLAN_G_C1_TXRX_FEM_M	252	IO252
GND	253	IO253
GND	254	IO254
GND	255	IO255
SHIELD_RFFE1_SCLK_SDR	256	IO256
GND	257	IO257
GND	258	IO258
GND	259	IO259
SHIELD_RFFE1_SDATA_SDR	260	IO260
GND	261	IO261
GND	262	IO262
GND	263	IO263
GND	264	IO264
GPIO_AOP_CANARY7_LEG3	265	IO265
GND	266	IO266
GPIO_AOP_CANARY7_LEG3	267	IO267
GND	268	IO268
GND	269	IO269
GND	270	IO270

ROOM=INTERPOSER

PAGE TITLE		
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Interposer Symbol (2/2)

J INT BOT
INTERPOSER-BOT-D53
SM
SYM 3 OF 4

998-20169			
GND	361	IO361	IO451
PP_APT_L_PA	362	IO362	IO452
PP_APT_L_PA	363	IO363	IO453
GND	364	IO364	IO454
GND	365	IO365	IO455
PP_QET1_PA	366	IO366	IO456
PP_QET1_PA	367	IO367	IO457
GND	368	IO368	IO458
GND	369	IO369	IO459
GND	370	IO370	IO460
50_RX_XCVR_PRX15_N79_RX_OUT2	371	IO371	IO461
GND	372	IO372	IO462
GND	373	IO373	IO463
GND	374	IO374	IO464
50_RX_XCVR_PRX17_UHB_RX_OUT2	375	IO375	IO465
50_RX_XCVR_PRX14_N79_RX_OUT1	376	IO376	IO466
GND	377	IO377	IO467
GND	378	IO378	
GND	379	IO379	IO469
GND	380	IO380	IO470
50_RX_XCVR_PRX16_UHB_RX_OUT1	381	IO381	IO471
SHIELD_RFFE2_SDATA_SDR	382	IO382	IO472
GND	383	IO383	IO473
SHIELD_RFFE2_SCLK_SDR	384	IO384	IO474
SHIELD_RFFE8_SDATA_SDR	385	IO385	IO475
SHIELD_RFFE8_SCLK_SDR	386	IO386	IO476
PP_VDD_BOOST	387	IO387	IO477
GND	388	IO388	IO478
GND	389	IO389	IO479
GND	390	IO390	IO480
50_TX_IN_TX2_UHB_PAD_TX_UHB_IN_1	391	IO391	IO481
GND	392	IO392	IO482
GND	393	IO393	IO483
GND	394	IO394	IO484
50_TX_IN_XCVR_TX1_LB_JB	395	IO395	IO485
SHIELD_RFFE3_SDATA_SDR	396	IO396	IO486
GND	397	IO397	IO487
SHIELD_RFFE3_SCLK_SDR	398	IO398	IO488
GND	399	IO399	IO489
GND	400	IO400	IO490
GND	401	IO401	IO491
PP_VDD_MAIN	402	IO402	IO492
GND	403	IO403	IO493
PP_VDD_MAIN	404	IO404	IO494
GND	405	IO405	IO495
PP_VDD_MAIN	406	IO406	IO496
GND	407	IO407	IO497
PP_VDD_MAIN	408	IO408	
GND	409	IO409	IO499
GND	410	IO410	IO500
GND	411	IO411	IO501
GND	412	IO412	IO502
GND	413	IO413	IO503
PP1V2_S2	414	IO414	IO504
GND	415	IO415	IO505
GND	416	IO416	IO506
GND	417	IO417	IO507
GND	418	IO418	IO508
GND	419	IO419	IO509
GPIO_SCM_AOP_FROM_JARVIS_INT	420	IO420	IO510
GND	421	IO421	IO511
I2C1_AOP_SCL	422	IO422	IO512
GND	423	IO423	IO513
I2C1_AOP_SDA	424	IO424	IO514
GND	425	IO425	IO515
GND	426	IO426	IO516
GND	427	IO427	IO517
90_PCIE_GP2_AP_TO_BB_TX_N	428	IO428	IO518
GND	429	IO429	IO519
90_PCIE_GP2_AP_TO_BB_TX_P	430	IO430	IO520
GND	431	IO431	IO521
GND	432	IO432	IO522
GND	433	IO433	IO523
90_PCIE_GP2_AP_FROM_BB_RX_P	434	IO434	IO524
GND	435	IO435	IO525
90_PCIE_GP2_AP_FROM_BB_RX_N	436	IO436	IO526
GND	437	IO437	IO527
GND	438	IO438	IO528
GND	439	IO439	IO529
90_PCIE_GP2_AP_TO_BB_REFCLK_N	440	IO440	IO530
GND	441	IO441	IO531
90_PCIE_GP2_AP_TO_BB_REFCLK_P	442	IO442	IO532
GND	443	IO443	IO533
GND	444	IO444	IO534
GND	445	IO445	IO535
PP_VDD_MAIN	446	IO446	
GND	447	IO447	IO537
PP_VDD_MAIN	448	IO448	IO538
GND	449	IO449	IO539
PP_VDD_MAIN	450	IO450	IO540

ROOM=INTERPOSER

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998-20169			
GND	541	IO541	IO631
GND	542	IO542	IO632
50_WLAN_A_C0_TXRX_FEM_M	543	IO543	IO633
GND	544	IO544	IO634
GND	545	IO545	IO635
50_WLAN_G_C0_TXRX_FEM_M	546	IO546	IO636
GND	547	IO547	IO637
GND	548	IO548	IO638
50_TRX_PAUHB_UHB_ANT1_TO_ANT4	549	IO549	IO639
GND	550	IO550	IO640
GND	551	IO551	IO641
50_TRX_PAUHB_N79_ANT1_TO_ANT4	552	IO552	IO642
GND	553	IO553	IO643
GND	554	IO554	IO644
GND	555	IO555	IO645
GND	556	IO556	IO646
GND	557	IO557	IO647
GND	558	IO558	IO648
GND	559	IO559	IO649
I0_NFC_TO_BBPMU_CLKREQ	560	IO560	IO650
GND	561	IO561	IO651
GND	562	IO562	IO652
GND	563	IO563	IO653
GND	564	IO564	IO654
GND	565	IO565	IO655
GND	566	IO566	IO656
GND	567	IO567	IO657
GND	568	IO568	IO658
GND	569	IO569	IO659
GND	570	IO570	IO660
GND	571	IO571	IO661
GND	572	IO572	IO662
GND	573	IO573	IO663
ANALOG_UAT_2G_FEM_TO_WLAN_PDTS	574	IO574	IO664
GND	575	IO575	IO665
GND	576	IO576	IO666
GND	577	IO577	IO667
GND	578	IO578	IO668
GND	579	IO579	IO669
GND	580	IO580	IO670
GND	581	IO581	IO671
50_TRX_CPLR_U_CPL2_ANT_LMB_MB_HB	582	IO582	IO672
GND	583	IO583	IO673
50_TRX_ANT8_UHB_N79	584	IO584	IO674
GND	585	IO585	IO675
GND	586	IO586	IO676
GND	587	IO587	IO677
GND	588	IO588	IO678
GND	589	IO589	IO679
GND	590	IO590	IO680
GPIO_AP_CANARY1	591	IO591	IO681
GND	592	IO592	IO682
GND	593	IO593	IO683
GPIO_AP_CANARY1_LEG1	594	IO594	IO684
GND	595	IO595	IO685
GND	596	IO596	IO686
GND	597	IO597	IO687
GND	598	IO598	IO688
GND	599	IO599	IO689
GND	600	IO600	IO690
GND	601	IO601	IO691
GND	602	IO602	IO692
GND	603	IO603	IO693
GND	604	IO604	IO694
GPIO_AP_CANARY1_LEG1	605	IO605	IO695
GND	606	IO606	IO696
GND	607	IO607	IO697
GND	608	IO608	IO698
GND	609	IO609	IO699
GND	610	IO610	IO700
GND	611	IO611	IO701
GND	612	IO612	IO702
GPIO_AP_CANARY1_LEG2	613	IO613	IO703
PP_LDO6_VIO_1V8_NORTH	614	IO614	IO704
GND	615	IO615	IO705
GPIO_BB_TO_R1_COEX	616	IO616	IO706
GND	617	IO617	IO707
I0_BT_TO_R1_TIME_SYNC	618	IO618	IO708
GND	619	IO619	IO709
GPIO_WLAN_TO_R1_COEX	620	IO620	IO710
GND	621	IO621	IO711
GND	622	IO622	IO712
GND	623	IO623	IO713
GND	624	IO624	IO714
50_TRX_ANT2_LB_LMB_MB_HB_L1	625	IO625	IO715
GND	626	IO626	IO716
50_LAA_RX_XCVR_DRX10	627	IO627	IO717
GND	628	IO628	IO718
GND	629	IO629	IO719
SHIELD_RFFE5_SCLK_SDR	630	IO630	IO720
			IO721
			IO722
			IO723
			IO724
			IO725
			IO726
			IO727

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PAGE TITLE		
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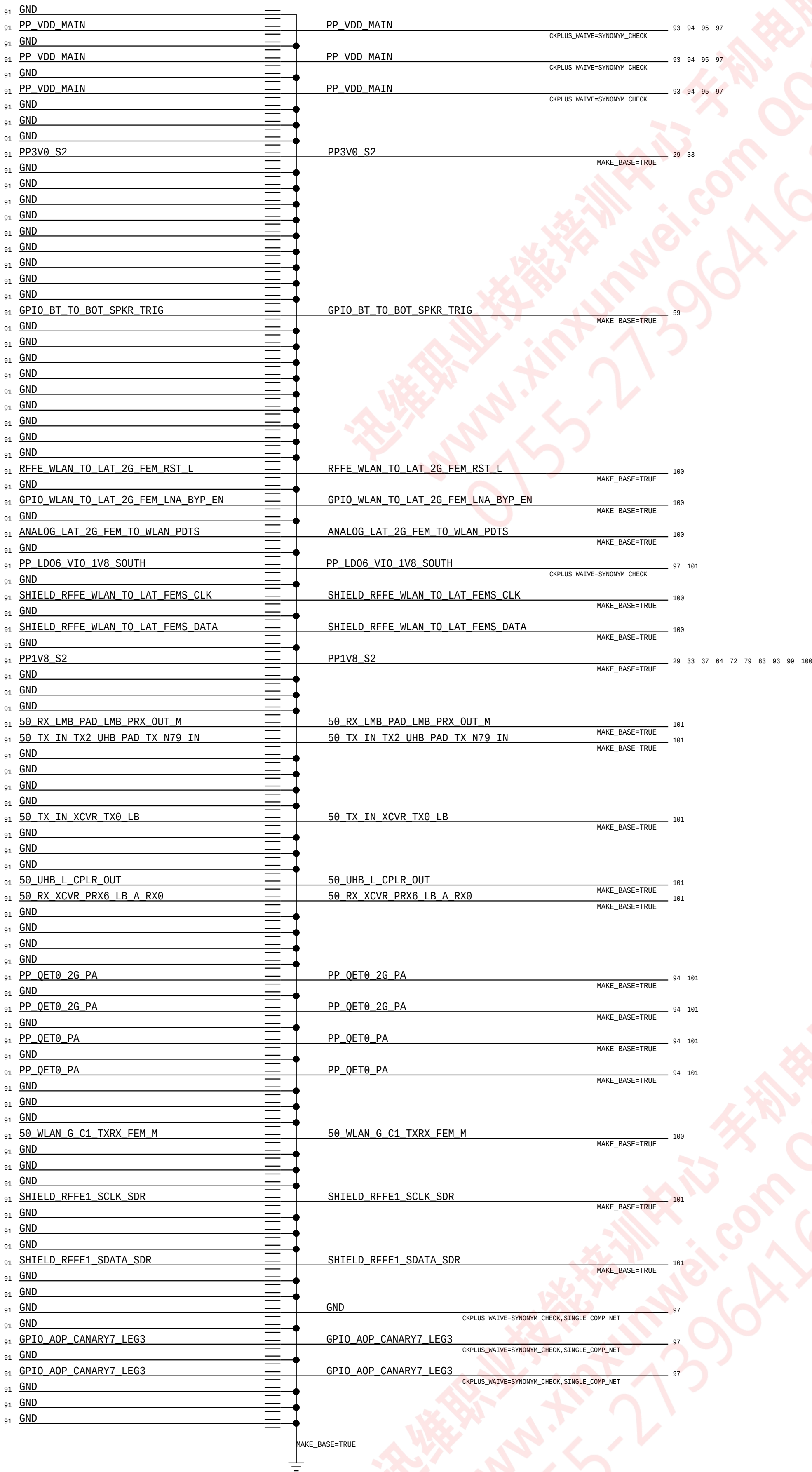
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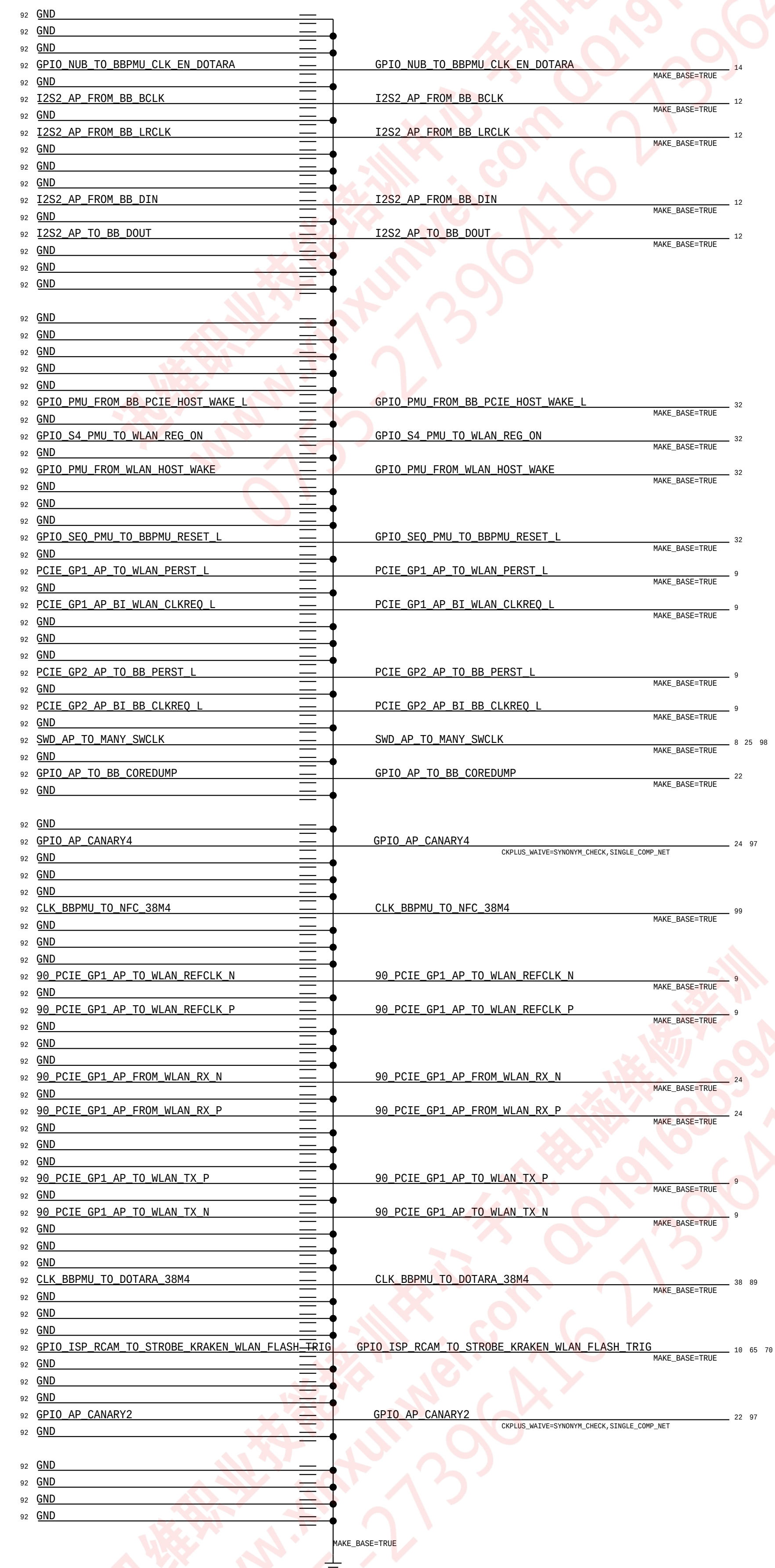
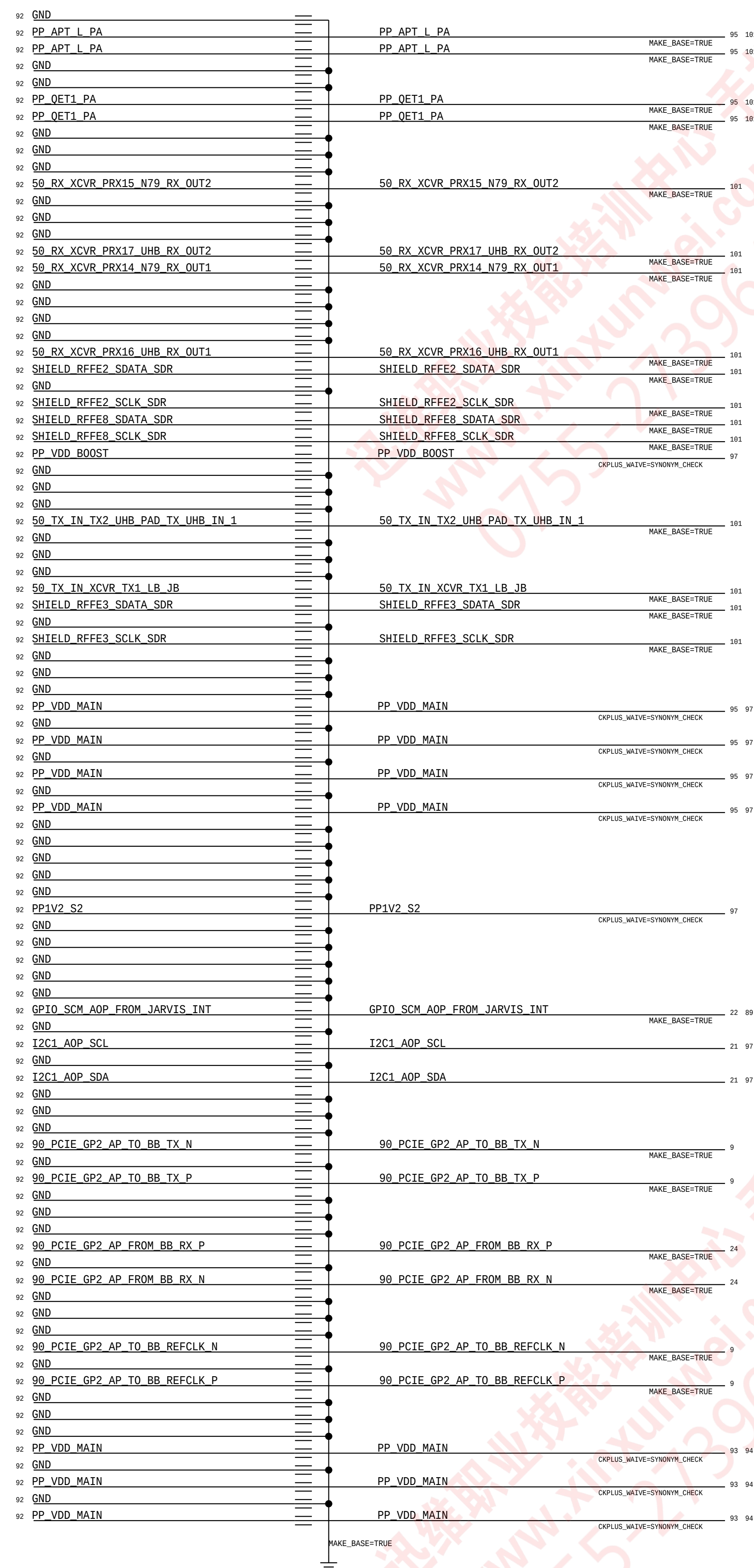
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Interposer Aliases (2/4)



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Interposer Aliases (3/4)



PAGE TITLE		
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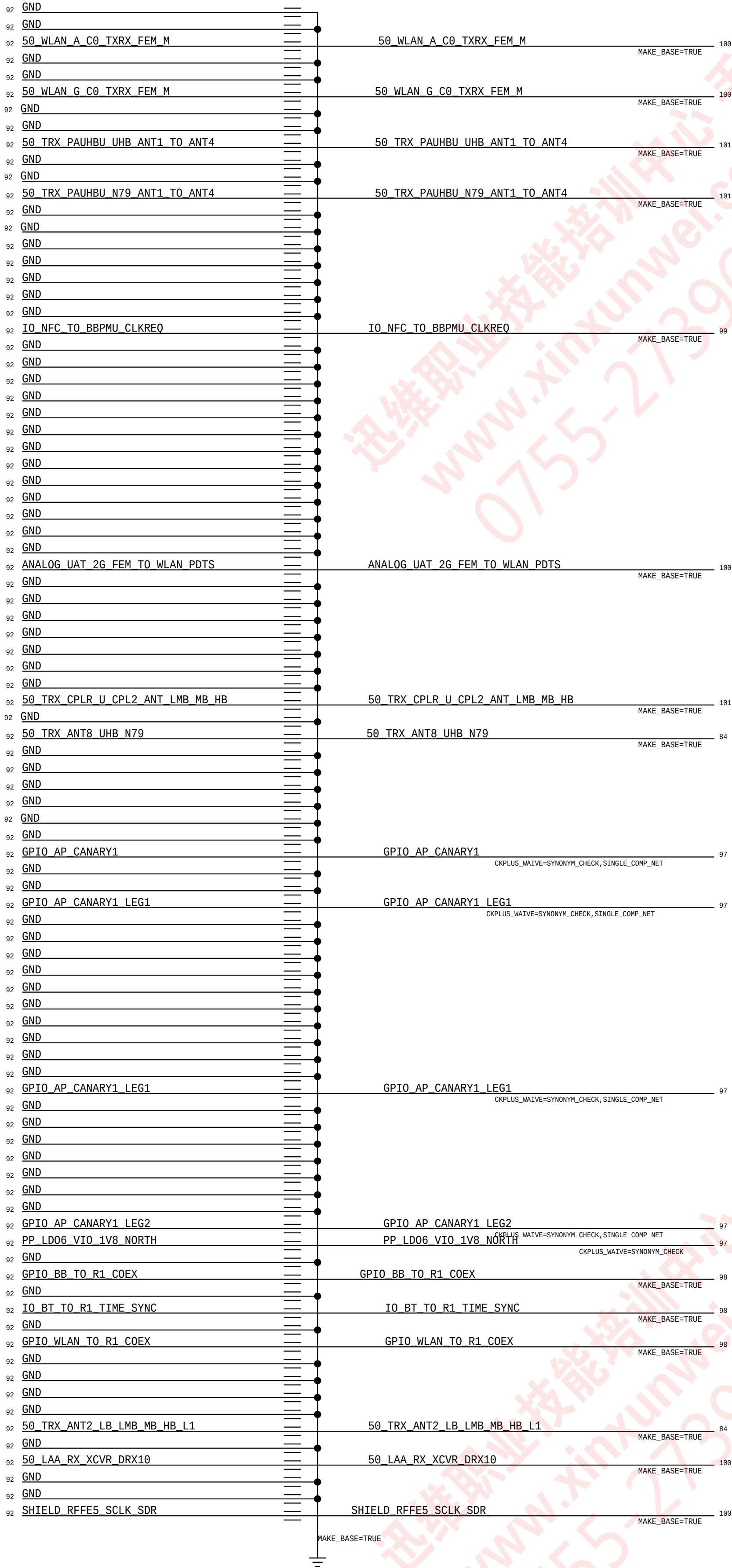
Interposer Aliases (4/4)

D

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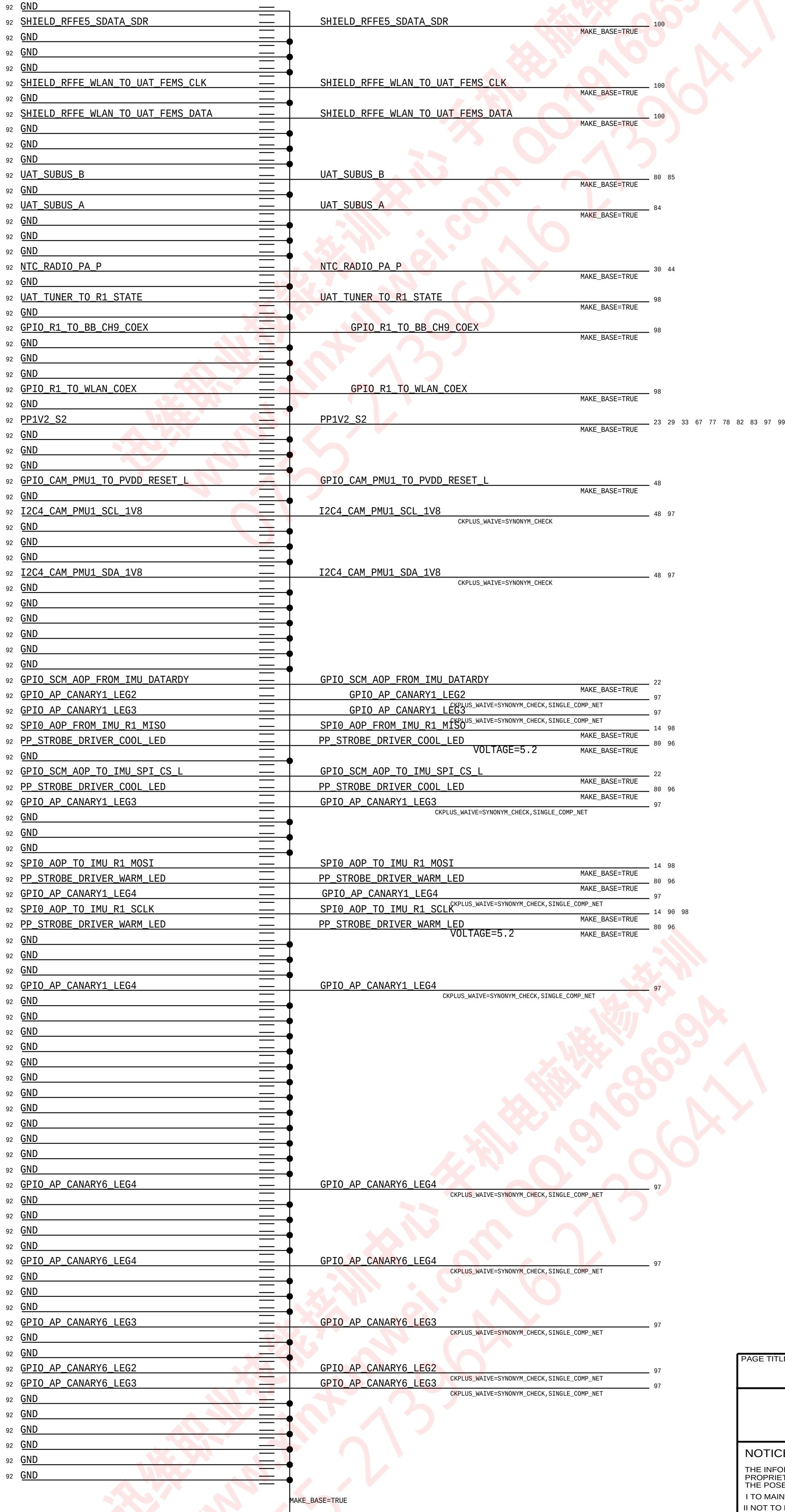


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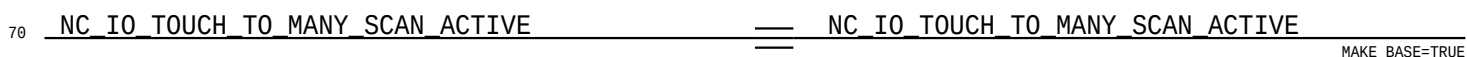
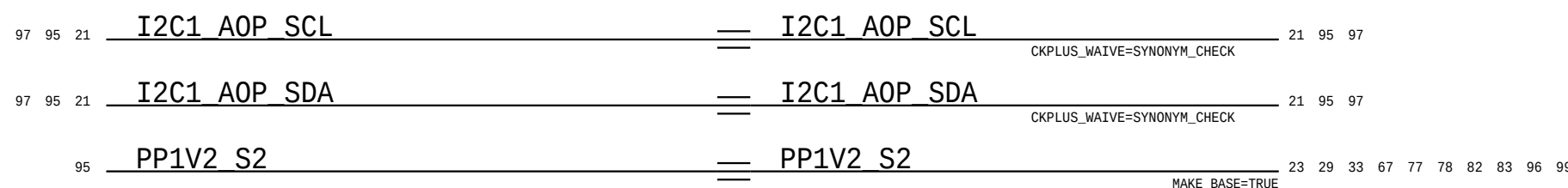
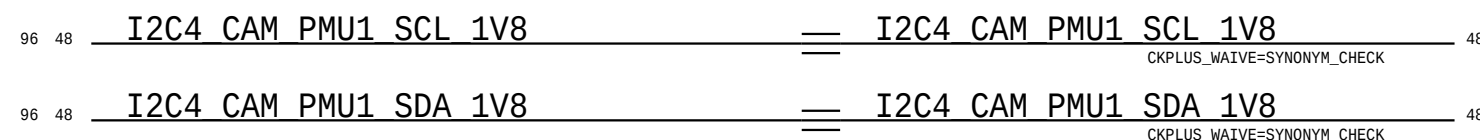
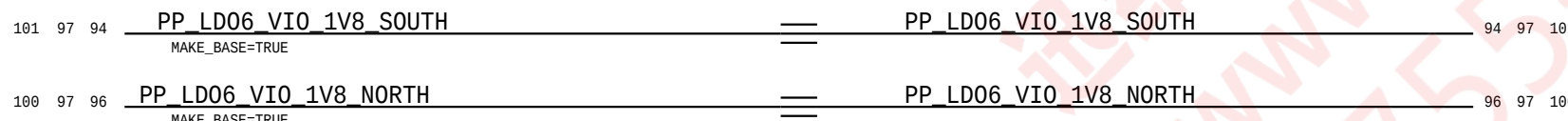
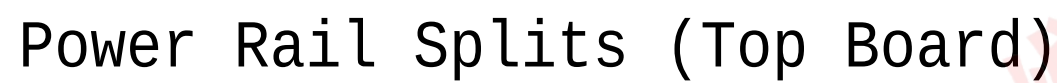
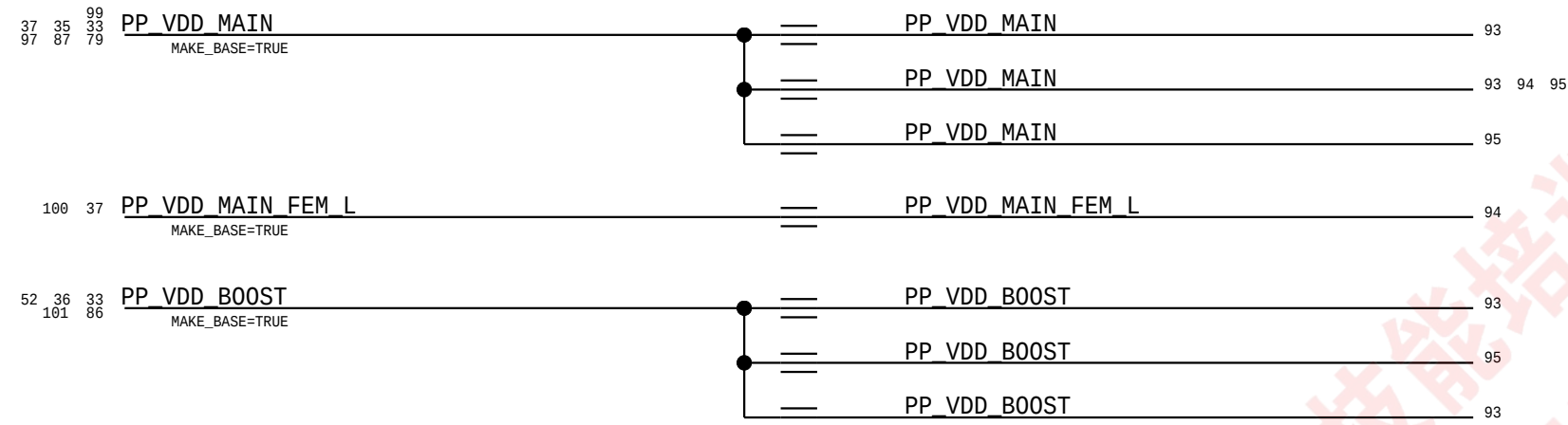
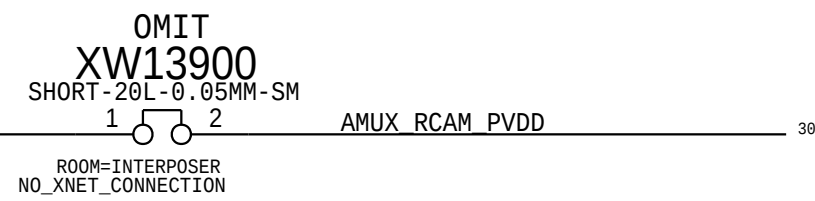
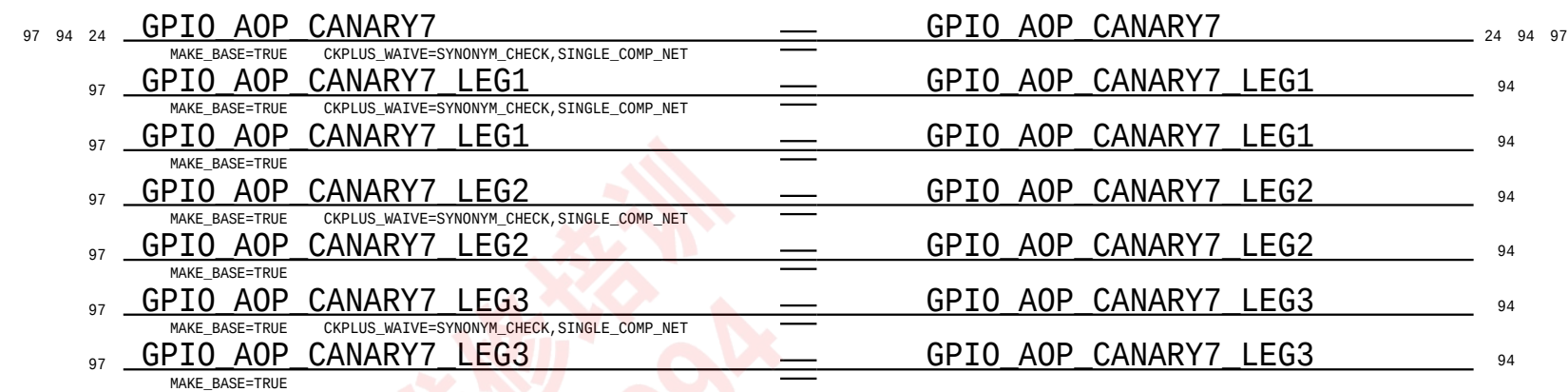
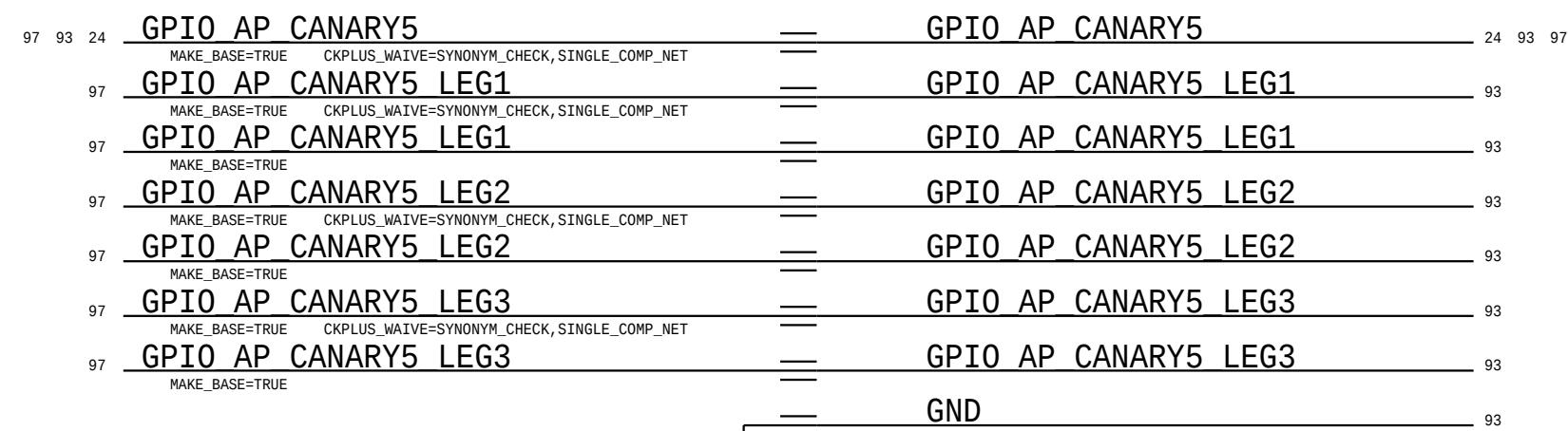
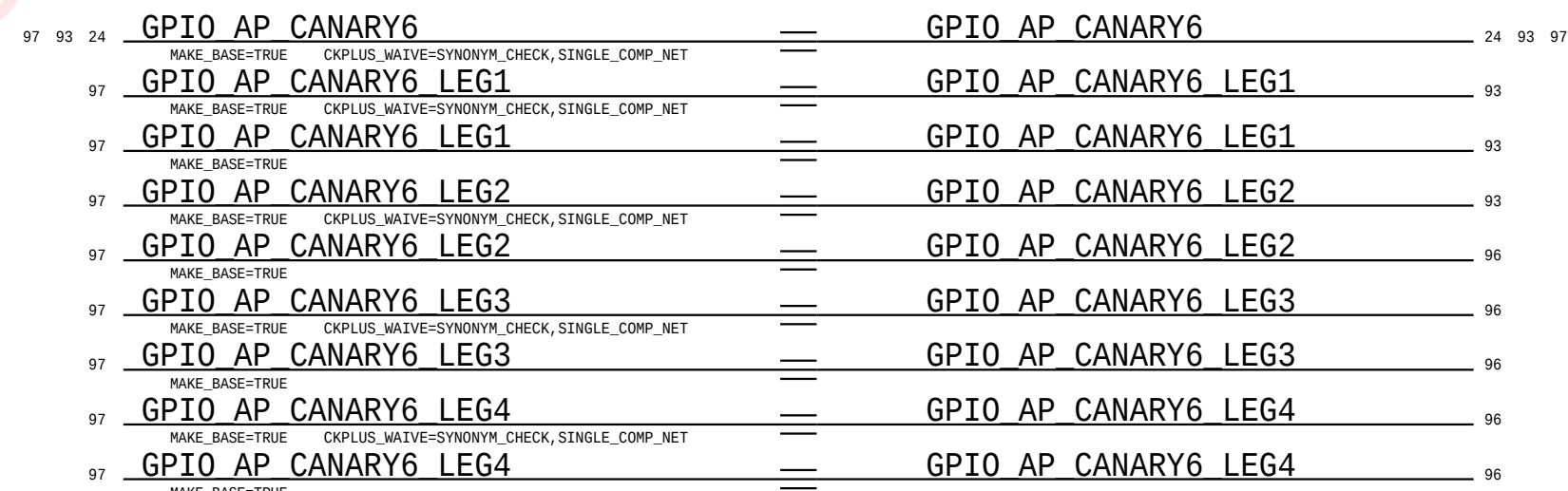
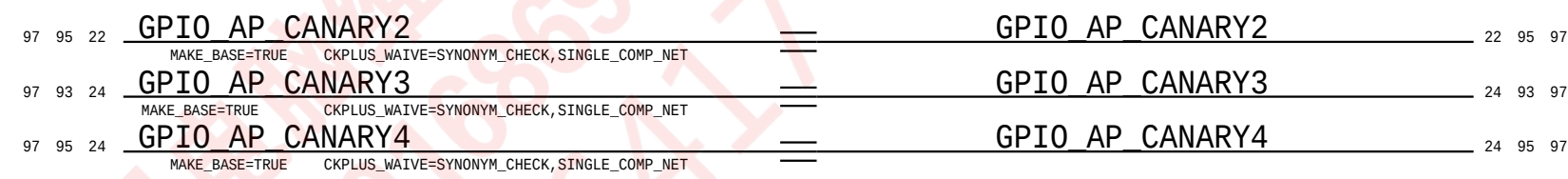
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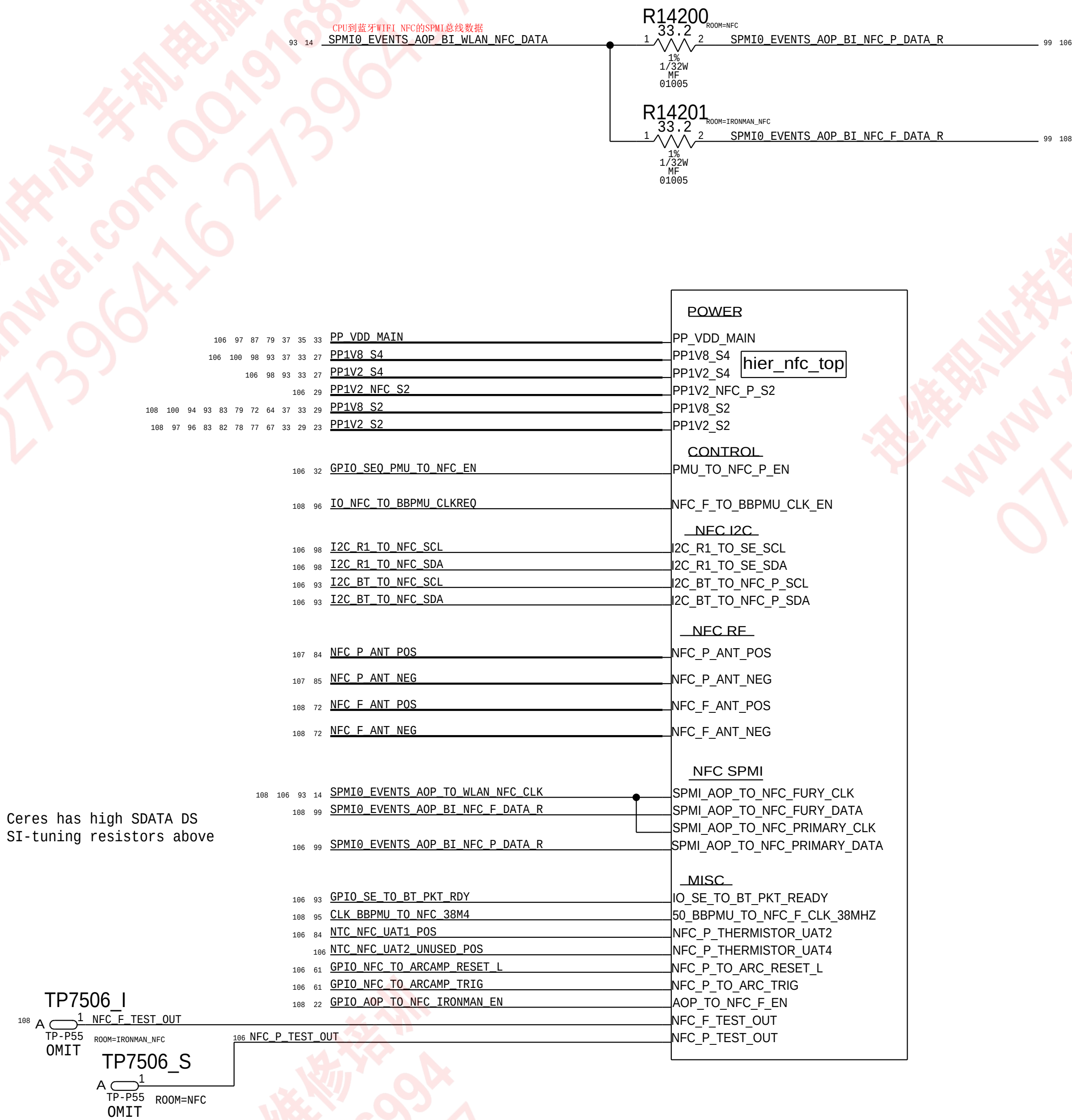
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Arrow Hierarchy

103	99	93	33	27	PP1V2_S4	PP1V2_S4		
103	100	99	93	37	33	27	PP1V8_S4	PP1V8_S4
103	29	PP1V0_R1_ANA_S4	PP1V0_R1_ANA					
103	29	PP1V0_S4	PP1V0_R1_SOC					
103	32	GPIO_S4_PMU_TO_R1_RESET_L	APPMU_TO_R1_RESET_L					
103	93	99	32	30	CLK_GPIO_S4_PMU_TO_WLAN_R1_32K	APPMU_TO_WLAN_32K_CLK		
103	22	GPIO_AOP_TO_R1_COREDUMP_TRIGGER	AOP_TO_R1_COREDUMP_TRIGGER					
103	22	GPIO_SCM_AOP_TO_R1_SPI_CS_L	SPI_AOP_TO_R1_CS_L					
103	96	98	14	SPI0_AOP_TO_IMU_R1_SCLK	SPI_AOP_TO_R1_SCLK			
103	96	14	SPI0_AOP_TO_IMU_R1_MOSI	SPI_AOP_TO_R1_MOSI				
103	96	14	SPI0_AOP_FROM_IMU_R1_MISO	SPI_R1_TO_AOP_MISO				
103	22	GPIO_SCM_AOP_FROM_R1_INT	R1_TO_AOP_INT					
103	99	I2C_R1_TO_NFC_SCL	I2C_R1_TO_SE_SCL					
103	99	I2C_R1_TO_NFC_SDA	I2C_R1_TO_SE_SDA	hier_arrow				
103	93	GPIO_BB_TO_R1_CH9_REQ	BB_TO_R1_CH9_REQ					
103	96	GPIO_WLAN_TO_R1_COEX	WLAN_TO_R1_COEX					
103	96	GPIO_R1_TO_BB_CH9_COEX	R1_TO_BB_GRANT					
103	96	GPIO_R1_TO_WLAN_COEX	R1_TO_WLAN_COEX					
103	96	GPIO_BB_TO_R1_COEX	BB_TO_R1_LAA_REQ					
103	22	GPIO_AOP_TO_R1_TIME_SYNC_L	AOP_TO_R1_TIME_SYNC					
103	96	IO_BT_TO_R1_TIME_SYNC	BT_TO_R1_TIME_SYNC					
103	93	IO_BT_TO_R1_WAKE	BT_TO_R1_WAKE					
103	8	SWD_AP_BI_R1_SWDIO CPU到U1的SWD总线数据	SWD_AOP_BI_R1_SWDIO					
103	95	25	8	SWD_AP_TO_MANY_SWCLK	SWD_AOP_TO_MANY_SWCLK			
104	84	50_TRX_R1_AOA1	50_R1_ANT0					
104	84	50_TRX_R1_AOA2	50_R1_ANT1					
104	84	50_TRX_R1_AOA3	50_R1_ANT3					
104	100	50_TRX_R1_ANT2_CH9	50_R1_ANT_CH9					
104	100	50_TRX_R1_ANT2_CH5	50_LAA_TO_R1					
103	96	UAT_TUNER_TO_R1_STATE	TUNER_TO_R1_STATE					
103	99	IO_R1_TO_BB_LAA_SW_CTRL	R1_TO_BB_LAA_SW_CTRL					

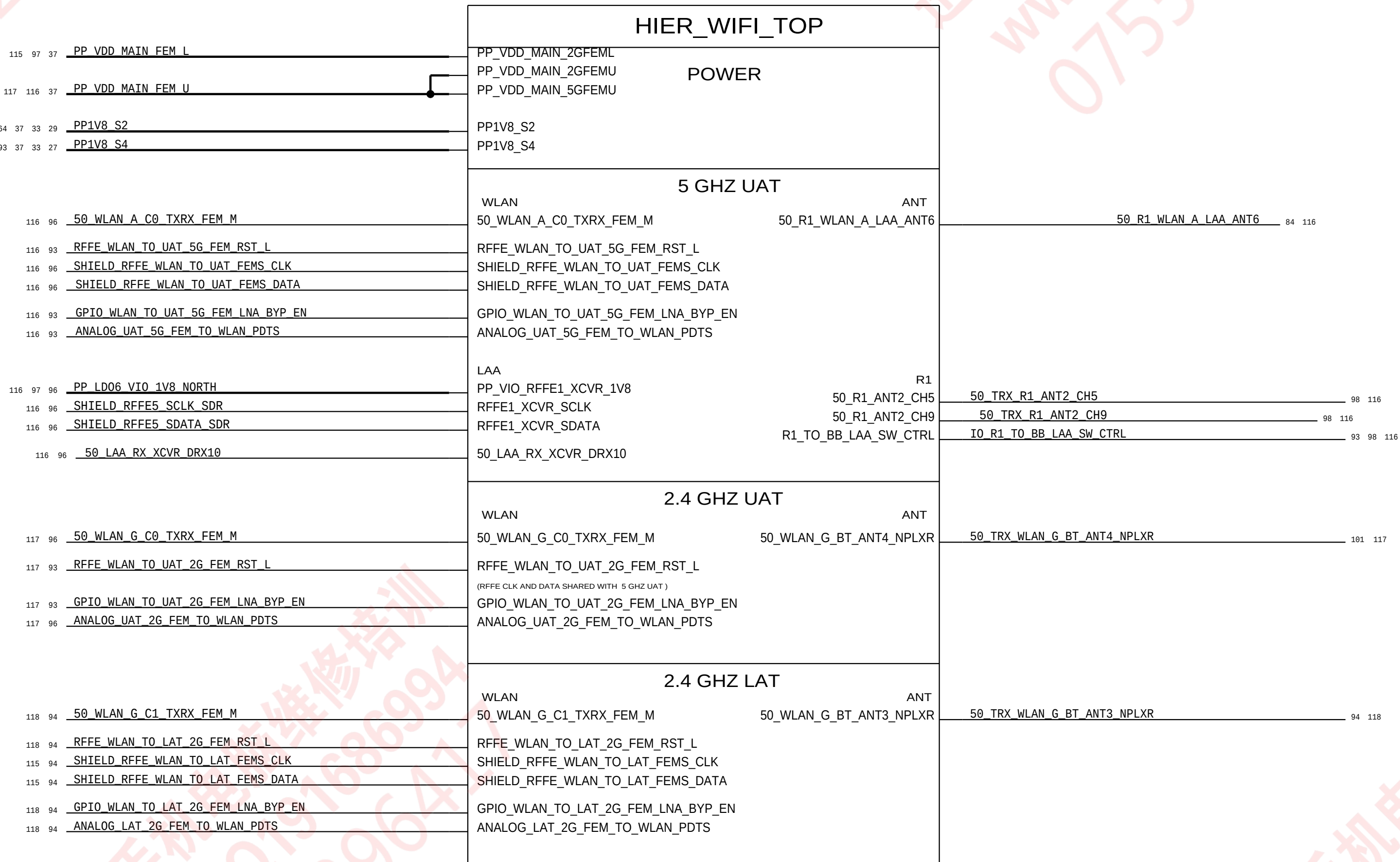
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	REVISION	4.6.0
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NFC Hierarchy



PAGE TITLE		
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	DRAWING NUMBER	051-05215
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WiFi Hierarchy



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Radio Hierarchy



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SCH #: 951-09582

PCB #: 920-05369

BOM #: 939-05686

POWER

IO PP1V2_S4

IO PP1V8_S4

IO PP1V0_R1 ANA S4

IO PP1V0_S4

HOST IPC

IO SPI0_AOP_TO_IMU_R1_SCLK

IO GPIO_SCM_AOP_TO_R1_SPI_CS_L

IO SPI0_AOP_FROM_IMU_R1_MISO

IO SPI0_AOP_TO_IMU_R1_MOSI

IO GPIO_AOP_TO_R1_COREDUMP_TRIGGER

IO GPIO_SCM_AOP_FROM_R1_INT

RESET/CLOCK

IO GPIO_S4_PMU_TO_R1_RESET_L

IO CLK_GPIO_S4_PMU_TO_WLAN_R1_32K

IO GPIO_AOP_TO_R1_TIME_SYNC_L

DEBUG

IO SWD_AP_TO_MANY_SWCLK

IO SWD_AP_BT_R1_SMDIO

SPMI/COEX GPIOs

IO GPIO_WLAN_TO_R1_COEX

IO GPIO_R1_TO_BB_CH9_COEX

IO GPIO_R1_TO_WLAN_COEX

IO GPIO_BB_TO_R1_COEX

IO GPIO_BB_TO_R1_CH9_REQ

SE IPC

IO I2C_R1_TO_NFC_SCL

IO I2C_R1_TO_NFC_SDA

BT TIME SYNC / WAKE

IO IO_BT_TO_R1_TIME_SYNC

IO IO_BT_TO_R1_WAKE

ANTENNA SWITCH CTRL

IO UAT_TUNER_TO_R1_STATE

IO IO_R1_TO_BB_LAA_SW_CTRL

ANTENNA

IO 50_TRX_R1_A0A1

IO 50_TRX_R1_A0A2

IO 50_TRX_R1_A0A3

IO 50_TRX_R1_ANT2_CH9

IO 50_TRX_R1_ANT2_CH5

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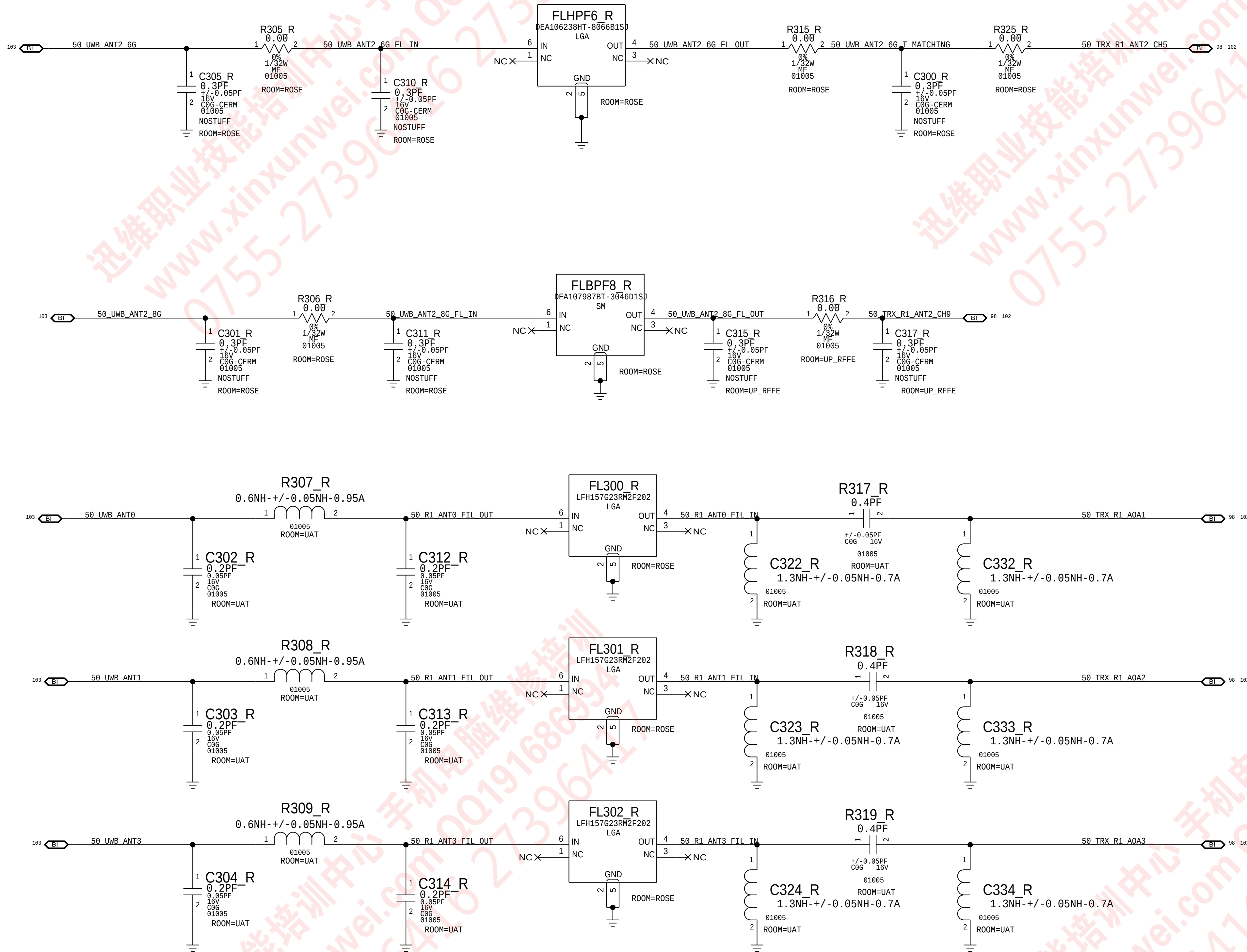
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FILTERS/MATCHING AND ANTENNA

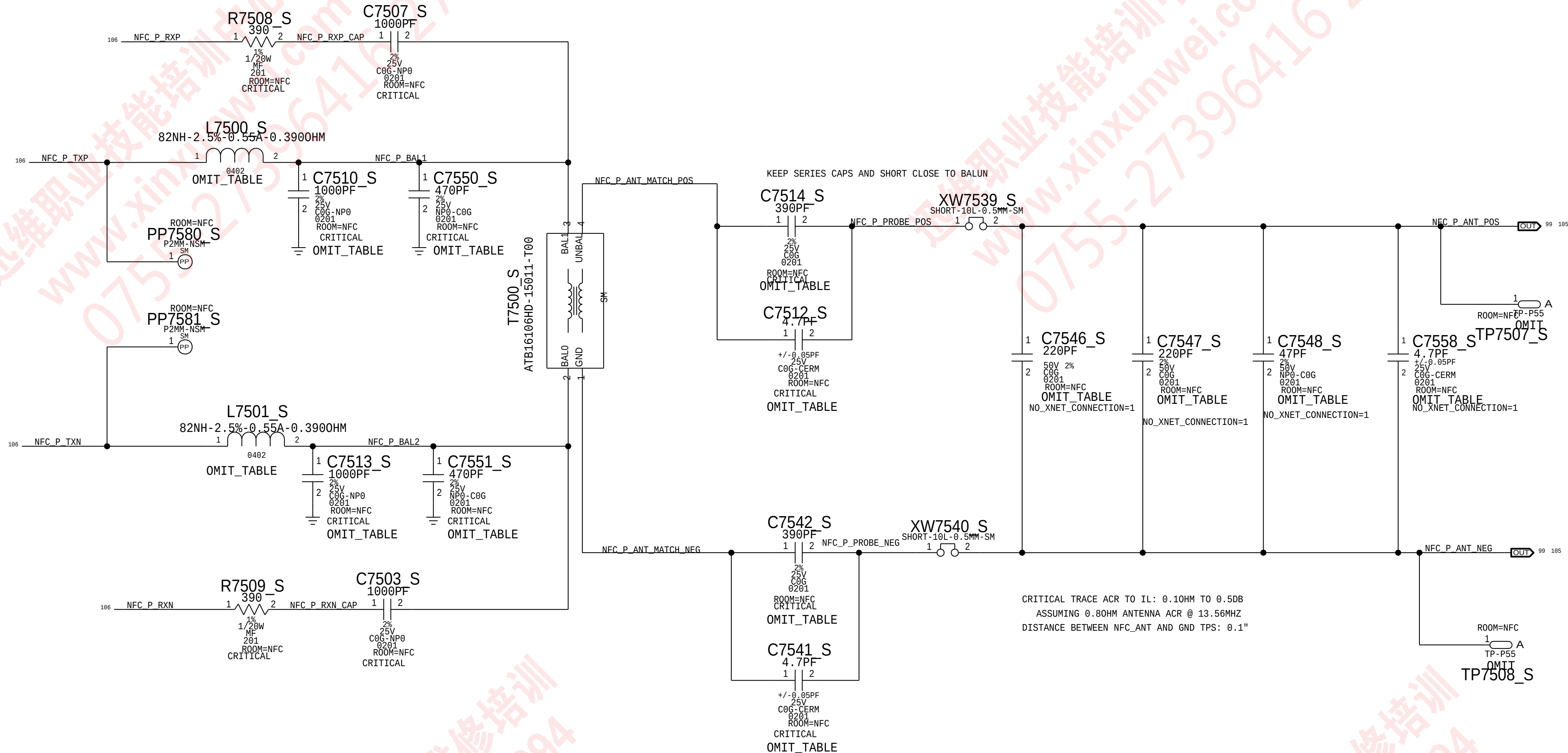


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PRIMARY NFC FRONT END



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POWER

PP_LD06_VI0_1V8_SOUTH

VOLTAGE=1.8

IN

101

110

111

112

PP_VDD_RF_1V2

VOLTAGE=1.2

IN

101

110

111

112

PP_VDD_BOOST

VOLTAGE=4.5

IN

101

110

111

112

ET/APT IC

PP_APT_L_PA

VOLTAGE=4.5

OUT

101

111

PP_OET0_PA

VOLTAGE=5.8V

OUT

101

112

PP_OET0_2G_PA

VOLTAGE=5.8V

OUT

101

112

PP_OET1_PA

VOLTAGE=5.8V

OUT

101

110

RF RFFE

SHIELD_RFFE1_SCLK_SDR

IN

101

112

SHIELD_RFFE1_SDATA_SDR

IO

101

112

SHIELD_RFFE2_SCLK_SDR

IN

101

111

SHIELD_RFFE2_SDATA_SDR

IO

101

111

SHIELD_RFFE3_SCLK_SDR

IN

101

110

111

112

SHIELD_RFFE3_SDATA_SDR

IO

101

110

111

112

SHIELD_RFFE8_SCLK_SDR

IN

101

110

SHIELD_RFFE8_SDATA_SDR

IO

101

110

LB PAD

50_RX_XCVR_PRX6_LB_A_RX0

OUT

101

110

50_RX_XCVR_PRX10_LB_PRX_OUT2

OUT

101

110

50_TX_IN_XCVR_TX1_LB_JB

IN

101

110

50_TX_IN_XCVR_TX0_LB

IN

101

110

50_TRX_LB_PORTB_RX1_M

OUT

101

110

50_TRX_LB_PORTB_TX_M

IO

101

110

50_TRX_LB_PAD_ANT1

OUT

101

110

UHB PA (LAT)

50_RX_XCVR_PRX14_N79_RX_OUT1

OUT

101

111

50_RX_XCVR_PRX15_N79_RX_OUT2

OUT

101

111

50_RX_XCVR_PRX16_UHB_RX_OUT1

OUT

101

111

50_RX_XCVR_PRX17_UHB_RX_OUT2

OUT

101

111

50_TX_IN_TX2_UHB_PAD_TX_N79_IN

IN

101

111

50_TX_IN_TX2_UHB_PAD_TX_UHB_IN_1

IN

101

111

50_UHB_L_CPLR_OUT

OUT

101

111

50_TRX_UHB_PAD_PORTB_TX_M

IO

101

111

50_TRX_UHB_PAD_N79_PORTB_TX_M

IO

101

111

50_TRX_PAUHBL_UHB_ANT2_TO_ANT9

IO

101

111

50_TRX_PAUHBL_N79_ANT2_TO_ANT9

IO

101

111

LMB/2G PAD

50_TX_IN_XCVR_TX0_LB2G

IN

101

112

50_TX_IN_XCVR_TX0_LMB

IN

101

112

50_TX_IN_XCVR_TX0_MB2G

IN

101

112

50_TRX_IN_LMB_PAD_LMB_ANT_M

IN

101

112

50_TRX_IN_LMB_PAD_2G_MB_OUT_M

IN

101

112

50_RX_LMB_PAD_LMB_PRX_OUT_M

IO

101

112

LAT

50_TRX_ANT7_UHB_N79

IO

101

113

UAT

50_TRX_ANT4_MB_HB_WLAN24

IO

101

114

50_TRX_CPLR_U_CPL2_ANT_LMB_MB_HB

IO

101

114

50_TRX_PAUHBU_UHB_ANT1_TO_ANT4

IO

101

114

50_TRX_PAUHBU_N79_ANT1_TO_ANT4

IO

101

114

50_TRX_WLAN_G_BT_ANT4_NPLXR

IO

101

114

SCH #：951-10915

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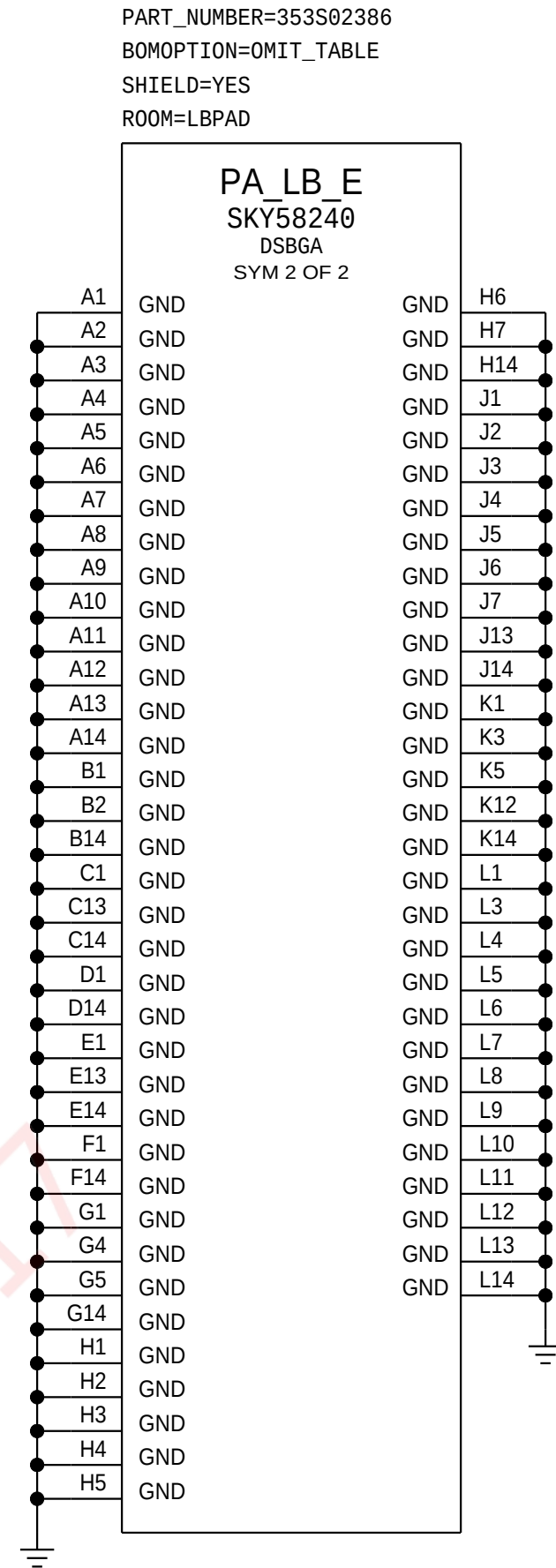
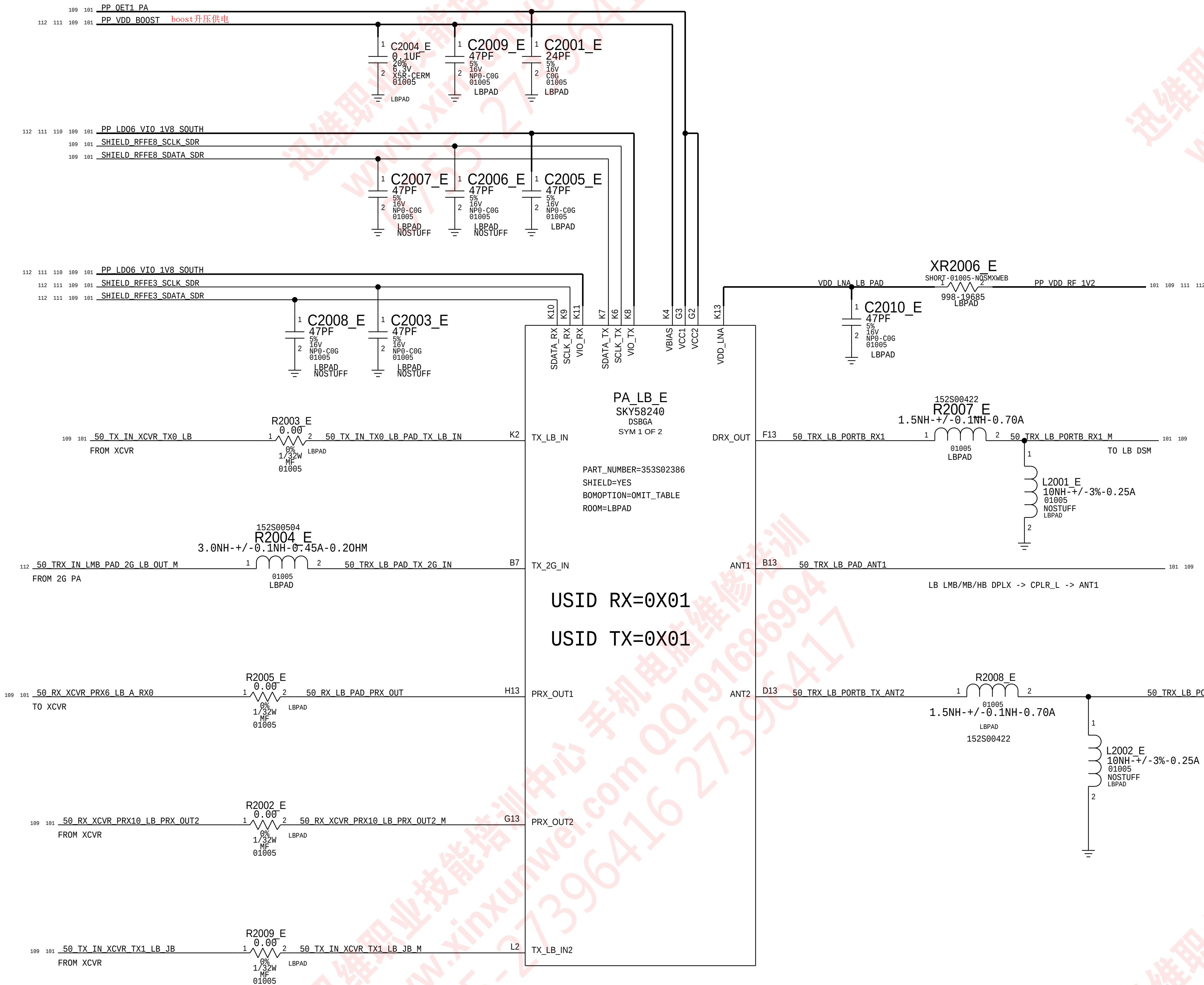
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LB PAD

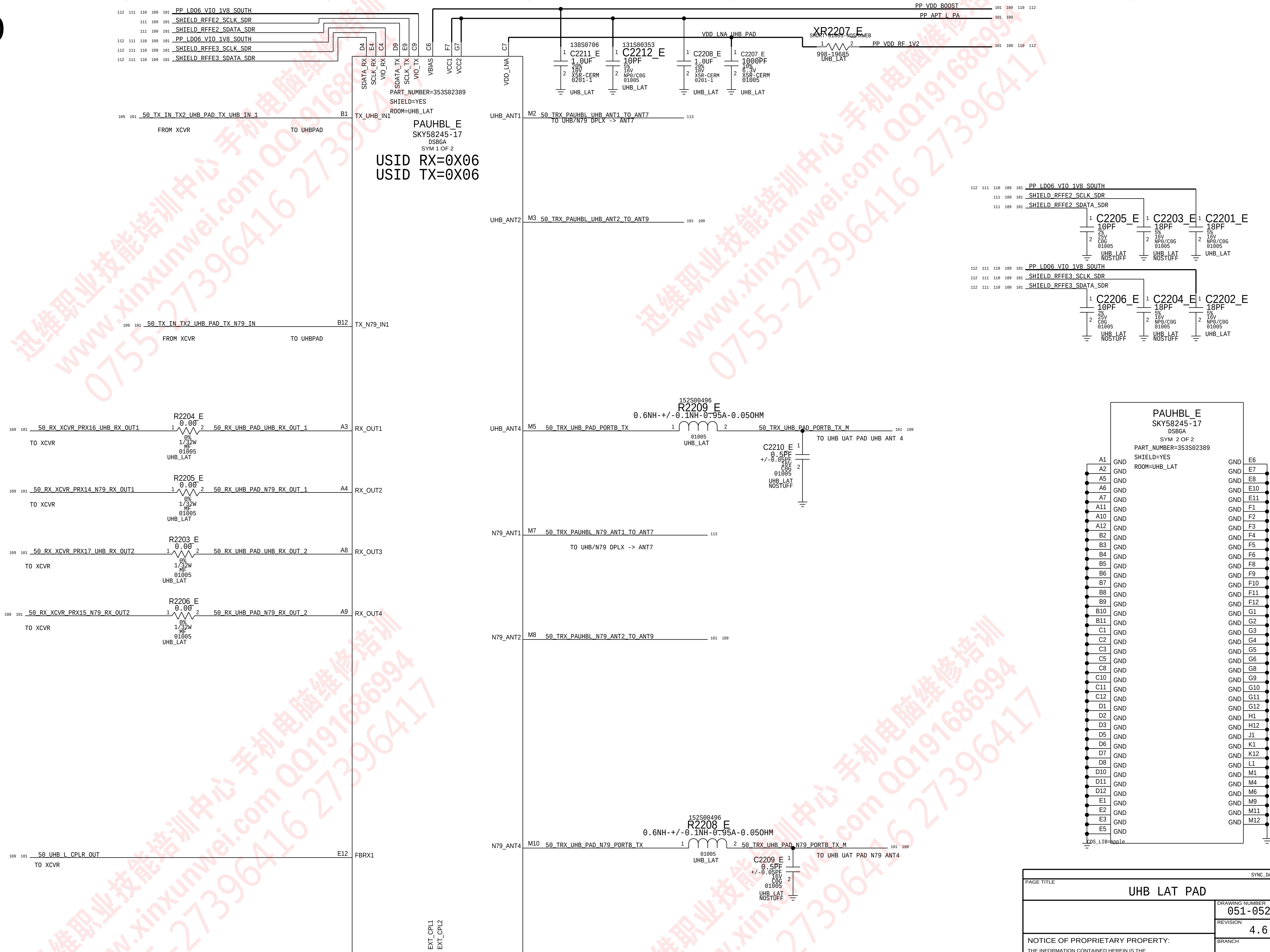
OMIT TABLE

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353S02387	1	IC, LB PAD, SKY58241, ROW, DSMBGA84	PA_LB_E	RF_SKU:ROW



PAGE TITLE			
LB PAD			
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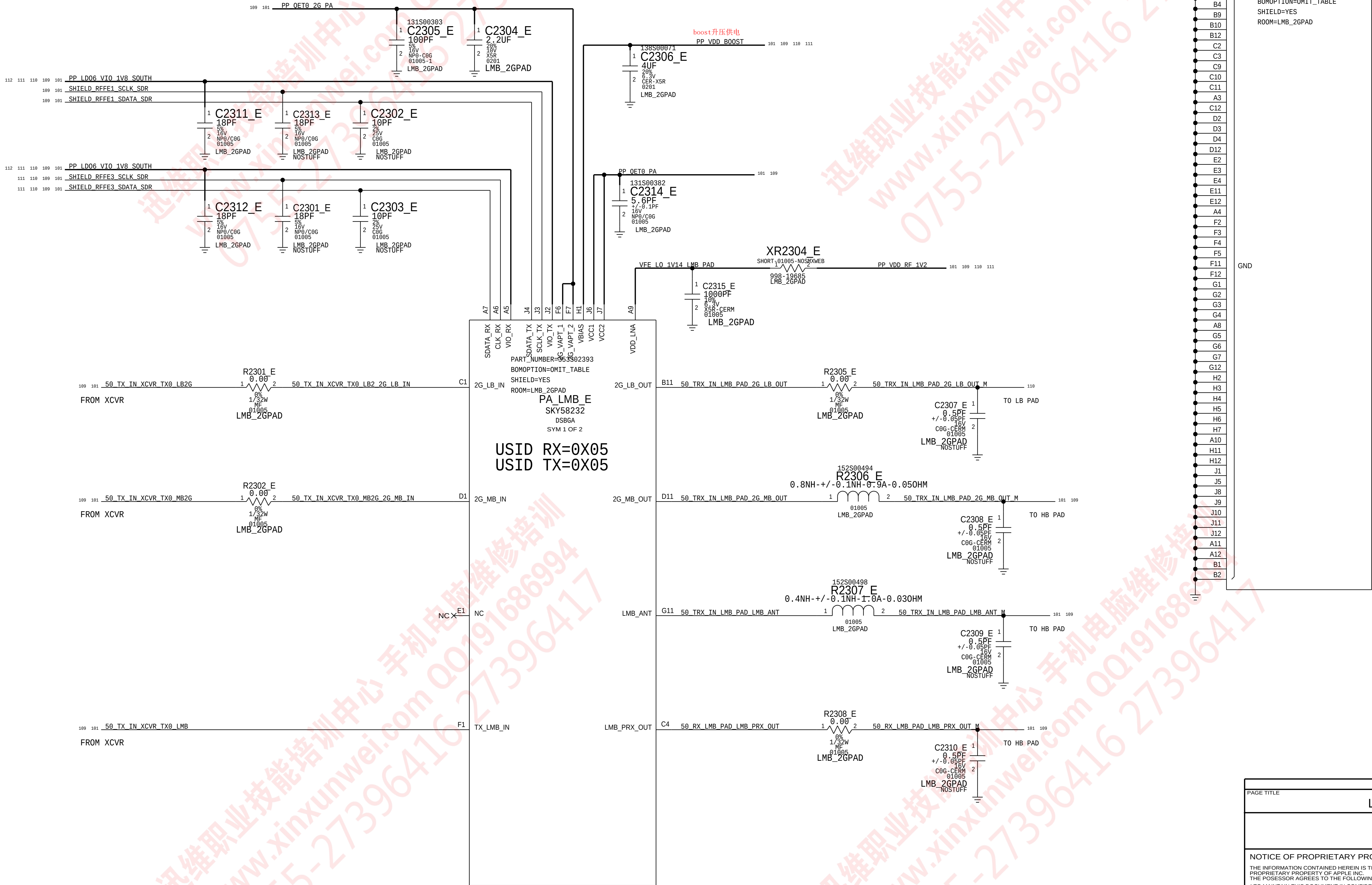
UHB LAT PAD



LMB_2G_PAD

OMIT TABLE

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353S02401	1	IC, 2G-LMB PAD, ROW, SKY58242, DSBGA79	PA_LMB_E	RF_SKU:ROW



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LMB_2G_PAD			051-05215		D	
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LOWER ANTENNA FEEDS: ANT7

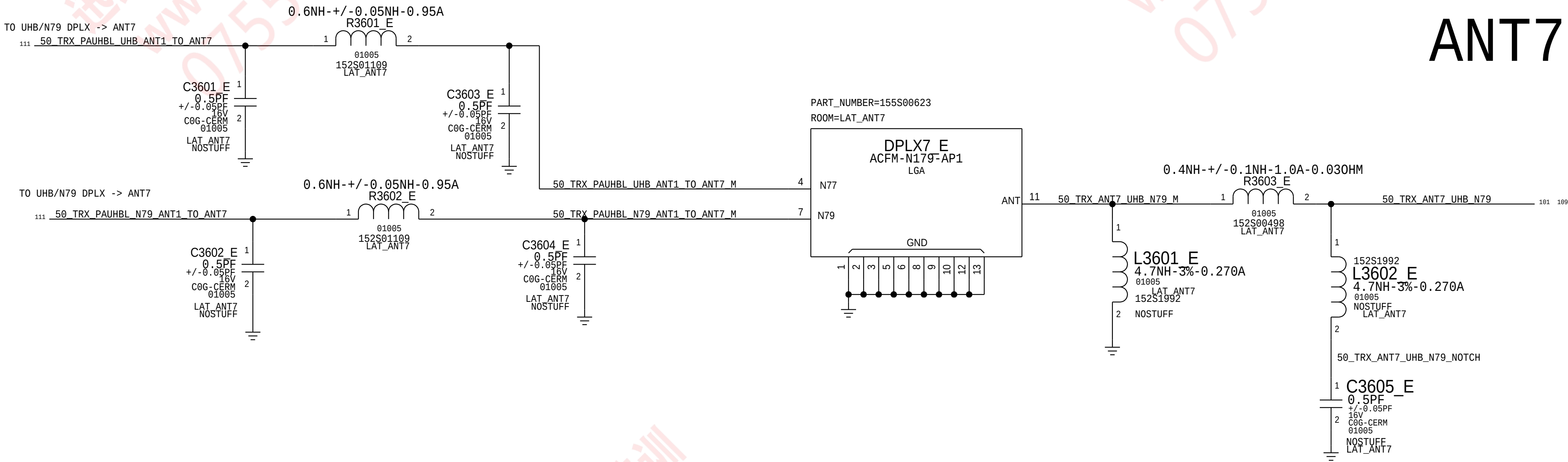
UAT ANTENNAS

ANT2	ANT4	ANT6	ANT8
LB	MB/HB	SAM	UHB
LMB/MB/HB	2.4GHZ	5GHZ	N79
L1 GNSS	UHB N79		

LAT ANTENNAS

ANT1	ANT3	ANT5	ANT7	ANT9
LB	MB/HB	5GHZ	UHB	UHB
LMB/MB/HB	2.4GHZ L5 GNSS		N79	N79

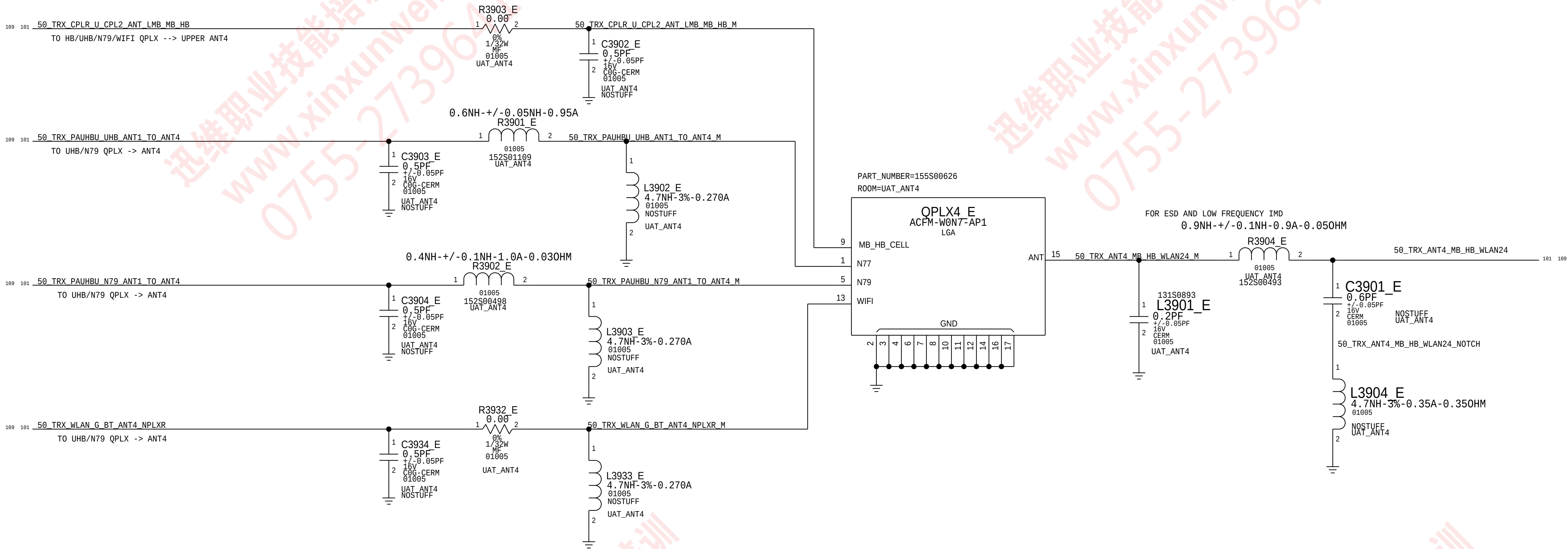
ANT7



5GHZ WLAN BPF IS ON LAA SCHEMATIC (ANT5)

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UPPER ANTENNA FEEDS



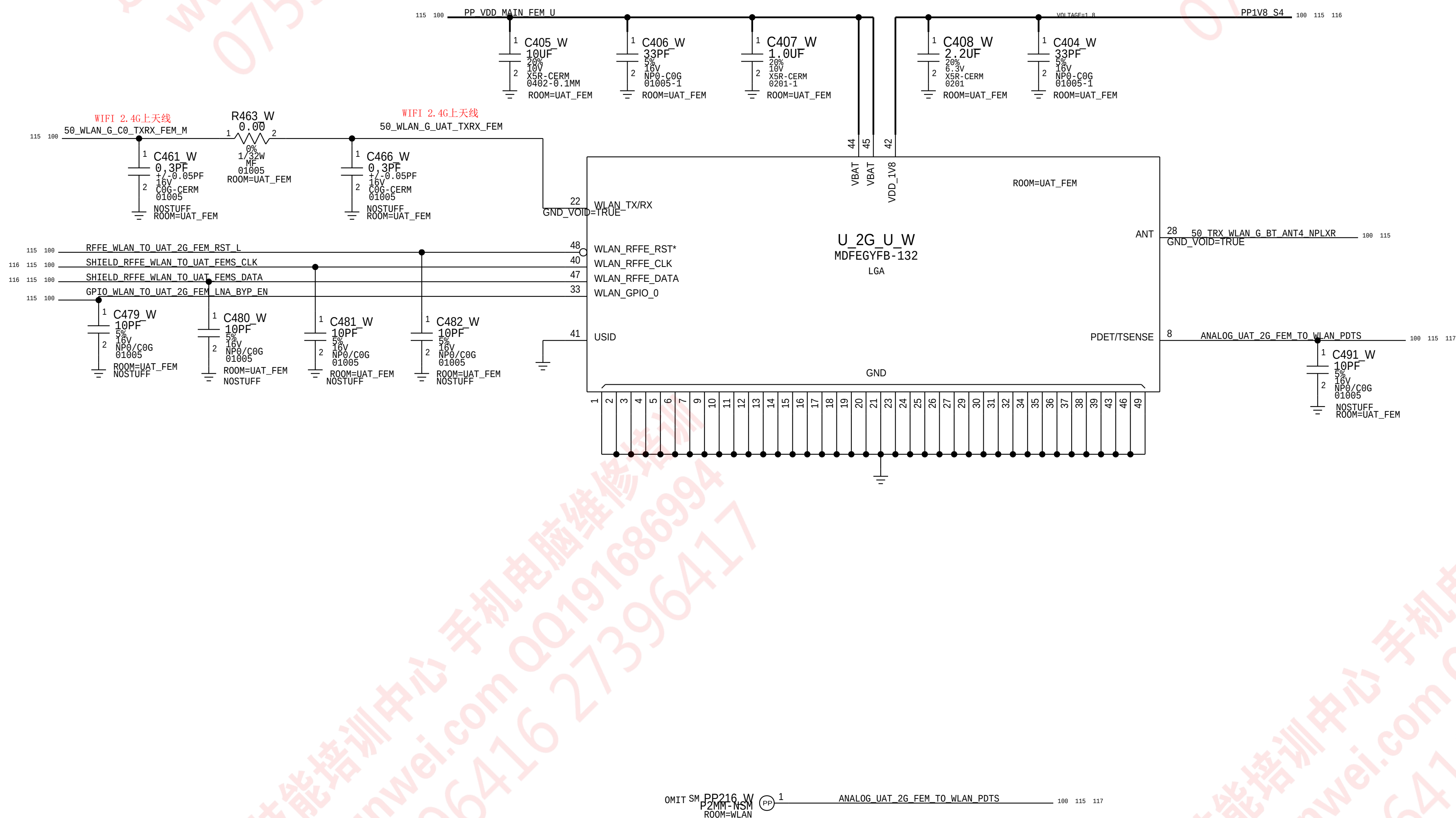
ANT4

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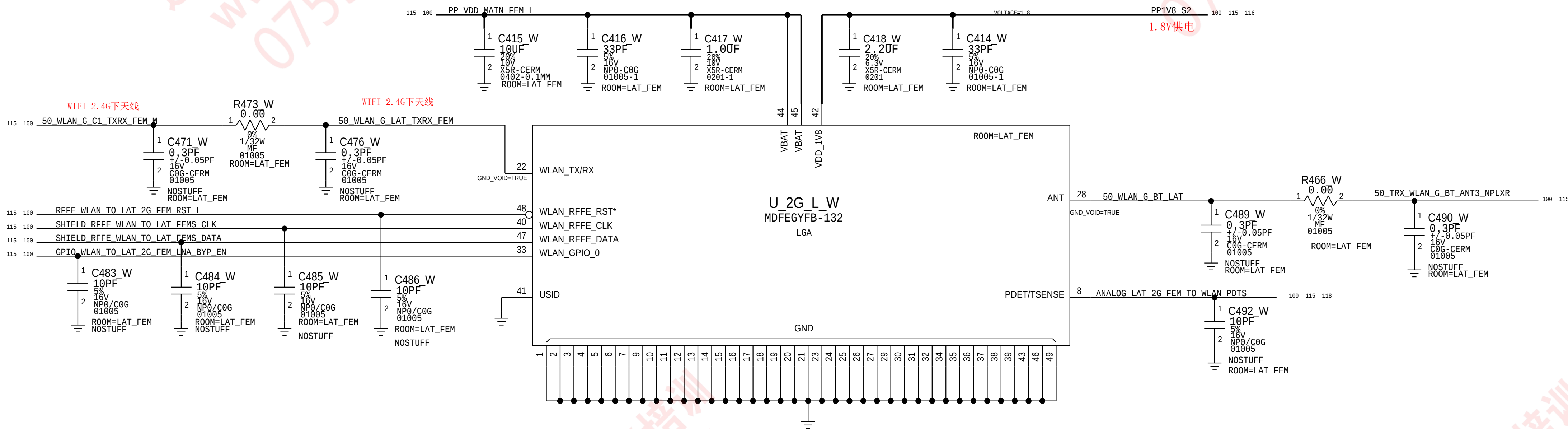
UAT 2.4 GHZ RFEM



OMIT SM PP216 W 1 ANALOG_UAT_2G_FEM_TO_WLAN_PDTS 100 115 117

PAGE TITLE		
2G4 rFEM (UAT)		
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LAT 2.4 GHZ RFEM



OMIT SW PP217 W P2MM-NSN ROOM=WLAN 1 ANALOG_LAT_2G_FEM_TO_WLAN_PDTS 100 115 118

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2G4 rFEM (LAT)		
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8

7

6

5

4

3

2

1

1. ALL RESISTANCE VALUES ARE IN OHMS, 0.1 WATT +/- 5%.

2. ALL CAPACITANCE VALUES ARE IN MICROFARADS.

3. ALL CRYSTALS & OSCILLATOR VALUES ARE IN HERTZ.

D53

INTERPOSER

EVT

LAST_MODIFICATION=Fri Mar 13 16:11:10 2020

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1	1	Table of Contents		
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3	3	Top Side: Symbol (2/2)		08/08/2019
4	4	Bottom Side: Symbol (1/2)		08/08/2019
5	5	Bottom Side: Symbol (2/2)		08/08/2019
6	6	INTERPOSER: Aliases (1/4)		08/08/2019
7	7	INTERPOSER: Aliases (2/4)		08/08/2019
8	8	INTERPOSER: Aliases (3/4)		08/08/2019
9	9	INTERPOSER: Aliases (4/4)		08/08/2019

APNs

BOM: 639-08623

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820-01957	1	PCBA,MLB,INT,D53	PCB	CRITICAL	?

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PART#	QTY	DESCRIPTION	REFERENCE DESIGNATOR(S)	CRITICAL	BOM OPTION
825-7691	1	EEEE FOR PCBA,MLB,INT,D53	MT10	CRITICAL	?

DRAWING TITLE

SCH,MLB,INT,D53

DRAWING NUMBER

051-05226

SIZE

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GND_OUTER_NORTHEAST_PENINSULA	1	IO1	IO91
GND_INNER_NORTHEAST	2	IO2	IO92
GND_OUTER_NORTHEAST_PENINSULA	3	IO3	IO93
GND_INNER_NORTHEAST	4	IO4	IO94
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GND_INNER_NORTHEAST	6	IO6	IO96
GPIO_AP_CANARY6_PIN_7	7	IO7	IO97
GND_INNER_NORTHEAST	8	IO8	IO98
GND_OUTER_NORTHEAST_PENINSULA	9	IO9	IO99
GPIO_AP_CANARY6_PIN_10	10	IO10	IO100
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GND_INNER_NORTHEAST	12	IO12	IO102
GND_OUTER_NORTHEAST_PENINSULA	13	IO13	IO103
GND_OUTER_NORTHEAST_PENINSULA	14	IO14	IO104
GPIO_AP_CANARY6_PIN_15	15	IO15	IO105
GPIO_AP_CANARY6	16	IO16	IO106
GND_OUTER_NORTHEAST_PENINSULA	17	IO17	IO107
PP1V8_S2_1	18	IO18	IO108
GND_OUTER_NORTHEAST_PENINSULA	19	IO19	IO109
GND_INNER_NORTHEAST	20	IO20	IO110
GND_OUTER_TAB_1	21	IO21	IO111
GND_INNER_NORTHEAST	22	IO22	IO112
GND_OUTER_TAB_1	23	IO23	IO113
GPIO_ISP_RCAM_TO_STROBE_TRIGGER_1V8	24	IO24	IO114
GND_OUTER_TAB_1	25	IO25	IO115
GND_INNER_NORTHEAST	26	IO26	IO116
GND_OUTER_TAB_1	27	IO27	IO117
IO_R1_TO_BB_LAA_SW_CTRL	28	IO28	IO118
GND_OUTER_TAB_1	29	IO29	IO119
IO_BT_TO_R1_WAKE	30	IO30	IO120
GND_OUTER_TAB_1	31	IO31	IO121
GND_INNER_NORTHEAST	32	IO32	IO122
GND_OUTER_TAB_1	33	IO33	IO123
PPVAR_RCAM_PVDD	34	IO34	IO124
GND_OUTER_TAB_1	35	IO35	IO125
GND_INNER_NORTHEAST	36	IO36	IO126
GND_OUTER_ARMPIT	37	IO37	IO127
GPIO_CAM_PMU1_TO_STROBE_HWEN_1V8	38	IO38	IO128
GND_OUTER_ARMPIT	39	IO39	IO129
GPIO_BB_TO_R1_CH9_REQ	40	IO40	IO130
GND_OUTER_ARMPIT	41	IO41	IO131
NTC_STROBE_MODULE	42	IO42	IO132
GND_OUTER_ARMPIT	43	IO43	IO133
GND_OUTER_ARMPIT	44	IO44	IO134
GPIO_AP_CANARY5	45	IO45	IO135
GND_OUTER_ARMPIT	46	IO46	IO136
GPIO_AP_CANARY5_PIN_47	47	IO47	IO137
GND_OUTER_ARMPIT	48	IO48	IO138
GND_OUTER_ARMPIT	49	IO49	IO139
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GPIO_AP_CANARY5_PIN_51	51	IO51	IO141
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GND_OUTER_ARMPIT	53	IO53	IO143
GPIO_AP_CANARY5_PIN_54	54	IO54	IO144
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GND_INNER_ARMPIT	60	IO60	IO150
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GND_INNER_ARMPIT	62	IO62	IO152
GPIO_AP_CANARY5_PIN_63	63	IO63	IO153
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GND_INNER_ARMPIT	65	IO65	IO155
GND_INNER_ARMPIT	66	IO66	IO156
GPIO_AP_CANARY5_PIN_67	67	IO67	IO157
GND_INNER_ARMPIT	68	IO68	IO158
GPIO_AP_CANARY5_PIN_69	69	IO69	IO159
GND_INNER_ARMPIT	70	IO70	IO160
GND_INNER_ARMPIT	71	IO71	IO161
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GND_INNER_ARMPIT	86	IO86	IO176
GPIO_AP_CANARY5_PIN_87	87	IO87	IO177
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PP_VDD_MAIN_8	186	IO186	IO276
GND_OUTER_EAST	187	IO187	IO277
GND_INNER_EAST	188	IO188	IO278
GND_OUTER_EAST	189	IO189	IO279
PP3V0_S2	190	IO190	IO280
GND_OUTER_EAST	191	IO191	IO281
GND_INNER_EAST	192	IO192	IO282
GND_OUTER_EAST	193	IO193	IO283
GND_INNER_EAST	194	IO194	IO284
GND_OUTER_EAST	195	IO195	IO285
GND_INNER_EAST	196	IO196	IO286
GND_OUTER_EAST	197	IO197	IO287
GND_INNER_EAST	198	IO198	IO288
GND_OUTER_EAST	199	IO199	
GPIO_BT_TO_BT_SPKR_TRIG	200	IO200	IO290
GND_OUTER_EAST	201	IO201	IO291
GND_INNER_EAST	202	IO202	IO292
GND_OUTER_EAST	203	IO203	IO293
GND_INNER_EAST	204	IO204	IO294
GND_OUTER_EAST	205	IO205	
GND_INNER_EAST	206	IO206	IO296
GND_OUTER_EAST	207	IO207	IO297
GND_INNER_EAST	208	IO208	IO298
GND_OUTER_EAST	209	IO209	IO299
RFFE_WLAN_TO_LAT_2G_FEM_RST_L	210	IO210	IO300
GND_OUTER_EAST	211	IO211	IO301
GPIO_WLAN_TO_LAT_2G_FEM_LNA_BYP_EN	212	IO212	IO302
GND_OUTER_EAST	213	IO213	IO303
ANALOG_LAT_2G_FEM_TO_WLAN_PDTS	214	IO214	IO304
GND_OUTER_EAST	215	IO215	IO305
PP_LDO6_VIO_1V8_SOUTH	216	IO216	IO306
GND_OUTER_EAST	217	IO217	IO307
SHIELD_RFFE_WLAN_TO_LAT_FEMS_CLK	218	IO218	IO308
GND_OUTER_EAST	219	IO219	IO309
SHIELD_RFFE_WLAN_TO_LAT_FEMS_DATA	220	IO220	IO310
GND_OUTER_EAST	221	IO221	IO311
PP1V8_S2_2	222	IO222	IO312
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GND_OUTER_EAST	224	IO224	IO314
GND_OUTER_EAST	225	IO225	IO315
50_RX_LMB_PAD_LMB_PRX_OUT_M	226	IO226	IO316
50_TX_IN_TX2_UHB_PAD_TX_N79_IN	227	IO227	IO317
GND_PA_CAVITY	228	IO228	IO318
GND_OUTER_EAST	229	IO229	IO319
GND_PA_CAVITY	230	IO230	IO320
GND_OUTER_EAST	231	IO231	IO321
50_TX_IN_XCVR_TX0_LB	232	IO232	IO322
GND_OUTER_EAST	233	IO233	IO323
GND_PA_CAVITY	234	IO234	IO324
GND_OUTER_EAST	235	IO235	IO325
50_UHB_L_CPLR_OUT	236	IO236	IO326
50_RX_XCVR_PRX6_LB_A_RX0	237	IO237	IO327
GND_PA_CAVITY	238	IO238	IO328
GND_OUTER_EAST	239	IO239	IO329
GND_PA_CAVITY	240	IO240	IO330
GND_OUTER_EAST	241	IO241	IO331
PP_QET0_2G_PA_1	242	IO242	IO332
GND_OUTER_EAST	243	IO243	IO333
PP_QET0_2G_PA_2	244	IO244	IO334
GND_OUTER_EAST	245	IO245	IO335
PP_QET0_PA_1	246	IO246	IO336
GND_OUTER_TAB_3	247	IO247	IO337
PP_QET0_PA_2	248	IO248	IO338
GND_OUTER_TAB_3	249	IO249	IO339
GND_PA_CAVITY	250	IO250	IO340
GND_OUTER_TAB_3	251	IO251	IO341
50_WLAN_G_C1_TXRX_FEM_M	252	IO252	IO342
GND_OUTER_TAB_3	253	IO253	IO343
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SHIELD_RFFE1_SCLK_SDR	256	IO256	IO346
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GND_OUTER_TAB_3	258	IO258	IO348
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GND_PA_CAVITY	262	IO262	IO352
GPIO_AOP_CANARY7_PIN_263	263	IO263	
GND_PA_CAVITY	264	IO264	IO354
GPIO_AOP_CANARY7_PIN_265	265	IO265	IO355
GND_PA_CAVITY	266	IO266	IO356
GPIO_AOP_CANARY7_PIN_267	267	IO267	IO357
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50_RX_XCVR_PRX16_UHB_RX_OUT1	381
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SHIELD_RFFE3_SDATA_SDR	396
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363	IO363	IO453	453	GND_OUTER_WEST
364	IO364	IO454	454	GPIO_NUB_TO_BBPMU_CLK_EN_DOTARA
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367	IO367	IO457	457	GND_OUTER_WEST
368	IO368	IO458	458	I2S2_AP_FROM_BB_LRCLK
369	IO369	IO459	459	GND_OUTER_WEST
370	IO370	IO460	460	GND_INNER_WEST
371	IO371	IO461	461	GND_OUTER_WEST
372	IO372	IO462	462	I2S2_AP_FROM_BB_DIN
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377	IO377	IO467	467	GND_OUTER_WEST
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379	IO379	IO469	469	GND_OUTER_WEST
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382	IO382	IO472	472	GND_INNER_WEST
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387	IO387	IO477	477	GND_OUTER_WEST
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389	IO389	IO479	479	GND_OUTER_WEST
390	IO390	IO480	480	GND_INNER_WEST
391	IO391	IO481	481	GND_OUTER_WEST
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393	IO393	IO483	483	GND_OUTER_WEST
394	IO394	IO484	484	PCIE_GP1_AP_TO_WLAN_PERST_L
395	IO395	IO485	485	GND_OUTER_WEST
396	IO396	IO486	486	PCIE_GP1_AP_B1_WLAN_CLKREQ_L
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398	IO398	IO488	488	GND_OUTER_WEST
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417	IO417	IO507	507	GND_OUTER_TAB_5
418	IO418	IO508	508	90_PCIE_GP1_AP_TO_WLAN_REFCLK_N
419	IO419	IO509	509	GND_OUTER_TAB_5
420	IO420	IO510	510	90_PCIE_GP1_AP_TO_WLAN_REFCLK_P
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437	IO437	IO527	527	GND_OUTER_NORTHWEST
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439	IO439	IO529	529	GND_OUTER_NORTHWEST
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442	IO442	IO532	532	GND_INNER_NORTHWEST
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		IO721	721
		IO722	722
		IO723	723
		IO724	724
		IO725	725
		IO726	726
		IO727	727

PAGE TITLE			SYNC_DATE=08/08/2019	
Bottom Side: Symbol (2/2)			DRAWING NUMBER	SIZE
			051-05226	D
			REVISION	4.0.0
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Interposer Aliases (1/4)

4	2	GND_OUTER_NORTHEAST_PENINSULA	GND_OUTER_NORTHEAST_PENINSULA	6
4	2	GND_INNER_NORTHEAST	GND_INNER_NORTHEAST	6
4	2	GND_OUTER_NORTHEAST_PENINSULA	GND_OUTER_NORTHEAST_PENINSULA	6
4	2	GND_INNER_NORTHEAST	GND_INNER_NORTHEAST	6
4	2	GND_OUTER_NORTHEAST_PENINSULA	GND_OUTER_NORTHEAST_PENINSULA	6
4	2	GND_INNER_NORTHEAST	GND_INNER_NORTHEAST	6
4	2	GPIO_AP_CANARY6_PIN_7	GPIO_AP_CANARY6_PIN_7	6
4	2	GND_INNER_NORTHEAST	GND_INNER_NORTHEAST	6
4	2	GND_OUTER_NORTHEAST_PENINSULA	GND_OUTER_NORTHEAST_PENINSULA	6
4	2	GPIO_AP_CANARY6_PIN_10	GPIO_AP_CANARY6_PIN_10	6
4	2	GND_OUTER_NORTHEAST_PENINSULA	GND_OUTER_NORTHEAST_PENINSULA	6
4	2	GND_INNER_NORTHEAST	GND_INNER_NORTHEAST	6
4	2	GND_OUTER_NORTHEAST_PENINSULA	GND_OUTER_NORTHEAST_PENINSULA	6
4	2	GND_OUTER_NORTHEAST_PENINSULA	GND_OUTER_NORTHEAST_PENINSULA	6
4	2	GPIO_AP_CANARY6_PIN_15	GPIO_AP_CANARY6_PIN_15	6
4	2	GPIO_AP_CANARY6	GPIO_AP_CANARY6	6
4	2	GND_OUTER_NORTHEAST_PENINSULA	GND_OUTER_NORTHEAST_PENINSULA	6
4	2	PP1V8_S2_1	PP1V8_S2_1	6
4	2	GND_OUTER_NORTHEAST_PENINSULA	GND_OUTER_NORTHEAST_PENINSULA	6
4	2	GND_INNER_NORTHEAST	GND_INNER_NORTHEAST	6
4	2	GND_OUTER_TAB_1	GND_OUTER_TAB_1	6
4	2	GND_INNER_NORTHEAST	GND_INNER_NORTHEAST	6
4	2	GND_OUTER_TAB_1	GND_OUTER_TAB_1	6
4	2	GPIO_ISP_RCAM_TO_STROBE_TRIGGER_1V8	GPIO_ISP_RCAM_TO_STROBE_TRIGGER_1V8	6
4	2	GND_OUTER_TAB_1	GND_OUTER_TAB_1	6
4	2	GND_INNER_NORTHEAST	GND_INNER_NORTHEAST	6
4	2	GND_OUTER_TAB_1	GND_OUTER_TAB_1	6
4	2	IO_R1_TO_BB_LAA_SW_CTRL	IO_R1_TO_BB_LAA_SW_CTRL	6
4	2	GND_OUTER_TAB_1	GND_OUTER_TAB_1	6
4	2	IO_BT_TO_R1_WAKE	IO_BT_TO_R1_WAKE	6
4	2	GND_OUTER_TAB_1	GND_OUTER_TAB_1	6
4	2	GND_INNER_NORTHEAST	GND_INNER_NORTHEAST	6
4	2	GND_OUTER_TAB_1	GND_OUTER_TAB_1	6
4	2	PPVAR_RCAM_PVDD	PPVAR_RCAM_PVDD	6
4	2	GND_OUTER_TAB_1	GND_OUTER_TAB_1	6
4	2	GND_INNER_NORTHEAST	GND_INNER_NORTHEAST	6
4	2	GND_OUTER_ARMPIT	GND_OUTER_ARMPIT	6
4	2	GPIO_CAM_PMU1_TO_STROBE_HWEN_1V8	GPIO_CAM_PMU1_TO_STROBE_HWEN_1V8	6
4	2	GND_OUTER_ARMPIT	GND_OUTER_ARMPIT	6
4	2	GPIO_BB_TO_R1_CH9_REQ	GPIO_BB_TO_R1_CH9_REQ	6
4	2	GND_OUTER_ARMPIT	GND_OUTER_ARMPIT	6
4	2	NTC_STROBE_MODULE	NTC_STROBE_MODULE	6
4	2	GND_OUTER_ARMPIT	GND_OUTER_ARMPIT	6
4	2	GND_OUTER_ARMPIT	GND_OUTER_ARMPIT	6
4	2	GPIO_AP_CANARY5	GPIO_AP_CANARY5	6
4	2	GND_OUTER_ARMPIT	GND_OUTER_ARMPIT	6
4	2	GPIO_AP_CANARY5_PIN_47	GPIO_AP_CANARY5_PIN_47	6
4	2	GND_OUTER_ARMPIT	GND_OUTER_ARMPIT	6
4	2	GND_OUTER_ARMPIT	GND_OUTER_ARMPIT	6
4	2	GND_OUTER_ARMPIT	GND_OUTER_ARMPIT	6
4	2	GPIO_AP_CANARY5_PIN_51	GPIO_AP_CANARY5_PIN_51	6
4	2	GND_OUTER_ARMPIT	GND_OUTER_ARMPIT	6
4	2	GND_OUTER_ARMPIT	GND_OUTER_ARMPIT	6
4	2	GPIO_AP_CANARY5_PIN_54	GPIO_AP_CANARY5_PIN_54	6
4	2	GND_OUTER_ARMPIT	GND_OUTER_ARMPIT	6
4	2	GND_INNER_ARMPIT	GND_INNER_ARMPIT	6
4	2	GND_OUTER_ARMPIT	GND_OUTER_ARMPIT	6
4	2	GND_INNER_ARMPIT	GND_INNER_ARMPIT	6
4	2	GND_OUTER_ARMPIT	GND_OUTER_ARMPIT	6
4	2	GND_INNER_ARMPIT	GND_INNER_ARMPIT	6
4	2	GPIO_AP_CANARY5_PIN_63	GPIO_AP_CANARY5_PIN_63	6
4	2	GND_INNER_ARMPIT	GND_INNER_ARMPIT	6
4	2	GND_INNER_ARMPIT	GND_INNER_ARMPIT	6
4	2	GND_INNER_ARMPIT	GND_INNER_ARMPIT	6
4	2	GPIO_AP_CANARY5_PIN_67	GPIO_AP_CANARY5_PIN_67	6
4	2	GND_INNER_ARMPIT	GND_INNER_ARMPIT	6
4	2	GPIO_AP_CANARY5_PIN_69	GPIO_AP_CANARY5_PIN_69	6
4	2	GND_INNER_ARMPIT	GND_INNER_ARMPIT	6
4	2	GND_INNER_ARMPIT	GND_INNER_ARMPIT	6
4	2	GND_INNER_ARMPIT	GND_INNER_ARMPIT	6
4	2	GND_OUTER_ARMPIT	GND_OUTER_ARMPIT	6
4	2	GND_INNER_ARMPIT	GND_INNER_ARMPIT	6
4	2	GND_OUTER_ARMPIT	GND_OUTER_ARMPIT	6
4	2	GND_INNER_ARMPIT	GND_INNER_ARMPIT	6
4	2	GND_OUTER_ARMPIT	GND_OUTER_ARMPIT	6
4	2	GND_INNER_ARMPIT	GND_INNER_ARMPIT	6
4	2	GND_OUTER_ARMPIT	GND_OUTER_ARMPIT	6
4	2	GND_INNER_ARMPIT	GND_INNER_ARMPIT	6
4	2	GPIO_AP_CANARY5_PIN_87	GPIO_AP_CANARY5_PIN_87	6
4	2	GND_INNER_ARMPIT	GND_INNER_ARMPIT	6
4	2	GND_OUTER_ARMPIT	GND_OUTER_ARMPIT	6
4	2	PP_VDD_MAIN_1	PP_VDD_MAIN_1	6
4	2	CKPLUS_WAIVE=SYNONYM_CHECK	CKPLUS_WAIVE=SYNONYM_CHECK	6

CPU到基带的SWD总线数据

4	2	GND_OUTER_ARMPIT	GND_OUTER_ARMPIT	6
4	2	PP_VDD_MAIN_2	PP_VDD_MAIN_2	6
4	2	GND_OUTER_ARMPIT	GND_OUTER_ARMPIT	6
4	2	PP_VDD_MAIN_3	PP_VDD_MAIN_3	6
4	2	GND_OUTER_ARMPIT	GND_OUTER_ARMPIT	6
4	2	PP_VDD_MAIN_4	PP_VDD_MAIN_4	6
4	2	GND_OUTER_ARMPIT	GND_OUTER_ARMPIT	6
4	2	GPIO_AP_CANARY3	GPIO_AP_CANARY3	6
4	2	GND_OUTER_ARMPIT	GND_OUTER_ARMPIT	6
4	2	GND_INNER_ARMPIT	GND_INNER_ARMPIT	6
4	2	GND_OUTER_TAB_2	GND_OUTER_TAB_2	6
4	2	CLK_GPIO_S4_PMU_TO_WLAN_R1_32K	CLK_GPIO_S4_PMU_TO_WLAN_R1_32K	6
4	2	GND_OUTER_TAB_2	GND_OUTER_TAB_2	6
4	2	PP_VDD_BOOST_NORTH_PA	PP_VDD_BOOST_NORTH_PA	6
4	2	GND_OUTER_TAB_2	GND_OUTER_TAB_2	6
4	2	PP1V2_S4	PP1V2_S4	6
4	2	GND_OUTER_TAB_2	GND_OUTER_TAB_2	6
4	2	GPIO_AOP_TO_WLAN_CONTEXT_A	GPIO_AOP_TO_WLAN_CONTEXT_A	6
4	2	GND_OUTER_TAB_2	GND_OUTER_TAB_2	6
4	2	PP1V8_S4	PP1V8_S4	6
4	2	GND_OUTER_TAB_2	GND_OUTER_TAB_2	6
4	2	RFFE_WLAN_TO_UAT_2G_FEM_RST_L	RFFE_WLAN_TO_UAT_2G_FEM_RST_L	6
4	2	GND_OUTER_TAB_2	GND_OUTER_TAB_2	6
4	2	RFFE_WLAN_TO_UAT_5G_FEM_RST_L	RFFE_WLAN_TO_UAT_5G_FEM_RST_L	6
4	2	GND_OUTER_TAB_2	GND_OUTER_TAB_2	6
4	2	GPIO_SE_TO_BT_PKT_RDY	GPIO_SE_TO_BT_PKT_RDY	6
4	2	GND_OUTER_EAST	GND_OUTER_EAST	6
4	2	GND_INNER_ARMPIT	GND_INNER_ARMPIT	6
4	2	GND_OUTER_EAST	GND_OUTER_EAST	6
4	2	I2C_BT_TO_NFC_SCL	I2C_BT_TO_NFC_SCL	6
4	2	GND_OUTER_EAST	GND_OUTER_EAST	6
4	2	ANALOG_UAT_5G_FEM_TO_WLAN_PDTS	ANALOG_UAT_5G_FEM_TO_WLAN_PDTS	6
4	2	GND_OUTER_EAST	GND_OUTER_EAST	6
4	2	GPIO_BB_TO_ALS_GSM_BURST_IND_1V8	GPIO_BB_TO_ALS_GSM_BURST_IND_1V8	6
4	2	GND_OUTER_EAST	GND_OUTER_EAST	6
4	2	GND_INNER_ARMPIT	GND_INNER_ARMPIT	6
4	2	GND_OUTER_EAST	GND_OUTER_EAST	6
4	2	I2C_BT_TO_NFC_SDA	I2C_BT_TO_NFC_SDA	6
4	2	GND_OUTER_EAST	GND_OUTER_EAST	6
4	2	GPIO_AP_FROM_WLAN_TIME_SYNC	GPIO_AP_FROM_WLAN_TIME_SYNC	6
4	2	GND_OUTER_EAST	GND_OUTER_EAST	6
4	2	GPIO_WLAN_TO_UAT_2G_FEM_LNA_BYP_EN	GPIO_WLAN_TO_UAT_2G_FEM_LNA_BYP_EN	6
4	2	GND_OUTER_EAST	GND_OUTER_EAST	6
4	2	GND_OUTER_EAST	GND_OUTER_EAST	6
4	2	GND_OUTER_EAST	GND_OUTER_EAST	6
4	2	GPIO_WLAN_TO_UAT_5G_FEM_LNA_BYP_EN	GPIO_WLAN_TO_UAT_5G_FEM_LNA_BYP_EN	6
4	2	GND_OUTER_EAST	GND_OUTER_EAST	6
4	2	GPIO_AP_TO_BB_PEAK_PWR_IND	GPIO_AP_TO_BB_PEAK_PWR_IND	6
4	2	GND_OUTER_EAST	GND_OUTER_EAST	6
4	2	SWD_AP_BT_BB_SWDIO	SWD_AP_BT_BB_SWDIO	6
4	2	GND_OUTER_EAST	GND_OUTER_EAST	6
4	2	GND_INNER_EAST	GND_INNER_EAST	6
4	2	GND_OUTER_EAST	GND_OUTER_EAST	6
4	2	GPIO_AP_FROM_BB_RESET_DETECT_L	GPIO_AP_FROM_BB_RESET_DETECT_L	6
4	2	GND_OUTER_EAST	GND_OUTER_EAST	6
4	2	SPMIO_EVENTS_AOP_BT_WLAN_NFC_DATA	SPMIO_EVENTS_AOP_BT_WLAN_NFC_DATA	6
4	2	GND_OUTER_EAST	GND_OUTER_EAST	6
4	2	SPMIO_EVENTS_AOP_TO_WLAN_NFC_CLK	SPMIO_EVENTS_AOP_TO_WLAN_NFC_CLK	6
4	2	GND_OUTER_EAST	GND_OUTER_EAST	6
4	2	GND_INNER_EAST	GND_INNER_EAST	6
4	2	GND_OUTER_EAST	GND_OUTER_EAST	6
4	2	GPIO_AP_FROM_BT_AUDIO_SYNC	GPIO_AP_FROM_BT_AUDIO_SYNC	6
4	2	GND_OUTER_EAST	GND_OUTER_EAST	6
4	2	GPIO_AP_TO_BB_TIME_MARK	GPIO_AP_TO_BB_TIME_MARK	6
4	2	GND_OUTER_EAST	GND_OUTER_EAST	6
4	2	NTC_RADIO_BB_P	NTC_RADIO_BB_P	6
4	2	GND_OUTER_EAST	GND_OUTER_EAST	6
4	2	GND_INNER_EAST	GND_INNER_EAST	6
4	2	GND_OUTER_EAST	GND_OUTER_EAST	6
4	2	UART1_AOP_FROM_BB_RXD	UART1_AOP_FROM_BB_RXD	6
4	2	GND_OUTER_EAST	GND_OUTER_EAST	6
4	2	UART1_AOP_TO_BB_TXD	UART1_AOP_TO_BB_TXD	6
4	2	GND_OUTER_EAST	GND_OUTER_EAST	6
4	2	GPIO_AOP_TO_BB_FORCE_PWM	GPIO_AOP_TO_BB_FORCE_PWM	6
4	2	GND_OUTER_EAST	GND_OUTER_EAST	6
4	2	GND_INNER_EAST	GND_INNER_EAST	6
4	2	GND_OUTER_EAST	GND_OUTER_EAST	6
4	2	PP_VDD_RF_1V2	PP_VDD_RF_1V2	6
4	2	GND_OUTER_EAST	GND_OUTER_EAST	6
4	2	GND_INNER_EAST	GND_INNER_EAST	6
4	2	GND_OUTER_EAST	GND_OUTER_EAST	6
4	2	GPIO_AOP_TO_WLAN_CONTEXT_B	GPIO_AOP_TO_WLAN_CONTEXT_B	6
4	2	GND_OUTER_EAST	GND_OUTER_EAST	6
4	2	GND_INNER_EAST	GND_INNER_EAST	6
4	2	GND_OUTER_EAST	GND_OUTER_EAST	6
4	2	PP_VDD_BOOST_BBPMU	PP_VDD_BOOST_BBPMU	6
4	2	GND_OUTER_EAST	GND_OUTER_EAST	6
4	2	GND_INNER_EAST	GND_INNER_EAST	6
4	2	GND_OUTER_EAST	GND_OUTER_EAST	6
4	2	PP_VDD_MAIN_5	PP_VDD_MAIN_5	6
4	2	CKPLUS_WAIVE=SYNONYM_CHECK	CKPLUS_WAIVE=SYNONYM_CHECK	6

PAGE TITLE			DRAWING NUMBER			SIZE		
INTERPOSER: Aliases (1/4)			051-05226			D		
REVISION			4.0.0					
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IV ALL RIGHTS RESERVED								

Interposer Aliases (2/4)

GND_OUTER_EAST	GND_OUTER_EAST	MAKE_BASE=TRUE	6-7
PP_VDD_MAIN_6	PP_VDD_MAIN_6	MAKE_BASE=TRUE	
GND_OUTER_EAST	GND_OUTER_EAST	MAKE_BASE=TRUE	6-7
PP_VDD_MAIN_7	PP_VDD_MAIN_7	MAKE_BASE=TRUE	
GND_OUTER_EAST	GND_OUTER_EAST	MAKE_BASE=TRUE	6-7
PP_VDD_MAIN_8	PP_VDD_MAIN_8	MAKE_BASE=TRUE	
GND_OUTER_EAST	GND_OUTER_EAST	MAKE_BASE=TRUE	6-7
GND_INNER_EAST	GND_INNER_EAST	MAKE_BASE=TRUE	6-7
GND_OUTER_EAST	GND_OUTER_EAST	MAKE_BASE=TRUE	6-7
PP3V0_S2	PP3V0_S2	MAKE_BASE=TRUE	
GND_OUTER_EAST	GND_OUTER_EAST	MAKE_BASE=TRUE	6-7
GND_INNER_EAST	GND_INNER_EAST	MAKE_BASE=TRUE	6-7
GND_OUTER_EAST	GND_OUTER_EAST	MAKE_BASE=TRUE	6-7
GND_INNER_EAST	GND_INNER_EAST	MAKE_BASE=TRUE	6-7
GND_OUTER_EAST	GND_OUTER_EAST	MAKE_BASE=TRUE	6-7
GND_INNER_EAST	GND_INNER_EAST	MAKE_BASE=TRUE	6-7
GND_OUTER_EAST	GND_OUTER_EAST	MAKE_BASE=TRUE	6-7
GND_INNER_EAST	GND_INNER_EAST	MAKE_BASE=TRUE	6-7
GND_OUTER_EAST	GND_OUTER_EAST	MAKE_BASE=TRUE	6-7
GND_INNER_EAST	GND_INNER_EAST	MAKE_BASE=TRUE	6-7
GND_OUTER_EAST	GND_OUTER_EAST	MAKE_BASE=TRUE	6-7
GND_INNER_EAST	GND_INNER_EAST	MAKE_BASE=TRUE	6-7
GND_OUTER_EAST	GND_OUTER_EAST	MAKE_BASE=TRUE	6-7
GND_INNER_EAST	GND_INNER_EAST	MAKE_BASE=TRUE	6-7
GND_OUTER_EAST	GND_OUTER_EAST	MAKE_BASE=TRUE	6-7
GND_INNER_EAST	GND_INNER_EAST	MAKE_BASE=TRUE	6-7
GND_OUTER_EAST	GND_OUTER_EAST	MAKE_BASE=TRUE	6-7
GND_INNER_EAST	GND_INNER_EAST	MAKE_BASE=TRUE	6-7
GND_OUTER_EAST	GND_OUTER_EAST	MAKE_BASE=TRUE	6-7
GND_INNER_EAST	GND_INNER_EAST	MAKE_BASE=TRUE	6-7
GND_OUTER_EAST	GND_OUTER_EAST	MAKE_BASE=TRUE	6-7
GND_INNER_EAST	GND_INNER_EAST	MAKE_BASE=TRUE	6-7
GND_OUTER_EAST	GND_OUTER_EAST	MAKE_BASE=TRUE	6-7
GND_INNER_EAST	GND_INNER_EAST	MAKE_BASE=TRUE	6-7
GND_OUTER_EAST	GND_OUTER_EAST	MAKE_BASE=TRUE	6-7
GND_INNER_EAST	GND_INNER_EAST	MAKE_BASE=TRUE	6-7
GND_OUTER_EAST	GND_OUTER_EAST	MAKE_BASE=TRUE	6-7
GND_INNER_EAST	GND_INNER_EAST	MAKE_BASE=TRUE	6-7
GND_OUTER_EAST	GND_OUTER_EAST	MAKE_BASE=TRUE	6-7
GND_INNER_EAST	GND_INNER_EAST	MAKE_BASE=TRUE	6-7
GND_OUTER_EAST	GND_OUTER_EAST	MAKE_BASE=TRUE	6-7
GND_INNER_EAST	GND_INNER_EAST	MAKE_BASE=TRUE	6-7
GND_OUTER_EAST	GND_OUTER_EAST	MAKE_BASE=TRUE	6-7
GND_INNER_EAST	GND_INNER_EAST	MAKE_BASE=TRUE	6-7
GND_OUTER_EAST	GND_OUTER_EAST	MAKE_BASE=TRUE	6-7
GND_INNER_EAST	GND_INNER_EAST	MAKE_BASE=TRUE	6-7
GND_OUTER_EAST	GND_OUTER_EAST	MAKE_BASE=TRUE	6-7
GND_INNER_EAST	GND_INNER_EAST	MAKE_BASE=TRUE	6-7
GND_OUTER_EAST	GND_OUTER_EAST	MAKE_BASE=TRUE	6-7
GND_INNER_EAST	GND_INNER_EAST	MAKE_BASE=TRUE	6-7
GND_OUTER_EAST	GND_OUTER_EAST	MAKE_BASE=TRUE	6-7
GND_INNER_EAST	GND_INNER_EAST	MAKE_BASE=TRUE	6-7
GND_OUTER_EAST	GND_OUTER_EAST	MAKE_BASE=TRUE	6-7
GND_INNER_EAST	GND_INNER_EAST	MAKE_BASE=TRUE	6-7
GND_OUTER_EAST	GND_OUTER_EAST	MAKE_BASE=TRUE	6-7
GND_INNER_EAST	GND_INNER_EAST	MAKE_BASE=TRUE	6-7
GND_OUTER_EAST	GND_OUTER_EAST	MAKE_BASE=TRUE	6-7
GND_INNER_EAST	GND_INNER_EAST	MAKE_BASE=TRUE	6-7
GND_OUTER_EAST	GND_OUTER_EAST	MAKE_BASE=TRUE	6-7
GND_INNER_EAST	GND_INNER_EAST	MAKE_BASE=TRUE	6-7
GND_OUTER_EAST	GND_OUTER_EAST	MAKE_BASE=TRUE	6-7
GND_INNER_EAST	GND_INNER_EAST	MAKE_BASE=TRUE	6-7
GND_OUTER_EAST	GND_OUTER_EAST	MAKE_BASE=TRUE	6-7
GND_INNER_EAST	GND_INNER_EAST	MAKE_BASE=TRUE	6-7
GND_OUTER_EAST	GND_OUTER_EAST	MAKE_BASE=TRUE	6-7
GND_INNER_EAST	GND_INNER_EAST	MAKE_BASE=TRUE	6-7
GND_OUTER_EAST	GND_OUTER_EAST	MAKE_BASE=TRUE	6-7
GND_INNER_EAST	GND_INNER_EAST	MAKE_BASE=TRUE	6-7
GND_OUTER_EAST	GND_OUTER_EAST	MAKE_BASE=TRUE	6-7
GND_INNER_EAST	GND_INNER_EAST	MAKE_BASE=TRUE	6-7
GND_OUTER_EAST	GND_OUTER_EAST	MAKE_BASE=TRUE	6-7
GND_INNER_EAST	GND_INNER_EAST	MAKE_BASE=TRUE	6-7
GND_OUTER_EAST	GND_OUTER_EAST	MAKE_BASE=TRUE	6-7
GND_INNER_EAST	GND_INNER_EAST	MAKE_BASE=TRUE	6-7
GND_OUTER_EAST	GND_OUTER_EAST	MAKE_BASE=TRUE	6-7
GND_INNER_EAST	GND_INNER_EAST	MAKE_BASE=TRUE	6-7
GND_OUTER_EAST	GND_OUTER_EAST	MAKE_BASE=TRUE	6-7
GND_INNER_EAST	GND_INNER_EAST	MAKE_BASE=TRUE	6-7
GND_OUTER_EAST	GND_OUTER_EAST	MAKE_BASE=TRUE	6-7
GND_INNER_EAST	GND_INNER_EAST	MAKE_BASE=TRUE	6-7
GND_OUTER_EAST	GND_OUTER_EAST	MAKE_BASE=TRUE	6-7
GND_INNER_EAST	GND_INNER_EAST	MAKE_BASE=TRUE	6-7
GND_OUTER_EAST	GND_OUTER_EAST	MAKE_BASE=TRUE	6-7
GND_INNER_EAST	GND_INNER_EAST	MAKE_BASE=TRUE	6-7
GND_OUTER_EAST	GND_OUTER_EAST	MAKE_BASE=TRUE	6-7
GND_INNER_EAST	GND_INNER_EAST	MAKE_BASE=TRUE	6-7
GND_OUTER_EAST	GND_OUTER_EAST	MAKE_BASE=TRUE	6-7
GND_INNER_EAST	GND_INNER_EAST	MAKE_BASE=TRUE	6-7
GND_OUTER_EAST	GND_OUTER_EAST	MAKE_BASE=TRUE	6-7
GND_INNER_EAST	GND_INNER_EAST	MAKE_BASE=TRUE	6-7
GND_OUTER_EAST	GND_OUTER_EAST	MAKE_BASE=TRUE	6-7
GND_INNER_EAST	GND_INNER_EAST	MAKE_BASE=TRUE	6-7
GND_OUTER_EAST	GND_OUTER_EAST	MAKE_BASE=TRUE	6-7
GND_INNER_EAST	GND_INNER_EAST	MAKE_BASE=TRUE	6-7
GND_OUTER_EAST	GND_OUTER_EAST	MAKE_BASE=TRUE	6-7
GND_INNER_EAST	GND_INNER_EAST	MAKE_BASE=TRUE	6-7
GND_OUTER_EAST	GND_OUTER_EAST	MAKE_BASE=TRUE	6-7
GND_INNER_EAST	GND_INNER_EAST	MAKE_BASE=TRUE	6-7
GND_OUTER_EAST	GND_OUTER_EAST	MAKE_BASE=TRUE	6-7
GND_INNER_EAST	GND_INNER_EAST	MAKE_BASE=TRUE	6-7
GND_OUTER_EAST	GND_OUTER_EAST	MAKE_BASE=TRUE	6-7
GND_INNER_EAST	GND_INNER_EAST	MAKE_BASE=TRUE	6-7
GND_OUTER_EAST	GND_OUTER_EAST	MAKE_BASE=TRUE	6-7
GND_INNER_EAST	GND_INNER_EAST	MAKE_BASE=TRUE	6-7
GND_OUTER_EAST	GND_OUTER_EAST	MAKE_BASE=TRUE	6-7
GND_INNER_EAST	GND_INNER_EAST	MAKE_BASE=TRUE	6-7
GND_OUTER_EAST	GND_OUTER_EAST	MAKE_BASE=TRUE	6-7
GND_INNER_EAST	GND_INNER_EAST	MAKE_BASE=TRUE	6-7
GND_OUTER_EAST	GND_OUTER_EAST	MAKE_BASE=TRUE	6-7
GND_INNER_EAST	GND_INNER_EAST	MAKE_BASE=TRUE	6-7
GND_OUTER_EAST	GND_OUTER_EAST	MAKE_BASE=TRUE	6-7
GND_INNER_EAST	GND_INNER_EAST	MAKE_BASE=TRUE	6-7
GND_OUTER_EAST	GND_OUTER_EAST	MAKE_BASE=TRUE	6-7
GND_INNER_EAST	GND_INNER_EAST	MAKE_BASE=TRUE	6-7
GND_OUTER_EAST	GND_OUTER_EAST	MAKE_BASE=TRUE	6-7
GND_INNER_EAST	GND_INNER_EAST	MAKE_BASE=TRUE	6-7
GND_OUTER_EAST	GND_OUTER_EAST	MAKE_BASE=TRUE	6-7
GND_INNER_EAST	GND_INNER_EAST	MAKE_BASE=TRUE	6-7
GND_OUTER_EAST	GND_OUTER_EAST	MAKE_BASE=TRUE	6-7
GND_INNER_EAST	GND_INNER_EAST	MAKE_BASE=TRUE	6-7
GND_OUTER_EAST	GND_OUTER_EAST	MAKE_BASE=TRUE	6-7
GND_INNER_EAST	GND_INNER_EAST	MAKE_BASE=TRUE	6-7
GND_OUTER_EAST	GND_OUTER_EAST	MAKE_BASE=TRUE	6-7
GND_INNER_EAST	GND_INNER_EAST	MAKE_BASE=TRUE	6-7
GND_OUTER_EAST	GND_OUTER_EAST	MAKE_BASE=TRUE	6-7
GND_INNER_EAST	GND_INNER_EAST	MAKE_BASE=TRUE	6-7
GND_OUTER_EAST	GND_OUTER_EAST	MAKE_BASE=TRUE	6-7
GND_INNER_EAST	GND_INNER_EAST	MAKE_BASE=TRUE	6-7
GND_OUTER_EAST	GND_OUTER_EAST	MAKE_BASE=TRUE	6-7
GND_INNER_EAST	GND_INNER_EAST	MAKE_BASE=TRUE	6-7
GND_OUTER_EAST	GND_OUTER_EAST	MAKE_BASE=TRUE	6-7
GND_INNER_EAST	GND_INNER_EAST	MAKE_BASE=TRUE	6-7
GND_OUTER_EAST	GND_OUTER_EAST	MAKE_BASE=TRUE	6-7
GND_INNER_EAST	GND_INNER_EAST	MAKE_BASE=TRUE	6-7
GND_OUTER_EAST	GND_OUTER_EAST	MAKE_BASE=TRUE	6-7
GND_INNER_EAST	GND_INNER_EAST	MAKE_BASE=TRUE	6-7
GND_OUTER_EAST	GND_OUTER_EAST	MAKE_BASE=TRUE	6-7
GND_INNER_EAST	GND_INNER_EAST	MAKE_BASE=TRUE	6-7
GND_OUTER_EAST</			

50_TRX_LB_PAD_ANT1	50_TRX_LB_PAD_ANT1	MAKE_BASE=TRUE	7
GND_OUTER_SOUTH	GND_OUTER_SOUTH	MAKE_BASE=TRUE	7
GND_PA_CAVITY	GND_PA_CAVITY	MAKE_BASE=TRUE	7
GND_PA_CAVITY	GND_PA_CAVITY	MAKE_BASE=TRUE	7
GND_OUTER_SOUTH	GND_OUTER_SOUTH	MAKE_BASE=TRUE	7
50_TRX_IN_LMB_PAD_2G_MB_OUT_M	50_TRX_IN_LMB_PAD_2G_MB_OUT_M	MAKE_BASE=TRUE	7
GND_PA_CAVITY	GND_PA_CAVITY	MAKE_BASE=TRUE	7
GND_OUTER_SOUTH	GND_OUTER_SOUTH	MAKE_BASE=TRUE	7
GND_PA_CAVITY	GND_PA_CAVITY	MAKE_BASE=TRUE	7
50_TRX_IN_LMB_PAD_LMB_ANT_M	50_TRX_IN_LMB_PAD_LMB_ANT_M	MAKE_BASE=TRUE	7
GND_OUTER_SOUTH	GND_OUTER_SOUTH	MAKE_BASE=TRUE	7
GND_PA_CAVITY	GND_PA_CAVITY	MAKE_BASE=TRUE	7
GND_PA_CAVITY	GND_PA_CAVITY	MAKE_BASE=TRUE	7
GND_OUTER_SOUTH	GND_OUTER_SOUTH	MAKE_BASE=TRUE	7
50_TRX_WLAN_G_BT_ANT3_NPLXR	50_TRX_WLAN_G_BT_ANT3_NPLXR	MAKE_BASE=TRUE	7
GND_PA_CAVITY	GND_PA_CAVITY	MAKE_BASE=TRUE	7
GND_OUTER_SOUTH	GND_OUTER_SOUTH	MAKE_BASE=TRUE	7
GND_PA_CAVITY	GND_PA_CAVITY	MAKE_BASE=TRUE	7
GND_OUTER_SOUTH	GND_OUTER_SOUTH	MAKE_BASE=TRUE	7
50_TX_IN_XCVR_TX0_MB2G	50_TX_IN_XCVR_TX0_MB2G	MAKE_BASE=TRUE	7
GND_PA_CAVITY	GND_PA_CAVITY	MAKE_BASE=TRUE	7
GND_OUTER_SOUTH	GND_OUTER_SOUTH	MAKE_BASE=TRUE	7
GND_PA_CAVITY	GND_PA_CAVITY	MAKE_BASE=TRUE	7
GND_OUTER_SOUTH	GND_OUTER_SOUTH	MAKE_BASE=TRUE	7
50_TRX_ANT7_UHB_N79	50_TRX_ANT7_UHB_N79	MAKE_BASE=TRUE	7
GND_PA_CAVITY	GND_PA_CAVITY	MAKE_BASE=TRUE	7
GND_OUTER_SOUTH	GND_OUTER_SOUTH	MAKE_BASE=TRUE	7
GND_PA_CAVITY	GND_PA_CAVITY	MAKE_BASE=TRUE	7
50_TX_IN_XCVR_TX0_LMB	50_TX_IN_XCVR_TX0_LMB	MAKE_BASE=TRUE	7
GPIO_AOP_CANARY7_PIN_302	GPIO_AOP_CANARY7_PIN_302	MAKE_BASE=TRUE	7
GND_PA_CAVITY	GND_PA_CAVITY	MAKE_BASE=TRUE	7
GND_PA_CAVITY	GND_PA_CAVITY	MAKE_BASE=TRUE	7
GND_OUTER_SOUTH	GND_OUTER_SOUTH	MAKE_BASE=TRUE	7
PP_VDD_MAIN_9	PP_VDD_MAIN_9	MAKE_BASE=TRUE	7
PP_VDD_MAIN_10	PP_VDD_MAIN_10	MAKE_BASE=TRUE	7
GND_OUTER_SOUTH	GND_OUTER_SOUTH	MAKE_BASE=TRUE	7
GND_OUTER_SOUTH	GND_OUTER_SOUTH	MAKE_BASE=TRUE	7
GPIO_AOP_CANARY7_PIN_310	GPIO_AOP_CANARY7_PIN_310	MAKE_BASE=TRUE	7
GND_SOUTHWEST	GND_SOUTHWEST	MAKE_BASE=TRUE	7
GND_OUTER_SOUTH	GND_OUTER_SOUTH	MAKE_BASE=TRUE	7
GND_SOUTHWEST	GND_SOUTHWEST	MAKE_BASE=TRUE	7
GND_SOUTHWEST	GND_SOUTHWEST	MAKE_BASE=TRUE	7
GND_PA_CAVITY	GND_PA_CAVITY	MAKE_BASE=TRUE	7
GND_SOUTHWEST	GND_SOUTHWEST	MAKE_BASE=TRUE	7
GND_PA_CAVITY	GND_PA_CAVITY	MAKE_BASE=TRUE	7
GND_SOUTHWEST	GND_SOUTHWEST	MAKE_BASE=TRUE	7
GND_SOUTHWEST	GND_SOUTHWEST	MAKE_BASE=TRUE	7
GPIO_AOP_CANARY7_PIN_321	GPIO_AOP_CANARY7_PIN_321	MAKE_BASE=TRUE	7
GND_PA_CAVITY	GND_PA_CAVITY	MAKE_BASE=TRUE	7
GND_SOUTHWEST	GND_SOUTHWEST	MAKE_BASE=TRUE	7
GND_SOUTHWEST	GND_SOUTHWEST	MAKE_BASE=TRUE	7
GND_SOUTHWEST	GND_SOUTHWEST	MAKE_BASE=TRUE	7
GPIO_AOP_CANARY7_PIN_326	GPIO_AOP_CANARY7_PIN_326	MAKE_BASE=TRUE	7
GND_SOUTHWEST	GND_SOUTHWEST	MAKE_BASE=TRUE	7
GPIO_AOP_CANARY7	GPIO_AOP_CANARY7	MAKE_BASE=TRUE	7
GND_OUTER_WEST	GND_OUTER_WEST	MAKE_BASE=TRUE	7
GND_PA_CAVITY	GND_PA_CAVITY	MAKE_BASE=TRUE	7
GND_OUTER_WEST	GND_OUTER_WEST	MAKE_BASE=TRUE	7
GND_PA_CAVITY	GND_PA_CAVITY	MAKE_BASE=TRUE	7
GND_OUTER_WEST	GND_OUTER_WEST	MAKE_BASE=TRUE	7
GND_PA_CAVITY	GND_PA_CAVITY	MAKE_BASE=TRUE	7
GND_OUTER_WEST	GND_OUTER_WEST	MAKE_BASE=TRUE	7
50_TRX_PAUHLB_N79_ANT2_TO_ANT9	50_TRX_PAUHLB_N79_ANT2_TO_ANT9	MAKE_BASE=TRUE	7
GND_OUTER_WEST	GND_OUTER_WEST	MAKE_BASE=TRUE	7
GND_PA_CAVITY	GND_PA_CAVITY	MAKE_BASE=TRUE	7
50_TRX_UHB_PAD_PORTB_TX_M	50_TRX_UHB_PAD_PORTB_TX_M	MAKE_BASE=TRUE	7
GND_PA_CAVITY	GND_PA_CAVITY	MAKE_BASE=TRUE	7
GND_OUTER_WEST	GND_OUTER_WEST	MAKE_BASE=TRUE	7
50_TRX_UHB_PAD_N79_PORTB_TX_M	50_TRX_UHB_PAD_N79_PORTB_TX_M	MAKE_BASE=TRUE	7
GND_OUTER_WEST	GND_OUTER_WEST	MAKE_BASE=TRUE	7
GND_PA_CAVITY	GND_PA_CAVITY	MAKE_BASE=TRUE	7
50_TRX_PAUHLB_UHB_ANT2_TO_ANT9	50_TRX_PAUHLB_UHB_ANT2_TO_ANT9	MAKE_BASE=TRUE	7
GND_PA_CAVITY	GND_PA_CAVITY	MAKE_BASE=TRUE	7
GND_OUTER_WEST	GND_OUTER_WEST	MAKE_BASE=TRUE	7
50_TRX_LB_PORTB_RX1_M	50_TRX_LB_PORTB_RX1_M	MAKE_BASE=TRUE	7
GND_OUTER_WEST	GND_OUTER_WEST	MAKE_BASE=TRUE	7
GND_SOUTHWEST	GND_SOUTHWEST	MAKE_BASE=TRUE	7
GND_PA_CAVITY	GND_PA_CAVITY	MAKE_BASE=TRUE	7
GND_INNER_WEST	GND_INNER_WEST	MAKE_BASE=TRUE	7
50_TX_IN_XCVR_TX0_LB2G	50_TX_IN_XCVR_TX0_LB2G	MAKE_BASE=TRUE	7
GND_INNER_WEST	GND_INNER_WEST	MAKE_BASE=TRUE	7
50_RX_XCVR_PRX10_LB_PRX_OUT2	50_RX_XCVR_PRX10_LB_PRX_OUT2	MAKE_BASE=TRUE	7
50_TRX_LB_PORTB_TX_M	50_TRX_LB_PORTB_TX_M	MAKE_BASE=TRUE	7
GND_PA_CAVITY	GND_PA_CAVITY	MAKE_BASE=TRUE	7
GND_INNER_WEST	GND_INNER_WEST	MAKE_BASE=TRUE	7

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Interposer Aliases (3/4)

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5 3	GND_PA_CAVITY		GND_PA_CAVITY	7 8
5 3	PP_APT_L_PA_1		PP_APT_L_PA_1	MAKE_BASE=TRUE
5 3	PP_APT_L_PA_2		PP_APT_L_PA_2	MAKE_BASE=TRUE
5 3	GND_INNER_WEST		GND_INNER_WEST	MAKE_BASE=TRUE
5 3	GND_PA_CAVITY		GND_PA_CAVITY	MAKE_BASE=TRUE
5 3	PP_QET1_PA_1		PP_QET1_PA_1	MAKE_BASE=TRUE
5 3	PP_QET1_PA_2		PP_QET1_PA_2	MAKE_BASE=TRUE
5 3	GND_INNER_WEST		GND_INNER_WEST	MAKE_BASE=TRUE
5 3	GND_PA_CAVITY		GND_PA_CAVITY	MAKE_BASE=TRUE
5 3	GND_INNER_WEST		GND_INNER_WEST	MAKE_BASE=TRUE
5 3	50_RX_XCVR_PRX15_N79_RX_OUT2		50_RX_XCVR_PRX15_N79_RX_OUT2	MAKE_BASE=TRUE
5 3	GND_INNER_WEST		GND_INNER_WEST	MAKE_BASE=TRUE
5 3	GND_PA_CAVITY		GND_PA_CAVITY	MAKE_BASE=TRUE
5 3	GND_INNER_WEST		GND_INNER_WEST	MAKE_BASE=TRUE
5 3	50_RX_XCVR_PRX17_UHB_RX_OUT2		50_RX_XCVR_PRX17_UHB_RX_OUT2	MAKE_BASE=TRUE
5 3	50_RX_XCVR_PRX14_N79_RX_OUT1		50_RX_XCVR_PRX14_N79_RX_OUT1	MAKE_BASE=TRUE
5 3	GND_PA_CAVITY		GND_PA_CAVITY	MAKE_BASE=TRUE
5 3	GND_INNER_WEST		GND_INNER_WEST	MAKE_BASE=TRUE
5 3	GND_PA_CAVITY		GND_PA_CAVITY	MAKE_BASE=TRUE
5 3	GND_INNER_WEST		GND_INNER_WEST	MAKE_BASE=TRUE
5 3	50_RX_XCVR_PRX16_UHB_RX_OUT1		50_RX_XCVR_PRX16_UHB_RX_OUT1	MAKE_BASE=TRUE
5 3	SHIELD_RFFE2_SDATA_SDR		SHIELD_RFFE2_SDATA_SDR	MAKE_BASE=TRUE
5 3	GND_PA_CAVITY		GND_PA_CAVITY	MAKE_BASE=TRUE
5 3	SHIELD_RFFE2_SCLK_SDR		SHIELD_RFFE2_SCLK_SDR	MAKE_BASE=TRUE
5 3	SHIELD_RFFE8_SDATA_SDR		SHIELD_RFFE8_SDATA_SDR	MAKE_BASE=TRUE
5 3	SHIELD_RFFE8_SCLK_SDR		SHIELD_RFFE8_SCLK_SDR	MAKE_BASE=TRUE
5 3	PP_VDD_BOOST_SOUTH_PA		PP_VDD_BOOST_SOUTH_PA	MAKE_BASE=TRUE
5 3	GND_INNER_EAST		GND_INNER_EAST	OKPLUS_WAIVE=SYNOWNM_CHECK
5 3	GND_PA_CAVITY		GND_PA_CAVITY	MAKE_BASE=TRUE
5 3	GND_INNER_EAST		GND_INNER_EAST	MAKE_BASE=TRUE
5 3	50_TX_IN_TX2_UHB_PAD_TX_UHB_IN_1		50_TX_IN_TX2_UHB_PAD_TX_UHB_IN_1	MAKE_BASE=TRUE
5 3	GND_INNER_EAST		GND_INNER_EAST	MAKE_BASE=TRUE
5 3	GND_PA_CAVITY		GND_PA_CAVITY	MAKE_BASE=TRUE
5 3	GND_INNER_EAST		GND_INNER_EAST	MAKE_BASE=TRUE
5 3	50_TX_IN_XCVR_TX1_LB_JB		50_TX_IN_XCVR_TX1_LB_JB	MAKE_BASE=TRUE
5 3	SHIELD_RFFE3_SDATA_SDR		SHIELD_RFFE3_SDATA_SDR	MAKE_BASE=TRUE
5 3	GND_PA_CAVITY		GND_PA_CAVITY	MAKE_BASE=TRUE
5 3	SHIELD_RFFE3_SCLK_SDR		SHIELD_RFFE3_SCLK_SDR	MAKE_BASE=TRUE
5 3	GND_PA_CAVITY		GND_PA_CAVITY	MAKE_BASE=TRUE
5 3	GND_INNER_WEST		GND_INNER_WEST	MAKE_BASE=TRUE
5 3	GND_OUTER_WEST		GND_OUTER_WEST	MAKE_BASE=TRUE
5 3	PP_VDD_MAIN_11		PP_VDD_MAIN_11	MAKE_BASE=TRUE
5 3	GND_OUTER_WEST		GND_OUTER_WEST	MAKE_BASE=TRUE
5 3	PP_VDD_MAIN_12		PP_VDD_MAIN_12	MAKE_BASE=TRUE
5 3	GND_OUTER_WEST		GND_OUTER_WEST	MAKE_BASE=TRUE
5 3	PP_VDD_MAIN_13		PP_VDD_MAIN_13	MAKE_BASE=TRUE
5 3	GND_OUTER_WEST		GND_OUTER_WEST	MAKE_BASE=TRUE
5 3	PP_VDD_MAIN_14		PP_VDD_MAIN_14	MAKE_BASE=TRUE
5 3	GND_OUTER_WEST		GND_OUTER_WEST	MAKE_BASE=TRUE
5 3	GND_INNER_WEST		GND_INNER_WEST	MAKE_BASE=TRUE
5 3	GND_OUTER_WEST		GND_OUTER_WEST	MAKE_BASE=TRUE
5 3	GND_INNER_WEST		GND_INNER_WEST	MAKE_BASE=TRUE
5 3	GND_OUTER_WEST		GND_OUTER_WEST	MAKE_BASE=TRUE
5 3	PP1V2_S2_1		PP1V2_S2_1	OKPLUS_WAIVE=SYNOWNM_CHECK
5 3	GND_OUTER_WEST		GND_OUTER_WEST	MAKE_BASE=TRUE
5 3	GND_INNER_WEST		GND_INNER_WEST	MAKE_BASE=TRUE
5 3	GND_OUTER_WEST		GND_OUTER_WEST	MAKE_BASE=TRUE
5 3	GND_INNER_WEST		GND_INNER_WEST	MAKE_BASE=TRUE
5 3	GND_OUTER_WEST		GND_OUTER_WEST	MAKE_BASE=TRUE
5 3	GPIO_SCM_AOP_FROM_JARVIS_INT		GPIO_SCM_AOP_FROM_JARVIS_INT	MAKE_BASE=TRUE
5 3	GND_OUTER_WEST		GND_OUTER_WEST	MAKE_BASE=TRUE
5 3	I2C1_AOP_SCL		I2C1_AOP_SCL	OKPLUS_WAIVE=I2C_PULLUP
5 3	GND_OUTER_WEST		GND_OUTER_WEST	MAKE_BASE=TRUE
5 3	I2C1_AOP_SDA		I2C1_AOP_SDA	OKPLUS_WAIVE=I2C_PULLUP
5 3	GND_OUTER_WEST		GND_OUTER_WEST	MAKE_BASE=TRUE
5 3	GND_INNER_WEST		GND_INNER_WEST	MAKE_BASE=TRUE
5 3	GND_OUTER_WEST		GND_OUTER_WEST	MAKE_BASE=TRUE
5 3	90_PCIE_GP2_AP_TO_BB_TX_N		90_PCIE_GP2_AP_TO_BB_TX_N	MAKE_BASE=TRUE
5 3	GND_OUTER_WEST		GND_OUTER_WEST	MAKE_BASE=TRUE
5 3	90_PCIE_GP2_AP_TO_BB_TX_P		90_PCIE_GP2_AP_TO_BB_TX_P	MAKE_BASE=TRUE
5 3	GND_OUTER_WEST		GND_OUTER_WEST	MAKE_BASE=TRUE
5 3	GND_INNER_WEST		GND_INNER_WEST	MAKE_BASE=TRUE
5 3	GND_OUTER_WEST		GND_OUTER_WEST	MAKE_BASE=TRUE
5 3	90_PCIE_GP2_AP_FROM_BB_RX_P		90_PCIE_GP2_AP_FROM_BB_RX_P	MAKE_BASE=TRUE
5 3	GND_OUTER_WEST		GND_OUTER_WEST	MAKE_BASE=TRUE
5 3	90_PCIE_GP2_AP_FROM_BB_RX_N		90_PCIE_GP2_AP_FROM_BB_RX_N	MAKE_BASE=TRUE
5 3	GND_OUTER_WEST		GND_OUTER_WEST	MAKE_BASE=TRUE
5 3	GND_INNER_WEST		GND_INNER_WEST	MAKE_BASE=TRUE
5 3	GND_OUTER_WEST		GND_OUTER_WEST	MAKE_BASE=TRUE
5 3	90_PCIE_GP2_AP_TO_BB_REFCLK_N		90_PCIE_GP2_AP_TO_BB_REFCLK_N	MAKE_BASE=TRUE
5 3	GND_OUTER_WEST		GND_OUTER_WEST	MAKE_BASE=TRUE
5 3	90_PCIE_GP2_AP_TO_BB_REFCLK_P		90_PCIE_GP2_AP_TO_BB_REFCLK_P	MAKE_BASE=TRUE
5 3	GND_OUTER_WEST		GND_OUTER_WEST	MAKE_BASE=TRUE
5 3	GND_INNER_WEST		GND_INNER_WEST	MAKE_BASE=TRUE
5 3	GND_OUTER_WEST		GND_OUTER_WEST	MAKE_BASE=TRUE
5 3	PP_VDD_MAIN_15		PP_VDD_MAIN_15	MAKE_BASE=TRUE
5 3	GND_OUTER_WEST		GND_OUTER_WEST	MAKE_BASE=TRUE
5 3	PP_VDD_MAIN_16		PP_VDD_MAIN_16	MAKE_BASE=TRUE
5 3	GND_OUTER_WEST		GND_OUTER_WEST	MAKE_BASE=TRUE
5 3	PP_VDD_MAIN_17		PP_VDD_MAIN_17	MAKE_BASE=TRUE

5 3	GND_OUTER_WEST		GND_OUTER_WEST	7 8
5 3	GND_INNER_WEST		GND_INNER_WEST	MAKE_BASE=TRUE
5 3	GND_OUTER_WEST		GND_OUTER_WEST	MAKE_BASE=TRUE
5 3	GPIO_NUB_TO_BBPMU_CLK_EN_DOTARA		GPIO_NUB_TO_BBPMU_CLK_EN_DOTARA	MAKE_BASE=TRUE
5 3	GND_OUTER_WEST		GND_OUTER_WEST	MAKE_BASE=TRUE
5 3	I2S2_AP_FROM_BB_BCLK		I2S2_AP_FROM_BB_BCLK	MAKE_BASE=TRUE
5 3	GND_OUTER_WEST		GND_OUTER_WEST	MAKE_BASE=TRUE
5 3	I2S2_AP_FROM_BB_LRCLK		I2S2_AP_FROM_BB_LRCLK	MAKE_BASE=TRUE
5 3	GND_OUTER_WEST		GND_OUTER_WEST	MAKE_BASE=TRUE
5 3	GND_INNER_WEST		GND_INNER_WEST	MAKE_BASE=TRUE
5 3	GND_OUTER_WEST		GND_OUTER_WEST	MAKE_BASE=TRUE
5 3	I2S2_AP_FROM_BB_DIN		I2S2_AP_FROM_BB_DIN	MAKE_BASE=TRUE
5 3	GND_OUTER_WEST		GND_OUTER_WEST	MAKE_BASE=TRUE
5 3	I2S2_AP_TO_BB_DOUT		I2S2_AP_TO_BB_DOUT	MAKE_BASE=TRUE
5 3	GND_OUTER_WEST		GND_OUTER_WEST	MAKE_BASE=TRUE
5 3	GND_INNER_WEST		GND_INNER_WEST	MAKE_BASE=TRUE
5 3	GND_OUTER_WEST		GND_OUTER_WEST	MAKE_BASE=TRUE
5 3	GND_OUTER_WEST		GND_OUTER_WEST	MAKE_BASE=TRUE
5 3	GND_INNER_WEST		GND_INNER_WEST	MAKE_BASE=TRUE
5 3	GND_OUTER_WEST		GND_OUTER_WEST	MAKE_BASE=TRUE
5 3	GND_INNER_WEST		GND_INNER_WEST	MAKE_BASE=TRUE
5 3	GND_OUTER_WEST		GND_OUTER_WEST	MAKE_BASE=TRUE
5 3	GPIO_PMU_FROM_BB_PCIE_HOST_WAKE_L		GPIO_PMU_FROM_BB_PCIE_HOST_WAKE_L	MAKE_BASE=TRUE
5 3	GND_OUTER_WEST		GND_OUTER_WEST	MAKE_BASE=TRUE
5 3	GPIO_S4_PMU_TO_WLAN_REG_ON		GPIO_S4_PMU_TO_WLAN_REG_ON	MAKE_BASE=TRUE
5 3	GND_OUTER_WEST		GND_OUTER_WEST	MAKE_BASE=TRUE
5 3	GPIO_PMU_FROM_WLAN_HOST_WAKE		GPIO_PMU_FROM_WLAN_HOST_WAKE	MAKE_BASE=TRUE
5 3	GND_OUTER_WEST		GND_OUTER_WEST	MAKE_BASE=TRUE
5 3	GND_INNER_WEST		GND_INNER_WEST	MAKE_BASE=TRUE
5 3	GND_OUTER_WEST		GND_OUTER_WEST	MAKE_BASE=TRUE
5 3	GPIO_SEQ_PMU_TO_BBPMU_RESET_L		GPIO_SEQ_PMU_TO_BBPMU_RESET_L	MAKE_BASE=TRUE
5 3	GND_OUTER_WEST		GND_OUTER_WEST	MAKE_BASE=TRUE
5 3	PCIE_GP1_AP_TO_WLAN_PERST_L		PCIE_GP1_AP_TO_WLAN_PERST_L	MAKE_BASE=TRUE
5 3	GND_OUTER_WEST		GND_OUTER_WEST	MAKE_BASE=TRUE
5 3	PCIE_GP1_AP_B1_WLAN_CLKREQ_L		PCIE_GP1_AP_B1_WLAN_CLKREQ_L	MAKE_BASE=TRUE
5 3	GND_OUTER_WEST		GND_OUTER_WEST	MAKE_BASE=TRUE
5 3	GND_OUTER_WEST		GND_OUTER_WEST	MAKE_BASE=TRUE
5 3	GND_OUTER_WEST		GND_OUTER_WEST	MAKE_BASE=TRUE
5 3	PCIE_GP2_AP_TO_BB_PERST_L		PCIE_GP2_AP_TO_BB_PERST_L	MAKE_BASE=TRUE
5 3	GND_OUTER_WEST		GND_OUTER_WEST	MAKE_BASE=TRUE
5 3	PCIE_GP2_AP_B1_BB_CLKREQ_L		PCIE_GP2_AP_B1_BB_CLKREQ_L	MAKE_BASE=TRUE
5 3	GND_OUTER_WEST		GND_OUTER_WEST	MAKE_BASE=TRUE
5 3	SWD_AP_TO_MANY_SWCLK_CPU输出的SWD总线时钟		SWD_AP_TO_MANY_SWCLK	MAKE_BASE=TRUE
5 3	GND_OUTER_WEST		GND_OUTER_WEST	MAKE_BASE=TRUE
5 3	GPIO_AP_TO_BB_COREDUMP		GPIO_AP_TO_BB_COREDUMP	MAKE_BASE=TRUE
5 3	GND_OUTER_WEST		GND_OUTER_WEST	MAKE_BASE=TRUE
5 3	GND_OUTER_WEST		GND_OUTER_WEST	MAKE_BASE=TRUE
5 3	GPIO_AP_CANARY4		GPIO_AP_CANARY4	MAKE_BASE=TRUE
5 3	GND_OUTER_TAB_5		GND_OUTER_TAB_5	MAKE_BASE=TRUE
5 3	GND_INNER_NORTHWEST		GND_INNER_NORTHWEST	MAKE_BASE=TRUE
5 3	GND_OUTER_TAB_5		GND_OUTER_TAB_5	MAKE_BASE=TRUE
5 3	CLK_BBPMU_TO_NFC_38M4		CLK_BBPMU_TO_NFC_38M4	MAKE_BASE=TRUE
5 3	GND_OUTER_TAB_5		GND_OUTER_TAB_5	MAKE_BASE=TRUE
5 3	GND_INNER_NORTHWEST		GND_INNER_NORTHWEST	MAKE_BASE=TRUE
5 3	GND_OUTER_TAB_5		GND_OUTER_TAB_5	MAKE_BASE=TRUE
5 3	90_PCIE_GP1_AP_TO_WLAN_REFCLK_N		90_PCIE_GP1_AP_TO_WLAN_REFCLK_N	MAKE_BASE=TRUE
5 3	GND_OUTER_TAB_5		GND_OUTER_TAB_5	MAKE_BASE=TRUE
5 3	90_PCIE_GP1_AP_TO_WLAN_REFCLK_P		90_PCIE_GP1_AP_TO_WLAN_REFCLK_P	MAKE_BASE=TRUE
5 3	GND_OUTER_TAB_5		GND_OUTER_TAB_5	MAKE_BASE=TRUE
5 3	GND_INNER_NORTHWEST		GND_INNER_NORTHWEST	MAKE_BASE=TRUE
5 3	GND_OUTER_TAB_5		GND_OUTER_TAB_5	MAKE_BASE=TRUE
5 3	90_PCIE_GP1_AP_FROM_WLAN_RX_N		90_PCIE_GP1_AP_FROM_WLAN_RX_N	MAKE_BASE=TRUE
5 3	GND_OUTER_TAB_5		GND_OUTER_TAB_5	MAKE_BASE=TRUE
5 3	90_PCIE_GP1_AP_FROM_WLAN_RX_P		90_PCIE_GP1_AP_FROM_WLAN_RX_P	MAKE_BASE=TRUE
5 3	GND_OUTER_NORTHWEST		GND_OUTER_NORTHWEST	MAKE_BASE=TRUE
5 3	GND_INNER_NORTHWEST		GND_INNER_NORTHWEST	MAKE_BASE=TRUE
5 3	GND_OUTER_NORTHWEST		GND_OUTER_NORTHWEST	MAKE_BASE=TRUE
5 3	90_PCIE_GP1_AP_TO_WLAN_TX_P		90_PCIE_GP1_AP_TO_WLAN_TX_P	MAKE_BASE=TRUE
5 3	GND_OUTER_NORTHWEST		GND_OUTER_NORTHWEST	MAKE_BASE=TRUE
5 3	90_PCIE_GP1_AP_TO_WLAN_TX_N		90_PCIE_GP1_AP_TO_WLAN_TX_N	MAKE_BASE=TRUE
5 3	GND_OUTER_NORTHWEST		GND_OUTER_NORTHWEST	MAKE_BASE=TRUE
5 3	GND_INNER_NORTHWEST		GND_INNER_NORTHWEST	MAKE_BASE=TRUE
5 3	GND_OUTER_NORTHWEST		GND_OUTER_NORTHWEST	MAKE_BASE=TRUE
5 3	CLK_BBPMU_TO_DOTARA_38M4		CLK_BBPMU_TO_DOTARA_38M4	MAKE_BASE=TRUE
5 3	GND_OUTER_NORTHWEST		GND_OUTER_NORTHWEST	MAKE_BASE=TRUE
5 3	GND_INNER_NORTHWEST		GND_INNER_NORTHWEST	MAKE_BASE=TRUE
5 3	GND_OUTER_NORTHWEST		GND_OUTER_NORTHWEST	MAKE_BASE=TRUE
5 3	GPIO_ISP_RCAM_TO_STROBE_KRAKEN_WLAN_FLASH_TRIG		GPIO_ISP_RCAM_TO_STROBE_KRAKEN_WLAN_FLASH_TRIG	MAKE_BASE=TRUE
5 3	GND_OUTER_NORTHWEST		GND_OUTER_NORTHWEST	MAKE_BASE=TRUE
5 3	GND_INNER_NORTHWEST		GND_INNER_NORTHWEST	MAKE_BASE=TRUE
5 3	GND_OUTER_NORTHWEST		GND_OUTER_NORTHWEST	MAKE_BASE=TRUE
5 3	GPIO_AP_CANARY2		GPIO_AP_CANARY2	MAKE_BASE=TRUE
5 3	GND_OUTER_NORTHWEST		GND_OUTER_NORTHWEST	MAKE_BASE=TRUE
5 3	GND_OUTER_NORTHWEST		GND_OUTER_NORTHWEST	MAKE_BASE=TRUE
5 3	GND_OUTER_NORTHWEST		GND_OUTER_NORTHWEST	MAKE_BASE=TRUE
5 3	GND_INNER_NORTHWEST		GND_INNER_NORTHWEST	MAKE_BASE=TRUE
5 3	GND_INNER_NORTHWEST		GND_INNER_NORTHWEST	MAKE_BASE=TRUE
5 3	GND_OUTER_NORTHWEST		GND_OUTER_NORTHWEST	MAKE_BASE=TRUE
5 3	GND_INNER_NORTHWEST		GND_INNER_NORTHWEST	MAKE_BASE=TRUE

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5	GND_OUTER_NORTHWEST	GND_OUTER_NORTHWEST	9
5	GND_INNER_NORTHWEST	GND_INNER_NORTHWEST	9
5	50_WLAN_A_C0_TXRX_FEM_M	50_WLAN_A_C0_TXRX_FEM_M	9
5	GND_INNER_NORTHWEST	GND_INNER_NORTHWEST	9
5	GND_OUTER_NORTHWEST	GND_OUTER_NORTHWEST	9
5	50_WLAN_G_C0_TXRX_FEM_M	50_WLAN_G_C0_TXRX_FEM_M	9
5	GND_OUTER_NORTHWEST	GND_OUTER_NORTHWEST	9
5	GND_INNER_NORTHWEST	GND_INNER_NORTHWEST	9
5	50_TRX_PAUHB_UHB_ANT1_TO_ANT4	50_TRX_PAUHB_UHB_ANT1_TO_ANT4	9
5	GND_INNER_NORTHWEST	GND_INNER_NORTHWEST	9
5	GND_OUTER_NORTHWEST	GND_OUTER_NORTHWEST	9
5	50_TRX_PAUHB_U79_ANT1_TO_ANT4	50_TRX_PAUHB_U79_ANT1_TO_ANT4	9
5	GND_OUTER_NORTHWEST	GND_OUTER_NORTHWEST	9
5	GND_INNER_NORTHWEST	GND_INNER_NORTHWEST	9
5	GND_OUTER_NORTHWEST	GND_OUTER_NORTHWEST	9
5	GND_INNER_NORTHWEST	GND_INNER_NORTHWEST	9
5	GND_OUTER_NORTHWEST	GND_OUTER_NORTHWEST	9
5	GND_INNER_NORTHWEST	GND_INNER_NORTHWEST	9
5	IO_NFC_TO_BBPMU_CLKREQ	IO_NFC_TO_BBPMU_CLKREQ	9
5	GND_OUTER_NORTHWEST	GND_OUTER_NORTHWEST	9
5	GND_INNER_NORTHWEST	GND_INNER_NORTHWEST	9
5	GND_OUTER_NORTHWEST	GND_OUTER_NORTHWEST	9
5	GND_INNER_NORTHWEST	GND_INNER_NORTHWEST	9
5	GND_OUTER_TAB_6	GND_OUTER_TAB_6	9
5	GND_OUTER_TAB_6	GND_OUTER_TAB_6	9
5	GND_OUTER_TAB_6	GND_OUTER_TAB_6	9
5	GND_INNER_NORTHWEST	GND_INNER_NORTHWEST	9
5	GND_OUTER_TAB_6	GND_OUTER_TAB_6	9
5	GND_INNER_NORTHWEST	GND_INNER_NORTHWEST	9
5	GND_OUTER_TAB_6	GND_OUTER_TAB_6	9
5	GND_INNER_NORTHWEST	GND_INNER_NORTHWEST	9
5	GND_OUTER_TAB_6	GND_OUTER_TAB_6	9
5	ANALOG_UAT_2G_FEM_TO_WLAN_PDTs	ANALOG_UAT_2G_FEM_TO_WLAN_PDTs	9
5	GND_OUTER_TAB_6	GND_OUTER_TAB_6	9
5	GND_INNER_NORTHWEST	GND_INNER_NORTHWEST	9
5	GND_OUTER_TAB_6	GND_OUTER_TAB_6	9
5	GND_INNER_NORTHWEST	GND_INNER_NORTHWEST	9
5	GND_OUTER_NORTH	GND_OUTER_NORTH	9
5	GND_INNER_NORTHWEST	GND_INNER_NORTHWEST	9
5	GND_OUTER_NORTH	GND_OUTER_NORTH	9
5	50_TRX_CPLR_U_CPL2_ANT_LMB_MB_HB	50_TRX_CPLR_U_CPL2_ANT_LMB_MB_HB	9
5	GND_OUTER_NORTH	GND_OUTER_NORTH	9
5	50_TRX_ANT8_UHB_N79	50_TRX_ANT8_UHB_N79	9
5	GND_OUTER_NORTH	GND_OUTER_NORTH	9
5	GND_OUTER_NORTH	GND_OUTER_NORTH	9
5	GND_INNER_NORTHWEST	GND_INNER_NORTHWEST	9
5	GND_OUTER_NORTH	GND_OUTER_NORTH	9
5	GND_INNER_NORTHWEST	GND_INNER_NORTHWEST	9
5	GPIO_AP_CANARY1_PIN_591	GPIO_AP_CANARY1_PIN_591	9
5	GND_INNER_NORTHWEST	GND_INNER_NORTHWEST	9
5	GND_NORTHWEST	GND_NORTHWEST	9
5	GPIO_AP_CANARY1_PIN_594	GPIO_AP_CANARY1_PIN_594	9
5	GND_NORTHWEST	GND_NORTHWEST	9
5	GND_OUTER_NORTH	GND_OUTER_NORTH	9
5	GND_NORTHWEST	GND_NORTHWEST	9
5	GND_OUTER_NORTH	GND_OUTER_NORTH	9
5	GND_NORTHWEST	GND_NORTHWEST	9
5	GND_OUTER_NORTH	GND_OUTER_NORTH	9
5	GND_INNER_NORTHWEST	GND_INNER_NORTHWEST	9
5	GND_NORTHWEST	GND_NORTHWEST	9
5	GND_INNER_NORTHWEST	GND_INNER_NORTHWEST	9
5	GND_NORTHWEST	GND_NORTHWEST	9
5	GND_INNER_NORTHWEST	GND_INNER_NORTHWEST	9
5	GND_NORTHWEST	GND_NORTHWEST	9
5	GPIO_AP_CANARY1_PIN_605	GPIO_AP_CANARY1_PIN_605	9
5	GND_INNER_NORTHWEST	GND_INNER_NORTHWEST	9
5	GND_OUTER_NORTH	GND_OUTER_NORTH	9
5	GND_INNER_NORTHWEST	GND_INNER_NORTHWEST	9
5	GND_OUTER_NORTH	GND_OUTER_NORTH	9
5	GND_INNER_NORTHWEST	GND_INNER_NORTHWEST	9
5	GND_OUTER_NORTH	GND_OUTER_NORTH	9
5	GND_INNER_NORTHWEST	GND_INNER_NORTHWEST	9
5	GND_OUTER_NORTH	GND_OUTER_NORTH	9
5	GPIO_AP_CANARY1_PIN_613	GPIO_AP_CANARY1_PIN_613	9
5	PP_LDO6_VIO_1V8_NORTH	PP_LDO6_VIO_1V8_NORTH	9
5	GND_OUTER_NORTH	GND_OUTER_NORTH	9
5	GPIO_BB_TO_R1_COEX	GPIO_BB_TO_R1_COEX	9
5	GND_OUTER_NORTH	GND_OUTER_NORTH	9
5	IO_BT_TO_R1_TIME_SYNC	IO_BT_TO_R1_TIME_SYNC	9
5	GND_OUTER_NORTH	GND_OUTER_NORTH	9
5	GPIO_WLAN_TO_R1_COEX	GPIO_WLAN_TO_R1_COEX	9
5	GND_OUTER_NORTH	GND_OUTER_NORTH	9
5	GND_INNER_NORTHEAST	GND_INNER_NORTHEAST	9
5	GND_OUTER_NORTH	GND_OUTER_NORTH	9
5	GND_INNER_NORTHEAST	GND_INNER_NORTHEAST	9
5	50_TRX_ANT2_LB_LMB_MB_HB_L1	50_TRX_ANT2_LB_LMB_MB_HB_L1	9
5	GND_INNER_NORTHEAST	GND_INNER_NORTHEAST	9
5	50_LAA_RX_XCVR_DRX10	50_LAA_RX_XCVR_DRX10	9
5	GND_OUTER_NORTH	GND_OUTER_NORTH	9
5	GND_OUTER_NORTH	GND_OUTER_NORTH	9
5	SHIELD_RFFE5_SCLK_SDR	SHIELD_RFFE5_SCLK_SDR	9

5	GND_OUTER_NORTH	GND_OUTER_NORTH	9
5	SHIELD_RFFE5_SDATa_SDR	SHIELD_RFFE5_SDATa_SDR	9
5	GND_OUTER_NORTH	GND_OUTER_NORTH	9
5	GND_INNER_NORTHEAST	GND_INNER_NORTHEAST	9
5	GND_OUTER_NORTH	GND_OUTER_NORTH	9
5	SHIELD_RFFE_WLAN_TO_UAT_FEMS_CLK	SHIELD_RFFE_WLAN_TO_UAT_FEMS_CLK	9
5	GND_OUTER_NORTH	GND_OUTER_NORTH	9
5	SHIELD_RFFE_WLAN_TO_UAT_FEMS_DATA	SHIELD_RFFE_WLAN_TO_UAT_FEMS_DATA	9
5	GND_OUTER_NORTH	GND_OUTER_NORTH	9
5	GND_INNER_NORTHEAST	GND_INNER_NORTHEAST	9
5	GND_OUTER_NORTH	GND_OUTER_NORTH	9
5	UAT_SUBUS_B	UAT_SUBUS_B	9
5	GND_OUTER_NORTH	GND_OUTER_NORTH	9
5	UAT_SUBUS_A	UAT_SUBUS_A	9
5	GND_OUTER_NORTH	GND_OUTER_NORTH	9
5	GND_INNER_NORTHEAST	GND_INNER_NORTHEAST	9
5	GND_OUTER_NORTH	GND_OUTER_NORTH	9
5	NTC_RADIO_PA_P	NTC_RADIO_PA_P	9
5	GND_OUTER_NORTH	GND_OUTER_NORTH	9
5	UAT_TUNER_TO_R1_STATE	UAT_TUNER_TO_R1_STATE	9
5	GND_OUTER_NORTH	GND_OUTER_NORTH	9
5	GPIO_R1_TO_BB_CH9_COEX	GPIO_R1_TO_BB_CH9_COEX	9
5	GND_OUTER_NORTH	GND_OUTER_NORTH	9
5	GND_INNER_NORTHEAST	GND_INNER_NORTHEAST	9
5	GND_OUTER_NORTH	GND_OUTER_NORTH	9
5	GPIO_R1_TO_WLAN_COEX	GPIO_R1_TO_WLAN_COEX	9
5	GND_OUTER_NORTH	GND_OUTER_NORTH	9
5	PP1V2_S2_2	PP1V2_S2_2	9
5	GND_OUTER_NORTH	GND_OUTER_NORTH	9
5	GND_INNER_NORTHEAST	GND_INNER_NORTHEAST	9
5	GND_OUTER_NORTH	GND_OUTER_NORTH	9
5	GPIO_CAM_PMU1_TO_PVDD_RESET_L	GPIO_CAM_PMU1_TO_PVDD_RESET_L	9
5	GND_OUTER_NORTH	GND_OUTER_NORTH	9
5	I2C4_CAM_PMU1_SCL_1V8	I2C4_CAM_PMU1_SCL_1V8	9
5	GND_OUTER_NORTH	GND_OUTER_NORTH	9
5	GND_INNER_NORTHEAST	GND_INNER_NORTHEAST	9
5	GND_OUTER_NORTH	GND_OUTER_NORTH	9
5	I2C4_CAM_PMU1_SDA_1V8	I2C4_CAM_PMU1_SDA_1V8	9
5	GND_OUTER_NORTH	GND_OUTER_NORTH	9
5	GND_INNER_NORTHEAST	GND_INNER_NORTHEAST	9
5	GND_OUTER_NORTH	GND_OUTER_NORTH	9
5	GND_OUTER_NORTH	GND_OUTER_NORTH	9
5	GND_OUTER_NORTH	GND_OUTER_NORTH	9
5	GPIO_SCM_AOP_FROM_IMU_DATARDY	GPIO_SCM_AOP_FROM_IMU_DATARDY	9
5	GPIO_AP_CANARY1_PIN_676	GPIO_AP_CANARY1_PIN_676	9
5	GPIO_AP_CANARY1_PIN_677	GPIO_AP_CANARY1_PIN_677	9
5	SPI0_AOP_FROM_IMU_R1_MISO	SPI0_AOP_FROM_IMU_R1_MISO	9
5	PP_STROBE_DRIVER_COOL_LED_1	PP_STROBE_DRIVER_COOL_LED_1	9
5	GND_OUTER_NORTH	GND_OUTER_NORTH	9
5	GPIO_SCM_AOP_TO_IMU_SPI_CS_L	GPIO_SCM_AOP_TO_IMU_SPI_CS_L	9
5	PP_STROBE_DRIVER_COOL_LED_2	PP_STROBE_DRIVER_COOL_LED_2	9
5	GPIO_AP_CANARY1_PIN_683	GPIO_AP_CANARY1_PIN_683	9
5	GND_OUTER_NORTH	GND_OUTER_NORTH	9
5	GND_INNER_NORTHEAST	GND_INNER_NORTHEAST	9
5	GND_OUTER_NORTH	GND_OUTER_NORTH	9
5	SPI0_AOP_TO_IMU_R1_MOSI	SPI0_AOP_TO_IMU_R1_MOSI	9
5	PP_STROBE_DRIVER_WARM_LED_1	PP_STROBE_DRIVER_WARM_LED_1	9
5	GPIO_AP_CANARY1_PIN_689	GPIO_AP_CANARY1_PIN_689	9
5	SPI0_AOP_TO_IMU_R1_SCLK	SPI0_AOP_TO_IMU_R1_SCLK	9
5	PP_STROBE_DRIVER_WARM_LED_2	PP_STROBE_DRIVER_WARM_LED_2	9
5	GND_OUTER_NORTH	GND_OUTER_NORTH	9
5	GND_OUTER_NORTH	GND_OUTER_NORTH	9
5	GND_INNER_NORTHEAST	GND_INNER_NORTHEAST	9
5	GPIO_AP_CANARY1_PIN_695	GPIO_AP_CANARY1_PIN_695	9
5	GND_INNER_NORTHEAST	GND_INNER_NORTHEAST	9
5	GND_OUTER_NORTH	GND_OUTER_NORTH	9
5	GND_INNER_NORTHEAST	GND_INNER_NORTHEAST	9
5	GND_OUTER_NORTH	GND_OUTER_NORTH	9
5	GND_INNER_NORTHEAST	GND_INNER_NORTHEAST	9
5	GND_OUTER_NORTH	GND_OUTER_NORTH	9
5	GND_INNER_NORTHEAST	GND_INNER_NORTHEAST	9
5	GND_OUTER_NORTH	GND_OUTER_NORTH	9
5	GPIO_AP_CANARY6_PIN_708	GPIO_AP_CANARY6_PIN_708	9
5	GND_OUTER_NORTH	GND_OUTER_NORTH	9
5	GND_NORTHEAST	GND_NORTHEAST	9
5	GND_INNER_NORTHEAST	GND_INNER_NORTHEAST	9
5	GPIO_AP_CANARY6_PIN_713	GPIO_AP_CANARY6_PIN_713	9
5	GND_INNER_NORTHEAST	GND_INNER_NORTHEAST	9
5	GND_NORTHEAST	GND_NORTHEAST	9
5	GND_INNER_NORTHEAST	GND_INNER_NORTHEAST	9
5	GPIO_AP_CANARY6_PIN_717	GPIO_AP_CANARY6_PIN_717	9
5	GND_INNER_NORTHEAST	GND_INNER_NORTHEAST	9
5	GND_NORTHEAST	GND_NORTHEAST	9
5	GPIO_AP_CANARY6_PIN_720	GPIO_AP_CANARY6_PIN_720	9
5	GPIO_AP_CANARY6_PIN_721	GPIO_AP_CANARY6_PIN_721	9
5	GND_NORTHEAST	GND_NORTHEAST	9
5	GND_NORTHEAST	GND_NORTHEAST	9
5	GND_INNER_NORTHEAST	GND_INNER_NORTHEAST	9
5	GND_INNER_NORTHEAST	GND_INNER_NORTHEAST	9
5	GND_INNER_NORTHWEST	GND_INNER_NORTHWEST	9
5	GND_INNER_NORTHWEST	GND_INNER_NORTHWEST	9

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